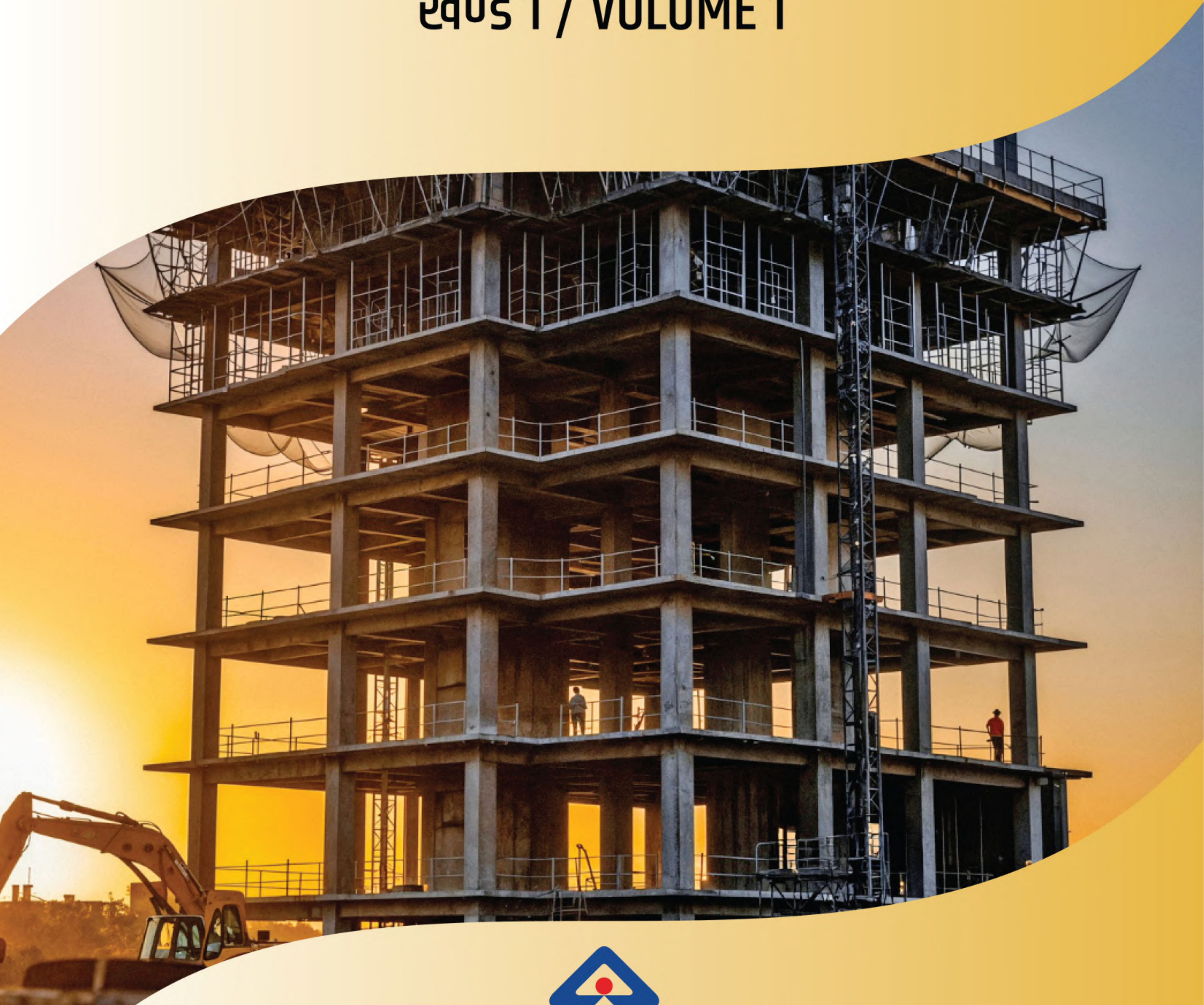
SP 7 : 2026

राष्ट्रीय भवन निर्माण मानक 2026
NATIONAL BUILDING CONSTRUCTION STANDARDS 2026

खण्ड 1 / VOLUME 1



भारतीय मानक ब्यूरो
BUREAU OF INDIAN STANDARDS

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खण्ड 1

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VOLUME 1

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BUREAU OF INDIAN STANDARDS

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National Building Code Sectional Committee, CED 46

FOREWORD

The Planning Commission, soon after the Third Five Year Plan, decided that the whole gamut of operations involved in construction, such as administrative, organizational, financial and technical aspects, be studied in depth. For this study, a Panel of Experts was appointed in 1965 by the Planning Commission and its recommendations are available in the 'Report on Economies in Construction Costs', published in 1968.

This study resulted in a recommendation that a National Building Code (NBC) be prepared to unify the building regulations throughout the country for use by government departments, municipal bodies and other construction agencies. The then Indian Standards Institution (now Bureau of Indian Standards) was entrusted by the Planning Commission with the responsibility of preparation of the National Building Code. Accordingly, Bureau of Indian Standard (then ISI) brought out first NBC in 1970. The same was subsequently revised in 1983, 2005 and 2016.

The third revision of NBC was published in 2016. This was a result of large scale changes in the building construction activities, such as change in nature of occupancies with prevalence of high rises and mixed occupancies, greater dependence and complicated nature of building services, development of new/innovative construction materials and technologies, greater need for preservation of environment and recognition of need for planned management of existing buildings and built environment, resulting in a paradigm shift in building construction scenario. A comprehensive revision was therefore brought out to address all these aspects and also reflect the changes incorporated in various standards which were considerably utilized in the 2016 revision.

Since the last revision, many things have changed, including per capita land availability, growing infrastructure requirements, socio-economic conditions, and technological advancements, to name a few. Hence, in these changing times, the requirements for development have also changed. The faster economic growth that India needs at present is possible if the ongoing reforms help the infrastructure sector to operate efficiently and compete cost-effectively by utilizing the available land optimally and reducing excessive compliance burden.

Based on the current realities, socio-economic scenario existing in the country, and the requirement of rapid urbanization and income growth, the total building stock is projected to increase exponentially in the coming years. Further, construction and infrastructure development are poised to determine the speed and scale of progress across every sector. These enable mobility, clean water, energy, and urban growth, which all directly contribute and shape economic opportunity and quality of life.

Construction, being a state subject, the contribution of the state government and local authorities are significant in infrastructure growth of the country. However, the land resources are limited. Maximum utilization of land resources is of prime importance in the current scenario. Further, India has come a long way from the time of first publication of the building code to the present day status. The requirements of standards and codes have also changed from a prescriptive regime, where the States and Local Authorities required hand holding, to now a more performance oriented outlook giving ample scope for innovation and decision making. The knowledge base in the country has also increased exponentially with exposure to the global best practices and adapting them to suit the country specific needs. Consequently, the contents of the current version, in which this publication has been renamed as the National Building Construction Standards (NBCS), have undergone substantial changes from those of the earlier NBC.

Thus in the current (fifth) version, this NBCS has been restructured substantially with a more safety and performance oriented approach rather than a prescriptive one. This includes leaving the adminstrative provisions land use requirements, like floor area ratio (FAR), development control rules (DCRs) and parking norms, which falls under the domain of local and state government, from the ambit of this NBCS. Open spaces, ground coverage, space utilization and land use related matters are primarily the state subject, and as such, these items have also not been included in this NBCS. As regards, other aspects covering construction practices, landscaping, sustainability aspects of built environment and asset and facility management, which are not directly linked to the design requirements vis-a-vis safety of the built environment have been taken out of this NBCS and will be published separately as guidelines of good practices mainly addressing management, sustainability, maintenance and practices to improve the working and efficiency of the built environment. The portions covering building materials, structural design aspects, building services and plumbing services are retained as part of this NBCS.

The fire and life safety aspect, a state subject, but due to its importance towards the safety and well-being of the occupants as well as the common public, has also been included in this NBCS.

The current version of this NBCS with the restructuring is as follows:

a) The current version consists of

Foreword

Part A Introduction

Part B Building Materials

Part C Structural Design

Part D Building Services

Part E Plumbing Services

Part F Fire and Life Safety

Index of Terms

b) The other Parts of NBC 2016, namely Part 7 'Construction Management, Practices and Safety'; Part 10 'Landscape Development, Signs and Outdoor Display Structures'; Part 11 'Approach to Sustainability'; and Part 12 'Asset and Facility Management', have also been taken out of this NBCS. However, these will be brought out separately as special publication in the form of handbooks of good practices.

Towards making this NBCS more comprehensive, forward-looking and easy to implement, various technical aspects have been introduced apart from updating the existing provisions. The key points that have been addressed in this version of NBCS are:

1) Innovative building materials.
2) Review of blast-resistant design of buildings.
3) Detailed review of ground improvement techniques and inclusion of combined piled-raft foundations.
4) Inclusion of design parameters of new and imported timber species.
5) Construction methodologies using various bamboo species.
6) Inclusion of earthquake resistant systems of masonry.
7) Inclusion of the structural safety aspect of tall concrete buildings.
8) Updating the provisions on structural steel based design; and inclusion of a new section on composite construction.
9) Detailed provisions on glazing systems covering therein bullet-resistant glass and blast-resistant glass.
10) Charging infrastructure for electric vehicles.
11) Inclusion of provisions relating to renewable energy sources for buildings, such as rooftop solar photovoltaic (PV) systems, building-integrated photovoltaic (BIPV) systems, and solar water heating systems.
12) Inclusion of new section on power-driven parking systems for vehicles in buildings.
13) Focus on water-efficient fixtures in buildings; bio-digesters and packaged sewage treatment plants or decentralized STPs.

Notwithstanding the above, the repeated onslaught of natural calamities such as earthquakes, cyclones, landslides, floods, faced by the country has brought home the ever-increasing need for a common guiding document to enable monitoring and implementation of sound technical provisions and comprehensive features built into such a document. This NBCS fulfills this need apart from bringing at one place comprehensive provisions relating to selection of building materials, resilient design principles, various building services and plumbing services, etc, all ensuring to follow a unified approach thereby promoting resource efficiency, environmental responsibility and long-term resilience. This NBCS also captures the global best practices and technological developments in setting a robust roadmap towards establishing a more efficient mechanism promoting ease of doing business.

As the regulation of land development and buildings is a State subject and is dealt by the States and local bodies such as urban (and rural) local bodies within their jurisdiction through building regulatory documents like building regulations, development control rules building byelaws, and fire regulations, this document is voluntary in nature and is non-binding. The implementation depends on adoption or adaptation by concerned parties or stipulations in a contract or by appropriate adoption by the concerned authorities.

COMMITTEE COMPOSITION

National Building Code Sectional Committee, CED 46

Chairperson

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Panel for Fire and Life Safety, CED 46:P2

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Panel for Building Materials, CED 46:P3

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AEON Design & Development LLP, Noida	MS SHEETAL RAKHEJA
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Builders' Association of India, Mumbai	DR KAARTHIK MANOHARAN
CSIR-Central Building Research Institute, Roorkee	DR NEERAJ JAIN SHRI RAJESH KUMAR (*Alternate*)
Central Public Works Department, New Delhi	SUPERINTENDING ENGINEER, DC-5 EXECUTIVE ENGINEER, CD-14 (*Alternate*)
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Delhi Development Authority, New Delhi	SHRI SANJAY KUMAR KHARE SHRI NARENDER KUMAR GOEL (*Alternate*)
Directorate General Border Roads, New Delhi	REPRESENTATIVE
Hindalco Industries Ltd, Mumbai	SHRI ANJAN PUTATUNDA SHRI AMRIT RAJ THAKUR (*Alternate*)

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Panel for Loads, Forces and Effects, CED 46:P4

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Panel for Soils and Foundations, CED 46:P5

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Indian Association of Structural Engineers, New Delhi	SHRI SUSHIL K. DHAWAN DR ABHAY GUPTA (*Alternate* I) DR ANIL JOSEPH (*Alternate* II)
Indian Geotechnical Society, New Delhi	DR A. P. SINGH DR NEELIMA SATYAM (*Alternate* I) DR ADITYA SINGH (*Alternate* II)
Indian Institute of Technology Delhi, New Delhi	DR J. T. SHAHU DR BAPPADITYA MANNA (*Alternate*)
Larsen & Toubro Limited, Chennai	DR G. D. RAJU SHRI VISAKAN R. (*Alternate*)
Military Engineer Services, Engineer-in-Chief's Branch, Army HQ, New Delhi	SHRI AJAY KUMAR SINGH SHRI AMAN GARG (*Alternate*)
NTPC Ltd, New Delhi	SHRI MANI BHUSHAN SHRI DEVENDER SINGH (*Alternate*)
National Capital Region Transport Corporation, New Delhi	SHRI JITENDRA KUMAR
National Council for Cement and Building Materials, Ballabgarh	SHRI P. N. OJHA SHRI AMIT TRIVEDI (*Alternate*)
National Institute of Disaster Management, New Delhi	DR AMIR ALI KHAN
The Institution of Engineers (India), Kolkata	ER VIJAY SHANKAR DR ALOK KUMAR (*Alternate*)
In Personal Capacity (*H-105, Casagrand Aristo, No. 5, Noble Street, Alandur, Chennai – 600016*)	SHRI A. VIJAYARAMAN

Panel for Timber and Bamboo, CED 46:P6

Organization	*Representative*(*s*)
In Personal Capacity [*Pratap Nursery Lane, (Near Gurudwara Sahib), Panditwari, Dehradun – 248007*]	SHRI K. S. PRUTHI (***Convener***)
Bamboo Society of India, Bengaluru	SHRI PUNATI SRIDHAR SHRI K. N. MURTHY (*Alternate*)

Organization	*Representative*(*s*)
Building Materials and Technology Promotion Council, New Delhi	DR SHAILESH KR AGRAWAL SHRI C. N. JHA (*Alternate*)
CEPT University, Ahmedabad	DR DHARA SHAH
CSIR-Central Building Research Institute, Roorkee	DR RAJESH KUMAR VERMA DR NEERAJ JAIN (*Alternate* I) SHRI PRASANTA KAR (*Alternate* II)
Central Public Works Department, New Delhi	SUPERINTENDING ENGINEER, DELHI CIRCLE-1 EXECUTIVE ENGINEER, L- DIV (*Alternate*)
Centre for Green Building Materials & Technology, Bengaluru	MS NEELAM MANJUNATH
Creative Design Consultants & Engineers Pvt Ltd, Greater Noida	SHRI AMAN DEEP
Forest Research Institute, Dehradun	SHRI RAJESH BHANDARI SHRI R. S. TOPWAL (*Alternate* I) SHRI ASHWATH HEGDE (*Alternate* II)
Forum of Scientists, Engineers and Technologists, Kolkata	SHRI RABI MUKHOPADHYAY DR PARTHASARATHI MUKHOPADHYAY (*Alternate*)
Housing and Urban Development Corporation Ltd, New Delhi	DR DEEPAK BANSAL MS POOJA NANDY (*Alternate* I) SHRIMATI NIRMAL BATRA (*Alternate* II)
Indian Association of Structural Engineers, New Delhi	DR DULAL GOLDAR DR VISALAKSHI TALAKOKULA (*Alternate*)
Indian Institute of Technology Delhi, New Delhi	DR SURESH BHALLA
ICFRE – Institute of Wood Science and Technology, Bengaluru	SHRI ANAND NANDANWAR SHRI AMITAVA SIL (*Alternate* I) DR NARASIMHAMURTHY (*Alternate* II)
Military Engineer Services, Engineer-in-Chief's Branch, Army HQ, New Delhi	SHRIMATI S. GAYATRI MUKHERJEE SHRI SHEKHAR GUPTA (*Alternate*)
Mutha Industries Pvt Ltd, Mumbai	SHRI ANIL MUTHA SHRI NEERAJ MUTHA (*Alternate* I) SHRI SUDIP CHAKRABORTY (*Alternate* II)
North East Centre for Technology Application and Reach, Shillong	SHRI KRISHNA KUMAR MS LYNGKSIAR KHONGWIR (*Alternate*)
The Institution of Engineers (India), Kolkata	SHRI MANOJ KUMAR SINGH ER KUMAR ARVIND (*Alternate*)
Windsor Wood (India) Pvt Ltd, Mumbai	SHRI PRAKASH NARMADASHANKAR SUTHAR
Wonder Grass Initiative Pvt Ltd, Nagpur	SHRI VAIBHAV KALEY
In Personal Capacity (*103/II*, *Vasant Vihar*, *P. O. New Forest*, *Dehradun – 248006*)	SHRI S. S. RAJPUT

Panel for Masonry, CED 46:P7

Organization	*Representative*(*s*)
In Personal Capacity (*131*, *Sector11D*, *Faridabad — 121006*)	SHRI V. V. ARORA (***Convener***)
Adlakha Associates Pvt Ltd, New Delhi	SHRI P. K. ADLAKHA SHRI V. K. SETHI (*Alternate*)

Organization	*Representative*(*s*)
Association of Consulting Civil Engineers (India), Bengaluru	SHRI UMESH B. RAO DR RAGHUNATH S. (*Alternate*)
Bhabha Atomic Research Centre, Trombay	SHRI K. SRINIVAS SHRI H. E. IYER (*Alternate*)
Building Materials & Technology Promotion Council, New Delhi	DR SHAILESH KR AGRAWAL SHRI PANKAJ GUPTA (*Alternate*)
CSIR-Central Building Research Institute, Roorkee	DR AJAY CHOURASIA MS HINA GUPTA GHOSH (*Alternate*)
CSIR-Structural Engineering Research Centre, Chennai	DR A. RAMACHANDRA MURTHY DR S. R. BALASUBRAMANIAN (*Alternate* I) SHRI FARVAZE AHMED A. K. (*Alternate* II)
Central Public Works Department, New Delhi	SUPERINTENDING ENGINEER & PROJECT DIRECTOR, DPC-1 EE CUM SM (CIVIL)-1 (*Alternate*)
Indian Association of Structural Engineers, New Delhi	DR DEEPAK BANSAL MS SANGEETA WIJ (*Alternate*)
Indian Institute of Science, Bengaluru	DR B. V. VENKATARAMA REDDY DR K. S. NANJUNDA RAO (*Alternate*)
Indian Institute of Technology Delhi, New Delhi	DR DIPTI RANJAN SAHOO DR SRINIVAS MOGILI (*Alternate*)
Indian Institute of Technology Guwahati, Guwahati	DR HEMANT B. KAUSHIK
Indian Institute of Technology Hyderabad, Kandi	DR KOLLURU V. L. SUBRAMANIAM
Indian Institute of Technology Madras, Chennai	DR ARUN MENON
Indian Institute of Technology Patna, Patna	DR VAIBHAV SINGHAL
Military Engineer Services, Engineer-in-Chief's Branch, Army HQ, New Delhi	SHRI SAMEER PANDEY SHRI M. V. SHANKAR (*Alternate*)
National Council for Cement and Building Materials, Ballabgarh	SHRI AMIT TRIVEDI SHRI AMIT PRAKASH (*Alternate*)
The Institution of Engineers (India), Kolkata	SHRI SATYA PRAKASH SHRI MASARRAT NOOR KHAN (*Alternate*)

Panel for Plain, Reinforced and Prestressed Concrete, CED 46:P8

Organization	*Representative*(*s*)
In Personal Capacity (*Grace Villa*, *Kadamankulam PO*, *Thiruvalla – 689583*)	SHRI JOSE KURIAN **(*Convener*)**
Association of Consulting Civil Engineers (India), Bengaluru	SHRI M. S. SUDHARSHAN
Central Public Works Department, New Delhi	CHIEF ENGINEER (CENTRAL VISTA PZ-1) SE, CPVC-5 (*Alternate*)
Creative Design Consultants & Engineers Pvt Ltd, Greater Noida	SHRI AMAN DEEP
CSIR-Central Building Research Institute, Roorkee	DR S. K. SINGH SHRI SUBASH CHANDRA BOSE GURRAM (*Alternate* I) SHRI GOVIND GAURAV (*Alternate* II)
CSIR-Central Road Research Institute, New Delhi	REPRESENTATIVE

Organization	*Representative(s)*
CSIR-Structural Engineering Research Centre, Chennai	DR B. H. BHARATKUMAR DR S. BHASKAR (*Alternate* I) SHRI THONDAMON V. (*Alternate* II)
Engineers India Limited, New Delhi	DR SUDIP PAUL MS RIYA DAS (*Alternate*)
Freyssinet Prestressed Concrete Company Limited, Mumbai	SHRI P. Y. MANJURE
Hindustan Construction Company Limited, Mumbai	SHRI SHAILESH THATTE
Indian Association of Structural Engineers, New Delhi	SHRI RAJEEV GOEL DR ABHAY GUPTA (*Alternate*)
Indian Concrete Institute, Chennai	SHRI A. K. JAIN SHRI K. P. ABRAHAM (*Alternate*)
Indian Institute of Technology Madras, Chennai	DR C. V. R. MURTY DR AMLAN K. SENGUPTA
Larsen & Toubro Limited, Chennai	SHRI V. SADASIVAM SHRI STHALADIPTI SAHA (*Alternate* I) SHRI J. RAJESH (*Alternate* II)
Military Engineer Services, Engineer-in-Chief's Branch, Army HQ, New Delhi	SHRI PRAMOD KUMAR SINGH SHRI NIRAJ KUMAR (*Alternate*)
Ministry of Road Transport and Highways, New Delhi	CHIEF ENGINEER, STANDARDS & RESEARCH (BRIDGES)
National Council for Cement and Building Materials, Ballabgarh	SHRI P. N. OJHA DR BRIJESH SINGH (*Alternate*)
Research, Designs and Standards Organization (Ministry of Railways), Lucknow	SHRI V. K. PANDEY SHRI SANDEEP SINGH (*Alternate*)
Tandon Consultants Pvt Ltd, New Delhi	SHRI MAHESH TANDON
The Institution of Engineers (India), Kolkata	SHRI MANOJ KUMAR SINGH DR M. A. CHAKRABARTI (*Alternate*)
VMS Consultants Private Limited, Mumbai	MS ALPA R. SHETH
In Personal Capacity (*131, Sector 11D, Faridabad – 121006*)	SHRI V. V. ARORA
In Personal Capacity (*M1F1 VGN Minerva Apartments, Guruswamy Road, Nolambur, Chennai – 600095*)	DR C. RAJKUMAR

Panel for Steel, CED 46:P9

Organization	*Representative(s)*
In Personal Capacity (*II, 2A, Rani Meyyammai Towers, MRC Nagar, R A Puram, Chennai – 600028*)	DR V. KALYANARAMAN ***(Convener)***
Association of Consulting Civil Engineers (India), Bengaluru	SHRI MANOJ KAWALKAR SHRI RAJKUMAR KACHARLA (*Alternate*)

Organization	*Representative*(*s*)
Central Public Works Department, New Delhi	CHIEF ENGINEER & ED, (LLT DELHI) ZONE EE CUM SM (CIVIL)-1 (*Alternate*)
Creative Design Consultants & Engineers Pvt Ltd, Greater Noida	SHRI AMAN DEEP
CSIR-Central Building Research Institute, Roorkee	DR AJAY CHOURASIA DR R. SIVA CHIDAMBARAM (*Alternate* I) DR CHANCHAL SONKAR (*Alternate* II)
CSIR-Structural Engineering Research Centre, Chennai	DR G. S. PALANI DR NAPA PRASAD RAO (*Alternate* I) DR M. SARVANAN (*Alternate* II)
Engineers India Limited, New Delhi	SHRIMATI PAPIA MANDAL SHRI CHANDRA SHEKHAR SHARMA (*Alternate* I) SHRI SAPTADIP SARKAR (*Alternate* II)
Indian Association of Structural Engineers, New Delhi	DR HARSHAVARDHAN SUBBARAO DR ABHAY GUPTA (*Alternate*)
Indian Institute of Technology Madras, Chennai	DR S. ARUL JAYACHANDRAN DR P. S. LAKSHMI PRIYA (*Alternate*)
Institute for Steel Development and Growth, Kolkata	SHRI PYDI LAKSHMANA RAO SHRI ARIJIT GUHA (*Alternate* I) SHRI M. M. GHOSH (*Alternate* II)
Interarch Building Products Pvt Ltd, Noida	SHRI GAUTAM SURI SHRI NAVAZ MANIKKAL (*Alternate* I) SHRI SUNIL PULIKKAL (*Alternate* II)
Jindal Steel and Power Limited, New Delhi	SHRI SANJAY NANDANWAR SHRI BIJU MAHIMA (*Alternate*)
Kirby Building Systems India Limited, New Delhi	DR PADMAJA GOKARAJU SHRI VENUMADHAVA REDDY (*Alternate*)
Larsen & Toubro Limited, Chennai	SHRI T. VENKATESH RAO
MECON Limited, Ranchi	SHRI A. KRISHNA RAO SHRI C. KRISHNAM RAJU (*Alternate*)
Military Engineer Services, Engineer-in-Chief's Branch, Army HQ, New Delhi	SHRI M. V. SHANKAR BRIG Y. K. AHUJA (*Alternate*)
PEB Manufacturers' Association, Navi Mumbai	SHRI MANISH GARG
Research, Designs and Standards Organization (Ministry of Railways), Lucknow	SHRI RAJESH KUMAR SRIVASTAVA SHRI SRIJAN TRIPATHI (*Alternate*)
Tata Steel Ltd, Jamshedpur	SHRI HARIHARAPUTHIRAN H.
The Institution of Engineers (India), Kolkata	DR S. SENTHIL SELVAN DR P. R. KANNAN RAJKUMAR (*Alternate*)

Panel for Prefabrication and Systems Buildings, CED 46:P10

Organization	*Representative*(*s*)
Larsen & Toubro Ltd, Chennai	SHRI K. SENOU **(*Convener*)**

Organization	*Representative(s)*
Adlakha Associates Pvt Ltd, New Delhi	SHRI P. K. ADLAKHA
Anna University, Chennai	DR K. P. JAYA
B.G. Shirke Construction Technology Pvt Ltd, Pune	SHRI YOGESH P. KAJALE COL SANJAY M. ADSAR (*Alternate*)
Building Materials and Technology Promotion Council, New Delhi	DR SHAILESH KUMAR AGRAWAL SHRI C. N. JHA (*Alternate*)
Central Public Works Department, New Delhi	CHIEF ENGINEER CUM ED, RPZ EE CUM SM (CIVIL)-1 (*Alternate*)
Creative Design Consultants & Engineers Pvt Ltd, Greater Noida	SHRI AMAN DEEP
CSIR-Central Building Research Institute, Roorkee	DR AJAY CHAURASIA SHRI ASHISH KAPOOR (*Alternate* I) SHRI SIDDHARTH BEHERA (*Alternate* II)
CSIR-Structural Engineering Research Centre, Chennai	DR J. PRABAKAR DR K. N. LAKSHMI KANDHAN (*Alternate* I) SHRI A. M. SARATH (*Alternate* II)
Indian Institute of Technology Hyderabad, Kandi	DR SURIYA PRAKASH S.
Indian Institute of Technology Madras, Chennai	DR RADHAKRISHNA G. PILLAI DR NIKHIL BUGALIA (*Alternate*)
Indian Association of Structural Engineers, New Delhi	SHRI MAHESH TANDON SHRI ALOK BHOWMICK (*Alternate*)
Innovela Building Solutions, Pune	SHRI SAURABH PURANDARE
Interarch Building Products Pvt Ltd, Noida	SHRI GAUTAM SURI SHRI MANISH KUMAR GARG (*Alternate*)
Kirby Building Systems India Ltd, Hyderabad	DR PADMAJA GOKARAJU SHRI G. VENUMADHAVA REDDY (*Alternate*)
Larsen & Toubro Ltd, Chennai	SHRI C. K. SOMARAJU SHRI AMIT BARDE (*Alternate* I) SHRI APARUP BISWAL (*Alternate* II)
Lloyd Insulations (India) Ltd, New Delhi	SHRI AJAY SINGH SHRI JITENDRA KUMAR JAIN (*Alternate*)
Military Engineer Services, Engineer-in-Chief's Branch, Army HQ, New Delhi	BRIG Y. K. AHUJA COL DEEPAK SINGH (*Alternate*)
National Council for Cement and Building Materials, Ballabgarh	SHRI AMIT TRIVEDI DR B. P. RANGA RAO (*Alternate*)
PEB Manufacturers' Association, Navi Mumbai	REPRESENTATIVE
Precast India, Pune	SHRI AJIT VASANT BHATE SHRI KAPILESH BHATE (*Alternate* I) DR K. SHANTHARAJU (*Alternate* II)
Precision Precast Solutions, Pune	SHRI DHANANJAY BHOSALE SHRI RAJENDRA R. KULKARNI (*Alternate*)
Tata Bluescope Steel Ltd, Pune	SHRI KRISHNAKANT RANE

Organization	*Representative(s)*
The Indian Institute of Architects, Mumbai	SHRI CHAMARTHI RAJENDRA RAJU SHRI KURIAN GEORGE (*Alternate* I) SHRIMATI VIBHA SHRIVASTAVA (*Alternate* II)
The Institution of Engineers (India), Kolkata	DR R. K. BHANDARI SHRI HEMANT S. VADALKAR (*Alternate*)
TRC Engineering India Pvt Ltd, Bengaluru	SHRI ARJUN MUDAMBI
In Personal Capacity (*132*, *Marigold*, *Serene County*, *Telecom Nagar*, *Gachibowli – 500032*)	SHRI M. P. NAIDU
In Personal Capacity (*303*, *Tower 4*, *Lotus Panache*, *Sector 110*, *Noida – 201304*)	SHRI J. K. PRASAD

Panel for Lighting and Natural Ventilation, CED 46:P12

Organization	*Representative(s)*
CEPT University, Ahmedabad	DR RAJAN RAWAL (***Convener***)
AB Initio, Thane	SHRI SHIRISH DESHPANDE
Ashok B Lall Architects, New Delhi	SHRI ASHOK B. LALL MS GIRISHA SETHI (*Alternate*)
Bureau of Energy Efficiency, Ministry of Power, New Delhi	MS PRAVATANALINI SAMAL SHRI MAYANKRAJ PRAJAPAT (*Alternate*)
Central Public Works Department, New Delhi	CHIEF ENGINEER (T&R)-II SE (T)-ELECT (*Alternate*)
CEPT University, Ahmedabad	DR YASH SHUKLA DR MINU AGARWAL (*Alternate*)
CSIR-Central Building Research Institute, Roorkee	SHRI NAGESH BABU BALAM SHRI NAVEEN NISHANT (*Alternate* I) SHRI ANUP KUMAR PRASAD (*Alternate* II)
CSIR-National Physical Laboratory, New Delhi	DR PARAG SHARMA SHRI V. K. JAISWAL (*Alternate*)
Directorate General Factory Advice Service and Labour Institutes, Mumbai	SHRI VIPUL MISHRA SHRI SOURISH DATTA CHOUDHURY (*Alternate*)
Edspring Technologies, Noida	SHRI ANKUR KHANDELWAL
Indian Institute of Technology Delhi, New Delhi	DR DIBAKAR RAKSHIT
Indian Institute of Technology Madras, Chennai	DR BENNY RAPHAEL DR SHANMUGARAJ S. (*Alternate*)
Indian Institute of Technology Roorkee, Roorkee	DR E. RAJASEKAR
Indian Society for Lighting Engineers, New Delhi	DR PRAKASH BARJATIA DR GIRJA SHANKAR (*Alternate* I) DR S. M. ALI (*Alternate* II)
Kaleidoscope, Noida	SHRI AMOR KOOL
Larsen & Toubro Ltd (Sustainability Group), Chennai	DR S. RAJKUMAR
Military Engineer Services, Engineer-in-Chief's Branch, Army HQ, New Delhi	SHRI V. K. TIWARI SHRI H. K. GUPTA (*Alternate*)

Organization	*Representative(s)*
Ministry of New and Renewable Energy, New Delhi	DR A. K. TRIPATHI SHRI H. C. BORAH (*Alternate*)
School of Planning and Architecture, New Delhi	DR V. K. PAUL DR ANIL DEWAN (*Alternate*)
Signify Innovations India Ltd, Hyderabad	SHRI NITISH POONIA SHRI SHRIKANT DATTATREYA PHANSE (*Alternate* I) MS MEGHA SONI (*Alternate* II)
Space Design Consultants, New Delhi	DR VINOD KUMAR GUPTA
The Indian Institute of Architects, Mumbai	SHRI GYANENDRA SINGH SHEKHAWAT MS RANEE VEDAMUTHU (*Alternate* I) SHRIMATI LEENA NIMBALKAR (*Alternate* II)
The Institution of Engineers (India), Kolkata	SHRI RAM NIWAS RAJPOOT
In Personal Capacity (*QRG Towers, 2D, Sec - 126, Expressway Noida – 201304*)	SHRIMATI SUDHESHNA MUKHOPADHYAY

Panel for Electrical Installations, CED 46:P13

Organization	*Representative(s)*
In Personal Capacity (*C-20, Sector 47, Noida – 201301*)	SHRI N. NAGARAJAN ***(Convener)***
Aeon Consultants, New Delhi	SHRI PUNEET GUPTA SHRI PRAVESH KUMAR PUNDIR (*Alternate*)
Bureau of Energy Efficiency, New Delhi	MS PRAVATANALINI SAMAL SHRI PANKAJ SHARMA (*Alternate*)
Cape Electric Pvt Ltd, Kanchipuram	SHRI S. GOPA KUMAR SHRI NANDU GOPAN (*Alternate*)
Central Electricity Authority, New Delhi	SHRI R. K. VERMA SHRI A. K. RAJPUT (*Alternate*)
Central Public Works Department, New Delhi	CHIEF ENGINEER CSQ (ELECT) SUPERINTENDING ENGINEER (ELECT), CSQ(E) (*Alternate*)
Chief Electrical Inspectorate, Govt of Tamil Nadu, Chennai	SHRI P. PALANI SHRI V. RAMAKRISHNAN (*Alternate*)
DEHN India Pvt Ltd, Gurugram	SHRI SAMIR SHANKAR SHRI MANU MUKUNDAN (*Alternate*)
Delhi Metro Rail Corporation, New Delhi	SHRI DEEPAK KUMAR GUPTA SHRI RAJEEV SAXENA (*Alternate*)
Energy Efficiency Services Limited, New Delhi	SHRI ASHISH SHARMA
Engineers India Ltd, New Delhi	SHRI HARISH KUMAR (*Alternate*)
Forum of Construction Utility Services, Bengaluru	SHRI MAXIM NORBET PINTO
International Copper Alliance, Mumbai	SHRI AMOL KALSEKAR SHRI MANAS KUNDU (*Alternate*)

Organization	*Representative*(*s*)
Larsen & Toubro Ltd, Chennai	SHRI D. MAHESWARAN SHRI K. K. JEMBU KAILAS (*Alternate*)
Maple Engineering Design Services, Bengaluru	SHRI B. S. ASWATHNARAYAN SHRI PRASANNA KUMAR (*Alternate*)
Military Engineer Services, Engineer-in-Chief's Branch, Army HQ, New Delhi	SHRI VIMAL KUMAR TIWARI SHRI ARVIND KUMAR (*Alternate*)
Ministry of New and Renewable Energy, New Delhi	SHRI H. R. KHAN SHRI G. UPADHYAY (*Alternate*)
National Federation of Engineers for Electrical Safety, Chennai	SHRI S. APPAVOO SHRI ABHISHEK SINGH (*Alternate*)
OBO Bettermann, Chennai	DR K. JANAKI RAMAN SHRI ARUL PRAKASH (*Alternate*)
Proion Consultants Private Limited, New Delhi	SHRIMATI SHRUTI GOEL
Public Works Department (Electrical Division), Govt of Maharashtra, Mumbai	SHRI HEMANT SALI
Schneider Electric India Pvt Ltd, Gurugram	SHRI C. S. KORE SHRI DEBDAS GOSWAMI (*Alternate*)
Siemens Ltd, Chennai	SHRI VIJAY WANWE
Solar Energy Corporation of India, New Delhi	SHRI C. K. SINGH
Tata Motors Ltd, Mumbai	SHRI SENTHILNATHAN THANGAVELU
The Institution of Engineers (India), Kolkata	DR G. MADHUSUDANAN DR S. SENTHILKUMAR (*Alternate*)
In Personal Capacity (*Cedar 102, SJR Park Vista Aptts, Haralur Road, Ambalipura, Bengaluru – 560012*)	SHRI J. N. BHAVANI PRASAD
In Personal Capacity (*J-403*, *Second Floor*, *New Rajinder Nagar*, *New Delhi – 110060*)	SHRI CHAITANYA KUMAR VARMA

Panel for Air Conditioning, Heating and Mechanical Ventilation, CED 46:P14

Organization	*Representative*(*s*)
Indian Society of Heating, Refrigerating and Air Conditioning Engineers, New Delhi	SHRI ASHISH RAKHEJA **(*Convener*)**
AEON Integrated Building Design Consultants LLP, Noida	SHRI SAURABH GOYEL SHRI ANURAG JOSHI (*Alternate*)
ASHRAE India Chapter, New Delhi	SHRI INDRAJIT BHATTACHARYA
Blowtech Air Devices Pvt Ltd, Noida	SHRI SUSHIL K. CHOUDHURY
Blue Star Limited, Chennai	SHRI B. THIAGARAJAN SHRI SRINIVAS REDDY (*Alternate*)
Bureau of Energy Efficiency, New Delhi	SHRI SAURABH DIDDI SHRI PANKAJ SHARMA (*Alternate*)

Organization	*Representative(s)*
Caire Consult, Greater Noida	SHRI ANKIT JAIN
Camfil India Pvt Ltd, Gurugram	SHRI ANIL CHOPRA
Carrier Air-conditioning and Refrigeration Ltd, Gurugram	SHRI BIMAL TANDON
CSIR-Central Building Research Institute, Roorkee	DR SHORAB JAIN DR TABISH ALAM (*Alternate* I) DR CHANDAN SWAROOP MEENA (*Alternate* II)
Central Public Works Department, New Delhi	CHIEF ENGINEER, CSQ (ELECT) SE (ELECT) CSQ (E) (*Alternate*)
CEPT University, Ahmedabad	DR RAJAN RAWAL DR YASH SHUKLA (*Alternate*)
Climaveneta Climate Technologies (P) Ltd, Bengaluru	SHRI ANIL DEV SHRI NANTHAGOPALAN T. K. (*Alternate* I) SHRI AKSHAY DEV (*Alternate* II)
Comfort Care System Pvt Ltd, New Delhi	SHRI YOGESH MALHOTRA
Daikin Air-conditioning India Pvt Ltd, Gurgram	SHRI KANWALJEET JAWA SHRI SANJAY GOYAL (*Alternate* I) SHRI GAURAV MEHTANI (*Alternate* II)
Delhi Fire Services, Govt of NCT of Delhi, New Delhi	DIRECTOR SHRI MUKESH VERMA (*Alternate*)
Delhi Metro Rail Corporation, New Delhi	SHRI ASHOK TEWARI SHRI RAJESH JAIN (*Alternate*)
Eskayem Consultants Pvt Ltd, Mumbai	SHRI GOUTHAM PITCHIKALA
Forum of Construction Utility Services, Bengaluru	SHRI K. D. SINGH SHRI RAMCHANDAR RAJU ADDHIPALLI (*Alternate* I) SHRI KAPIL KAPUR (*Alternate* II)
Indian Institute of Technology Delhi, New Delhi	DR MUKESH KHARE
Indian Society for Heating, Refrigeration & Air-Conditioning Engineers, New Delhi	SHRI K. RAMACHANDRAN SHRIMATI NIVEDITA JADHAV (*Alternate*)
International Ground Source Heat Pump Association, India Chapter, New Delhi	SHRI RICHIE MITTAL SHRI MADHUSUDHAN RAPOLE (*Alternate*)
Kanwal Industries Corporation, Noida	SHRI PAWANDEEP SINGH SHRI JAIVEER SINGH (*Alternate*)
Kirloskar Chillers Pvt Ltd, Pune	SHRI AVINASH MANJUL SHRI GAURANG DABHOLKAR (*Alternate* I) SHRI AMEY MAJGAONKAR (*Alternate* II)
Malaviya National Institute of Technology, Jaipur	DR JYOTIRMAY MATHUR
Military Engineer Services, Engineer-in-Chief's Branch, Army HQ, New Delhi	SHRI H. K. GUPTA SHRI ARVIND KUMAR (*Alternate*)
Proion Consultants Private Limited, New Delhi	SHRI SANDEEP GOEL
Schneider Electric India Pvt Ltd, Gurgram	SHRI HIMANSHU NIGAM

Organization	*Representative(s)*
Shinryo Suvidha Engineers India Private Limited, Noida	SHRI ASHOK K. VIRMANI SHRI SANJAY MOOKERJEE (*Alternate*)
Sterling India Consulting Engineers, New Delhi	SHRI G. C. MODGIL MS KHUSHBOO MODGIL (*Alternate*)
The Institution of Engineers (India), Kolkata	SHRI PRADEEP CHATURVEDI SHRI ANIRBAN DATTA (*Alternate*)
The Chemours Company, Gurugram	DR AVINASH SHALIGRAM SHRI VIKAS MEHTA (*Alternate*)
UL India Pvt Ltd, Bengaluru	SHRI V. MANJUNATH SHRI SATISH KUMAR (*Alternate* I) SHRIMATI PALLAVI TRIPATHI (*Alternate* II)
Voltas Limited, New Delhi	SHRI ASHWANI SHARMA SHRI PANKAJ GOEL (*Alternate*)
In Personal Capacity (*C-9*, *9511*, *Vasant Kunj*, *New Delhi – 110070*)	DR R. S. AGARWAL
In Personal Capacity (*E 203*, *Belaire*, *Golf Course Road*, *Gurugram – 122022*)	SHRI PRABHAT GOEL
In Personal Capacity (*262*, *Solanipuram*, *Roorkee – 247667*)	DR P. K. BHARGAVA

Panel for Acoustics, Sound Insulation and Noise Control, CED 46:P15

Organization	*Representative(s)*
Interarch Building Products Private Limited, Noida	SHRI GAUTAM SURI (***Convener***)
All India Radio, New Delhi	MS ANURADHA AGGARWAL SHRI RAJESH KUMAR (*Alternate*)
Central Public Works Department, New Delhi	SENIOR ARCHITECT (ADG PROJECTS DELHI) ARCHITECT (ADG PROJECTS DELHI) (*Alternate*)
CSIR-Central Building Research Institute, Roorkee	SHRI SORAJ KUMAR PANIGARHI SHRI ANUP KUMAR PRASAD (*Alternate* I) SHRI KANTI LAL SOLANKI (*Alternate* II)
CSIR-National Physical Laboratory, New Delhi	DR NAVEEN GARG DR CHITRA GAUTAM (*Alternate*)
Indian Institute of Science, Bengaluru	DR M. L. MUNJAL
Indian Institute of Technology Kharagpur, Kharagpur	DR A. R. MOHANTY DR AMARNATH REDDY (*Alternate*)
Indian Institute of Technology Madras, Chennai	DR P. CHANDRAMOULI DR ABHIJIT SARKAR (*Alternate*)
Indian Institute of Technology Roorkee, Roorkee	DR E. RAJASEKAR

Organization	*Representative(s)*
Military Engineer Services, E-in-C's Branch, Army HQ, New Delhi	SHRI SAMEER PANDEY SHRI NIRAJ KUMAR (*Alternate*)
P. S. Subramanian Associates, Chennai	DR S. KANDASWAMY DR P. SENTHIL KUMAR (*Alternate*)
School of Planning and Architecture, New Delhi	DR SHUVOJIT SARKAR
The Indian Institute of Architects, Mumbai	SHRI B. SUDHIR SHRI VIKAS DUBEY (*Alternate* I) SHRI SHITAL PRASHANT NEMANE (*Alternate* II)
The Institution of Engineers (India), Kolkata	SHRI P. K. ADLAKHA
In Personal Capacity (*H109/2*, *7th Avenue*, *Besant Nagar*, *Chennai – 600090*)	DR S. NARAYANAN

Panel for Installation of Lifts, Escalators and Moving Walks, CED 46:P16

Organization	*Representative(s)*
TAK Consulting Pvt Ltd, Mumbai	SHRI T. A. K. MATHEWS (***Convener***)
Central Public Works Department, New Delhi	CHIEF ENGINEER (T&R)-LL SE (T) - ELECT (*Alternate*)
Chief Electrical Inspectorate, Govt of Tamil Nadu, Chennai	SHRI G. JOSEPH AROCKIADOSS SHRI P. PALANI (*Alternate*)
Delhi Development Authority, New Delhi	SHRI SANJAY KUMAR KHARE SHRI K. P. YADAV (*Alternate*)
Delhi Fire Services, Govt of NCT of Delhi, New Delhi	DIRECTOR
Directorate of Maharashtra Fire & Emergency Services, Mumbai	SHRI SANTOSH WARICK SHRI KIRAN HATYAL (*Alternate*)
ECE Industries Limited, Ghaziabad	SHRI AMIT KACHROO SHRI ANURAG MANGLIK (*Alternate*)
Elevator and Escalator Component Manufacturing Association of India (EECMAI), Chennai	SHRI HARISH A. NAKMAN SHRI ABEY GEORGE (*Alternate*)
Electrical Inspectorate, Govt of NCT of Delhi, New Delhi	SHRI MUKESH KUMAR SHARMA SHRI JOGENDER SINGH (*Alternate*)
Forum of Construction Utility Services, Bengaluru	SHRI V. JAGADISH KUMAR
Fujitec India Pvt Ltd, Chennai	SHRI R. RAJESH SHRI S. MANOKAR (*Alternate*)
Indian Electrical and Electronics Manufacturers' Association (Elevator Division), Mumbai	SHRI P. K. DIXIT SHRI RUSHAD DIVECHA (*Alternate* I) SHRI UTTAM KUMAR (*Alternate* II)

Organization	*Representative(s)*
Johnson Lifts Pvt Ltd, Chennai	SHRI T. SUBRAMANIAN SHRI V. KARTHIKEYAN (*Alternate* I) SHRI SACHIN MORE (*Alternate* II)
Kone Elevator India Limited, Chennai	SHRI U. VISWANATHAN SHRI BALAJI K. (*Alternate*)
Military Engineer Services, Engineer-in- Chief's Branch, Army HQ, New Delhi	SHRI H. K. GUPTA SHRI ARVIND KUMAR (*Alternate*)
NBCC (India) Ltd, New Delhi	SHIR RAM BACHAN SHRI TEJ PRAKASH DHABHAI (*Alternate*)
Otis Elevator Company (India) Limited, Bengaluru	SHRI ABHIJIT DANDEKAR SHRI SHRIHARI VISPUTE (*Alternate* I) SHRI PRAVEENA SIDDARAMANNA (*Alternate* II)
Public Works Department, Govt of Maharashtra, Mumbai	SHRI SANDEEP A. PATIL SHRI UDAY U. DAMBE (*Alternate*)
Schindler India Pvt Limited, Mumbai	SHRI NIMISH DESHPANDE SHRI NITIN VITHAL KADAM (*Alternate*)
TAK Consulting Pvt Ltd, Mumbai	SHRI GAURAV GHAG MS PRANALI PAWASKAR (*Alternate*)
TK Elevator India Pvt Ltd, New Delhi	SHRI VISHNU PARASHAR SHRI DEEPAK BALANI (*Alternate* I) SHRI SACHIN AGASHE (*Alternate* II)
The Indian Institute of Architects, New Delhi	SHRI PRAKASH DESHMUKH SHRI M. R. KUMARCHANDRA (*Alternate* I) MS ANAGHA VISHWAS KOTKAR (*Alternate* II)
The Institution of Engineers (India), Kolkata	DR R. K. DAVE
Wohr Parking Systems Pvt Ltd, Pune	SHRI SANDEEP KULKARNI SHRI VIKAS PATHAK (*Alternate*)
In Personal Capacity (*43, Juhu Gaurav, Gulmohar Cross Road No. 11, J.V.P.D. Scheme, Mumbai – 400049*)	SHRI P. M. TIPNIS

Panel for Plumbing Services, CED 46:P17

Organization	*Representative(s)*
Indian Plumbing Association, New Delhi	SHRI B. S. ASWATHNARAYAN (***Convener***)
Brihan Mumbai Licenced Plumbers' Association, Mumbai	SHRI BIJAL M. SHAH SHRI H. G. GANDHI (*Alternate*)
Central Ground Water Board, Faridabad	MEMBER (SML) REGIONAL DIRECTOR (*Alternate*)
Central Pollution Control Board, New Delhi	DR A. B. AKOLKAR DR SANJEEV AGRAWAL (*Alternate*)

Organization	*Representative*(*s*)
Central Public Health and Environmental Engineering Organisation, New Delhi	DR RAMAKANT SHRI VIPIN KUMAR PATEL (*Alternate*)
Central Public Works Department, New Delhi	CHIEF ENGINEER (WORKS CUM TLQA) SE & PD, CIRCLE-1 (*Alternate*)
CSIR-Central Building Research Institute, Roorkee	DR KISHOR KULKARNI DR NAVEEN NISHANT (*Alternate* I) DR VEENA CHAUDHARY (*Alternate* II)
CSIR-National Environmental Engineering Research Institute, Nagpur	SHRI A. S. DUBEY SHRI M. KARTHIK (*Alternate*)
Delhi Development Authority, New Delhi	SHRI K. S. MEENA SHRI PANKAJ KUMAR VERMA (*Alternate*)
Delhi Jal Board, New Delhi	CHIEF ENGINEER (CENTRAL & NORTH)
Envirotech Instruments Pvt Ltd, New Delhi	SHRI S. K. GUPTA
Indian Plumbing Association, New Delhi	SHRI MUKESH ASIJA DR ANANTHA SIVA IYER (*Alternate*)
Indraprastha Gas Ltd, New Delhi	SHRI PANKAJ SHARMA SHRI AVIJIT NARAYAN (*Alternate*)
Mahanagar Gas Ltd, Mumbai	SHRIMATI PRATIBHA BIJAY
Maple Engineering Design Services (India) Pvt Ltd, Bengaluru	SHRI S. L. PRASANNA KUMAR
MKG Consultants, New Delhi	SHRI M. K. GUPTA SHRI NISHANT GUPTA (*Alternate* I) SHRI HIMANSHU PARIHAR (*Alternate* II)
Military Engineer Services, Engineer-in-Chief's Branch, Army HQ, New Delhi	SHRI H. K. GUPTA COL PANKAJ BANIYAL (*Alternate*)
Ministry of Jal Shakti, New Delhi	SHRI SAURABH GUPTA SHRI SANTOSH KUMAR (*Alternate*)
Municipal Corporation of Greater Mumbai, Mumbai	DY HYDRAULIC ENGINEER (PLANNING & CONTROL) EXECUTIVE ENGINEER (PLANNING & RESEARCH) (*Alternate*)
Nous Hospital Consultants Pvt Ltd, New Delhi	DR K. B. SOOD
Pratham Envirotech Solutions, Navi Mumbai	SHRI PURSHOTTAM YEMUL
Proion Consultants Private Limited, New Delhi	SHRI SANDEEP GOEL
Tata Consulting Engineers Limited, Mumbai	SHRI DHEERAJ KAUSHIK
The Indian Institute of Architects, New Delhi	SHRI UDAYA SHANKAR DONI SHRI RAJ KUMAR PRAJAPATI (*Alternate* I) MS SWOPNADUTTA MOHANTY (*Alternate* II)
The Institution of Engineers (India), Kolkata	DR D. ELANGO DR V. S. PRIYA (*Alternate*)
Water Management & Plumbing Skill Council, New Delhi	SHRI VINAY GUPTA SHRIMATI SHRUTI GOEL (*Alternate*)

Organization	*Representative*(*s*)
In Personal Capacity (*Flat No 514*, *Charms Solitaire Apartment*, *Plot No 14*, *Mall Road*, *AK-2*, *Indirapuram*, *Ghaziabad – 201014*)	DR BHARAT BHUSHAN NAGAR

Panel for Glass and Glazing, CED 46:P20

Organization	*Representative*(*s*)
Central Public Works Department, New Delhi	SHRI PRADEEP GARG **(*Convener*)**
All India Glass Manufacturers' Federation, New Delhi	SHRI SHAILESH RANJAN SHRI RUPINDER SHELLY (*Alternate*)
Bureau of Energy Efficiency Ministry of Power, New Delhi	MS PRAVATANALINI SAMAL SHRI PANKAJ SHARMA (*Alternate*)
BES Consultants Pvt Ltd, Mumbai	SHRI RAJAN GOVIND SHRI VENUGOPAL B. (*Alternate*)
Central Public Works Department, New Delhi	SENIOR ARCHITECT (NATIONAL CPWD ACADEMY) ARCHITECT (NCA) (*Alternate*)
Creative Design Consultants & Engineers Pvt Ltd, Greater Noida	SHRI AMAN DEEP SHRI BARJINDER SINGH GHAI (*Alternate*)
CSIR-Central Building Research Institute, Roorkee	DR CHANDAN SWAROOP MEENA DR NAGESH BABU BALAM (*Alternate* I) DR KISHOR S. KULKARNI (*Alternate* II)
Dow Chemical International Private Limited, Mumbai	SHRI RAVISHANKAR SUBRAMANIAN
Facade India Testing Inc, Thane	SHRI V. S. RAVI
Glazing Society of India, Chennai	SHRI G. N. GOHUL DEEPAK MS DILNA SUBRAMANIAN (*Alternate* I) MS RANJANI. A (*Alternate* II)
GSC Glass Ltd, Greater Noida	SHRI C. J. SINGH
Hilti India Private Limited, New Delhi	SHRI SHOUNAK MITRA SHRI RAGHAVENDRA KUMAR VAKKALAGADDA (*Alternate*)
Indian Association of Structural Engineers, New Delhi	SHRI MAHESH TANDON DR S. P. ANCHURI (*Alternate*)
International Fenestration Forum, Noida	SHRI ARUN SHARMA
Indian Institute of Technology Madras (Structural Glass Research & Testing Facility), Chennai	DR S. ARUL JAYACHANDRAN DR ALAGAPPAN PONNALAGU (*Alternate*)
Indian Institute of Technology Roorkee, Roorkee	DR AVLOKITA AGRAWAL DR E. RAJASEKAR (*Alternate*)
Larsen & Toubro Ltd, Chennai	SHRI STHALADIPTI SAHA SHRI SURYA PRAKASH KARRI (*Alternate*)
Meinhardt Facade Technology India Pvt Ltd, Chennai	SHRI MAHESH ARUMUGAM
Municipal Corporation of Greater Mumbai (Mumbai Fire Brigade), Mumbai	CHIEF FIRE OFFICER DY CHIEF FIRE OFFICER (*Alternate*)

Organization	*Representative*(*s*)
NBCC (India) Limited, New Delhi	SHRI ANUJ KUMAR GOYAL SHRI VINAY KUMAR KHULLAR (*Alternate*)
Procural Private Limited, Telangana	SHRI SATISH KUMAR SHRI SANTHANAM RENGASWAMY (*Alternate*)
Saint-Gobain Glass India Pvt Ltd, Chennai	SHRI SHOAIB SHAIKH SHRI U. GOPINATH (*Alternate* I) MS KAARNIKA M. (*Alternate* II)
Schueco India Pvt Ltd, Bengaluru	SHRI ANTONY JOHN V. S. SHRI KUSHAGRA AWASTHI (*Alternate*)
The Indian Institute of Architects, New Delhi	SHRI RUPINDER AHUJA SHRI ASHUTOSH KUMAR AGARWAL (*Alternate* I) SHRIMATI JYOTI RAJEEV PANSE (*Alternate* II)
The Institution of Engineers (India), Kolkata	ER M. NAGARAJ ER INDERDEEP SINGH OBEROI (*Alternate*)
UPVC Windows and Doors Manufacturers Association, New Delhi	SHRI ULLAS GULIANI SHRI MARIO SCHMIDT (*Alternate* I) SHRI SATISH KUMAR (*Alternate* II)
Winwall Technology India Pvt Ltd, Chennai	SHRI P. JOTHI RAMALINGAM

Panel for Information and Communication Enabled Installations, CED 46:P21

Organization	*Representative*(*s*)
In Personal Capacity (*C501*, *Shivam Apartments*, *Plot 14*, *Sector 12*, *Dwarka*, *New Delhi – 110078*)	SHRI A. K. MITTAL **(*Convener*)**
Broadband India Forum, New Delhi	SHRI T. V. RAMACHANDRAN BRIG ANIL TANDAN (*Alternate*)
Building Industry Consulting Service International, Mumbai	SHRI NINAD MOHAN DESAI
Cellular Operators Association of India, New Delhi	SHRI VIKRAM TIWATHIA SHRIMATI VERTIKA MISRA (*Alternate*)
Central Industrial Security Force, New Delhi	SHRI NARESH KUMAR SHRI SANJEEV KUMAR SADDI (*Alternate*)
Central Public Works Department, New Delhi	CHIEF ENGINEER (T&R)-LL SE (T)-ELECT (*Alternate*)
CMAI Association of India, New Delhi	DR N. K. GOYAL
Department of Telecommunication, New Delhi	SHRI ASHOK KUMAR SHRI RAJAT JAIN (*Alternate*)
Dept of Electronics & Information Technology, Ministry of Electronics and Information Technology, New Delhi	DR ANIL KUMAR KAUSHIK SHRI PRAKASH KUMAR (*Alternate*)
Digital Infrastructure Providers Association, New Delhi	SHRI MANOJ KUMAR SINGH SHRI RAMAKANT GUPTA (*Alternate* I) SHRI GREESH KUMAR (*Alternate* II)

Organization	*Representative(s)*
Gujarat International Finance Tec City Company Limited, Gandhinagar	SHRI RAMAKANT JHA DR NILESH KUMAR PUREY (*Alternate*)
Military Engineer Services, Engineer-in-Chief's Branch, Army HQ, New Delhi	SHRI S. K. GUPTA SHRI BIJENDRA PAL SINGH (*Alternate*)
Ministry of Railways, New Delhi	SHRIMATI VINITA NARERA
Proion Consultants Private Limited, New Delhi	SHRIMATI SHRUTI GOEL
Roads and Buildings Department, Govt of Telangana, Hyderabad	SHRI G. RAMAKRISHNA KUMAR DEPUTY CHIEF ENGINEER (*Alternate*)
Shyam Spectra Private Limited, Gurugram	SHRI HEM CHAND RAJPUT
Sterlite Technologies Limited, Pune	SHRI S. N. GUPTA MD SHAHID KHAN (*Alternate*)
Telecom Equipment Manufacturers Association of India, New Delhi	SHRI BHUPESH YADAV SHRI RAHUL DUBEY (*Alternate*)
Telecom Engineering Centre, New Delhi	SHRI JASVEER SINGH PANESAR
The Indian Institute of Architects, Mumbai	DR JYOTI PANSE
The Institution of Engineers (India), Kolkata	DR LAXMI KANT MISHRA ER RAKESH KUMAR RATHORE (*Alternate*)
Town and Country Planning Organization, New Delhi	SHRI NARESH KUMAR DHIRAN SHRI PARESH KUMAR DURIA (*Alternate* I) SHRI R. SRINIVAS (*Alternate* II)
In Personal Capacity (*Cedar 102, SJR Park Vista Aptts, Haralur Road, Ambalipura, Bengaluru – 560012*)	SHRI J. N. BHAVANI PRASAD

WORKING GROUPS UNDER THE PANEL, CED 46:P2

Working Group for Smoke Management, CED 46:P2:WG1

Organization	*Representative(s)*
Directorate of Maharashtra Fire Services, Mumbai	SHRI SANTOSH S. WARICK (***Convener***)
Fire and Security Association of India, Chennai	SHRI SRINIVAS VALLURI
Indian Society of Heating, Refrigerating and Air Conditioning Engineers, New Delhi	SHRI K. RAMACHANDRAN

Working Group for Electrical Safety, CED 46:P2:WG2

Organization	*Representative(s)*
Fire and Security Association of India, Chennai	SHRI SRINIVAS VALLURI (***Convener***)
National Federation of Engineers for Electrical Safety, Chennai	SHRI S. GOPA KUMAR
In Personal Capacity (*27A, Tapovan Senior Citizens Foundation, Coimbatore – 641010*)	SHRI T. R. A. KRISHNAN

Drafting Group of the Fire and Life Safety Chapter, CED 46:P2:WG3

Organization	*Representative*(*s*)
Fire Safe India Foundation, Mumbai	SHRI M. V. DESHMUKH (***Convener***)
Fire and Security Association of India, Chennai	SHRI SRINIVAS VALLURI
Kapse Consulting Engineers Private Limited, Thane	SHRI VINOD KAPSE
Directorate of Maharashtra Fire Services, Mumbai	SHRI SANTOSH S. WARICK SHRI KIRAN HATYAL (*Alternate*)
Proion Consultants Private Limited, New Delhi	SHRI SANDEEP GOEL
TTS Consultant, Kolkata	SHRI TARAK CHAKRABORTY
In Personal Capacity (*Shop No. 15*, *Natraj Building*, *Shiv Srishti Complex*, *M.G.Link Road Mulund* (*W*) *Mumbai – 400080*)	SHRI SHASHIKANT L. JADHAV
In Personal Capacity (*Flat No. 9221*, *ATS Pristine*, *Sector 150*, *Noida – 201 310*)	DR G. C. MISRA
In Personal Capacity (*Ganeshkunj A-10 Chanakyapuri*, *New Sama Road*, *Baroda – 390008*)	SHRI ABHAY D. PURANDARE

Advisory Group for Revision of Part 4, CED 46:P2:WG4

Organization	*Representative*(*s*)
In Personal Capacity (*K-33-A*, *Green Park*, (*First Floor*), *New Delhi – 110016*)	SHRI S. K. DHERI (***Convener***)
Delhi Fire Services, New Delhi	DIRECTOR
In Personal Capacity (*27A*, *Tapovan Senior Citizens Foundation*, *Coimbatore – 641010*)	SHRI T. R. A. KRISHNAN
In Personal Capacity (*D-317*, *IInd Floor*, *Nirman Vihar*, *New Delhi – 110 092*)	SHRI R. C. SHARMA
In Personal Capacity (*'Isa Vasyam' TC-15/1675*, *B-8/2-Lakshmi Nagar*, *Kesavadaspuram – 695004*)	SHRI V. SURESH

Metro Related Provisions, CED 46:P2/WG5

Organization	*Representative*(*s*)
In Personal Capacity (*Flat No. 9221*, *ATS Pristine*, *Sector 150*, *Noida – 201310*)	DR G. C. MISRA (***Convener***)
Chennai Metro Rail Limited, Chennai	SHRIMATI RAJALAKSHMI. S
Delhi Metro Rail Corporation Limited, Delhi	SHRI ASHOK TEWARI SHRIMATI RASHMI BHARDWAJ (*Alternate* I) MS NABILA MARYAM (*Alternate* II)
National Capital Region Transport Corporation, New Delhi	SHRI GAUTAM BHARDWAJ
In Personal Capacity (*27A*, *Tapovan Senior Citizens Foundation*, *Coimbatore – 641010*)	SHRI T. R. A. KRISHNAN

WORKING GROUP UNDER THE PANEL, CED 46:P3

Expert Group for Revision, CED 46:P3/WG1

Organization	*Representative*(*s*)
Central Public Works Department, New Delhi	SHRI DINESH K. UJJAINIA (***Convener***)
AEON Design & Development LLP, Noida	MS SHEETAL RAKHEJA
Builders' Association of India, Mumbai	REPRESENTATIVE
CSIR - Central Building Research Institute, Roorkee	DR NEERAJ JAIN
Indian Institute of Technology Madras, Chennai	DR AMLAN KUMAR SENGUPTA
Larsen and Toubro Construction, Chennai	SHRI PRAVEEN KUMAR RAI
Military Engineer Services, Engineer-in-Chief's Branch, Army HQ, New Delhi	SHRIMATI S. GAYATRI MUKHERJEE
National Council for Cement and Building Materials, Faridabad	SHRI P. N. OJHA
The Indian Institute of Architects, New Delhi	DR VENU SHREE

WORKING GROUPS UNDER THE PANEL, CED 46:P6

Working Group for Bamboo, CED 46:P6/WG1

Organization	*Representative*(*s*)
Wonder Grass Initiative Private Limited, Nagpur	SHRI VAIBHAV KALEY (***Convener***)
Bamboo Society of India, Bengaluru	SHRI PUNATI SRIDHAR
Centre for Green Building Materials & Technology, Bengaluru	MS NEELAM MANJUNATH
CSIR - Central Building Research Institute, Roorkee	ER MICKEY MECON DALBEHERA
Forest Research Institute, Dehradun	DR AJAY THAKUR

Working Group for Engineered Bamboo, CED 46:P6/WG2

Organization	*Representative*(*s*)
ICFRE-Institute of Wood Science and Technology, Bengaluru	SHRI ANAND NANDANWAR (***Convener***)
CEPT University, Ahmedabad	DR DHARA SHAH
Forest Research Institute, Dehradun	DR AJAY THAKUR
Indian Institute of Technology Delhi, New Delhi	SHRI SURESH BHALLA
ICFRE-Institute of Wood Science and Technology, Bengaluru	DR NARASIMHAMURTHY
Mutha Industries Pvt Ltd, Mumbai	SHRI ANIL MUTHA
North East Centre for Technology Application and Research, Shillong	SHRI KRISHNA KUMAR

Working Group for Timber, CED 46:P6/WG3

Organization	*Representative(s)*
Forestry Innovation Consulting India Pvt Ltd, Mumbai	DR JIMMY THOMAS
Forest Research Institute, Dehradun	SHRI RAJESH BHANDARI
Forum of Scientists, Engineers and Technologists, Kolkata	DR PARTHASARATHI MUKHOPADHYAY
ICFRE-Institute of Wood Science and Technology, Bengaluru	SHRI ANAND NANDANWAR SHRI AMITAVA SIL
In Personal Capacity (*P.O. New Forest, Dehradun – 248006*)	DR SADHNA TRIPATHI

Working Group for Engineered Timber, CED 46:P6/WG4

Organization	*Representative(s)*
ICFRE-Institute of Wood Science and Technology, Bengaluru	SHRI ANAND NANDANWAR (***Convener***)
CEPT University, Ahmedabad	DR DHARA SHAH
Forest Research Institute, Dehradun	SHRI RAJESH BHANDARI
North East Centre for Technology Application and Research, Shillong	SHRI KRISHNA KUMAR
Windsor Wood (India) Pvt Ltd, Mumbai	SHRI PRAKASH NARMADASHANKAR SUTHAR

WORKING GROUPS UNDER THE PANEL, CED 46:P12

Working Group for Building Orientation, CED 46:P12/WG1

Organization	*Representative(s)*
Ashok B Lall Architects, New Delhi	SHRI ASHOK B. LALL (***Convener***)
Ministry of New and Renewable Energy, New Delhi	DR A. K. TRIPATHI

Working Group for Building Lightning, CED 46:P12/WG2

Organization	*Representative(s)*
CEPT University, Ahmedabad	DR MINU AGARWAL (***Convener***)
CSIR-Central Building Research Institute, Roorkee	SHRI ANUP KUMAR PRASAD

Working Group for Ventilation, CED 46:P12/WG3

Organization	*Representative(s)*
CEPT University, Ahmedabad	DR RAJAN RAWAL (***Convener***)
Military Engineer Services, Engineer-in-Chief's Branch, Army HQ, New Delhi	SHRI ASHOK KUMAR

WORKING GROUPS UNDER THE PANEL, CED 46:P13

Working Group for Electric Vehicles, Battery Storage, and Load Factor, CED 46:P13/WG2

Organization	*Representative(s)*
In Personal Capacity (*Cedar 102, SJR Park Vista Aptts, Haralur Road, Ambalipura, Bengaluru – 560012*)	SHRI J. N. BHAVANI PRASAD (***Convener***)
Automotive Research Association of India, Pune	SHRI A. A. DESHPANDE
Cape Electric Private Limited, Kancheepuram	SHRI S. GOPA KUMAR
Central Public Works Department, New Delhi	SHRI VIKAS RANA
Directorate of Maharashtra Fire Services, Mumbai	SHRI SANTOSH S. WARICK
India Energy Storage Alliance, Pune	SHRI DEBI PRASAD DASH
Indian Institute of Technology Madras, Chennai	DR ASHOK JHUNJHUNWALA
Tata Motors Limited, Pune	SHRI SENTHILNATHAN THANGAVELU
In Personal Capacity (*J-403 Second Floor, New Rajinder Nagar, New Delhi – 110060*)	SHRI CHAITANYA KUMAR VARMA

Working Group for Revision of the Chapter, CED 46:P13/WG3

Organization	*Representative(s)*
Cape Electric Private Limited, Kancheepuram	SHRI S. GOPA KUMAR (***Convener***)
Aeon Integrated Building Design Consultants, Noida	SHRI PUNEET GUPTA
Central Electricity Authority, New Delhi	SHRI ASHOK KUMAR RAJPUT
Central Public Works Department, New Delhi	SHRI VIKAS RANA
Government of Tamil Nadu, Chief Electrical Inspectorate, Chennai	SHRI P. PALANI
Ministry of New and Renewable Energy, New Delhi	SHRI A. K. TRIPATHI
National Federation of Engineers for Electrical Safety, Chennai	SHRI ABHISHEK SINGH
Obo Bettermann India Private Limited, Oragadam	SHRI K. JANAKI RAMAN
ProtegoPlus Electrotech Private Limited, Mumbai	SHRI KRISH THEOBALD
Proion Consultants Private Limited, New Delhi	SHRIMATI SHRUTI GOEL
Public Works Department, Government of Maharashtra, Mumbai	SHRI HEMANT M. SALI
Schneider Electric India Private Limited, Gurugram	SHRI DEBDAS GOSWAMI

Working Group for Solar Energy and Power from Multiple Sources and Related Protection and Earthing, CED 46:P13/WG4

Organization	*Representative(s)*
Proion Consultants Private Limited, New Delhi	SHRIMATI SHRUTI GOEL (***Convener***)
Cape Electric Private Limited, Kancheepuram	SHRI S. GOPA KUMAR
Central Public Works Department, New Delhi	SHRI VIKAS RANA
Government of Tamil Nadu, Chief Electrical Inspectorate, Chennai	SHRI P. PALANI
Ministry of New and Renewable Energy, New Delhi	DR A. K. TRIPATHI
Obo Bettermann India Private Limited, Oragadam	SHRI K. JANAKI RAMAN
Public Works Department, Government of Maharashtra, Mumbai	SHRI HEMANT M SALI
Solar Energy Corporation of India Limited, New Delhi	SHRI C. K. SINGH
The Institution of Engineers (India), Kolkata	DR S. SENTHILKUMAR

WORKING GROUP UNDER THE PANEL, CED 46:P14

Core Group for Revision of HVAC Chapter, CED 46:P14/WG1

Organization	*Representative(s)*
ACR Project Consultants Private Limited, Pune	SHRI HARSHAL SURANGE
Adani Energy Solution Pvt Ltd, Ahmedabad	SHRI PRAMOD PRABHAKAR
Anna University, Chennai	DR R. SARAVANAN
Blowtech Air Devices Private Limited, Noida	SHRI SUSHIL K. CHOUDHURY
Breathe Easy Labs, New Delhi	SHRI BARUN AGARWAL
CBRE India, Gurugram	SHRI ANEESH KADYAN
CEPT University, Ahmedabad	DR YASH SHUKLA
Degree Day Private Limited, Indore	SHRI NISHANT GUPTA
Dynacraft Air Controls, Mumbai	SHRI V. KRISHNAN
ELM Enterprises Private Limited, Delhi	SHRI PRASHANT DESAI
Enova HVAC & MEP Consulting, Mumbai	MS SANGITA JHANGIANI
Eskayem Consultants Private Limited, Mumbai	SHRI GOUTHAM PITCHIKALA
Envirocon Engineering Services, New Delhi	SHRIMATI SAMTA BAJAJ

Organization	*Representative(s)*
Genex Consultants, Hyderabad	SHRI CHANDRASEKAR NARAYANAN
Godrej & Boyce Manufacturing Company Limited, Mumbai	SHRI MUKESH SUTHAR
Impec Filters Private Limited, Chennai	DR SHANKAR RAJASEKARAN
Indian Society of Heating, Refrigerating and Air Conditioning Engineers, New Delhi	SHRI ASHISH RAKHEJA SHRI K. RAMACHANDRAN SHRIMATI NIVEDITA JADHAV
International Ground Source Heat Pump Association, New Delhi	SHRI MADHUSUDAN RAPOLE
Lead Consultancy and Engineering Services (India) Private Limited, Bengaluru	SHRI M. SELAVARASU
Malaviya National Institute of Technology, Jaipur	DR JYOTIRMAY MATHUR
Overdrive Engineering Private Limited, New Delhi	SHRI RICHIE MITTAL
Paharpur Cooling Towers Limited, Kolkata	SHRI VARUN SWARUP
Proion Consultants Private Limited, New Delhi	SHRI SANDEEP GOEL
Reliance Industries Limited, Bengaluru	SHRI AJAJ KAZI
3S Green SSS Consultants, Bengaluru	SHRI C. SUBRAMANIAM
Sanelac Consultants Private Limited, New Delhi	SHRI VARUN JAIN
Sterling India Consulting Engineers, New Delhi	SHRI G. C. MODGIL
The Chemours India Private Limited, Gurugram	SHRI VIKAS MEHTA
UL India Private Limited, Bengaluru	SHRI V. MANJUNATH
Univac Environment Systems Private Limited, Mumbai	SHRI VIKRAM MURTHY
Vision Electro Mechanical Consultants Private Limited, Pune	DR NITIN DEODHAR
In Personal Capacity (*E-203, Belaire, Golf Course Road, Gurugram – 122022*)	SHRI PRABHAT GOEL
In Personal Capacity (*Experiqs Pvt Ltd*, *7022*, *RBTIC Building*, *IIT Bombay*, *Powai*, *Mumbai – 400076*)	SHRI AVINASH G. SHALIGRAM

WORKING GROUP UNDER THE PANEL, CED 46:P15

Expert Group for Revision, CED 46:P15/WG1

Organization	*Representative(s)*
Interarch Building Products Private Limited, Noida	SHRI GAUTAM SURI (***Convener***)
CSIR-National Physical Laboratory, New Delhi	DR NAVEEN GARG

Organization	*Representative(s)*
Indian Institute of Science, Bengaluru	DR M. L. MUNJAL
Indian Institute of Technology Madras, Chennai	DR S. NARAYANAN
Indian Institute of Technology Roorkee, Roorkee	DR E. RAJASEKAR
P. S. Subramanian Associates, Chennai	DR S. KANDASWAMY
The Indian Institute of Architects, New Delhi	SHRI B. SUDHIR

WORKING GROUP UNDER THE PANEL, CED 46:P16

Working Group for Revision of the Chapter, CED 46:P16/WG1

Organization	*Representative(s)*
TAK Consulting Private Limited, Mumbai	SHRI T. A. K. MATHEWS (***Convener***)
Directorate of Maharashtra Fire Services, Mumbai	SHRI SANTOSH S. WARICK
ECE Industries Limited, New Delhi	SHRI AMIT KACHROO SHRI ANURAG MANGLIK (*Alternate*)
Elevator and Escalator Component Manufacturers' Association of India, Chennai	SHRI HARISH A. NAKMAN SHRI ABEY GEORGE (*Alternate*)
Energy Department, Government of Maharashtra, Mumbai	SHRI SANDEEP A. PATIL SHRI UDAY U. DAMBE (*Alternate*)
Fujitec India Private Limited, Gurugram	SHRI R. RAJESH SHRI MANOKAR S. (*Alternate* I) SHRI ARUN SANKAR (*Alternate* II)
IEEMA Elevator Division, Mumbai	SHRI P. K. DIXIT SHRI RUSHAD DIVECHA (*Alternate* I) SHRI UTTAM KUMAR (*Alternate* II)
Johnson Lifts Pvt Limited, Chennai	SHRI V. KARTHIKEYAN SHRI SACHIN MORE (*Alternate*)
Kone Elevator India Private Limited, Chennai	SHRI U. VISWANATHAN SHRI BALAJI K. (*Alternate*)
Otis Elevator Company (India) Limited, Bengaluru	SHRI SHRIHARI VISPUTE SHRI ABHIJIT DANDEKAR (*Alternate* I) SHRI PRAVEENA SIDDARAMANNA (*Alternate* II)
Schindler India Private Limited, Mumbai	SHRI NIMISH DESHPANDE SHRI RAJAGOPALAN RENGANATHAN (*Alternate* I) SHRI NITIN VITHAL KADAM (*Alternate* II)
TAK Consulting Private Limited, Mumbai	SHRI GAURAV GHAG
TK Elevator India Private Limited, New Delhi	SHRI VISHNU PARASHAR SHRI DEEPAK BALANI (*Alternate*)
Wohr Parking Systems Pvt Ltd, Pune	SHRI SANDEEP KULKARNI

Organization	*Representative(s)*
In Personal Capacity (*43, Juhu Gaurav, Gulmohar Cross Road No. 11, J.V.P.D. Scheme, Mumbai – 400 049*)	SHRI P. M. TIPNIS

WORKING GROUPS UNDER THE PANEL, CED 46:P17

Working Group for Water Supply, CED 46:P17/WG1

Organization	*Representative(s)*
Indian Plumbing Association, New Delhi	SHRI B. S. ASWATHNARAYAN (***Convener***)
CSIR-Central Building Research Institute, Roorkee	SHRI CHANDAN SWAROOP MEENA
D' Plumbing Consultants, Mumbai	SHRI DEEPAK
Indian Plumbing Association Maple Engineering - Design Services India Private Limited, Bengaluru	SHRI S. L. PRASANNAKUMAR
Indian Plumbing Association, New Delhi	SHRI MUKESH ASIJA
Ministry of Heavy Industries and Public Enterprises, New Delhi	SHRI VIJAY PRAKASH
Municipal Corporation of Greater Mumbai, Mumbai	DR DEEPAK NAGARSOGE
Nous Hospital Consultants Pvt Ltd, New Delhi	DR K. B. SOOD
Proion Consultants Private Limited, New Delhi	SHRI SANDEEP GOEL
Tata Consulting Engineers Limited, Navi Mumbai	SHRI DHEERAJ KAUSHIK

Working Group for Drainage and Sanitation, CED 46: P17/WG2

Organization	*Representative(s)*
MKG Consultants, New Delhi	SHRI M. K. GUPTA (***Convener***)
Brihan Mumbai Licensed Plumbers' Association, Mumbai	SHRI H. G. GANDHI
Central Public Works Department, New Delhi	SHRI DINESH K. UJJAINIA
Maple Engineering Design Services (India) Private Limited, Bengaluru	SHRI S. L. PRASANNA KUMAR
Military Engineer Services, Engineer-in-Chief's Branch, Army HQ, New Delhi	SHRI N. K. GOEL
The Indian Institute of Architects, New Delhi	SHRI RAJ KUMAR PRAJAPATI
In Personal Capacity (*610, Technology Apartments, 24, Patparganj, New Delhi – 110002*)	SHRI SUBIR PAUL

Working Group for Solid Waste Management, CED 46: P17/WG3

Organization	*Representative(s)*
Central Public Health Environmental Engineering Organisation, New Delhi	DR RAMAKANT (***Convener***)

Organization	*Representative*(*s*)
CSIR-Central Building Research Institute, Roorkee	DR AJAY CHOURASIA
Central Pollution Control Board, New Delhi	DR A. B. AKOLKAR DR SANJEEV AGRAWAL (*Alternate*)
Indian Institute of Technology Bombay, Mumbai	DR HARIHARAN S. SHANKAR
The Institution of Engineers (India), Kolkata	SHRI D. ELANGO
In Personal Capacity (*377*, *Church View Apartment*, *Sector 29*, *Noida – 201303*)	CAPT MANJU MINHAS

Working Group for Gas Supply, CED 46:P17/WG4

Organization	*Representative*(*s*)
Indian Plumbing Association, New Delhi	SHRI B. S. ASWATHNARAYAN (***Convener***)
Indraprastha Gas Limited, New Delhi	SHRI PANKAJ SHARMA
CSIR-National Environmental Engineering Research Institute, Nagpur	SHRI A. S. DUBEY
Ministry of Heavy Industries and Public Enterprises, New Delhi	SHRI VIJAY PRAKASH
Tata Consulting Engineers Limited, Navi Mumbai	SHRI GANESH KUMAR SINGH

WORKING GROUPS UNDER THE PANEL, CED 46:P20

Working Group for Energy and Thermal Performance, CED 46:P20/WG1

Organization	*Representative*(*s*)
Indian Institute of Technology Roorkee, Roorkee	DR E. RAJASEKAR (***Convener***)
Bureau of Energy Efficiency, New Delhi	SHRI SAURABH DIDDI
Glazing Society of India, Chennai	SHRI GOHUL DEEPAK G. N.
International Fenestration Forum, New Delhi	SHRI ARUN SHARMA
Larsen & Toubro Ltd, Chennai	SHRI SURYA PRAKASH KARRI
Schueco India Private Limited, Bengaluru	SHRI KUSHAGRA AWASTHI

Working Group for Special Glass, CED 46:P20/WG2

Organization	*Representative*(*s*)
Meinhardt Facade Technology India Private Limited, Chennai	SHRI MAHESH ARUMUGAM (***Convener***)
Central Public Works Department, New Delhi	SHRI PRADEEP GARG

Organization	*Representative*(*s*)
Defence Research and Development Organisation, Terminal Ballistics Research Laboratory, Chandigarh	SHRIMATI SAKSHI ARORA DR PREETI JAIN (*Alternate*)
Glazing Society of India, Chennai	SHRI GOHUL DEEPAK G. N.
International Fenestration Forum, New Delhi	SHRI ARUN SHARMA
Larsen & Toubro Ltd, Chennai	SHRI SURYA PRAKASH KARRI
Saint-Gobain Glass India Private Limited, Chennai	SHRI AVINASH GOWRA
Schueco India Private Limited, Bengaluru	SHRI ANTONY JOHN V. S.

Working Group for Fire Safety Aspects, CED 46:P20/WG3

Organization	*Representative*(*s*)
Indian Institute of Technology Gandhinagar, Gandhinagar	DR GAURAV SRIVASTAVA (***Convener***)
Aadit Fire Testing Laboratory Private Limited, Gummidipoondi	SHRI P. JOTHI RAMALINGAM DR MANIGANDAN SHRI AKHIL CHACKO
All India Glass Manufacturers' Federation, New Delhi	SHRI NAGENDRA KUMAR
BES Consultants Private Limited, Mumbai	SHRI RAJAN GOVIND
CSIR-Central Building Research Institute, Roorkee	SHRI S. K. NEGI
Glazing Society of India, Chennai	SHRI GOHUL DEEPAK G. N.
Hilti India Private Limited, New Delhi	SHRI RAGHAVENDRA KUMAR VAKKALAGADDA
Indian Institute of Technology Tirupati, Yerpedu	DR JASHNAV PANCHETI

Working Group for Structural Design Aspects, CED 46:P20/WG4

Organization	*Representative*(*s*)
BES Consultants Private Limited, Mumbai	SHRI RAJAN GOVIND (***Convener***)
Glazing Society of India, Chennai	SHRI GOHUL DEEPAK G. N.
Hilti India Private Limited, New Delhi	SHRI SHOUNAK MITRA
Indian Association of Structural Engineers, New Delhi	SHRI MAHESH TANDON
Indian Institute of Technology Madras, Chennai	DR ALAGAPPAN PONNALAGU
Larsen & Toubro Ltd, Chennai	SHRI SURYA PRAKASH KARRI
Schueco India Private Limited, Bengaluru	SHRI ANTONY JOHN V. S.

Working Group for Performance and Testing, CED 46:P20/WG5

Organization	*Representative*(*s*)
Schueco India Private Limited, Bengaluru	SHRI ANTONY JOHN V. S. (***Convener***)
Eminent International Testing Centre Private Limited, Hyderabad	SHRI REJI BHAMI
Facade India Testing Inc, Thane	SHRI V. S. RAVI
Glazing Society of India, Chennai	SHRI GOHUL DEEPAK G. N.
Winwall Technology India Private Limited, Chennai	SHRI P. JOTHI RAMALINGAM

WORKING GROUPS UNDER THE PANEL, CED 46:P21

Working Group for Indoor and Outdoor Plants, CED 46:P21/WG1

Organization	*Representative*(*s*)
ITS India Forum, New Delhi	DR SHIV KUMAR (***Convener***)
Broadband India Forum, New Delhi	SHRI ATUL JAIN
Building Industry Consulting Service International, Mumbai	SHRI NINAD MOHAN DESAI
Ranext Technologies Private Limited, Gurugram	SHRI JAGMOHAN SINGH
Sterlite Technologies Limited, Pune	SHRI SHAHID KHAN
Telecom Regulatory Authority of India, New Delhi	SHRI VIRENDRA SINGH RAJPUT

Working Group for Wireless Systems, CED 46:P21/WG2

Organization	*Representative*(*s*)
In Personal Capacity (*C501*, *Shivam Apartments*, *Plot 14*, *Sector 12*, *Dwarka*, *New Delhi – 110078*)	SHRI A. K. MITTAL (***Convener***)
Broadband India Forum, New Delhi	SHRI DEBASHISH BHATTACHARYA
Cellular Operators Association of India, New Delhi	SHRI VIKRAM TIWATHIA SHRI DHANANJAY GAWANDE
Crest Digitel Private Limited, Gurugram	SHRI VINAY KUMAR
Digital Infrastructure Providers Association, New Delhi	SHRI MANOJ KUMAR SINGH
Telecom Engineering Centre, New Delhi	SHRI JASVEER SINGH PANESAR
Telecom Regulatory Authority of India, New Delhi	SHRI TEJPAL SINGH

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART A INTRODUCTION

BUREAU OF INDIAN STANDARDS

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART A INTRODUCTION

This Part covers guidelines to be followed for judicious implementation of the provisions of various Parts/Sections of the National Building Construction Standards (NBCS), following a systematic approach.

In order to provide safe and resilient habitat, careful consideration needs to be paid to the building construction activity. Building planning, designing and construction activities have developed over the centuries. Large number of ancient monuments and historical buildings all over the world bear testimony to the growth of civilization from the prehistoric era with the extensive use of manual labour and simple systems as appropriate to those ages to the present day mechanized and electronically controlled operations for designing and constructing buildings and for operating and maintaining systems and services. Technological and socio-economic developments in recent times have led to remarkable increase in demand for more and more sophistication in buildings resulting in ever increasing complexities. For achieving these performance demand high levels of inputs from professionals of different disciplines such as architecture, civil engineering, structural engineering, functional and life safety services including special aspects relating to utilities, landscaping in conceptualization, spatial planning, design and construction of buildings of various material and technology streams, and various services including operation, maintenance, repairs and rehabilitation aspects throughout the service life of the building.

Present day construction scenario enables freedom of action to adopt appropriate practices and provides for building planning, designing and construction for absorbing traditional practices as well as latest developments in knowledge in the various disciplines as relevant to a building including computer aided and/or other modern sensors aided activities in the various stages of conceptualization, planning, designing, constructing, maintaining and repairing the buildings. India being a large country with substantial variations from region to region, this document has endeavoured to meet the requirements of different regions of the country, both urban and rural, by taking into consideration factors such as climatic and environmental conditions, geographical terrain, vulnerability to natural disasters, ecologically appropriate practices, use of eco-friendly materials, use of appropriate conventional and alternative technologies, and also socio-economic considerations, towards the creation of sustainable human settlements and built environment.

Buildings can be classified based on occupancy and types of construction as below:

Occupancy classification

a) Residential;
b) Educational;
c) Institutional;
d) Assembly;
e) Business;
f) Mercantile;
g) Industrial;
h) Storage; and
j) Hazardous.

Types of construction — Based on fire resistance rating for different set of components (suitably covered under Part F — Fire and Life Safety) are:

i) Type 1;
ii) Type 2;
iii) Type 3; and
iv) Type 4.

Buildings have various parts (such as plinth, habitable rooms, kitchen, bathroom and water closets, ledges, mezzanine, store room, garage, basement, parapet, cabin, boundary wall, wells, septic tanks, meter rooms, staircases, roofs) and specific requirements such as essential habitable sizes/ heights, distance between well and septic tanks, to name a few that ensure a liveable environment thereby addressing health, safety and public safety. The safe distance between buildings and electric lines is also essential towards occupant safety. Accessibility for persons with disabilities and the elderlies in specific public buildings and low income housing are also essential towards inclusivity in the built spaces. Open spaces within a plot also play a crucial part for harnessing natural lighting and ventilation, apart from movement space for fire tenders/emergency vehicles. Commensurate vehicular parking facilities within a built environment are also important so as to not burden the public spaces like roads/right of way. Such aspects may vary from place to place be it in the plains, hills and coastal areas; for which the state/local authorities should specify the specific criteria.

Based on their complexity, buildings may be classified as high rise buildings or special buildings in which the number of inmates and the requirement for their safe and habitable living are to be addressed. Importantly, the constructed buildings need to be well maintained and periodically audited covering structural audit, electrical audit, fire audit, accessibility audit to be carried out over the useful life of a building, say every 3 to 5 years.

A building project and the built facility in each stage necessarily requires professionals of many disciplines who should work together as a well-coordinated team to achieve the desired product delivery with quality, safety and other objectives, in an effective manner.

In large/complex projects, appropriate multi-disciplinary teams may be constituted to successfully meet the requirements of different stages. These may be design team, project management/execution team and operation and maintenance teams. Each team may comprise need-based professionals out of the following depending upon the nature, magnitude and complexity of the project:

1) architect;
2) civil engineer;
3) structural engineer;
4) geotechnical engineer;
5) electrical engineer;
6) plumbing engineer;
7) fire protection engineer;
8) heating, ventilation and air conditioning engineer;
9) lift, escalator and moving walk specialist;
10) acoustics specialist;
11) information/communication technology engineer;
12) health, safety and environment specialist;
13) environment/sustainability specialist;
14) town planner;
15) urban designer;
16) landscape architect;
17) security system specialist;
18) interior designer;

19) quantity surveyor;

20) project/construction manager;

21) project planning and construction digitization specialist;

22) accessibility and universal design specialist; and

23) other subject specialist(s).

This NBCS (Part A) gives an overall direction for understanding the provisions of different types of buildings and built environment, and proposes guidance for utilizing appropriate knowledge and experience of qualified professionals, right from the conceptualization through construction and completion stages of a building project and indeed during the entire life cycle.

This part has been so formulated as to necessitate guidance to ensure quality and safety in the whole gamut of activities, primarily divided into planning, design and execution of a building or built environment. This may be ensured by obtaining assistance from the registered professionals involved in planning, design, execution, and supervision. Apart from these, various other resource groups are involved whose contribution may be duly taken into account depending on the type, nature, magnitude and complexity of the project, such as security system specialist, environment specialist/sustainability and green building specialist, accessibility specialist, project management consultant, interior designer in proper construction/erection, commissioning and operation of buildings and built environment. In this endeavour, a guide for qualification and competence of the associated professionals, is included in Annex A.

The aim of this Part is to get the maximum benefit from the building and its services in terms of objectives such as quality, timeliness and cost-effectiveness by following a team approach. In the team approach, which is an essential pre-requisite for a unified approach, the aim clearly is to maximize the efficiency of the total system through appropriate optimization of each of its sub-systems, and ensuring coordination among various disciplines and agencies that are involved during planning, design, construction, operation and maintenance of buildings and the associated infrastructure. In other words, in the team, the timely inputs from each of the professional disciplines have to be so optimized that the total system's efficiency becomes maximum. These teams may be design team, construction/execution team, and operation and maintenance team.

It is important that leaders and members of the teams, depending on the size and complexity of the project, are carefully selected considering their qualification, experience and expertise in these fields.

Tools like building information modelling (BIM) may be adopted in all major and complex building projects as they help in managing and sharing information throughout the life cycle of the project enabling better collaboration, planning and decision making. In such projects, BIM may be strategically adopted to enhance the multi-disciplinary approach in design and execution and as a fundamental tool in the planning and development stages.

After the useful service life of a building is over, or for other reasons such as redevelopment and proposed change in use of the land and built facility, it may be required to deconstruct a building. Such a deconstruction is preceded by an organized decommissioning. The decommissioning and deconstruction needs to be well planned and coordinated among concerned building professionals so as to ensure safety during such operations, as also retrieval of appropriate components and systems, for their possible reuse or recycling, or disposal as may be appropriate. This may in turn require a comprehensive decommissioning and deconstruction (including demolition) plan, which may be prepared during the initial stages of the project and kept available for use at the end of life cycle of the same.

ANNEX A

GUIDE FOR THE QUALIFICATIONS AND COMPETENCE OF REGISTERED BUILDING PROFESSIONALS

A-1 ESSENTIAL REQUIREMENTS

Every building/development work should be planned, designed and supervised by registered professionals. The registered professionals for carrying out the various activities should be: (a) architect, (b) engineer, (c) structural engineer, (d) geotechnical engineer, (e) supervisor, (f) town planner, (g) landscape architect, (h) urban designer, and (j) utility services engineer. Requirements of registration for various professionals by the Authority or by the body governing such profession and constituted under a statute, as applicable to practice within the local body's jurisdiction, are given in **A-2.1** to **A-2.8**. The competence of such registered personnel to carry out various activities is also indicated in **A-2.1.1** to **A-2.8.1**.

The qualification and competence of the engineers for utility services and of builder/constructor is prescribed in **A-2.9** and **A-2.10**, respectively. The qualification and competence of the professionals involved in peer reviewing/ proof checking with respect to structural design is prescribed in **A-2.11.**

A-2 REQUIREMENTS FOR REGISTRATION AND COMPETENCE OF BUILDING PROFESSIONALS

A-2.1 Architect

The minimum qualifications for an architect should be the qualifications as provided for in the *Architects Act*, 1972 for registration with the Council of Architecture.

A-2.1.1 *Competence*

The registered architect should be competent to carry out the work related to the building/development permit as given below:

a) Preparation of all plans and information connected with building permit except engineering services of high rise/special buildings;
b) Issuing certificate of supervision and completion of all buildings pertaining to architectural aspects;
c) Preparation of subdivision/layout plans and related information connected with development permit of area up to 1 ha for metro-cities, and 2 ha for other places; and
d) Issuing certificate of supervision for development of land of area up to 1 ha for metro-cities, and 2 ha for other places.

A-2.2 Engineer

The minimum qualifications for an engineer should be graduate in civil engineering/architectural engineering of recognized Indian or foreign university, or the Corporate Member of Civil Engineering Division/Architectural Engineering Division of the Institution of Engineers (India) or the member of the statutory body governing such profession, as and when established.

A-2.2.1 *Competence*

The registered engineer should be competent to carry out the work related to the building/development permit as given below:

a) Preparation of all plans and information connected with building permit;
b) Structural details and calculations of buildings including sub-surface investigation on plot up to 500 m^2 and up to 5 storeys or 16 m in height;
c) Issuing certificate of supervision and completion for all buildings;
d) Preparation of subdivision/layout plans and related information connected with development permit of area up to 1 ha for metro-cities, and 2 ha for other places;

e) Preparation of all service plans and related information connected with development permit; and

f) Issuing certificate of supervision for development of land for all area.

A-2.3 Structural Engineer

The minimum qualifications for a structural engineer should be graduate in civil engineering of recognized Indian or foreign university, or Corporate Member of Civil Engineering Division of Institution of Engineers (India), and with minimum 3 years' experience in structural engineering practice with designing and field work.

NOTE — The 3 years' experience may be relaxed to 2 years in the case of post graduate degree of recognized Indian or foreign university in the branch of structural engineering. In case of doctorate in structural engineering, the experience required would be one year.

A-2.3.1 *Competence*

The registered structural engineer should be competent to prepare the structural design, calculations and details for all buildings and carry out supervision.

A-2.3.1.1 In case of buildings having special structural features, as decided by the Authority, which are within the horizontal areas and vertical limits specified in **A-2.2.1**(b) and **A-2.5.1**(a) should be designed only by structural engineers.

A-2.4 Geotechnical Engineer

The minimum qualifications for a geotechnical engineer should be graduate in civil engineering of recognized Indian or foreign university, or Corporate Member of Civil Engineering Division of Institution of Engineers (India), and with minimum 3 years' experience in geotechnical engineering practice with designing and field work.

NOTE — The 3 years' experience may be relaxed to 2 years in the case of post graduate degree of recognized Indian or foreign university in the branch of geotechnical engineering. In case of doctorate in geotechnical engineering, the experience required would be one year.

A-2.4.1 *Competence*

The registered geotechnical engineer should be competent to carry out sub-surface investigations and give report thereof. These may *inter-alia* include performing various tests required to determine engineering properties of sub-strata and ground water and making recommendations about the type of foundation, soil bearing capacity and the depth at which the foundations shall be placed, considering the structural system and loads provided by the engineer/structural engineer.

A-2.4.1.1 In case of buildings to be founded on soils having special geotechnical conditions, as decided by the Authority, which are within the horizontal areas and vertical limits specified in **A-2.2.1**(b) and **A-2.5.1**(a) should be designed only by geotechnical engineers.

A-2.4.1.2 For specific information on pre-construction services and construction stage services to be provided by the geotechnical engineers, reference may be made to IS 19235 : 2025 'Geotechnical engineering services — Requirements'.

A-2.5 Supervisor

The minimum qualifications for a supervisor should be diploma in civil engineering or architectural assistantship, or the qualification in architecture or engineering equivalent to the minimum qualification prescribed for recruitment to non-gazetted service by the Government of India plus 5 years' experience in building design, construction and supervision.

A-2.5.1 *Competence*

The registered supervisor should be competent to carry out the work related to the building permit as given below:

a) All plans and related information connected with building permit for residential buildings on plot up to 200 m^2 and up to two storeys or 7.5 m in height; and

b) Issuing certificate of supervision for buildings as per (a).

A-2.6 Town Planner

The minimum qualification for a town planner should be the Associate Membership of the Institute of Town Planners or graduate or post-graduate degree in town and country planning.

A-2.6.1 *Competence*

The registered town planner should be competent to carry out the work related to the development permit as given below:

a) Preparation of plans for land subdivision/layout and related information connected with development permit for all areas; and

b) Issuing of certificate of supervision for development of land of all areas.

NOTE — However, for land layouts for development permit above 5 ha in area, landscape architect may also be associated, and for land development infrastructural services for roads, water supplies, sewerage/drainage, electrification, the registered engineers for utility services may be associated.

A-2.7 Landscape Architect

The minimum qualification for a landscape architect should be the bachelor master's degree in landscape architecture or equivalent from recognized Indian or foreign university.

A-2.7.1 *Competence*

The registered landscape architect should be competent to carry out the work related to landscape design for building/development permit for land areas 5 ha and above. In case of metro-cities, this limit of land area may be 2 ha and above.

NOTE — For smaller areas below the limits indicated above, association of landscape architect may also be considered from the point of view of desired landscape development.

A-2.8 Urban Designer

The minimum qualification for an urban designer should be the master's degree in urban design or equivalent from recognized Indian or foreign university.

A-2.8.1 *Competence*

The registered urban designer should be competent to carry out the work related to the building permit for urban design for land areas more than 5 ha and campus area more than 2 ha. He/she should also be competent to carry out the work of urban renewal for all areas.

NOTE — For smaller areas below the limits indicated above, association of urban designer may be considered from the point of view of desired urban design.

A-2.9 Engineers for Utility Services

The work of building and plumbing services should be executed under the planning, design and supervision of competent personnel. The qualification for registered mechanical engineer (including HVAC), electrical engineer and plumbing engineers for carrying out the work of air conditioning, heating and mechanical ventilation, electrical installations, lifts and escalators and water supply, drainage, sanitation and gas supply installations respectively should be as given in Part D 'Building Services' and Part E 'Plumbing Services' of this NBCS or as decided by taking into account practices of the national professional bodies dealing with the specialist engineering services.

Such an approach may be followed for association of other/multi-disciplinary professionals for taking inputs and associating with their areas of specialization. For example, specific to fire (as covered under Part F 'Fire and Life Safety' of this NBCS), the role of fire protection engineer is important, who could be, any of the following:

a) a graduate in fire protection engineering (or a related discipline like fire engineering, fire safety engineering, fire technology and safety engineering, of recognized Indian or foreign university);

b) a graduate in mechanical/civil/electrical/electronics engineering and having a post graduate degree in fire protection engineering;

c) a graduate in mechanical/civil/electrical/electronics engineering and having at least 10 years of experience in the field of planning, designing, and commissioning of fire protection work;

d) an advance diploma in fire engineering (Divisional Officers Course) from National Fire Service College, Nagpur and having atleast five years' experience in the field of fire prevention, fire protection and fire safety matters; and

e) a diploma in fire engineering from National Fire Service College, Nagpur and having atleast ten years' experience in the field of fire prevention, fire protection and fire safety matters.

NOTE — The existing fire protection professionals having graduation in science or diploma in engineering having experience of at least 10 years may be allowed for the activity.

A-2.10 BUILDER/CONSTRUCTOR

The minimum qualification for the builder/constructor or his representative for execution of respective works should be as given in **A-2.1**, **A-2.2**, **A-2.3**, **A-2.4**, **A-2.5**, **A-2.6**, **A-2.7**, **A-2.8** and **A-2.9** for the concerned professional.

A-2.10.1 Competence

The qualified builder/constructor or his representative should be competent to carry out execution of work, which should have the same extent as for supervision by such professional as prescribed in **A-2.1.1**, **A-2.2.1**, **A-2.3.1**, **A-2.4.1**, **A-2.5.1**, **A-2.6.1**, **A-2.7.1**, **A-2.8.1** and **A-2.9**.

A-2.11 PEER REVIEWER/PROOF CHECKER (FOR STRUCTURAL DESIGN)

The individual peer reviewer/proof checker or the team leader of the peer review/proof checking organization should have the following qualification, experience and competence:

a) *For buildings up to the height of 15 m*

   B.E./B.Tech (Civil) with 7 years of experience in structural engineering practice with designing and field work of relevant structures. The 7 years of experience should comprise a minimum of 5 years exclusively in structural designing.

b) *For buildings of height more than 15 m and up to 50 m*

   B.E./B.Tech (Civil) with 10 years of experience in structural engineering practice with designing and field work of relevant structures. The 10 years of experience should comprise a minimum of 7 years exclusively in structural designing.

c) *For buildings of height more than 50 m and specialized structures*

   1) Master's degree with major in structural engineering and 10 years of experience in structural engineering practice with designing and field work of relevant structures. The 10 years of experience should comprise a minimum of 7 years exclusively in structural designing; or

   2) B.E./B.Tech (Civil) with 15 years of experience in structural engineering practice with designing and field work of relevant structures. The 15 years of experience should comprise a minimum of 10 years exclusively in structural designing.

d) Peer review may also be carried out by reputed academic institutions/research institutions (such as IITs, NITs, IISc, leading engineering colleges, CSIR-SERC, CBRI, CRRI).

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART B BUILDING MATERIALS

BUREAU OF INDIAN STANDARDS

CONTENTS

National Building Code Sectional Committee, CED 46

FOREWORD

This NBCS (Part B) lists Indian Standard specifications for materials used in building construction. The Indian Standards for methods of tests, to ensure the requirements demanded of the materials in the various situations, are also included. This is with a view to ensuring the quality and effectiveness of building materials used in the construction and of their storage, which are as important as other aspects of building activity like planning, designing and constructing the building itself.

Historically, choice of building materials was determined by what was locally available, appropriateness to geo-climatic conditions and affordability of users. In the recent past, different initiatives have been taken in the areas of research and development, standardization, and development and promotion of innovative materials. A review of the recent trends indicates that the growth in the area of building materials covers emerging trends and latest developments in the use of wastes, mineral admixtures in cement and concrete, substitutes to conventional timber, composite materials and recycling of wastes, at the same time ensuring desired response of materials in relation to fire safety, long term performance and durability. In addition to these, nowadays, development of specific materials is being researched which may be structured and designed to meet needs to specially developed construction technologies in time to come, such as, for disaster prone areas or aggressive climatic and industrial situations.

To encourage use of appropriate materials, it may be desirable to have to the largest extent possible, performance oriented approach for specifications rather than prescriptive approach. The approach has been already adopted in some cases in development of standards, wherever found possible.

Indian Standards cover the requirements for most of the materials in use. However, there may be a gap between development of new materials and techniques of application and formulation of standards. It, therefore, becomes necessary for the NBCS to be flexible to recognize building materials other than those for which Indian Standards are available. This Part of NBCS, therefore, since its first version, duly takes care of this aspect and explicitly provides for use of new or alternate building materials, provided it is proved by authentic tests that the new or alternative material is effective and suitable for the purpose intended. However, it is worthwhile that more and more emphasis is given to the satisfaction of performance requirements expected of a building material, so that a wide range of such new or alternate materials can be evaluated and used, if found appropriate.

As already emphasized, quality of material is quite important for their appropriate usage, whether it is a material for which an Indian Standard is available or a new or alternative material as defined in **4** of this Part of the NBCS. While continuing to emphasize on conformity of building materials to available Indian Standards, this Part recommends that the building regulating authority may also recognize the use of building materials conforming to other specifications and test methods (*see* **4.2**), in case Indian Standards are not available for particular materials. Third party certification schemes available in the country for quality assurance of above materials can be used with advantage to ensure the appropriateness of these materials.

It is well recognized that buildings during their construction and subsequently in their operation consume lot of resources. The holistic approach for ensuring proper and sustainable developments is to take care of the concerned aspects affecting sustainability during the conceptualization, planning, design, construction, and operation and maintenance stages. The building material production, transportation and utilization assume a substantial component of the embodied energy that goes into a building. It is therefore important that sustainability of building materials is well understood and ensured in a building project. This Part of the NBCS recognizes this aspect and covers appropriate provision therefore.

This Part of the Code was first published in 1970 as part of the National Building Code of India and subsequently revised in 1983, 2005 and 2016 as Part 5 of NBC 2016. The first and second revision of this Part incorporated an updated version of the list of Indian Standards given at the end of this Part of the Code. Also the list of building materials was completely revised to make it more user-friendly. The third revision of this Part included a new clause on sustainable building materials, expanded the clause on new/alternative materials to provide examples of some potential new/alternative materials, reorganized some of the existing categories and added new categories either under the existing principal categories or as separate principal categories. Additionally, the list of Indian Standards was updated.

In this revision, this publication has been renamed as National Building Construction Standards (NBCS).

This fourth revision of this Part, while basically retaining the structure of previous version, incorporates the following modifications:

a) The clause on new/alternative materials has been updated with examples of some potential new/alternative materials;

b) A clause on recycled materials has been added to address/encourage the usage of recycled materials in building construction; and

c) The list at the end of this Part has been updated to reflect the latest available Indian Standard specifications and methods of test.

A reference to SP 21 : 2005 'Summaries of Indian Standards for Building Materials' may be useful. This publication gives the summaries of Indian Standards covering various building materials, fittings and components except standards relating to paints.

All standards cross-referred in the main text of this Part, are subject to revision. The parties to agreement based on this Part are encouraged to investigate the possibility of applying the most recent editions of the standards.

As the regulation of land development and buildings is a State subject and is dealt by the States and local bodies such as urban local bodies within their jurisdiction through building regulatory documents like building regulations, building byelaws, and fire regulations, this document is voluntary in nature and is non-binding. The implementation depends on adoption by concerned parties or stipulations in a contract or by appropriate adoption by the concerned authorities.

For the purpose of deciding whether a particular requirement of this Part is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 2022 'Rules for rounding off numerical values (*second revision*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this Part.

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART B BUILDING MATERIALS

1 SCOPE

This NBCS (Part B) covers the requirements of building materials and components, and criteria for accepting new or alternative building materials and components.

2 MATERIALS

Every material used in fulfilment of the requirements of this Part, unless otherwise specified in this NBCS or approved, shall conform to the relevant Indian Standards. A list of Indian Standard specifications as the 'accepted standards' is given at the end of this Part of the NBCS. At the time of publication of this NBCS, the editions indicated were valid. All standards are subject to amendments and revisions. The Authority shall take cognizance of such amendments and revisions. The latest version of a specification shall, as far as possible, be adopted at the time of enforcement of this Part of the NBCS.

3 SUSTAINABLE MATERIALS

Choice of building materials is important in sustainable design because of the extensive network of activities such as extraction, processing and transportation steps required for making a material, and activities involved thereafter till building construction and even thereafter. These activities may pollute the air, soil and water, as well as destroy natural habitats and deplete natural resources. One of the most effective strategies for minimizing the environmental impacts of material usage is to reuse existing buildings. Rehabilitation of existing building, and their shell and non-shell components, not only reduces the volume of solid waste generated and its subsequent diversion to landfills but also the environmental impacts associated with the production, delivery and use or installation of new building materials. However, the use of sustainable building materials is one of the best strategies in the pursuit to sustainable buildings.

An ideal sustainable building material is not only environment friendly but also causes no adverse impact on health of occupants, is readily available, can be reclaimed and recycled, and is not only made from renewable raw material, but also uses predominantly renewable energy in its extraction, production, transportation, fixing and ultimate disposal. Practically, this kind of ideal material may not be available and hence, it may be best to choose materials which fulfil most of these criteria. Sustainable building materials also offer specific benefits to the building owner and building occupants, such as, reduced maintenance/replacement costs over the life of the building; energy conservation; improved occupant health and productivity; lower costs associated with changing space configurations; and greater design flexibility.

The use of used materials (*see* **6**) and recycled and waste materials (*see* **7**) contribute substantially towards sustainability of building material usage.

The selection of sustainable building materials may be made in accordance with relevant guidelines/best practices (Handbook on 'Sustainability in Buildings').

4 NEW OR ALTERNATIVE MATERIALS

4.1 The provisions of this Part are not intended to prevent the use of any material not specifically prescribed under **2**. Any such material may be approved by the Authority or an agency appointed by them for the purpose, provided it is established that the material is satisfactory for the purpose intended and the equivalent of that required in this Part or any other specification issued or approved by the Authority. The Authority or an agency appointed by them shall take into account the following parameters, as applicable to the concerned new or alternative building material:

a) Requirements of the material specified/expected in terms of the provisions given in the standards on its usage, including its applicability in geo-climatic condition;

b) General appearance;

c) Dimension and dimensional stability;

d) Structural stability including strength properties;

e) Fire safety;

f) Durability;

g) Thermal properties;

h) Mechanical properties;

j) Acoustical properties;

k) Optical properties;

m) Biological effect;

n) Environmental aspects;

p) Working characteristics;

q) Ease of handling;

r) Consistency and workability;

s) UV resistance; and

t) Toxicity.

Some of such new/alternative materials may be decorative concrete, polymer concrete, plastic waste bricks, recycled plastic lumber, micro-concrete repair materials, dry-mix mortar, non-shrink grout, special materials for bunkers/blast resistant structures, artificial stones, nanotechnology based advanced materials, vermiculite based boards, exfoliated perlite, pervious concrete, high-pressure laminate sheet, aluminium honeycomb panels, zinc composite panels, stainless steel insulated water tanks, etc.

For establishing the performance of the material/component, laboratory/field tests, and field trials, as required, and study of historical data are recommended. For sampling and frequency of tests for new or alternate materials, similar product standards shall be referred to.

The above materials would no longer be treated as new/alternative materials as soon as Indian Standards for the same are established.

4.2 Approval in writing of the Authority or an agent appointed by them for the purpose of approval of material, shall be obtained by the owner or his agent before any new, alternative or equivalent material is used. The Authority or their agent shall base such approval on the principle set forth in **4.1** and shall require that tests be made (*see* **9.1**) or sufficient evidence or proof be submitted, at the expense of the owner or his agent, to substantiate any claim for the proposed material.

5 THIRD PARTY CERTIFICATION

For ensuring the conformity of materials for which Indian Standards exist and for new or alternative building materials, to requisite quality parameters the services under the third party certification schemes of the Government, may be utilized with advantage.

6 USED MATERIALS

Utilization of used materials and recycled materials may not be precluded provided these meet the requirements of this Part for new materials.

7 RECYCLED MATERIALS AND WASTE MATERIALS

7.1 A recycled building material refers to any product or material that has been previously utilized in another construction context. Various materials, such as brick, steel, timber, and even entire components like doors, windows and tiles, can be reused and repurposed. This term also encompasses building materials produced from waste, such as recycled plastic bricks, or concrete made from recycled aggregate from buildings, structures and pavements. Using suitable technological interventions, the natural and industrial by-products/ wastes can be gainfully utilized in the construction activity, not merely as fillers, but in combination with traditional materials or through repurposing. Some such potential materials include: fly ash, bottom ash, red mud, flue gas desulfurization (FGD) gypsum, phosphogypsum, fluorogypsum, blast furnace slag, Linz-Donawitz (LD) slag, textile sludge, marble waste (from cutting and processing), granite waste (from cutting and processing), iron tailings, mine tailings, copper tailings, copper slag, coal mining waste (coal tailings), rice husk ash, jarosite, broken glass and ceramics, kiln dust, galvanizing waste, tannery waste, stone waste (from crusher units), metal plating industry sludge, lime sludge, vermiculite, perlite and bagasse and other agro-wastes. One of such recycled/waste derived materials is aggregate derived from construction and demolition (C&D) waste, as described in **7.1.1** and **7.1.2**.

7.1.1 *Recycled Mixed Aggregate and Recycled Concrete Aggregate*

Recycled mixed aggregate (RMA) is made from construction and demolition (C&D) waste which may comprise concrete, brick, tiles, stones, etc. C&D waste is generated from construction, renovation, repair, and demolition of buildings and other structures. Recycled concrete aggregate (RCA) on the other hand is derived from concrete waste after requisite processing. Recycled concrete aggregate (RCA) comprises not only the original aggregate, but also the hydrated cement paste adhering to its surface.

7.1.2 *Reclaimed Asphalt Pavement Aggregates*

Milling or demolition of distressed bituminous/flexible pavements generates reclaimed asphalt pavement (RAP) aggregates. These aggregates consist of natural stones fully coated with 2 percent to 7.5 percent (by mass) of bitumen/asphalt binder and are typically used in the construction of new bituminous roads. The use of RAP in cement concrete pavements is rather a new approach and has been found to enhance the workability and toughness characteristics. The strength properties should also be evaluated before permitting for use. Suitably, the coarser fraction of RAP can be used as part replacement (in the order of 10 percent to 20 percent) of natural aggregates for non-structural lean-concrete applications such as paver blocks, kerb stones, etc.

8 STORAGE OF MATERIALS

All building materials shall be stored on the building site in such a way as to prevent deterioration or the loss or impairment of their structural and other essential properties [see relevant guidelines/best practices (Handbook on ‘Construction Management, Practices and Safety in Buildings’)].

9 METHODS OF TEST

9.1 Every test of material as required in this Part or by the Authority shall be carried out in accordance with the methods of test prescribed in relevant Indian Standards. In cases of methods of tests for which Indian Standards are not available, the same shall conform to the methods of tests issued by the Authority or their agent. A list of Indian Standard methods of test is given at the end of this Part of the NBCS as the ‘accepted standards’. Laboratory tests shall be conducted by recognized laboratories acceptable to the Authority.

9.1.1 The manufacturer/supplier shall satisfy himself that materials conform to the requirements of the specifications and if requested shall supply a certificate to this effect to the purchaser or his representative. When such test certificates are not available, the specimen of the material shall be tested.

LIST OF STANDARDS

Following are the Indian Standards for various building materials and components, to be complied with in fulfilment of the requirements of this NBCS.

In the following list, while enlisting the Indian Standards, the materials have been categorized in such a way as to make the list user friendly. In the process, if so required, some of the standards have been included even in more than one category of products, such as in the category based on composition as well as on end application of the materials. The list has been arranged in alphabetical order of their principal category as given below:

1 Aluminium and other light metals and their alloys
2 Bitumen and tar products
3 Bricks, blocks and other masonry building units
4 Builder’s hardware
5 Building chemicals
6 Building lime and products
7 Clay and stabilized soil products
8 Cement and concrete (including steel reinforcement for concrete)
9 Composite matrix products (including cement and resin matrix products)
10 Conductors and cables
11 Doors, windows and ventilators
12 Electrical wiring, fittings and accessories
13 Fillers, stoppers and putties
14 Floor covering, roofing and other finishes
15 Glass
16 Gypsum based materials
17 Mortar (including sand for mortar)
18 Paints and allied products
19 Polymers, plastics and geosynthetics/geotextiles
20 Sanitary appliances and water fittings
21 Steel and its alloys
22 Stones
23 Structural sections
24 Thermal insulation materials
25 Threaded fasteners, rivets and nails
26 Timber, bamboo and other lignocellulosic building materials
27 Unit weights of building materials
28 Waterproofing and damp-proofing materials
29 Welding electrodes and wires
30 Wire ropes and wire products (including wire for fencing)

1 ALUMINIUM AND OTHER LIGHT METALS AND THEIR ALLOYS

IS No.	*Title*
IS 733 : 2025	Wrought aluminium and aluminium alloy bars, rods, sections, extruded tubes, hollow sections and profiles for general engineering purposes — Specification (*fourth revision*)
IS 734 : 1975	Specification for wrought aluminium and aluminium alloy forging stock and forgings (for general engineering purposes) (*second revision*)
IS 736 : 1986	Specification for wrought aluminium and aluminium alloys plate for general engineering purposes (*third revision*)
IS 737 : 2024	Wrought aluminium and aluminium alloys sheet and strip for general engineering purposes — Specification (*fifth revision*)
IS 738 : 1994	Wrought aluminium and aluminium alloys — Drawn tube for general engineering purposes — Specification (*third revision*)
IS 739 : 1992	Wrought aluminium and aluminium alloys — Wire for general engineering purposes — Specification (*third revision*)
IS 740 : 1977	Specification for wrought aluminium and aluminium alloys rivet stock for general engineering purposes (*second revision*)
IS 1254 : 2025	Corrugated aluminium sheet (*fifth revision*)
IS 1284 : 2024	Wrought aluminium alloys bolt and screw stock (for general engineering purposes) — Specification (*third revision*)
IS 1500	Metallic materials — Brinell hardness test:
(Part 1) : 2019	Test method (*fifth revision*)
(Part 2) : 2021	Verification and calibration of testing machines (*fifth revision*)
(Part 3) : 2019	Calibration of reference blocks (*fifth revision*)
(Part 4) : 2019	Table of hardness values (*fifth revision*)
IS 1501	Metallic materials — Vickers hardness test:
(Part 1) : 2025	Test method (*sixth revision*)
(Part 2) : 2020	Verification and calibration of testing machines (*fifth revision*)
(Part 3) : 2020	Calibration of reference blocks (*fifth revision*)
(Part 4) : 2020	Tables of hardness values (*fifth revision*)
IS 1586	Metallic materials — Rockwell hardness test:
(Part 1) : 2025/ ISO 6508-1 : 2023	Test method (*sixth revision*)
(Part 2) : 2025/ ISO 6508-2 : 2023	Verification and calibration of testing machines and indenters (*sixth revision*)

IS No.	*Title*
IS 1586 (Part 3) : 2025/ ISO 6508-3 : 2023	Metallic materials — Rockwell hardness test: Part 3 Calibration of reference blocks (*sixth revision*)
IS 1716 : 2023	Metallic materials wire reverse bend test (*third revision*)
IS 1717 : 2018	Metallic materials — Wire — Simple torsion test (*fourth revision*)
IS 1757	Metallic materials — Charpy pendulum impact test:
(Part 1) : 2020	Test method (*fourth revision*)
(Part 2) : 2020	Verification of testing machines (*fourth revision*)
(Part 3) : 2020	Preparation and characterization of charpy V-notch test pieces for indirect verification of pendulum impact machines (*fourth revision*)
IS 1772 : 2024	Electroplated coatings of copper — Specification (*second revision*)
IS 1828 (Part 1) : 2022	Metallic materials — Calibration and verification of static uniaxial testing machines: Part 1 Tension/Compression testing machines — Calibration and verification of the force-measuring system (*fifth revision*)
IS 2258 : 2023	Rolled zinc plate, sheet and strip — Specification (*third revision*)
IS 2328 : 2018	Metallic materials — Tube — Flattening test (*third revision*)
IS 2479 : 1981	Colour code for the identification of aluminium and aluminium alloys for general engineering purposes (*second revision*)
IS 2605 : 2023	Zinc anodes for electroplating — Specification (*second revision*)
IS 2676 : 1981	Dimensions for wrought aluminium and aluminium alloys, sheet and strip (*first revision*)
IS 2677 : 1979	Dimensions for wrought aluminium and aluminium alloys, plates and hot rolled sheets (*first revision*)
IS 2782 : 2023	Refined nickel — Specification (*first revision*)
IS 3577 : 1992	Wrought aluminium and its alloys — Rivet, bolt and screw stock — Dimensions and tolerances (*first revision*)
IS 3711 : 2020	Steel and steel products — Location and preparation of samples and test pieces for mechanical testing (*third revision*)
IS 3965 : 1981	Dimensions for wrought aluminium and aluminium alloys bar, rod and section (*first revision*)
IS 4258 : 2018	Metallic materials — Conversion of hardness values (*third revision*)
IS 5075 : 2023	Metallic materials — Rotating bar bending fatigue testing (*second revision*)
IS 5528 : 2024/ ISO 9227 : 2022	Corrosion tests in artificial atmospheres — Salt spray tests (*second revision*)
IS 6477 : 1983	Dimensions for wrought aluminium and aluminium alloys extruded hollow sections (*first revision*)

IS No.	*Title*
IS 6885	Metallic materials — Knoop hardness test:
(Part 1) : 2025/ ISO 4545-1 : 2023	Test method (*third revision*)
(Part 2) : 2020	Verification and calibration of testing machines (*second revision*)
(Part 3) : 2020	Calibration of reference blocks (*second revision*)
(Part 4) : 2020	Tables of hardness values (*second revision*)
IS 7811 : 2019	Phosphor bronze rods and bars (*second revision*)
IS 8632 : 2023	Metallic materials — Designation of test specimen axes in relation to product texture (*first revision*)
IS 10175 : 2018	Metallic materials — Sheet and strip — Erichsen cupping test (*third revision*)
IS 11999 : 2022	Metallic materials — Sheet and strip — Determination of plastic strain ratio (*second revision*)
IS 12260 : 2018	Metallic materials — Tube — Ring —Expanding test (*first revision*)
IS 12278 : 2018	Metallic materials — Tube — Ring tensile test (*second revision*)
IS 12444 : 2020	Copper wire rods for electrical applications — Specification (*first revision*)
IS 12871 : 2022	Copper, lead, zinc and nickel concentrates — Sampling procedures for determination of metal and moisture content (*second revision*)
IS 14712 : 2025	Wrought aluminium and its alloys — Chequered/tread sheets for general engineering purposes — Specification (*first revision*)
IS 15756 : 2022	Metallic materials — Sheet and strip — Determination of tensile strain hardening exponent (*first revision*)
IS 15965 : 2012	Pre-painted aluminium zinc alloy metallic coated steel strip and sheet (Plain)
IS 16842 : 2022	Metallic materials — Fatigue testing — Fatigue crack growth method (*first revision*)
IS 16843 : 2018	Metallic materials — Fatigue testing — Strain-controlled thermomechanical fatigue testing method
IS 17143 : 2023	Metallic materials torque — Controlled fatigue testing (*first revision*)
IS 17144	Metallic materials — Instrumented indentation test for hardness and materials parameters:
(Part 1) : 2019	Test method
(Part 2) : 2019	Verification and calibration of testing machines
(Part 3) : 2019	Calibration of reference blocks
(Part 4) : 2019	Test method for metallic and non-metallic coatings

IS No.	*Title*
IS 17145	Metallic materials — Fatigue testing — Variable amplitude fatigue testing:
(Part 1) : 2019	General principles, test method and reporting requirements
(Part 2) : 2019	Cycle counting and related data reduction methods
IS 17149	Metallic materials — Leeb hardness test:
(Part 1) : 2019	Test method
(Part 2) : 2019	Verification and calibration of the testing devices
(Part 3) : 2019	Calibration of reference test blocks
IS 17151 : 2023	Metallic materials unified method of test for the determination of quasistatic fracture toughness (*first revision*)
IS 17413	Metallic materials — Tensile testing at high strain rates:
(Part 1) : 2020	Elastic-bar-type systems
(Part 2) : 2020	Servo-hydraulic and other systems
IS 17415 : 2023	Metallic materials torsion test at room temperature (*first revision*)
IS 17416 : 2024/ ISO 14556 : 2023	Metallic materials — Charpy V-Notch pendulum impact test — Instrumented test method (*first revision*)
IS 17418 : 2020	Metallic materials — Sheet and strip — Biaxial tensile testing method using a cruciform test piece
IS 17146 (Part 1) : 2023	Metallic materials determination of forming-limit curves for sheet and strip: Part 1 Measurement and application of forming-limit diagrams in the press shop (*first revision*)
IS 17417	Metallic materials — Dynamic force calibration for uniaxial fatigue testing:
(Part 1) : 2020	Testing systems
(Part 2) : 2020	Dynamic calibration device (DCD) instrumentation
IS 17419 : 2025	Metallic materials — Ductility testing — High speed compression test for porous and cellular metals (*first revision*)
IS 17679 : 2021	Metallic materials — Method of test for the determination of quasistatic fracture toughness of welds
IS 17682 : 2021	Aluminium composite panel specification
IS 17795 : 2025	Metallic materials — Uniaxial creep testing in tension — Method of test (*first revision*)
IS 17915 : 2022	Method for evaluation of tensile properties of metallic superplastic materials
IS 17937 : 2022	Mechanical testing of metals ductility testing compression test for porous and cellular metals
IS 18649 : 2024/	Corrosion of metals and alloys — Alternate immersion test in salt solution

IS No.	*Title*
ISO 11130 : 2017	
IS 18650 : 2024/ ISO 26146 : 2012	Corrosion of metals and alloys — Method for metallographic examination of samples after exposure to high-temperature corrosive environments

2 BITUMEN AND TAR PRODUCTS

IS No.	*Title*
IS 73 : 2013	Paving bitumen — Specification (*fourth revision*)
IS 212 : 2021	Crude coal tar for general use (*third revision*)
IS 216 : 2006	Coal tar pitch — Specification (*second revision*)
IS 217 : 2018	Cutback bitumen — Specification (*third revision*)
IS 218 : 1983	Specification for creosote oil for use as wood preservatives (*second revision*)
IS 334 : 2023	Glossary of terms relating to bitumen and tar (*fourth revision*)
IS 702 : 2022	Specification for industrial bitumen (*third revision*)
IS 1201 to IS 1220	Methods for testing tar and bituminous materials:
IS 1201 : 2021	Sampling (*second revision*)
IS 1202 : 2021	Determination of specific gravity (*second revision*)
IS 1203 : 2022	Determination of penetration (*second revision*)
IS 1204 : 2022	Determination of residue of specified penetration (*second revision*)
IS 1205 : 2022	Determination of softening point — Ring and ball apparatus (*second revision*)
IS 1206	Determination of viscosity:
(Part 1) : 2023	By volumetric flow rate method
(Part 2) : 2022	Absolute viscosity
(Part 3) : 2021	Kinematic viscosity
IS 1207 : 2022	Determination of equiviscous temperature EVT (*second revision*)
IS 1208	Methods for Testing Tar and Bituminous Materials
(Part 1) : 2023	Determination of ductility (*second revision*)
(Part 2) : 2023	Determination of elastic recovery (*second revision*)
IS 1209 : 2021	Determination of flash point and fire point (*second revision*)
IS 1210 : 2020	Float test (*second revision*)
IS 1211 : 2022	Determination of water content dean and stark method (*second revision*)
IS 1212 : 2022	Determination of loss on heating (*second revision*)

IS No.	*Title*
IS 1213 : 2025	Tar and bituminous materials — Distillation test — Methods of test (*third revision*)
IS 1215 : 2021	Determination of matter insoluble in toluene (*second revision*)
IS 1216 : 2023	Determination of solubility in trichloroethylene (*second revision*)
IS 1217 : 2024	Tar and bituminous materials — Determination of mineral matter (ash) — Methods of test (*second revision*)
IS 1218 : 2024	Tar and bituminous materials — Methods for test — Determination of phenols (*second revision*)
IS 1219 : 2024	Tar and bituminous materials — Methods for test — Determination of naphthalene (*second revision*)
IS 1220 : 2024	Tar and bituminous materials — Methods for test — Determination of volatile matter content *(second revision)*
IS 3117 : 2004	Bitumen emulsion for roads and allied applications (anionic type) — Specification (*first revision*)
IS 8887 : 2018	Bitumen emulsion for roads cationic type — Specification *(third revision)*
IS 9381 : 2024	Tar and bituminous materials — Determination of Fraass breaking point of bitumen — Methods of test (*first revision*)
IS 9382 : 2022	Methods for testing tar and bituminous materials — Determination of effect of heat and air by thin film oven test (*first revision*)
IS 10511 : 2024	Method for determination of asphaltenes in bitumen by precipitation with normal heptane (*first revision*)
IS 10512 : 2003	Method for determination of wax content in bitumen (*first revision*)
IS 13758	Coal tar pitch — Methods of test:
(Part 1) : 1993	Determination of matter insoluble in quinoline
(Part 2) : 1993	Determination of coking value
IS 14982 : 2022	Anti-Stripping agents for bitumen addition — Specification (*second revision*)
IS 15172 : 2002	Methods for testing tar and bituminous materials — Determination of curing index for cutback bitumens
IS 15173 : 2002	Methods for testing tar and bituminous materials — Determination of breaking point for cationic bitumen emulsion
IS 15174 : 2002	Methods for testing tar and bituminous material — Determination of breaking point for anionic bitumen emulsion
IS 15462 : 2019	Polymer modified bitumen (PMB) specification (*first revision*)
IS 15799 : 2021	Methods for testing tar and bituminous materials — Ageing of bitumen (RTFOT and PAV) (*first revision*)
IS 15808 : 2008	Multigrade bitumen for use in pavement construction — Specification
IS 17016 : 2018	Cationic modified bitumen emulsion — Specification

IS No.	*Title*
IS 17052 : 2018	Rejuvenating agents for hot mix recycling of bituminous surfacing — Specification
IS 17079 : 2019	Rubber modified bitumen (RMB) — Specification
IS 17116 : 2019	Determination of resistance of compacted bituminous mixture to moisture — Induced damage
IS 17124 : 2019	Crack sealants — Specification
IS 17127 : 2019	Determination of stability, flow and bulk density of bituminous mixes by Marshall method
IS 17335 : 2020	Determination of apparent viscosity of bituminous binders at elevated temperature by rotational viscometer
IS 18104 : 2022	Methods for testing tar and bituminous materials — Determination of draindown characteristics of uncompacted bitumen mixtures
IS 18130 : 2023	Method for recovery of bitumen from solution using rotary evaporator
IS 18883 : 2025	Trackless emulsion — Specification
IS 19008 : 2023	Determination of bitumen content in bituminous paving mixtures

3 BRICKS, BLOCKS AND OTHER MASONRY BUILDING UNITS

IS No.	*Title*
IS 1077 : 2025	Common burnt clay building bricks — Specification (*sixth revision*)
IS 1725 : 2023	Stabilized soil blocks used in general building construction — Specification (*third revision*)
IS 2180 : 1988	Specification for heavy duty burnt clay building bricks (*third revision*)
IS 2185	Specification for concrete masonry units:
(Part 1) : 2005	Hollow and solid concrete blocks (*third revision*)
(Part 2) : 1983	Hollow and solid lightweight concrete blocks (*first revision*)
(Part 3) : 1984	Autoclaved cellular (aerated) concrete blocks (*first revision*)
(Part 4) : 2008	Preformed foam cellular concrete blocks
IS 2222 : 1991	Specification for burnt clay perforated building bricks (*third revision*)
IS 2691 : 2017	Burnt clay facing bricks — Specification (*third revision*)
IS 2849 : 1983	Specification for non-load bearing gypsum partition blocks (solid and hollow types) (*first revision*)
IS 3115 : 1992	Specification for lime based blocks (*second revision*)
IS 3316 : 1974	Specification for structural granite (*first revision*)
IS 3495	Burnt clay building bricks method of tests:

IS No.	*Title*
(Part 1) : 2019	Determination of compressive strength (*fourth revision*)
(Part 2) : 2019	Determination of water absorption (*fourth revision*)
(Part 3) : 2019	Determination of efflorescence (*fourth revision*)
(Part 4) : 2019	Determination of warpage (*fourth revision*)
(Part 5) : 2021	Determination of initial rate of absorption
(Part 6) : 2022	Determination of modulus of rupture
IS 3583 : 1988	Specification for burnt clay paving bricks (*second revision*)
IS 3620 : 1979	Specification for laterite stone block for masonry (*first revision*)
IS 3952 : 2013	Burnt clay hollow bricks and blocks for walls and partitions — Specification (*third revision*)
IS 4139 : 1989	Specification for calcium silicate bricks (*second revision*)
IS 4860 : 1968	Specification for acid resistant bricks
IS 4885 : 1988	Specification for sewer bricks (*first revision*)
IS 5454 : 2024	Burnt clay bricks and burnt clay tiles — Methods of sampling (*second revision*)
IS 5751 : 1984	Specification for precast concrete coping blocks (*first revision*)
IS 5779 : 1986	Specification for burnt clay soling bricks (*first revision*)
IS 6165 : 1992	Dimensions for special shapes of clay bricks (*first revision*)
IS 9893 : 1981	Specification for precast concrete blocks for lintels and sills
IS 10360 : 1982	Specification for lime pozzolana concrete blocks for paving
IS 12440 : 1988	Specification for precast concrete stone masonry blocks
IS 12894 : 2002	Specification for pulverized fuel ash lime bricks (*first revision*)
IS 13757 : 1993	Burnt clay fly ash building bricks — Specification
IS 15658 : 2021	Concrete paving blocks — Specification (*first revision*)
IS 16720 : 2018	Pulverized fuel ash-cement bricks — Specification
IS 17165 : 2020	Manufacture of stabilized soil blocks — Guidelines

4 BUILDER’S HARDWARE

IS No.	*Title*
IS 204	Specification for tower bolts:
(Part 1) : 1991	Ferrous metals (*fifth revision*)
(Part 2) : 1992	Non-ferrous metals (*fifth revision*)

IS No.	*Title*
IS 205 : 1992	Specification for non-ferrous metal butt hinges (*fourth revision*)
IS 206 : 2010	Specification for tee and strap hinges (*fifth revision*)
IS 208 : 2020	Door handles — Specification (*sixth revision*)
IS 281 : 2009	Specification for mild steel sliding door bolts for use with padlocks (*fourth revision*)
IS 362 : 1991	Specification for parliament hinges (*fifth revision*)
IS 363 : 1993	Specification for hasps and staples (*fourth revision*)
IS 364 : 1993	Specification for fanlight catch (*third revision*)
IS 452 : 1973	Specification for door springs, rat-tail type (*second revision*)
IS 453 : 1993	Specification for double-acting spring hinges (*third revision*)
IS 729 : 2021	Drawer locks cupboard locks and box locks — Specification (*fourth revision*)
IS 1019 : 1974	Specification for rim latches (*second revision*)
IS 1341 : 2018	Steel butt hinges — Specification (*sixth revision*)
IS 1823 : 1980	Specification for floor door stoppers (*third revision*)
IS 1837 : 1966	Specification for fanlight pivots (*first revision*)
IS 2681 : 1993	Specification for nonferrous metal sliding door bolts for use with padlocks (*third revision*)
IS 3564 : 2025	Hydraulically regulated door closers — Specification (*fifth revision*)
IS 3818 : 1992	Specification for continuous (piano) hinges (*third revision*)
IS 3828 : 1966	Specification for ventilator chains
IS 3843 : 1995	Steel back flap hinges — Specification (*second revision*)
IS 4621 : 1975	Specification for indicating bolts for use in public baths and lavatories (*first revision*)
IS 4948 : 2020	Welded steel wire fabric for general use — Specification (*third revision*)
IS 4992 : 2024	Door handles for mortice locks — Specification (*second revision*)
IS 5187 : 1972	Specification for flush bolts (*first revision*)
IS 5899 : 1970	Specification for bathroom latches
IS 6315 : 2025	Hydraulically regulated floor springs — Specification (*third revision*)
IS 6318 : 1971	Specification for plastic window stays and fasteners
IS 6343 : 1982	Specification for door closers (pneumatically regulated) for light door weighing up to 40 kg (*first revision*)
IS 7196 : 1974	Specification for hold fast

IS No.	*Title*
IS 7197 : 1974	Specification for double action floor springs (without oil check) for heavy doors
IS 7534 : 1985	Specification for sliding locking bolts for use with padlocks (*first revision*)
IS 7881	Glossary of terms relating to builders’ hardware:
(Part 1) : 2023	Locks (*first revision*)
(Part 2) : 2022	Latches (*first revision*)
IS 8756 : 1978	Specification for ball catches for use in wooden almirah
IS 9106 : 1979	Specification for rising butt hinges
IS 9131 : 2021	Rim locks and latches mechanically operated — Specification (*first revision*)
IS 9460 : 1980	Specification for flush drop handle for drawer
IS 9899 : 1981	Specification for hat coat and wardrobe hooks
IS 10019 : 1981	Specification for mild steel stays and fasteners
IS 10090 : 1982	Specification for numericals
IS 10342 : 1982	Specification for curtain rail system
IS 12817 : 2020	Stainless steel butt hinges — Specification (*third revision*)
IS 12867 : 1989	Specification for PVC hand rails covers
IS 14912 : 2001	Specification for door closers concealed type (hydraulically regulated)
IS 15833 : 2009	Stainless steel tower bolts — Specification
IS 15834 : 2020	Stainless steel sliding door bolts (Aldrops) for use with padlocks — Specification *(first revision)*
IS 16015 : 2022	Mortise locks and latches mechanically operated — Specification (*first revision*)
IS 16016 : 2013	Cylindrical locks with pin tumbler mechanism — Specification
IS 17296 : 2020	Stainless steel door stoppers — Specification
IS 17706 : 2022	Pull handles — Specification
IS 17954 : 2023	Telescopic ball bearing drawer slide — Specification
IS 18297 : 2023	Cabinet hinges — Specification

5 BUILDING CHEMICALS:

a) Anti-termite Chemicals

IS No.	*Title*
IS 5549 : 2023	Warfarin bait concentrates — Specification (*first revision*)
IS 7168 : 2023	Warfarin sodium salt water soluble powders — Specification (*first revision*)

IS No.	*Title*
IS 8703 : 2023	Diuron water dispersible powder concentrates — Specification (*first revision*)
IS 8944 : 2005	Chlorpyrifos emulsifiable concentrates — Specification (*first revision*)
IS 16706 : 2018	Carpropamid, technical — Specification
IS 16955 : 2019	Spiromesifen, suspension concentrate (SC) — Specification

b) Chemical Admixture/Water Proofing Compounds

IS No.	*Title*
IS 2645 : 2003	Integral waterproofing compounds for cement mortar and concrete — Specification (*second revision*)
IS 6925 : 1973	Methods of test for determination of water soluble chlorides in concrete admixtures
IS 9103 : 1999	Specification for concrete admixtures (*first revision*)

c) Sealants/Fillers

IS No.	*Title*
IS 1834 : 1984	Specification for hot applied sealing compound for joint in concrete (*first revision*)
IS 1838	Specification for preformed fillers for expansion joint in concrete pavements and structures (non extruding and resilient type):
(Part 1) : 1983	Bitumen impregnated fibre (*first revision*)
(Part 2) : 1984	CNSL aldehyde resin and coconut pith
(Part 3) : 2011	Polymer based
IS 10566 : 1983	Methods of tests for preformed fillers for expansion joint in concrete paving and structural construction
IS 10959 : 2024/ ISO 6927 : 2021	Sealants for building purposes — Glossary of terms (*second revision*)
IS 11433	Specification for one-part gun — Grade polysulphide-based joint sealants:
(Part 1) : 1985	General requirements
(Part 2) : 1986	Methods of test
IS 12118	Specification for two parts polysulphide-based sealants:
(Part 1) : 1987	General requirements
(Part 2) : 1987	Methods of test
IS 13923 : 2023	Lead seal — Specification (*first revision*)

d) Adhesives

IS No.	*Title*
IS 848 : 2025	Synthetic resin adhesives for plywood (phenolic, aminoplastic and bio-materials based) — Specification (*third revision*)
IS 851 : 1978	Specification for synthetic resin adhesives for construction work (non-structural) in wood (*first revision*)
IS 852 : 1994	Animal glue for general wood-working purposes — Specification (*second revision*)
IS 1508 : 1972	Specification for extenders for use in synthetic resin adhesives (urea-formaldehyde) for plywood (*first revision*)
IS 4835 : 1979	Specification for polyvinyl acetate dispersion-based adhesives for wood (*first revision*)
IS 5666 : 2024	Etch primer — Specification (*first revision*)
IS 9188 : 1979	Performance requirements for adhesive for structural laminated wood products for use under exterior exposure condition
IS 12830 : 1989	Rubber based adhesives for fixing PVC tiles to cement — Specification
IS 12994 : 1990	Epoxy adhesives, room temperature curing general purpose — Specification
IS 13238 : 2021	Epoxy based zinc phosphate primer (two pack) — Specification (*first revision*)
IS 15477 : 2019	Adhesives for use with ceramic tiles and mosaics (*first revision*)
IS 16239 : 2021	Polyurethane matt finish (two pack) — Specification
IS 16676 : 2024	Solventless liquid epoxy system for application on interior and exterior surfaces of steel water pipeline — Specification (*first revision*)
IS 16943 : 2019	Two components high build epoxy micaceous iron oxide (MIO) pigmented intermediate coat — Specification
IS 16944 : 2018	Two components high build epoxy titanium dioxide (TiO_2) pigmented intermediate coat — Specification
IS 17218 : 2019	Solar thermo-reflective exterior coating for cooling of roof top
IS 17545 : 2021	White cement based polymeric putty
IS 18837 : 2024	Solvent cements for chlorinated poly (vinyl chloride) (CPVC) plastic pipe and fittings

6 BUILDING LIME AND PRODUCTS

IS No.	*Title*
IS 712 : 1984	Specification for building limes (*third revision*)
IS 1624 : 1986	Method of field testing of building lime (*second revision*)
IS 2686 : 1977	Specification for cinder as fine aggregates for use in lime concrete (*first revision*)

IS No.	*Title*
IS 3068 : 1986	Specification for broken brick (burnt-clay) coarse aggregates for use in lime concrete (*second revision*)
IS 3115 : 1992	Specification for lime based blocks (*second revision*)
IS 3182 : 1986	Specification for broken bricks (burnt clay) fine aggregates for use in lime mortar (*second revision*)
IS 4098 : 1983	Specification for lime-pozzolana mixture (*first revision*)
IS 4139 : 1989	Specification for calcium silicate bricks (*second revision*)
IS 6932	Method of tests for building limes:
(Part 1) : 2025	Determination of insoluble residue in dilute acid and alkali, loss on ignition, insoluble residue in hydrochloric acid, silicon dioxide, ferric and aluminium oxide, calcium oxide and magnesium oxide (*first revision*)
(Part 2) : 2025	Determination of carbon dioxide content (*first revision*)
(Part 3) : 2025	Determination of residue on slaking of quicklime (*first revision*)
(Part 4) : 2025	Determination of fineness of hydrated lime (*first revision*)
(Part 5) : 2025	Determination of unhydrated oxide (*first revision*)
(Part 6) : 2025	Determination of volume yield of quicklime (*first revision*)
(Part 7) : 2025	Determination of compressive and transverse strength (*first revision*)
(Part 8) : 2025	Determination of workability (*first revision*)
(Part 9) : 2025	Determination of soundness (*first revision*)
(Part 10) : 2025	Determination of popping and pitting of hydrated lime (*first revision*)
(Part 11) : 2025	Determination of setting time of hydrated lime (*first revision*)
IS 10360 : 1982	Specification for lime pozzolana concrete blocks for paving
IS 10772 : 1983	Specification for quick setting lime pozzolana mixture
IS 12894 : 2002	Specification for pulverized fuel ash lime bricks (*first revision*)
IS 15648 : 2006	Pulverized fuel ash for lime-pozzolana mixture applications — Specification

7 CLAY AND STABILIZED SOIL PRODUCTS

a) Blocks

IS No.	*Title*
IS 3952 : 2013	Burnt clay hollow bricks and blocks for walls and partitions — Specification (*third revision*)

b) Stabilized Soil Products

IS No.	*Title*
IS 1725 : 2023	Stabilized soil blocks used in general building construction — Specification (*third revision*)

c) Bricks

IS No.	*Title*
IS 1077 : 2025	Common burnt clay building bricks — Specification (*sixth revision*)
IS 2180 : 1988	Specification for heavy duty burnt clay building bricks (*third revision*)
IS 2222 : 1991	Specification for burnt clay perforated building bricks (*third revision*)
IS 2691 : 2017	Burnt clay facing bricks — Specification (*third revision*)
IS 3495	Burnt clay building bricks — Method of tests
(Part 1) : 2019	Determination of compressive strength (*fourth revision*)
(Part 2) : 2019	Determination of water absorption (*fourth revision*)
(Part 3) : 2019	Determination of efflorescence (*fourth revision*)
(Part 4) : 2019	Determination of warpage (*fourth revision*)
(Part 5) : 2021	Determination of initial rate of absorption
(Part 6) : 2022	Determination of modulus of rupture
IS 3583 : 1988	Specification for burnt clay paving bricks (*second revision*)
IS 3952 : 2013	Burnt clay hollow bricks and blocks for walls and partitions — Specification (*third revision*)
IS 4885 : 1988	Specification for sewer bricks (*first revision*)
IS 5454 : 2024	Burnt clay bricks and burnt clay tiles — Methods of sampling (*second revision*)
IS 5779 : 1986	Specification for burnt clay soling bricks (*first revision*)
IS 6165 : 1992	Dimensions for special shapes of clay bricks (*first revision*)
IS 13757 : 1993	Burnt clay fly ash building bricks — Specification

d) Jallies

IS No.	*Title*
IS 7556 : 1988	Specification for burnt clay jallies (*first revision*)

e) Tiles

IS No.	*Title*
IS 654 : 2023	Clay roofing tiles Mangalore pattern — Specification (*fourth revision*)

IS No.	*Title*
IS 1464 : 2024	Clay ridge and ceiling tiles — Specification (*third revision*)
IS 1478 : 2023	Clay flooring tiles — Specification (*third revision*)
IS 2690	Burnt clay flat terracing tiles — Specification
(Part 1) : 2023	Machine made (*third revision*)
(Part 2) : 2023	Hand made (*second revision*)
IS 3367 : 2023	Burnt clay tiles for use in lining irrigation and drainage works — Specification (*third revision*)
IS 3951	Hollow clay tiles for floor and roofs — Specification
(Part 1) : 2023	Filler type (*third revision*)
(Part 2) : 2023	Structural type (*third revision*)
IS 8920 : 1978	Methods for sampling of burnt clay tiles
IS 13317 : 2023	Clay roofing county tiles half round and flat tiles — Specification (*first revision*)
IS 17190 : 2020	Grouts for filling of joints of tiles and stones — Specification
IS 17897 : 2022	Stone-polymer composite flooring tiles and planks — Specification
IS 18387 : 2023	Gypsum ceiling tiles — Specification

8 CEMENT AND CONCRETE (INCLUDING STEEL REINFORCEMENT FOR CONCRETE)

a) Aggregates

IS No.	*Title*
IS 383 : 2016	Coarse and fine aggregate for concrete — Specification (*third revision*)
IS 4961 : 2022	Determination of particle size of powders by air elutriation methods (*first revision*)
IS 650 : 1991	Specification for standard sand for testing of cement (*second revision*)
IS 1542 : 1992	Specification for sand for plaster (*second revision*)
IS 2116 : 1980	Specification for sand for masonry mortars (*first revision*)
IS 2386	Methods of test for aggregates for concrete:
(Part 1) : 1963	Particle size and shape
(Part 2) : 1963	Estimation of deleterious materials and organic impurities
(Part 3) : 1963	Specific gravity, density, voids, absorption and bulking
(Part 4) : 1963	Mechanical properties
(Part 5) : 1963	Soundness

IS No.	*Title*
(Part 6) : 1963	Measuring mortar making properties of fine aggregates
(Part 7) : 1963	Alkali aggregate reactivity
(Part 8) : 1963	Petrographic examination
IS 2430 : 1986	Methods of sampling of aggregates of concrete (*first revision*)
IS 4879 : 2023	Particulate materials — Sampling and sample splitting technique (*first revision*)
IS 5282	Determination of particle size distribution by gravitational liquid sedimentation methods:
(Part 1) : 2023	General principles and guidelines (*first revision*)
(Part 2) : 2023	Fixed pipette method (*first revision*)
(Part 3) : 2023	X-Ray gravitational technique *(first revision)*
(Part 4) : 2023	Balance method *(first revision)*
IS 6579 : 1981	Specification for coarse aggregate for water bound macadam (*first revision*)
IS 9142	Artificial light-weight aggregates for concrete — Specification:
(Part 1) : 2018	For concrete masonry blocks and for applications other than for structural concrete (*first revision*)
(Part 2) : 2018	Sintered fly ash coarse aggregate (*first revision*)
IS 10086 : 2021	Specification for moulds for use in tests of cement and concrete (*first revision*)
IS 18657	Representation of results of particle size analysis:
(Part 1) : 2024/ ISO 9276-1 : 1998	Graphical representation
(Part 2) : 2024/ ISO 9276-2 : 2014	Calculation of average particle sizes/diameters and moments from particle size distributions
(Part 3) : 2024/ ISO 9276-3 : 2008	Adjustment of an experimental curve to a reference model
(Part 4) : 2024/ ISO 9276-4 : 2001	Characterization of a classification process
(Part 5) : 2024/ ISO 9276-5 : 2005	Methods of calculation relating to particle size analyses using logarithmic normal probability distribution
(Part 6) : 2024/ ISO 9276-6 : 2008	Descriptive and quantitative representation of particle shape and morphology

b) Cement

IS No.	*Title*
IS 269 : 2015	Ordinary portland cement — Specification (*sixth revision*)
IS 455 : 2015	Portland slag cement — Specification (*fifth revision*)

IS 1489	Portland pozzolana cement — Specification:
(Part 1) : 2015	Flyash based (*fourth revision*)
(Part 2) : 2015	Calcined clay based (*fourth revision*)
IS 3466 : 2026	Masonry cement — Specification (*third revision*)
IS 6452 : 1989	Specification for high alumina cement for structural use (*first revision*)
IS 6909 : 2026	Supersulphated cement — Specification (*second revision*)
IS 8041 : 2026	Rapid hardening portland cement — Specification (*third revision*)
IS 8042 : 2015	White portland cement — Specification (*third revision*)
IS 8043 : 2026	Hydrophobic portland cement — Specification (*third revision*)
IS 12330 : 1988	Specification for sulphate resisting portland cement
IS 12600 : 1989	Portland cement, low heat — Specification
IS 16415 : 2015	Composite cement — Specification
IS 16993 : 2018	Microfine ordinary portland cement — Specification
IS 17908 : 2023	Fibre cement sheet — Lightweight concrete sandwich panels specification
IS 18189 : 2023	Portland calcined clay limestone cement — Specification

c) Supplementary Cementitious Materials (Mineral Admixtures and Pozzolanas)

IS No.	*Title*
IS 1344 : 1981	Specification for calcined clay pozzolana (*second revision*)
IS 1727 : 1967	Methods of test for pozzolanic materials (*first revision*)
IS 3812	Pulverized fuel ash — Specification:
(Part 1) : 2013	For use as pozzolana in cement, cement mortar and concrete (*third revision*)
(Part 2) : 2013	For use as admixture in cement mortar and concrete (*third revision*)
IS 6491 : 1972	Method of sampling of flyash
IS 12089 : 1987	Specification for granulated slag for manufacture of portland slag cement
IS 12870 : 1989	Methods of sampling calcined clay pozzolana
IS 15388 : 2003	Specification for silica fume
IS 16354 : 2015	Metakaolin for use in cement, cement mortar and concrete — Specification
IS 16714 : 2018	Ground granulated blast furnace slag for use in cement mortar and concrete — Specification
IS 16715 : 2018	Ultrafine ground granulated blast furnace slag — Specification
IS 19058 : 2024	Ultrafine fly ash — Specification

d) Chemical Admixtures

IS No.	*Title*
IS 6925 : 1973	Methods of test for determination of water soluble chlorides in concrete admixtures
IS 9103 : 1999	Specification for admixtures for concrete (*first revision*)

e) Concrete

IS No.	*Title*
IS 456 : 2000	Code of practice for plain and reinforced concrete (*fourth revision*)
IS 1343 : 2012	Code of practice for prestressed concrete (*second revision*)
IS 4926 : 2003	Code of practice for ready-mixed concrete (*second revision*)

f) Cement and Concrete Sampling and Methods of Test

IS No.	*Title*
IS 516	Hardened concrete methods of test:
(Part 1/Sec 1) : 2021	Testing of strength of hardened concrete, Section 1 Compressive, flexural and split tensile strength (*first revision*)
(Part 2)	Properties of hardened concrete other than strength,
(Sec 1) : 2018	Density of hardened concrete and depth of water penetration under pressure (*first revision*)
(Sec 2) : 2020	Initial surface absorption (*first revision*)
(Sec 3) : 2022	Oxygen permeability index (*first revision*)
(Sec 4) : 2021	Determination of the carbonation resistance by accelerated carbonation method (*first revision*)
(Part 4) : 2018	Sampling preparing and testing of concrete cores (*first revision*)
(Part 5)	Non-destructive testing of concrete,
(Sec 1) : 2018	Ultrasonic pulse velocity testing (*first revision*)
(Sec 2) : 2021	Half-cell potentials of uncoated reinforcing steel in concrete (*first revision*)
(Sec 3) : 2021	Carbonation depth test (*first revision*)
(Sec 4) : 2020	Rebound hammer test (*first revision*)
(Part 6) : 2020	Determination of drying shrinkage and moisture movement of concrete samples (*first revision*)
(Part 8/Sec 1) : 2020	Determination of modulus of elasticity, Section 1 Static modulus of elasticity and poissons ratio in compression (*first revision*)
(Part 11) : 2020	Determination of portland cement content of hardened hydraulic cement concrete (*first revision*)

IS No.	*Title*
IS 1199	Fresh concrete — Methods of sampling testing and analysis:
(Part 1) : 2018	Sampling of fresh concrete (*first revision*)
(Part 2) : 2018	Determination of consistency of fresh concrete (*first revision*)
(Part 3) : 2018	Determination of density of fresh concrete (*first revision*)
(Part 4) : 2018	Determination of air content of fresh concrete (*first revision*)
(Part 5) : 2018	Making and curing of test specimens (*first revision*)
(Part 6) : 2018	Tests on fresh self-compacting concrete (*first revision*)
(Part 7) : 2018	Determination of setting time of concrete by penetration resistance (*first revision*)
IS 2770 (Part 1) : 1967	Methods of testing bond in reinforced concrete: Part 1 Pullout test
IS 3085 : 1965	Methods of test for permeability of cement mortar and concrete
IS 3535 : 1986	Methods of sampling hydraulic cement (*first revision*)
IS 4031	Methods of physical tests for hydraulic cement:
(Part 1) : 1996	Determination of fineness by dry sieving (*second revision*)
(Part 2) : 1999	Determination of fineness by specific surface by Blaine air permeability method (*second revision*)
(Part 3) : 1988	Determination of soundness (*first revision*)
(Part 4) : 1988	Determination of consistency of standard cement paste (*first revision*)
(Part 5) : 1988	Determination of initial and final setting times (*first revision*)
(Part 6) : 1988	Determination of compressive strength of hydraulic cement (other than masonry cement) (*first revision*)
(Part 7) : 1988	Determination of compressive strength of masonry cement (*first revision*)
(Part 8) : 1988	Determination of transverse and compressive strength of plastic mortar using prism (*first revision*)
(Part 9) : 1988	Determination of heat of hydration (*first revision*)
(Part 10) : 1988	Determination of drying shrinkage (*first revision*)
(Part 11) : 1988	Determination of density (*first revision*)
(Part 12) : 1988	Determination of air content of hydraulic cement mortar (*first revision*)
(Part 13) : 1988	Measurement of water retentivity of masonry cement (*first revision*)
(Part 14) : 1989	Determination of false set
(Part 15) : 1991	Determination of fineness by wet sieving
IS 4032 : 1985	Methods of chemical analysis for hydraulic cement (*first revision*)
IS 5816 : 1999	Method of test for splitting tensile strength of concrete (*first revision*)

IS No.	*Title*
IS 9013 : 1978	Method of making, curing and determining compressive strength of accelerated cured concrete test specimens
IS 9284 : 1979	Method of test for abrasion resistance of concrete
IS 12423 : 1988	Methods for colorimetric analysis of hydraulic cement
IS 12803 : 1989	Methods of analysis of hydraulic cement by X-ray fluorescence spectrometer
IS 12813 : 1989	Method of analysis of hydraulic cement by atomic absorption spectrophotometer
IS 13311	Methods of non-destructive testing of concrete:
(Part 2) : 1992	Rebound hammer
IS 14959	Method of test for determination of water soluble and acid soluble chlorides in mortar and concrete
(Part 1) : 2001	Fresh mortar and concrete
(Part 2) : 2001	Hardened mortar and concrete
IS 17161 : 2020	Flexural strength and toughness parameters of fibre reinforced concrete — Method of test
IS 17725 : 2022	Precast concrete circular manhole — Specification

g) Treatment of Concrete Joints

IS No.	*Title*
IS 1834 : 1984	Specification for hot applied sealing compound for joint in concrete (*first revision*)
IS 1838	Specification for preformed fillers for expansion joint in concrete pavements and structures (non extruding and resilient type):
(Part 1) : 1983	Bitumen impregnated fibre (*first revision*)
(Part 2) : 1984	CNSL aldehyde resin and coconut pith
(Part 3) : 2011	Polymer based
IS 10566 : 1983	Methods of test for preformed fillers for expansion joints in concrete paving and structural construction
IS 11433	Specification for one grade polysulphide based joint sealant:
(Part 1) : 1985	General requirements
(Part 2) : 1986	Methods of test
IS 12118	Specification for two parts polysulphide based sealants:
(Part 1) : 1987	General requirements
(Part 2) : 1987	Methods of test

h) Steel Reinforcement for Concrete

IS No.	*Title*
IS 432	Specification for mild steel and medium tensile steel bars and hard drawn steel wire for concrete reinforcement:
(Part 1) : 1982	Mild steel and medium tensile steel bars (*third revision*)
(Part 2) : 1982	Hard drawn steel wire (*third revision*)
IS 1566 : 1982	Specification for hard drawn steel wire fabric for concrete reinforcement (*second revision*)
IS 1599 : 2023/ ISO 7438 : 2020	Metallic materials — Bend test (*fifth revision*)
IS 1608	Metallic materials — Tensile testing:
(Part 1) : 2022/ ISO 6892-1 : 2019	Method of test at room temperature (*fifth revision*)
(Part 2) : 2020/ ISO 6892-2 : 2018	Method of test at elevated temperature (*fourth revision*)
(Part 3) : 2018/ ISO 6892-3 : 2015	Method of test at low temperature
IS 1785	Specification for plain hard drawn steel wire for pre-stressed concrete:
(Part 1) : 1983	Cold drawn stress-relieved wire (*second revision*)
(Part 2) : 1983	As drawn wire (*first revision*)
IS 1786 : 2008	High strength deformed steel bars and wires for concrete reinforcement Specification (*fourth revision*)
IS 2090 : 1983	Specification for high tensile steel bars used in prestressed concrete (*first revision*)
IS 6003 : 2010	Indented wire for prestressed concrete — Specification (*second revision*)
IS 6006 : 2014	Uncoated stress-relieved strand for prestressed concrete — Specification (*second revision*)
IS 9417 : 2018	Welding of high strength steel bars for reinforced concrete construction — Recommendations (*second revision*)
IS 10790	Methods of sampling of steel for prestressed and reinforced concrete:
(Part 1) : 1984	Prestressing steel
(Part 2) : 1984	Reinforcing steel
IS 12594 : 1988	Hot-dip zinc coating on structural steel bars for concrete reinforcement — Specification
IS 13620 : 1993	Specification for fusion bonded epoxy coated reinforcing bars
IS 14268 : 2022	Uncoated stress relieved low relaxation seven-wire ply strand for prestressed concrete — Specification (*second revision*)

IS No.	*Title*
IS 16172 : 2023	Reinforcement couplers for mechanical splices of steel bars in concrete — Specification (*first revision*)
IS 16644 : 2018	Stress-Relieved low relaxation steel wire for prestressed concrete — Specification
IS 16651 : 2017	High strength deformed stainless steel bars and wires for concrete reinforcement — Specification
IS 18255 : 2023	Fibre-Reinforced Polymer FRP bars for concrete reinforcement — Methods of test
IS 18256 : 2023	Solid round glass fibre reinforced polymer GFRP bars for concrete reinforcement — Specification
IS 19068 : 2025	Continuous Hot-dip galvanized steel bars for concrete reinforcement — Specification

9 COMPOSITE MATRIX PRODUCTS

a) Cement Matrix Products

i) *Precast Concrete Products*

IS No.	*Title*
IS 2174 : 1962	Specification for reinforced concrete dust bins
IS 2185	Specification for concrete masonry units:
(Part 1) : 2005	Hollow and solid concrete blocks (*third revision*)
(Part 2) : 1983	Hollow and solid lightweight concrete blocks (*first revision*)
(Part 3) : 1984	Autoclaved cellular (aerated) concrete blocks (*first revision*)
(Part 4) : 2008	Preformed foam cellular concrete blocks
IS 4996 : 1984	Specification for reinforced concrete fence posts (*first revision*)
IS 5751 : 1984	Specification for precast concrete coping blocks (*first revision*)
IS 5758 : 2020	Precast concrete kerbs channels edging quadrants and other associated fittings — Specification (*second revision*)
IS 5820 : 2024	Precast concrete cable covers — Specification (*first revision*)
IS 6072 : 2023	Autoclaved reinforced cellular concrete wall slabs — Specification (*first revision*)
IS 6073 : 2006	Specification for autoclaved reinforced cellular concrete floor and roof slabs (*first revision*)
IS 6441	Methods of test for autoclaved cellular concrete products:
(Part 1) : 1972	Determination of unit weight or bulk density and moisture content
(Part 2) : 1972	Determination of drying shrinkage

IS No.	*Title*
(Part 4) : 1972	Corrosion protection of steel reinforcement in autoclaved cellular concrete
(Part 5) : 1972	Determination of compressive strength
(Part 6) : 1973	Strength, deformation and cracking of flexural members subject to bending-short duration loading test
(Part 7) : 1973	Strength, deformation and cracking of flexural members subject to bending-sustained loading test
(Part 8) : 1973	Loading tests for flexural members in diagonal tension
(Part 9) : 1973	Jointing of autoclaved cellular concrete elements
IS 6523 : 1983	Specification for precast reinforced concrete door and window frames (*first revision*)
IS 9375 : 1979	Specification for precast reinforced concrete plant guards
IS 9872 : 1981	Specification for precast concrete septic tanks
IS 9893 : 1981	Specification for precast concrete blocks for lintels and sills
IS 12440 : 1988	Specification for precast concrete stone masonry blocks
IS 12592 : 2002	Specification for precast concrete manhole covers and frames (*first revision*)
IS 13356 : 1992	Specification for precast ferrocement water tanks (250 litres to 10 000 litres capacity)
IS 13990 : 1994	Specification for precast reinforced concrete planks and joists for flooring and roofing
IS 14143 : 1994	Specification for prefabricated brick panel and partially precast concrete joist for flooring and roofing
IS 14201 : 1994	Specification for precast reinforced concrete channel unit for construction of floors and roofs
IS 14241 : 1995	Specification for precast L-Panel units for roofing
IS 18661 : 2024	Precast concrete grating — Specification
IS 18889 : 2024	Precast concrete paving flags — Specification
IS 18894 : 2024	Precast concrete paving grids and grass pavers — Specification

ii) *Asbestos Fibre Cement Products*

IS No.	*Title*
IS 459 : 1992	Specification for corrugated and semi-corrugated asbestos cement sheets (*third revision*)
IS 1592 : 2003	Specification for asbestos cement pressure pipes and joints (*fourth revision*)
IS 1626	Specification for asbestos cement building pipes and pipe fittings, gutters and gutter fittings and roofing fittings:

IS No.	*Title*
(Part 1) : 1994	Pipes and pipe fittings (*second revision*)
(Part 2) : 1994	Gutters and gutter fittings (*second revision*)
(Part 3) : 1994	Roofing fittings (*second revision*)
IS 2096 : 1992	Specification for asbestos cement flat sheets (*first revision*)
IS 2098 : 1997	Specification for asbestos cement building boards (*first revision*)
IS 5913 : 2003	Methods of test for asbestos cement products (*second revision*)
IS 6908 : 1991	Specification for asbestos cement pipes and fittings for sewerage and drainage (*first revision*)
IS 7639 : 1975	Method of sampling asbestos cement products
IS 9627 : 1980	Specification for asbestos cement pressure pipes (light duty)
IS 13000 : 1990	Silica-asbestos-cement flat sheets — Specification
IS 13008 : 1990	Specification for shallow corrugated asbestos cement sheets

iii) *Other Fibre Cement Products*

IS No.	*Title*
IS 14862 : 2000	Specification for fibre cement flat sheets
IS 14871 : 2000	Specification for products in fibre reinforced cement long corrugated or asymmetrical section sheets and fittings for roofing and cladding

iv) *Concrete Pipes and Pipes Lined/Coated with Concrete or Mortar*

IS No.	*Title*
IS 458 : 2021	Precast concrete pipes with and without reinforcement (*fifth revision*)
IS 784 : 2019	Prestressed concrete pipes including specials — Specification (*third revision*)
IS 1916 : 2018	Steel cylinder pipes with concrete lining and coating — Specification (*second revision*)
IS 3597 : 1998	Methods of test for concrete pipes (*second revision*)
IS 4350 : 1967	Specification for concrete porous pipes for under drainage
IS 7319 : 1974	Specification for perforated concrete pipes
IS 7322 : 1985	Specification for specials for steel cylinder reinforced concrete pipes (*first revision*)
IS 15155 : 2020	Barwire wrapped steel cylinder pipes with mortar lining and coating including specials specification (*first revision*)

b) Resin Matrix Products

IS No.	*Title*
IS 1998 : 1962	Methods of test for thermosetting synthetic resin bonded laminated sheets
IS 2036 : 1995	Specification for phenolic laminated sheets (*second revision*)
IS 2046	Decorative thermosetting synthetic resin bonded laminated sheets — Specification:
(Part 1) : 2025/ ISO 4586-1 : 2018	Introduction and general information (*third revision*)
(Part 2) : 2025/ ISO 4586-2 : 2018	Determination of properties (*third revision*)
(Part 3) : 2025/ ISO 4586-3 : 2018	Classification and specifications for laminates less than 2 mm thick and intended for bonding to supporting substrates (*third revision*)
(Part 4) : 2025 ISO 4586-4 : 2018	Classification and specifications for compact laminates of thickness 2 mm and greater (*third revision*)
(Part 5) : 2025 ISO 4586-5 : 2018	Classification and specifications for flooring grade laminates less than 2 mm thick intended for bonding to supporting substrates (*third revision*)
(Part 6) : 2025 ISO 4586-6 : 2018	Classification and specifications for exterior-grade compact laminates of thickness 2 mm and greater (*third revision*)
(Part 7) : 2025 ISO 4586-7 : 2018	Classification and specifications for design laminates (*third revision*)
(Part 8) : 2025 ISO 4586-8 : 2018	Classification and specifications for alternative core laminates (*third revision*)

10 CONDUCTORS AND CABLES

IS No.	*Title*
IS 694 : 2010	Polyvinyl chloride insulated unsheathed and sheathed cables/cords with rigid and flexible conductor for rated voltages up to and including 450/750 V (*fourth revision*)
IS 1554	Specification for PVC insulated (heavy duty) electric cables:
(Part 1) : 1988	For working voltages up to and including 1 100 volts (*third revision*)
(Part 2) : 1988	For working voltages from 3.3 kV up to and including 11 kV (*second revision*)
IS 3961	Recommended current ratings for cables:
(Part 2) : 2017	PVC insulated and PVC sheathed heavy duty cables (*first revision*)
(Part 3) : 1968	Rubber insulated cables
(Part 5) : 1968	PVC insulated light duty cables
(Part 6) : 2016	Crosslinked polyethylene insulated PVC sheathed cables
(Part 7) : 2017	Crosslinked polyethylene insulated thermoplastic sheathed cables

IS No.	*Title*
IS 4289	Specification for flexible cables for lifts and other flexible connections:
(Part 1) : 1984	Elastomer insulated cables (*first revision*)
(Part 2) : 2000	PVC insulated circular cables
IS 7098	Cross-linked polyethylene insulated sheathed cables — Specification
(Part 1) : 2025	For working voltage up to and including 1 100 V (*second revision*)
(Part 2) : 2011	For working voltages form 3.3 kV up to and including 33 kV (*second revision*)
(Part 3) : 1993	For working voltages from 66 kV upto and including 220 kV
IS 9968	Elastomer insulated cables — Specification:
(Part 1) : 2025	For working voltages up to and including 1 100 V (*second revision*)
(Part 2) : 2002	For working voltages from 3.3 kV up to and including 33 kV (*first revision*)
IS 10810	Methods of test for cables:
(Part 0) : 1984	General
(Part 1) : 1984	Annealing test for wires used in conductors
(Part 2) : 1984	Tensile test for aluminium wires
(Part 3) : 1984	Wrapping test for aluminium wires
(Part 4) : 1984	Persulphate test of conductor
(Part 5) : 1984	Conductor resistance test
(Part 6) : 1984	Thickness of thermoplastic and elastomeric insulation and sheath
(Part 7) : 1984	Tensile strength and elongation at break of thermoplastic and elastomeric insulation and sheath
(Part 8) : 1984	Breaking strength and elongation at break for impregnated paper insulation
(Part 9) : 1984	Tear resistance for paper insulation
(Part 10) : 1984	Loss of mass test
(Part 11) : 1984	Thermal ageing in air
(Part 12) : 1984	Shrinkage test
(Part 13) : 1984	Ozone resistance test
(Part 14) : 1984	Heat shock test
(Part 15) : 1984	Hot deformation test
(Part 16) : 1986	Accelerated ageing test by oxygen pressure method
(Part 17) : 1986	Tear resistance test for heavy duty sheath

IS No.	*Title*
(Part 19) : 1984	Bleeding and blooming test
(Part 20) : 1984	Cold bend test
(Part 21) : 1984	Cold impact test
(Part 22) : 1984	Vicat softening point
(Part 23) : 1984	Melt-flow index
(Part 24) : 1984	Water soluble impurities test of insulating paper
(Part 25) : 1984	Conductivity of water extract test of insulating paper
(Part 26) : 1984	*p*H value of water extract test of insulating paper
(Part 27) : 1984	Ash content test of insulating paper
(Part 28) : 1984	Water absorption test (Electrical)
(Part 29) : 1984	Environmental stress cracking test
(Part 30) : 1984	Hot set test
(Part 31) : 1984	Oil resistance test
(Part 32) : 1984	Carbon content test for polyethylene
(Part 33) : 1984	Water absorption test (Gravimetric)
(Part 34) : 1984	Measurement of thickness of metallic sheath
(Part 35) : 1984	Determination of tin in lead alloy for sheathing
(Part 36) : 1984	Dimensions of armouring material
(Part 37) : 1984	Tensile strength and elongation at break of armouring materials
(Part 38) : 1984	Torsion test on galvanized steel wires for armouring
(Part 39) : 1984	Winding test on galvanized steel strips for armouring
(Part 40) : 1984	Uniformity of zinc coating on steel armour
(Part 41) : 1984	Mass of zinc coating on steel armour
(Part 42) : 1984	Resistivity test of armour wires and strips and conductance test of armour (wires/strips)
(Part 43) : 1984	Insulation resistance
(Part 44) : 1984	Spark test
(Part 45) : 1984	High voltage test
(Part 46) : 1984	Partial discharge test
(Part 47) : 1984	Impulse test

IS No.	*Title*
(Part 48) : 1984	Dielectric power factor test
(Part 49) : 1984	Heating cycle test
(Part 50) : 1984	Bending test
(Part 51) : 1984	Dripping test
(Part 53) : 1984	Flammability test
(Part 54) : 1984	Static flexibility test
(Part 55) : 1986	Abrasion test
(Part 56) : 1987	Accelerated ageing by the air-pressure method
(Part 57) : 1987	Flexing test
(Part 58) : 1998	Oxygen index test
(Part 59) : 1988	Determination of the amount of halogen acid gas evolved during combustion of polymeric materials taken from cables
(Part 60) : 1988	Thermal stability of PVC insulation and sheath
(Part 61) : 1988	Flame retardant test
(Part 62) : 1993	Fire resistance test for bunched cables
(Part 63) : 1993	Smoke density of electric cables under fire conditions
(Part 64) : 2003	Measurement of temperature index
IS 12943 : 1990	Brass glands for PVC cables — Specification
IS 14763 : 2022	Conduit systems for cable management outside diameters of conduits for electrical installations and threads for conduits and fittings
IS 16205	Conduit systems for cable management:
(Part 1) : 2017	General requirements
(Part 21) : 2017	Particular requirements rigid conduit systems
(Part 22) : 2017	Particular requirements — Pliable conduit systems
(Part 23) : 2017	Particular requirements flexible conduit systems
(Part 24) : 2017	Particular requirements conduit systems buried under ground
IS 16783 : 2018	Cable cleats for electrical installations
IS 17048 : 2018	Halogen free flame retardant HFFR cables for working voltages up to and including 1 100 Volts — Specification
IS 17293 : 2020	Electric cables for photovoltaic systems for rated voltage 1 500 V d. c.

IS No.	*Title*
IS 17505 (Part 1) : 2021	Specification for thermosetting insulated fire survival cables for fixed installation having low emission of smoke and corrosive gases when affected by fire for working voltages upto and including 1 100 V a.c. and 1 500 V d.c.: Part 1 Requirements for armoured cables
IS 18833 : 2024/ IEC 62895 : 2017	High voltage direct current (HVDC) power transmission — Cables with extruded insulation and their accessories for rated voltages up to 320 kV for land applications — Test methods and requirements
IS/IEC 60371-3-7 : 1995	Specification for insulating materials based on MICA: Part 3 Specifications for Individual materials, Section 7 Polyester film mica paper with an epoxy resin binder for single conductor taping
IS/IEC 60371-3-8 : 1995	Specification for insulating materials based on MICA: Part 3 Specification for individual materials, Section 8 Mica paper tapes for flame resistant security cables
IS/IEC 62275 : 2018	Cable management systems — Cable ties for electrical installations (*first revision*)

11 DOORS, WINDOWS AND VENTILATORS

a) Wooden Doors, Windows and Ventilators

IS No.	*Title*
IS 1003	Specification for timber panelled and glazed shutters
(Part 1) : 2003	Door shutters (*fourth revision*)
(Part 2) : 1994	Window and ventilator shutters (*third revision*)
IS 2191	Wooden flush door shutters (cellular, hollow and tubular core type) — Specification
(Part 1) : 2022	Plywood face panels (*fifth revision*)
(Part 2) : 2022	Particle board high density fibre board medium density fibre board and fibre hardboard face panels (*fourth revision*)
IS 2202	Wooden flush door shutters solid core type — Specification
(Part 1) : 2023	Plywood face panels (*seventh revision*)
(Part 2) : 2022	Particle board high density fibre board medium density fibre board and hardboard face panels (*fourth revision*)
IS 4020	Method of tests for door shutters:
(Part 1) : 1998	General (*third revision*)
(Part 2) : 1998	Measurement of dimensions and squareness (*third revision*)
(Part 3) : 1998	Measurement of general flatness (*third revision*)
(Part 4) : 1998	Local planeness test (*third revision*)
(Part 5) : 1998	Impact indentation test (*third revision*)

IS No.	*Title*
(Part 6) : 1998	Flexure test (*third revision*)
(Part 7) : 1998	Edge loading test (*third revision*)
(Part 8) : 1998	Shock resistance test (*third revision*)
(Part 9) : 1998	Buckling resistance test (*third revision*)
(Part 10) : 1998	Slamming test (*third revision*)
(Part 11) : 1998	Misuse test (*third revision*)
(Part 12) : 1998	Varying humidity test (*third revision*)
(Part 13) : 1998	End immersion test (*third revision*)
(Part 14) : 1998	Knife test (*third revision*)
(Part 15) : 1998	Glue adhesion test (*third revision*)
(Part 16) : 1998	Screw withdrawal resistance test (*third revision*)
IS 4021 : 1995	Specification for timber door, window and ventilator frames (*third revision*)
IS 6198 : 1992	Specification for ledged, braced and battened timber door shutters (*second revision*)

b) Metal Doors, Windows Frames and Ventilators

IS No.	*Title*
IS 1038 : 1983	Specification for steel doors, windows and ventilators (*third revision*)
IS 1361 : 1978	Specification for steel windows for industrial buildings (*first revision*)
IS 1948 : 2024	Aluminium framed doors, windows and sliders — Specification (*first revision*)
IS 4351 : 2024	Steel Frames for Doors — Specification (*third revision*)
IS 6248 : 1979	Specification for metal rolling shutters and rolling grills (*first revision*)
IS 7452 : 1990	Specification for hot rolled steel sections for doors, windows and ventilators (*second revision*)
IS 10451 : 1983	Specification for steel sliding shutters (top hung type)
IS 10521 : 1983	Specification for collapsible gates
IS 18648 : 2024	Classification and performance requirements for doors, windows and sliders — Specification

c) Plastic Doors and Windows

IS No.	*Title*
IS 14856 : 2000	Specification for glass fibre reinforced plastic (GRP) panel type door shutters for internal use

IS 15380 : 2023	Moulded high density fibre HDF panelled door shutters — Specification (*first revision*)
IS 15931 : 2012	Solid panel foam UPVC door shutters — Specification
IS 17953 : 2023	uPVC profiles for windows and doors — Specification
IS 18648 : 2024	Classification and performance requirements for doors, windows and sliders — Specification

d) Concrete Doors and Windows

IS No.	*Title*
IS 6523 : 1983	Specification for precast reinforced concrete door and window frames (*first revision*)

e) Other Composite Material Doors and Windows

IS No.	*Title*
IS 16073 : 2013	Bamboo-jute composite panel door shutter — Specification
IS 16074 : 2024	Steel flush door shutters — Specification (*first revision*)
IS 16096 : 2013	Bamboo-Jute composite hollow core door shutter — Specification

f) Fire Check Doors

IS No.	*Title*
IS 3614 : 2021	Fire doors and doorsets — Specification (*first revision*)

g) Mesh/Net for Mosquito/Vector Prevention

IS No.	*Title*
IS 1568 : 1970	Specification for wire cloth for general purposes (*first revision*)
IS 3150 : 1982	Specification for hexagonal wire netting for general purposes (*second revision*)
IS 11199 : 2024	Textiles — High density polyethylene (HDPE) monofilament twine door nets — Specification (*first revision*)

h) Doors and Windows — Method of tests

IS No.	*Title*
IS 4020	Method of tests for door shutters:
(Part 1) : 1998	General (*third revision*)
(Part 2) : 1998	Measurement of dimensions and squareness (*third revision*)
(Part 3) : 1998	Measurement of general flatness (*third revision*)
(Part 4) : 1998	Local planeness test (*third revision*)
(Part 5) : 1998	Part 5 Impact indentation test (*third revision*)

IS No.	*Title*
(Part 6) : 1998	Flexure test (*third revision*)
(Part 7) : 1998	Edge loading test (*third revision*)
(Part 8) : 1998	Shock resistance test (*third revision*)
(Part 9) : 1998	Buckling resistance test (*third revision*)
(Part 10) : 1998	Slamming test (*third revision*)
(Part 11) : 1998	Misuse test (*third revision*)
(Part 12) : 1998	Varying humidity test (*third revision*)
(Part 13) : 1998	End immersion test (*third revision*)
(Part 14) : 1998	Knife test (*third revision*)
(Part 15) : 1998	Glue adhesion test (*third revision*)
(Part 16) : 1998	Screw withdrawal resistance test (*third revision*)
IS 12458 : 2019	Fire resistance of through penetration firestops — Method of test (*first revision*)
IS 17518 (Part 1) : 2022/ ISO 3008-1 : 2019	Fire resistance tests — Door and shutter assemblies: Part 1 General requirements
IS 17909 : 2022/ ISO 8274 : 2005	Windows and doors — Resistance to repeated opening and closing — Test method
IS 17910	Thermal performance of windows and doors — Determination of thermal transmittance by the hot-box method:
(Part 1) : 2022/ ISO 12567-1:2010	Complete windows and doors
(Part 2) : 2022/ ISO 12567-2 : 2005	Roof windows and other projecting windows
IS 17911 : 2022/ ISO 15099 : 2003	Thermal performance of windows, doors and shading devices — Detailed calculations
IS 17920 : 2022/ ISO 10077-2 : 2017	Thermal performance of windows, doors and shutters — Calculation of thermal transmittance — Numerical method for frames
IS 18268 : 2023/ ISO 9379 : 2005	Operating forces — Test methods — Doors
IS 18434 : 2023	Field measurement of air permeability and water penetration through installed curtain wall, windows, doors, sliders and skylights under static air pressure difference — Method of test
IS 18459 : 2024	Water penetration of curtain walls, windows, sliders, doors and skylights by uniform static air pressure difference — Method of test
IS 18473 : 2024	Wind resistance for curtain walls, windows, sliders, doors and skylights — Method of test

IS No.	*Title*
IS 18647 : 2024	Water penetration of curtain walls, windows, sliders and doors by dynamic air pressure — Method of test
IS 18648 : 2024	Classification and performance requirements for doors, windows and sliders — Specification
IS 19006 : 2023/ISO 8248 : 1985	Windows and door height windows — Mechanical tests

12 ELECTRICAL WIRING, FITTINGS AND ACCESSORIES

IS No.	*Title*
IS 302 (Part 1) : 2024/ IEC 60335-1 : 2020	Household and similar electrical appliances — Safety: Part 1 General requirements (*seventh revision*)
IS 302	Safety of household and similar electrical appliances:
(Part 2)	Particular requirements,
(Sec 21) : 2024/ IEC 60335-2-21 : 2022	Storage water heaters (*third revision*)
(Sec 35) : 2017	Electric instantaneous water heaters (*second revision*)
(Sec 74) : 2024/ IEC 60335-2-74 : 2021	Portable immersion heaters
IS 302-2-80 : 2017	Household and similar electrical appliances — Safety: Part 2-80 Particular requirements for fans (*first revision*)
IS 369 : 2019	Household electric direct — Acting room heaters — Performance requirements (*fourth revision*)
IS 371 : 1999	Specification for ceiling roses (*third revision*)
IS 374 : 2019	Specification for electric ceiling type fans and regulators (*third revision*)
IS 418 : 2004	Specification for tungsten filament lamp for domestic and similar general lighting purposes (*fourth revision*)
IS 1258 : 2024/ IEC 61184 : 2017	Bayonet lampholders (*fifth revision*)
IS 1293 : 2019	Plugs and socket-outlets of rated voltage up to and including 250 volts and rated current up to and including 16 amperes — Specification (*fourth revision*)
IS 1554	PVC insulated (heavy duty) electric cables:
(Part 1) : 1988	For working voltages upto and including 1 100 V (*third revision*)
(Part 2) : 1988	For working voltages from 3.3 kV upto and including 11 kV (*second revision*)
IS 2082 : 2018	Stationary storage type electric water heaters — Specification (*fifth revision*)
IS 2086 : 1993	Specification for carriers and bases used in re-wirable type electric fuses up to 650 volts (*third revision*)
IS 2215 : 2006	Specification for starters for fluorescent lamps (*third revision*)
IS 2418	Tubular fluorescent lamps for general lighting service:

IS No.	*Title*
(Part 1) : 2018	Safety requirements (*second revision*)
(Part 2) : 2018	Performance requirements (*second revision*)
(Part 3) : 1977	Dimensions of G-5 and G-13 bi-pin caps (*first revision*)
(Part 4) : 1977	Go and no-go gauges for G-5 and G-13 bi-pin caps (*first revision*)
IS 3323 : 1980	Specification for bi-pin lamp holders for tubular fluorescent lamps (*first revision*)
IS 3324 : 1982	Specification for holders for starters for tubular fluorescent lamps (*first revision*)
IS 3419 : 1988	Specification for fittings for rigid non-metallic conduits (*second revision*)
IS 3480 : 2024	Flexible steel conduits for electrical wiring — Specification (*first revision*)
IS 3724 : 2017	Cartridge type heating elements (non-embedded type) — Specification (*first revision*)
IS 3837 : 1976	Specification for accessories for rigid steel conduits for electrical wiring (*first revision*)
IS 3854 : 2023	Specification for switches for domestic and similar purposes (*third revision*)
IS 4159 : 2021	Mineral filled sheathed heating elements (*fourth revision*)
IS 4160 : 2005	Specification for interlocking switch socket outlets (*first revision*)
IS 4649 : 1968	Specification for adaptors for flexible steel conduits
IS 9385 (Part 1) : 2024/ IEC 60282-1 : 2020	High -voltage fuses: Part 1 Current-limiting fuses (*second revision*)
IS 9385 (Part 2) : 2018	High-voltage fuses: Part 2 Expulsion fuses (*first revision*)
IS 9385 (Part 6) : 2025/ IEC 60282-4 : 2020	High-voltage fuses: Part 6 Additional testing requirements for high-voltage expulsion fuses utilizing polymeric insulators
IS 9537	Specification for conduits for electrical installations:
(Part 1) : 1980	General requirements
(Part 2) : 1981	Rigid steel conduits
(Part 3) : 1983	Rigid plain conduits for insulating materials
(Part 4) : 1983	Pliable self-recovering conduits for insulating materials
(Part 5) : 2000	Pliable conduits of insulating materials
(Part 6) : 2000	Pliable conduits of metal or composite materials
(Part 8) : 2003	Rigid non threadable conduits of aluminium alloy
IS 9926 : 1981	Specification for fuse wires used in rewirable type electric fuses up to 650 volts

IS No.	*Title*
IS 10027 : 2018	Composite units of air-break switches and rewirable type fuses for voltages not exceeding 650 V a.c. — Specification (*second revision*)
IS 10322	Specification for luminaires:
(Part 1) : 2014	General requirements and tests (*first revision*)
(Part 5)	Particular requirements,
(Sec 1) : 2012	General purpose luminaires (*first revision*)
(Sec 2) : 2012	Recessed luminaires (*first revision*)
(Sec 3) : 2012	Luminaires for road and street lighting (*first revision*)
(Sec 4) : 1987	Portable general purpose luminaires
(Sec 5) : 2013	Flood light (*first revision*)
(Sec 6) : 2013	Hand lamps
(Sec 7) : 2017/ IEC 60598- 2-20 : 2014	Lighting chains (*first revision*)
(Sec 8) : 2013	Emergency lighting
(Sec 9) : 2017/ IEC 60598-2-21 : 2014	Rope lights
IS 11037 : 2019	Electronic type fan regulators specification (*first revision*)
IS 12640	Residual current operated circuit-breakers without integral overcurrent protection for household and similar uses (RCCBs):
(Part 1) : 2024/ IEC 61008-1 : 2013	General rules (*third revision*)
(Part 2) : 2016/ IEC 61009-1 : 2012	General rules (*second revision*)
(Part 3) : 2018/ IEC 1008-2-1 : 1990	Applicability of the general rule to RCCB's functionally independent of line voltage
(Part 4) : 2018/ IEC 61008-2-2 : 1990	Applicability of the general rules to RCCB's functionally dependent on line voltage
IS 12776 : 2002	Specification for galvanized strand for earthing (*first revision*)
IS 13010 : 2002	AC watt-hour meters, Class 0.5, 1 and 2 (*first revision*)
IS 13779 : 2020	a.c. static watthour meters class 1 and 2-Specefication (*second revision*)
IS 13947 (Part 5/Sec 2) : 2004/ IEC 60947-5-2(1999)	Low-voltage switchgear and controlgear — Specification: Part 5 Control circuit devices and switching elements, Section 2 Proximity switches
IS 14614 : 2023	Residual current — operated protective devices (RCDs) for household and similar use — Electromagnetic compatibility (*first revision*)

IS No.	*Title*
IS 14697 : 2021	a.c. Static transformer operated watthour meters (Class 0.2 S and 0.5 S) and Var-hour meters (Class 0.2 S, 0.5 S and 1 S) — Specification (*first revision*)
IS 14763 : 2022	Conduit systems for cable management outside diameters of conduits for electrical installations and threads for conduits and fittings (*first revision*)
IS 14768	Conduit fittings for electrical installations:
(Part 1) : 2000	General requirements
(Part 2) : 2003	Metal conduit fittings
IS 14772 : 2020	Boxes and enclosures for electrical accessories for household and similar fixed electrical installations general requirements (*first revision*)
IS 14897 : 2021	Glass bulb designation system for lamps (*first revision*)
IS 14927	Cable trunking systems and cable ducting systems for electrical installations:
(Part 1) : 2023	General requirements (*first revision*)
(Part 2/Sec 1) : 2023/ IEC 61084-2-1 : 2017	Cable trunking systems and cable ducting systems intended for mounting on walls and ceilings, Section 1 Particular requirements (*first revision*)
IS 15111	Self-Ballasted lamps for general lighting services:
(Part 1) : 2002	Safety requirements
(Part 2) : 2002	Performance requirements
IS 15368 : 2003	Cable reels for household and similar purposes
IS 15518 (Part 1) : 2004	Safety requirements for incandescent lamps: Part 1 Tungsten filament lamps for domestic and similar general lighting purposes
IS 15787 : 2025	Switched socket-outlets without interlock for fixed installations — Particular requirements (*first revision*)
IS 15968 : 2013	Ballasts for tubular fluorescent lamps — Performance requirements
IS 15974 : 2013	Auxiliaries for lamps — Starting devices (other than glow starters) — Performance requirements
IS 16102	Self-ballasted LED-lamps for general lighting services:
(Part 1) : 2026	Safety requirements (*first revision*)
(Part 2) : 2026	Performance requirements (*second revision*)
IS 16103	LED modules for general lighting:
(Part 1) : 2025/ IEC 62031 : 2018	Safety — Specifications (*first revision*)
(Part 2) : 2025/ IEC 62717 : 2019 CSV	Performance requirements (*first revision*)
IS 16148 : 2014/ IEC 61167 : 2011	Metal halide lamps — Performance specification

IS No.	*Title*
IS 16444 (Part 2) : 2017	a.c. Static transformer operated watthour and var-hour smart meters, class 0.2 s, 0.5 s and 1.0 s: Part 2 Specification transformer operated smart meters
IS 16463 (Part 12) : 2026/ IEC 61643-12 : 2020	Low-voltage surge protective devices: Part 12 Surge protective devices connected to low-voltage power distribution systems — Selection and application principles (*first revision*)
IS 16614	Double-capped LED linear lamps:
(Part 1) : 2026	Safety requirements (*first revision*)
(Part 2) : 2022	Performance requirements
IS 17119 : 2019	Residual current devices with or without overcurrent protection for socket-outlets for household and similar uses
IS 17121 : 2019/IEC 62606 : 2017	General requirements for arc fault detection devices
IS 17122 : 2019	Electrical accessories — Circuit-Breakers and similar equipment for household use — Auxiliary contact units
IS 17345	Power track system:
(Part 1) : 2020	General requirement
(Part 21) : 2020	Particular requirements for power track systems intended for wall and ceiling mounting
IS 18751 : 2024	Polymer coated inner tanks for storage water heater — Specification
IS 18830 : 2024	Cord extension sets — Specifications
IS/IEC 60079	Explosive atmospheres:
(Part 0) : 2017	General requirements (*third revision*)
(Part 1) : 2014	Equipment protection by flameproof enclosures “d” (*first revision*)
(Part 7) : 2017	Equipment protection by increased safety “e” (*second revision*)
(Part 13) : 2017	Equipment protection by pressurized room “p” and artificially ventilated room “v” (*first revision*)
(Part 15) : 2017	Equipment protection by type of protection “n” (*second revision*)
(Part 19) : 2019	Equipment repair overhaul and reclamation (*second revision*)
(Part 25) : 2020	Intrinsically safe electrical systems (*second revision*)
(Part 26) : 2021	Equipment with separation elements or combined levels of protection (*second revision*)
(Part 31) : 2022	Equipment dust ignition protection by enclosure “t” (*second revision*)
(Part 39) : 2015	Intrinsically safe systems with electronically controlled spark duration limitation

IS No.	*Title*
(Part 42) : 2019	Electrical safety devices for the control of potential ignition sources from ex-equipment
(Part 43) : 2017	Equipment in adverse service conditions
(Part 46) : 2017	Equipment assemblies
(Part 47) : 2021	Equipment Protection by 2-wire intrinsically safe ethernet concept (2-WISE)
IS/IEC 60127	Miniature fuses:
(Part 3) : 2015	Sub-miniature fuse-links
(Part 10) : 2001	User guide for miniature fuses
IS/IEC 60309	Plugs fixed or portable socket-outlets and appliance inlets for industrial purposes
(Part 1) : 2021	General requirements (*second revision*)
(Part 2) : 2021	Dimensional compatibility requirements for pin and contact-tube accessories (*second revision*)
IS/IEC 60320	Appliance couplers for household and similar general purposes
(Part 1) : 2021	General requirements (*first revision*)
(Part 2/Sec 3) : 2018	Particular requirement, Section 3 Appliance couplers with a degree of protection higher than IPX0 (*first revision*)
IS/IEC 60669 (Part 2/Sec 1) : 2021	Switches for household and similar fixed electrical installations: Part 2-1: Particular requirements electronic control devices (*first revision*)
IS/IEC 60794	Optical fibre cables:
(Part 1)	Generic specification,
(Sec 1) : 2015	General (*first revision*)
(Sec 2) : 2017	Basic optical cable test procedures — General guidance (*first revision*)
(Part 2) : 2017	Indoor cables, Sectional specification (*first revision*)
(Part 2/Sec 11) : 2020	Indoor cables, Section 11 Detailed specification for simplex and duplex cables for use in premises cabling (*first revision*)
(Part 4/Sec 20) : 2018	Sectional specification, Section 20 Aerial optical cables along electrical power lines family specification for ADSS all dielectric self-supported optical cables
IS/IEC 60898	Electrical accessories — Circuit-breakers for overcurrent protection for household and similar installations
(Part 1) : 2015	Circuit-breakers for a.c. operation (*first revision*)
(Part 2) : 2016	Circuit-breakers for a.c. and d.c. operation (*first revision*)
(Part 3) : 2019	Circuit-breakers for d.c. operation
IS/IEC 60947	Low-voltage switchgear and controlgear:

IS No.	*Title*
(Part 1) : 2020	General rules (*second revision*)
(Part 2) : 2016	Circuit-breakers (*first revision*)
(Part 3) : 2020	Switches, disconnectors, switch-disconnectors and fuse-combination units (*first revision*)
(Part 4)	Contactors and motor-starters,
(Sec 1) : 2023	Electromechanical contactors and motor-starters (*third revision*)
(Sec 2) : 2020	Semiconductor motor controllers, starters and soft-starters (*second revision*)
(Sec 3) : 2020	Semiconductor controllers and semiconductor contactors for non-motor loads *(third revision*)
(Part 5)	Control circuit devices and switching elements,
(Sec 1) : 2024	Electromechanical control circuit devices (*third revision*)
(Sec 2) : 2019	Proximity switches (*first revision*)
(Sec 5) : 2016	Electrical emergency stop devices with mechanical latching function
(Part 6)	Multiple function equipment,
(Sec 1) : 2021	Transfer switching equipment (*first revision*)
(Sec 2) : 2020	Control and protective switching devices or equipment (CPS) (*first revision*)
(Part 7)	Ancillary equipment,
(Sec 1) : 2009	Terminal blocks for copper conductors
(Sec 2) : 2009	Protective conductor terminal blocks for copper conductors
IS/IEC 61058	Switches for appliances:
(Part 1)	Particular requirements,
(Sec 1) : 2016	Mechanical switches
(Sec 2) : 2016	Particular requirements for electronics switches
IS/IEC 61439	Low-voltage switchgear and controlgear assemblies:
(Part 1) : 2020	General rules (*first revision*)
(Part 2) : 2020	Power switchgear and controlgear assemblies (*first revision*)
IS/IEC 61757 : 2018	Fibre optic sensors — Generic specification
IS/IEC/TR 62655 : 2013	Tutorial and application guide for high-voltage fuses
IS/IEC 63008 : 2020	Household and similar electrical appliances accessibility of control elements, doors, lids, drawers and handles

IS No.	*Title*
IS/ISO/IEC 80079	Explosive atmospheres:
(Part 20)	Material characteristics for gas and vapour classification,
(Sec 1) : 2017	Test methods and data
(Sec 2) : 2016	Combustible dusts test methods
(Part 36) : 2016	Non-electrical equipment for explosive atmospheres — Basic method and requirements
(Part 37) : 2016	Non-electrical equipment for explosive atmospheres — Non electrical type of protection constructional safety "c", control of ignition source "b", liquid immersion "k"

13 FILLERS, STOPPERS AND PUTTIES

IS No.	*Title*
IS 110 : 2017	Ready mixed paint brushing grey filler for enamels for use over primers — Specification (*third revision*)
IS 419 : 1967	Specification for putty for use on window frames (*first revision*)
IS 419 : 2023	Putty for use on window frames — Specification (*second revision*)
IS 423 : 1961	Specification for plastic wood, for joiner's filler (*revised*)
IS 3709 : 1966	Specification for mastic cement for bedding of metal windows
IS 7164 : 1973	Specification for stopper
IS 13184 : 1991	Specification for mastic filler, epoxy based

14 FLOOR COVERING, ROOFING AND OTHER FINISHES

a) Concrete Flooring

IS No.	*Title*
IS 1237 : 2012	Specification for cement concrete flooring tiles (*second revision*)
IS 6073 : 2006	Autoclaved reinforced cellular concrete floor and roof slabs — Specification (*first revision*)
IS 13801 : 2013	Chequered cement concrete tiles — Specification (*first revision*)
IS 13990 : 1994	Specification for precast reinforced concrete planks and joists for flooring and roofing
IS 14201 : 1994	Precast reinforced concrete channel units for construction of floors and roofs — Specification
IS 15658 : 2021	Concrete paving blocks — Specification (*first revision*)

b) Flooring Compositions

IS No.	*Title*
IS 657 : 1982	Specification for materials for use in the manufacture of magnesium oxychloride flooring compositions (*second revision*)
IS 9162 : 1979	Methods of tests for epoxy resin, hardeners and epoxy resin composition for floor topping
IS 9197 : 1979	Specification for epoxy resin, hardness and epoxy resin compositions for floor topping
IS 10132 : 1982	Method of test for materials for use in the preparation of magnesium oxychloride flooring composition

c) Linoleum Flooring

IS No.	*Title*
IS 653 : 1992	Specification for linoleum sheets and tiles (*third revision*)
IS 9704 : 1980	Methods of tests for linoleum sheets and tiles

d) Rubber Flooring

IS No.	*Title*
IS 809 : 1992	Specification for rubber flooring materials for general purposes (*second revision*)

e) Stone Flooring

IS No.	*Title*
IS 1128 : 1974	Specification for limestone (slab and tiles) (*first revision*)
IS 1130 : 1969	Specification for marble (blocks, slabs and tiles)
IS 3316 : 1974	Specification for structural granite (*first revision*)
IS 3622 : 1977	Specification for sand stone (slabs and tiles) (*first revision*)
IS 14223 (Part 1) : 2023	Polished building stones specification: Part 1 Granite and similar stones (*first revision*)

f) Bituminous Flooring

IS No.	*Title*
IS 1195 : 2002	Specification for bitumen mastic for flooring (*third revision*)
IS 5317 : 2002	Specification for pitch-mastic for bridge decking and roads (*second revision*)
IS 8374 : 1977	Specification for bitumen mastic, anti-static and electrically conducting grade
IS 9510 : 1980	Specification for bitumen mastic acid resisting grade
IS 13026 : 1991	Specification for bitumen mastic for flooring for industries handling LPG and other light hydrocarbon products

IS 15194 : 2002	Specification for pitch-mastic flooring for industries handling heavy hydrocarbon products like kerosene, diesel and furnace oil

g) Plastic Flooring

IS No.	*Title*
IS 3461 : 1980	Specification for PVC asbestos floor tiles (*first revision*)
IS 3462 : 1986	Specification for unbacked flexible PVC flooring (*second revision*)
IS 3464 : 1986	Methods of test for plastic flooring and wall tiles (*second revision*)

h) Ceramic/Vitreous Flooring and Wall Finishing

IS No.	*Title*
IS 2333 : 2025	Plaster of Paris for ceramic and glass industry — Specification (*third revision*)
IS 4457 : 2007	Ceramic unglazed vitreous acid resisting tile — Specification (*second revision*)
IS 13630	Ceramic tiles methods of test sampling and basis for acceptance:
(Part 1) : 2019	Determination of dimensions and surface quality (*second revision*)
(Part 2) : 2019	Determination of water absorption and bulk density (*second revision*)
(Part 3) : 2019	Determination of moisture expansion using boiling water (*second revision*)
(Part 4) : 2019	Determination of linear thermal expansion (*second revision*)
(Part 5) : 2019	Determination of resistance to thermal shock (*second revision*)
(Part 6) : 2019	Determination of modulus of rupture and breaking strength (*second revision*)
(Part 7) : 2019	Determination of stain and chemical resistance of unglazed tiles (*second revision*)
(Part 8) : 2019	Determination of stain and chemical resistance of glazed tiles (*second revision*)
(Part 9) : 2019	Determination of crazing resistance of glazed tiles (*second revision*)
(Part 10) : 2019	Determination of frost resistance (*second revision*)
(Part 11) : 2019	Determination of resistance to surface abrasion of glazed tiles (*second revision*)
(Part 12) : 2019	Determination of resistance to deep abrasion of unglazed tiles (*second revision*)
(Part 13) : 2019	Determination of scratch hardness of surface according to Mohs scale (*second revision*)
(Part 14) : 2019	Determination of impact resistance by measurement of coefficient of restitution (*first revision*)
(Part 15) : 2019	Sampling and basis for acceptance (*first revision*)
(Part 16) : 2019	Determination of lead and cadmium given off by glazed tiles

IS No.	*Title*
IS 13712 : 2019	Ceramic tiles definitions classifications characteristics and marking (*second revision*)
IS 15622 : 2017	Pressed ceramic tiles — Specification (*first revision*)
IS 17933 : 2022	Fine ceramics (advanced ceramics, advanced technical ceramics) — Test method for flexural strength of monolithic ceramics at room temperature
IS 17935 : 2022	Fine ceramics (advanced ceramics, advanced technical ceramics) — Determination of density and apparent porosity
IS 17975 : 2022	Fine ceramics (advanced ceramics, advanced technical ceramics) — test method for fracture resistance of silicon nitride materials for rolling bearing balls at room temperature by indentation fracture (IF) method
IS 17981 : 2022	Fine ceramics (advanced ceramics, advanced technical ceramics) — Test method for elastic moduli of monolithic ceramics at room temperature by sonic resonance
IS 17991 : 2022	Fine ceramics (advanced ceramics, advanced technical ceramics) — Test method for hardness of monolithic ceramics at room temperature
IS 18221 : 2023	Fine ceramics (Advanced ceramics, advanced technical ceramics) — Test method for linear thermal expansion of monolithic ceramics by push-rod technique

j) Clay Flooring

IS No.	*Title*
IS 1478 : 2023	Clay flooring tiles specification (*third revision*)
IS 3951	Hollow clay tiles for floors and roofs specification:
(Part 1) : 2023	Filler type (*third revision*)
(Part 2) : 2023	Structural type (*third revision*)

k) Roofing

IS No.	*Title*
IS 277 : 2018	Galvanized steel strips and sheets plain and corrugated — Specification (*seventh revision*)
IS 459 : 1992	Specification for corrugated and semi-corrugated asbestos cement sheets (*third revision*)
IS 654 : 2023	Clay roofing tiles mangalore pattern specification (*fourth revision*)
IS 1464 : 2024	Clay ridge and ceiling tiles — Specification (*third revision*)
IS 2690	Burnt clay flat terracing tiles — Specification:
(Part 1) : 2023	Machine made (*third revision*)
(Part 2) : 2023	Handmade (*third revision*)

IS No.	*Title*
IS 3951	Hollow clay tiles for floors and roofs specification:
(Part 1) : 2023	Filler type (*third revision*)
(Part 2) : 2023	Structural type (*third revision*)
IS 6073 : 2006	Autoclaved reinforced cellular concrete floor and roof slabs (*first revision*)
IS 6250 : 1981	Specification for roofing slate tiles (*first revision*)
IS 10388 : 1982	Specification for corrugated coir wood wool cement roofing sheets
IS 12583 : 1988	Specification for corrugated bitumen roofing sheets
IS 12866 : 2021	Plastic translucent sheets made from thermo-setting polyester resin (glass fibre reinforced) — Specification (*first revision*)
IS 13317 : 2023	Clay roofing country tiles half round and flat tiles Specification (*first revision*)
IS 13990 : 1994	Specification for precast reinforced concrete planks and joists for flooring and roofing
IS 14201 : 1994	Precast reinforced concrete channel units for construction of floors and roofs — Specification
IS 14241 : 1995	Specification for precast L-Panel units for roofing
IS 14246 : 2024	Continuously pre-painted galvanized steel sheets and strips — Specification (*second revision*)
IS 15965 : 2012	Pre-Painted aluminium zinc alloy metallic coated steel strip and sheet (plain)
IS 18385 : 2023	Hot-Dip galvanized/galvannealed steel sheet, plate and strip for automotive applications — Specification

m) Other Floorings and Roofings

IS No.	*Title*
IS 4456	Methods of test for chemical resistant mortars:
(Part 1) : 2026	Silicate type and resin type (*first revision*)
(Part 2) : 2026	Sulphur type (*first revision*)
IS 4457 : 2007	Ceramic unglazed vitreous acid resisting tile — Specification (*second revision*)
IS 4832	Chemical resistant mortars specification:
(Part 1) : 2023	Silicate type (*first revision*)
(Part 2) : 2023	Resin type (*first revision*)
(Part 3) : 2023	Sulphur type (*first revision*)
IS 4860 : 1968	Specification for acid resistant bricks

IS No.	*Title*
IS 14862 : 2000	Specification for fibre cement flat sheets
IS 14143 : 1994	Specification for prefabricated brick panel and partially precast concrete joist for flooring and roofing
IS 14871 : 2000	Products in fibre reinforced cement — Long corrugated or asymmetrical section sheets and fittings for roofing and cladding — Specification
IS 18433	Measurement of surface frictional properties — Methods of test:
(Part 1) : 2023	Pendulum test
(Part 2) : 2023	Ramp test

n) Wall Coverings/Finishing

IS No.	*Title*
IS 1542 : 1992	Specification for sand for plaster (*second revision*)
IS 3952 : 2013	Burnt clay hollow bricks and blocks for walls and partitions — Specification (*third revision*)
IS 4456	Methods of test for chemical resistant mortars:
(Part 1) : 1967	Silicate type and resin type
(Part 2) : 1967	Sulphur type
IS 4832	Chemical resistant mortars specification:
(Part 1) : 2023	Silicate type (*first revision*)
(Part 2) : 2023	Resin type (*first revision*)
(Part 3) : 2023	Sulphur type (*first revision*)
IS 15418 : 2003	Wall coverings in roll form for finished wall papers, wall vinyls and plastic wall coverings — Specification

15 GLASS

IS No.	*Title*
IS 2553 (Part 1) : 2018	Safety glass — Specification: Part 1 Architectural building and general uses (*fourth revision*)
IS 2835 : 1987	Specification for flat transparent sheet glass (*third revision*)
IS 3438 : 2023	Silvered glass mirrors for general purposes specification (*third revision*)
IS 5437 : 2024	Rolled glass patterned extra clear patterned wired and wired-patterned glass specification (*second revision*)
IS 9154 : 2020	Method for determination of alkali resistance of glass (*first revision*)
IS 12869	Methods for determination of viscosity and viscometric fixed points of glass:
(Part 1) : 2022	Standard test methods (*first revision*)

IS No.	*Title*
(Part 2) : 2022	Determination of softening point (*first revision*)
IS 14900 : 2018	Transparent float glass — Specification (*first revision*)
IS 16526 : 2017	Atactic polypropylene (APP) modified bituminous waterproofing and damp-proofing membrane with glass fibre reinforcement — Specification
IS 16945 : 2018	Fire resistance test for glass walls
IS 16947 : 2018	Fire resistance tests for doors with glass panes, openable glass windows and sliding glass doors
IS 16978	Glass in building — Forced-entry security glazing:
(Part 1) : 2022	Test and classification by repetitive ball drop (*first revision*)
(Part 2) : 2018	Test and classification by repetitive impact of a hammer and axe at room temperature
(Part 3) : 2018	Test and classification by manual attack
(Part 4) : 2018	Test and classification by pendulum impact under thermally and fire stressed conditions
IS 16982 : 2018	Heat strengthened glass — Specification
IS 17004 : 2018	Testing methods for processed glass
IS 17346 : 2020	Insulating glazing unit — Specification
IS 18222 : 2023	Glass-Resistance to attack by hydrochloric acid at 100 °C — Flame emission or flame atomic absorption spectrometric method
IS 18518 : 2023	Bullet resistant security glass — Quality and performance requirements
IS 19328 : 2025	Fire resistant glass — Specification
IS 19331 : 2025	Tinted float glass — Specification

16 GYPSUM BASED MATERIALS

IS No.	*Title*
IS 2095	Specification for gypsum plaster boards:
(Part 1) : 2023	Plain gypsum plaster boards (*fourth revision*)
(Part 2) : 2022	Coated laminated gypsum plaster boards (*third revision*)
(Part 3) : 2022	Reinforced gypsum plaster boards tiles cornices and mouldings (*third revision*)
IS 2542	Gypsum plaster concrete and products methods of test:
(Part 1)	Plaster and concrete,
(Sec 1) : 2023	Determination of normal consistency of gypsum plaster (*second revision*)

IS No.	*Title*
(Sec 2) : 2023	Determination of normal consistency of gypsum concrete (*second revision*)
(Sec 3) : 2023	Determination of setting time of plaster and concrete (*second revision*)
(Sec 4) : 2023	Determination of transverse strength of gypsum plaster (*second revision*)
(Sec 5) : 2023	Determination of compressive strength and dry set density of gypsum plaster (*second revision*)
(Sec 6) : 2023	Determination of soundness of gypsum plaster (*second revision*)
(Sec 7) : 2023	Determination of impact resistance of gypsum plaster by dropping ball test (*second revision*)
(Sec 8) : 2023	Determination of mass of coarse particles (*second revision*)
(Sec 9) : 2023	Determination of expansion of gypsum plaster (*second revision*)
(Sec 10) : 2023	Determination of sand in set gypsum plaster (*second revision*)
(Sec 11) : 2023	Determination of wood fibre content in wood fibre gypsum plaster (*second revision*)
(Sec 12) : 2023	Determination of dry bulk density (*second revision*)
(Sec 13) : 2023	Determination of free water (*second revision*)
(Sec 14) : 2023	Determination of fineness (*second revision*)
(Part 2)	Gypsum products,
(Sec 1) : 1981	Measurement of dimensions (*first revision*)
(Sec 2) : 1981	Determination of mass (*first revision*)
(Sec 3) : 1981	Determination of mass and thickness of paper surfacing (*first revision*)
(Sec 4) : 1981	Transverse strength (*first revision*)
(Sec 5) : 1981	Compressive strength (*first revision*)
(Sec 6) : 1981	Gypsum products, Section 6 Water absorption (*first revision*)
(Sec 7) : 1981	Moisture content (*first revision*)
(Sec 8) : 1981	Nail retention of precast reinforced gypsum slabs (*first revision*)
IS 2547	Specification for gypsum building plaster:
(Part 1) : 1976	Excluding premixed lightweight plaster (*first revision*)
(Part 2) : 1976	Premixed lightweight plaster (*first revision*)
IS 2849 : 1983	Specification for non-load bearing gypsum partition blocks (solid and hollow types) (*first revision*)
IS 8272 : 1984	Specification for gypsum plaster for use in the manufacture of fibrous plasterboards (*first revision*)

IS No.	*Title*
IS 9498 : 1980	Specification for inorganic aggregates for use in gypsum plaster
IS 12679 : 2023	By-Product gypsum for construction — Specification (*second revision*)
IS 17400 : 2021	Glass fibre reinforced gypsum building panels – Specification
IS 18387 : 2023	Gypsum ceiling tiles — Specification

17 MORTAR (Including Sand for Mortar)

IS No.	*Title*
IS 2116 : 1980	Sand for masonry mortars — Specification (*first revision*)
IS 2250 : 1981	Code of practice for preparation and use of masonry mortars (*first revision*)
IS 3085 : 1965	Method of test for permeability of cement mortar and concrete
IS 4098 : 1983	Specification for lime-pozzolana mixture (*first revision*)
IS 4456	Methods of test for chemical resistant mortars:
(Part 1) : 1967	Silicate type and resin type
(Part 2) : 1967	Sulphur type
IS 4832	Chemical resistant mortars specification:
(Part 1) : 2023	Silicate type (*first revision*)
(Part 2) : 2023	Resin type (*first revision*)
(Part 3) : 2023	Sulphur type (*first revision*)
IS 13077 : 1991	Guide for preparation and use of mud mortar in masonry
IS 14959	Method of test determination of water soluble and acid soluble chlorides in mortar and concrete
(Part 1) : 2001	Fresh mortar and concrete
(Part 2) : 2001	Hardened mortar and concrete

18 PAINTS AND ALLIED PRODUCTS

a) Water Based Paints and Pigments

IS No.	*Title*
IS 427 : 2013	Distemper, dry, colour as required — Specification (*third revision*)
IS 428 : 2013	Washable distemper — Specification (*third revision*)
IS 5410 : 2013	Specification for cement paint (*second revision*)
IS 15489 : 2024	Paint, plastic emulsion — Specification (*second revision*)

b) Ready Mixed Paints, Enamels and Powder Coatings

IS No.	*Title*
IS 101	Methods of sampling and test for paints, varnishes and related products:
(Part 1)	Tests on liquid paints general and physical,
(Sec 1) : 2023	Sampling (*fourth revision*)
(Sec 2) : 2023/ ISO 1513 : 2010	Preliminary examination and preparation of samples for testing (*fourth revision*)
(Sec 3) : 1986	Preparation of panels (*third revision*)
(Sec 4) : 2024	Brushing test (*fourth revision*)
(Sec 5) : 2024	Consistency (*fourth revision*)
(Sec 5) : 2024	Consistency (*fourth revision*)
(Sec 6) : 1987	Flash point (*third revision*)
(Sec 7) : 2020/ ISO 2811-1	Mass per 10 litres determination of density pycnometer method (*fourth revision*)
(Sec 8) : 2022/ ISO 787-9 : 2019	Pigments and extenders determination of *p*H value of an aqueous suspension (*first revision*)
(Part 2)	Test on liquid paints chemical examination,
(Sec 1) : 2018	Water content (*fourth revision*)
(Sec 2) : 2025	Determination of non-volatile matter content (*fourth revision*)
(Sec 3) : 2015/ ISO 11890-1	Determination of volatile organic compound VOC content — Difference method
(Sec 4) : 2022/ ISO 11890-2 : 2020	Determination of volatile organic compound VOC and or semi volatile organic compounds SVOC content gas-chromatographic (*first revision*)
(Sec 5) : 2017/ ISO 17895 : 2005	Determination of volatile organic compound content of low-VOC emulsion paints (In-Can VOC)
(Part 3)	Tests on paint film formation,
(Sec 1) : 1986	Drying time (*third revision*)
(Sec 2) : 2025/ ISO 2808 : 2019	Determination of film thickness (*fourth revision*)
(Sec 4) : 1987	Finish (*third revision*)
(Sec 5) : 2022/ ISO 1524 : 2020	Determination of fineness of grind (*fifth revision*)
(Sec 6) : 2024/ ISO 19403-3 : 2017	Determination of the surface tension of liquid using the pendant drop method
(Part 4)	Optical test,

IS No.	*Title*
(Sec 1) : 1988	Opacity (*third revision*)
(Sec 2) : 2021/ ISO 3668 : 2017	Colour-visual comparison of colour of paints (*fourth revision*)
(Sec 3) : 1988	Light fastness test (*third revision*)
(Sec 4) : 2020/ ISO 2813 : 2014	Gloss determination of gloss value at 20°, 60° and 85° (*fourth revision*)
(Part 5)	Mechanical test on paint films,
(Sec 1) : 1988	Hardness tests (*third revision*)
(Sec 2) :1988	Flexibility and adhesion (*third revision*)
(Sec 3) : 2019/ ISO 6272-1 : 2011	Impact resistance — Falling-weight test large-area indenter (*fifth revision*)
(Sec 4) : 1988	Print free test (*third revision*)
(Sec 5) : 2019/ ISO 6272-2 : 2011	Impact resistance — Falling-weight test small-area indenter
(Sec 6) : 2024/ ISO 7784-1 : 2023	Determination of resistance to abrasion by abrasive-paper covered wheels and rotating test specimen
(Sec 7) : 2024/ ISO 7784-2 : 2023	Determination of resistance to abrasion by abrasive rubber wheels and rotating test specimen
(Part 6)	Durability tests,
(Sec 1) : 1988	Resistance to humidity under conditions of condensation (*third revision*)
(Sec 2) : 1989	Keeping properties (*third revision*)
(Sec 3) : 1990	Moisture vapour permeability (*third revision*)
(Sec 4) : 1991	Degradation of coatings (pictorial aids for evaluation)
(Sec 5) : 1997	Accelerated weathering test (*third revision*)
(Part 7)	Environmental tests on paint films,
(Sec 1) : 1989	Resistance to water (*third revision*)
(Sec 2) : 1990	Resistance to liquids (*third revision*)
(Sec 3) : 2020/ ISO 3248 : 2016	Determination of the effect of heat (*fourth revision*)
(Sec 4) : 1990	Resistance to bleeding of pigments (*third revision*)
(Part 8)	Tests for pigments and other solids,
(Sec 1) : 1989	Residue on sieve (*third revision*)
(Sec 2) : 1990	Pigments and non –volatile matter (*third revision*)

IS No.	*Title*
(Sec 3):1993	Ash content (*third revision*)
(Sec 4) : 2015	Phthalic anhydride (*fourth revision*)
(Sec 5) : 2022	Lead restriction test (*fourth revision*)
(Sec 6) : 1993	Volume solids
(Part 9)	Tests for lacquers and varnish,
(Sec 1) : 1993	Acid value (*third revision*)
(Sec 2) : 1993	Rosin test (*third revision*)
(Part 10)	Instrumental analysis,
(Sec 1) : 2022/ ISO 13320 : 2020	Particle size analysis — Laser diffraction methods (*first revision*)
(Sec 2) : 2018/ ISO 16773-1 : 2016	Electrochemical impedance spectroscopy EIS on coated and uncoated metallic specimens — Terms and definitions
(Sec 3) : 2018/ ISO 16773-2 : 2016	Electrochemical impedance spectroscopy EIS on coated and uncoated metallic specimens — Collection of data
(Sec 4) : 2018/ ISO 16773-3 : 2016	Electrochemical impedance spectroscopy EIS on coated and uncoated metallic specimens — Processing and analysis of data from dummy cells
(Sec 5) : 2018/ ISO 16773-4	Electrochemical impedance spectroscopy EIS on coated and uncoated metallic specimens — Examples of spectra of polymer — Coated and uncoated specimens
(Part 11)	Evaluation of degradation of coatings — Designation of quantity and size of defects and of intensity of uniform changes in appearance,
(Sec 2) : 2024/ ISO 4628-2 : 2016	Assessment of degree of blistering
(Sec 5) : 2024/ ISO 4628-5 : 2022	Assessment of degree of flaking
IS 104 : 2017	Specification for ready mixed paint brushing zinc chrome priming (*third revision*)
IS 109 : 2017	Specification for ready mixed paint finishing priming plaster to Indian Standard colour no 361 light stone and no 631 light grey white and off — white (*second revision*)
IS 137 : 2017	Ready mixed paint brushing matt or eggshell flat finishing interior to Indian standard colour as required — Specification (*second revision*)
IS 151 : 2017	Ready mixed paint, spraying, finishing, stoving, enamel for general purposes, colour as required — Specification (*second revision*)
IS 158 : 2015	Ready mixed paint brushing bituminous black acid alkali and heat resisting — Specification (*fourth revision*)
IS 159 : 2024	Ready mixed paint, brushing, acid resisting — Specification (*second revision*)
IS 164 : 2023	Ready mixed paint for road marking — Specification (*third revision*)

IS No.	*Title*
IS 168 : 2024	Ready mixed paint air drying for general purpose — Specification (*fifth revision*)
IS 341 : 2016	Black japan, types A, B and C — Specification (*second revision*)
IS 486 : 2023	Brushes, sash tool for paints and varnishes — Specification (*fifth revision*)
IS 2074 : 2023	Ready mixed paint air drying red oxide zinc chrome priming — Specification (*fourth revision*)
IS 2075 : 2017	Ready mixed paint stoving red oxide zinc chrome priming — Specification (*third revision*)
IS 2339 : 2013	Specification for aluminium paint for general purposes (*first revision*)
IS 2932 : 2025	Air drying enamel paint for architectural and industrial applications — Specification (*fifth revision*)
IS 3536 : 2016	Ready mixed paint, brushing, wood primer — Specification (*second revision*)
IS 3537 : 2017	Ready mixed paint, finishing, interior, for general purposes, to Indian Standard colours — Specification (*first revision*)
IS 3539 : 1966	Specification for ready mixed paint, undercoating, for use under oil finishes, to Indian Standard colours, as required
IS 3585 : 2018	Ready mixed paint, aluminium, brushing, priming, water resistant for woodwork — Specification (*first revision*)
IS 3678 : 2016	Ready mixed paint, thick white, for lettering — Specification (*first revision*)
IS 8687	Vitreous enamels and frits — Method of test:
(Part 1) : 2023	Sieve analysis (*first revision*)
(Part 2) : 2020	Fusion flow test (*first revision*)
IS 8709 : 2023	Colour retention of vitreous enamel coatings — Method of test (*first revision*)
IS 8767 : 2020	Calcium carbonate, precipitated, and activated, for paints —Specification (*first revision*)
IS 9862 : 2017	Specification for ready mixed paint brushing bituminous black acid alkali water and chlorine resisting (*first revision*)
IS 11883 : 2017	Ready mixed paint, brushing, red oxide, priming for metals – Specification (*first revision*)
IS 12744 (Part 1) : 2013	Ready mixed paint, air drying, red oxide — Zinc phosphate, priming – Specification: Part 1 For domestic and decorative applications (*fourth revision*)
IS 13183 : 1991	Specification for aluminium paints, heat resistant
IS 13213 : 2018	Polyurethane full gloss enamel (two pack) — Specification (*first revision*)
IS 13607 : 1992	Specification for ready mixed paint, finishing, general purposes, synthetic
IS 13871 : 2021	Powder coating — Specification (*first revision*)
IS 14506 : 1998	Epoxy redoxide zinc phosphate weldable primer, two component — Specification

IS No.	*Title*
IS 14589 : 1999	Zinc priming paint, epoxy based, two pack — Specification
IS 16719 : 2022	Solvent free polyurethane coatings for exterior of steel pipe and fittings — Specification (*first revision*)
IS 16942 : 2018	Silicone soya alkyd exterior paint — Specification
IS 17044 : 2018	Intumescent fire retardant paint (water based) — Specification
IS 17680 (Part 1) : 2021/ ISO 18314-1 : 2015	Analytical colorimetry: Part 1 Practical colour measurement
IS 18563 : 2024	Nanomodified paints and coatings — Code of practice
IS 19003 : 2022	Re-dispersible polymer powder — Specification

c) Thinners and Solvents

IS No.	*Title*
IS 82:1973	Methods of sampling and test for thinners and solvents for paints (*first revision*)
IS 384	Flat brushes for paints and varnishes — Specification:
(Part 1) : 2023	Heavy duty (*seventh revision*)
(Part 2) : 2023	Household purposes *(seventh revision)*
IS 430 : 1972	Specification for paint remover, solvent type, non-flammable (*second revision*)
IS 431 : 1972	Specification for Paint remover, solvent type, flammable (*second revision*)
IS 486 : 2023	Brushes, sash tool for paints and varnishes — Specification (*fifth revision*)
IS 487 : 2023	Brushes, paints and varnishes – Oval, ferrule bound; and round, ferrule bound (*sixth revision*)
IS 533:2007	Specification for gum spirit of turpentine (oil of turpentine) (*third revision*)
IS 644 : 2023	Dipentene for paints — Specification (*third revision*)
IS 5667 : 1970	Specification for thinner for cellulose nitrate based paints and lacquers
IS 14314 : 1995	Specification for thinner general purposes for synthetic paints and varnishes

d) Varnishes and Lacquers

IS 344 : 2023	Varnish stoving — Specification (*second revision*)
IS 347 : 2023	Varnish shellac for general purposes — Specification (*second revision*)
IS 348 : 1968	Specification for French polish (*first revision*)
IS 524 : 2024	Varnish for general purpose — Specification (*third revision*)
IS 642 : 2024	Varnish medium for aluminium paint — Specification *(second revision)*

IS 5691 : 2024	Lacquer, cellulose nitrate, pigmented, finishing, glossy — Specification (*first revision*)
IS 6127 : 2023	Varnish, spar and fungicidal — Specification (*first revision*)
IS 10018:1981	Specification for lacquer, cellulose, nitrate clear, finishing, glossy for wood

19 POLYMERS, PLASTICS AND GEOSYNTHETICS/GEOTEXTILES

IS No.	*Title*
IS 1998 : 1962	Methods of test for thermosetting synthetic resin bonded laminated sheets
IS 2036 : 1995	Specification for phenolic laminated sheets (*second revision*)
IS 2046 : 2025/ ISO 4586 : 2018	Decorative thermosetting synthetic resin bonded laminated sheets — Specification:
(Part 1) : 2025	Introduction and general information (*third revision*)
(Part 2) : 2025	Determination of properties (*third revision*)
(Part 3) : 2025	Classification and specifications for laminates less than 2 mm thick and intended for bonding to supporting substrates (*third revision*)
(Part 4) : 2025	Classification and specifications for compact laminates of thickness 2 mm and greater (*third revision*)
(Part 5) : 2025	Classification and specifications for flooring grade laminates less than 2 mm thick intended for bonding to supporting substrates (*third revision*)
(Part 6) : 2025	Classification and specifications for exterior-grade compact laminates of thickness 2 mm and greater (*third revision*)
(Part 7) : 2025	Classification and specifications for design laminates (*third revision*)
(Part 8) : 2025	Classification and specifications for alternative core laminates (*third revision*)
IS 2076 : 1981	Specification for unsupported polyvinyl chloride sheeting (*first revision*)
IS 2508 : 2024	Polyethylene films and sheets — Specification (*fourth revision*)
IS 6307 : 2023	Rigid PVC sheets — Specification (*second revision*)
IS 10889 : 2004	Specification for high density polyethylene films (*first revision*)
IS 12830:1989	Specification for rubber based adhesives for fixing PVC tiles to cement
IS 13162	Geotextiles — Methods of test:
(Part 2) : 2024	Determination of resistance to the exposure of ultraviolet light, moisture and heat (xenon-arc type apparatus) (*first revision*)
(Part 3) : 2021	Single layers (*first revision*)
(Part 4) : 2025	Dynamic perforation test (cone drop test) (*first revision*)
IS 13262 : 1992	Specification for pressure sensitive adhesive tapes with plastic base

IS No.	*Title*
IS 13325 : 1992	Method of test for the determination to tensile properties of extruded polymer geogrids using the wide strip
IS 13326 (Part 1) : 2025/ ISO 12957-1 : 2018	Geosynthetics — Determination of friction characteristics: Part 1 Direct shear test (*first revision*)
IS 14182 : 1994	Specification for solvent cement for use with unplasticized polyvinylchloride plastic pipe and fittings
IS 14293 : 2024	Geotextiles — Method for determination of trapezoid tearing strength (*first revision*)
IS 14294 : 2024	Geotextiles — Method for determination of apparent opening size by dry sieving technique (*first revision*)
IS 14324 : 2025/ ISO 11058 : 2019	Geotextiles and geotextile-related products — Determination of water permeability characteristics normal to the plane, without load (*first revision*)
IS 14443 : 1997	Specification for polycarbonate sheets
IS 14500 : 1998	Linear low-density polyethylene (LLDPE) films — Specification
IS 14643 : 1999	Specification for unsintered polytetrafluoroethylene (PTFE) Tape for thread sealing applications
IS 14706 : 2024/ ISO 9862 : 2023	Geosynthetics — Sampling and preparation of test specimens (*first revision*)
IS 14714 : 2024	Geotextiles — Determination of abrasion resistance (*first revision*)
IS 14715	Jute geotextiles:
(Part 1) : 2016	Strengthening of sub-grade in roads — Specification (*second revision*)
(Part 2) : 2016	Control of bank erosion in rivers and waterways — Specification (*second revision*)
IS 14716 : 2021	Geosynthetics — Test method for the determination of mass per unit area of geotextiles and geotextile-related products (*first revision*)
IS 14739 : 2026/ ISO 13431 : 1999	Geotextiles and geotextile-related products: Determination of tensile creep and creep rupture behaviour (*second revision*)
IS 14753 : 1999	Specification for poly(methyl) methacrylate (PMMA) (acrylic) sheets
IS 14986 : 2001	Guidelines for application of jute geotextile for rain water erosion control in road and railway embankments and hill slopes
IS 15060 : 2018/ ISO 10321 : 2008	Geosynthetics tensile test for joint seams by wide-width strip method (*first revision*)
IS 15351 : 2015	Agro textiles laminated high density polyethylene HDPE woven geomembrane for water proof lining — Specification (*second revision*)
IS 15869 : 2020	Textiles-Open weave coir Bhoovastra — Specification (*first revision*)
IS 15909 : 2020	PVC geomembranes for lining — Specification (*second revision*)

IS No.	*Title*
IS 16352 : 2020	Geosynthetics — High density polyethylene (HDPE) geomembranes for lining-specification (*first revision*)
IS 16362 : 2024	Geosynthetics — Geotextiles for subsurface drainage, subgrade separation, subgrade stabilization, filtration and erosion control (in hard armor systems) applications — Specification (*second revision*)
IS 16380 : 2020	Geosynthetics — Method of test for measuring pullout resistance of geosynthetics in soil (*first revision*)
IS 16483 : 2017	Geosynthetics — Method for microscopic evaluation of the dispersion of carbon black in polyolefin geosynthetics
IS 16493 : 2017	Geosynthetics — Method of test for determination of pyramid puncture resistance of unprotected and protected geomembranes
IS 16627 : 2017	Agro textiles — High density polyethylene (HDPE) laminated woven lay flat tube for use in mains and submains of drip irrigation system — Specification
IS 16635 : 2026	Geosynthetics — Wide-width tensile test (*first revision*)
IS 16653 : 2017	Geosynthetics — Needle punched nonwoven geobags for coastal and waterways protection — Specification
IS 16654 : 2017	Geosynthetics — Polypropylene multifilament woven geobags for coastal and waterways protection — Specification
IS 17179 : 2025/ ISO 12958 : 2020	Geotextiles and geotextile-related products — Determination of water flow capacity in their plane:
(Part 1)	Index text
(Part 2)	Performance test (*first revision*)
IS 17360 : 2020	Geosynthetics — Screening test method for determining the resistance of geotextiles and geotextile-related products to oxidation
IS 17363 : 2025	Geotextiles and geotextile-related products — Screening test method for determining the resistance to acid and alkaline liquids (*first revision*)
IS 17365 : 2025/ ISO/TS 20432 : 2022	Determination of the long-term strength of geosynthetics for soil reinforcement — Guidelines (*first revision*)
IS 17368 : 2020	Geosynthetics — Determination of damage to geosynthetic caused during installation
IS 17369	Geotextiles and geotextile-related products — Strength of internal structural junctions:
(Part 1) : 2020/ ISO 13426-1 : 2019	Geocells
(Part 2) : 2025/ ISO 13426-2 : 2005	Geocomposites (*first revision*)
IS 17371 : 2020	Geosynthetics — Geogrids for flexible pavements — Specification
IS 17372 : 2020	Geosynthetics — Polymeric strip/geostrip used as soil reinforcement in retaining structures — Specification

IS No.	*Title*
IS 17373 : 2020	Geosynthetics — Geogrids used in reinforced soil retaining structures — Specification
IS 17374 : 2020	Geosynthetics — Reinforced HDPE membrane for effluents and chemical resistance lining — Specification
IS 17420 : 2020	Geosynthetics — Index test procedure for the evaluation of mechanical damage under repeated loading-damage caused by granular material (laboratory test method)
IS 17483	Geosynthetics — Geocells — Specification:
(Part 1) : 2020	Load bearing application
(Part 2) : 2020	Slope erosion protection application
IS 17544 : 2022	Multilayer (PE-AL-PE) plastic piping system for indoor gas installations with a maximum operating pressure up to and including 500 KPA (5 Bar) — Specification
IS 17728 : 2021	Agro textiles — High density polyethylene (HDPE) laminated woven lay flat tube and fittings for use in rain irrigation system — Specification
IS 17880 : 2022	Geosynthetics — Synthetic polymer rope gabions for coastal and waterways protection — Specification
IS 18309 : 2023	Geosynthetics — Prefabricated vertical drains for quick consolidation of very soft plastic soil-Specification
IS 18998 : 2025/ ISO 24187 : 2023	Principles for the analysis of microplastics present in the environment
IS 19050 : 2025/ ISO 17541 : 2014	Plastics — Quantitative evaluation of scratch-induced damage and scratch visibility
IS 19169 : 2025/ ISO 21475 : 2019	Plastics — Methods of exposure to determine the wavelength dependent degradation using spectrally dispersed radiation
IS 19104 : 2025	Geosynthetics — Geotextile tubes for coastal and waterways protection — Specification
IS 19161 : 2025/ ISO 16869 : 2008	Plastics — Assessment of the effectiveness of fungistatic compounds in plastics formulations
IS 19162 : 2025/ ISO 15314 : 2018	Plastics — Methods for marine exposure
IS 19165 : 2025/ ISO 22196 : 2011	Measurement of antibacterial activity on plastics and other non-porous surfaces
IS 19166 : 2025/ ISO 10640 : 2011	Plastics — Methodology for assessing polymer photoageing by FTIR and UV/visible spectroscopy
IS 19167 : 2025/ ISO 21702 : 2019	Measurement of antiviral activity on plastics and other non-porous surfaces
IS 19168 : 2025/ ISO 29664 : 2010	Plastics — Artificial weathering including acidic deposition

IS No.	*Title*
IS 19218 : 2025	Textiles — Coir non-woven stitched composite geotextiles for erosion control applications — Specification

20 SANITARY APPLIANCES AND WATER FITTINGS

a) General

IS No.	*Title*
IS 804 : 1967	Specification for rectangular pressed steel tanks (*first revision*)
IS 906 : 2024	Portable and fixed type revolving branch for firefighting — Specification (*fourth revision*)
IS 1726 : 1991	Specification for cast iron manhole covers and frames (*third revision*)
IS 2963 : 2013	Copper alloy waste fittings and waste plug for wash-basins and sinks — Specification (*second revision*)
IS 5219 : 2013	Cast copper alloy traps — Specification (*first revision*)
IS 5455 : 1969	Specification for cast-iron steps for manholes
IS 5961 : 1970	Specification for cast iron gratings for drainage purposes
IS 9140 : 1996	Method of sampling of vitreous and fire clay sanitary appliances (*second revision*)
IS 9872 : 1981	Specification for precast concrete septic tanks
IS 12592 : 2002	Specification for precast concrete manhole covers and frames (*first revision*)
IS 12701 : 1996	Specification for rotational moulded polyethylene water storage tanks (*first revision*)
IS 13356 : 1992	Specification for precast ferrocement water tanks (250 to 10 000 litres capacity)
IS 14399	Hot press moulded thermosetting glass fibre reinforced (GRP) sectional water storage tanks:
(Part 1) : 1996	Specification for panels
(Part 2) : 1996	Guidelines for assembly, installation and testing
IS 17650	Water efficient plumbing products — Requirements:
(Part 1) : 2021	Sanitaryware
(Part 2) : 2021	Sanitary fittings
IS 18666 : 2024	Rotationally moulded polyethylene septic tanks — Specification
IS 18797 : 2024	Packaged sewage treatment plant — Specification
IS 19315 : 2025	Ball valves for water works purposes — Specification

b) Pipes and Fittings excluding valves

i) *Brass and Copper Pipes and Fittings*

IS No.	*Title*
IS 407 : 1981	Specification for brass tubes for general purposes (*third revision*)
IS 2501 : 1995	Specification for solid drawn copper tubes for general engineering purposes (*third revision*)

ii) *Cast Iron Pipes and Fittings*

IS No.	*Title*
IS 1536 : 2001	Specification for centrifugally cast (spun) iron pressure pipes for water, gas and sewage (*fourth revision*)
IS 1536 : 2023	Centrifugally cast spun iron pressure pipes for water gas and sewage specification (*fifth revision*)
IS 1537 : 1976	Specification for vertically cast iron pressure pipes for water, gas and sewage (*first revision*)
IS 1538 : 1993	Specification for cast iron fittings for pressure pipes for water, gas and sewage (*third revision*)
IS 1729 : 2023	Sand cast iron spigot and socket pipes, fittings and accessories — Specification (*third revision*)
IS 1729 : 2023	Sand cast iron spigot and socket pipes fittings and accessories — Specification (*third revision*)
IS 1879 : 2010	Malleable cast iron pipe fittings — Specification (*third revision*)
IS 3989 : 2024	Centrifugally cast (spun) iron spigot and socket pipes, fittings and accessories — Specification (*fourth revision*)
IS 5531 : 2014	Cast iron specials for asbestos cement pressure pipes for water and sewage — Specification (*third revision*)
IS 6418 : 1971	Specification for cast iron and malleable cast iron flanges for general engineering purposes
IS 7181 : 1986	Specification for horizontally cast iron double flanged pipes for water, gas and sewage (*first revision*)
IS 8329 : 2000	Specification for centrifugally cast (spun) ductile iron pressure pipes for water, gas and sewage (*third revision*)
IS 8794 : 1988	Specification for cast iron detachable joints for use with asbestos cement pressure pipes (*first revision*)
IS 9523 : 2000	Ductile iron fittings for pressure pipes for water gas and sewage — Specification (*first revision*)
IS 10292 : 1988	Dimensional requirements for rubber sealing rings for C.I.D. Joints in asbestos cement piping (*first revision*)

IS No.	*Title*
IS 10299 : 1982	Cast iron saddle pieces for service connection from asbestos cement pressure pipes
IS 11606 : 1986	Methods of sampling cast iron pipes & fittings
IS 12820 : 2004	Dimensional requirements of rubber gaskets for mechanical joints and push-on joints for use with cast iron pipes and fittings for carrying water gas and sewage (*first revision*)
IS 12987 : 1991	Cast iron detachable joints for use with asbestos cement pressure pipes (light duty) — Specification
IS 12988 : 1991	Rubber sealing rings for CID joints for light duty AC pipes — Dimensional requirements
IS 13382 : 2018	Cast iron specials for mechanical and push-on flexible joints for pressure pipes for water gas and sewage — Specification (*second revision*)
IS 15905 : 2024	Hubless centrifugally cast (spun) iron pipes, fittings and accessories — Spigot series — Specification (*first revision*)

iii) *Lead Pipes and Fittings*

IS No.	*Title*
IS 404 (Part 1) : 1993	Specification for lead pipes: Part 1 For other than chemical purpose (*third revision*)

iv) *Fibre Pipes and Fittings*

IS No.	*Title*
IS 11925 : 1986	Specification for pitch-impregnated fibre pipes and fittings for drainage purposes

v) *Plastic Pipes and Fittings*

IS No.	*Title*
IS 4984 : 2016	Polyethylene pipes for water supply — Specification (*fifth revision*)
IS 4985 : 2021	Unplasticized PVC pipes for potable water supplies — Specification (*fourth revision*)
IS 7834	Specification for injection moulded PVC socket fittings with solvent cement joints for water supplies:
(Part 1) : 1987	General requirements (*first revision*)
(Part 2) : 1987	Specific requirements for 45° elbows (*first revision*)
(Part 3) : 1987	Specific requirements for 90° elbows (*first revision*)
(Part 4) : 1987	Specific requirements for 90° tees (*first revision*)
(Part 5) : 1987	Specific requirements for 45° tees (*first revision*)

IS No.	*Title*
(Part 6) : 1987	Specific requirements for sockets (*first revision*)
(Part 7) : 1987	Specific requirements for unions (*first revision*)
(Part 8) : 1987	Specific requirements for caps (*first revision*)
IS 8008 : 2022	Injection moulded machined polyethylene fittings for water supply — Specification (*second revision*)
IS 8360 : 2022	Fabricated polyethylene fittings for water supply — Specification (*first revision*)
IS 10124	Specification for fabricated PVC fittings for potable water supplies:
(Part 1) : 2009	General requirements (*second revision*)
(Part 2) : 2009	Specific requirements for sockets (*second revision*)
(Part 3) : 2009	Specific requirements of straight reducers (*second revision*)
(Part 4) : 2009	Specific requirements for caps (*second revision*)
(Part 5) : 2009	Specific requirements for equal tees (*second revision*)
(Part 6) : 2009	Specific requirements for flanged tail piece with metallic flanges (*second revision*)
(Part 7) : 2009	Specific requirements for threaded adaptors (*second revision*)
(Part 8) : 2009	Specific requirements for 90° bends (*second revision*)
(Part 9) : 2009	Specific requirements for 60° bends (*second revision*)
(Part 10) : 2009	Specific requirements for 45° bends (*second revision*)
(Part 11) : 2009	Specific requirements for 30° bends (*second revision*)
(Part 12) : 2009	Specific requirements for 22½° bends (*second revision*)
(Part 13) : 2009	Specific requirements for 11¼° bends (*second revision*)
IS 12235	Methods of test for unplasticised PVC pipes for potable water supplies:
(Part 1) : 2004	Measurement of dimensions (*first revision*)
(Part 2) : 2004	Determination of Vicat softening temperature (*first revision*)
(Part 3) : 2004	Test for opacity (*first revision*)
(Part 4) : 2004	Determining the detrimental effect on the composition of water (*first revision*)
(Part 5)	Longitudinal reversion:
(Sec 1) : 2004	Determination methods (*first revision*)
(Sec 2) : 2004	Determination parameters (*first revision*)
(Part 6) : 2004	Stress relief test (*first revision*)
(Part 7) : 2004	Resistance to sulphuric acid (*first revision*)

IS No.	*Title*
(Part 8)	Resistance to internal hydrostatic pressure,
(Sec 1) : 2004	Resistance to internal hydrostatic pressure at constant internal water pressure (*first revision*)
(Sec 2) : 2004	Leak-tightness of elastomeric sealing ring type socket joints under positive internal pressure and with angular deflection (*first revision*)
(Sec 3) : 2004	Leak-tightness of elastomeric sealing ring type socket joints under negative internal pressure and with angular deflection (*first revision*)
(Sec 4) : 2004	Leak-tightness of elastomeric sealing ring type socket joints under positive internal pressure without angular deflection (*first revision*)
(Part 9) : 2004	Resistance to external blows (impact resistance) at 0 °C (Round-the-clock method)
(Part 10) : 2004	Determination of organotin as tin aqueous solution (*first revision*)
(Part 11) : 2004	Resistance to dichloromethane at specified temperature (*first revision*)
(Part 12) : 2004	Determination of titanium dioxide content
(Part 13) : 2004	Determination of tensile strength and elongation
(Part 14) : 2004	Determination of density/relative density (specific gravity)
(Part 15) : 2004	Determination of vinyl chloride monomer content
(Part 16) : 2004	High temperature test
(Part 17) : 2004	Determination of ash content and sulphated ash content
(Part 18) : 2004	Determination of ring stiffness
(Part 19) : 2004	Flattening test
IS 12709 : 1994	Specification for glass-fibre reinforced plastic (GRP) pipes joints and fittings for use for potable water supply (*first revision*)
IS 12818 : 2010	Specification for unplasticized PVC screen and casing pipes for bore/tubewell (*second revision*)
IS 13592:2013	Unplasticized polyvinyl chloride (PVC-U) pipes for soil and waste discharge systemfor inside and outside buildings including ventilation and rainwater system — Specification (*first revision*)
IS 14333 : 2022	Polyethylene pipes for sewerage and industrial chemicals and effluent — Specification (*first revision*)
IS 14402 : 1996	Specification for GRP pipes, joints and fittings for use for sewerage, industrial waste and water (other than potable)
IS14735 : 1999	Specification for unplasticized polyvinyl chloride (UPVC) injection moulded fittings for soil and waste discharge system for inside buildings including ventilation and rain water system

IS No.	*Title*
IS 14885 : 2022	Polyethylene pipes for the supply of gaseous fuels — Specification (*first revision*)
IS 15328:2003	Specification for unplasticized non-pressure polyvinyl chloride (PVC-U) pipes for use in underground drainage and sewerage system
IS 15450 : 2022	Polyethylene-aluminium-polyethylene composite pressure pipes for hot and cold water supplies — Specification (*first revision*)
IS 15778 : 2007	Chlorinated polyvinyl chloride (CPVC) pipes for potable hot and cold water distribution supplies — Specification
IS 15801 : 2008	Polypropylene-random copolymer pipes for hot and cold water supplies — Specification
IS 15927	Specification for polyethylene fittings for use with polyethylene pipes for the supply of gaseous fuels:
(Part 1) : 2012	Fittings for sockets using heated tools
(Part 2) : 2012	Spigot fittings for butt fusion jointing or socket fusion using heated tools, spigot fittings for use with electro-fusion fittings
(Part 3) : 2011	Electro-fusion fittings
IS 16098	Structured — wall plastics piping systems for non-pressure drainage and sewerage — Specification
(Part 1) : 2013	Pipes and fittings with smooth external surface, Type A
(Part 2) : 2013	Pipes and fittings with non-smooth external surface, Type B
IS 16534 : 2017	Chlorinated polyvinyl chloride (CPCV) Pipe fittings for automatic sprinkler fire extinguishing system — Specification
IS 16647 : 2017	Oriented unplasticized polyvinyl chloride (PVC-O) pipes for water supply — Specification
IS 17546 : 2021	Chlorinated polyvinyl chloride (CPVC) fittings for potable hot and cold water distribution supplies — Specification

vi) *Steel Tubes, Pipes and Fittings*

IS No.	*Title*
IS 1239	Steel tubes tubulars and other wrought steel fittings — Specification:
(Part 1) : 2004	Steel tubes (*sixth revision*)
(Part 2) : 2011	Steel pipe fittings (*fifth revision*)
IS 3589 : 2001	Steel pipes for water and sewage 1 683 mm to 2 540 mm outside diameter — Specification (*third revision*)
IS 4270 : 2001	Steel tubes used for water wells — Specification (*third revision*)
IS 5504 : 2025	Spiral welded pipes — Specification (*second revision*)

IS 6286 : 1971	Specification for seamless and welded steel pipe for sub-zero temperature service
IS 6392 : 2020	Steel pipes flanges — Specification (*first revision*)

vii) *Stoneware Pipes and Fittings*

IS No.	*Title*
IS 651 : 2007	Glazed stoneware pipes and fittings — Specification (*sixth revision*)
IS 3006 : 1979	Specification for chemically resistant glazed stoneware pipes and fittings (*first revision*)

viii) *Asbestos Cement Pipes*

[*See* 9 (a) (ii) under the category 'Composite Matrix Products']

ix) *Concrete Pipes and Pipes lined/coated with Concrete or Mortar*

[*See* 9 (a) (iv) under the category 'Composite Matrix Products']

c) Kitchen and Sanitary Appliances

IS No.	*Title*
IS 774 : 2021	Ceramic vitreous china flushing cisterns for water closets and urinals — Specification (*sixth revision*)
IS 2548	Specification for plastic seats and covers for water-closets:
(Part 1) : 1996	Thermoset seats and covers (*fifth revision*)
(Part 2) : 1996	Thermoplastic seats and covers (*fifth revision*)
IS 2556	Vitreous china sanitary appliances — Specification:
(Part 1) : 2021	General requirements (*fourth revision*)
(Part 2) : 2024	Specific requirements of washdown water closets (*sixth revision*)
(Part 3) : 2024	Specific requirements of squatting pans (*sixth revision*)
(Part 4) : 2004	Specific requirements of wash basins (*fourth revision*)
(Part 5) : 1994	Specific requirements of laboratory sinks (*third revision*)
(Part 6) : 2021	Specific requirements of urinal and partition plates (*fifth revision*)
(Part 7) : 1995	Specific requirements of accessories for sanitary appliances (*third revision*)
(Part 8) : 2021	Specific requirements of close coupled and one-piece pedestal washdown and syphonic water closets (*sixth revision*)
(Part 9) : 2004	Specific requirements of pedestal type bidets (*fifth revision*)
(Part 14) : 2024	Specific requirements of integrated squatting pans (*second revision*)
(Part 15) : 2024	Specific requirements of universal water closets (*third revision*)

IS No.	*Title*
(Part 16) : 2024	Specific requirements of washdown wall mounted water closets (*first revision*)
(Part 17) : 2001	Specific requirements of wall mounted bidets
IS 6411 : 1985	Specification for gel-coated glass fibre reinforced polyester resin bath tubs (*first revision*)
IS 7231 : 2021	Plastic flushing cisterns for water closets and urinals — Specification (*third revision*)
IS 11246 : 1992	Specification for glass fibre reinforced polyester resins (GRP) squatting pans (*first revision*)
IS 13983 : 1994	Specification for stainless steel sinks for domestic purposes

d) Valves and Water Fittings (Including Ferrules)

IS No.	*Title*
IS 778 : 1984	Specification for copper alloy gate, globe, and check valves for water works purposes (*fourth revision*)
IS 781 : 1984	Specification for cast copper alloy screw-down bib taps and stop valves for water services (*third revision*)
IS 1701 : 1960	Specification for mixing valves for ablutionary and domestic purposes
IS 1703 : 2000	Water fittings — Copper alloy float valves (horizontal plunger type) — Specification (*fourth revision*)
IS 1711 : 1984	Specification for self-closing taps for water supply purposes (*second revision*)
IS 1795 : 1982	Specification for pillar taps for water supply purposes (*second revision*)
IS 2692 : 1989	Specification for ferrules for water services (*second revision*)
IS 3004 : 1979	Specification for plug cocks for water supply purposes (*first revision*)
IS 3042 : 1965	Specification for single faced sluice gates (200 mm to1 200 mm size)
IS 3950 : 1979	Specification for surface boxes for sluice valves (*first revision*)
IS 4038 : 1986	Specification for foot valves for water works purposes (*second revision*)
IS 4346 : 1982	Specification for washers for use with fittings for water services (*first revision*)
IS 5312	Specification for swing check type reflux (non-return) valves:
(Part 1) : 2004	Single door pattern (*second revision*)
(Part 2) : 2013	Multi-door pattern (*first revision*)
IS 8931 : 2021	Copper alloy single taps combination tap assemblies stop valves and single lever mixers for water services — Specification (*second revision*)
IS 9338 : 2013	Cast iron/S.G. iron, cast steel screw-down stop valves for water works purposes — Specification (*second revision*)

IS No.	*Title*
IS 9739 : 1981	Specification for pressure reducing valves for domestic water supply systems
IS 9758 : 1981	Specification for flush valves and fittings for water-closets and urinals
IS 9762 : 1994	Specification for polyethylene floats (spherical) for float valves (*first revision*)
IS 9763 : 2000	Plastic bib taps, pillar taps, angle valves, hot and cold water services — Specification (*second revision*)
IS 12234 : 2021	Plastic equilibrium float valves for cold water services — Specification (*first revision*)
IS 13049 : 1991	Specification for diaphragm type (plastic body) float operated valves for cold water services
IS 13114 : 1991	Specification for forged brass gate, globe and check valves for water works purposes
IS 13349 : 1992	Specification for single faced cast iron thimble mounted sluice gates
IS 14845 : 2000	Resilient seated cast iron air relief valves for water works purposes — Specification
IS 14846 : 2000	Sluice valves for water works purposes (50 mm to 1 200 mm) — Specification
IS 17585 : 2021	Beta (Outflow) valves for gravity type flushing cisterns — Specification

e) Water Meters

IS No.	*Title*
IS 779 : 1994	Specification for water meters (domestic type) (*sixth revision*)
IS 2373 : 1981	Specification for water meters (bulk type) (*third revision*)
IS 6784 : 1996	Method for performance testing of water meters (domestic type) (*second revision*)

21 STEEL AND ITS ALLOYS (*See* 23 FOR STRUCTURAL SECTIONS)

a) General

IS No.	*Title*
IS 1030 : 1998	Carbon steel castings for general engineering purposes (*fifth revision*)
IS 1136 : 2008	Preferred sizes for wrought metal products (*third revision*)
IS 1762 (Part 1) : 1974	Code for designation of steels: Part 1 Based on letter symbols (*first revision*)
IS 2049 : 1978	Colour code for the identification of wrought steel for general engineering purposes (*first revision*)
IS 2644 : 1994	High strength steel castings for general engineering and structural purposes — Specification (*fourth revision*)
IS 2708 : 1993	1.5 percent manganese steel castings for general engineering purposes — Specification (*third revision*)

IS No.	*Title*
IS 7598 : 1990	Classification of steels (*first revision*)
IS 10461	Resistance to inter-granular corrosion of austenitic stainless steels — Method for determination:
(Part 1) : 2025	Corrosion test in nitric acid medium by measurement of loss in mass (Huey Test) (*second revision*)
(Part 2) : 1994	Corrosion test in a sulphuric acid/copper sulphate medium in the presence of copper turnings (Monypenny strauss test) (*first revision*)
IS/ISO 11951 : 2016	Cold-reduced tinmill products — Blackplate *(first revision)*
IS 12591 : 2018	Cold-reduced tinmill products — Electrolytic chromium/chromium oxide — Coated steel (*second revision*)
IS 14126 : 2019	Metallic coatings — Measurement of coating thickness — Coulometric method by anodic dissolution (*first revision*)
IS 16733 : 2018	Ferritic-pearlitic engineering steels for precipitation hardening from hot-working temperatures
IS 16854 : 2018	Single pack zinc rich coating for cold application — Specification
IS 17111 : 2025/ ISO 683-17 : 2023	Heat-Treatable steels, alloy steels and free-cutting steels — Ball and roller bearing steels (*first revision*)
IS 18435 (Part 11) : 2023	Corrosion of metals and alloys — Stress corrosion testing: Part 11 Testing the resistance of metals and alloys to hydrogen embrittlement and hydrogen-assisted cracking — Guidelines
IS 18436 : 2023	Metallic and other inorganic coatings — Post-coating treatments of iron or steel to reduce the risk of hydrogen embrittlement
IS 18437 : 2023/ ISO 15324 : 2000	Corrosion of metals and alloys — Evaluation of stress cracking by the drop evaporation test
IS 18438 : 2023/ ISO 18069 : 2015	Corrosion of metals and alloys — Method for determination of the uniform corrosion rate of stainless steels and nickel based alloys in liquids
IS 18463 : 2023/ ISO 9587 : 2007	Metallic and other inorganic coatings — Pretreatment of iron or steel to reduce the risk of hydrogen embrittlement

b) Structural Steel

IS No.	*Title*
IS 1161 : 2014	Steel tubes for structural purposes — Specification (*fifth revision*)
IS 2062	Structural steel
(Part 1) : 2025	Hot rolled medium and high tensile steel (*eight revision*)
(Part 2) : 2025	Hot rolled quenched and tempered steel plates, sheets, strips and wide flats
IS 2830 : 2012	Specification for carbon steel cast billet ingots, billets, blooms and slabs for re-rolling into steel for general structural purposes (*third revision*)

IS No.	*Title*
IS 2831 : 2012	Specification for carbon steel billets ingots, blooms and slabs for re-rolling into structural steel (ordinary quality) (*fourth revision*)
IS 4923 : 2017	Hollow steel sections for structural use — Specification (*third revision*)
IS 8052 : 2006	Specification for steel ingots, billets and blooms for the production of springs, rivets and screws for general engineering applications (*second revision*)
IS 8952 : 1995	Steel ingots, blooms and billets for production of mild steel wire rods for general engineering purposes (*first revision*)
IS 11587 : 1986	Specification for structural weather resistant steels
IS 15103 : 2002	Fire resistant steel — Specification
IS 15911 : 2010	Structural steel (Ordinary quality) — Specification
IS 15962 : 2012	Structural steels for buildings and structures with improved seismic resistance
IS 16732 : 2019	Galvanized structural steel — Specification

c) Sheet and Strip

IS No.	*Title*
IS 277 : 2018	Galvanized steel strips and sheets plain and corrugated — Specification (*seventh revision*)
IS 412 : 1975	Specification for expanded metal steel sheets for general purposes (*second revision*)
IS 513	Cold reduced carbon steel sheet and strip:
(Part 1) : 2016	Cold forming and drawing purpose (*sixth revision*)
(Part 2) : 2016	High tensile and multi-phase steel (*sixth revision*)
IS 1079 : 2017	Hot rolled carbon steel sheet plate and strip — Specification (*seventh revision*)
IS 6911 : 2026	Stainless steel plate sheet and strip — Specification (*third revision*)
IS 7226 : 1974	Specification for cold rolled medium, high carbon and low alloy steel strip for general engineering purposes
IS 14246 : 2024	Continuously pre-painted galvanized steel sheets and strips — Specification (*second revision*)
IS 15961 : 2025	Hot-dip alumnium zinc alloy metallic coated steel strip and sheet — Specification (*first revision*)
IS 15965 : 2012	Pre-painted aluminium zinc alloy metallic coated steel strip & sheet (plain)
IS 18513 : 2023	Hot-dip zinc-aluminium-magnesium alloy coated steel sheets plates and strips — Specification
IS 18733 : 2024/ ISO 11531 : 2022	Metallic materials — Sheet and strip — Earing test

IS No.	*Title*
IS 19210 : 2025	Pre-Painted Zinc-aluminium-magnesium alloy coated steel strip and sheet
IS 19474 : 2026	Hot-dip aluminium and aluminium-silicon coated steel sheet and strip — Specification
IS 19608 : 2026	Hot-rolled carbon and alloy steel plates, sheets and strips — Specification
IS 19626 : 2026	Electrolytic Zinc-nickel alloy coated steel sheet and strip — Specification

d) Bars, Rods, Wire and Wire Rods

IS No.	*Title*
IS 280 : 2006	Mild steel wire for general engineering purposes — Specification (*fourth revision*)
IS 1148 : 2009	Steel rivet bars medium and high tensile — For structural purposes — For structural purposes (*fourth revision*)
IS 1673 : 1984	Specification for mild steel wire cold heading quality (*second revision*)
IS 1812 : 1982	Specification for carbon steel wire for the manufacture of wood screw (*second revision*)
IS 1835 : 1976	Specification for round steel wire for ropes (*third revision*)
IS 1875 : 2025	Steel ingots, billets, blooms, slabs, and bars for forging — Specification (*sixth revision*)
IS 1921 : 2025	Solder wire — Solid and flux cored — Specification (*third revision*)
IS 2591 : 1982	Dimensions for hot rolled bars for threaded components (*second revision*)
IS 3150 : 1982	Specification for hexagonal wire netting for general purposes
IS 4826 : 2023	Hot-dip galvanized coatings on round steel wires requirements (*second revision*)
IS 6528 : 2024	Stainless steel wire — Specification (*second revision*)
IS 6603 : 2024	Stainless steel semi-finished products, bars, wire rods and bright bars — Specification (*second revision*)
IS 7887:1992	Specification for mild steel wire rods for general engineering purposes (*first revision*)
IS 7904 : 2018	High carbon steel wire rods — Specification (*second revision*)
IS 10631 : 1983	Stainless steel for welding electrode core wire
IS/ISO 16124 : 2015/ ISO 16124 : 2015	Steel wire rod — Dimensions and tolerances (*first revision*)

e) Plates and Studs

IS No.	*Title*
IS 1862 : 1975	Specification for studs (*second revision*)
IS 3502 : 2009	Specification for steel chequered plates (*third revision*)

f) Tubes and Tubulars

IS No.	*Title*
IS 1161 : 2014	Specification for steel tubes for structural purposes (*fifth revision*)
IS 4923 : 2017	Hollow steel sections for structural use — Specification (*third revision*)
IS 17875 : 2022	Stainless steel seamless pipes and tubes for general services — Specification
IS 17876 : 2022	Stainless steel welded pipes and tubes for general services — Specification

g) Slotted Sections

IS No.	*Title*
IS 8081 : 1976	Specification for slotted sections

22 STONES

IS No.	*Title*
IS 1121	Methods of test for determination of strength properties of natural building stones:
(Part 1) : 2023	Compressive strength (*third revision*)
(Part 2) : 2023	Transverse strength (*third revision*)
(Part 3) : 2023	Indirect tensile strength (*third revision*)
(Part 4) : 2013	Shear strength (*second revision*)
(Part 5) : 2023	Flexural modulus of elasticity
IS 1122 : 2023	Determination of true specific gravity of natural building stones — Method of test (*second revision*)
IS 1123 : 1975	Method of identification of natural building stones (*first revision*)
IS 1124 : 1974	Method of test for determination of water absorption, apparent specific gravity and porosity of natural building stones (*first revision*)
IS 1125 : 2013	Method of test for determination of weathering of natural building stones (*second revision*)
IS 1126 : 2013	Method of test for determination of durability of natural building stones (*second revision*)
IS 1127 : 1970	Recommendations for dimensions and workmanship of natural building stones for masonry work (*first revision*)
IS 1128 : 1974	Specification for limestone (slab and tiles) (*first revision*)
IS 1129 : 1972	Recommendation for dressing of natural building stones (*first revision*)
IS 1130 : 1969	Specification for marble (blocks, slabs and tiles)

IS No.	*Title*
IS 1706 : 2024	Determination of resistance to wear by abrasion of natural building stones — Method of test (*second revision*)
IS 3316 : 1974	Specification for structural granite (*first revision*)
IS 3620 : 1979	Specification for laterite stone block for masonry (*first revision*)
IS 3622 : 1977	Specification for sand stone (slabs and tiles) (*first revision*)
IS 4121 : 1967	Method of test for determination of water transmission rate by capillary action through natural building stones
IS 4122 : 1967	Method of test for surface softening of natural building stones by exposure to acidic atmospheres
IS 4348 : 2024	Determination of permeability of natural building stones — Methods of test (*second revision*)
IS 5218 : 1969	Method of test for toughness of natural building stones
IS 5640 : 1970	Method of test for determining the aggregate impact value of soft coarse aggregates
IS 6241 : 2024	Determination of stripping value of road aggregates — Method of test (*first revision*)
IS 6250 : 1981	Specification for roofing slate tiles (*first revision*)
IS 6579 : 1981	Specification for coarse aggregate for water bound macadam (*first revision*)
IS 7779	Schedule for properties and availability of stones for construction purposes:
(Part 1)	Gujarat State,
(Sec 1) : 1975	Availability of stones
(Sec 2) : 1975	Engineering properties of building stones
(Sec 3) : 1975	Engineering properties of stone aggregates
(Part 2)	Maharashtra state,
(Sec 1) : 1979	Availability of stones
(Sec 2) : 1979	Engineering properties of building stones
(Sec 3) : 1979	Engineering properties of stone aggregates
(Part 3)	Tamil Nadu state,
(Sec 2) : 1980	Engineering properties of building stones
(Sec 3) : 1980	Engineering properties of stone aggregates
(Part 4/Sec 1 to 3) : 1996	Karnataka state, Sections (1 to 3)
(Part 5)	Andhra Pradesh,
(Sec 1) : 1997	Availability of stones

IS No.	*Title*
(Sec 2) : 1997	Engineering properties of building stones
(Sec 3):1997	Engineering properties of stone aggregates
IS 8759 : 2024	Maintenance and preservation of stone in buildings — Code of practice (*first revision*)
IS 9394 : 2024	Stone lintels — Specification (*first revision*)
IS 14223	Polished building stones specification:
(Part 1) : 2023	Granite and similar stones (*first revision*)
(Part 2) : 2025	Marble and similar stones

23 STRUCTURAL SECTIONS

a) Structural Shapes

IS No.	*Title*
IS 811 : 1987	Specification for cold formed light gauge structural steel sections (*second revision*)
IS 1173 : 2025	Hot rolled and slit steel tee bars — Dimensions and properties (*third revision*)
IS 1863 : 1979/ ISO 657-19	Rolled steel bulb flats (*first revision*)
IS 2314	Steel sheet piling section specification:
(Part 1) : 2023	Hot rolled sheet pile (*second revision*)
(Part 2) : 2023	Cold formed sheet pile (*second revision*)
IS 3443 : 1980	Specification for crane rail sections (*first revision*)
IS 3908 : 2025	Aluminium beam, channel and angle sections — Dimensions and properties (*second revision*)
IS 3964 : 1980	Specification for light rails (*first revision*)
IS 6445 : 1985	Specification for aluminium tee sections (*first revision*)

b) Dimensional Standards

IS No.	*Title*
IS 808 : 2021	Hot rolled steel beam column channel and angle sections — Dimensions and properties (*fourth revision*)
IS 1730 : 1989	Dimensions for steel plates, sheets strips and flats for general engineering purposes (*second revision*)
IS 1732 : 1989	Dimensions for round and square steel bars for structural and general engineering purposes (*second revision*)
IS 1852 : 1985	Rolling and cutting tolerances for hot rolled steel products (*fourth revision*)

IS No.	*Title*
IS 2525 : 1982	Dimensions for wrought aluminium and aluminium alloy wire (*first revision*)
IS 2591 : 1982	Dimensions for hot rolled steel bars for threaded components (*second revision*)
IS 2673 : 2002	Dimensions for wrought aluminium and aluminium alloys, extruded round tube (*second revision*)
IS 2676 : 1981	Dimensions for wrought aluminium and aluminium alloys, sheet and strip (*first revision*)
IS 2677 : 1979	Dimensions for wrought aluminium and aluminium alloys, plates and hot rolled sheets (*first revision*)
IS 2678 : 1987	Dimensions and tolerances for wrought aluminium and aluminium alloys, drawn round tubes (*second revision*)
IS 3577 : 1992	Dimensions and tolerances for wrought aluminium and aluminium alloys rivet, bolt and screw stock (*first revision*)
IS 3954 : 1991	Hot rolled steel channel sections for general engineering purposes — Dimensions (*first revision*)
IS 3965 : 1981	Dimensions for wrought aluminium and aluminium alloys, bar, rod and section (*first revision*)
IS 6477 : 1983	Dimensions for wrought aluminium and aluminium alloys, extruded hollow sections (*first revision*)
IS 12779 : 1989	Rolling and cutting tolerances for hot rolled parallel flange beam and column sections
IS/ISO 16124 : 2015	Steel wire rod — Dimensions and tolerances (*first revision*)
IS/ISO 16160 : 2012	Hot-rolled steel sheet products dimensional and shape tolerances (*first revision*)
IS/ISO 16162 : 2012	Cold-rolled steel sheet products — Dimensional and shape tolerances (*first revision*)
IS/ISO 16163 : 2012	Continuously hot-dipped coated steel sheet products — Dimensional and shape tolerances (*first revision*)
IS 16998 : 2018	Hot-rolled steel plates — Tolerances on dimensions and shape

24 THERMAL INSULATION MATERIALS

IS No.	*Title*
IS 3144 : 2024	Mineral wool thermal insulation materials — Methods of test (*third revision*)
IS 3346 : 1980	Methods for the determination of thermal conductivity of thermal insulation materials (two slab, guarded hot-plate method (*first revision*)

IS No.	*Title*
IS 3677 : 2026	Unbonded rock and slag wool for thermal insulation — Specification (*third revision*)
IS 4671 : 2018	Expanded polystyrene for thermal insulation purposes (*second revision*)
IS 5688 : 2018	Methods of test for preformed block-type and pipe-covering type thermal insulation (*second revision*)
IS 5724 : 2018	Thermal insulating cements — Methods of test (*first revision*)
IS 6598 : 2018	Cellular concrete for thermal insulation — Specification (*first revision*)
IS 7509 : 2024	Thermal insulating cements — Specification (*second revision*)
IS 8154 : 2024	Preformed calcium silicate insulation for temperatures up to 650 °C — Specification (*second revision*)
IS 8183 : 2024	Bonded mineral wool — Specification (*second revision*)
IS 9403 : 2018	Method of test for thermal conductance and transmittance of built-up Section by means of guarded hot box (*first revision*)
IS 9489 : 2018	Method of test for thermal conductivity of thermal insulation materials by means of heat flow meter (*first revision*)
IS 9490 : 2018	Method for determination of thermal conductivity of thermal insulation materials water calorimeter method (*first revision*)
IS 9742 : 2024	Sprayed mineral wool thermal insulation — Specification (*second revision*)
IS 9743 : 2020	Thermal insulation finishing cement — Specification (*second revision*)
IS 9842 : 2024	Preformed fibrous pipe insulation – Specification (*second revision*)
IS 11128 : 2018	Spray applied hydrated calcium silicate thermal insulation (*first revision*)
IS 11129 : 2018	Method of test for tumbling friability of preformed block-type thermal insulation (*first revision*)
IS 11239	Methods of test for rigid cellular thermal insulation materials:
(Part 1) : 2009/ ISO 1923 : 1981	Dimensions (*first revision*)
(Part 2) : 2019/ ISO 845 : 2006	Apparent density (*second revision*)
(Part 3) : 2009/ ISO 2796 : 1986	Dimensional stability (*first revision*)
(Part 4) : 2025/ ISO 1663 : 2023	Water vapour transmission rate (*second revision*)
(Part 5) : 2019/ ISO 4590 : 2016	Volume percent of open and closed cells (*second revision*)
(Part 6) : 1985	Heat distortion temperature
(Part 7) : 1985	Coefficient of linear thermal expansion at low temperatures

IS No.	*Title*
(Part 8) : 2024	Flame height, time of burning and loss of mass (*first revision*)
(Part 9) : 1988	Water absorption
(Part 10) : 1985	Flexural strength
(Part 11) : 2024	Compressive strength (*first revision*)
(Part 12) : 1988	Horizontal burning characteristics
(Part 14) : 1992	Determination of flammability by oxygen index
IS 11307 : 2025	Specification for cellular glass block and pipe thermal insulating (*first revision*)
IS 11308 :2024	Hydraulic setting thermal insulating castables for temperatures up to 1 250 °C — Specification (*first revision*)
IS 12436 : 1988	Specification for preformed rigid polyurethane (PUR) and polyisocyanurate (PIR) foams for thermal insulation
IS 13204 : 2024	Specification for rigid phenolic foams for thermal insulation (*first revision*)
IS 13286 : 2025	Methods of test for surface spread of flame for thermal insulation materials (*first revision*)
IS 14656 : 2026	Ceramic fibre products — Methods of test (*first revision*)

25 THREADED FASTENERS, RIVETS AND NAILS

IS No.	*Title*
IS 207 : 1964	Specification for gate and shutter hooks and eyes (*revised*)
IS 451 : 1999	Specification for technical supply conditions for wood screws (*third revision*)
IS 554 : 1999	Specification for pipe threads where pressure-tide joints are made on the threads — Dimensions, tolerances and designation (*fourth revision*)
IS 723 : 2023	Steel countersunk head wire nails — Specification (*second revision*)
IS 724 : 2023	Mild steel and brass cup, ruler and square hooks and screw eyes — Specification (*second revision*)
IS 725 : 1961	Specification for copper wire nails (*revised*)
IS 730 : 2021	Specification for hook bolts for corrugated sheet roofing (*third revision*)
IS 1120 : 1975	Specification for coach screws (*first revision*)
IS 1363	Hexagon head bolts, screws and nuts of product grade C:
(Part 1) : 2023/ ISO 4016 : 2022	Hexagon head bolts (size range M5 to M64) (*sixth revision*)
(Part 2) : 2023/ ISO 4018 : 2022	Hexagon head screws (size range M5 to M64) (*sixth revision*)

IS No.	*Title*
(Part 3) : 2018/ ISO 4034 : 2012	Style 1 Hexagon nuts (size range M5 to M64) (*fifth revision*)
IS 1364	Hexagon head bolts, screws and nuts of property grades A and B:
(Part 1) : 2023/ ISO 4014 : 2022	Hexagon head bolts (size range M1.6 to M64) (*sixth revision*)
(Part 2) : 2023/ ISO 4017 : 2022	Hexagon head screws (size range M1.6 to M64) (*sixth revision*)
(Part 3) : 2026/ ISO 4032 : 2023	Hexagon nuts, style 1 (size range M5 to M39) (*sixth revision*)
(Part 4) : 2025/ ISO 4035 : 2023	Hexagon nuts, style 0 (size range M1.6 to M64) (*fifth revision*)
(Part 5) : 2002/ ISO 4036 : 1999	Hexagon thin nuts — Product grade B (unchamfered) (size range M 1.6 to M10) (*fourth revision*)
(Part 6) : 2025/ ISO 4033 : 2023	Hexagon nuts, style 2 (Size Range M5 to M39) (*second revision*)
IS 1365 : 2022/ ISO 2009 : 2011	Slotted countersunk flat head screws — Product grade A (*fifth revision*)
IS 1366 : 2022/ ISO 1207 : 2011	Slotted cheese head screws — Product grade A (*fourth revision*)
IS 1367	Technical supply conditions for threaded steel fasteners:
(Part 1) : 2014/ ISO 8992 : 2005	General requirements for bolts, screws, studs and nuts (*fourth revision*)
(Part 2) : 2002/ ISO 4759-1 : 2000	Tolerances for fasteners — Bolts, screws, studs and nuts — Product grades a b and c (*third revision*)
(Part 3) : 2017/ ISO 898-1 : 2013	Mechanical properties of fasteners made of carbon steel and bolts, screws and studs (*fifth revision*)
(Part 5) : 2018 ISO 898-5 : 2012	Mechanical properties of fasteners made of carbon steel and alloy steel — Set screws and similar threaded fasteners with specified hardness classes-Coarse thread and fine pitch thread (*fourth revision*)
(Part 6) : 2025/ ISO 898-2:2022	Mechanical properties of fasteners made of carbon steel and alloy steel — Nuts with specified property classes (*fifth revision*)
(Part 7) : 1980	Mechanical properties and test methods for nuts without specified proof loads (*second revision*)
(Part 8) : 2020/ ISO 2320 : 2015	Prevailing torque type steel nuts functional properties (*fifth revision*)
(Part 9)	Surface discontinuities,
(Sec 1) : 1993/ ISO 6157-1 : 1988	Bolts screws and studs for general applications (*third revision*)

IS No.	*Title*
(Sec 2) : 1993/ ISO 6167-3 : 1988	Bolts screws and studs for special applications (*third revision*)
(Part 10) : 2002/ ISO 6157-2 : 1995	Surface discontinuities — Nuts (*third revision*)
(Part 11) : 2024/ ISO 4042 : 2022	Electroplated coating systems (*fifth revision*)
(Part 12) : 1981	Phosphate coatings on threaded fasteners (*second Revision*)
(Part 13) : 2020/ ISO 10684 : 2004	Hot dip galvanized coatings on threaded fasteners (*third Revision*)
(Part 14)	Mechanical properties of corrosion-resistant stainless steel fasteners,
(Sec 1) : 2023/ ISO 3506-1 : 2020	Bolts screws and studs with specified grades and property classes (*fifth revision*)
(Sec 2) : 2023/ ISO 3506-2 : 2020	Nuts with specified grades and property classes (*fifth revision*)
(Sec 3) : 2018/ ISO 3506-3 : 2009	Set screws and similar fasteners not under tensile stress (*fourth revision*)
(Sec 4) : 2023/ ISO 3506-4 : 2009	Tapping screws
(Part 16) : 2002/ ISO 8991 : 1986	Designation system for fasteners (*third revision*)
(Part 17) : 2023/ ISO 3269 : 2019	Inspections, sampling and acceptance procedure (*fifth revision*)
IS 1367	Industrial fasteners — Threaded steel fasteners — Technical supply conditions:
(Part 18) : 1996	Packaging (*third revision*)
(Part 19) : 1997/ ISO 3800 : 1993	Axial load fatigue testing of bolts screws and studs
(Part 20) : 1996/ ISO 896-7 : 1992	Torsional test and minimum torques for bolts and screws with nominal diameters 1 mm to 10 mm
IS 1929 : 1982	Specification for hot forged steel rivets for hot closing (12 mm to 36 mm diameter) (*first revision*)
IS 2016 : 1967	Specification for plain washers (*first revision*)
IS 2155 : 1982	Specification for cold forged solid steel rivets for hot closing (6 mm to 16 mm diameter) (*first revision*)
IS 2269 : 2006	Hexagon socket head cap screws (*fifth revision*)
IS 2585 : 2006	Square head bolts, screws and square nuts of product Grade C — Specification (*second revision*)

IS No.	*Title*
IS 2609 : 1972	Specification for coach bolts (*first revision*)
IS 2643 : 2005	Pipe threads where pressure-tight joints are not made on the threads — Dimensions, tolerances and designation (*third revision*)
IS 2687 : 1991	Specification for cap nuts (*second revision*)
IS 2769 : 1969	Sizes for squares and square holes for general engineering purposes (*first revision*)
IS 2907 : 1998	Specification for non-ferrous rivets (*first revision*)
IS 2998 : 1982	Specification for cold forged steel rivets for cold closing (1 mm to 16 mm diameter) (*first revision*)
IS 3063 : 1994	Specification for fasteners single coil rectangular section spring lock washers (*second revision*)
IS 3121 : 2023	Rigging screws and stretching screws — Specification (*second revision*)
IS 3468:1991	Specification for pipe nuts (*second revision*)
IS 3640 : 1982	Specification for hexagon fit bolts (*first revision*)
IS 3757 : 1985	Specification for high strength structural bolts (*second revision*)
IS 4206 : 2018/ ISO 888 : 2012	Dimensions for nominal lengths and thread lengths for bolts screws and studs *(second revision)*
IS 4762 : 2002	Fasteners — Worm drive hose clamps for general purposes — Specification (*second revision*)
IS 5370 : 1969	Specification for plain washers with outside diameter ≈ 3 × inside diameter
IS 5372 : 1975	Specification for taper washer for channels (ISMC) (*first revision*)
IS 5373 : 1969	Specification for square washers for wood fastenings
IS 5374 : 1975	Specification for taper washers for l-beam (ISMB) (*first revision*)
IS 5624 : 2021	Foundation bolts – Specification (*second revision*)
IS 6101 : 2018	Slotted pan head screws — Product Grade A (*third revision)*
IS 6113 : 1970	Specification for aluminium fasteners for building purposes
IS 6610 : 1972	Specification for heavy washers for steel structures
IS 6623 : 2004	High strength structural nuts — Specification (*second revision*)
IS 6639 : 1972	Specification for hexagon bolts for steel structures
IS 6649 : 1985	Specification for hardened and tempered washers for high strength structural bolts and nuts (*first revision*)
IS 6733 : 1972	Specification for wall and roofing nails
IS 6736 : 1972	Specification for slotted raised countersunk head wood screws

IS No.	*Title*
IS 6739 : 1972	Specification for slotted round head wood screws
IS 6760 : 1972	Specification for slotted countersunk head wood screws
IS 6761 : 2023	Fasteners — Hexagon socket countersunk head screws with reduced loadability (*third revision*)
IS 7002 : 2018	Prevailing torque type hexagon regular nuts (with non-metallic insert) — Property classes 5,8 and 10 (*third revision*)
IS 7178 : 2018	Heat-Treated steel tapping screws — Mechanical properties (*fourth revision*)
IS 7483 : 2018/ ISO 7045 : 2011	Pan head screws with type H or type Z cross recess — Product grade A (*third revision*)
IS 7485	Countersunk flat head screws (common head style) with Type H or Type Z cross recess — Product grade A:
(Part 1) : 2018/ ISO 7046-1 : 2011	Steel screws of property class 4.8 (*third revision*)
(Part 2) : 2018/ ISO 7046-2 : 2011	Steel screws of property class 8.8, stainless steel screws and non- ferrous metal screws (*third revision*)
IS 7486 : 2018/ ISO 7047 : 2011	Raised countersunk head screws (common head style) with type H or type Z cross recess — Product grade A (*third revision*)
IS 8033 : 1976	Specification for washers with square hole for wood fastenings
IS 8412 : 1977	Specification for slotted countersunk head bolts for steel structures
IS 8536 : 2021/ ISO 225 : 2010	Fasteners — Bolts, screws, studs and nuts — Symbols and descriptions of dimensions (*second revision*)
IS 8822 : 1978	Specification for slotted mushroom head roofing bolts
IS 8869 : 1978	Specification for washers for corrugated sheet roofing
IS 8911 : 2018/ ISO 2010 : 2011	Slotted raised countersunk head screws — Product grade A (*second revision*)
IS 10102 : 1982	Specification for technical supply conditions for rivets
IS 10238 : 2001	Fasteners — Threaded steel fasteners — Step bolts for steel structures — Specification (*first revision*)
IS/ISO 10683 : 2018	Fasteners — Non electrolytically applied zinc flake coating systems
IS 12427 : 2001	Threaded steel fasteners — hexagon head transmission tower bolts — Specification (*first revision*)
IS 13722 : 2025/ ISO 8673 : 2023	Hexagon regular nuts (Style 1) with metric fine pitch thread — Product grades A and B — Specification (*third revision*)
IS 13723 : 2025/ ISO 8674 : 2023	Hexagon high nuts (Style 2) with metric fine pitch thread — Product grades A and B — Specification (*third revision*)
IS 13724 : 2025/ ISO 8675 : 2023	Hexagon thin nuts (Style 0) with metric fine pitch thread — Product grades A and B — Specification (*third revision*)

IS No.	*Title*
IS 13725 : 2023/ ISO 8676 : 2022	Hexagon head screws with metric fine pitch thread –Product grades A and B (*third revision*)
IS 13726 : 2023/ ISO 8765 : 2022	Hexagon head bolts with metric fine pitch thread — Product grades A and B (*third revision*)
IS 14894 : 2023/ ISO 4015 : 2022	Fasteners-Hexagon head bolts of product grade B — Reduced shank (Shank diameter ≈ Pitch diameter) — Specification (*first revision*)
IS 15274 : 2021/ ISO 4759-3 : 2016	Tolerances for fasteners – Washers for bolts, screws and nuts — Product grades A, C and F (*first revision*)
IS 15581 : 2018	Hexagon nuts with flange, style 2 — Coarse thread (*first revision*)
IS 15582 : 2018	Hexagon bolts with flange – Small series — Product grade A with driving feature of product grade B (*first revision*)
IS 15628 : 2022/ ISO 4766 : 2011	Slotted set screws with flat point — Specification (*first revision*)
IS 17445 : 2020/ ISO 15330 : 1999	Fasteners — Preloading test for the detection of hydrogen embrittlement — Parallel bearing surface method
IS 17894 : 2022/ ISO 22081 : 2021	Geometrical product specifications (GPS) –Geometrical tolerancing — General geometrical specifications and general size specifications
IS 18471 : 2025	Fasteners — Drilling screws with tapping screw thread — Specification (*first revision*)
IS 18476 : 2023	Fasteners — Hexagon washer head drilling screws with tapping screw thread
IS 18480 : 2025	Self-tapping screws — Specification (*first revision*)
IS 18507 : 2026	Drywall screws — Specification (*first revision*)
IS 18508 : 2026	Chipboard screws — Specification (*first revision*)
IS 18509 : 2024	Cross recessed countersunk head wood screws — Specification
IS 18741 : 2024	Concrete nails — Specification
IS 19078 : 2025/ ISO 10666 : 1999	Drilling screws with tapping screw thread — Mechanical and functional properties
IS 19406 : 2026/ ISO 10587 : 2000	Metallic and other inorganic coatings — Test for residual embrittlement in both metallic coated and uncoated externally threaded articles and rods — Inclined wedge method

26 TIMBER, BAMBOO AND OTHER LIGNOCELLULOSIC BUILDING MATERIALS

a) Timber and Bamboo

i) *Timber Classification*

IS No.	*Title*
IS 287 : 1993	Permissible moisture content for timber used for different purposes — Recommendations (*third revision*)
IS 399 : 1963	Classification of commercial timbers and their zonal distribution (*revised*)
IS 1150 : 2000	Trade names and abbreviated symbols *for timber species* (*third revision*)
IS 4970 : 1973	Key for identification of commercial timber (*first revision*)

ii) *Timber Conversion and Grading*

IS No.	*Title*
IS 190 : 1991	Specification for coniferous sawn timber (baulks and scantlings) (*fourth revision*)
IS 1326 : 2023	Non-Coniferous sawn timber baulks and scantling — Specification (*third revision*)
IS 1331 : 1971	Specification for cut sizes of timber (*second revision*)
IS 2377 : 1967	Tables for volumes of cut sizes of timber (*first revision*)
IS 3337 : 1978	Specification for ballies for general purposes (*first revision*)
IS 3386 : 1979	Specification for wooden fence posts (*first revision*)
IS 5966 : 2023	Non-coniferous timber in converted form for general purposes — Specification (*second revision*)
IS 14960 : 2001	Specification for preservative treated and seasoned sawn timber from rubberwood (*Hevea brasiliensis*)

iii) *Timber Testing*

IS No.	*Title*
IS 1708	Methods of testing small clear specimens of timber:
(Part 1) : 1986	Determination of moisture content (*second revision*)
(Part 2) : 1986	Determination of specific gravity (*second revision*)
(Part 3) : 1986	Determination of volumetric shrinkage (*second revision*)
(Part 4) : 1986	Determination of radial and tangential shrinkage and fibre saturation point (*second revision*)
(Part 5) : 1986	Determination of static bending strength (*second revision*)
(Part 6) : 1986	Determination of static bending strength under two point loading (*second revision*)

IS No.	*Title*
(Part 7) : 1986	Determination of impact bending strength (*second revision*)
(Part 8) : 1986	Determination of compressive strength parallel to grain (*second revision*)
(Part 9) : 1986	Determination of compressive strength perpendicular to grain (*second revision*)
(Part 10) : 1986	Determination of hardness under static indentation (*second revision*)
(Part 11) : 1986	Determination of shear strength parallel to grain (*second revision*)
(Part 12) : 1986	Determination of tensile strength parallel to grain (*second revision*)
(Part 13) : 1986	Determination of tensile strength perpendicular to grain (*second revision*)
(Part 14) : 1986	Determination of cleavage strength parallel to grain (*second revision*)
(Part 15) : 1986	Determination of nail and screw holding power (*second revision*)
(Part 16) : 1986	Determination of brittleness by izod impact (*second revision*)
(Part 17) : 1986	Determination of brittleness by charpy impact (*second revision*)
(Part 18) : 1986	Determination of torsional strength (*second revision*)
IS 2408 : 1963	Methods of static tests of timbers in structural sizes
IS 2455 : 1990	Method of sampling of model trees and logs for timber testing and their conversion (*second revision*)
IS 2753	Methods for estimation of preservatives in treated timber and treating solutions:
(Part 1) : 1991	Determination of copper, arsenic, chromium, zinc, boron, creosote and fuel oil (*first revision*)
(Part 2) : 2014	Determination of copper in copper organic preservative salt (*second revision*)
IS 4907 : 2004	Method of testing timber connector joints (*first revision*)
IS 8292 : 1992	Methods for evaluation of working quality of timber under different wood operations — Method of test (*first revision*)
IS 8720 : 1978	Methods of sampling of timber scantlings from depots and their conversion for testing
IS 8745 : 1994	Methods of presentation of data of physical and mechanical properties of timber (*first revision*)
IS 10420 : 1982	Method of determination of sound absorption coefficient of timber by standing wave method
IS 10754 : 1983	Method of determination of thermal conductivity of timber
IS 11215 : 1991	Methods for determination of moisture content of timber and timber products (*first revision*)
IS 13621 : 1993	Method of test for determination of dielectric constant of wood under microwave frequencies

iv) *Structural Timber and Test*

IS No.	*Title*
IS 2372 : 2021	Timber for cooling towers — Specification (*third revision*)
IS 3629 : 1986	Specification for structural timber in building (*first revision*)
IS 4891 : 1988	Specification for preferred cut sizes of structural timber (*first revision*)
IS 4924	Method of test for nail jointed timber trusses:
(Part 1) : 1968	Destructive test
(Part 2) : 1968	Proof test

v) *Logs*

IS No.	*Title*
IS 656 : 2023	Logs for plywood — Specification (*fourth revision*)
IS 3364	Method of measurement and evaluation of defects in timber:
(Part 1) : 1976	Logs (*first revision*)
(Part 2) : 1976	Converted timber (*first revision*)
IS 4895:1985	Specification for teak logs (*first revision*)
IS 5246:2000	Specification for coniferous logs (*first revision*)
IS 7308:1999	Specification for non-coniferous logs (*first revision*)

vi) *Bamboo*

IS No.	*Title*
IS 6874 : 2008	Method of tests for bamboos (*first revision*)
IS 8242 : 1976	Methods of tests for split bamboos
IS 17571 : 2021	Flattened bamboo board — Specification
IS 17572 : 2022	BWP grade bamboo mat-veneer composite board — Specification
IS 17573 : 2021	BWP grade bamboo mat board – Specification

(b) Reconstituted Products

i) *Plywood*

IS No.	*Title*
IS 303 : 2024	Plywood for general purposes — Specification (*fourth revision*)
IS 710 : 2024	Marine plywood — Specification (*third revision*)
IS 1328 : 2025	Veneered decorative plywood — Specification (*fourth revision*)

IS No.	*Title*
IS 1734	Methods of test for plywood:
(Part 1) : 1983	Determination of density and moisture content (*second revision*)
(Part 2) : 1983	Determination of resistance of dry heat (*second revision*)
(Part 3) : 1983	Determination of fire resistance (*second revision*)
(Part 4) : 1983	Determination of glue shear strength (*second revision*)
(Part 5) : 1983	Test for adhesion of plies (*second revision*)
(Part 6) : 1983	Determination of water resistance (*second revision*)
(Part 7) : 1983	Mycological test (*second revision*)
(Part 8) : 1983	Determination of *p*H value (*second revision*)
(Part 9) : 1983	Determination of tensile strength (*second revision*)
(Part 10) : 1983	Determination of compressive strength (*second revision*)
(Part 11) : 1983	Determination of static bending strength (*second revision*)
(Part 12) : 1983	Determination of scarf joint strength (*second revision*)
(Part 13) : 1983	Determination of panel shear strength (*second revision*)
(Part 14) : 1983	Determination of plate shear strength (*second revision*)
(Part 15) : 1983	Central loading of plate test (*second revision*)
(Part 16) : 1983	Vibration of plywood plate test (*second revision*)
(Part 17) : 1983	Long time loading test of plywood strips (*second revision*)
(Part 18) : 1983	Impact resistance test on the surface of plywood (*second revision*)
(Part 19) : 1983	Part 19 Determination of nail and screw holding power (*second revision*)
(Part 20) : 1983	Part 20 Acidity and alkalinity resistance test (*second revision*)
IS 4990 : 2024	Plywood for concrete shuttering works — Specification *(fourth revision)*
IS 5509 : 2021	Fire retardant plywood – Specification *(third revision)*
IS 5539:1969	Specification for preservative treated plywood
IS 7316:1974	Specification for decorative plywood using plurality of veneers for decorative faces
IS 10701:2012	Structural plywood — Specification (*first revision*)
IS 13957:1994	Specification for metal faced plywood
IS 15791:2007	Museum plywood — Specification

ii) *Blockboards, Particle Boards and Fibre Boards*

IS No.	*Title*
IS 1658 : 2006	Fibre hardboards — Specification (*third revision*)
IS 1659 : 2004	Block boards — Specification *(fourth revision)*
IS 2380	Methods of test for wood particle boards and boards from other lignocellulosic materials:
(Part 1) : 1977	Preparation and conditioning of test specimens (*first revision*)
(Part 2) : 1977	Accuracy of dimensions of boards (*first revision*)
(Part 3) : 1977	Determination of moisture content and density (*first revision*)
(Part 4) : 1977	Determination of static bending strength (*first revision*)
(Part 5) : 1977	Determination of tensile strength perpendicular to surface (*first revision*)
(Part 6) : 1977	Determination of tensile strength parallel to surface (*first revision*)
(Part 7) : 1977	Determination of compression-perpendicular to plane of the board (*first revision*)
(Part 8) : 1977	Compression parallel to surface test (*first revision*)
(Part 9) : 1977	Determination of resistance to shear in plane of the board (*first revision*)
(Part 10) : 1977	Falling hammer impact test (*first revision*)
(Part 11) : 1977	Surface hardness (*first revision*)
(Part 12) : 1977	Central loading of plate test (*first revision*)
(Part 13) : 1977	Long time loading bending test (*first revision*)
(Part 14) : 1977	Screw and nail withdrawal test (*first revision*)
(Part 15) : 1977	Lateral nail resistance (*first revision*)
(Part 16) : 1977	Determination of water absorption (*first revision*)
(Part 17) : 1977	Determination of swelling in water (*first revision*)
(Part 18) : 1977	Determination of mass and dimensional changes caused by moisture changes (*first revision*)
(Part 19) : 1977	Durability cyclic test for interior use (*first revision*)
(Part 20) : 1977	Accelerated weathering cyclic test for exterior use (*first revision*)
(Part 21) : 1977	Planeness test under uniform moisture content (*first revision*)
(Part 22) : 1981	Determination of surface glueability test
(Part 23) : 1981	Vibration test for particle boards
IS 3087 : 2025	Medium density particle boards of wood and other lignocellulosic materials for general purpose — Specification (*third revision*)

IS No.	*Title*
IS 3097 : 2025	Veneered medium density particle boards from wood and other lignocellulosic material — Specification (*third revision*)
IS 3129 : 1985	Specification for low density particle boards (*first revision*)
IS 3308 : 1981	Specification for wood wool building slabs (*first revision*)
IS 3348 : 1965	Specification for fibre insulation boards
IS 3478 : 1966	Specification for high density wood particle boards
IS 12406 : 2025	Medium density fibre boards of wood and other lignocellulosic materials for general purpose — Specification (*third revision*)
IS 12823 : 2025	Prelaminated medium density particle boards from wood and other lignocellulosic material — Specification (*second revision*)
IS 13745 : 2020	Method for determination of formaldehyde content in wood-based panels by extraction method called perforator method (*first revision*)
IS 14276 : 2016	Cement bonded particle boards — Specification (*first revision*)
IS 14587 : 2025	Prelaminated medium density fibre boards of wood and other lignocellulosic materials — Specification (*second revision*)
IS 15786 : 2008	Prelaminated cement bonded particle boards — Specification
IS 19267 : 2025	Oriented strand boards — Specification

iii) *Wood-Based Laminates and Lumber*

IS No.	*Title*
IS 3513	Specification for resin treated compressed wood laminates (compregs):
(Part 3) : 1989	For General Purposes (*first revision*)
(Part 4) : 1966	Sampling and tests
IS 7638 : 1999	Methods of sampling for wood/lignocellulosic based panel products
IS 9307	Methods of tests for wood-based structural sandwich construction:
(Part 1) : 1979	Flexure test
(Part 2) : 1979	Edgewise compression test
(Part 3) : 1979	Flatwise compression test
(Part 4) : 1979	Shear test
(Part 5) : 1979	Flatwise tension test
(Part 6) : 1979	Flexure creep test
(Part 7) : 1979	Cantilever vibration test
(Part 8) : 1979	Weathering test

IS No.	*Title*
IS 14315 : 1995	Specification for commercial veneers
IS 14616 : 1999	Specification for laminated veneer lumber
IS 16171 : 2014	Veneer laminated lumber – Specification

iv) *Bamboo and Coir Board Products*

IS No.	*Title*
IS 13958 : 1994	Specification for bamboo mat board for general purposes
IS 14588 : 1999	Specification for bamboo mat veneer composite for general purposes
IS 14842 : 2000	Specification for coir veneer board for general purposes
IS 15476 : 2004	Specification for bamboo mat corrugated sheets
IS 15491 : 2004	Specification for medium density coir boards for general purposes
IS 15877 : 2010	Coir faced block boards — Specification
IS 15878 : 2010	Coir hardboard for general purposes — Specification
IS 15972 : 2012	Bamboo-jute corrugated and semi-corrugated sheets — Specification

v) *Adhesives*

IS No.	*Title*
IS 848 : 2025	Synthetic resin adhesives for plywood (phenolic, aminoplastic and bio-materials based) — Specification (*third revision*)
IS 851 : 1978	Specification for synthetic resin adhesives for construction work (non-structural) in wood (*first revision*)
IS 852 : 1994	Animal glue for general wood-working purposes Specification (*second revision*)
IS 1508 : 1972	Specification for extenders for use in synthetic resin adhesives (urea-formaldehyde) for plywood (*first revision*)
IS 4835 : 1979	Specification for polyvinyl acetate dispersion-based adhesives for wood (*first revision*)
IS 9188 : 1979	Performance requirements for adhesive for structural laminated wood products for use under exterior exposure condition

27 UNIT WEIGHTS OF BUILDING MATERIALS

IS No.	*Title*
IS 875 (Part 1) : 1987	Code of practice for design loads (other than earthquake) for buildings and structures: Part 1 Dead loads — Unit weights of building material and stored materials (*second revision*)

28 WATERPROOFING AND DAMP-PROOFING MATERIALS

IS No.	*Title*
IS 1322 : 1993	Specification for bitumen felts for waterproofing and damp-proofing (*fourth revision*)
IS 1580 : 1991	Specification for bituminous compound for waterproofing and caulking purposes (*second revision*)
IS 3037 : 1986	Specification for bitumen mastic for use in waterproofing of roofs (*first revision*)
IS 3384 : 1986	Specification for bitumen primer for use in waterproofing and damp-proofing (*first revision*)
IS 5871 : 1987	Specification for bitumen mastic for tanking and damp-proofing (*first revision*)
IS 7193 : 2013	Glass fibre base bitumen felts — Specification (*second revision*)
IS 12027 : 1987	Silicone-based water repellents — Specification
IS 12027 (Part 2) : 2025	Water repellents — Specification: Part 2 Silane based (*first revision*)
IS 13435	Acrylic based polymer waterproofing material — Method of tests:
(Part 1) : 2021	Determination of solid content (*first revision*)
(Part 2) : 1992	Determination of coarse particles
(Part 3) : 1992	Determination of capillary water take-up
(Part 4) : 1992	Determination of *p*H value
IS 13826	Bitumen based felts — Method of test:
(Part 1) : 1993	Breaking strength test
(Part 2) : 2022	Pliability test (*first revision*)
(Part 3) : 1993	Storage sticking test
(Part 4) : 1993	Pressure head test
(Part 5) : 1994	Heat resistance test
(Part 6) : 1993	Water absorption test
(Part 7) : 1993	Determination of binder content
IS 14695 : 1999	Glass fibre base coal tar pitch outer wrap — Specification
IS 16525 : 2017	Styrene butadiene styrene (SBS) modified bituminous waterproofing and damp-proofing membrane with polyester reinforcement — Specification
IS 16526 : 2017	Atactic polypropylene (APP) modified bituminous waterproofing and damp-proofing membrane with glass fibre reinforcement — Specification

IS No.	*Title*
IS 16532 : 2017	APP modified bituminous waterproofing and damp-proofing membrane with polyester reinforcement — Specification
IS 16540 : 2017	Atactic polypropylene (APP) modified high molecular high density polyethylene (HMHDPE) bituminous waterproofing and damp-proofing membrane — Specification
IS 17856 : 2022	Vulcanized single layer rubber sheets for use as membrane — Specification

29 WELDING ELECTRODES AND WIRES

IS No.	*Title*
IS 814 : 2004	Covered electrodes for manual metal arc welding of carbon and carbon manganese steel — Specification (*sixth revision*)
IS 1278 : 1972	Specification for filler rods and wires for gas welding (*second revision*)
IS 1395 : 1982	Specification for low and medium alloy steel covered electrodes for manual metal arc welding (*third revision*)
IS 2879 : 1998	Mild steel for metal arc welding electrodes (*third revision*)
IS 4972	Resistance spot welding:
(Part 1) : 2022/ ISO 8430-1 : 2016	Electrode holders — Taper fixing 1 : 10 (*first revision*)
(Part 2) : 2022/ ISO 8430-2 : 2016	Electrode holders — Morse taper fixing (*first revision*)
(Part 3) : 2022/ ISO 8430-3 : 2016	Electrode holders parallel shank fixing for end thrust *(first revision)*
(Part 4) : 2022/ ISO 5183 – 1 : 1998	Electrode adaptors, male taper 1 : 10 — Conical fixing, taper 1 : 10 (*first revision*)
(Part 5) : 2022/ ISO 5183-2 : 2000	Electrode adaptors, male taper 1 : 10 — Parallel shank fixing for end-thrust electrodes (*first revision*)
(Part 6) : 2022/ ISO 5184 : 1979	Straight resistance spot welding electrodes (*first revision*)
(Part 7) : 2022/ ISO 1089 : 2023	Electrode taper fits for spot welding equipment — Dimensions (*second revision*)
(Part 8) : 2022/ ISO 5182 : 2016	Materials for electrodes and ancillary equipment (*first revision*)
IS 5206 : 1983	Covered electrodes for manual arc welding of stainless steel and other similar high alloy steels (*first revision*)
IS 5511 : 1991	Specification for covered electrodes for manual metal arc welding of cast iron (*first revision*)
IS 5856 : 2022/ ISO 14343 : 2017	Welding consumables — Wire electrodes, strip electrodes, wires and rods for arc welding of stainless and heat resisting steels — Classification (*third revision*)

IS No.	*Title*
IS 5897 : 2026	Welding consumables — Wire electrodes, wires and rods for welding of aluminium and aluminium alloys — Specification (*second revision*)
IS 5898 : 1970	Specification for copper and copper alloy bare solid welding rods and electrodes
IS 6419 : 1996	Specification for welding rods and bare electrodes for gas shielded arc welding of structural steel (*first revision*)
IS 6560 : 2017/ ISO 21952 : 2012	Welding consumables — Wire electrodes, wires, rods and deposits for gas shielded arc welding of creep-resisting steels — Classification (*second revision*)
IS 8363 : 1976	Specification for bare wire electrodes for electroslag welding of steels
IS 10631 : 1983	Stainless steel for welding electrode core wire
IS 13007 : 2018	Arc welding and cutting — Non consumable tungsten electrodes — Classification (*first revision*)
IS 15977 : 2013	Classification and acceptance tests for bare solid wire electrodes and wire flux combination for submerged arc welding of structural steel — Specification
IS 18641 : 2024/ ISO 544 : 2017	Welding consumables — Technical delivery conditions for filler materials and fluxes — Type of product, dimensions, tolerances and markings
IS 18734 : 2024/ ISO 6847 : 2020	Welding consumables — Deposition of a weld metal pad for chemical analysis
IS 18738	Welding consumables — Methods of test:
(Part 1) : 2024/ ISO 15792-1 : 2020	Preparation of all-weld metal test pieces and specimens in steel, nickel and nickel alloys
(Part 2) : 2024/ ISO 15792-2 : 2020	Preparation of single-run and two-run technique test pieces and specimens in steel
(Part 3) : 2024/ ISO 15792-3 : 2011	Classification testing of positional capacity and root penetration of welding consumables in a fillet weld
IS 18783 : 2024/ ISO 14175 : 2008	Welding consumables — Gases and gas mixtures for fusion welding and allied processes
IS 8666 : 1977	Specification for copper and copper alloy covered electrodes for manual metal arc welding

30 WIRE ROPES AND WIRE PRODUCTS (Including wire for fencing)

IS No.	*Title*
IS 278 : 2009	Galvanized steel barbed wire for fencing — Specification (*fourth revision*)
IS 1755 : 2018	Metallic materials — Wire-wrapping test (*second revision*)
IS 2140 : 2022	Specification for stranded galvanized steel wire for fencing (*second revision*)
IS 2266 : 2024	Steel wire ropes for general engineering purpose — Specification (*sixth revision*)

IS No.	*Title*
IS 2315 : 2023	Thimbles for wire ropes — Specification (*second revision*)
IS 2365 : 2024/ ISO 4344 : 2022	Steel wire ropes for lifts — Minimum requirements (*third revision*)
IS/ISO 2408 : 2017	Steel wire ropes — Requirements
IS 2485 : 2022	Drop forged sockets for wire ropes for general engineering purposes — Specification (*second revision*)
IS 2721 : 2025	Galvanized polyvinyl chloride (PVC), polyolefin and other polymer-coated steel chain link fence fabric — Specification (*third revision*)
IS 6594 : 2024	Technical supply conditions for steel wire ropes and strands (*fourth revision*)
IS 13902	Sockets for wire ropes for general purposes:
(Part 1) : 2022	General characteristics and conditions of acceptance (*first revision*)
(Part 2) : 2022	Special requirements for sockets produced by forging or machined from the solid (*first revision*)
IS 16013 : 2018	Welded wire gabions (metallic-coated or metallic-coated with PVC coating) — Specification (*first revision*)
IS 16014 : 2024	Mechanically woven, Double-Twisted, hexagonal wire mesh gabions, revet mattresses, rock fall netting and other products for civil engineering purposes (galvanized steel wire or galvanized steel wire with polymer coating) — Specification (*second revision*)
IS/ISO 17745 : 2016	Steel wire ring net panels — Definitions and specifications
IS/ISO 17746 : 2016	Steel wire rope net panels and rolls — Definitions and specifications
IS 17929	Fibre ropes for offshore station keeping:
(Part 1) : 2022/ ISO 18692-1 : 2018	General specification
(Part 2) : 2022/ ISO 18692-2 : 2019	Polyester
(Part 3) : 2022/ ISO 18692-3 : 2020	High modulus polyethylene (HMPE)
(Part 4) : 2024/ ISO 18692-4 : 2023	Polyarylate
(Part 5) : 2024/ ISO 18692-5 : 2024	Aramid
IS 18759 : 2024/ ISO 10325 : 2018	Fibre ropes — High modulus polyethylene — 8-Strand braided ropes, 12-Strand braided ropes and covered ropes
IS 19363 : 2025	Metallic or polymer coated — Welded wire mesh fencing — Specification

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART C STRUCTURAL DESIGN

SECTION 1 LOADS AND LOAD EFFECTS

BUREAU OF INDIAN STANDARDS

CONTENTS

National Building Code Sectional Committee, CED 46

FOREWORD

This NBCS (Part C/Section 1) covers the various loads, forces and effects which are to be taken into account for structural design of buildings. The various loads that are covered under this Section are dead load, imposed load, wind load, seismic force, snow load, special loads and load combinations.

This Section was first published in 1970 and subsequently revised in 1983, 2005 and 2016 as a chapter of the National Building Code of India. The first revision of this Section was modified in 1987 through Amendment No. 2 to the 1983 version to bring this Section in line with the latest revised loading code. Thereafter, in view of the revision of an important Indian Standard related to earthquake resistant design of structures, that is IS 1893, a need arose to revise this part of NBC. Therefore, the second revision of 2005 was formulated to consider the revised standard, IS 1893 (Part 1) : 2002 'Criteria for earthquake resistant design of structures: Part 1 General provisions and buildings (*fifth revision*)' and to incorporate latest information on additional loads, forces and effects and the details regarding multi-hazard risk in various districts of India.

The significant changes incorporated in the second revision of 2005 included: the revision of the seismic zone map providing only four zones, instead of five, with erstwhile Zone I being merged into Zone II; changing of the values of seismic zone factors to reflect more realistic values of effective peak ground acceleration considering maximum considered earthquake (MCE) and service life of structure in each seismic zone; specifying of response spectra for three types of founding strata, namely rock and hard soil, medium soil and soft soil; the revision of empirical expression for estimating the fundamental natural period T_a of multi-storeyed buildings with regular moment resisting frames; adoption of the procedure of first calculating the actual force that may be experienced by the structure during the probable maximum earthquake, if it were to remain elastic, also bringing in the concept of response reduction due to ductile deformation and associated energy dissipation by introducing the 'response reduction factor' in place of the earlier performance factor; specifying a lower bound for the design base shear of buildings, based on empirical estimate of the fundamental natural period T_a; deletion of the soil-foundation system factor and instead, introduction of a clause to restrict the use of foundations vulnerable to differential settlements in severe seismic zones; revision of torsional eccentricity values upwards in view of serious damages observed in buildings with irregular plans; revision of modal combination rule in dynamic analysis of buildings; inclusion of a new clause on multi-hazard risk in various districts of India and a list of districts identified as multi-hazard prone districts; incorporation of latest amendments issued to IS 875 introduction of a clause on vibration in buildings for general guidance; and inclusion of reference to the Indian Standards on landslide control and design of retaining walls which were formulated after the first revision of the Section.

The significant changes incorporated in the third revision of 2016 included: enhanced provisions for parapets and balustrades, incorporating concentrated horizontal loads and adjustments in horizontal load per unit length; new areas of application, such as appurtenances fixed to structures; aerodynamic roughness heights for different terrain categories were explicitly addressed, aiding in the determination of turbulence intensity and wind speed profiles; a new modification factor k_4 was introduced for cyclonic regions; simplified empirical expressions for variations in hourly mean wind speed and turbulence intensity across different terrains were suggested, along with provisions for directionality, area averaging and correlation of pressures for design wind pressure; wind-induced interference for both tall and low-rise buildings was accounted for, with recommendations for wind tunnel or CFD studies for final designs of critical structures; updates to the wind speed map of India were included; and new provisions for design loads for helipads, emergency vehicles and collisions with structural elements in parking areas. For earthquake effects, the design spectra was defined for natural periods up to 6 s, applicable to all buildings regardless of construction material; temporary structures and buildings with flat slabs were brought under the scope of the standard (though with some conditions), with guidelines for addressing structural irregularities and estimating the in-plane stiffness of masonry infill walls; additional revisions including simplified torsional provisions, methods for computing natural periods in various structures, new provisions for evaluating liquefaction potential and blast loads; and inputs were also derived from the Heliport Manual, 1995 of the International Civil Aviation Organization for the design of roof top helipad, etc.

In the current revision, this publication has been renamed as National Building Construction Standards (NBCS).

In this revision of this Part C/Section 1, load provisions have been updated in alignment with the existing IS 875 series and IS 4991. The IS 875 framework continues and the changes may be seen as its evolution. The present deterministic approach with safety margins will gradually be replaced by a unified approach of load classification and quantification. Loads will be grouped as natural or man-made and further as force-based or displacement-based. For natural hazards, return period–based probabilistic treatment will apply, while man-made loads will be based on statistical measures such as mean and standard deviation. This philosophy may, in due course, be reflected in future revisions of various Indian Standards on loading, following which their safety factors may also be recalibrated.

Further, considering the recent developments and R&D efforts nationally and internationally, the following significant modifications and inclusions have been made:

1) The provision relating to structural design basis report (SDBR) is introduced.

Dead Loads:

2) Unit weights of construction materials and stored materials have been included in Annex A.
3) Provisions relating to dead load for gardening and landscaping have been introduced.

Imposed Loads:

4) Imposed loads for dwelling units planned and executed in accordance with IS 8888 : 2020 'Requirements of low income housing for Urban areas guide (*second revision*)' have been brought at par with other residential buildings.
5) Imposed loads for datacenters have been included.
6) Imposed loads for roof top solar panels (with or without frames and with anchor blocks) and roof top HVAC equipment, have been included.
7) Imposed load for garage floors with and without stack parking have been updated.
8) Imposed load for refuge area and disaster shelter have been provided.

Wind Loads:

9) Provision relating to testing of building fabrics against wind borne debris has been included.
10) Table of basic wind speed for various cities of India has been updated.

Snow Loads:

11) A new term 'characteristic ground snow load' has been introduced and defined.
12) Characteristic ground snow load maps for union territory of Jammu and Kashmir, union territory of Ladakh and states of Himachal Pradesh, Uttarakhand and Sikkim have been introduced, resulting from studies conducted by Defence Geoinformatics Research Establishment (DRDO), Chandigarh.
13) Equation for calculation of characteristic ground snow load has been suggested in Annex H.
14) Clause relating to partial loading due to melting, sliding, snow redistribution and snow removal has been included.
15) Clause relating to ponding instability has been included.

Earthquake loads:

16) The Indian Standard IS 1893, on which the clause of earthquake load is largely based, is under revision at the time of publication of this NBCS. Major changes have been envisaged in the revision of IS 1893 which include apart from others, the introduction of probabilistic seismic hazard map of India having updated seismic zones. In the absence of availability of finalized version of revised IS 1893 at the time of revision of this Section, in this revision of the NBCS, the provisions of design as per existing IS 1893 (Part 1) : 2016 'Criteria for earthquake resistant design of structures: Part 1 General provisions and buildings' have been continued through app ropriate reference to the same.

Blast Loads:

17) Detailed provisions relating to design for blast loads have been included in Annex L. This annex provides prescriptive requirements for estimation of blast loads and simplified methods for structural analysis.
18) General requirements for blast-protection and categorization of structures have been provided.
19) Threat assessment requirements for determination of design explosive charge mass and scaled distance have been given.
20) Blast load estimation methods due to above-ground explosion of high-explosive under three different detonation scenarios: surface burst, free-air burst and air burst have been specified.
21) Evaluation of blast loads effects on structures of regular geometry and shape has been specified.
22) Prescriptive simplified single degree of freedom (SDOF) analysis procedure for peak response evaluation of structural elements based on flexural mode has been provided.
23) Guidelines for the application of multi-degree of freedom (MDOF) and detailed finite element analysis have been provided.

The provisions outlined in this Chapter are intended to assist structural designers in making informed engineering decisions when dealing with diverse loading scenarios, including special loads and unique load combinations that may arise in practical applications. It is expected that this Chapter will enable designers to adopt a rational and scientific approach in assessing structural loads and their combinations, thereby enhancing the resilience of structures against various external forces and ensuring their long-term performance. The Chapter serves as a fundamental reference for structural engineers, ensuring that the design approach aligns with safety requirements, reliability principles and advancements in engineering practices. It integrates knowledge from past experiences, research developments and international best practices while considering the specific conditions prevalent in India.

Structures are subjected to various loads, environmental effects and material degradation over time, which can impact their performance and safety. Hence, monitoring of structures has been gaining attention. Monitoring of structures is seen now along three directions, namely health monitoring, performance monitoring and safety monitoring. Structural health monitoring (SHM) involves monitoring of the condition of the materials of the structure, structural performance monitoring (SPM) involves monitoring of the response of the structure (without any damage induced in it) to daily loads the structure, such as vibrations of the structure under wind loads and structural safety monitoring involves monitoring the response of the structure when subjected to loads that damage the structure, such as, strong ground shaking of the structure during an earthquake, or air pressure induced under blast. In all cases, the engineer needs to assess the structure to identify actions needed towards repairing, correcting, protecting or retrofitting the structure, as the case be.

Monitoring of structures is a crucial process that enables the continuous or periodic assessment of structural integrity, ensuring early detection of potential damage and preventing catastrophic failures. It provides real-time insights into the behaviour of buildings and infrastructure under different conditions by utilizing advanced sensors, data acquisition systems and analytical models. The need for monitoring the structures arises from the increasing complexity of modern structures, aging infrastructure and the demand for enhanced safety and serviceability. It helps in identifying structural anomalies, degradation patterns and potential failure modes before they become critical. The system of monitoring of structures should be adopted to validate analytical models and evaluate techniques to ensure accurate assessment and decision-making. Implementing monitoring systems in buildings and structures where necessary enhances durability, reduces maintenance costs and supports proactive intervention strategies, ultimately improving structural resilience and public safety.

There has been a growing awareness among the consultants, academicians, researchers and practice engineers for design and construction of wind sensitive structures. In order to augment the available limited good quality meteorological wind data and structural response data, it is necessary to conduct full scale measurements in the field. Thus, as emphasized in the previous revision, all individuals and organizations responsible for putting-up of tall structures are encouraged to provide instrumentation in their existing and new structures at different elevations (at least at two levels) to continuously measure and monitor wind data. The instruments are required to collect data on wind direction, wind speed and structural response of the structure due to wind (with the help of accelerometer, strain gauges, etc). Such instrumentation in tall structures shall not in any way affect or alter the functional behaviour of such structures. The data so collected shall be very valuable in evolving more accurate wind loading of structures.

The information contained in this Section is largely based on the following Indian Standards:

IS 875	Code of practice for design loads (other than earthquake) for buildings and structures:
Part 1 : 2026	Dead loads-unit weights of building material and stored materials (*third revision*) (*under preparation*)
Part 2 : 1987	Imposed loads (*second revision*)
Part 3 : 2015	Wind loads (*third revision*)
Part 4 : 2021	Snow loads (*third revision*)
Part 5 : 1987	Special loads and load combinations (*second revision*)
IS 4991 (Part 1) : 2024	Criteria for blast resistant design of structures: Part 1 Above-ground explosions (*first revision*)

This Section shall be read together with Sections 2 to 8 of Part C 'Structural Design' of this NBCS.

A special publication, SP 64 (S&T) : 2001 'Explanatory Handbook on Indian Standard Code of Practice for Design Loads (Other than Earthquake) for Buildings and Structures IS 875 (Part 3) : 1987' is also available. This publication gives detailed background information on the provisions for wind loads and the use of these provisions for arriving at the wind loads on buildings and structures while evaluating their structural safety. But, in view of revision of IS 875 (Part 3) in 2015, applicability of this revised standard and this Section of the NBCS shall be taken as final in case of conflict of provisions with SP 64.

Reference may also be made to the Vulnerability Atlas of India, 2011 and Landslide Hazard Zonation Map of India, 2003, published by Building Materials and Technology Promotion Council, Ministry of Urban Development and Poverty Alleviation, Government of India. The Vulnerability Atlas contains information pertaining to each State and Union Territory of India, on (a) seismic hazard map, (b) cyclone and wind map, (c) flood prone area map and (d) housing stock vulnerability table for each district indicating for each house type the level of risk to which it could be subjected. The Atlas can be used to identify areas in each district of the country which are prone to high risk from more than one hazard. The information will be useful in establishing the need of developing housing designs to resist the combination of such hazards.

All standards, whether given herein above or cross-referred to in the main text of this Section, are subject to revision. The parties to agreement based on this Section are encouraged to investigate the possibility of applying the most recent editions of the standards.

As the regulation of land development and buildings is a State subject and is dealt by the States and local bodies such as urban local bodies within their jurisdiction through building regulatory documents like building regulations, building byelaws and fire regulations, this document is voluntary in nature and is non-binding. The implementation depends on adoption by concerned parties or stipulations in a contract or by appropriate adoption by the concerned authorities.

For the purpose of deciding whether a particular requirement of this section is complied with, the final value observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 2022 'Rules for rounding of numerical values (*second revision*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this section.

Important Explanatory Note for Users of the NBCS

In any Part/Section of this NBCS, where reference is made to **'good practice'** in relation to design, constructional procedures or other related information and where reference is made to **"accepted standard"** in relation to material specification, testing, or other related information, the Indian Standards listed at the end of the Part/Section shall be used as a guide to the interpretation.

At the time of publication, the editions indicated in the standards were valid. All standards are subject to revision and parties to agreements based on any Part/ Section are encouraged to investigate the possibility of applying the most recent editions of the standards.

In the list of standards given at the end of a Part/Section, the number appearing within parentheses in the first column indicates the number of the reference of the standard in the Part/Section. For example:

a) Good practices [C-1(1)] refers to the Indian Standard(s) give at serial number (1) of the list of standards given at the end of this Part/Section 1, that is, 8888 : 2020 'Requirements of low income housing for urban areas — Guide (second revision)'

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART C STRUCTURAL DESIGN

SECTION 1 LOADS AND LOAD EFFECTS

1 SCOPE

This NBCS (Part C/Section 1) covers basic design loads to be assumed in the design of buildings. The loads specified in this section are minimum loads to be considered for design purposes in the design of buildings.

2 DEAD LOAD

2.1 Assessment of Dead Load

The dead load in a building shall comprise the weight of all walls, partitions, floors and roofs and shall include the weights of all other permanent constructions (including built-in partitions, finishes, cladding and other similarly incorporated architectural and structural items, weight of fixed service equipment) in the building.

2.2 The list of dead loads of various construction materials and stored materials are given in Annex A.

2.3 Gardening and Landscaping

Elements on roof gardening or landscaping, including soil, plants and drainage layers, along with other features like walkways, fences and walls, are fixed in place and thus classified as dead loads. While the weight of walkways, fences and walls remains constant. The weight of soil and drainage layers, which support plant growth, can vary significantly due to their ability to absorb and retain water. For load calculations, when the weight adds to other structural loads, the dead load should be based on fully saturated soil and drainage layers to account for maximum weight. If the weight is used to counteract uplift forces, calculations should assume the dry weight of soil and drainage layers. Vegetative or landscaped roof areas may retain even more water than fully saturated soil, either from rainfall or irrigation. This additional water retention should be included in the design to accurately assess load variations.

In the design calculation of overturning, uplifting and sliding effects, the beneficial effects of soil shall not be considered. When the soil is used as dead weight and locked permanently, the same may be considered.

3 IMPOSED LOAD

3.1 This clause covers imposed loads to be assumed in the design of buildings. The imposed loads specified herein are minimum loads which should be taken into consideration for the purpose of structural safety of buildings.

NOTE — This Section does not cover detailed provisions for loads incidental to construction and special cases of vibration, such as moving machinery, heavy acceleration from cranes, hoists and the like. Such loads shall be dealt with individually in each case.

3.2 Terminology

3.2.1 For the purpose of imposed loads specified herein, the following definitions shall apply:

3.2.1.1 *Assembly buildings* — These shall include any building or part of a building where groups of people congregate or gather for amusement, recreation, social, religious, patriotic, civil, travel and similar purposes; for example, theatres, motion picture houses, assembly halls, conference/meeting halls, city halls, marriage halls, town halls, auditoria, exhibition halls, museums, skating rinks, gymnasiums, restaurants (also used as assembly halls), place of worship, dance halls, club rooms, passenger stations and terminals of air, surface and other public transportation services, recreation piers and stadia, etc.

3.2.1.2 *Business buildings* — These shall include any building or part of a building, which is used for transaction of business (other than that covered by mercantile buildings); for keeping of accounts and records for similar purposes; offices, banks, professional establishments, court houses and libraries shall be classified in this group so far as principal function of these is transaction of public business and the keeping of books and records.

NOTE — The office buildings are the buildings primarily to be used as an office or for office purposes; 'office purposes' include the purpose of administration, clerical work, handling money, telephone and telegraph operating and operating computers, calculating machines, 'clerical work' includes writing, book-keeping, sorting papers, typing, filing, duplicating, drawing of matter for publication and the editorial preparation of matter for publication, etc.

3.2.1.3 *Dwellings* — These shall include any building or part occupied by members of single/multi-family units with independent cooking facilities. These shall also include apartment houses (flats).

3.2.1.4 *Educational buildings* — These shall include any building used for school, college or day-care purposes involving assembly for instruction, education or recreation and which is not covered by assembly buildings.

3.2.1.5 *Imposed load* — The load assumed to be produced by the intended use or occupancy of a building including the weight of movable partitions, distributed and concentrated loads, loads due to impact and vibration and dust loads but excluding wind, seismic, snow and other loads due to temperature changes, creep, shrinkage, differential settlement, etc.

3.2.1.6 *Industrial buildings* — These shall include any building or a part of a building or structure, in which products or materials of various kinds and properties are fabricated, assembled, manufactured or processed, for example, assembly plants, industrial laboratories, dry cleaning plants, power plants, generating units, pumping stations, fumigation chambers, laundries, buildings or structures in gas plants, refineries, dairies and saw-mills, etc.

3.2.1.7 *Institutional buildings* — These shall include any building or a part thereof, which is used for purposes such as medical or other treatment in case of persons suffering from physical and mental illness, disease or infirmity; care of infants, convalescents or aged persons and for penal or correctional detention in which the liberty of the inmates is restricted. Institutional buildings ordinarily provide sleeping accommodation for the occupants. It includes hospitals, sanatoria, custodial institutions or penal institutions like jails, prisons and reformatories.

3.2.1.8 *Occupancy or use group* — The principal occupancy for which a building or part of a building is used or intended to be used; for the purpose of classification of a building according to occupancy, an occupancy shall be deemed to include subsidiary occupancies which are contingent upon it.

3.2.1.9 *Mercantile buildings* — These shall include any building or a part of a building which is used as shops, stores, market for display and sale of merchandize either wholesale or retail. Office, storage and service and facilities incidental to the sale of merchandize and located in the same building shall be included under this group.

3.2.1.10 *Residential buildings* — These shall include any building in which sleeping accommodation is provided for normal residential purposes with or without cooking or dining or both facilities (except buildings under institutional buildings). It includes lodging or rooming houses, one or two-family private dwellings, dormitories, apartment houses (flats) and hotels.

3.2.1.11 *Storage buildings* — These shall include any building or part of a building used primarily for the storage or sheltering of goods, wares or merchandize, like warehouses, cold storages, freight depots, transit sheds, store houses, garages, hangers, truck terminals, grain elevators, barns and stables.

3.3 Imposed Loads on Floors Due to Use and Occupancy

3.3.1 *Imposed Loads*

The imposed loads to be assumed in the design of buildings shall be the greatest loads that probably will be produced by the intended use or occupancy, but shall not be less than the equivalent minimum loads specified in Table 1 subject to any reductions permitted in **3.3.2**.

Floors shall be investigated for both the uniformly distributed load (UDL) and the corresponding concentrated load specified in Table 1 and designed for the most adverse effects but they shall not be considered to act simultaneously. The concentrated loads specified in Table 1 may be assumed to act over an area of 0.3 m × 0.3 m. However, the concentrated loads need not be considered where the floors are capable of effective lateral distribution of this load.

All other structural elements shall be investigated for the effects of uniformly distributed loads on the floors specified in Table 1.

NOTES

1 Where, in Table 1, no values are given for concentrated load, it may be assumed that the tabulated distributed load is adequate for design purposes.

2 The loads specified in Table 1 are equivalent uniformly distributed loads on the plan area and provide for normal effects of impact and acceleration. They do not take into consideration special concentrated loads and other loads.

3 Where the use of an area or floor is not provided in Table 1, the imposed load due to the use and occupancy of such an area shall be determined from the analysis of loads resulting from:

a) weight of the probable assembly of persons;

b) weight of the probable accumulation of equipment and furnishing;

c) weight of the probable storage materials; and

d) impact factor, if any.

4 While selecting a particular loading, the possible change in use or occupancy of the building should be kept in view. Designers should not necessarily select in every case the lower loading appropriate to the first occupancy. In doing this they might introduce considerable restrictions in the use of the building at a later date and thereby reduce its utility.

5 The loads specified herein, which are based on estimations, may be considered as the characteristic loads for the purpose of limit state method of design till such time statistical data are established based on load surveys to be conducted in the country.

6 When an existing building is altered by an extension in height or area, all existing structural parts affected by the addition shall be strengthened where necessary and all new structural parts shall be designed to meet the requirements for building hereafter erected.

7 The loads specified in the Section does not include loads incidental to construction. Therefore, close supervision during construction is essential to ensure that overloading of the building due to loads by way of stacking of building materials or use of equipment (for example, cranes and trucks) during construction or loads which may be induced by floor to floor propping in multi-storeyed construction, does not occur. However, if construction loads were of short duration, permissible increase in stresses in the case of working stress method or permissible decrease in load factors in limit state method, as applicable to relevant design codes, may be allowed for.

8 The loads in Table 1 are grouped together as applicable to buildings having separate principal occupancy or use. For a building with multiple occupancies, the loads appropriate to the occupancy with comparable use shall be chosen from other occupancies.

9 Regarding loading on lift machine rooms including storage space used for repairing lift machines, designers should go by the recommendations of lift manufacturers for the present. Regarding loading due to false ceiling, the same should be considered as imposed loads on the roof/floor to which it is fixed.

Table 1 Imposed Floor Loads for Different Occupancies

(*Clauses* 3.3.1, 3.3.1.1 *and* 3.4.1)

Sl No.	Occupancy Classification	Uniformly Distributed Load (UDL) kN/m^2	Concentrated Load kN
(1)	(2)	(3)	(4)
i)	*Residential Buildings*		
	a) Dwelling houses:		
	1) All rooms and kitchens	2.0	1.8
	2) Toilets and bath rooms	2.0	–
	3) Corridors, passages, staircases including fire escapes and store rooms	3.0	4.5
	4) Balconies	3.0	1.5 per metre run concentrated at the outer edge
	b) Dwelling units planned and executed in accordance with the accepted standard [C-1(1)] only:		
	1) Habitable rooms, kitchens and toilets and bath rooms	2.0	1.8
	2) Corridors, passages and staircases including fire escapes	3.0	4.5
	3) Balconies	3.0	1.5 per metre run concentrated at the outer edge
	c) Hotels, hostels, boarding houses, lodging houses, dormitories and residential clubs:		
	1) Living rooms, bed rooms and dormitories	2.0	1.8
	2) Kitchen and laundries	3.0	4.5

Table 1 (*Continued*)

Sl No.	Occupancy Classification	Uniformly Distributed Load (UDL) kN/m²	Concentrated Load kN
(1)	(2)	(3)	(4)
	3) Billiards room and public lounges	3.0	2.7
	4) Store rooms	5.0	4.5
	5) Dining rooms, cafeterias and restaurants	4.0	2.7
	6) Office rooms	2.5	2.7
	7) Rooms for indoor games	3.0	1.8
	8) Baths and toilets	2.0	–
	9) Corridors, passages staircases including fire escapes and lobbies – as per the floor serviced (excluding stores and the like) but not less than	3.0	4.5
	10) Balconies	Same as rooms to which they give access but with a minimum of 4.0	1.5 per metre run concentrated at the outer edge
	d) Boiler rooms and plant rooms – to be calculated but not less than	5.0	6.7
	e) Garages:		
	1) Garage floors (including parking area and repair workshops) for passenger cars and vehicles not exceeding 2.5 tonne gross weight, including access ways and ramps – to be calculated but not less than	2.5	9.0

Table 1 (*Continued*)

Sl No.	Occupancy Classification	Uniformly Distributed Load (UDL) kN/m²	Concentrated Load kN
(1)	(2)	(3)	(4)
	i) Without stack	5.0	9.0
	ii) With 2 level stack		
	2) Garage floors for vehicles not exceeding 4.0 tonne gross weight (including access ways and ramps) – to be calculated but not less than	5.0	9.0
	f) Roof top solar panels (with or without frames and with anchor blocks)	1.75	1.75
	g) Roof top HVAC loading	9.0	9.0
ii)	*Educational Buildings*		
	a) Class rooms and lecture rooms (not used for assembly purposes)	3.0	2.7
	b) Dining rooms, cafeterias and restaurants	3.0[1)]	2.7
	c) Offices, lounges and staff rooms	2.5	2.7
	d) Dormitories	2.0	2.7
	e) Projection rooms (with audio-visual facilities)	5.0	–
	f) Kitchens	3.0	4.5
	g) Toilets and bath rooms	2.0	–
	h) Store rooms	5.0	4.5
	j) Libraries and archives:		
	1) Stack room/stack area	6.0 kN/m² for a minimum height of 2.2 m + 2.0 kN/m² per metre height beyond 2.2 m	4.5
	2) Reading rooms (without separate storage)	4.0	4.5
	3) Reading rooms (with separate storage)	3.0	4.5
	k) Boiler rooms and plant rooms – to be	4.0	4.5

Table 1 (*Continued*)

Sl No.	**Occupancy Classification**	**Uniformly Distributed Load (UDL)** kN/m^2	**Concentrated Load** kN
(1)	(2)	(3)	(4)
	calculated but not less than		
	m) Corridors, passages, lobbies, staircases including fire escapes – as per the floor serviced (without accounting for storage and projection rooms) but not less than	4.0	4.5
	n) Balconies	Same as rooms to which they give access but with a minimum of 4.0	1.5 per metre run concentrated at the outer edge
iii)	*Institutional Buildings*		
	a) Bed rooms, wards, dressing rooms, dormitories and lounges	2.0	1.8
	b) Kitchens, laundries and laboratories	3.0	4.5
	c) Dining rooms, cafeterias and restaurants	3.0[1)]	2.7
	d) Toilets and bathrooms	2.0	–
	e) X-ray rooms, operating rooms and general storage areas – to be calculated but not less than	3.0	4.5
	f) Office rooms and O.P.D. rooms	2.5	2.7
	g) Corridors, passages, lobbies, staircases including fire escapes – as per the floor serviced (without accounting for storage and projection rooms) but not less than	4.0	4.5
	j) Boiler rooms and plant rooms – to be	5.0	4.5

Table 1 (*Continued*)

Sl No.	Occupancy Classification	Uniformly Distributed Load (UDL) kN/m²	Concentrated Load kN
(1)	(2)	(3)	(4)
	calculated but not less than		
	k) Balconies	Same as rooms to which they give access but with a minimum of 4.0	1.5 per metre run concentrated at the outer edge
iv)	*Assembly Buildings*		
	a) Assembly areas :		
	1) With fixed seats[2)]	4.0	–
	2) Without fixed seats	5.0	3.6
	c) Restaurants (subject to assembly), museums and art galleries and gymnasia	4.0	4.5
	d) Projection rooms (with audio-visual facilities)	5.0	–
	e) Stages	5.0	4.5
	f) Office rooms, kitchens and laundries	3.0	4.5
	g) Dressing rooms	2.0	1.8
	h) Lounges and billiards rooms	2.0	2.7
	j) Toilets and bathrooms	2.0	–
	k) Corridors, passages and staircases including fire escapes	4.0	4.5
	m) Balconies	Same as rooms to which they give access but with a minimum of 4.0	1.5 per metre run concentrated at the outer edge
	n) Boiler rooms and plant rooms including weight of machinery	7.5	4.5
	p) Corridors, passages, subject to loads greater than from crowds, such as wheeled vehicles, trolleys and the like corridors, staircases and passages in grandstands	5.0	4.5

Table 1 (*Continued*)

Sl No.	Occupancy Classification	Uniformly Distributed Load (UDL) kN/m²	Concentrated Load kN
(1)	(2)	(3)	(4)
v)	*Business and Office Buildings* (*see also* 3.2.1)		
	a) Rooms for general use with separate storage	2.5	2.7
	b) Rooms without separate storage	4.0	4.5
	c) Banking halls	3.0	2.7
	d) Business computing machine rooms (with fixed computers or similar	3.0	4.0
	e) equipment)		
	f) UPS, server and electrical panel	5.0	4.5
	g) Records/files store rooms and storage space	5.0	4.5
	h) Vaults and strong rooms – to be calculated but not less than	5.0	4.5
	j) Cafeterias and dining rooms	3.0[1)]	2.7
	k) Kitchens	3.0	2.7
	m) Corridors, passages, lobbies, staircases including fire escapes – as per the floor serviced (excluding stores) but not less than	4.0	4.5
	n) Bath and toilets rooms	2.0	–
	p) Balconies	Same as rooms to which they give access but with a minimum of 4.0	1.5 per metre run concentrated at the outer edge
	q) Stationary stores	4.0 for each metre of storage height	9.0
	r) Boiler rooms and plant rooms – to be calculated but not less than	5.0	6.7
	s) Libraries	*See* Sl No. (ii)	

Table 1 (*Continued*)

Sl No.	**Occupancy Classification**	**Uniformly Distributed Load (UDL)** kN/m²	**Concentrated Load** kN
(1)	(2)	(3)	(4)
vi)	*Mercantile Buildings*		
	a) Retail shops	4.0	3.6
	b) Wholesale shops – to be calculated but not less than	6.0	4.5
	c) Office rooms	2.5	2.7
	d) Dining rooms, restaurants and cafeterias	3.0[1)]	2.7
	e) Toilets	2.0	–
	f) Kitchens and laundries	3.0	4.5
	g) Boiler rooms and plant rooms – to be calculated but not less than	5.0	6.7
	h) Corridors, passages, stair-cases including fire escapes and lobbies	4.0	4.5
	j) Corridors, passages, staircases subject to loads greater than from crowds, such as wheeled vehicles, trolleys and the like	5.0	4.5
	k) Balconies	Same as rooms to which they give access but with a minimum of 4.0	1.5 per metre run concentrated at the outer edge
vii)	*Industrial Buildings*[3)]		
	a) Work areas without machinery/ equipment	2.5	4.5
	a) Work areas with machinery/equipment[4)] (To be calculated but not less than)		
	1) Light duty	5.0	4.5
	2) Medium duty	7.0	4.5
	3) Heavy duty	10.0	4.5

Table 1 (*Continued*)

Sl No.	**Occupancy Classification**	**Uniformly Distributed Load (UDL)** kN/m²	**Concentrated Load** kN
(1)	(2)	(3)	(4)
	b) Boiler rooms and plant rooms – to be calculated but not less than	5.0	6.7
	c) Cafeterias and dining rooms	3.0[1]	2.7
	d) Corridors, passages, stair cases including fire escapes	4.0	4.5
	e) Corridors, passages, lobbies, staircases subject to machine loads and wheeled vehicles – to be calculated but not less than	5.0	4.5
	f) Kitchens	3.0	4.5
	g) Toilets and bathrooms	2.0	–
viii)	*Storage Buildings*[4]		
	a) Storage rooms (other than cold storage) and warehouses – to be calculated based on the bulk density of materials stored but not less than	2.4 kN/m² per metre of storage height with a minimum of 7.5 kN/m²	7.0
	b) Cold storage – to be calculated but not less than	5.0 kN/m² per metre of storage height with a minimum of 15 kN/m²	9.0
	c) Corridors, passages, staircases including fire escapes – as per the floor serviced but not less than	4.0	4.5
	d) Corridors, passages subject to loads greater than from crowds, such as wheeled vehicles, trolleys and the like	5.0	4.5
	e) Boiler rooms and plant rooms	7.5	4.5
ix)	Refuge area and disaster shelter	3.0	-

Table 1 (*Concluded*)

Sl No.	Occupancy Classification	Uniformly Distributed Load (UDL) kN/m²	Concentrated Load kN
(1)	(2)	(3)	(4)
x)	Datacentres		
	a) Data hall Load (With or without base isolation)		
	1) Tier 1	7.2	
	2) Tier 2	8.4	
	3) Tier 3	12	
	4) Tier 4	12	
	b) Data hall floor Hanging load (With or without base isolation)		
	1) Tier 1	1.2	
	2) Tier 2	1.2	
	3) Tier 3	2.4	
	4) Tier 4	2.4	

1) Where unrestricted assembly of persons is anticipated, the value of UDL should be increased to 4.0 kN/m^2.

2) With fixed seats implies that the removal of the seating and the use of the space for other purposes is improbable. The maximum likely load in this case is, therefore, closely controlled.

3) The loading in industrial buildings (workshops and factories) varies considerably and so three loadings under the terms 'light', 'medium' and 'heavy' are introduced in order to allow for more economical designs but the terms have no special meaning in themselves other than the imposed load for which the relevant floor is designed. It is, however important particularly in the case of heavy weight loads, to assess the actual loads to ensure that they are not in excess of 10 kN/m^2; in case where they are in excess, the design shall be based on the actual loadings.

4) For various mechanical handling equipment which are used to transport goods, as in warehouses, workshops, store rooms, etc, the actual load coming from the use of such equipment shall be ascertained and design should cater to such loads.

3.3.1.1 *Load application*

The uniformly distributed loads specified in Table 1 shall be applied as static loads over the entire floor area under consideration or a portion of the floor area, whichever arrangement produces critical effects on the structural elements as provided in respective design codes.

In the design of floors, the concentrated loads are considered to be applied in the positions which produce the maximum stresses and where deflection is the main criterion in the positions which produce the maximum deflections. Concentrated load, when used for the calculation of bending and shear, are assumed to act at a point. When used for the calculation of local effects such as crushing or punching, they are assumed to act over an actual area of application of 0.3 m × 0.3 m.

3.3.1.2 *Loads due to light partitions*

In office and other buildings, where actual loads due to light partitions cannot be assessed at the time of planning the floors and the supporting structural members shall be designed to carry, in addition to other loads, uniformly distributed loads per square metre of not less than 33.33 percent of weight per metre run of finished partitions, subject to a minimum of 1 kN/m^2, provided total weight of partition walls per m^2 of the wall area does not exceed 1.5 kN/m^2 and the total weight per metre length is not greater than 4.0 kN.

3.3.2 *Reduction in Imposed Loads on Floors*

3.3.2.1 *For members supporting floors*

Except as provided for in **3.3.2.1**(a), the following reductions in assumed total imposed loads on the floors may be made in designing columns, load bearing walls, piers, their supports and foundations:

a) No reduction shall be made for any plant or machinery which is specifically allowed for, or for buildings for storage purposes, warehouses and garages. But, for other buildings, where the floor is designed for an imposed floor load of 5.0 kN/m^2 or more, the reductions shown in **3.3.2.1** may be taken provided that the loading assumed is not less than it would have been, if all the floors had been designed for 5.0 kN/m^2 with no reductions; and

NOTE — In case, if the reduced load in the lower floor is lesser than the reduced load in the upper floor, then the reduced load of the upper floor will be adopted.

b) An example is given in Annex B illustrating the reduction of imposed loads in a multi-storeyed building in the design of column members.

Number of Floors (Including the Roof) to be Carried by Member Under Consideration	*Reduction in Total Distributed Imposed Load on All Floors to be Carried by the Member Under Consideration* Percent
1	0
2	10
3	20
4	30
5 to 10	40
Over 10	50

3.3.2.2 *For beams in each floor level*

Where a single span of beam, girder or truss supports not less than 50 m^2 of floor at one general level, the imposed floor load may be reduced in the design of the beams, girders or trusses by 5 percent for each 50 m^2 area supported subject to a maximum reduction of 25 percent. But, no reduction shall be made in any of the following types of loads:

a) Any superimposed moving load;

b) Any actual load due to machinery or similar concentrated loads;

c) The additional load in respect of partition walls; and

d) Any impact or vibration.

NOTE — The above reduction does not apply to beams, girders or trusses supporting roof loads.

3.3.3 *Posting of Floor Capacities*

Where a floor or part of a floor of a building has been designed to sustain a uniformly distributed load exceeding 3.0 kN/m^2 and in assembly, business, mercantile, industrial or storage buildings, a permanent notice in the form shown below indicating the actual uniformly distributed and/or concentrated loadings for which the floor has been structurally designed shall be posted in a conspicuous place in a position adjacent to such floor or on such part of a floor.

NOTES

1 The lettering of such notice shall be embossed or cast suitably on a tablet whose least dimension shall not be less than 0.25 m and located not less than 1.5 m above floor level with lettering of a minimum size of 25 mm.

2 If a concentrated load or a bulk load has to occupy a definite position on the floor, the same could also be indicated in the table.

DESIGNED IMPOSED FLOOR LOADING
Distributed……………..kN/m^2
Concentrated…………kN

Label Indicating Designed Imposed Floor Loading

3.4 Imposed Loads on Roofs

3.4.1 *Imposed Loads on Various Types of Roofs*

On flat roofs, sloping roofs and curved roofs, the imposed loads due to use and occupancy of the buildings and the geometry of the types of roofs shall be as given in Table 2.

Table 2 Imposed Loads on Various Types of Roofs

(*Clauses* 3.4.1 *and* 3.4.3)

Sl No.	Type of Roof	Imposed Load Measured on Plan Area	Minimum Imposed Load Measured on Plan
(1)	(2)	(3)	(4)
i)	Flat, sloping or curved roof with slopes up to and including 10 degrees		
	a) Access provided	1.5 kN/m^2	3.75 kN uniformly distributed over any span of one metre width of the roof slab and 9 kN uniformly distributed over the span of any beam or truss or wall
	b) Access not provided except for maintenance	0.75 kN/m^2	1.9 kN uniformly distributed over any span of one metre width of the roof slab and 4.5 kN uniformly distributed over the span of any beam or truss or wall
ii)	Sloping roof with slope greater than 10 degrees	For roof membrane sheets or purlins – 0.75 kN/m^2 less 0.02 kN/m^2 for every degree increase in slope over 10 degrees	Subject to a minimum of 0.4 kN/m^2
iii)	Curved roof with slope of line obtained by joining springing point to the crown with the horizontal, greater than 10 degrees	$(0.75 – 0.52\ \alpha^2)$ kN/m^2 where $\alpha = h/l$	Subject to a minimum of 0.4 kN/m^2

Table 2 (*Concluded*)

Sl No.	Type of Roof	Imposed Load Measured on Plan Area	Minimum Imposed Load Measured on Plan
(1)	(2)	(3)	(4)
		h = height of the highest point of the structure measured from its springing; and *l* = chord width of the roof if singly curved and shorter of the two sides if doubly curved. Alternatively, where structural analysis can be carried out for curved roofs of all slopes in a simple manner applying the laws of statistics, the curved roofs shall be divided into minimum 6 equal segments and for each segment imposed load shall be calculated appropriate to the slope of the chord of each segment as given in (i) and (ii).	

NOTES

1 The loads given above do not include loads due to snow, rain, dust collection, etc. The roof shall be designed for imposed loads given above or for snow/rain load, whichever is greater.

2 For special types of roofs with highly permeable and absorbent material, the contingency of roof material increasing in weight due to absorption of moisture shall be provided for.

Roofs of buildings used for promenade or incidental to assembly purposes shall be designed for the appropriate imposed floor loads given in Table 1 for the occupancy.

3.4.2 *Concentrated Load on Roof Coverings*

To provide for loads incidental to maintenance, unless otherwise specified by the Engineer-in-Charge, all roof coverings (other than glass or transparent sheets made of fibre glass) shall be capable of carrying an incidental load of 0.90 kN concentrated on an area of 1 250 mm^2 so placed as to produce maximum stresses in the covering. The intensity of the concentrated load may be reduced with the approval of the Engineer-in-Charge, where it is ensured that the roof coverings would not be traversed without suitable aids. In any case, the roof coverings shall be capable of carrying the loads in accordance with **3.4.1**, **3.4.3** and **3.4.4** and wind load.

3.4.3 *Loads Due to Rain*

On surfaces whose positioning, shape and drainage system are such as to make accumulation of rain water possible, loads due to such accumulation of water and the imposed loads for the roof as given in Table 2 shall be considered separately and the more critical of the two shall be adopted in the design.

3.4.4 *Dust Loads*

In areas prone to settlement of dust on roofs (example, steel plants, cement plants), provision for dust load equivalent to probable thickness of accumulation of dust may be made.

3.4.5 *Loads on Members Supporting Roof Coverings*

Every member of the supporting structure which is directly supporting the roof covering(s) shall be designed to carry the more severe of the following loads except as provided in **3.4.5.1**:

a) The load transmitted to the members from the roof covering(s) in accordance with **3.4.1**, **3.4.3** and **3.4.4**; and

b) An incidental concentrated load of 0.90 kN concentrated over a length of 125 mm placed at the most critical positions on the member.

NOTE — Where it is ensured that the roofs would be traversed only with the aid of planks and ladders capable of distributing the loads on them to two or more supporting members, the intensity of concentrated load indicated in **3.4.5**(**b**) may be reduced to 0.5 kN with the approval of the Engineer-in-Charge.

3.4.5.1 In case of sloping roofs with slope greater than 10º, members supporting the roof purlins, such as trusses, beams, girders, etc, may be designed for two-thirds of the imposed load on purlin or roofing sheets.

3.5 Imposed Horizontal Loads on Parapets, Balustrades and Other Appurtenances Fixed to Structure and on Grandstands

3.5.1 *Parapets, Parapet Walls, Balustrades and other Appurtenances Fixed to the Structure such as Grab Bars, Fixed Ladders and Guardrails in Stilt Parking*

Parapets, parapet walls, balustrades and other appurtenances fixed to the structure such as grab bars, fixed ladders and guardrails in stilt parking, together with the members which give them structural support, shall be designed for the minimum loads given in Table 3. These are expressed as horizontal forces acting at handrail or coping level. These loads shall be considered to act vertically also but not simultaneously with the horizontal forces. The values given in Table 3 are minimum values and where values for actual loadings are available, they shall be used instead.

Table 3 Horizontal Loads on Parapets, Parapet Walls and Balustrades

(*Clause* 3.5.1)

Sl No.	Usage Area	Intensity of Horizontal Load kN/m Run	Concentrated Load (For each 3 m Length) kN
(1)	(2)	(3)	(4)
i)	Light access stairs, gangways and like, not more than 600 mm wide	0.25	—
ii)	Light access stairs, gangways and like, more than 600 mm wide; stairways, landings, balconies and parapet walls (private and part of dwellings)	0.35	—
iii)	a) All other stairways, landings and balconies and all parapets and handrails to roofs [except those subject to overcrowding covered under (iv)]	0.75	0.89
	b) Panel fillers	0.20	0.22

Table 3 (*Concluded*)

Sl No.	Usage Area	Intensity of Horizontal Load kN/m Run	Concentrated Load (For each 3 m Length) kN
(1)	(2)	(3)	(4)
iv)	Parapets and balustrades:		
	a) In places of assembly, such as restaurants and bars, retail and public areas, not likely to be overcrowded	1.5	1.5
	b) In places of assembly, such as retail areas, theatres, cinemas, bars, auditoria, shopping malls, discothèques, places of worship, likely to be overcrowded	3.0	1.5
v)	Grab bars	—	1.11
vi)	Fixed ladders	—	1.33
vii)	Guardrail systems and handrail assemblies	0.73	0.89

NOTE — In the case of guard parapets on a floor of multi-storeyed car park or crash barriers provided in certain buildings for fire escape, the value of imposed horizontal load (together with impact load) may be determined. For (vii), this load need not be considered for (a) one- and two- family dwellings and (b) factory, industrial and storage occupancies, in areas that serve occupant load not greater than 22 kN.

3.5.2 *Grandstands and the Like*

Grandstands, stadia, assembly platforms, reviewing stands and the like shall be designed to resist a horizontal force applied to seats of 0.35 kN per linear metre along the line of seats and 0.15 kN per linear metre perpendicular to the line of the seats. These loadings need not be applied simultaneously. Platforms without seats shall be designed to resist a minimum horizontal force of 0.25 kN/m^2 of plan area.

3.6 Loading Effects Due to Impact and Vibration

The crane loads to be considered under imposed loads shall include the vertical loads, eccentricity effects induced by vertical loads, impact factors, lateral and longitudinal braking forces acting across and along the crane rails, respectively.

3.6.1 *Impact Allowance for Lifts, Hoists and Machinery*

The imposed loads specified in 3.3.1 shall be assumed to include adequate allowance for ordinary impact conditions. However, for structures carrying loads which induce impact or vibration, as far as possible, calculations shall be made for increase in the imposed load due to impact or vibration. In the absence of sufficient data for such calculation, the increase in the imposed loads shall be as follows:

Structures	*Impact Allowance* Percent *Min*
a) For frames supporting lifts and hoists	100
b) For foundations, footings and piers supporting lifts and hoisting apparatus	40
c) For supporting structures and foundations for light machinery, shaft or motor units	20
d) For supporting structures and foundations for reciprocating machinery or power units	50

3.6.2 *Concentrated Imposed Loads with Impact and Vibration*

3.6.2.1 Concentrated imposed loads with impact and vibration which may be due to installed machinery shall be considered and provided for in the design. The impact factor shall not be less than 20 percent which is the amount allowable for light machinery.

3.6.2.2 Provision shall also be made for carrying any concentrated equipment loads while the equipment is being installed or moved for servicing and repairing.

3.6.3 *Impact Allowance for Crane Girders*

For crane gantry girders and supporting columns, the impact allowances (given below) shall be deemed to cover all forces set up by vibration, shock from slipping of slings, kinetic action of acceleration and retardation and impact of wheel loads.

Type of Load	*Additional Load*
a) Vertical loads for electric overhead cranes	25 percent of maximum static loads for crane girders for all class of cranes 25 percent for columns supporting Class III and Class IV cranes 10 percent for columns supporting Class I and Class II cranes No additional load for design of foundations
b) Vertical loads for hand operated cranes	10 percent of maximum wheel loads for crane girders only
c) Horizontal forces transverse to rails:	
1) For electric overhead cranes with trolley having rigid mast for suspension of lifted weight (such as soaker crane, stripper crane, etc)	10 percent of weight of crab and the weight lifted by the cranes, acting on any one crane track rail, acting in either direction and equally distributed amongst all the wheels on one side of rail track For frame analysis, this force, calculated as above, shall be applied on one side of the frame at a time in either direction
2) For all other electric overhead cranes and hand operated cranes	5 percent of weight of crab and the weight lifted by the cranes, acting on any one crane track rail, acting in either direction and equally distributed amongst the wheels on one side of rail track For the frame analysis, the force, calculated as above, shall be applied on one side of the frame at a time in either direction

Type of Load	*Additional Load*
d) Horizontal traction forces along the rails for overhead cranes, either electrically operated or hand operated	5 percent of all static wheel loads

Forces specified in (c) and (d) shall be considered as acting at the rail level and being appropriately transmitted to the supporting system. Gantry girders and their vertical supports shall be designed on the assumption that either of the horizontal forces in (c) and (d) may act at the same time as the vertical load.

NOTE — *See* as per good practice [C-1(2)] for classification (Class I to Class IV) of cranes.

3.6.3.1 *Overloading factors in crane supporting structures*

For all ladle cranes and charging cranes where there is possibility of overloading from production considerations, an overloading factor of 10 percent of the maximum wheel loading shall be taken.

3.6.4 *Crane Load Combinations*

In the absence of any specific indications, the load combinations shall be as indicated below.

3.6.4.1 *Vertical loads*

In an aisle, where more than one crane is in operation or has provision for more than one crane in future, the following load combinations shall be taken for vertical loading:

a) two adjacent cranes working in tandem with full load and with overloading according to **3.6.3.1**; and

b) for long span gantries, where more than one crane can come in the span, the girder shall be designed for one crane fully loaded with overloading according to **3.6.3.1** plus as many loaded cranes as can be accommodated on the span but without taking into account overloading according to **3.6.3**(**a**) to give the maximum effect.

3.6.4.2 *Lateral surge*

For design of columns and foundations, supporting crane girders, the following crane combinations shall be considered:

a) *For single bay frames* — Effect of one crane in the bay giving the worst effect shall be considered for calculation of surge force; and

b) *For multi-bay frames* — Effect of two cranes working, one each in any of two bays in the cross section to give the worst effect shall be considered for calculation of surge force.

3.6.4.3 *Tractive force*

3.6.4.3.1 Where one crane is in operation with no provision for future crane, tractive force from only one crane shall be taken.

3.6.4.3.2 Where more than one crane is in operation or there is provision for future crane, tractive force from two cranes giving maximum effect shall be considered.

NOTE — Lateral surge force and longitudinal tractive force acting across and along the crane rail respectively shall not be assumed to act simultaneously. However, if there is only one crane in the bay, the lateral and longitudinal forces may act together simultaneously with vertical loads.

3.7 Rooftop Helipad

Elevated helipad structures shall be designed as per the latest guidelines of Directorate General of Civil Aviation and International Civil Aviation Organization's Heliport Manual.

3.7.1 *Structural Design*

Elevated helipads may be designed for a specific helicopter type though greater operational flexibility will be obtained from a classification system of design. The final approach and take-off area (FATO) should be designed for the largest or heaviest type of helicopter that is anticipated to use the helipad and account taken of other types of loading such as personnel, freight, snow, refueling equipment, etc. For the purpose of design, it is to be assumed that the helicopter will land on two main wheels, irrespective of the actual number of wheels in the undercarriage, or on two skids, if they are fitted. The loads imposed on the structure should be taken as point loads at the wheel centres as shown below:

Sl No.	*Helicopter Category*	*Maximum Take-off Mass*		*Point Load for Each Wheel*	*Under-Carriage Wheel Centres*	*Super-imposed Load* S_{Ha}	*Super-imposed Load* S_{Hb}
		kg	kN	kN	M	kN/m^2	kN/m^2
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	1	Up to 2300	Up to 22.6	12.0	1.75	0.5	1.5
ii)	2	2 301 to 5000	22.6 to 49.2	25.0	2.0	0.5	2.0
iii)	3	5001 to 9000	49.2 to 88.5	45.0	2.5	0.5	2.5
iv)	4	9001 to 13500	88.5 to 133.0	67.0	3.0	0.5	3.0
v)	5	13501 to 19500	133.0 to 192.0	96.0	3.5	0.5	3.0
vi)	6	19501 to 27000	192.0 to 266.0	133.0	4.5	0.5	3.0

3.7.2 The FATO should be designed for the worse condition derived from consideration of the following two cases.

3.7.3 *Case A — Helicopter on Landing*

When designing a FATO on an elevated helipad and in order to cover the bending and shear stresses that result from a helicopter touching down, the following should be taken into account:

a) *Dynamic load due to impact on touchdown* — This should cover the normal touchdown, with a rate of descent of 1.8 m/s, which equates to the serviceability limit state. The impact load is then equal to 1.55 times the maximum take-off mass of the helicopter;

   The emergency touchdown should also be covered at a rate of descent of 3.6m/s (12 ft/s), which equates to the ultimate limit state. The partial safety factor in this case should be taken as 1.66. Therefore,
   Ultimate design load = 1.66 × service load

   = (1.66 × 1.5) maximum take-off mass

   = 2.5 maximum take-off mass

   The sympathetic response factor discussed at 3.7.3(b) below should be applied.

b) *Sympathetic response on the FATO* — The dynamic load should be increased by a structural response factor dependent upon the natural frequency of the platform slab when considering the design of supporting beams and columns. This increase in loading will usually apply only to slabs with one or more freely supported edges. It is recommended that the average structural response factor (R) of 1.3 should be used in determining the ultimate design load;

c) *Over-all superimposed load on the FATO* (S_{Ha}) — To allow for snow load, personnel, freight and equipment loads, etc, in addition to wheel loads, an allowance of 0.5 KN/m^2 should be included in the design;

d) *Lateral load on the platform supports* — The supports of the platform should be designed to resist a horizontal point load equivalent to 0.5 maximum take-off mass of the helicopter, together with the wind loading [*see* (*f*)], applied in the direction which will provide the greater bending moments;

e) *Dead load of structural members* — The partial safety factor to be used for the dead load should be taken as 1.4;

f) *Wind loading* — Shall be as per 4; and

g) *Punching shear* — Check for the punching shear of an undercarriage wheel or skid using the ultimate design load with a contact area of 64.5×10^3 mm².

3.7.4 *Case B — Helicopter at Rest*

When designing a FATO on an elevated helipad and in order to cover the bending and shear stresses from a helicopter at rest, the following should be taken into account:

a) *Dead load of the helicopter* — Each structural element must be designed to carry the point load, in accordance with the table under 3.7.1, from the two main wheels or skids applied simultaneously in any position on the FATO so as to produce the worst effect from both bending and shear;

b) *Over-all superimposed load* (S_{Hb}) — In addition to wheel loads, an allowance for over-all superimposed load given in the table under 3.7.1, over the area of the FATO, should be included in the design; and

c) *Dead load on structural members and wind loading* — The same factors should be included in the design for these items as given for Case A.

NOTE — The above design loads for helicopters at rest are summarized below:

Design Load for Helicopter on Landing — Case A

i) *Superimposed loads*	
a) Helicopter	2.5 L_H distributed as two point loads at the wheel centres for the helicopter category given in the informal table under 3.7.1. Average value for R = 1.3.
b) Lateral load	1.6 $\frac{L_H}{2}$ applied horizontally in any direction.
c) Over-all superimposed load	Load at platform level together with the maximum wind loading. 1.4 S_{Ha} over the whole area of the platform (S_{Ha} given in the informal table under 3.7.1).
ii) Dead load	1.4G
iii) Wind loading	1.4W
iv) Punching shear check	2.5 $L_H R$ load over tyre skid contact area of 64.5×10^3 mm^2.

Design Load For Helicopter at Rest — Case B

i) *Superimposed loads*	
a) Helicopter	1.6 L_H distributed as two point loads at the wheel centres for the helicopter category given in the informal table under 3.7.1.
b) Over-all superimposed load (Personnel, freight, etc)	1.6 S_{Hb} over the whole area of the platform. S_{Hb} given in the informal table under 3.7.1.
ii) Shear check	Check as appropriate

Symbol	*Meaning*
L_H	Maximum take-off mass of helicopter
G	Dead load of structure
W	Wind loading
R	Structural response factor
S_{Ha}	Superimposed load — Case A
S_{Hb}	Superimposed load — Case B

Load Type	*Partial Load Factors*
Dynamic load (ultimate design load)	2.5
Imposed load	1.6
Dead load	1.4
Wind loading	1.4

3.7.5 Normally, the upper load limit of the selected helicopter category should be used for design purposes; however, to avoid over-design, the upper limit in any band may be exceeded by up to 10 percent if the maximum take-off mass of a helicopter falls just into the next higher category.

3.8 Fire Tenders and Emergency Vehicles

Where a structure or portions of a structure are accessed and loaded by fire department access vehicles and other similar emergency vehicles, the structure shall be designed for the greater of the following loads:

a) the actual operational loads, including outrigger reactions and contact areas of the vehicles as stipulated and provided by the building official;

b) the contact area of wheel dimensions to be used considering 45° distribution at the reinforcement of slab considering the bear coat on the slab and any soil filling.

or

in absence of any data, the bare slab loading may be considered as 15 kN/m^2 to 20 kN/m^2.

c) when the fire tender load is combined with gravity loads and the same can be considered as accidental load with the partial load factor of 1.05; and

d) fire tender load need not be combined with lateral loads.

4 WIND LOAD

4.1 General

4.1.1 This clause gives wind forces and their effects (static and dynamic) that should be taken into account when designing buildings, structures and components thereof.

4.1.2 Wind speeds vary randomly both in time and space and hence assessment of wind loads and response predictions are very important in the design of several buildings and structures. A large majority of structures met with in practice do not however, suffer wind induced oscillations and generally do not require to be examined for the dynamic effects of wind. For such normal, short and heavy structures, estimation of loads using static wind analysis has proved to be satisfactory. The details of this method involving important wind characteristics such as the basic wind speeds, terrain categories, modification factors, wind pressure and force coefficients, etc, are given in **4.4** and **4.5**.

4.1.3 Nevertheless, there are various types of structures or their components such as some tall buildings which require investigation of wind induced oscillations. The influence of dynamic velocity fluctuations on the along wind loads (drag loads) for these structures shall be determined using Gust Factor Method, included in **4.8**. A method for calculation of across wind response of tall buildings is included in **4.8.3**.

4.1.4 This Section also applies to buildings or other structures during erection/construction and the same shall be considered carefully during various stages of erection/construction. In locations where the strongest winds and icing may occur simultaneously, loads on structural members, cables and ropes shall be calculated by assuming an ice covering based on climatic and local experience.

4.1.5 Wind is air in motion relative to the surface of the earth. The primary cause of wind is traced to earth's rotation and differences in terrestrial radiation. The radiation effects are primarily responsible for convection either upwards or downwards. The wind generally blows horizontal to the ground at high wind speeds. Since vertical components of atmospheric motion are relatively small, the term 'wind' denotes almost exclusively the horizontal wind; vertical winds are always identified as such. The wind speeds are assessed with the aid of anemometers or anemographs which are installed at meteorological observatories at heights generally varying from 10 m to 30 m above ground.

4.1.6 Very strong winds (more than 80 km/h) are generally associated with cyclonic storms, thunderstorms, dust storms or vigorous monsoons. A feature of the cyclonic storms over the Indian area is that they rapidly weaken after crossing the coasts and move as depressions/lows inland. The influence of a severe storm after striking the coast does not; in general exceed about 60 km, though sometimes, it may extend even up to 120 km. Very short duration hurricanes of very high wind speeds called Kal Baisaki or Norwesters occur fairly frequently during summer months over North East India.

4.1.7 The wind speeds recorded at any locality are extremely variable and in addition to steady wind at any time, there are effects of gusts which may last for a few seconds. These gusts cause increase in air pressure but their effect on stability of the building may not be so important; often, gusts affect only part of the building and the increased local pressures may be more than balanced by a momentary reduction in the pressure elsewhere. Because of the inertia of the building, short period gusts may not cause any appreciable increase in stress in main components of the building although the walls, roof sheeting and individual cladding units (glass panels) and their supporting members such as purlins, sheeting rails and glazing bars may be more seriously affected. Gusts can also be extremely important for design of structures with high slenderness ratios.

4.1.8 The liability of a building to high wind pressures depends not only upon the geographical location and proximity of other obstructions to air flow but also upon the characteristics of the structure itself.

4.1.9 The effect of wind on the structure as a whole is determined by the combined action of external and internal pressures acting upon it. In all cases, the calculated wind loads act normal to the surface to which they apply.

4.1.10 The stability calculations as a whole shall be done considering the combined effect, as well as separate effects of imposed loads and wind loads on vertical surfaces, roofs and other part of the building above general roof level.

4.1.11 Buildings shall also be designed with due attention to the effects of wind on the comfort of people inside and outside the buildings.

4.1.12 Wind borne debris present in cyclone or high speed wind has been causing a significant amount of damage to building envelopes. The actual in-service performance of fenestration assemblies and impact protective systems in areas prone to severe high speed wind depends on many factors. The main factors that affect the actual loading on building surfaces during a severe windstorm include varying wind direction, duration of the wind event, height above ground, building shape, terrain and surrounding structures. The resistance of fenestration or protective systems assemblies to wind loading after impact depends upon the product design, installation, load magnitude, duration and repetition. The method for determination of performance relating to safety of external building envelope like glazed curtain walls, exterior doors, exterior windows, impact protective systems (shutters, screens), skylight, wall cladding system, etc, impacted by wind borne debris, for buildings and other structures located in geographic regions that are prone to cyclone or extreme wind events shall be in accordance with [C-1(3)].

NOTES

1 This Section does not apply to buildings or structures with unconventional shapes, unusual locations and abnormal environmental conditions that have not been covered in this NBCS. Special investigations are necessary in such cases to establish wind loads and their effects. Wind tunnel studies may also be required in such situations.

2 In the case of tall structures with unsymmetrical geometry, the designs may have to be checked for torsional effects due to wind pressure.

3 The provisions given in good practice [C-1(4)] may be suitably referred by the design engineer/planner for reference.

4.2 Notations

The notations to be followed unless otherwise specified in relevant clauses under wind loads are given in **C-1**.

4.3 Terminology

4.3.1 For the purpose of wind loads, the following definitions shall apply.

4.3.1.1 *Angle of attack* — An angle between the direction of wind and a reference axis of the structure.

4.3.1.2 *Breadth* — It means horizontal dimension of the building measured normal to the direction of wind.

4.3.1.3 *Depth* — It means the horizontal dimension of the building measured in the direction of the wind.

NOTE — Breadth and depth are dimensions measured in relation to the direction of wind, whereas length and width are dimensions related to the plan.

4.3.1.4 *Developed height* — It is the height of upward penetration of the velocity profile in a new terrain. At large fetch lengths, such penetration reaches the gradient height, above which the wind speed may be taken to be constant. At lesser fetch lengths, a velocity profile of a smaller height but similar to that of the fully developed profile of that terrain category has to be taken, with the additional provision that the velocity at the top of this shorter profile equal to that of the un-penetrated earlier velocity profile at that height.

4.3.1.5 *Effective frontal area* — The projected area of the structure normal to the direction of wind.

4.3.1.6 *Element of surface area* — The area of surface over which the pressure coefficient is taken to be constant.

4.3.1.7 *Force coefficient* — A non-dimensional coefficient such that the total wind force on a body is the product of the force coefficient, the dynamic pressure of the incident design wind speed and the reference area over which the force is required.

NOTE — When the force is in the direction of the incident wind, the non-dimensional coefficient will be called as 'drag coefficient'. When the force is perpendicular to the direction of incident wind, the non-dimensional coefficient will be called as 'lift coefficient'.

4.3.1.8 *Ground roughness* — The nature of the earth's surface as influenced by small scale obstructions such as trees and buildings (as distinct from topography) is called ground roughness.

4.3.1.9 *Gust* — A positive or negative departure of wind speed from its mean value, lasting for not more than, say, 2 min over a specified interval of time.

4.3.1.10 *Peak gust* — A peak gust or peak gust speed is the wind speed associated with the maximum amplitude.

4.3.1.11 *Fetch length* — It is the distance measured along the wind from a boundary at which a change in the type of terrain occurs. When the changes in terrain types are encountered (such as, the boundary of a town or city, forest, etc), the wind profile changes in character but such changes are gradual and start at ground level, spreading or penetrating upwards with increasing fetch length.

4.3.1.12 *Gradient height* — It is the height above the mean ground level at which the gradient wind blows as a result of balance among pressure gradient force, Coriolis force and centrifugal force. For the purpose of this Section, the gradient height is taken as the height above the mean ground level, above which the variation of wind speed with height need not be considered.

4.3.1.13 *Tall building* — A building with a height more than or equal to 50 m or having a height to smaller lateral dimension more than 6.

4.3.1.14 *Low rise building* — A building having its height less than 20 m.

4.3.1.15 *Mean ground level* — The mean ground level is the average horizontal plane of the area enclosed by the boundaries of the structure.

4.3.1.16 *Pressure coefficient* — It is the ratio of the difference between the pressure acting at a point on the surface and the static pressure of the incident wind to the design wind pressure, where the static and design wind pressures are determined at the height of the point considered after taking into account the geographical location, terrain conditions and shielding effect. The pressure coefficient is also equal to $[1 - (V_p/V_z)^2]$, where V_p is the actual wind speed at any point on the structure at a height corresponding to that of V_z.

NOTE — Positive sign of the pressure coefficient indicates pressure acting towards the surface and negative sign indicates pressure acting away from the surface.

4.3.1.17 *Return period* — It is the number of years, reciprocal of which gives the probability of extreme wind exceeding a given wind speed in anyone year.

4.3.1.18 *Shielding effect* — Shielding effect or shielding refers to the condition where wind has to pass along some structure(s) or structural element(s) located on the upstream wind side, before meeting the structure or structural element under consideration. A factor called 'shielding factor' is used to account for such effects in estimating the force on the shielded structures.

4.3.1.19 *Suction* — It means pressure less than the atmospheric (static) pressure and is taken to act away from the surface.

4.3.1.20 *Solidity ratio* — It is equal to the effective area (projected area of all the individual elements) of a frame normal to the wind direction divided by the area enclosed by the boundary of the frame normal to the wind direction.

NOTE — Solidity ratio is to be calculated for individual frames.

4.3.1.21 *Terrain category — It means the characteristics of the surface irregularities of an area which arise from natural or constructed features. The categories are numbered in increasing order of roughness.

4.3.1.22 *Topography* — The nature of the earth's surface as influenced by the hill and valley configurations.

4.3.1.23 *Velocity profile* — The variation of the horizontal component of the atmospheric wind speed at different heights above the mean ground level is termed as velocity profile.

4.4 Wind Speed

4.4.1 *Nature of Wind in Atmosphere*

In general, wind speed in the atmospheric boundary layer increases with height from zero at ground level to maximum at a height called the gradient height. There is usually a slight change in direction (Ekman effect) but this is ignored in this Section. The variation with height depends primarily on the terrain conditions. However, the wind speed at any height never remains constant and it has been found convenient to resolve its instantaneous magnitude into an average or mean value and a fluctuating component around this average value. The average value depends on the average time employed in analyzing the meteorological data and this averaging time varies from few seconds to several minutes. The magnitude of fluctuating component of the wind speed, which is called gust, depends on the averaging time. In general, smaller the averaging interval more is the magnitude of the gust speed.

4.4.2 *Basic Wind Speed*

Figure 1 gives basic wind speed map of India, as applicable to 10 m height above mean ground level for different zones of the country. Basic wind speed is based on peak gust velocity averaged over a short time interval of about 3 s and corresponds to mean heights above ground level in an open terrain (Category 2). Basic wind speeds presented in Fig. 1 have been worked out for a 50 year return period. Basic wind speed for some important cities/towns is also given in Annex D.

4.4.3 *Design Wind Speed* (V_z)

The basic wind speed (V_b) for any site shall be obtained from Fig. 1 and shall be modified to include the following effects to get design wind speed, V_z at any height z, for the chosen structure:

a) Risk level;

b) Terrain roughness and height of structure;

c) Local topography; and

d) Importance factor for the cyclonic region.

It can be mathematically expressed as follows:

$$V_z = V_b \, k_1 \, k_2 \, k_3 \, k_4$$

where

V_z = design, in m/s, wind speed at height z;

k_1 = probability factor (risk coefficient) (*see* **4.4.3.1**);

k_2 = terrain roughness and height factor (*see* **4.4.3.2**);

k_3 = topography factor (*see* **4.4.3.3**); and

k_4 = importance factor for the cyclonic region (*see* **4.4.3.4**).

NOTE — Wind speed may be taken as constant up to a height of 10 m. However, pressures for buildings less than 10 m high may be reduced by 20 percent for evaluating stability and design of the framing.

4.4.3.1 *Risk coefficient* (k_1 factor)

Figure 1 gives basic wind speeds for terrain Category 2 as applicable at 10 m above ground level based on 50 years mean return period. The suggested life period to be assumed in design and the corresponding k_1 factors for different class of structures for the purpose of design are given in Table 4. In the design of buildings and structures, a regional basic wind speed having a mean return period of 50 years shall be used except as specified in the Note of Table 4.

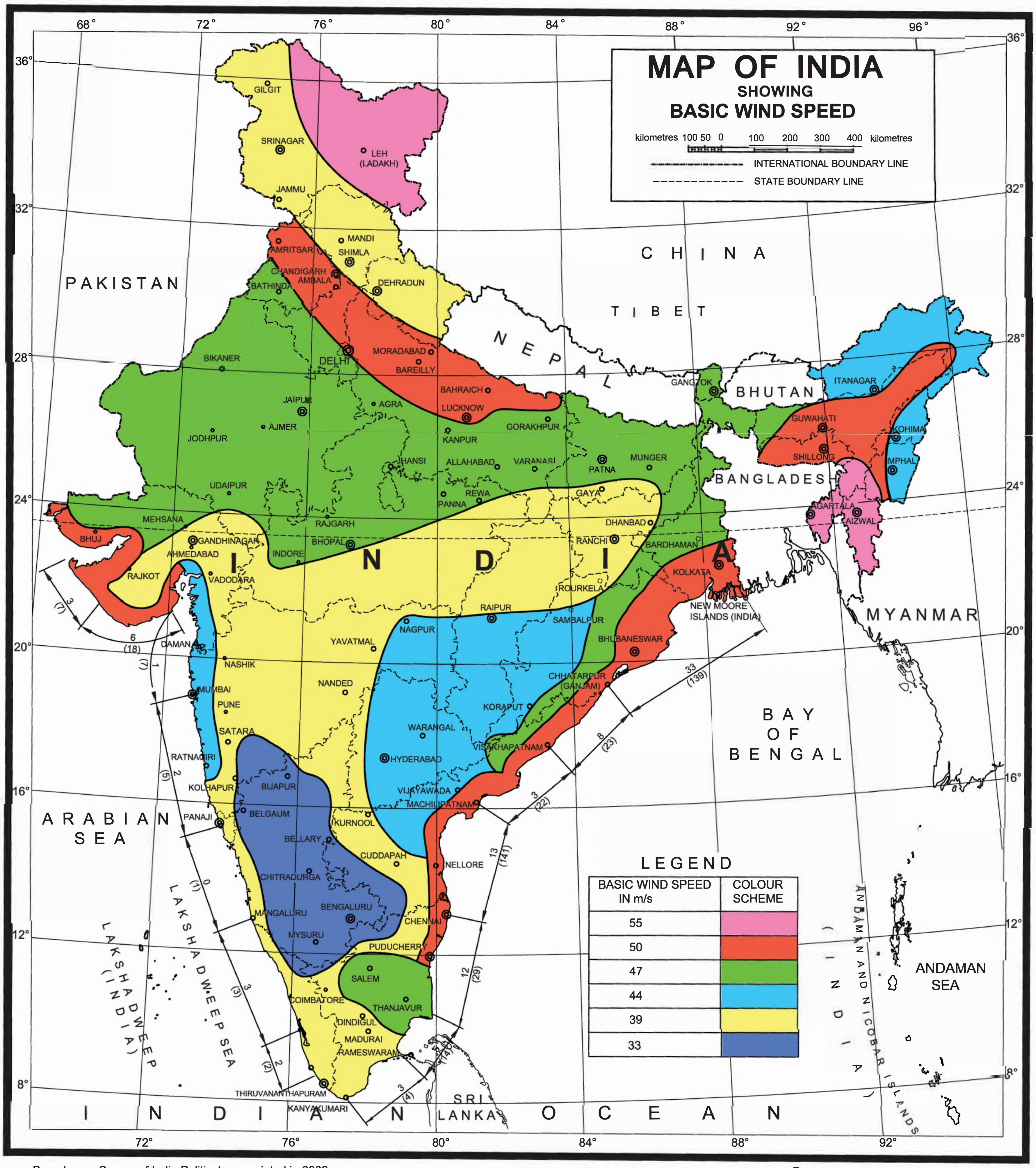
4.4.3.2 *Terrain, height factor* (k_2 factor)

4.4.3.2.1 *Terrain*

Selection of terrain categories shall be made with due regard to the effect of obstructions which constitute the ground surface roughness. The terrain category used in the design of a structure may vary depending on the direction of wind under consideration. Wherever sufficient meteorological information is available about the nature of wind direction, the orientation of any building or structure may be suitably planned.

Terrain in which a specific structure stands shall be assessed as being one of the following terrain categories:

a) *Category* 1 — Exposed open terrain with few or no obstructions and in which the average height of any



Based upon Survey of India Political map printed in 2002.

The territorial waters of India extend into the sea to a distance of twelve nautical miles measured from the appropriate baseline.

The interstate boundaries between Arunachal Pradesh, Assam and Meghalaya shown on this map are as interpreted from the North-Eastern Areas (Reorganization) Act, 1971, but have yet to be verified.

The external boundaries and coastlines of India agree with the Record/Master Copy certified by Survey of India.

The responsibility for the correctness of internal details rest with the publisher.

NOTES

1 The occurrence of a tornado is possible in virtually any part of India. They are particularly more severe in the northern parts of India. The recorded number of these tornados is too small to assign any frequency. The devastation caused by a tornado is due to exceptionally high winds about its periphery, and the sudden reduction in atmospheric pressure at its centre, resulting in an explosive outward pressure on the elements of the structure. the regional basic wind speeds do not include any specific allowance for tornados. It is not the usual practice to allow for the effect of tornados unless special requirements are called for as in the case of important structures such as, nuclear power reactors and satellite communication towers.

2 The total number of cyclonic storms that have struck different sections of east and west coasts are included in Fig. 1, based on available records for the period from 1877 to 1982. The figures above the lines (between the stations) indicate the total number of severe cyclonic storms with or without a core of hurricane winds (speeds above 87 km/h) and the figures in the brackets below the lines indicate the total number of cyclonic storms. Their effect on land is already reflected in the basic wind speeds specified in Fig. 1. These have been included only as additional information.

FIG. 1 BASIC WIND SPEED IN m/s (BASED ON 50-YEARS RETURN PERIOD)

object surrounding the structure is less than 1.5 m. The equivalent aerodynamic roughness height, ($z_{0,1}$) for this terrain is 0.002 m. Typically this category represents open sea-coasts and flat plains without trees;

b) *Category* 2 — Open terrain with well scattered obstructions having heights generally between 1.5 m and 10 m. The equivalent aerodynamic roughness height, ($z_{0,2}$) for this terrain is 0.02 m. This is the criterion for measurement of regional basic wind speeds and represents airfields, open park lands and undeveloped sparsely built-up outskirts of towns and suburbs. Open land adjacent to sea coast may also be classified as Category 2 due to roughness of large sea waves at high winds;

c) *Category* 3 — Terrain with numerous closely spaced obstructions having the size of buildings/structures up to 10 m in height with or without a few isolated tall structures. The equivalent aerodynamic roughness height, ($z_{0,3}$) for this terrain is 0.2 m. This category represents well wooded areas and shrubs, towns and industrial areas full or partially developed. It is likely that the, next higher category than this will not exist in most design situations and that selection of a more severe category will be deliberate; and

d) *Category* 4 — Terrain with numerous large high closely spaced obstructions. The equivalent aerodynamic roughness height, ($z_{0,4}$) for this terrain is 2.0 m. This category represents large city centres, generally with obstructions above 25 m and well developed industrial complexes.

Table 4 Risk Coefficients for Different Classes of Structures in Different Wind Speed Zones

(*Clause* 4.4.3.1)

Sl No.	Class of Structure	Mean Probable Design Life of Structure in Years	k_1 Factor for Basic Wind Speed (in m/s) of					
			33	39	44	47	50	55
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
i)	All general buildings and structures and ground based free standing tower up to 50 m	50	1.0	1.0	1.0	1.0	1.0	1.0
ii)	Temporary sheds, structures such as those used during construction operations (for example forL-work and false work), structures during construction stages and boundary walls	5	0.82	0.76	0.73	0.71	0.70	0.67
iii)	Buildings and structures presenting a low degree of hazard to life and property in the event of failure, such as isolated towers in wooded areas, farm buildings other than residential buildings	25	0.94	0.92	0.91	0.90	0.90	0.89
iv)	Important buildings and structures, such as hospitals, communication buildings/towers and power plant structures	100	1.05	1.06	1.07	1.07	1.08	1.08

NOTE — The factor k_1 is based on statistical concepts which take into account the degree of reliability required and period of time in years during which there will be exposure to wind, that is, life of the structure. Whatever wind speed is adopted for design purposes, there is always a probability (however small) that it may be exceeded in a storm of exceptional violence; more the period of years over which there is exposure to the wind, more is the probability. Larger return periods ranging from 100 to 1 000 years (implying lower risk level) in association with larger periods of exposure may have to be selected for exceptionally important structures, such as, nuclear power reactors and satellite communication towers. Equation given below may be used in such cases to estimate k_1 factors for different periods of exposure and chosen probability of exceedance (risk level). The probability level of 0.63 is normally considered sufficient for design of buildings and structures against wind effects and the values of k_1 corresponding to this risk level are given above.

Table 4 (*Concluded*)

$$k_1 = \frac{X_{\mathrm{N,P}}}{X_{50,0.63}} = \frac{A - B\left[l_n\left\{-\frac{1}{N}l_n\left(1-P_N\right)\right\}\right]}{A+4B}$$

where

N = mean probable design life of structure in years;

P_{N} = risk level in N consecutive years (probability that the design wind speed is exceeded;

$X_{\mathrm{N,P}}$ = at least once in N successive years), nominal value = 0.63; and

$X_{50,0.63}$ = extreme wind speed for given values of N and P_{N}; .

A and *B* have the following values for different basic wind speed zones:

Zone m/s	***A**** m/s	***B**** m/s
33	23.1 (83.2)	2.6 (9.2)
39	23.3 (84.0)	3.9 (14.0)
44	24.4 (88.0)	5.0 (18.0)
47	24.4 (88.0)	5.7 (20.5)
50	24.7 (88.8)	6.3 (22.8)
55	25.2 (90.8)	7.6 (27.3)

* Values of *A* and *B* in 'km/h' are given within bracket

4.4.3.2.2 *Variation of wind speed with height in different terrains* (k_2)

Table 5 gives multiplying factors (k_2) by which the basic wind speed given in Fig. 1 shall be multiplied to obtain the wind speed at different heights, in each terrain category.

Table 5 Factors to Obtain Design Wind Speed Variation with Height in Different Terrains

(*Clause* 4.4.3.2.2)

Sl No.	**Height (z),** m	**Terrain and Height Multiplier (k_2)**			
		Terrain Category 1	Terrain Category 2	Terrain Category 3	Terrain Category 4
(1)	(2)	(3)	(4)	(5)	(6)
i)	10	1.05	1.00	0.91	0.80
ii)	15	1.09	1.05	0.97	0.80
iii)	20	1.12	1.07	1.01	0.80
iv)	30	1.15	1.12	1.06	0.97
v)	50	1.20	1.17	1.12	1.10
vi)	100	1.26	1.24	1.20	1.20
vii)	150	1.30	1.28	1.24	1.24
viii)	200	1.32	1.30	1.27	1.27
ix)	250	1.34	1.32	1.29	1.28
x)	300	1.35	1.34	1.31	1.30

Table 5 (*Concluded*)

Sl No.	Height (z), m	Terrain and Height Multiplier (k_2)			
		Terrain Category 1	Terrain Category 2	Terrain Category 3	Terrain Category 4
(1)	(2)	(3)	(4)	(5)	(6)
xi)	350	1.35	1.35	1.32	1.31
xii)	400	1.35	1.35	1.34	1.32
xiii)	450	1.35	1.35	1.35	1.33
xiv)	500	1.35	1.35	1.35	1.34

NOTE — For intermediate values of height *z* in a given terrain category, use linear interpolation.

4.4.3.2.3 *Terrain categories in relation to the direction of wind*

The terrain category used in the design of a structure may vary depending on the direction of wind under consideration. Where sufficient meteorological information is available, the basic wind speed may be varied for specific wind direction.

4.4.3.2.4 *Changes in terrain categories*

The velocity profile for a given terrain category does not develop to full height immediately with the commencement of that terrain category but develop gradually to height (h_x) which increases with the fetch or upwind distance (x).

a) *Fetch and developed height relationship* — The relation between the developed height (h_x) and the fetch (x) for wind-flow over each of the four terrain categories may be taken as given in Table 6; and

b) For structures of heights more than the developed height (h_x) in Table 6, the velocity profile may be determined in accordance with the following:

   1) The less or least rough terrain; or

   2) The method described in Annex E.

Table 6 Fetch and Developed Height Relationship

(*Clause* 4.4.3.2.4)

Sl No.	Fetch (x) km	Developed Height (h_x) m			
		Terrain Category 1	Terrain Category 2	Terrain Category 3	Terrain Category 4
(1)	(2)	(3)	(4)	(5)	(6)
i)	0.2	12	20	35	60
ii)	0.5	20	30	35	95
iii)	1	25	45	80	130
iv)	2	35	65	110	190
v)	5	60	100	170	300
vi)	10	80	140	250	450

Table 6 (*Concluded*)

Sl No.	Fetch (x) km	Developed Height (h_x) m			
		Terrain Category 1	Terrain Category 2	Terrain Category 3	Terrain Category 4
(1)	(2)	(3)	(4)	(5)	(6)
vii)	20	120	200	350	500
viii)	50	180	300	400	500

4.4.3.3 *Topography* (k_3 *factor*)

The basic wind speed V_b given in Fig. 1 takes into account the general level of site above sea level. This does not allow for local topographic features such as hills, valleys, cliffs, escarpments, or ridges which can significantly affect wind speed in their vicinity. The effect of topography is to accelerate wind near the summits of hills or crests of cliffs, escarpments or ridges and decelerate the wind in valleys or near the foot of cliffs, steep escarpments, or ridges.

The effect of topography shall be significant at a site when the upwind slope (θ) is more than about 3° and below that, the value of k_3 may be taken to be equal to 1.0. The value of k_3 is confined in the range of 1.0 to 1.36 for slopes more than 3°. A method of evaluating the value of k_3 for values more than 1.0 is given in Annex F. It may be noted that the value of k_3 varies with height above ground level, with a maximum near the ground and reducing to 1.0 at higher levels.

4.4.3.4 *Importance factor for cyclonic region* (k_4 *factor*)

The east coast of India is relatively more vulnerable for occurrences of severe cyclones. On the west coast, Gujarat is vulnerable for severe cyclones. Studies of wind speed and damage to buildings and structures point to the fact that the speeds given in the basic wind speed map are often exceeded during the cyclones. The effect of cyclonic storms is largely felt in a belt of approximately 60 km width at the coast. In order to ensure better safety of structures in this region (60 km wide on the east coast as well as on the Gujarat coast), the values of k_4 are stipulated as applicable according to the importance of the structure:

	k_4
Structures of post-cyclone importance for emergency services (such as cyclone shelters, hospitals, schools, communication towers, etc)	1.30
Industrial structures	1.15
All other structures	1.00

NOTE — For guidelines related to estimation of cyclonic factor (k4) as given in good practice [C-1(5)] may be referred.

4.4.4 *Hourly Mean Wind Speed*

The hourly mean wind speed at height *z*, for different terrains can be obtained as:

$$\overline{V}_{z,H} = \overline{k}_{2,i} V_b$$

where

$\overline{k}_{2,i}$ = hourly mean wind speed factor for terrain category 'i'.

$$= 0.1423 \left[\ln\left(\frac{z}{z_{0,i}}\right) \right] \left(z_{0,i}\right)^{0.0706}$$

The design hourly mean wind speed at height z can be obtained as:

$$\overline{V}_{z,d} = \overline{V}_{z,H} k_1 k_3 k_4 \quad = V_b k_1 \overline{k}_{2,i} k_3 k_4$$

4.4.5 *Turbulence Intensity*

The turbulence intensity variations with height for different terrains can be obtained using the relations given below:

a) *Terrain category* 1

$$I_{z,1} = 0.3507 - 0.0535 \log_{10}\left(\frac{z}{z_{0,1}}\right)$$

b) *Terrain category* 2

$$I_{z,2} = I_{z,1} + \frac{1}{7}\left(I_{z,4} - I_{z,1}\right)$$

c) *Terrain category* 3

$$I_{z,3} = I_{z,1} + \frac{3}{7}\left(I_{z,4} - I_{z,1}\right)$$

d) *Terrain category* 4

$$I_{z,4} = 0.466 - 0.1358 \log_{10}\left(\frac{z}{z_{0,4}}\right)$$

4.4.6 *Off-shore Wind Velocity*

Cyclonic storms form far away from the sea coast and gradually reduce in speed as they approach the sea coast. Cyclonic storms generally extend up to about 60 km inland after striking the coast. Their effect on land is already reflected in basic wind speeds specified in Fig. 1. The influence of wind speed off the coast up to a distance of about 200 km may be taken as 1.15 times the value on the nearest coast in the absence of any definite wind data. The factor 1.15 shall be used in addition to k_4.

4.5 Wind Pressures and Forces on Buildings/Structures

4.5.1 *General*

The wind load on a building/structure shall be calculated for:

a) the building/structure as a whole;

b) individual structural elements as roofs and walls; and

c) individual cladding units including glazing and their fixings.

4.5.2 *Design Wind Pressure*

The wind pressure at any height above mean ground level shall be obtained by the following relationship between wind pressure and wind speed:

$$p_z = 0.6 V_z^{\,2}$$

where

p_z = wind pressure at height z, in N/m^2; and

V_z = design wind speed at height z, in m/s.

The design wind pressure p_d can be obtained as:

$$p_d = K_d K_a K_c p_z$$

where

K_d = wind directionality factor;

K_a = area averaging factor; and

K_c = combination factor [*see* **4.5.3.3.13**].

The value of p_d, however shall not be taken as less than 0.70 p_z.

NOTES

1 The coefficient 0.6 (in SI units) in the above formula depends on a number of factors and mainly on the atmospheric pressure and air temperature. The value chosen corresponds to the average Indian atmospheric conditions.

2 K_d should be taken as 1.0 when considering local pressure coefficients.

4.5.2.1 *Wind directionality factor, K_d*

Considering the randomness in the directionality of wind and recognizing the fact that pressure or force coefficients are determined for specific wind directions, it is specified that for buildings, solid signs, open signs, lattice frameworks and trussed towers (triangular, square, rectangular) a factor of 0.90 may be used on the design wind pressure. For circular or near-circular forms this factor may be taken as 1.0.

For the cyclone affected regions also the factor K_d shall be taken as 1.0.

4.5.2.2 *Area averaging factor, K_a*

Pressure coefficients given in **4.5.3** are a result of averaging the measured pressure values over a given area. As the area becomes larger, the correlation of measured values decrease and *vice-versa*. The decrease in pressures due to larger areas may be taken into account as given in Table 7.

Table 7 Area Averaging Factor (K_a)

(*Clause* 4.5.2.2)

Sl No.	**Tributary Area (A)** m^2	**Area Averaging Factor (K_a)**[1)]
(1)	(2)	(3)
i)	≤ 10	1.0
ii)	25	0.9
iii)	≥ 100	0.8

[1)] Linear interpolation for intermediate values of tributary area, A is permitted.

4.5.2.2.1 *Tributary area*

a) *Overall structure* — For evaluating loads on frames the tributary area shall be taken as the centre to centre distances between frames multiplied by the individual panel dimension in the other direction together with overall pressure coefficients;

b) *Individual elements* — For beam type elements, purlins, etc, the tributary area shall be taken as effective span multiplied by spacing. The effective span is the actual span for mid span and cantilever load effects; and half the sum of adjacent spans for support moments and reactions;

c) For plate type elements, the area of individual plates between supports is taken as the tributary area; and

d) For glass cladding, individual pane area of glass is the tributary area.

4.5.3 *Pressure Coefficients*

The pressure coefficients are always given for a particular surface or part of the surface of a building/structure. The wind load acting normal to a surface is obtained by multiplying the area of that surface or its appropriate portion by the pressure coefficient (C_p) and the design wind pressure at the height of the surface from the ground. The average values of these pressure coefficients for some building shapes are given in **4.5.3.2** and **4.5.3.3**.

Average values of pressure coefficients are given for critical wind directions in one or more quadrants. In order to determine the maximum wind load on the building/structure, the total load should be calculated for each of the critical directions shown from all quadrants. Where considerable variation of pressure occurs over a surface, it has been sub-divided and mean pressure coefficients given for each of its several parts.

In addition, areas of high local suction (negative pressure concentration) frequently occurring near the edges of walls and roofs are separately shown. Coefficients for the local effects should only be used for calculation of forces on these local areas affecting roof sheeting, glass panels and individual cladding units including their fixtures. They should not be used for calculating force on entire structural elements such as roof, walls or structure as a whole.

NOTES

1 The pressure coefficients given in different tables have been obtained mainly from measurements on models in wind tunnels and the great majority of data available has been obtained in conditions of relatively smooth flow. Where sufficient field data exists as in the case of rectangular buildings, values have been obtained to allow for turbulent flow.

2 In recent years, wall glazing and cladding design has been a source of major concern. Although of less consequence than the collapse of main structures, damage to glass can be hazardous and cause considerable financial losses.

3 For pressure coefficients for structures not covered herein, reference may be made to specialist literature on the subject or advice may be sought from specialists in the subject.

4.5.3.1 *Wind load on individual members*

When calculating the wind load on individual structural elements such as roofs and walls and individual cladding units and their fittings, it is essential to take account of the pressure difference between opposite faces of such elements or units. For clad structures, it is, therefore, necessary to know the internal pressure as well as the external pressure. Then the wind load, F (in N), acting in a direction normal to the individual structural element or cladding unit is:

$$F = (C_{pe} - C_{pi})\, A\, p_d$$

where

C_{pe} = external pressure coefficient;

C_{pi} = internal pressure coefficient;

A = surface area of structural element or cladding unit, in m^2; and

p_d = design wind pressure, in N/ m^2.

NOTES

1 If the surface design pressure varies with height, the surface areas of the structural element may be sub-divided so that the specified pressures are taken over appropriate areas.

2 Positive wind load indicates the force acting towards the structural element and negative away from it.

4.5.3.2 *Internal pressure coefficients*

Internal air pressure in a building depends upon the degree of permeability of cladding to the flow of air. The internal air pressure may be positive or negative depending on the direction of flow of air in relation to openings in the buildings.

a) In the case of buildings where the claddings permit the flow of air with openings not more than about 5 percent of the wall area but where there are no large openings, it is necessary to consider the possibility of the internal pressure being positive or negative. Two design conditions shall be examined, one with an internal pressure coefficient of + 0.2 and another with an internal pressure coefficient of - 0.2.

The internal pressure coefficient is algebraically added to the external pressure coefficient and the analysis which indicates greater distress of the member shall be adopted. In most situations a simple inspection of the sign of external pressure will at once indicate the proper sign of the internal pressure coefficient to be taken for design.

NOTE — The term normal permeability relates to the flow of air commonly afforded by claddings not only through open windows and doors, but also through the slits round the closed windows and doors and through chimneys, ventilators and through the joints between roof coverings, the total open area being less than 5 percent of area of the walls having the openings.

b) *Buildings/structures with medium and large openings* — Buildings/structures with medium and large openings may also exhibit either positive or negative internal pressure depending upon the direction of wind. Buildings/structures with medium openings between about 5 percent to 20 percent of wall area shall be examined for an internal pressure coefficient of + 0.5 and later with an internal pressure coefficient of - 0.5 and the analysis which produces greater distress of the member shall be adopted. Buildings/structures with large openings, that is, openings larger than 20 percent of the wall area shall be examined once with an internal pressure coefficient of + 0.7 and again with an internal pressure coefficient of - 0.7 and the analysis which produces greater distress of the member shall be adopted.

Buildings/structures with one open side or opening exceeding 20 percent of wall area may be assumed to be subjected to internal positive pressure or suction similar to those of buildings with large openings. A few examples of buildings with one side openings are shown in Fig. 2 indicating values of internal pressure coefficients with respect to the direction of wind.

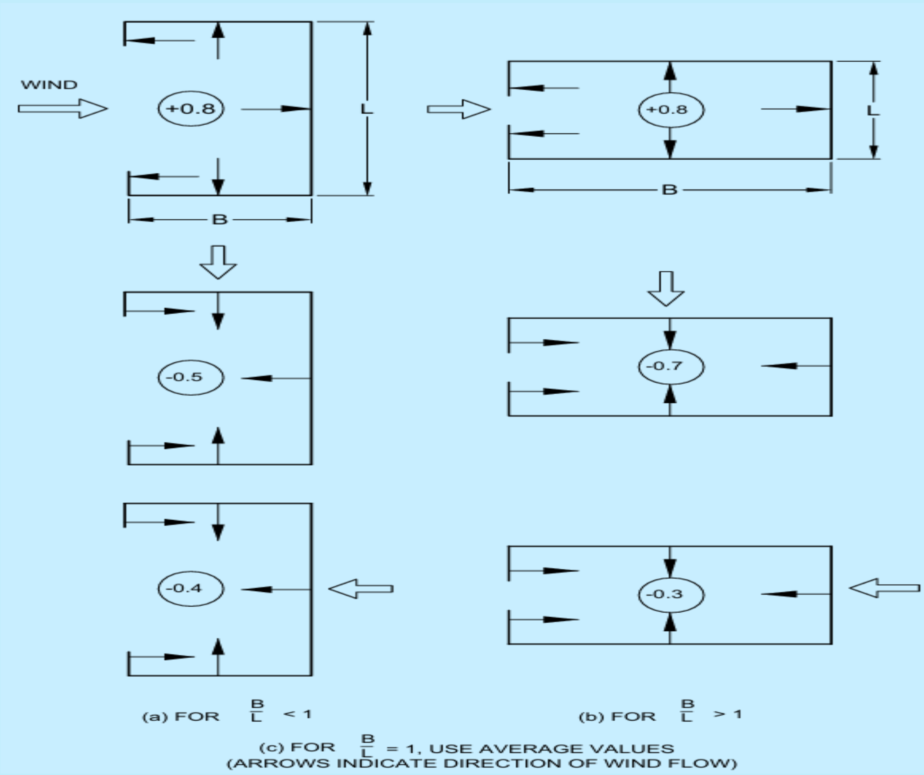


FIG. 2 VALUES OF INTERNAL PRESSURE COEFFICIENTS WITH RESPECT TO THE DIRECTION OF WIND FOR BUILDINGS WITH ONE SIDE OPENINGS

4.5.3.3 *External pressure coefficients*

4.5.3.3.1 *Walls*

The average external pressure coefficient for the walls of clad buildings of rectangular plan shall be as given in Table 8. In addition, local pressure concentration coefficients are also given.

4.5.3.3.2 *Pitched, hipped and monoslope roofs of clad buildings*

The average external pressure coefficients and pressure concentration coefficients for pitched roofs of rectangular clad building shall be as given in Table 9. Where no pressure concentration coefficients are given, the average coefficients shall apply. The pressure coefficients on the underside of any overhanging roof shall be taken in accordance with **4.5.3.3.5**.

NOTES

1 The pressure concentration shall be assumed to act outward (suction pressure) at the ridges, eaves, cornices and 90° corners of roofs.

2 The pressure concentration shall not be included with the net external pressure when computing overall loads.

3 For hipped roofs, pressure coefficients (including local values) may be taken on all the four slopes, as appropriate from Table 9 and be reduced by 20 percent for the hip slope.

For mono slope roofs of rectangular clad buildings, the average pressure coefficient and pressure concentration coefficient for mono slope (lean-to) roofs of rectangular clad buildings shall be as given in Table 10.

4.5.3.3.3 *Canopy roofs with* ($1/4 < h/w < 1$ *and* $1 < L/w < 3$)

The pressure coefficients are given in Table 11 and Table 12 separately for mono pitch and double pitch canopy roofs such as open-air parking garages, shelter areas, outdoor areas, railway platforms, stadiums and theatres. The coefficients take into account of the combined effect of the wind exerted on and under the roof for all wind directions; the resultant is to be taken normal to the canopy. Where the local coefficients overlap, the greater of the two given values should be taken. However, the effect of partial closures of one side and or both sides, such as those due to trains, buses and stored materials shall be foreseen and taken into account.

The solidity ratio, ϕ, is equal to the area of obstructions under the canopy divided by the gross area under the canopy, both areas normal to the wind direction. $\phi = 0$ represents a canopy with no obstructions underneath. $\phi = 1$ represents the canopy fully blocked with contents to the downwind eaves. Values of C_p for intermediate solidities may be linearly interpolated between these two extremes and apply upwind of the position of maximum blockage only. For downwind of the position of maximum blockage, the coefficients for $\phi = 0$ may be used.

In addition to the forces due to the pressures normal to the canopy, there will be horizontal loads on the canopy due to the wind pressure on any fascia and to friction over the surface of the canopy. For any wind direction, only the greater of these two forces need to be taken into account. Fascia loads should be calculated on the area of the surface facing the wind, using a force coefficient of 1.3. Frictional drag should be calculated using the coefficients given in **4.5.4.1**.

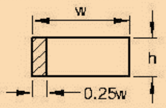
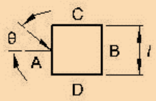
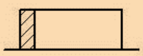
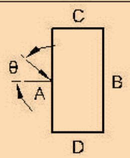
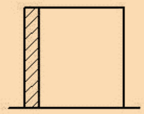
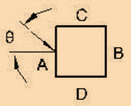
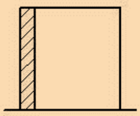
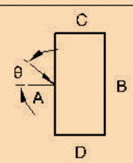
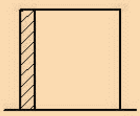
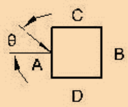
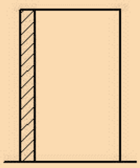
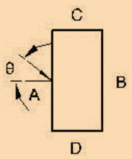
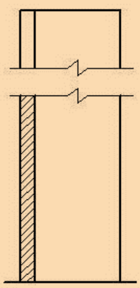
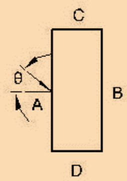
NOTE — Table 13 to Table 18 may be used to get internal and external pressure coefficients for pitches and troughed free roofs for some specific cases for which aspect ratios and roof slopes have been specified. However, while using Table 13 to Table 18 any significant departure from it should be investigated carefully. No increase shall be made for local effects except as indicated.

4.5.3.3.4 *Pitched and saw-tooth roofs of multi-span buildings*

For pitched and saw-tooth roofs of multi-span buildings, the external average pressure coefficients shall be as given in Table 19 and Table 20, respectively provided that all the spans shall be equal and the height to the eaves shall not exceed the span.

Table 8 External Pressure Coefficients (C_{pe}) for Walls of Rectangular Clad Buildings

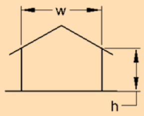
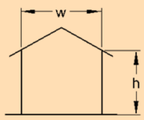
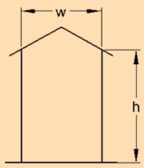
(*Clause* 4.5.3.3.1)

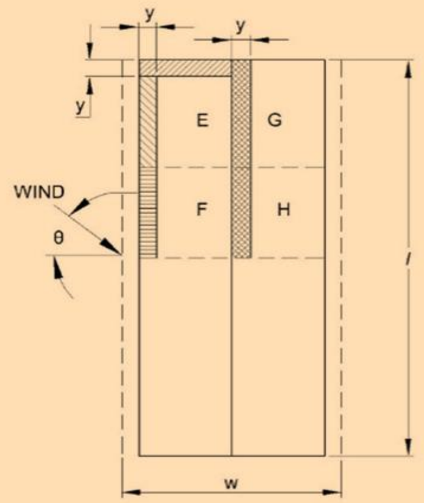
BUILDING HEIGHT RATIO	BUILDING PLAN RATIO	ELEVATION	PLAN	WIND ANGLE θ (Degrees)	C_{pe} FOR SURFACE A	C_{pe} FOR SURFACE B	C_{pe} FOR SURFACE C	C_{pe} FOR SURFACE D	LOCAL C_{pe}
$\frac{h}{w} \le \frac{1}{2}$	$1 < \frac{l}{w} \le \frac{3}{2}$			0 90	+0.7 −0.5	−0.2 −0.5	−0.5 +0.7	−0.5 −0.2	−0.8
	$\frac{3}{2} < \frac{l}{w} < 4$			0 90	+0.7 −0.5	−0.25 −0.5	−0.6 +0.7	−0.6 −0.1	−1.0
$\frac{1}{2} < \frac{h}{w} \le \frac{3}{2}$	$1 \le \frac{l}{w} \le \frac{3}{2}$			0 90	+0.7 −0.6	−0.25 −0.6	−0.6 +0.7	−0.6 −0.25	−1.1
	$\frac{3}{2} \le \frac{l}{w} < 4$			0 90	+0.7 −0.5	−0.3 −0.5	−0.7 +0.7	−0.7 −0.1	−1.1
$\frac{3}{2} < \frac{h}{w} < 6$	$1 < \frac{l}{w} \le \frac{3}{2}$			0 90	+0.8 −0.8	−0.25 −0.8	−0.8 +0.8	−0.8 −0.25	−1.2
	$\frac{3}{2} \le \frac{l}{w} < 4$			0 90	+0.7 −0.5	−0.4 −0.5	−0.7 +0.8	−0.7 −0.1	−1.2
$\frac{h}{w} \ge 6$	$\frac{l}{w} = \frac{3}{2}$			0 90	+0.95 −0.8	−1.85 −0.8	−0.9 +0.9	−0.9 −0.85	1.25
	$\frac{l}{w} = 1.0$			0 90	+0.95 −0.7	−1.25 −0.7	−0.7 +0.95	−0.7 −1.25	1.25
	$\frac{l}{w} = 2$			0 90	+0.85 −0.75	−0.75 −0.75	−0.75 +0.85	−0.75 −0.75	1.25

NOTE : *h* is the height to eaves or parapet, *l* is the greater horizontal dimension of a building and *w* is the lesser horizontal dimension of a building.

Table 9 External Pressure Coefficients (C_{pe}) for Pitched Roofs of Rectangular Clad Buildings

(*Clause* 4.5.3.3.2)

BUILDING HEIGHT RATIO	ROOF ANGLE α (Degrees)	WIND ANGLE θ 0° EF	WIND ANGLE θ 0° GH	WIND ANGLE θ 90° EG	WIND ANGLE θ 90° FH	LOCAL COEFFICIENTS			
$\frac{h}{w} \le \frac{1}{2}$ 	0	- 0.8	- 0.4	- 0.8	- 0.4	- 2.0	- 2.0	- 2.0	-----
	5	- 0.9	- 0.4	- 0.8	- 0.4	- 1.4	- 1.2	- 1.2	- 1.0
	10	- 1.2	- 0.4	- 0.8	- 0.6	- 1.4	- 1.4		- 1.2
	20	- 0.4	- 0.4	- 0.7	- 0.6	- 1.0			- 1.2
	30	0	- 0.4	- 0.7	- 0.6	- 0.8			- 1.1
	45	+ 0.3	- 0.5	- 0.7	- 0.6				- 1.1
	60	+ 0.7	- 0.6	- 0.7	- 0.6				- 1.1
$\frac{1}{2} < \frac{h}{w} \le \frac{3}{2}$ 	0	- 0.8	- 0.6	- 1.0	- 0.6	- 2.0	- 2.0	- 2.0	-----
	5	- 0.9	- 0.6	- 0.9	- 0.6	- 2.0	- 2.0	- 1.5	- 1.0
	10	- 1.1	- 0.6	- 0.8	- 0.6	- 2.0	- 2.0	- 1.5	- 1.2
	20	- 0.7	- 0.5	- 0.8	- 0.6	- 1.5	- 1.5	- 1.5	- 1.0
	30	- 0.2	- 0.5	- 0.8	- 0.8	- 1.0			- 1.0
	45	+ 0.2	- 0.5	- 0.8	- 0.8				
	60	+ 0.6	- 0.5	- 0.8	- 0.8				
$\frac{3}{2} < \frac{h}{w} < 6$ 	0	- 0.7	- 0.6	- 0.9	- 0.7	- 2.0	- 2.0	- 2.0	-----
	5	- 0.7	- 0.6	- 0.8	- 0.8	- 2.0	- 2.0	- 1.5	- 1.0
	10	- 0.7	- 0.6	- 0.8	- 0.8	- 2.0	- 2.0	- 1.5	- 1.2
	20	- 0.8	- 0.6	- 0.8	- 0.8	- 1.5	- 1.5	- 1.5	- 1.2
	30	- 1.0	- 0.5	- 0.8	- 0.7	- 1.5			
	40	- 0.2	- 0.5	- 0.8	- 0.7	- 1.0			
	45	+ 0.2	- 0.5	- 0.8	- 0.7				
	60	+ 0.5	- 0.5	- 0.8	- 0.7				



KEY PLAN

$y = h$ or $0.15\,w$

Whichever is the lesser.

NOTE - 1 *h* is the height to eaves or parapet and *w* is the lesser horizontal dimension of a building.

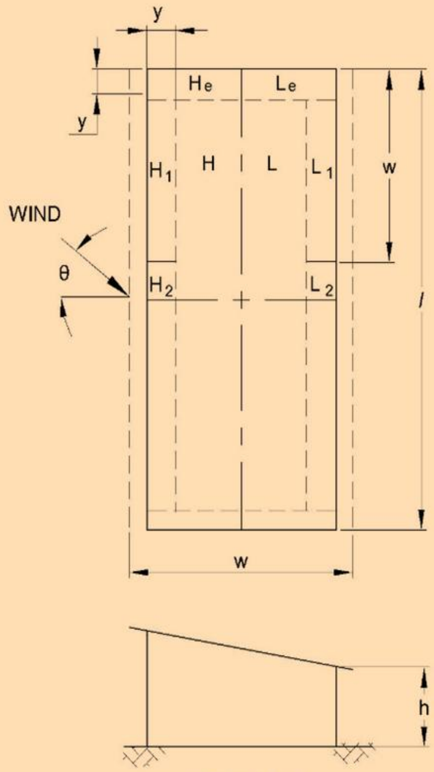
NOTE - 2 Where no local coefficients are given, the overall coefficient apply.

NOTE - 3 For hipped roofs the local coefficient for the hip ridge may be conservatively taken as the the appropriate ridge value.

NOTE - 4 *w* and *l* are dimensions between the walls excluding overhangs.

Table 10 External Pressure Coefficients (C_{pe}) for Monoslope Roofs for Rectangular Clad Buildings $\frac{h}{w} < 2$

(*Clause* 4.5.3.3.2)



y = *h* or 0.15 *w*
Whichever is the lesser.

NOTE :- Area *H* and area *L* refer to the whole quadrant.

ROOF ANGLE α	WIND ANGLE θ										LOCAL COEFFICIENTS (C_{pe})					
	0°		45°		90°		135°		180°							
	H	*L*	*H*	*L*	*H* & *L* *	*H* & *L* **	*H*	*L*	*H*	*L*	H_1	H_2	L_1	L_2	H_e	L_e
Degrees																
5	- 1.0	- 0.5	- 1.0	- 0.9	- 1.0	- 0.5	- 0.9	- 1.0	- 0.5	- 1.0	- 2.0	- 1.5	- 2.0	- 1.5	- 2.0	- 2.0
10	- 1.0	- 0.5	- 1.0	- 0.8	- 1.0	- 0.5	- 0.8	- 1.0	- 0.4	- 1.0	- 2.0	- 1.5	- 2.0	- 1.5	- 2.0	- 2.0
15	- 0.9	- 0.5	- 1.0	- 0.7	- 1.0	- 0.5	- 0.6	- 1.0	- 0.3	- 1.0	- 1.8	- 0.9	- 1.8	- 1.4	- 2.0	- 2.0
20	- 0.8	- 0.5	- 1.0	- 0.6	- 0.9	- 0.5	- 0.5	- 1.0	- 0.2	- 1.0	- 1.8	- 0.8	- 1.8	- 1.4	- 2.0	- 2.0
25	- 0.7	- 0.5	- 1.0	- 0.6	- 0.8	- 0.5	- 0.3	- 0.9	- 0.1	- 0.9	- 1.8	- 0.7	- 0.9	- 0.9	- 2.0	- 2.0
30	- 0.5	- 0.5	- 1.0	- 0.6	- 0.8	- 0.5	- 0.1	- 0.6	0	- 0.6	- 1.8	- 0.5	- 0.5	- 0.5	- 2.0	- 2.0

* Applied to length *w* / 2 from wind-ward end.

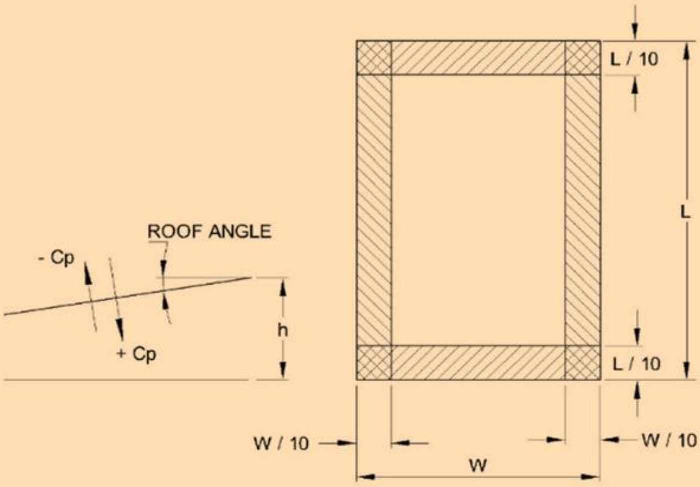
** Applies to remainder

NOTE :- 1 *h* is the height of eaves at lower side, *l* is the greater horizontal dimensions of a building and *w* is the lesser horizontal dimension of a building.

NOTE :- 2 l and w are overall length and width including overhangs,

Table 11 Pressure Coefficients for Monoslope Free Roofs

(*Clause* 4.5.3.3.3)



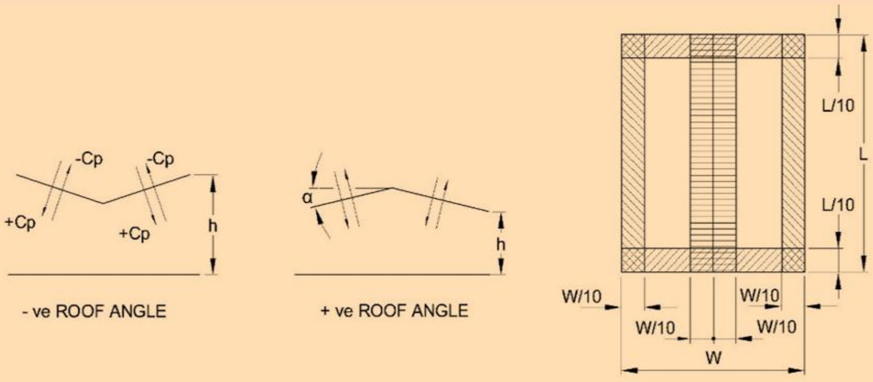
ROOF ANGLE (Degrees) α	SOLIDITY RATIO Φ	MAXIMUM (LARGEST + ve) AND MINIMUM (LARGEST - ve) PRESSURE COEFFICIENTS			
		OVERALL COEFFICIENTS	LOCAL COEFFICIENTS		
0	All values of Φ	+ 0.2	+ 0.5	+ 1.8	+ 1.1
5		+ 0.4	+ 0.8	+ 2.1	+ 1.3
10		+ 0.5	+ 1.2	+ 2.4	+ 1.6
15		+ 0.7	+ 1.4	+ 2.7	+ 1.8
20		+ 0.8	+ 1.7	+ 2.9	+ 2.1
25		+ 1.0	+ 2.0	+ 3.1	+ 2.3
30		+ 1.2	+ 2.2	+ 3.2	+ 2.4
0	Φ = 0	- 0.5	- 0.6	- 1.3	- 1.4
	Φ = 1	- 1.0	- 1.2	- 1.8	- 1.9
5	Φ = 0	- 0.7	- 1.1	- 1.7	- 1.8
	Φ = 1	- 1.1	- 1.6	- 2.2	- 2.3
10	Φ = 0	- 0.9	- 1.5	- 2.0	- 2.1
	Φ = 1	- 1.3	- 2.1	- 2.6	- 2.7
15	Φ = 0	- 1.1	- 1.8	- 2.4	- 2.5
	Φ = 1	- 1.4	- 2.3	- 2.9	- 3.0
20	Φ = 0	- 1.3	- 2.2	- 2.8	- 2.9
	Φ = 1	- 1.5	- 2.6	- 3.1	- 3.2
25	Φ = 0	- 1.6	- 2.6	- 3.2	- 3.2
	Φ = 1	- 1.7	- 2.8	- 3.5	- 3.5
30	Φ = 0	- 1.8	- 3.0	- 3.8	- 3.6
	Φ = 1	- 1.8	- 3.0	- 3.8	- 3.6

NOTE 1: For monopitch canopies the centre of pressure should be taken to act at 0.3 w from the windward edge.

NOTE 2: W and L are overall width and length including overhangs,

Table 12 Pressure Coefficients for Free Standing Double Sloped Roofs

(*Clause* 4.5.3.3.3)



KEY PLAN

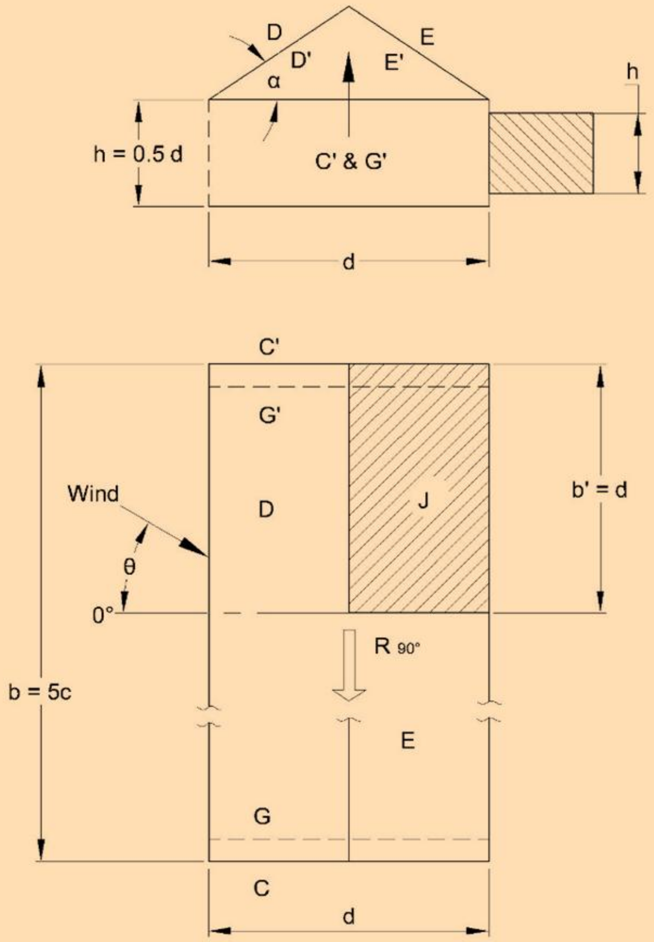
ROOF ANGLE (Degrees) α	SOLIDITY RATIO Φ	MAXIMUM (LARGEST + ve) AND MINIMUM (LARGEST - ve) PRESSURE COEFFICIENTS				
		OVERALL COEFFICIENTS	LOCAL COEFFICIENTS			
			▭	▨	▧	▤
- 20	All values of Φ	+ 0.7	+ 0.8	+ 1.6	+ 0.6	+ 1.7
- 15		+ 0.5	+ 0.6	+ 1.5	+ 0.7	+ 1.4
- 10		+ 0.4	+ 0.6	+ 1.4	+ 0.8	+ 1.1
- 5		+ 0.3	+ 0.5	+ 1.5	+ 0.8	+ 0.8
+ 5		+ 0.3	+ 0.6	+ 1.8	+ 1.3	+ 0.4
+ 10		+ 0.4	+ 0.7	+ 1.8	+ 1.4	+ 0.4
+ 15		+ 0.4	+ 0.9	+ 1.9	+ 1.4	+ 0.4
+ 20		+ 0.6	+ 1.1	+ 1.9	+ 1.5	+ 0.4
+ 25		+ 0.7	+ 1.2	+ 1.9	+ 1.6	+ 0.5
+ 30		+ 0.9	+ 1.3	+ 1.9	+ 1.6	+ 0.7
- 20	Φ = 0	- 0.7	- 0.9	- 1.3	- 1.6	- 0.6
	Φ = 1	- 0.9	- 1.2	- 1.7	- 1.9	- 1.2
- 15	Φ = 0	- 0.6	- 0.8	- 1.3	- 1.6	- 0.6
	Φ = 1	- 0.8	- 1.1	- 1.7	- 1.9	- 1.2
- 10	Φ = 0	- 0.6	- 0.8	- 1.3	- 1.5	- 0.6
	Φ = 1	- 0.8	- 1.1	- 1.7	- 1.9	- 1.3
- 5	Φ = 0	- 0.5	- 0.7	- 1.3	- 1.6	- 0.6
	Φ = 1	- 0.8	- 1.5	- 1.7	- 1.9	- 1.4
+ 5	Φ = 0	- 0.6	- 0.6	- 1.4	- 1.4	- 1.1
	Φ = 1	- 0.9	- 1.3	- 1.8	- 1.8	- 2.1
+ 10	Φ = 0	- 0.7	- 0.7	- 1.5	- 1.4	- 1.4
	Φ = 1	- 1.1	- 1.4	- 2.0	- 1.8	- 2.4
+ 15	Φ = 0	- 0.8	- 0.9	- 1.7	- 1.4	- 1.8
	Φ = 1	- 1.2	- 1.5	- 2.2	- 1.9	- 2.8
+ 20	Φ = 0	- 0.9	- 1.2	- 1.8	- 1.4	- 2.0
	Φ = 1	- 1.3	- 1.7	- 2.3	- 1.9	- 3.0
+ 25	Φ = 0	- 1.0	- 1.4	- 1.9	- 1.4	- 2.0
	Φ = 1	- 1.4	- 1.9	- 2.4	- 2.1	- 3.0
+ 30	Φ = 0	- 1.0	- 1.4	- 1.9	- 1.4	- 2.0
	Φ = 1	- 1.4	- 2.1	- 2.6	- 2.2	- 3.0

Each slope of a duopitch canopy should be able to withstand forces using both the maximum and the minimum coefficients, and the whole canopy should be able to support forces using one slope at the maximum coefficient with the other slope at the minimum coefficient. For duoptich canopies the centre of pressure should be taken to act at the centre of each slope.

Note :- W and L are overall width and length including overhangs

Table 13 Pressure Coefficients (Top and Bottom) for Pitched Roofs, Roof Slope,

(*Clause* 4.5.3.3.3)



Roof slope $\alpha = 30°$

$\theta = 0° - 45°$, *D, D′, E, E′* full length.

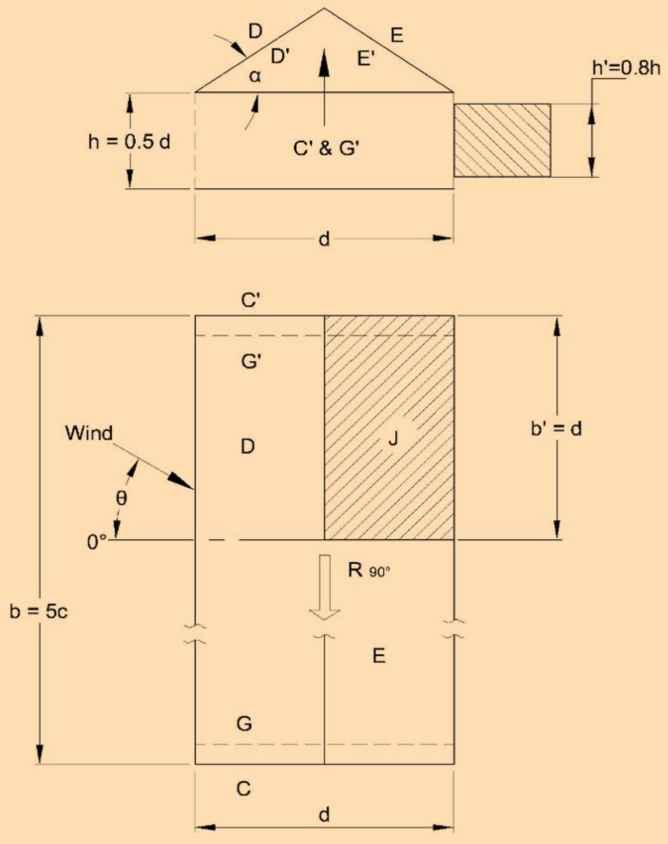
$\theta = 90°$, *D, D′, E, E′* part length.

b′, thereafter $C_p = 0$

θ	PRESSURE COEFFICIENTS, C_p							
	D	*D′*	*E*	*E′*	END SURFACES			
					C	*C′*	*G*	*G′*
0°	+ 0.6	- 1.0	- 0.5	- 0.9				
45°	+ 0.1	- 0.3	- 0.6	- 0.3				
90°	- 0.3	- 0.4	- 0.3	- 0.4	- 0.3	- 0.8	- 0.3	- 0.4
For all value of θ	For J : C_p Top = 1.0, C_p bottom = - 0.2 Tangentially acting friction : $R_{90°} = 0.05\, p_d\, bd$							

Table 14 Pressure Coefficients (Top and Bottom) for Pitched Free Roofs, $\alpha = 30°$ With Effects of Train or Stored Materials

(*Clause* 4.5.3.3.3)

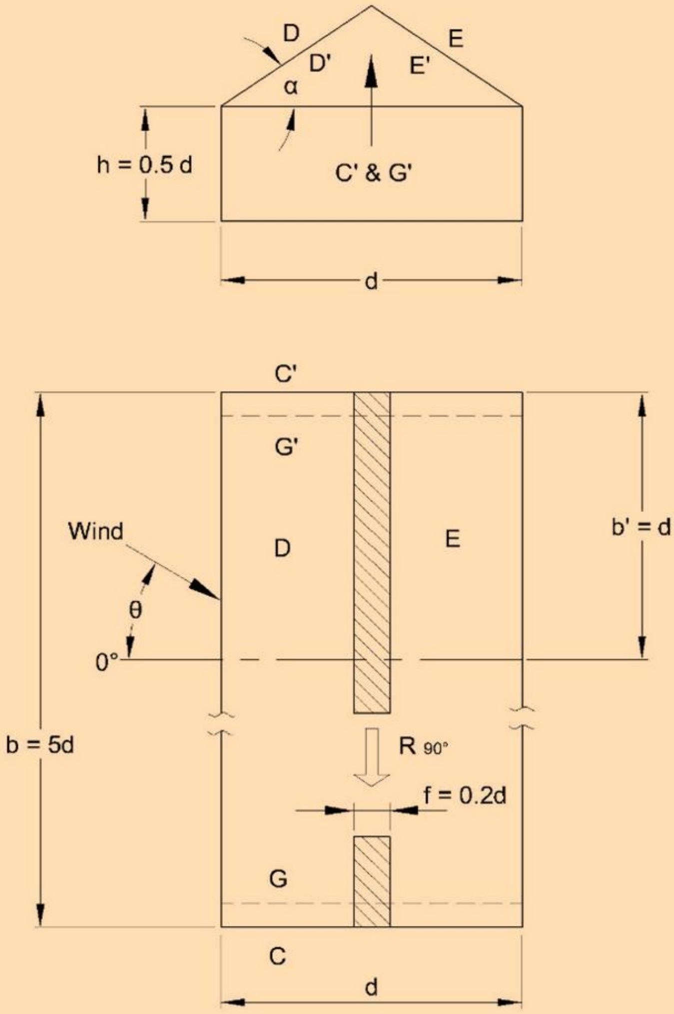


Roof slope $\alpha = 30°$
Effects of trains or stored materials.
$\theta = 0° - 45°$, or $135° - 180°$, *D, D', E, E'* full length.
$\theta = 90°$, *D, D', E, E'* part length.
b', thereafter $C_p = 0$

θ	PRESSURE COEFFICIENTS, C_p							
					END SURFACES			
	D	*D'*	*E*	*E'*	*C*	*C'*	*G*	*G'*
0°	+ 0.1	+ 0.8	- 0.7	+ 0.9				
45°	- 0.1	+ 0.5	- 0.8	+ 0.5				
90°	- 0.4	- 0.5	- 0.4	- 0.5	- 0.3	+ 0.8	+ 0.3	- 0.4
180°	- 0.3	- 0.6	+ 0.4	- 0.6				
For all value of θ	For J : C_p Top = -1.5, C_p bottom = 0.5 Tangentially acting friction : $R_{90°} = 0.05\, p_d\, bd$							

Table 15 Pressure Coefficients (Top and Bottom) for Pitched Free Roofs, $\alpha = 10^\circ$

(*Clause* 4.5.3.3.3)



Roof slope $\alpha = 10^\circ$

$\theta = 0^\circ - 45^\circ$, *D, D′, E, E′* full length.

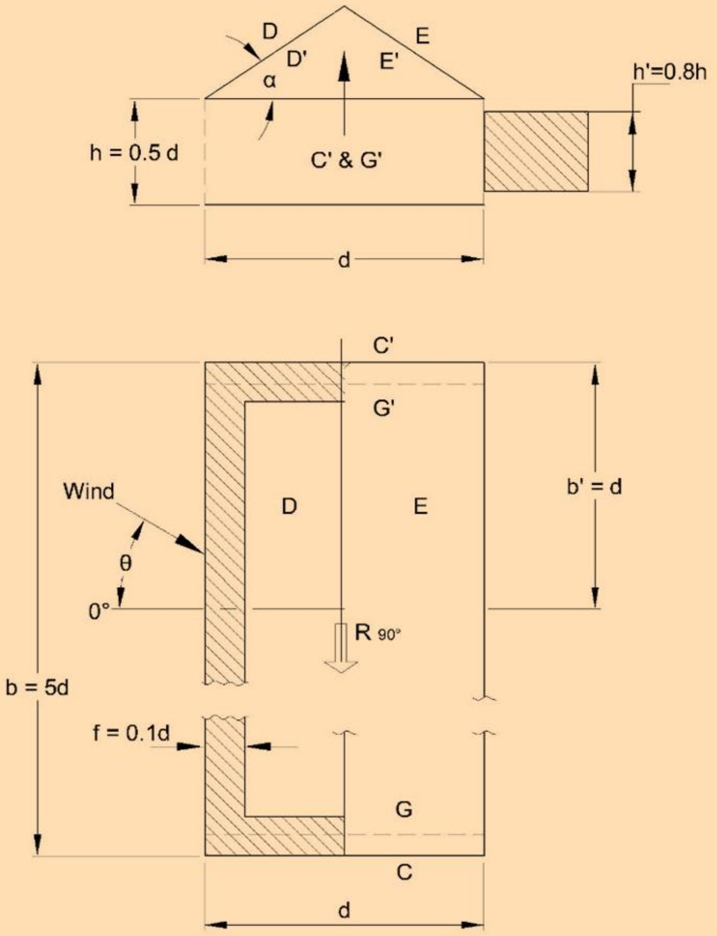
$\theta = 90^\circ$, *D, D′, E, E′* part length.

b′, thereafter $C_p = 0$

θ	PRESSURE COEFFICIENTS, C_p							
					END SURFACES			
	D	*D′*	*E*	*E′*	*C*	*C′*	*G*	*G′*
0°	- 1.0	+ 0.3	- 0.5	+ 0.2				
45°	- 0.3	+ 0.1	- 0.3	+ 0.1				
90°	- 0.3	0	- 0.3	0	- 0.4	+ 0.8	+ 0.3	- 0.6
For all value of θ	For *f* : C_p Top = - 1.0, C_p bottom = 0.4 Tangentially acting friction : $R_{90^\circ} = 0.1\, p_d\, bd$							

Table 16 Pressure Coefficients (Top and Bottom) for Pitched Free Roofs, $\alpha = 10°$ With Effects of Train or Stored Materials

(*Clause* 4.5.3.3.3)



Roof slope $\alpha = 10°$

Effects of trains or stored materials.

$\theta = 0° - 45°$, or $135° - 180°$, *D, D′, E, E′* full length.

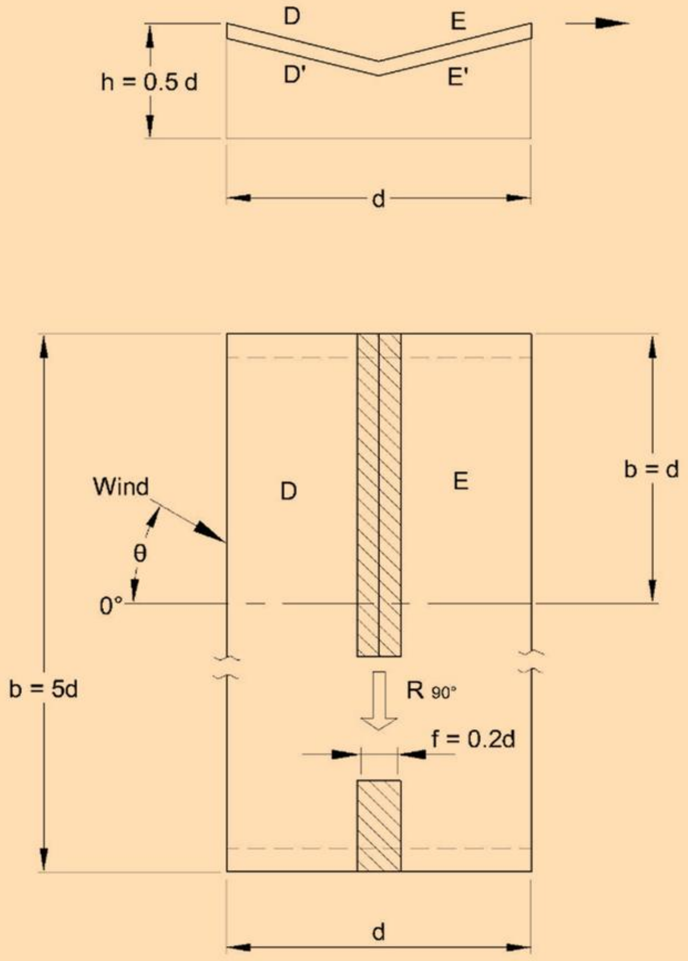
$\theta = 90°$, *D, D′, E, E′* part length.

b′, thereafter $C_p = 0$

θ	PRESSURE COEFFICIENTS, C_p							
					END SURFACES			
	D	*D′*	*E*	*E′*	*C*	*C′*	*G*	*G′*
0°	- 1.3	+ 0.8	- 0.6	0.7				
45°	- 0.5	+ 0.4	- 0.3	+ 0.3				
90°	- 0.3	0	- 0.3	0	- 0.4	+ 0.8	+ 0.3	- 0.6
180°	- 0.4	- 0.3	- 0.6	- 0.3				
For all value of θ	For *f* : C_p Top = -1.6, C_p bottom = - 0.9 Tangentially acting friction : $R_{90°} = 0.1\, p_d\, bd$							

Table 17 Pressure Coefficients for Troughed Free Roofs, $\alpha = 10^{\circ}$

(*Clause* 4.5.3.3.3)



Roof slope $\alpha = 10^{\circ}$

$\theta = 0^{\circ}$ - 45°, *D, D', E, E'* full length.

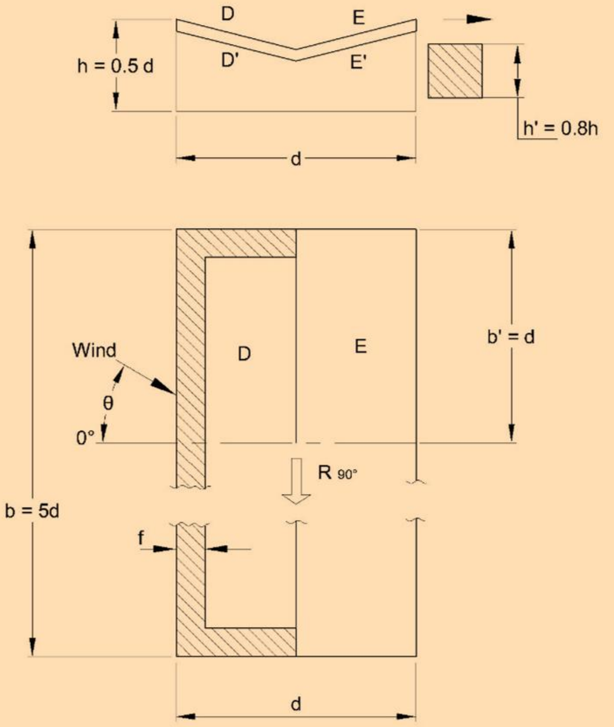
$\theta = 90^{\circ}$, *D, D', E, E'* part length.

b', thereafter $C_p = 0$

θ	PRESSURE COEFFICIENTS, C_p			
	D	*D'*	*E*	*E'*
0°	+ 0.3	- 0.7	+ 0.2	- 0.9
45°	0	- 0.2	+ 0.1	- 0.3
90°	- 0.1	0.1	- 0.1	+ 0.1
For all value of θ	For *f* : C_p Top = 0.4, C_p bottom = - 1.5 Tangentially acting friction : $R_{90^{\circ}} = 0.1\, p_d\, bd$			

Table 18 Pressure Coefficients (Top and Bottom) for Troughed Free Roofs, $\alpha = 10^\circ$ With Effects of Train or Stored Material

(*Clause* 4.5.3.3.3)

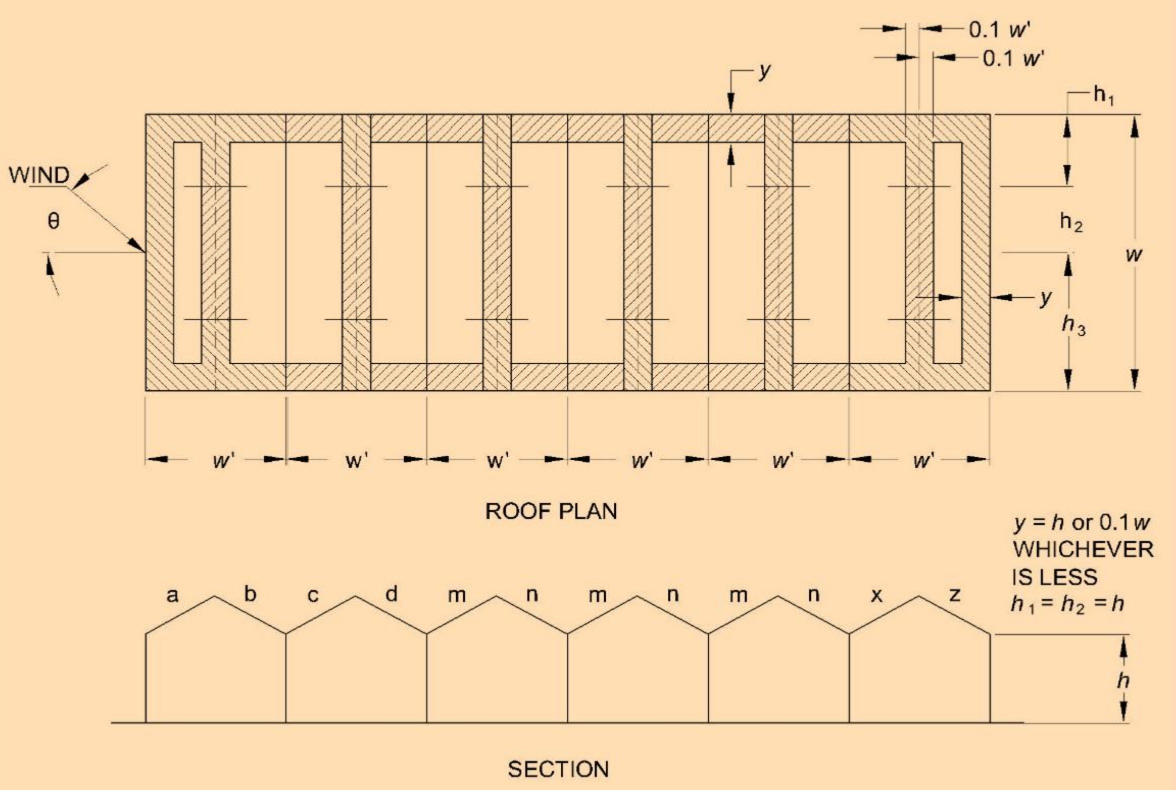


Roof slope α = 10°
Effects of trains or stored materials.
θ = 0°- 45°, or 135° - 180°, *D, D', E, E'* full length.
θ = 90°, *D, D', E, E'* part length.
b', thereafter C_p = 0

θ	PRESSURE COEFFICIENTS, C_p			
	D	*D'*	*E*	*E'*
0°	- 0.7	+ 0.8	- 0.6	+ 0.6
45°	- 0.4	+ 0.3	- 0.2	+ 0.2
90°	- 0.1	+ 0.1	- 0.1	+ 0.1
180°	- 0.4	- 0.2	- 0.6	- 0.3
For all value of θ	For *f* : C_p Top = -1.1, C_p bottom = 0.9 Tangentially acting friction : $R_{90^\circ} = 0.1\, p_d\, bd$			

Table 19 External Pressure Coefficients (C_{pe}) for Pitched Roofs of Multi-span Buildings (All Spans Equal) with $h < w'$

(*Clause* 4.5.3.3.4)



ROOF ANGLE α Degrees	WIND ANGLE θ Degrees	FIRST SPAN a	FIRST SPAN b	FIRST INTERME-DIATE SPAN c	FIRST INTERME-DIATE SPAN d	OTHER INTERME-DIATE SPANS m	OTHER INTERME-DIATE SPANS n	END SPAN x	END SPAN z	LOCAL COEFFICIENT (hatched ＼)	LOCAL COEFFICIENT (hatched ／)
5	0	- 0.9	- 0.6	- 0.4	- 0.3	- 0.3	- 0.3	- 0.3	- 0.3		
10		- 1.1	- 0.6	- 0.4	- 0.3	- 0.3	- 0.3	- 0.3	- 0.4		
20		- 0.7	- 0.6	- 0.4	- 0.3	- 0.3	- 0.3	- 0.3	- 0.5	- 2.0	- 1.5
30		- 0.2	- 0.6	- 0.4	- 0.3	- 0.2	- 0.3	- 0.2	- 0.5		
45		+ 0.3	- 0.6	- 0.6	- 0.4	- 0.2	- 0.4	- 0.2	- 0.5		

DISTANCE				
ROOF ANGLE α DEGREES	WIND ANGLE θ DEGREES	h_1	h_2	h_3
UP TO 45	90	- 0.8	- 0.6	- 0.2

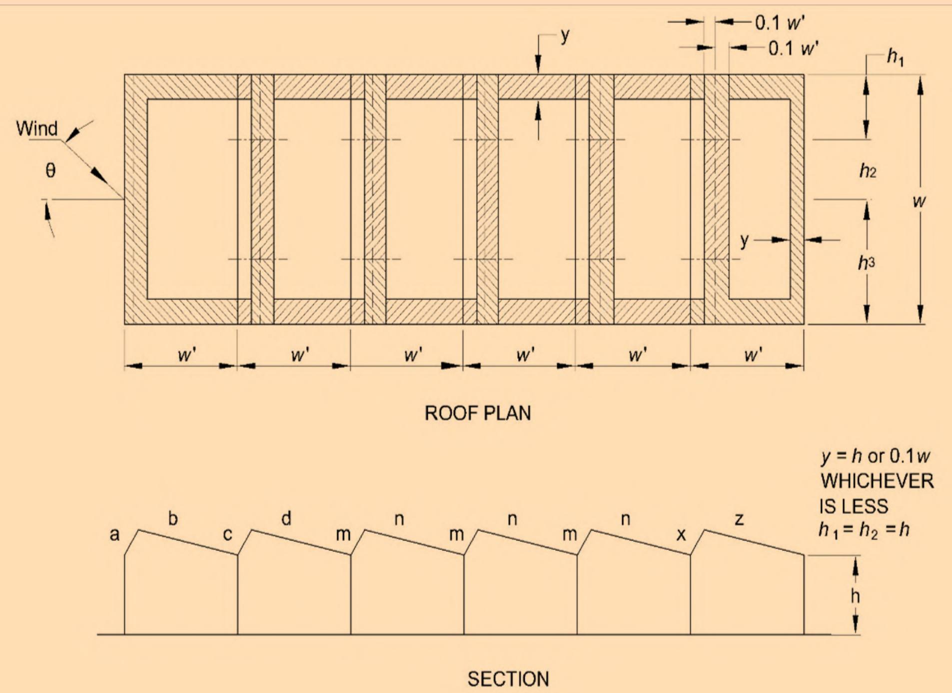
Frictional drag : When wind angle θ = 0°, horizontal forces due to frictional drag are allowed for in the above values, and

When wind angle θ = 90°, allow for frictional drag in accordance with **4.5.4.1**

NOTE — Evidence on these buildings is fragmentary and any departure from the cases given should be investigated separately.

Table 20 External Pressure Coefficients (Cpe) for Saw Tooth Roofs of Multi-Span Buildings (All Spans Equal) with h < w'

(*Clause* 4.5.3.3.4)



WIND ANGLE	FIRST SPAN		FIRST INTERME-DIATE SPAN		OTHER INTERME-DIATE SPANS		END SPAN		LOCAL COEFFICIENT	
θ Degrees	a	b	c	d	m	n	x	z	(hatched area, cross-hatching)	(hatched area, diagonal hatching)
0	+ 0.6	- 0.7	- 0.7	- 0.4	- 0.3	- 0.2	- 0.1	- 0.3	- 2.0	- 1.5
180	- 0.5	- 0.3	- 0.3	- 0.3	- 0.4	- 0.6	- 0.6	- 0.1		

DISTANCE			
WIND ANGLE θ DEGREES	h_1	h_2	h_3
90	- 0.8	- 0.6	- 0.2
270	Similar to 90°, h_1, h_2, h_3, are needed to be reckoned from the windword edge in the same order		

Frictional drag : When wing angle θ = 0°, horizontal forces due to frictional drag are allowed for in the above values, and

When wind angle θ = 90˚, allow for frictional drag in accordance with **4.5.4.1**

NOTE — Evidence on these buildings is fragmentary and any departure from the cases given should be investigated separately.

4.5.3.3.5 *Pressure coefficients on overhangs from roofs*

The pressure coefficients on the top overhanging portion of the roofs shall be taken to be the same as that of the nearest top portion of the non-overhanging portion of the roofs. The pressure coefficients for the underside surface of the overhanging portions shall be taken as follows and shall be taken as positive, if the overhanging portion is on the windward side:

a) 1.25, if the overhanging slopes downwards;

b) 1.00, if the overhanging is horizontal; and

c) 0.75, if the overhanging slopes upwards.

For overhanging portions on sides other than windward side, the average pressure coefficients on adjoining walls may be used.

4.5.3.3.6 *Curved roofs*

For curved roofs the external pressure coefficients shall be as given in Table 21. Allowance for local effects shall be made in accordance with Table 9. Two values of C_2 have been given for elevated curved roofs. Both the load cases have to be analysed and critical load effects are to be considered in design.

4.5.3.3.7 *Cylindrical structures*

For the purpose of calculating the wind pressure distribution around a cylindrical structure of circular cross section, the value of external pressure coefficients given in Table 22 may be used, provided that the Reynolds number is more than 10 000. They may be used for wind blowing normal to the axis of cylinders having axis normal to the ground plane and cylinders having their axis parallel to the ground plane. 'h' is height of a vertical cylinder or length of a horizontal cylinder. Where there is a free flow of air around both ends, h is to be taken as half the length when calculating h/D ratio.

In the calculation of resultant load on the periphery of the cylinder, the value of C_{pi} shall be taken into account. For open ended cylinders, C_{pi} shall be taken as follows:

a) - 0.8, where h/D is more than or equal to 0.3; and

b) - 0.5, where h/D is less than 0.3.

4.5.3.3.8 *Roofs and bottoms of cylindrical elevated structures*

The external pressure coefficients for roofs and bottoms of cylindrical elevated structures shall be as given in Table 23. Alternately, the pressure distribution given in Fig. 3 can be used together with the force coefficients given in Table 28 for the cylindrical portion.

4.5.3.3.9 *Combined roofs*

The average external pressure coefficients for combined roofs are shown in Table 24.

4.5.3.3.10 *Roofs with skylight*

The average external pressure coefficients for roofs with skylight are shown in Table 25.

4.5.3.3.11 *Grandstands*

The pressure coefficients on the roof (top and bottom) and rear wall of a typical grandstand roof which is open on three sides are given in Table 26. The pressure coefficients are valid for a particular ratio of dimensions as specified in Table 24 but may be used for deviations up to 20 percent. In general, the maximum wind load occurs when the wind is blowing into the open front of the stand, causing positive pressure under the roof and negative pressure on the roof.

4.5.3.3.12 *Spheres*

The external pressure coefficients for spheres shall be as given in Table 27.

4.5.3.3.13 *Frames*

When taking wind loads on frames of clad buildings it is reasonable to assume that the pressures or suctions inside and outside the structure shall not be fully correlated. Therefore, when taking the combined effect of wind loads on the frame, a reduction factor of $K_c = 0.90$ may be used over the building envelope when roof is subjected to pressure and internal pressure is suction, or *vice-versa*.

4.5.4 *Force Coefficients*

The value of force coefficients (C_f) apply to a building or structure as a whole and when multiplied by the effective frontal area (A_e) of the building or structure and design wind pressure, p_d gives the total wind load (F) on that particular building or structure, expressed as:

$$F = C_f A_e p_d$$

where

F = force acting in a direction specified in the respective tables; and

C_f = force coefficient for the building.

NOTES

1 The value of the force coefficient differs for the wind acting on different faces of a building or structure. In order to determine the critical load, the total wind load should be calculated for each wind direction.

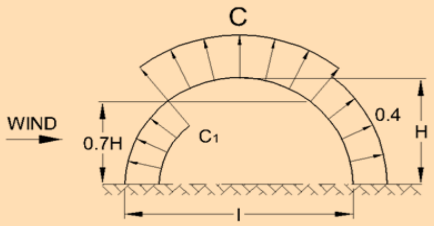
2 If surface design pressure varies with height, the surface area of the building/structure may be sub-divided so that specified pressures are taken over appropriate areas.

3 In tapered buildings/structures, the force coefficients shall be applied after sub-dividing the building/structure into suitable number of strips and the load on each strip calculated individually, taking the area of each strip as A_e.

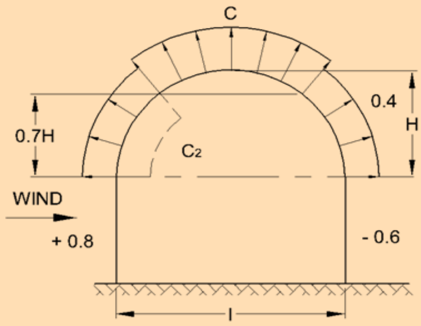
4 For force coefficients for structures not covered herein, reference may be made to specialist literature on the subject or advice may be sought from specialist in the subject.

Table 21 External Pressure Coefficients (Cpe) for Curved Roofs

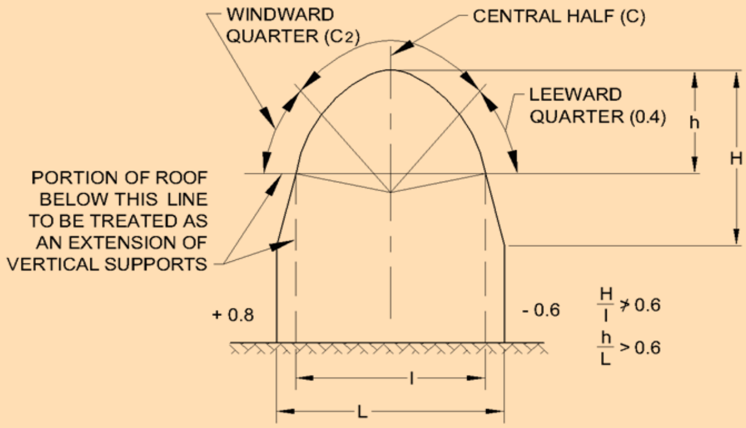
(*Clause* 4.5.3.3.6)



a) ROOF SPRINGING FROM GROUND LEVEL



b) ROOF ON ELEVATED STRUCTURE



c) DOUBLE CURVED ROOFS

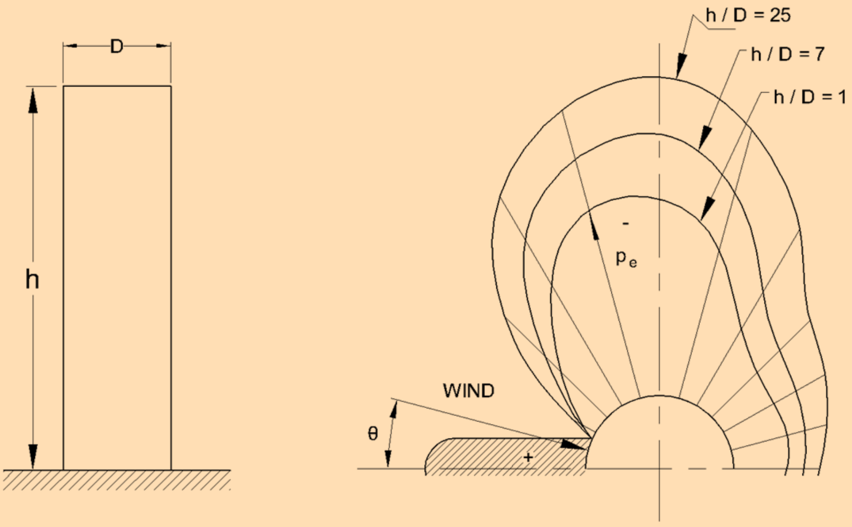
VALUES OF C, C_1 and C_2

H/l	C	C_1	C_2	C_2
0.1	- 0.8	+ 0.1	- 0.8	+ 0.05
0.2	- 0.9	+ 0.3	- 0.7	+ 0.1
0.3	- 1.0	+ 0.4	- 0.3	+ 0.15
0.4	- 1.1	+ 0.6	+ 0.4	-
0.5	- 1.2	+ 0.7	+ 0.7	-

Note :- When the wind is blowing normal to gable ends, C_{pe} may be taken as equal to - 0.7 for the full width of the roof over a length of $l/2$ from the gable ends and - 0.5 for the remaining portion.

Table 22 External Pressure Coefficients (C_{pe}) Around Cylindrical Structures

(*Clause* 4.5.3.3.7)



POSITION OF PERIPHERY, θ IN DEGREES	PRESSURE COEFFICIENTS C_{pe}		
	h / *D* = 25	*h* / *D* = 7	*h* / *D* = 1
0	1.0	1.0	1.0
15	0.8	0.8	0.8
30	0.1	0.1	0.1
45	- 0.9	- 0.8	- 0.7
60	- 1.9	- 1.7	- 1.2
75	- 2.5	- 2.2	- 1.6
90	- 2.6	- 2.2	- 1.7
105	- 1.9	- 1.7	- 1.2
120	- 0.9	- 0.8	- 0.7
135	- 0.7	- 0.6	- 0.5
150	- 0.6	- 0.5	- 0.4
165	- 0.6	- 0.5	- 0.4
180	- 0.6	- 0.5	- 0.4

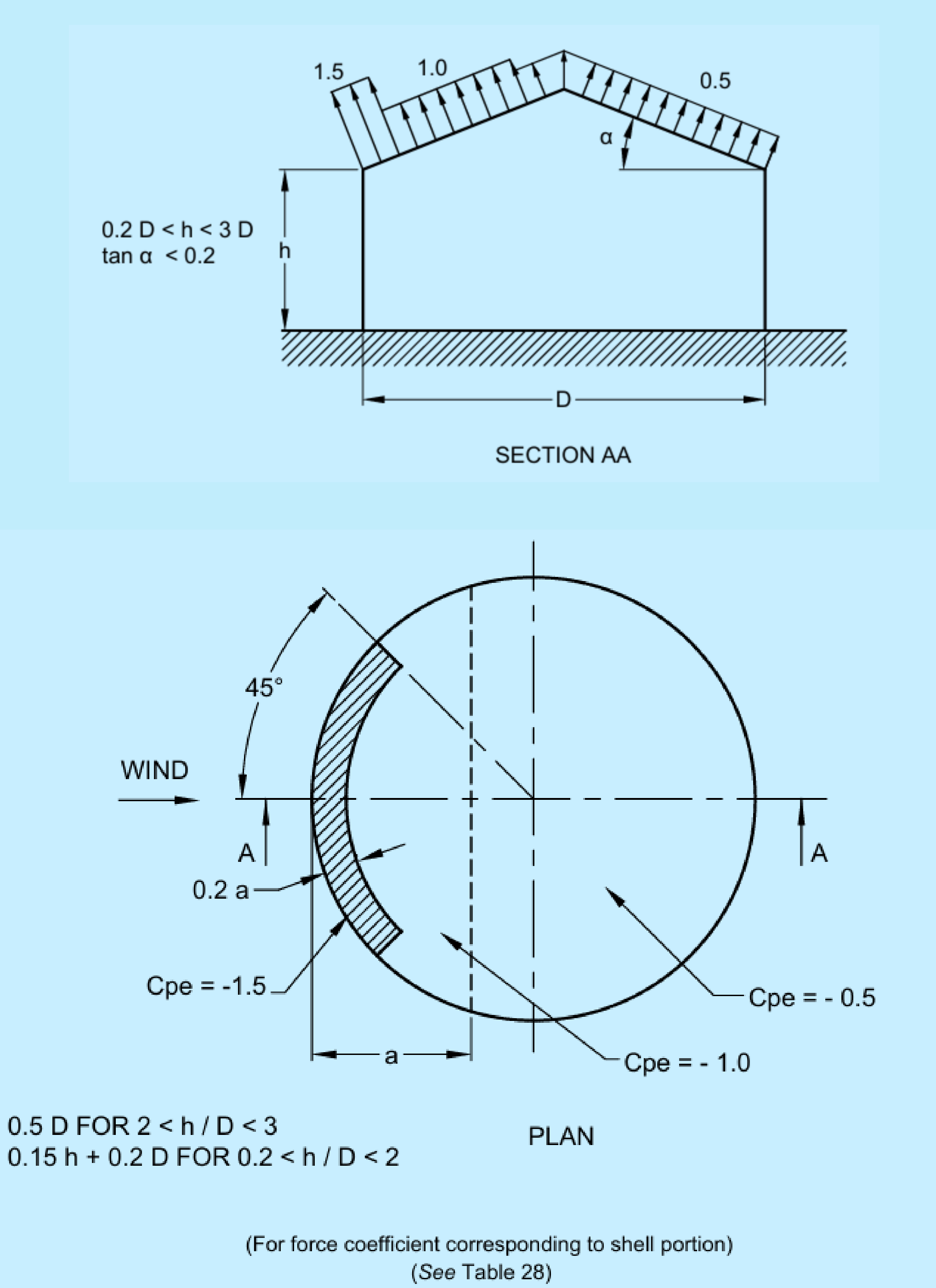
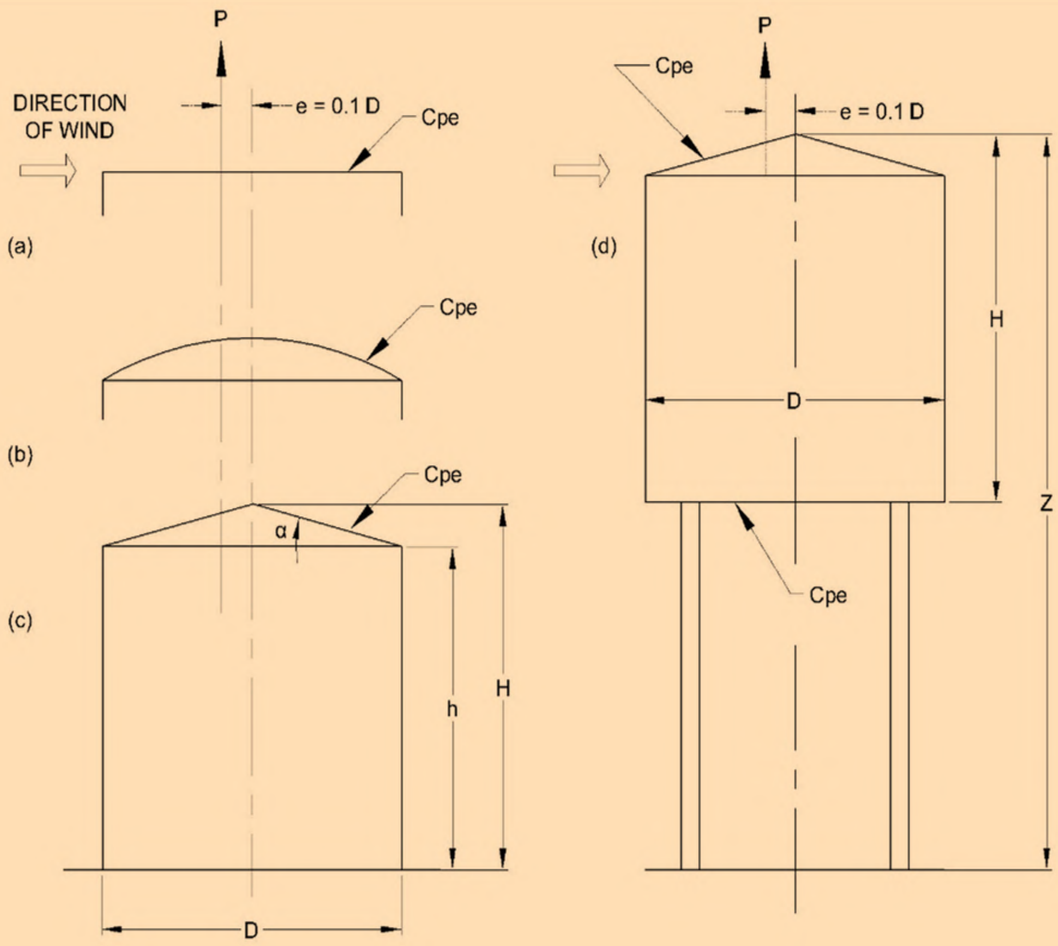


FIG. 3 EXTERNAL PRESSURE COEFFICIENTS ON THE UPPER ROOF SURFACE OF CYLINDRICAL STRUCTURES STANDING ON THE GROUND

Table 23 External Pressure Coefficients (C_{pe}) for Roofs and Bottoms of Cylindrical Structures

(*Clause* 4.5.3.3.8)



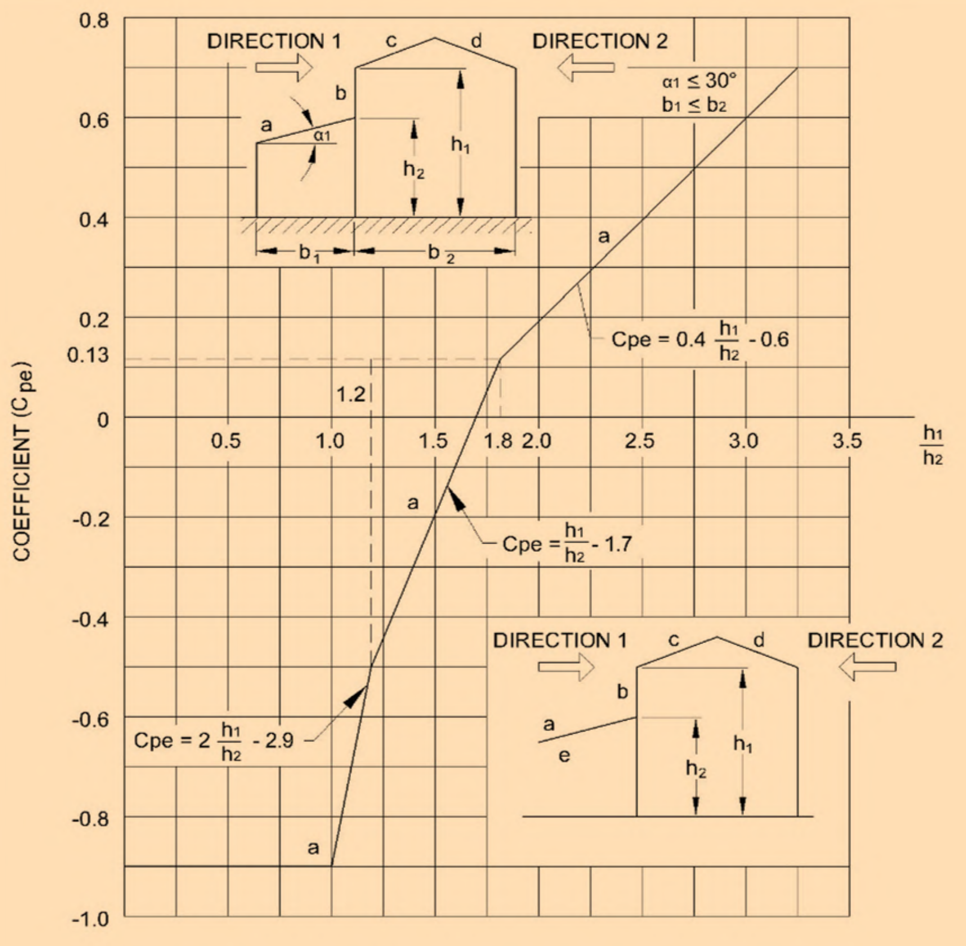
COEFFICIENTS OF EXTERNAL PRESSURE C_{pe}				
STRUCTURE ACCORDING TO SHAPE				
a, b and c		d		
H / D	ROOF	(z / H) - 1	ROOF	BOTTOM
0.5	- 0.65	1.00	- 0.75	- 0.8
1.00	- 1.00	1.25	- 0.75	- 0.7
2.00	- 1.00	1.50	- 0.75	- 0.6

Total force acting on the roof of the structure, $P = 0.785\, D^2 (C_{pi} - C_{pe})\, p_d$

The resultant of P lies eccentrically, e = 0.1D

Table 24 External Pressure Coefficients (C_{pe}) for Combined Roofs

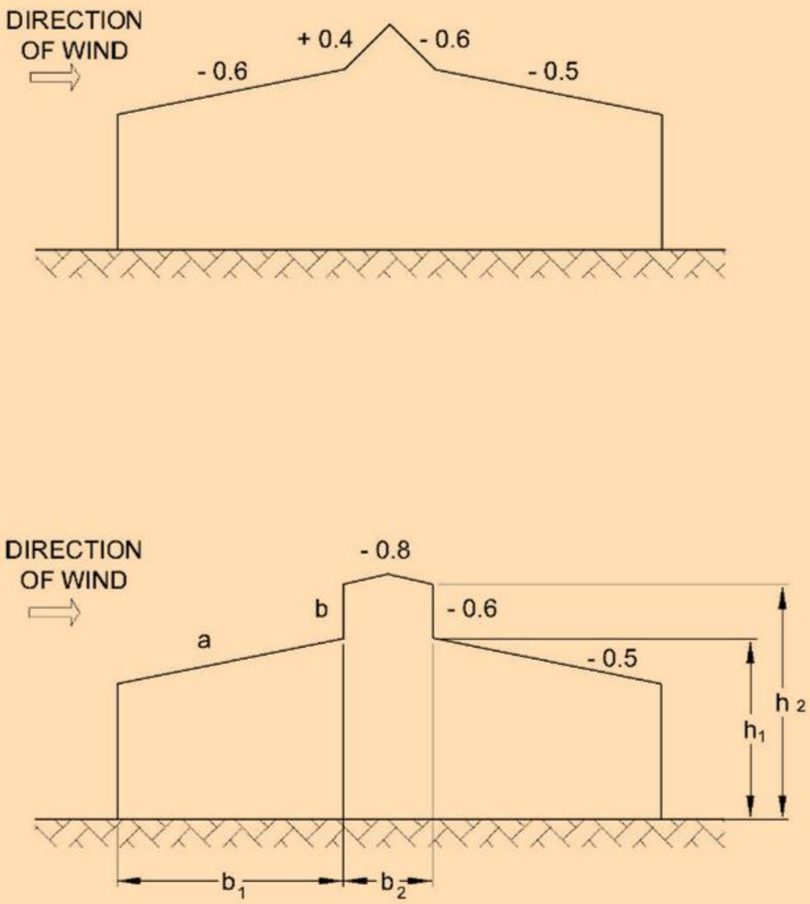
(*Clause* 4.5.3.3.9)



VALUES OF COEFFICIENTS (C_{pe})		
PORTION	DIRECTION 1	DIRECTION 2
a	FROM THE DIAGRAM	- 0.4
b	$C_{pe} = -0.5, \frac{h_1}{h_2} \le 1.5$ $C_{pe} = +0.7; \frac{h_1}{h_2} > 1.5$	
c and d	See Table 6	
e	See Clause 6.3.3.5	

Table 25 External Pressure Coefficients (C_{pe}) for Roofs with a Sky Light

(*Clause* 4.5.3.3.10)

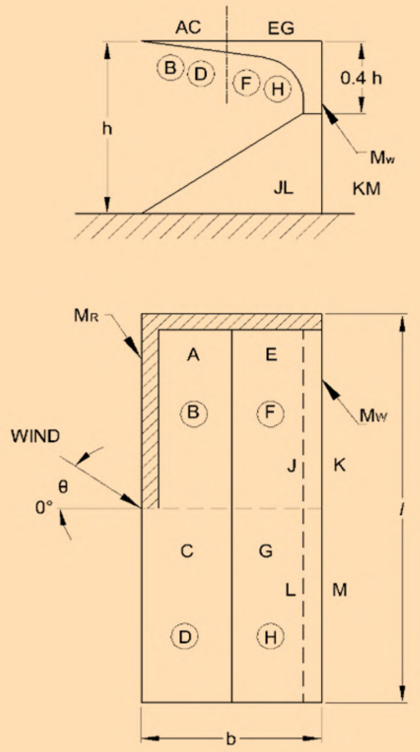


b) ROOFS WITH A SKY LIGHT

VALUES OF COEFFICIENTS (C_{pe})			
PORTION	$b_1 > b_2$		$b_1 \leq b_2$
	a	*b*	*a* and *b*
C_{pe}	- 0.6	+ 0.7	See Table for combined roofs

Table 26 Pressure Coefficients at Top and Bottom Roof of Grand Stands Open Three Sides (Roof Angle Up to 3°)

(*Clause* 4.5.3.3.11)



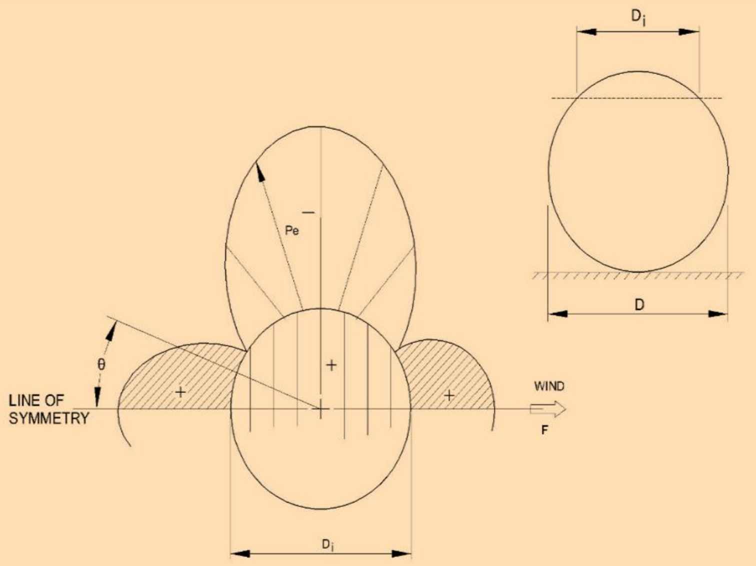
(Shaded area to scale)

FRONT AND BACK OF WALL				
θ	*J*	*K*	*L*	*M*
0°	+ 0.9	- 0.5	+ 0.9	- 0.5
45°	+ 0.8	- 0.6	+ 0.4	- 0.4
135°	- 1.1	+ 0.6	- 1.0	+ 0.4
180°	- 0.3	+ 0.9	- 0.3	+ 0.9
60°		*Mw* - Cp of K = - 1.0		
60°		*Mw* - Cp of J = + 1.0		

TOP AND BOTTOM ROOF								
θ	*A*	*B*	*C*	*D*	*E*	*F*	*G*	*H*
0°	- 1.0	+ 0.9	- 1.0	+ 0.9	- 0.7	+ 0.9	+ 0.7	+ 0.9
45°	- 1.0	+ 0.7	- 0.7	+ 0.4	- 0.5	+ 0.8	- 0.5	+ 0.3
135°	- 0.4	- 1.1	- 0.7	- 1.0	- 0.9	- 1.1	- 0.9	- 1.0
180°	- 0.6	- 0.3	- 0.6	- 0.3	- 0.6	- 0.3	- 0.6	- 0.3
45°	'M_R' - Cp (top) = - 2.0							
45°	'M_R' - Cp (bottom) = + 1.0							

Table 27 External Pressure Coefficients (C_{pe}) Around Spherical Structures

(*Clause* 4.5.3.3.12)



POSITION OF PERIPHERY, θ IN DEGREES	C_{pe}
0	+ 1.0
15	+ 0.9
30	+ 0.5
45	- 0.1
60	- 0.7
75	- 1.1
90	- 1.2
105	- 1.0
120	- 0.6
135	- 0.2
150	+ 0.1
165	+ 0.3
180	+ 0.4

4.5.4.1 *Frictional drag*

In certain buildings of special shape, a force due to frictional drag shall be taken into account in addition to those loads specified in **4.5.3**. For rectangular clad buildings, this addition is necessary only where the ratio d/h or d/b is more than 4. The frictional drag force, F', in the direction of the wind is given by the following formulae:

a) If $h \leq b$, $F' = C_f' (d - 4h)\, b\, p_d + C'_f (d - 4h)\, 2h\, p_d$; and

b) If $h > b$, $F' = C_f' (d - 4b)\, b\, p_d + C'_f (d - 4b)\, 2h\, p_d$.

The first term in each case gives the drag on the roof and the second on the walls. The C_f' has the following values:

a) 0.01 for smooth surfaces without corrugations or ribs across the wind direction;

b) 0.02 for surfaces with corrugations across the wind direction; and

c) 0.04 for surfaces with ribs across the wind direction.

For other buildings, the frictional drag has been indicated, where necessary, in the tables of pressure coefficients and force coefficients.

4.5.4.2 *Force coefficients for clad buildings*

4.5.4.2.1 *Clad buildings of uniform section*

The overall force coefficients for rectangular clad buildings of uniform section with flat roofs in uniform flow shall be as given in Fig. 4 and for other clad buildings of uniform section (without projections, except where otherwise shown) shall be as given in Table 28.

NOTE — Structures that are in the supercritical flow regime, because of their size and design wind velocity, may need further calculation to ensure that the greatest loads do not occur at some wind speed below the maximum, when the flow will be sub-critical.

The coefficients are for buildings without projections, except where otherwise shown.

In Table 28, $\overline{V}_d b$ is used as an indication of the airflow regime.

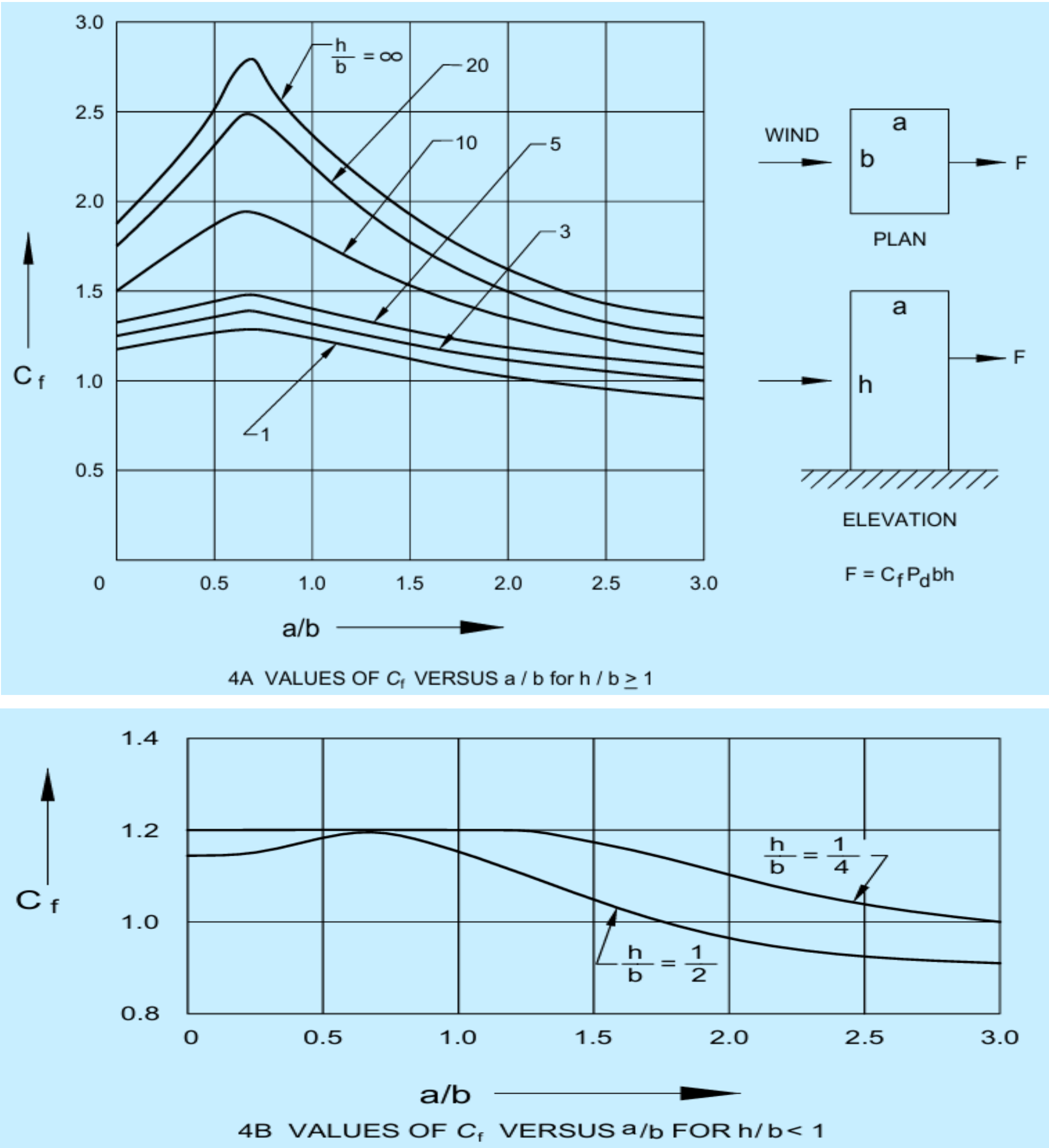


FIG. 4 FORCE COEFFICIENT FOR RECTANGULAR CLAD BUILDING IN UNIFORM FLOW

Table 28 Force Coefficients, C_f for Clad Buildings of Uniform Section (Acting in the Direction of Wind)

(*Clause* 4.5.4.2.1)

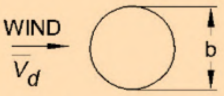
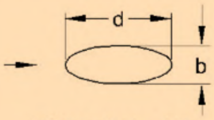
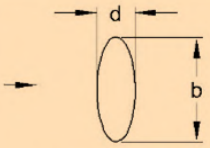
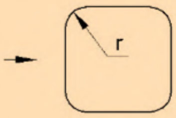
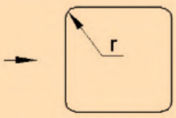
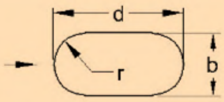
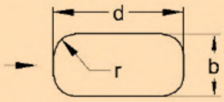
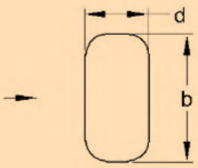
PLAN SHAPE	$\bar{V}_d b$ m^2/s	C_f FOR HEIGHT / BREADTH RATIO						
		UPTO 1/2	1	2	5	10	20	∞
 WIND $\bar{V}_d$ b	ALL SURFACES < 6 / ROUGH or WITH PROJECTION ≥ 6	0.7	0.7	0.7	0.8	0.9	1.0	1.2
See also Appendix D	SMOOTH ≥ 6	0.5	0.5	0.5	0.5	0.5	0.6	0.6
 d b	< 10	0.5	0.5	0.5	0.5	0.6	0.6	0.7
Ellipse b/d = 1/2	≥ 10	0.2	0.2	0.2	0.2	0.2	0.2	0.2
 d b	< 8	0.8	0.8	0.9	1.0	1.1	1.3	1.7
Ellipse b/d = 2	≥ 8	0.8	0.8	0.9	1.0	1.1	1.3	1.5
 r	< 4	0.6	0.6	0.6	0.7	0.8	0.8	1.0
b/d = 1 r/b = 1/3	≥ 4	0.4	0.4	0.4	0.4	0.5	0.5	0.5
 r	< 10	0.7	0.8	0.8	0.9	1.0	1.0	1.3
b/d = 1 r/b = 1/6	≥ 10	0.5	0.5	0.5	0.5	0.6	0.6	0.6
 d r b	< 3	0.3	0.3	0.3	0.3	0.3	0.3	0.4
b/d = 1/2 r/b = 1/2	≥ 3	0.2	0.2	0.2	0.2	0.3	0.3	0.3
 d r b b/d = 1/2 r/b = 1/6	All values	0.5	0.5	0.5	0.5	0.6	0.6	0.7
 d b b/d = 2 r/b = 1/12	All values	0.9	0.9	1.0	1.1	1.2	1.5	1.9

Table 28 (*Continued*)

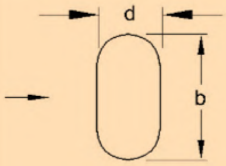
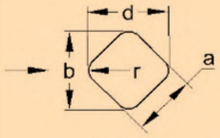
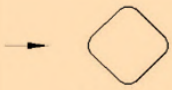
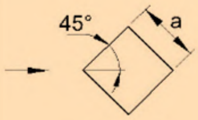
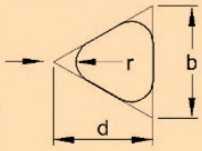
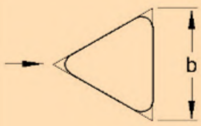
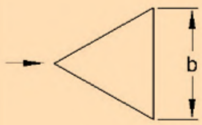
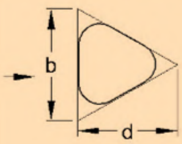
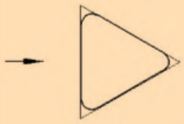
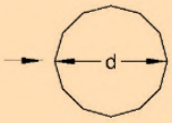
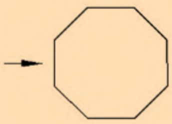
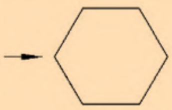
PLAN SHAPE	$\bar{V}_d\, b$ / m^2/s	C_f FOR HEIGHT / BREADTH RATIO						
		UPTO 1/2	1	2	5	10	20	∞
 b/d = 2 r/b = 1/4	< 6	0.7	0.8	0.8	0.9	1.0	1.2	1.6
	≥ 6	0.5	0.5	0.5	0.5	0.5	0.6	0.6
 r/a = 1/3	< 10	0.8	0.8	0.9	1.0	1.1	1.3	1.5
	≥ 10	0.5	0.5	0.5	0.5	0.5	0.6	0.6
 r/a = 1/12	All values	0.9	0.9	0.9	1.1	1.2	1.3	1.6
 r/a = 1/48	All values	0.9	0.9	0.9	1.1	1.2	1.3	1.6
 r/b = 1/4	< 11	0.7	0.7	0.7	0.8	0.9	1.0	1.2
	≥ 11	0.4	0.4	0.4	0.4	0.5	0.5	0.5
 r/b = 1/12	All values	0.8	0.8	0.8	1.0	1.1	1.2	1.4
 r/b = 1/48	All values	0.7	0.7	0.8	0.9	1.0	1.1	1.3
 r/b = 1/4	< 8	0.7	0.7	0.8	0.9	1.0	1.1	1.3
	≤ 8	0.4	0.4	0.4	0.4	0.5	0.5	0.5

Table 28 (*Concluded*)

PLAN SHAPE	$\bar{V}_d b$ / m^2/s	C_f FOR HEIGHT / BREADTH RATIO						
		UPTO 1/2	1	2	5	10	20	∞
 1/48 < r/b < 1/12	All values	1.2	1.2	1.2	1.4	1.6	1.7	2.1
 12 SIDED POLYGON	< 12	0.7	0.7	0.8	0.9	1.0	1.1	1.3
	≥ 12	0.7	0.7	0.7	0.7	0.8	0.9	1.1
 OCTAGON	All values	1.0	1.0	1.1	1.2	1.2	1.3	1.4
 HEXAGON	All values	1.0	1.1	1.2	1.3	1.4	1.4	1.5

4.5.4.2.2 *Buildings of circular shapes*

Force coefficients for buildings of circular cross section shapes shall be as given in Table 28. However, more precise estimation of force coefficients for circular shapes of infinite length can be obtained from Fig. 5 taking into account the average height of surface roughness ε. When the length is finite the values obtained from Fig. 5 shall be reduced by the multiplication factor K (*See also* Table 31 and Annex G).

4.5.4.2.3 *Free standing walls and hoardings*

Force coefficients for free standing walls and hoardings shall be as given in Table 29.

To allow for oblique winds, the design shall also be checked for net pressure normal to the surface varying linearly from a maximum of 1.7 C_f at the windward edge to 0.44 C_f at the leeward edge.

The wind load on appurtenances and supports for hoardings shall be accounted for separately by using the appropriate net pressure coefficients. Allowance shall be made for shielding effects of one element on another.

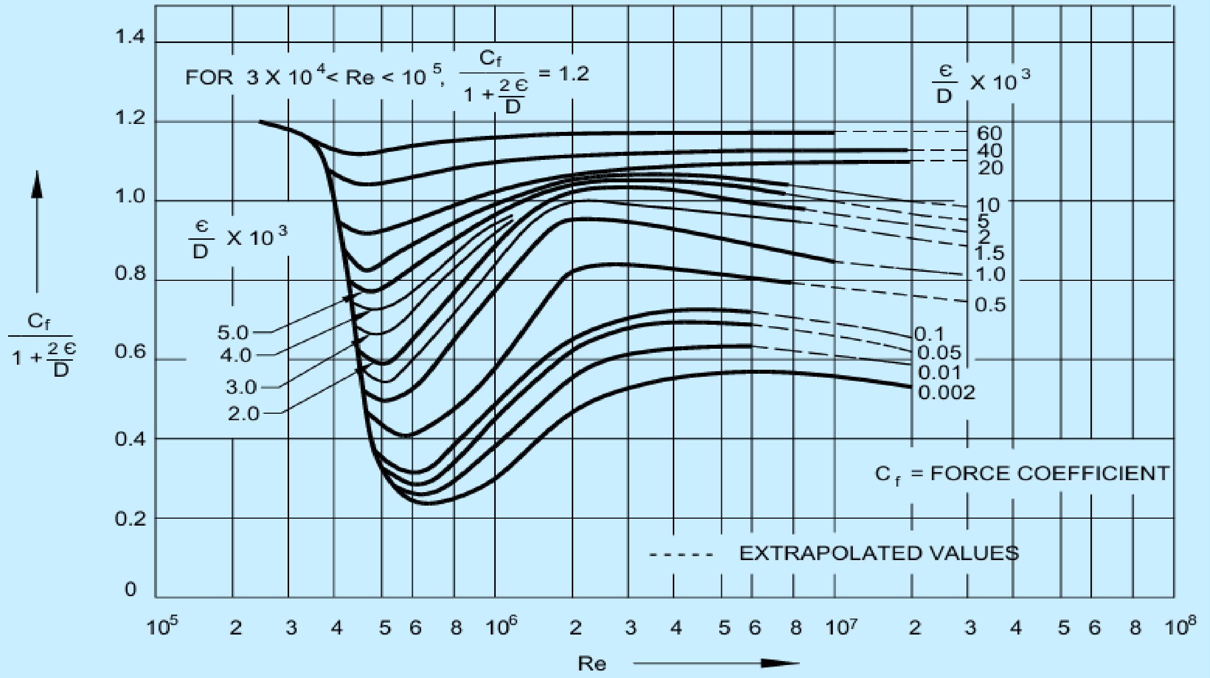
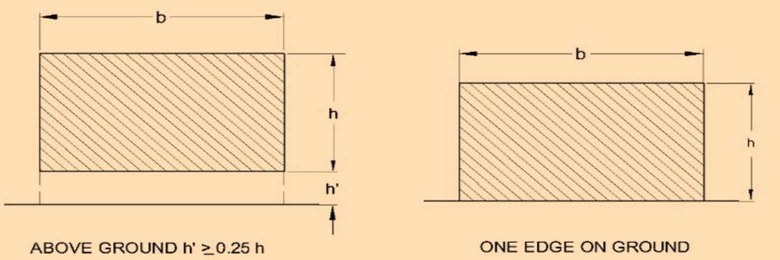


Fig. 5 VARIATION OF $\frac{C_f}{1+\frac{2\epsilon}{D}}$ WITH Re $< 3 \times 10^4$ FOR CIRCULAR SECTIONS

Table 29 Force Coefficients, C_f for Low Walls or Hoardings (< 15m High)

(*Clause* 4.5.4.2.3)

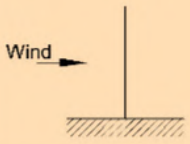
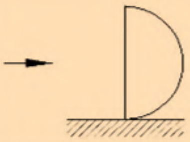
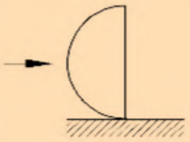
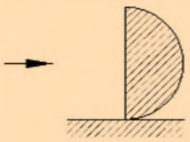
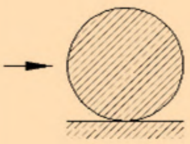


(WIND NORMAL TO FACE)

WIDTH TO HEIGHT RATIO, *b/h*		FORCE COEFFICIENT
WALL ABOVE GROUND	WALL ON GROUND	DRAG COEFFICIENT C_f
FROM 0.5 TO 6	FROM 1 TO 12	1.2
10	20	1.3
16	32	1.4
20	40	1.5
40	80	1.75
60	120	1.8
80 OR MORE	160 OR MORE	2.0

Table 30 Force Coefficients, C_f for Solid Shapes Mounted on a Surface

(*Clause* 4.5.4.2.4)

SIDE ELEVATION	DESCRIPTION OF SHAPE	C_f
	CIRCULAR DISC	1.2
	HEMISPHERICAL BOWL	1.4
	HEMISPHERICAL BOWL	0.4
	HEMISPHERICAL SOLID	1.2
	SPHERICAL SOLID	0.5 FOR $V_z D < 7$ 0.2 FOR $V_z D \geq 7$

4.5.4.2.4 *Solid circular shapes mounted on a surface*

The force coefficients for solid circular shapes mounted on a surface shall be as given in Table 30.

4.5.4.3 *Force coefficients for unclad buildings*

4.5.4.3.1 This section applies to permanently unclad buildings and to frameworks of buildings while temporarily unclad. In the case of buildings whose surfaces are well-rounded, such as those with elliptic, circular or oval cross sections, the total force can be more at a wind speed much less than maximum due to transition in the nature of boundary layer on them. Although this phenomenon is well known in the case of circular cylinders, the same phenomenon exists in the case of many other well-rounded structures and this possibility must be checked.

4.5.4.3.2 *Individual members*

a) The force coefficient given in Table 32 refers to members of infinite length. For members of finite length, the coefficients should be multiplied by a factor K that depends on the ratio l/b where l is the length of the member and b is the width across the direction of wind. Table 31 gives the required values of K. The following special cases must be noted while estimating K:

   1) when any member abuts on to a plate or wall in such a way that free flow of air around that end of the member is prevented, then the ratio of l/b shall be doubled for the purpose of determining K; and

   2) when both ends of a member are so obstructed, the ratio shall be taken as infinity for the purpose of determining K.

b) *Flat-sided members* — Force coefficients for wind normal to the longitudinal axis of flat-sided structural members shall be as given in Table 32.

The force coefficients are given for two mutually perpendicular directions relative to a reference axis on the structural member. They are denoted by C_{fn} and C_{ft} and give the forces normal and transverse, respectively to the reference plane as shown in Table 32.

Normal force $F_n = (C_{fn}\, p_d\, K)/b$

Transverse force $F_t = (C_{ft}\, p_d\, K)/b$

c) *Circular sections* — Force coefficients for members of circular section shall be as given in Table 28 (*See also* Annex G).

d) Force coefficients for wires and cables shall be as given in Table 33 according to the diameter (D), the design wind speed (V_d) and the surface roughness.

Table 31 Reduction Factor *K* for Individual Members

[*Clauses* 4.5.4.2.2, 4.5.4.3.2(a) *and* 4.5.4.3.3(b)(2)]

Sl No.	***l/b* or *l/D***	**2**	**5**	**10**	**20**	**40**	**50**	**100**	**∞**
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
i)	Circular cylinder, sub-critical flow	0.58	0.62	0.68	0.74	0.82	0.87	0.98	1.00
ii)	Circular cylinder, supercritical flow ($DV_d \geq 6$ m²/s)	0.80	0.80	0.82	0.90	0.98	0.99	1.00	1.00
iii)	For plate perpendicular to wind ($bV_d \geq 6$ m²/s)	0.62	0.66	0.69	0.81	0.87	0.90	0.95	1.00

Table 32 Force Coefficients, C_f for Individual Structural Members of Infinite Length

[*Clause* 4.5.4.3.2(b)]

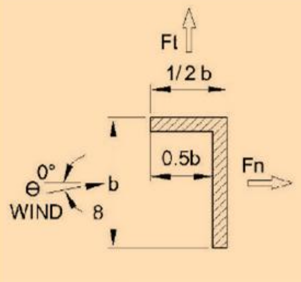
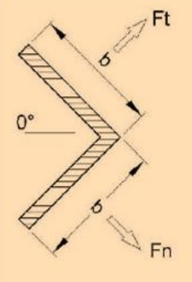
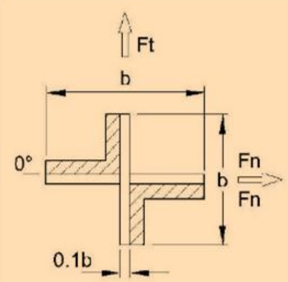
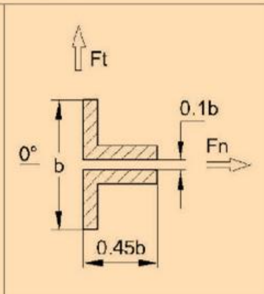
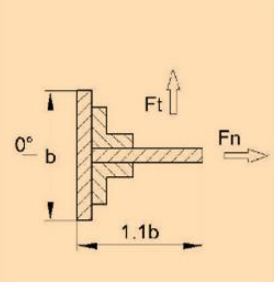
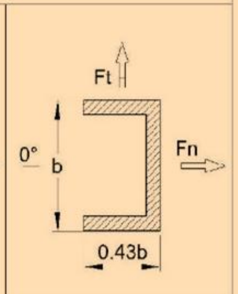
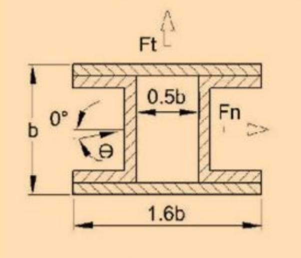
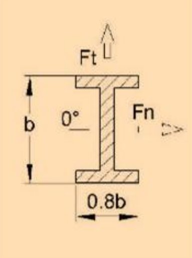
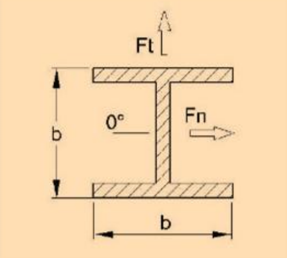
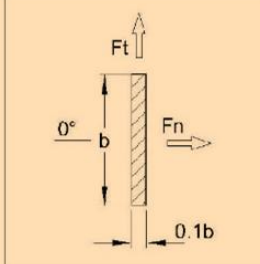
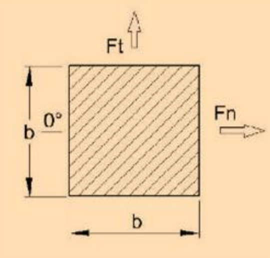
												
θ DEGREE	C_{fn}	C_{ft}	C_{fn}	C_{ft}	C_{fn}	C_{ft}	C_{fn}	C_{ft}	C_{fn}	C_{ft}	C_{fn}	C_{ft}
0	+ 1.9	+ 0.95	+ 1.8	+ 1.8	+ 1.75	+ 0.1	+ 1.6	0	+ 2.0	0	+ 2.05	0
45	+ 1.8	+ 0.8	+ 2.1	+ 1.8	+ 0.85	+ 0.85	+ 1.5	- 0.1	+ 1.2	+ 0.9	+ 1.85	+ 0.6
90	+ 2.0	+ 1.7	- 1.9	- 1.0	+ 0.1	+ 1.75	- 0.95	+ 0.7	- 1.6	+ 2.15	0	+ 0.6
135	- 1.8	- 0.1	- 2.0	+ 0.3	- 0.75	+ 0.75	- 0.5	+ 1.05	- 1.1	+ 2.4	- 1.6	+ 0.4
180	- 2.0	+ 0.1	- 1.4	- 1.4	- 1.75	- 0.1	- 1.5	0	- 1.7	+ 2.1	- 1.8	0
												
θ DEGREE	C_{fn}	C_{ft}	C_{fn}	C_{ft}	C_{fn}	C_{ft}	C_{fn}	C_{ft}	C_{fn}	C_{ft}		
0	+ 1.4	0	+ 2.05	0	+ 1.6	0	+ 2.0	0	+ 2.0	0		
45	+ 1.2	+ 1.6	+ 1.95	+ 0.6	+ 1.5	+ 1.5	+ 1.8	+ 0.1	+ 1.55	+ 1.55		
90	0	+ 2.2	+ 0.5	+ 0.9	0	+ 1.9	0	+ 0.1	0	+ 2.0		

Table 33 Force Coefficients, C_f for Wires and Cables (l/D = 100)

[*Clause* 4.5.4.3.2(d)]

Sl No.	Flow Regime	Force Coefficient, C_f for			
		Smooth Surface	Moderately Smooth Wire (Galvanized or Painted)	Fine Stranded Cables	Thick Stranded Cables
(1)	(2)	(3)	(4)	(5)	(6)
i)	$D\overline{V}_d < 6\ m^2/s$	1.2	1.2	1.2	1.3
ii)	$D\overline{V}_d \geq 6\ m^2/s$	0.5	0.7	0.9	1.1

4.5.4.3.3 *Single frames*

Force coefficients for a single frame having either:

a) all flat sided members; or

b) all circular members in which all the members of the frame have either:

1) $D\overline{V}_d$ less than 6 m^2/s; or

2) $D\overline{V}_d$ more than or equal to 6 m^2/s. shall be as given in Table 31 according to the type of the member, the diameter (D), the design hourly mean wind speed ($\overline{V}_d$) and the solidity ratio (Φ). Force coefficients for a single frame not complying with the above requirements shall be calculated as follows:

$$C_f = \gamma\, C_{f\,super} + (1-\gamma)\frac{A_{circ\,sub}}{A_{sub}} C_{f\,sub} + (1-\gamma)\frac{A_{flat}}{A_{sub}} C_{f\,flat}$$

where

$C_{f\ super}$ = force coefficient for the supercritical circular members as given in Table 34 or Annex G;

γ = $\dfrac{\text{Area of the frame in a supercritical flow}}{A_e}$;

A_e = effective frontal area;

$A_{circ\ sub}$ = effective area of subcritical circular members;

$C_{f\ sub}$ = force coefficient for subcritical circular members as given in Table 34 or Annex G;

$C_{f\ flat}$ = force coefficient for the flat sided members as given in Table 34;

A_{flat} = effective area of flat-sided members; and

A_{sub} = Acirc sub + Aflat.

Table 34 Force Coefficients for Single Frames

(*Clause* 4.5.4.3.3)

Sl No.	Solidity Ratio, Φ	Force Coefficient, C_f For		
		Flat Sided Members	Circular Sections	
			Subcritical Flow ($D\overline{V}_d < 6\ m^2/s$)	Super Critical Flow ($D\overline{V}_d \geq 6\ m^2/s$)
(1)	(2)	(3)	(4)	(5)
i)	0.1	1.9	1.2	0.7
ii)	0.2	1.8	1.2	0.8
iii)	0.3	1.7	1.2	0.8
iv)	0.4	1.7	1.1	0.8
v)	0.5	1.6	1.1	0.8
vi)	0.75	1.6	1.5	1.4
vii)	1.00	2.0	2.0	2.0

NOTE — Linear interpolation between the values is permitted.

4.5.4.3.4 *Multiple frame buildings*

This Section applies to structures having two or more parallel frames where the windward frames may have a shielding effect upon the frames to leeward side. The windward frame and any unshielded parts of other frames shall be calculated in accordance with **4.5.4.3.3**, but the wind load on the parts of frames that are sheltered should be multiplied by a shielding factor which is dependent upon the solidity ratio of the windward frame, the types of members comprising the frame and the spacing ratio of the frames. The values of the shielding factors are given in Table 35.

Where there are more than two frames of similar geometry and spacing, the wind load on the third and subsequent frames should be taken as equal to that on the second frame. The loads on the various frames shall be added to obtain total load on the structure.

a) The frame spacing ratio is equal to the centre to centre distance between the frames, beams or girders divided by the least overall dimension of the frames, beam or girder measured in a direction normal to the direction of wind. For triangular framed structures or rectangular framed structures diagonal to the wind, the spacing ratio should be calculated from the mean distance between the frames in the direction of the wind.

b) Effective solidity ratio, Φ_e:

$\Phi_e = \Phi$ for flat-sided members.

Φ_e is to be obtained from Fig. 6 for members of circular cross sections.

Table 35 Shielding Factor, η for Multiple Frames

(*Clause* 4.5.4.3.4)

Sl No.	Effective Solidity Ratio, Φ_e	Frame Spacing Ratio				
		< 0.5	1.0	2.0	4.0	> 8.0
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	0	1.0	1.0	1.0	1.0	1.0
ii)	0.1	0.9	1.0	1.0	1.0	1.0
iii)	0.2	0.8	0.9	1.0	1.0	1.0
iv)	0.3	0.7	0.8	1.0	1.0	1.0
v)	0.4	0.6	0.7	1.0	1.0	1.0
vi)	0.5	0.5	0.6	0.9	1.0	1.0
vii)	0.7	0.3	0.6	0.8	0.9	1.0
viii)	1.0	0.3	0.6	0.6	0.8	1.0

NOTE — Linear interpolation between the values is permitted.

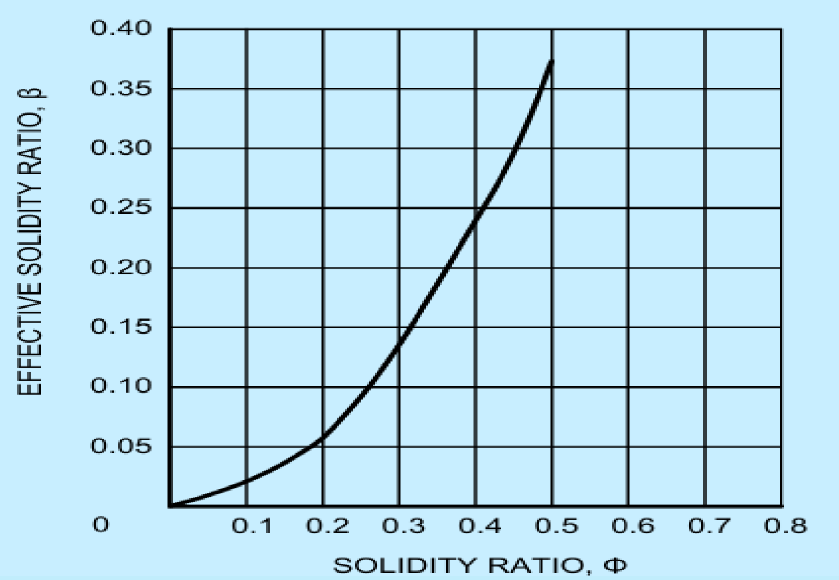


FIG. 6 EFFECTIVE SOLIDITY RATIO FOR CIRCULAR SECTION MEMBERS

4.5.4.3.5 *Lattice towers*

a) Force coefficient for lattice towers of square or equilateral triangle section with flat sided members for wind blowing against any face shall be as given in Table 36;

b) For square lattice towers with flat-sided members the maximum load, which occurs when the wind blows into a corner, shall be taken as 1.2 times the load for the wind blowing against a face;

c) For equilateral triangle lattice towers with flat-sided members, the load may be assumed to be constant for any inclination of wind to a face;

d) Force coefficients for lattice towers of square section with circular members, all in the same flow regime, may be as given in Table 37; and

e) Force coefficients for lattice towers of equilateral-triangle section with circular members all in the same flow regime may be as given in Table 38.

4.5.4.3.6 *Tower appurtenances*

The wind loading on tower appurtenances, such as ladders, conduits, lights, elevators, etc shall be calculated using appropriate net pressure coefficients for these elements. Allowance may be made for shielding effect from other elements.

Table 36 Overall Force Coefficients for Towers Composed of Flat Sided Members
[*Clause* 4.5.4.3.5(a)]

Sl No.	Solidity Ratio, Φ	Force Coefficient	
		Square Towers	Equilateral Triangular Towers
(1)	(2)	(3)	(4)
i)	< 0.1	3.8	3.1
ii)	0.2	3.3	2.7
iii)	0.3	2.8	2.3
iv)	0.4	2.3	1.9
v)	0.5	2.1	1.5

Table 37 Overall Force Coefficients for Square Towers Composed of Circular Members
[*Clause* 4.5.4.3.5(d)]

Sl No.	Solidity Ratio of Front Face Φ	Force Coefficient			
		Subcritical Flow ($D\overline{V}_d < 6$ m²/s)		Super Critical Flow ($D\overline{V}_d \geq 6$ m²/s)	
		Onto Face	Onto Corner	Onto Face	Onto Corner
(1)	(2)	(3)	(4)	(5)	(6)
i)	< 0.05	2.4	2.5	1.1	1.2
ii)	0.1	2.2	2.3	1.2	1.3
iii)	0.2	1.9	2.1	1.3	1.6
iv)	0.3	1.7	1.9	1.4	1.6
v)	0.4	1.6	1.9	1.4	1.6
vi)	0.5	1.4	1.9	1.4	1.6

Table 38 Overall Force Coefficients for Equilateral Triangular Towers Composed of Circular Members

[*Clause* 4.5.4.3.5(e)]

Sl No.	Solidity Ratio of Front Face Φ	Force Coefficient	
		Subcritical flow ($D\overline{V}_d$ < 6 m^2/s)	Super critical flow ($D\overline{V}_d \geq$ 6 m^2/s)
		All wind directions	All wind directions
(1)	(2)	(3)	(4)
i)	< 0.05	1.8	0.8
ii)	0.1	1.7	0.8
iii)	0.2	1.6	1.1
iv)	0.3	1.5	1.1
v)	0.4	1.5	1.1
vi)	0.5	1.4	1.2

4.6 Interference Effects

4.6.1 *General*

Wind interference is caused by modification in the wind characteristics produced by the obstruction caused by an object or a building/structure in the path of the wind. If such wind strikes another structure, the wind pressures usually get enhanced, though there can also be some shielding effect between two very closely spaced buildings/structures. The actual phenomenon is too complex to justify generalization of the wind forces/pressures produced due to interference which can only be ascertained by detailed wind tunnel/CFD studies. However, some guidance can be provided for the purpose of preliminary design. To account for the effect of interference, a wind interference factor (IF) has been introduced as a multiplying factor to be applied to the design wind pressure/force. Interference effects can be more significant for tall buildings. The interference factor is defined as the ratio between the enhanced pressure/force in the grouped configuration to the corresponding pressure/force in isolated configuration. Since the values of IF can vary considerably based on building geometry and location, the given values of IF are a kind of median values and are meant only for preliminary design estimates. The designer is advised that for assigning values of IF for final design particularly for tall buildings, specialist literature be consulted or a wind tunnel study carried out.

4.6.2 *Roof of Low Rise Buildings*

Maximum increase in wind force on the roof due to interference from similar buildings in case of closely spaced low rise buildings with flat roofs may be up to 25 percent for c/c distance (x) between the buildings of 5 times the dimension (b) of the interfering building normal to the direction of wind (*see* Fig. 7). Interference effect beyond 20b may be considered to be negligible. For intermediate spacing linear interpolation may be used.

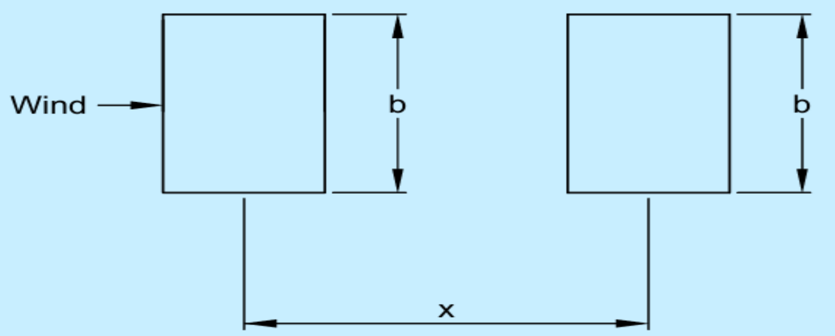


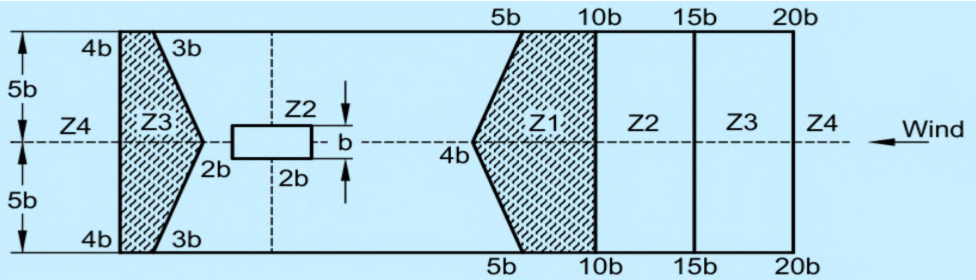
FIG. 7 LOW-RISE BUILDING IN TANDEM CAUSING INTERFERENCE EFFECT

4.6.3 *Tall Buildings*

Based on studies on tall rectangular buildings, Fig. 8 gives various zones of interference. The interference factor (IF), which needs to be considered as a multiplication factor for wind loads corresponding to isolated building, may be assumed as follows for preliminary estimate of the wind loads under interference caused by another interfering tall building of same or more height located at different zone Z1 to zone Z4 as shown in Fig. 8:

Zone	Z1	Z2	Z3	Z4
IF	1.35	1.25	1.15	1.07

The interference effect due to buildings of height less than one-third of the height of the building under consideration may be considered to be negligible while for interference from a building of intermediate height, linear interpolation may be used between one-third and full height.



Z1 Zone of high interference
Z2 Zone of moderate interference
Z3 Zone of low interference
Z4 Zone of insignificant interference

FIG. 8 INTERFERENCE ZONES FOR TALL RECTANGULAR BUILDINGS OF SAME OR GREATER HEIGHT

4.7 Dynamic Effects

4.7.1 *General*

Flexible slender structures and structural elements shall be investigated to ascertain the importance of wind induced oscillations or excitations in along wind and across wind directions.

In general, the following guidelines may be used for examining the problems of wind-induced oscillations:

a) Buildings and closed structures with a height to minimum lateral dimension ratio of more than about 5.0, or

b) Buildings and structures whose natural frequency in the first mode is less than 1.0 Hz.

Any building or structure which satisfies either of the above two criteria shall be examined for dynamic effects of wind.

NOTES

1 The fundamental time period (T), in second, may either be established by experimental observations on similar buildings or calculated by any rational method of analysis. In the absence of such data, T may be determined as follows for multi-storied buildings:

a) For moment resistant frames without bracings or shear walls resisting the lateral loads:

$$T = 0.1\ n$$

where

n = number of storeys, including basement storeys.

b) For all others:

$$T = \frac{0.09H}{\sqrt{d}}$$

where

H = total height of the main structures of the building, in m; and

d = maximum base dimension of building in metres in a direction parallel to the applied wind force.

2 If preliminary studies indicate that wind-induced oscillations are likely to be significant, investigations should be pursued with the aid of analytical methods or if necessary, by means of wind tunnel tests on models.

3 Across-wind motions may be due to lateral gustiness of the wind, unsteady wake flow (for example, vortex shedding), negative aerodynamic damping or due to a combination of these effects. These cross-wind motions may become critical in the design of tall buildings/structures.

4 Motions in the direction of wind (also known as buffeting) are caused by fluctuating wind force associated with gust. The excitation depends on gust energy available at the resonant frequency.

5 The eddies shed from an upstream body may intensify motion in the direction of the wind and may also affect cross-wind motion.

6 The designer should also be aware of the following three forms of wind-induced motion which are characterized by increasing amplitude of oscillation with the increase of wind speed.

a) *Galloping* — Galloping is transverse oscillations of some structures due to the development of aerodynamic forces which are in phase with the motion. It is characterized by the progressively increasing amplitude of transverse vibration with increase of wind speed. The cross sections which are particularly prone to this type of excitation include the following:

1) All structures with non-circular cross sections, such as triangular, square, polygons, as well as angles, crosses and T sections.
2) Twisted cables and cables with ice encrustations.

b) *Flutter* — Flutter is unstable oscillatory motion of a structure due to coupling between aerodynamic force and elastic deformation of the structure. Perhaps the most common form is oscillatory motion due to combined bending and torsion. Although oscillatory motion in each degree of freedom may be damped, instability can set in due to energy transfer from one mode of oscillation to another and the structure is seen to execute sustained or divergent oscillations with a type of motion which is a combination of the individual modes of vibration. Such energy transfer takes place when the natural frequencies of modes taken individually are close to each other (ratio being typically less than 2.0). Flutter can set in at wind speeds much less than those required for exciting the individual modes of motion. Long span suspension bridge decks or any member of a structure with large values of d/t (where d is the length of the member and t is its dimension parallel to wind stream) are prone to low speed flutter. Wind tunnel testing is required to determine critical flutter speeds and the likely structural response. Other types of flutter are single degree of freedom stall flutter, torsional flutter, etc.

c) *Ovalling* — Thin walled structures with open ends at one or both ends such as oil storage tanks and natural draught cooling towers in which the ratio of the diameter or minimum lateral dimension to the wall thickness is of the order of 100 or more are prone to ovalling oscillations. These oscillations are characterized by periodic radial deformation of the hollow structure.

7 Buildings and structures that may be subjected to significant wind excited oscillations require careful investigations. It is to be noted that wind induced oscillations may occur at wind speeds lower than the design wind speed.

8 Analytical methods for the evaluation of response of dynamic structures to wind loading can be found in the special publications.

9 In assessing wind loads due to such dynamic phenomenon as galloping, flutter and ovalling, in the absence of the required information either in the special publications or other literature, expert advice should be sought including experiments on models in boundary layer wind tunnels.

4.7.2 *Motion due to Vortex Shedding*

4.7.2.1 *Slender structures*

For a structure, the vortex shedding frequency, fs, shall be determined by the following formula:

$$fs = \frac{S_t \overline{V}_{z,H}}{b}$$

where

S_t = strouhal number

$\overline{V}_{z,H}$ = hourly mean wind speed at height z; and

b = breadth of a structure or structural member normal to the wind direction in the horizontal plane.

a) *Circular structures* — For structures of circular cross section:

S_t = 0.20 for $D\overline{V}_{z,H}$ less than 6 m^2/s; and

= 0.25 for $D\overline{V}_{z,H}$ more than or equal to 6 m^2/s.

b) *Rectangular structures* — For structures of rectangular cross section:

$$S_t = 0.10$$

NOTES

1 Significant cross wind motions may be produced by vortex shedding, if the natural frequency of the structure or structural element is equal to the frequency of the vortex shedding within the range of expected wind speeds. In such cases, further analysis should be carried out on the basis of special publications.

2 Unlined welded steel cylindrical structures are prone to excitations by vortex shedding.

3 Intensification of the effects of periodic vortex shedding has been reported in cases where two or more similar structures are located in close proximity, for example at less than $20b$ apart, where b is the dimension of the structure normal to the wind.

4 The formulae given in 4.7.2.1(a) is valid for infinitely long cylindrical structures. The value of S_t decreases slowly as the ratio of length to maximum transverse width decreases, the reduction being up to about half the value, if the structure is only three times higher than its width. Vortex shedding need not be considered, if the ratio of length to maximum transverse dimension is less than 2.0.

4.8 Dynamic Wind Response

4.8.1 *General*

Tall buildings/structures which are 'wind sensitive' shall be designed for dynamic wind loads. Hourly mean wind speed is used as a reference wind speed to be used in dynamic wind analysis. For calculation of along wind loads and response (bending moments, shear forces, or tip deflections) the Gust factor (GF) method is used as specified in 4.8.2. The across wind design peak base overturning moment and tip deflection shall be calculated using 4.8.3.

4.8.2 *Along Wind Response*

For calculation of along-wind load effects at a level s on a building/structure, the design hourly mean wind pressure at height *z* shall be multiplied by the Gust factor (*G*). This factor is dependent on both the overall height *h* and the level *s* under consideration (*see* Fig. 9). For calculation of base bending moment and deflection at the top of the building/structure, '*s*' should be taken as zero.

Note that $0 < s < h$ and $s < z < h$.

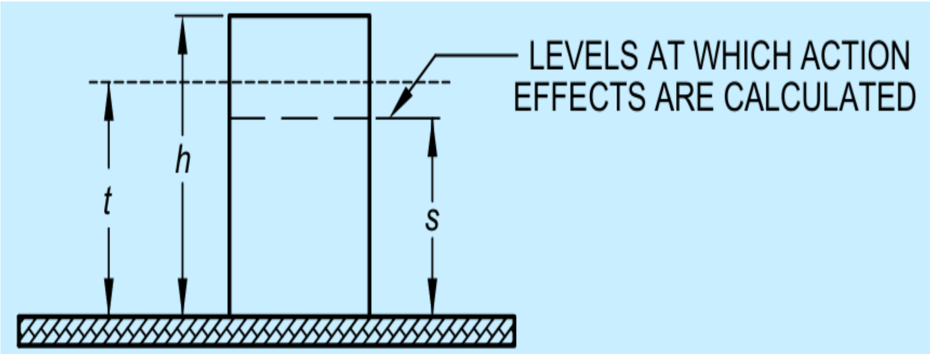


FIG. 9 NOTATIONS FOR HEIGHTS

The design peak along wind base bending moment (M_a) shall be obtained by summing the moments resulting from design peak along wind loads acting at different heights, z, along the height of the building/structure and can be obtained from:

$$M_a = \sum F_z \, z$$

$$F_z = C_{f,z} A_z \overline{p}_d G$$

where

F_z	=	design peak along wind load on the building/structure at any height *z;*
A_z	=	the effective frontal area of the building/structure at any height *z*, in m^2;
$\overline{p}_d$	=	design hourly mean wind pressure corresponding to $\overline{V}_{z,d}$ and obtained as $0.6\overline{V}_{z,d}^2$ (N/m^2);
$\overline{V}_{z,d}$	=	design hourly mean wind speed at height *z*, in m/s (*see* **4.4.4**);
$C_{f,z}$	=	the drag force coefficient of the building/structure corresponding to the area A_z;
G	=	Gust Factor and is given by:

$$1 + r\sqrt{\left[g_v^2 B_s (1+\phi)^2 + \frac{H_s g_R^2 SE}{\beta}\right]}$$

r	=	roughness factor which is twice the longitudinal turbulence intensity, $I_{h,i}$ (*see* **4.4.5**);
g_v	=	peak factor for upwind velocity fluctuation
		= 3.0 for category 1 and 2 terrains; and
		= 4.0 for category 3 and 4 terrains;

B_s = background factor indicating the measure of slowly varying component of fluctuating wind load cused by the lower frequency wind speed variations

$$= \frac{1}{\left[1+\frac{\sqrt{0.26(h-s)^2+0.46b_{sh}^2}}{L_h}\right]}$$

where

b_{sh} = average breadth of the building/structure between heights s and h;

L_h = measure of effective turbulence length scale at the height, h, in m;

$= 85\left(\frac{h}{10}\right)^{0.25}$ for terrain category 1 to 3; and

$= 70\left(\frac{h}{10}\right)^{0.25}$ for terrain category 4

ϕ = factor to account for the second order turbulence intensity;

$= \frac{g_v I_{h,i}\sqrt{B_s}}{2}$

$I_{h,i}$ = turbulence intensity at height h in terrain category i;

H_s = height factor for resonance response;

$$= 1+\left(\frac{s}{h}\right)^2$$

S = size reduction factor given by

$$= \frac{1}{\left[1+\frac{3.5 f_a h}{\overline{V}_{h,d}}\right]\left[1+\frac{4 f_a b_{0h}}{\overline{V}_{h,d}}\right]}$$

where

b_{0h} = average breadth of the building/structure between 0 and h;

E = spectrum of turbulence in the approaching wind stream:

$$\frac{\pi N}{\left(1+70.8N^2\right)^{5/6}}$$

where

N = an effective reduced frequency

$$= \frac{f_a L_h}{\overline{V}_{h,d}}$$

f_a = first mode natural frequency of the building/structure in along wind direction, in Hz

$\overline{V}_{h,d}$ = design hourly mean wind speed at height, h in m/s (*see* **4.4.4**);

β = damping coefficient of the building/structure (*see* Table 39); and

g_R = peak factor for resonant response.

$$= \sqrt{\left[2\ln(3600 f_a)\right]}$$

Table 39 Suggested Values of Structural Damping Coefficients

(*Clauses* 4.8.2 *and* 4.8.3.1)

Sl No.	Kind of Structure	Damping Coefficient, β
(1)	(2)	(3)
i)	Welded steel structures	0.010
ii)	Bolted steel structures/RCC structures	0.020
iii)	Prestressed concrete structures	0.016

4.8.2.1 *Peak acceleration in along wind direction*

The peak acceleration at the top of the building/structure in along wind direction ($\hat{\ddot{x}}$ in m/s^2) is given by the following equation:

$$\hat{\ddot{x}} = (2\pi f_a)^2 \bar{x}\, g_R\, r \sqrt{\frac{SE}{\beta}}$$

where

$\bar{x}$ = mean deflection at the position where the acceleration is required. Other notations are same as given in 9.2. For computing the peak acceleration in the along wind direction, a mean wind speed at the height of the building/structure, $\bar{V}_h$ corresponding to a 5 year mean return period shall be used. A reduced value of 0.011 is also suggested for the structural damping, β for reinforced concrete structures.

4.8.3 *Across Wind Response*

This section gives method for determining equivalent static wind load and base overturning moment in the across wind direction for tall enclosed buildings and towers of rectangular cross section. Calculation of across wind response is not required for lattice towers.

The across wind design peak base bending moment M_c for enclosed buildings and towers shall be determined as follows:

$$M_c = 0.5 g_h \bar{p}_h b h^2 (1.06 - 0.06k)\sqrt{\frac{\pi C_{fs}}{\beta}}$$

where

g_h = a peak factor;

= $\sqrt{[2\ln(3600 f_c)]}$ in cross wind direction;

$\bar{p}_h$ = hourly mean wind pressure at height h, in Pa;

b = breadth of the structure normal to the wind, in m;

h = height of the structure, in m;

k = a mode shape power exponent for representation of the fundamental mode shape as represented by:

$$\psi(z) = \left(\frac{z}{h}\right)^k \text{; and}$$

f_c = first mode natural frequency of the building/structure in across wind direction, in Hz.

The across wind load distribution on the building/structure can be obtained from M_c using linear distribution of loads as given below:

$$F_{z,c} = \left(\frac{3M_c}{h^2}\right)\left(\frac{z}{h}\right)$$

where

$F_{z,c}$ = across wind load per unit height at height *z*.

4.8.3.1 *Peak acceleration in across wind direction*

The peak acceleration at the top of the building/structure in across-wind direction ($\hat{\ddot{y}}$ in m/s^2) with approximately constant mass per unit height shall be determined as follows:

$$\hat{\ddot{y}} = 1.5\frac{g_h\,\bar{p}_h\,b}{m_0}(0.76+0.24k)\sqrt{\left(\frac{\pi C_{fs}}{\beta}\right)}$$

where

C_{fs} = across wind force spectrum coefficient generalized for a linear mode, (*see* Fig. 10);

β = damping coefficient of the building/structure (*see* Table 39); and

m_0 = the average mass per unit height of the structure, in kg/m.

Typical values of the mode shape power exponent, k are as follows:

a) uniform cantilever, k = 1.5

b) slender framed structure (moment resisting), k = 0.5

c) building with a central core and moment resisting façade, k = 1.0

d) lattice tower decreasing in stiffness with height, or a tower with a large mass at the top, k = 2.3

4.8.4 *Combination of Along Wind and Across Wind Load Effects*

The along wind and across wind loads have to be applied simultaneously on the building/structure during design.

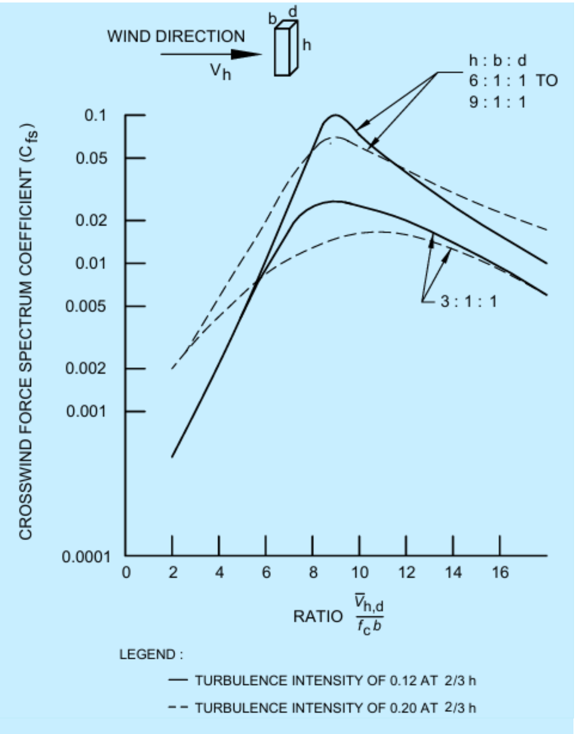


FIG. 10(A) VALUES OF THE CROSS WIND FORCE SPECTRUM COEFFICIENT FOR SQUARE SECTION BUILDINGS

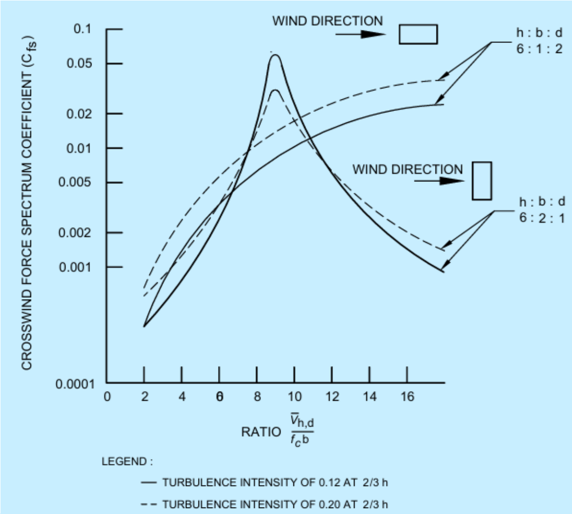


FIG. 10(B) VALUES OF THE CROSS WIND FORCE SPECTRUM COEFFICIENT FOR 2 : 1 AND 1 : 2 RECTANGULAR SECTION BUILDINGS

FIG. 10 WIND FORCE SPECTRUM COEFFICIENTS

5 EARTHQUAKE EFFECTS — GENERAL

For assessment of earthquake loads, determination of seismic zones and earthquake resistant design of buildings, good practice [C-1(6)] shall be referred to. Additionally, it outlines the principles of earthquake-resistant design, desirable and unacceptable features of earthquake-resistant structural system configurations and methods of analysis and treatment of geotechnical aspects for all types of structures.

6 EARTHQUAKE EFFECTS — BUILDINGS

6.1 *Masonry Buildings*

6.1.1 Provisions given in Part C 'Structural Design/Section 4 Masonry' of this NBCS deals with the selection of materials, special aspects related to design and construction of new earthquake-resistant masonry buildings, including masonry buildings using rectangular masonry units with cast in-situ or prefabricated flooring and roofing elements.

6.1.2 Guidelines for earthquake-resistant design of buildings constructed using masonry of low strength and earthen buildings are covered in as per good practice [C-1(7)].

6.2 *Concrete Buildings*

Provisions given in Part C 'Structural Design/ Section 5 Structural Concrete' of this NBCS and good practice [C-1(8)] deal with the selection of materials and special aspects related to design and construction for cast in-situ earthquake-resistant reinforced concrete (RC) buildings.

6.3 *Steel Buildings*

Provisions given in Part C 'Structural Design/Sec 6A Steel' of this NBCS and good practice [C-1(9)] deal with the selection of materials, special aspects related to design and construction for earthquake-resistant steel buildings.

7 SNOW LOAD

7.1 This clause deals with snow loads on roofs of buildings. Roofs should be designed for the actual load due to snow or for the imposed loads specified in 3, whichever is more severe.

NOTE — Mountainous regions in northern and eastern parts of India are subjected to snow-fall. In northern India, parts of union territory of Jammu and Kashmir (J and K), union territory of Ladakh, Himachal Pradesh, Uttarakhand and in eastern India, parts of Arunachal and Sikkim experience snow-fall of varying depths two to three times in a year.

7.2 Notations

a) μ (Dimensionless) — Nominal values of the shape coefficients, taking into account snow drifts, sliding snow, etc, with subscripts, if necessary;

b) l_i (m) — Horizontal dimension with numerical subscripts, if necessary;

c) h_i (m) — Vertical dimensions with numerical subscripts, if necessary;

d) β_i (degree) — Roof and other surface slope;

e) s_0 (kPa) — Characteristic value of snow load on the ground with a specified annual exceedance probability;

f) s (kPa) — Snow load on roofs other surfaces; and

g) A (m) — Site altitude above sea level.

7.3 Snow Load in Roof(s)

7.3.1 The minimum design snow load on a roof area or any other area above ground which is subjected to snow accumulation is obtained by multiplying the snow load on ground, s_o by the shape coefficient μ, as applicable to the particular roof area considered, expressed as:

$$s = \mu\, s_o$$

where

s	=	design snow load on plan area of roof, in Pa;
μ	=	shape coefficient (*see* 7.4); and
s_o	=	ground snow load in Pa (1 Pa=1 N/m^2).

NOTE — Ground snow load at any place depends on the critical combination of the maximum depth of undisturbed aggregate cumulative snow fall and its average density. In due course the characteristic snow load on ground for different regions will be included based on studies. Till such time the users of this NBCS are advised to contact either Defence Geoinformatics Research Establishment (DGRE), Chandigarh of Defence Research and Development Organization (DRDO), or India Meteorological Department (IMD), Pune in the absence of any specific information for any location.

7.3.2 *Characteristic Ground Snow Load* (s_0)

The characteristic ground snow load (s_0) at any site is defined based on an annual probability of exceedance of 0.034 and corresponds to a mean return period of 30 years. The value of s_0 (kPa) for any site in snow-bound regions of Indian Himalayas should be obtained using the procedure described in Annex H.

NOTE — Annex H gives the characteristic ground snow load map for union territory of Jammu and Kashmir, union territory of Ladakh, Himachal Pradesh, Uttarakhand and Sikkim resulting from studies conducted by DGRE, Chandigarh.

7.3.3 *Partial Loading Due to Melting, Sliding, Snow Redistribution and Snow Removal*

A loading corresponding to severe imbalances resulting from snow removal, redistribution, sliding, melting, etc (for example, zero snow load on specific parts of the roof) should always be considered. Such considerations are particularly important for structures which are sensitive to unbalance loading (for example, curved roofs, arches, domes, collar beam roofs, continuous beam systems). Water load due to ponding in flat roof as well as valley of pitched roof shall be taken as not less than 0.24 kPa.

7.3.4 *Ponding Instability*

7.3.4.1 Roofs shall be designed to preclude ponding.

7.3.4.2 For flat roofs (or with a small slope), roof deflections caused by snow loads shall be investigated when determining the likelihood of ponding from rain-on-snow or from snow meltwater. The angle for snow melting is 15º.

7.4 Shape Coefficients

7.4.1 *General Principles*

In perfectly calm weather, falling snow would cover roofs and the ground with a uniform blanket of snow and the design snow load could be considered as a uniformly distributed load. Truly uniform loading conditions, however, are rare and have usually only been observed in areas that are sheltered on all sides by high trees, buildings, etc. In such a case, the shape coefficient would be equal to unity.

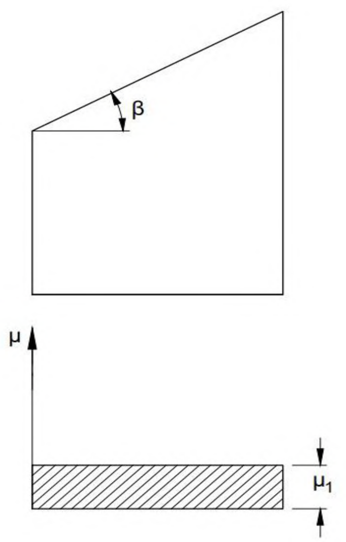
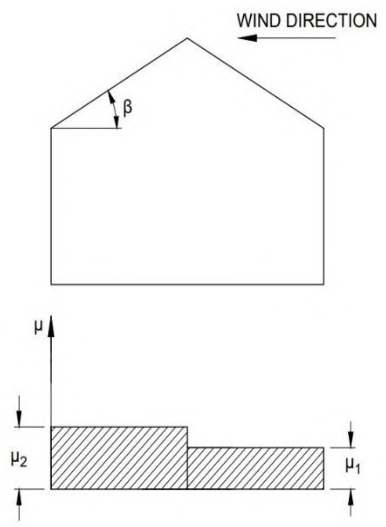
In most regions, snow-falls are accompanied or followed by winds. The winds will redistribute the snow and on some roofs especially multilevel roofs, the accumulated drift load may reach a multiple of the ground load. Roofs which are sheltered by other buildings, vegetation, etc, may collect more snow load than the ground level. The phenomenon is of the same nature as that illustrated for multilevel roofs in 7.4.2.4.

So far, sufficient data are not available to determine the shape coefficient on a statistical basis. Therefore, a nominal value is given. A representative sample of roofs is shown in 7.4.2. However, in special cases such as strip loading, cleaning of the roof periodically by deliberate heating of the roof, etc, have to be treated separately. However, no reduction in load of snow on roof having underside insulation and/or internal heating system shall be done.

The distribution of snow in the direction parallel to the caves is assumed to be uniform.

7.4.2 *Shape Coefficients for Selected Types of Roofs*

7.4.2.1 *Monopitched and simple pitched roofs*

	 Simple Flat and Monopitch Roofs	 Simple Pitched Roofs (Positive Roof Slope)*
$0° < \beta \leq 15°$		$\mu_2 = \mu_1 = 0.80$

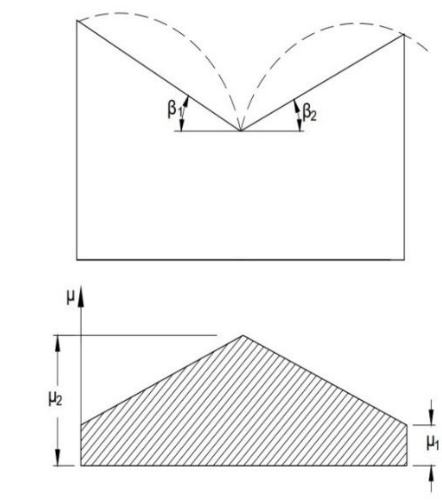
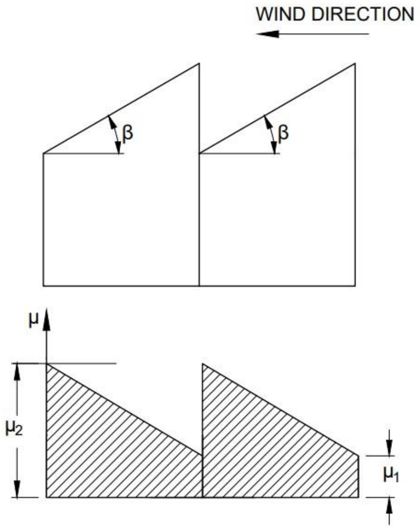
$15° < \beta \leq 30°$	$\mu_1 = 0.80$	$\mu_2 = 0.80 + 0.40\left(\frac{\beta - 15}{15}\right)$ $\mu_1 = 0.80$
$30° < \beta \leq 60°$	$\mu_1 = 0.80\left(\frac{60 - \beta}{30}\right)$	$\mu_2 = 1.20\left(\frac{60 - \beta}{30}\right)$ $\mu_1 = 0.80\left(\frac{60 - \beta}{30}\right)$
$\beta > 60°$	$\mu_1 = 0$	$\mu_2 = \mu_1 = 0$
*For asymmetrical simple pitched roofs, each side of the roof shall be treated as one half of corresponding symmetrical roofs.		

NOTES

1 For monopitch roof, wind direction will not affect the value of μ.

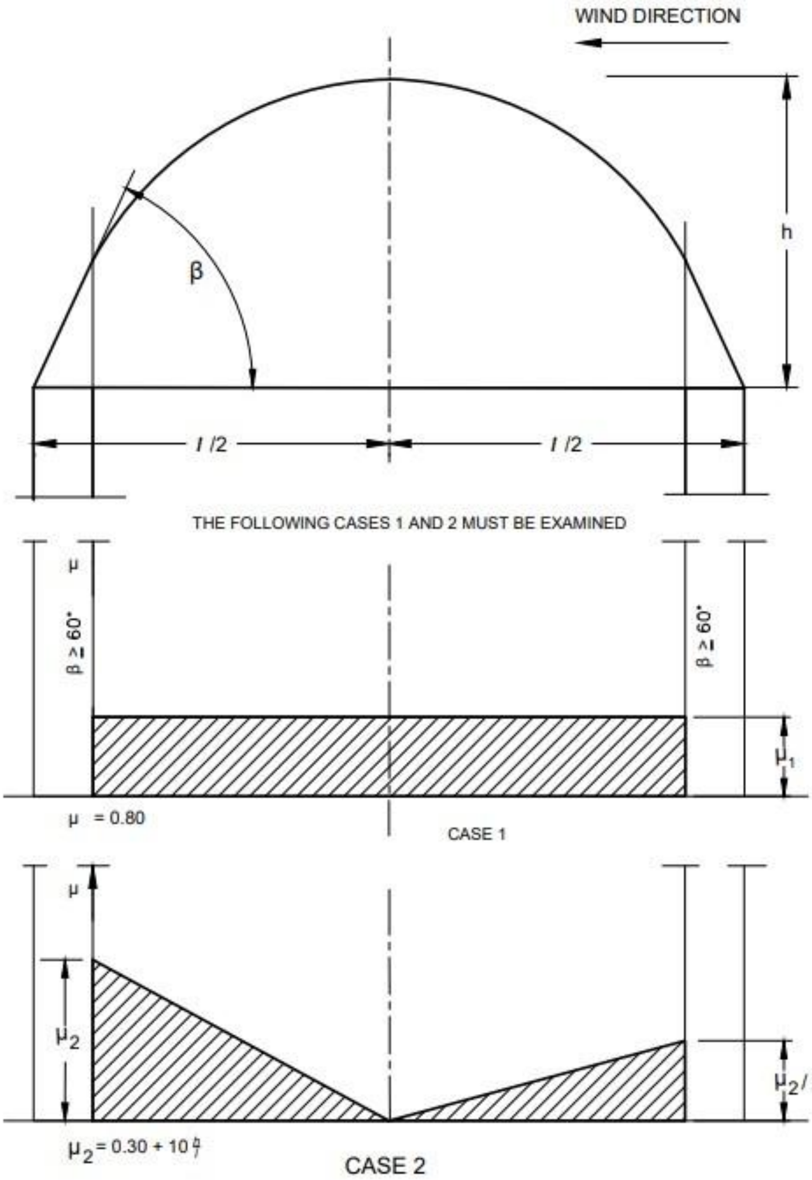
2 The values of $\mu 1$ and $\mu 2$ shall be reversed in case the wind direction changes from left to right for simple pitched roof.

7.4.2.2 *Multiple pitched roofs*

	 Simple on Multiple Pitched Roofs (Negative Roof Slope)	 Two – Span or Multispan Roofs
$0° < \beta \leq 30°$	$\mu_2 = 0.80\left(\frac{30 + \beta}{30}\right)$ $\mu_1 = 0.80$	$\mu_2 = 0.80\left(\frac{30 + \beta}{30}\right)$ $\mu_1 = 0.80$
$30° < \beta \leq 60°$	$\mu_1 = 1.60$ $\mu_1 = 0.80\left(\frac{60 - \beta}{30}\right)$	$\mu_2 = 1.60$ $\mu_1 = 0.80\left(\frac{60 - \beta}{30}\right)$

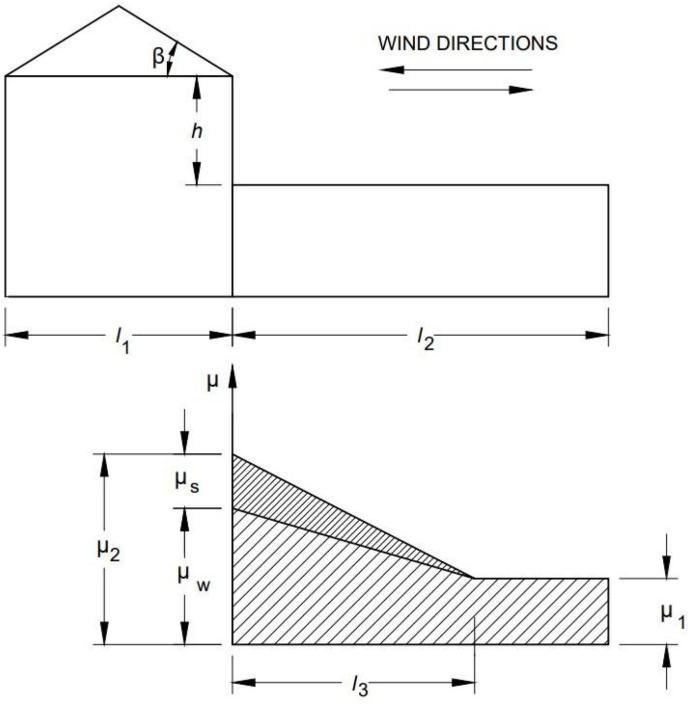
$\beta > 60°$	$\mu_1 = 1.60$ $\mu_1 = 0$	$\mu_2 = 1.60$ $\mu_1 = 0$

7.4.2.3 *Simple curved roofs*



Restrictions:
$\mu_2 \leq 2.3$
$\mu = 0\ if\ \beta \geq 60^o$

7.4.2.4 *Multilevel roofs*[1)]



$$\mu_1 = 0.80$$

$$\mu_2 = \mu_s + \mu_w$$

where

μ_s = due to sliding; and

μ_w = due to wind.

$l_3 = 2h$ [2)] but is restricted as follows:

$$5\ m \leq l_3 \leq 15\ m$$

$$\mu_w = \frac{l_1 + l_2}{2h} \leq \frac{kh}{s_0} \text{ with the restriction } 0.80 \leq \mu_w \leq 4.0$$

where

h is in metres

s_0 is in kPa (kN/m^2)

$k = 2$ kN/m^2

NOTES

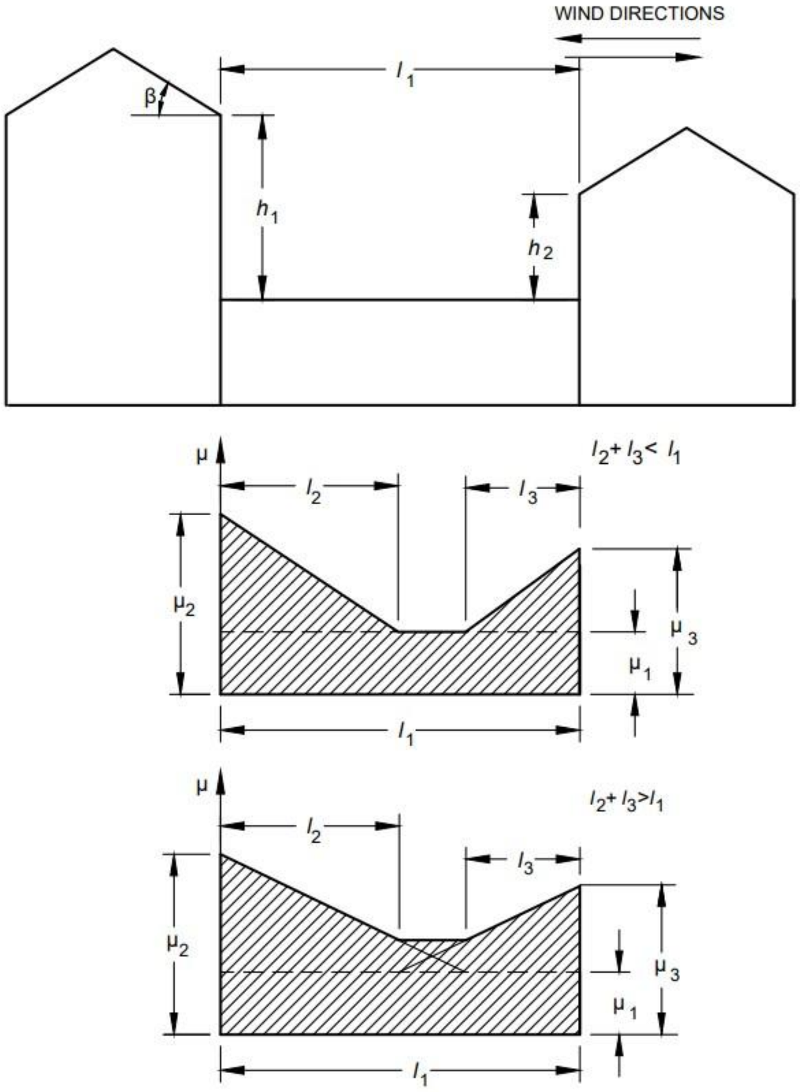
1 A more extensive formula for μ_w is described in Annex J.

2 If $l_2 < l_3$, the coefficient μ is determined by interpolation between μ_1 and μ_2.

If $\beta > 15^\circ$: μ_s is determined from an additional load amounting to 50 percent of the maximum total load on the adjacent slope of the upper roof and is distributed linearly as shown in the figure. The load on the upper roof is calculated according to 7.4.2.1 or 7.4.2.2.

If $\beta \leq 15^\circ$: $\mu_s = 0$

7.4.2.5 *Complex multilevel roofs*



$l_2 = 2h_1$; $l_3 = 2h_2$; $\mu_1 = 0.80$;

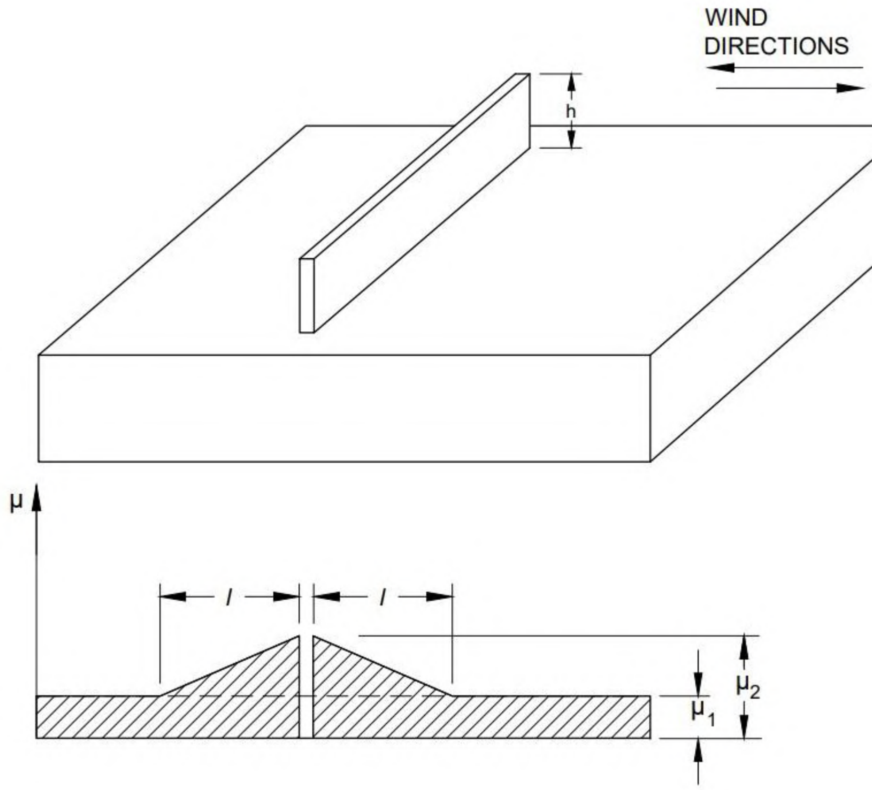
Restrictions:

$5\,m \leq l_2 \leq 15\,m$

$5\,m \leq l_3 \leq 15\,m$

μ_2 and $\mu_3 = \mu_s + \mu_w$, are calculated according to **7.4.2.1**, **7.4.2.2** and **7.4.2.4**

7.4.2.6 *Roofs with local projections and obstructions*



$$\mu_2 = \frac{kh}{s_0}$$

$$\mu_1 = 0.80$$

where

h is in metres

s_0 is in kPa $(\mathrm{kN/m^2})$

$k = 2\,\mathrm{kN/m^2}$

$l = 2\,h$

Restriction:

$0.80 \leq \mu_2 \leq 2.0$

$5\,m \leq l \leq 15\,m$

7.4.3 *Shape Coefficients in Areas Exposed to Wind*

The shape coefficients given in **7.4.2** and Annex J may be reduced by 15 percent, provided the designer has demonstrated that the following conditions are fulfilled:

a) The building is located in an exposed location such as open level terrain with only scattered buildings, trees or other obstructions so that the roof is exposed to the winds on all sides and is not likely to become shielded in the future by obstructions higher than the roof within a distance from the building equal to ten times the height of the obstruction above the roof level; and

b) The roof does not have any significant projections such as parapet walls which may prevent snow from being blown off the roof.

NOTE — In some areas, winter climate may not be of such a nature as to produce a significant reduction of roof loads from the snow load on the ground. This typically occurs in winter calm valleys in mountainous regions, where layers of snow may accumulate on roofs without any appreciable removal by wind, as well as in relatively higher temperature areas where the maximum snow load may result from a single snowstorm, occasionally without significant wind removal. In such conditions, the determination of shape coefficients should be based on local experience, with due consideration given to the likelihood of wind drifting and sliding.

8 SPECIAL LOADS

8.1 This clause gives guidance on loads and load effects due to temperature changes, soil and hydrostatic pressures, internally generating stresses (due to creep, shrinkage, differential settlement, etc), accidental loads, etc, to be considered in the design of buildings as appropriate. This clause also includes guidance on load combinations. The nature of loads to be considered for a particular situation is to be based on engineering judgment (*See also* **3.6**)

8.2 Temperature Effects

8.2.1 Expansion and contraction due to changes in temperature of the materials of a structure shall be considered in design. Provision shall be made either to relieve the stress by the provision of expansion/contraction joints in accordance with good practice [C-1(10)] or design the structure to carry additional stresses due to temperature effects as appropriate to the problem.

8.2.1.1 The temperature range varies for different regions and under different diurnal and seasonal conditions. The absolute maximum and minimum temperature which may be expected in different localities in the country are indicated in good practice [C-1(11)]. These figures may be used for guidance in assessing the maximum variations of temperature.

8.2.1.2 The temperatures indicated are the air temperatures in the shade. The range of variation in temperature of the building materials may be appreciably greater or less than the variation of air temperature and is influenced by the condition of exposure and the rate at which the materials composing the structure absorb or radiate heat. This difference in temperature variations of the material and air should be given due consideration.

8.2.1.3 The structural analysis must take account of changes of the mean (through the section) temperature in relation to the initial (st) and the temperature gradient through the section:

a) it should be borne in mind that the changes of mean temperature in relation to the initial are liable to differ as between one structural element and another in buildings or structures, as for example, between the external walls and the internal elements of a building. The distribution of temperature through section of single-leaf structural elements may be assumed linear for the purpose of analysis; and

b) the effect of mean temperature changes t_1 and t_2 and the temperature gradients v_1 and v_2 in the hot and cold seasons for single-leaf structural elements shall be evaluated on the basis of analytical principles.

NOTES

1 For portions of the structure below ground level, the variation of temperature is generally insignificant. However, during the period of construction, when the portions of the structure are exposed to weather elements, adequate provision should be made to encounter adverse effects, if any.

2 If it can be shown by engineering principles, or if it is known from experience, that neglect of some or all the effects of temperature do not affect the structural safety and serviceability, they need not be considered in design.

8.3 Hydrostatic and Soil Pressure

8.3.1 In the design of structures or parts of structures below ground level such as retaining walls and other walls in basement floors, the pressure exerted by the soil or water or both shall be duly accounted for on the basis of established theories. Due allowance shall be made for possible surcharge from stationary or moving loads. When a portion or whole of the soil is below the free water surface, the lateral earth pressure shall be evaluated for weight of soil diminished by buoyancy and the full hydrostatic pressure.

8.3.1.1 All foundation slabs and other footings subjected to water pressure shall be designed to resist a uniformly distributed uplift equal to the full hydrostatic pressure. Checking of overturning of foundation under submerged condition shall be done considering buoyant weight of foundation.

8.3.2 While determining the lateral soil pressure on column like structural members such as pillars which rest in sloping soils, the width of the member shall be taken as follows (*see* Fig. 11).

Sl No. (1)	*Actual Width of Member* (2)	*Ratio of Effective Width to Actual Width* (3)
i)	Less than 0.5 m	3.0
ii)	Beyond 0.5 m and up to 1 m	3.0 to 2.0
iii)	Beyond 1 m	2.0

The relieving pressure of soil in front of the structural member concerned may generally not be taken into account.

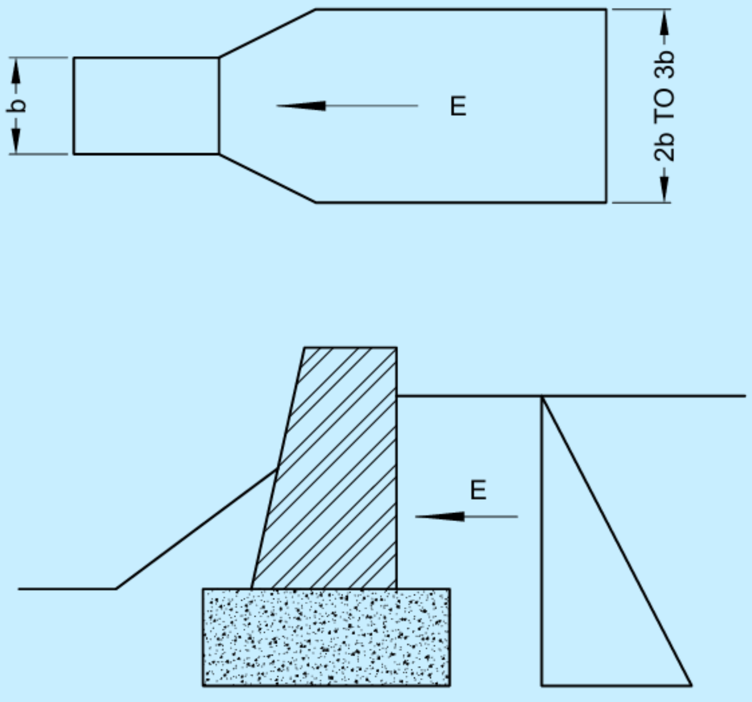


FIG. 11 SKETCH SHOWING EFFECTIVE WIDTH OF PILLAR FOR CALCULATING SOIL PRESSURE

8.3.3 Safe-guarding of structures and structural members against overturning and horizontal sliding shall be verified. Imposed loads having favourable effect shall be disregarded for the purpose. Due consideration shall be given to the possibility of soil being permanently or temporarily removed.

8.4 Fatigue

8.4.1 *General*

Fatigue cracks are usually initiated at points of high stress concentration. These stress concentrations may be caused by or associated with holes (such as bolt or rivet holes in steel structures), welds including stray or fusions in steel structures, defects in materials and local and general changes in geometry of members. The cracks usually propagate if the loading is cyclic and repetitive.

Where there is such cyclic and repetitive loading, sudden changes of shape of a member or part of a member, especially in regions of tensile stress and/or local secondary bending, shall be avoided. Suitable steps shall be taken to avoid critical vibrations due to wind and other causes.

8.4.2 Where necessary, permissible stresses shall be reduced to allow for the effects of fatigue. Allowance for fatigue shall be made for combinations of stresses due to dead load and imposed load. Stresses due to wind and earthquakes may be ignored, when fatigue is being considered, unless otherwise specified in relevant Codes of practice.

Each element of the structure shall be designed for the number of stress cycles of each magnitude to which it is estimated that the element is liable to be subjected during the expected life of the structure. The number of cycles of each magnitude shall be estimated in the light of available date regarding the probable frequency of occurrence of each type of loading.

NOTE — Apart from the general observations made herein, the section is unable to provide any precise guidance in estimating the probabilistic behaviour and response of structures of various types arising out of repetitive loading approaching fatigue conditions in structural members, joints, materials, etc.

8.5 Structural Safety During Construction

All loads required to be carried by the structures or any part of it due to storage or positioning of construction materials and erection equipment including all loads due to operation of such equipment, shall be considered as erection loads. Proper provision shall be made, including temporary bracings, to take care of all stresses due to erection loads. The conjunction with the temporary bracings shall be capable of sustaining these erection loads, without exceeding the permissible stresses specified in respective Codes of practice. Dead load, wind load and such parts of imposed load, as would be imposed on the structure during the period of erection, shall be taken as acting together with erection loads.

8.6 Accidental Loads

The occurrence of which, with a significant value, is unlikely on a given structure over the period of time under consideration and also in most cases, is of short duration. The occurrence of an accidental load could, in many cases, be expected to cause severe consequences, unless special measures are taken.

The accidental loads arising out of human action include the following:

a) Impacts and collisions;
b) Explosions; and
c) Fire.

Characteristic of the above stated loads are that they are not a consequence of normal use and that they are undesired and that extensive effects are made to avoid them. As a result, the probability of occurrence of an accidental load is small whereas the consequences may be severe.

The causes of accidental loads may be:

a) inadequate safety of equipment (due to poor design or poor maintenance); and
b) wrong operation (due to insufficient teaching or training, indisposition, negligence or unfavourable external circumstances).

In most cases, accidental loads only develop under a combination of several unfavourable occurrences. In practical applications, it may be necessary to neglect the most unlikely loads. The probability of occurrence of accidental loads, which are neglected, may differ for different consequences of a possible failure. A data base for a detailed calculation of the probability will seldom be available.

8.6.1 *Impact and Collisions*

8.6.1.1 *General*

During an impact, the kinetic impact energy has to be absorbed by the vehicle hitting the structure and by the structure itself. In an accurate analysis, the probability of occurrence of an impact with a certain energy object hitting the structure and the structure itself at the actual place must be considered. Impact energies for dropped object should be based on the actual loading capacity and lifting height.

Common sources of impact are:

a) vehicles;

b) dropped objects from cranes, fork lifts, etc;

c) cranes out of control, crane failures; and

d) flying fragments.

The codal requirements regarding impact from vehicles and cranes are given in **8.6.1.2** and **8.6.1.3**.

8.6.1.2 The requirements for loads as a result of collision between vehicle and structural or non-structural elements shall be as per **8.6.1.2.1** and **8.6.1.2.2**.

8.6.1.2.1 *Collisions between vehicles and structural elements*

In road traffic, the requirement that a structure shall be able to resist collision may be assumed to be fulfilled, if it is demonstrated that the structural element is able to stop a fictitious vehicle, as described below. It is assumed that the vehicle strikes the structural element at a height of 1.2 m in any possible direction and at a speed of 10 m/s (36 km/h).

The fictitious vehicle shall be considered to consist of two masses m_1 and m_2 which during compression of the vehicle, produce an impact force increasing uniformly from zero, corresponding to the rigidities C_1 and C_2. It is assumed that the mass m_1 is broken completely before the breaking of mass m_2 begins.

The following numerical values should be used:

a) m_1 = 400 kg, C_1 = 10 000 kN/m, the vehicle is compressed; and

b) m_2 = 12 000 kg, C_2 = 300 kN/m, the vehicle is compressed.

NOTE — The described fictitious collision corresponds in the case of a non-elastic structural element to a maximum static force of 630 kN for the mass m_1 and 600 kN for the mass m_2 irrespective of the elasticity, it will therefore be on the safe side to assume the static force to be 630 kN.

In addition, breaking of the mass m_1 will result in an impact wave, the effect of which will depend, to a great extent, on the kind of structural element concerned. Consequently, it will not always be sufficient to design for the static force.

8.6.1.2.1.1 *For structural elements in stilts and car parking levels*

For structural elements in stilts and car parking levels, where driveways and car parking are provided in the buildings which have limited vehicle speeds, the design shall take care of the following considerations:

a) Speed of the vehicle to be 20 km/h maximum (since within a building and inside the car parks, speed of the vehicle may not generally be more than 20 km/h); and

b) Impact to be considered at a height of 450 mm to 600 mm.

Structural elements such as columns, RC walls or vertical load resisting elements shall be designed for loads due to the above unless protected by separate resisting system.

8.6.1.2.2 *Collision between vehicles and non-structural elements:*

Non-structural elements, such as vehicle barriers for car parks shall be designed to resist the impact and the momentum as outlined below:

a) The horizontal force F (in kN), normal to and uniformly distributed over any length of 1.5 m of a barrier for a car park, required to withstand the impact of a vehicle shall be as given below:

$$F = \frac{0.5mv^2}{\delta_c + \delta_b}$$

where

m = gross mass of the vehicle, in kg;

v = velocity of the vehicle normal to the barrier, in m/s;

δ_c = deformation of the vehicle, in mm; and

δ_b = deflection of the barrier, in mm.

b) Where the car park has been designed on the basis that the gross mass of the vehicles using it will not exceed 2500 kg, the following values shall be used to determine the force F:

m = 1500 kg;
v = 4.5 m/s;
δ_c = 100 mm;

For a rigid barrier, for which δ_b may be taken as zero, the force, F, appropriate to vehicles up to 2500 kg gross mass, shall be taken as 150 kN.

c) Where the car park has been designed for vehicles whose gross mass exceeds 2500 kg, the following values shall be used to determine the force F:

m = actual mass of the vehicle for which the car park is designed, in kg;

v = 4.5 m/s;

δ_c = 100 mm;

d) The force determined as in (b) or (c) may be considered to act at bumper height. In the case of car parks intended for motor cars whose gross mass does not exceed 2500 kg, this height may be taken as 375 mm above the floor level.

e) Barriers to access ramps of car parks have to withstand one-half of the force determined in (b) or (c) acting at a height of 610 mm above the ramp.

Opposite to the ends of straight ramps intended for downward travel, which exceed 20 m in length, the barrier shall be designed to withstand twice the force determined in (b) or (c) acting at a height of 610 mm above the ramp.

NOTE – The mass of 1500 kg is taken as being more representative of the vehicle population than the extreme value of 2500 kg.

8.6.1.3 *Safety railings*

With regard to safety, railings put up to protect structures against collision due to road traffic, it should be shown that the railings are able to resist the impact as described in 8.6.1.2.

NOTE — When a vehicle collides with safety railings, the kinetic energy of the vehicle will be absorbed partly by the deformation of the railings and partly by the deformation of the vehicle. The part of the kinetic energy which the railings should be able to absorb without breaking down may be determined on the basis of the assumed rigidity of the vehicle during compression.

8.6.1.4 *Crane impact load on buffer stop*

The basic horizontal load P_y (tonne), acting along the crane track produced by impact of the crane on the buffer stop, is calculated by the following formula:

$$P_y = MV^2/F$$

where

V = speed at which the crane is traveling at the moment of impact (assumed equal to half the nominal value) (m/s).

F = maximum shortening of the buffer, assumed equal to 0.1 m for light duty, mediuL-duty and heavy-duty cranes with flexible load suspension and loading capacity not exceeding 50 t and 0.2 m in every other cranes.

M = reduced crane mass, (t.s^2/m); and is obtained by the formula:

$$M = \frac{1}{g}\left[\frac{P_h}{2} + \left(P_t + kQ\right)\frac{L_k - l}{L_k}\right]$$

where

P_h = crane bridge weight (t);

P_t = crab bridge weight (t);

k = coefficient, assumed equal to zero for cranes with flexible load suspension and to one for cranes with rigid suspension;

Q = crane loading capacity (t);

L_k = crane span (m); and

ℓ = nearness of crab (m); and

g = acceleration due to gravity (9.81 m/s^2).

8.6.2 *Explosions*

8.6.2.1 *General*

Explosions may cause impulsive loading on a structure. The following types of explosions are particularly relevant:

a) Internal gas explosions which may be caused by leakage of gas piping (including piping outside the room), evaporation from volatile liquids or unintentional evaporation from surface material (for example, fire);

b) Internal dust explosions;

c) Boiler failure;

d) External gas cloud explosions; and

e) External explosions of high explosives (TNT, dynamite).

The codal requirement regarding internal gas explosions is given in 8.6.2.2.

8.6.2.2 *Explosion effect in closed rooms*

Gas explosion may be caused, for example by leaks in gas pipes (inclusive of pipes outside for room), evaporation from volatile liquids or unintentional evaporation of gas from wall sheathings (for example, caused by fire).

NOTES

1 The effect of explosions depends on the exploding medium, the concentration of the explosion, the shape of the room, possibilities of ventilation of the explosion and the ductility and dynamic properties of the structure. In rooms with little possibility for relief of the pressure from the explosion, very large pressures may occur.

2 Internal over pressure from an internal gas explosion in rooms of sizes comparable to residential rooms and with ventilation areas consisting of window glass breaking at a pressure of 4 kN/m^2 (3 mm to 4 mm machine made glass) may be calculated from the following method:

a) The over pressure is assumed to depend on a factor A/V, where A is the total windows area, in m^2 and V is the volume, in m^3, of the room considered;

b) The internal pressure is assumed to act simultaneously upon all walls and floors in one closed room; and

c) The action q_o may be taken as static action.

3 If account is taken of the time curve of the action, the schematic correspondence between pressure and time is assumed (*see* Fig. 12), where t_1 is the time from the start of combustion until maximum pressure is reached and t_2 is the time from maximum pressure to the end of combustion. For t_1 and t_2, the most unfavourable values should be chosen in relation to the dynamic properties of the structures. However, the values should be chosen within the intervals as given in Fig. 13.

Figure 12 is based on tests with gas explosions in room corresponding to ordinary residential flats and should, therefore, not be applied to considerably different conditions. The figure corresponds to an explosion caused by town gas and it might, therefore, be somewhat on the safe side in rooms where there is only the possibility of gases with a lower rate of combustion.

The pressure may be applied solely in one room or in more rooms at the same time. In the latter case, all rooms are incorporated in the volume V. Only windows or other similarly weak and light weight structural elements may be taken to be ventilation areas even though certain limited structural parts break at pressures less than q_o.

Figure 12 is given purely as guide and probability of occurrence of an explosion should be checked in each case using appropriate values.

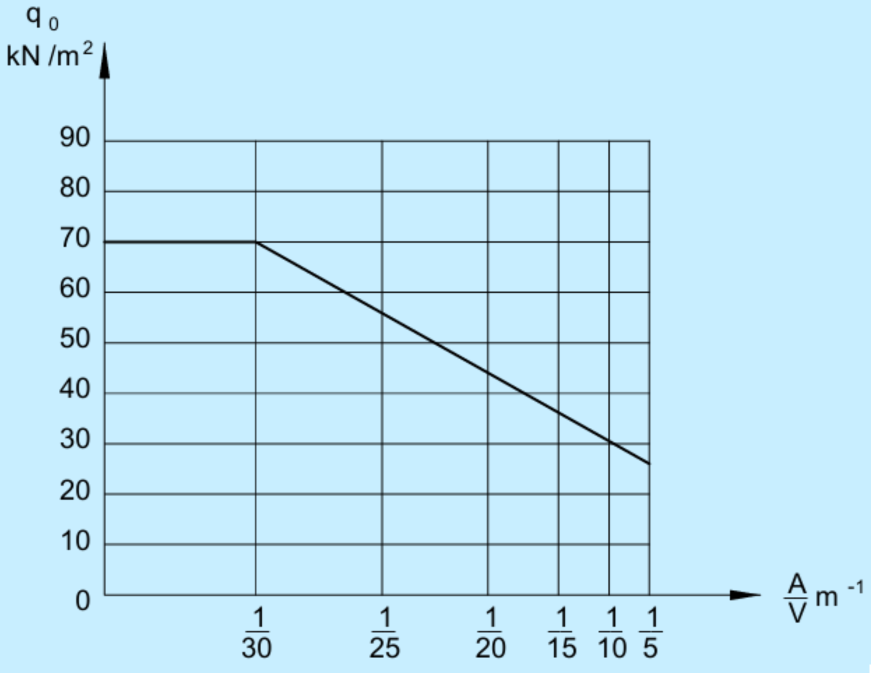


FIG. 12 SKETCH SHOWING RELATION BETWEEN PRESSURE AND TIME

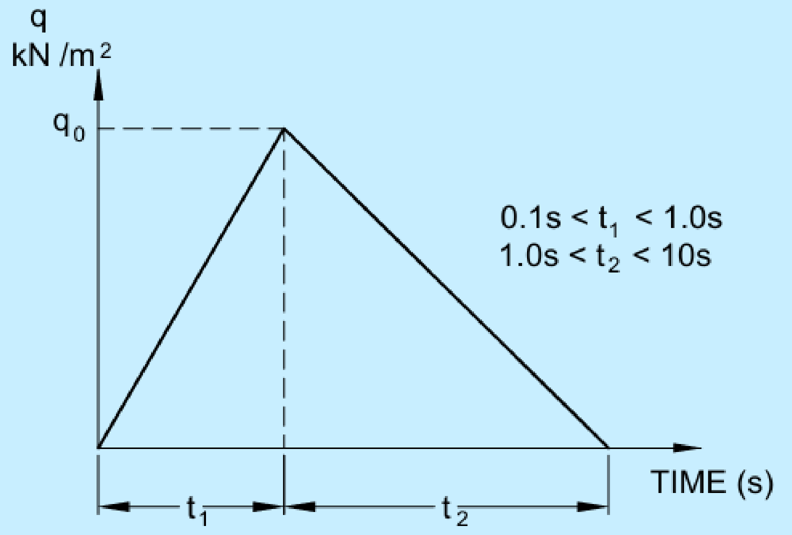


FIG. 13 SKETCH SHOWING TIME INTERVAL AND PRESSURE

8.6.3 *Vertical Load on Air Raid Shelters*

8.6.3.1 *Characteristic values*

As regards buildings in which the individual floors are acted upon by a total characteristic imposed action of up to 5.0 kN/m^2, vertical actions on air raid shelters generally located below ground level, for example, basement, etc, should be considered to have the following characteristic values:

a)	Buildings with up to 2 storeys	:	28 kN/m^2
b)	Buildings with 3-4 storeys	:	34 kN/m^2
c)	Buildings with more than 4 storeys	:	41 kN/m^2
d)	Buildings of particularly stable construction irrespective of the number of storeys	:	28 kN/m^2

In the case of buildings with floors that are acted upon by a characteristic imposed action larger than 5.0 kN/m^2, the above values should be increased by the difference between the average imposed action on all storeys above the one concerned and 5.0 kN/m^2.

NOTES

1 By storeys it is understood, every utilizable storey above the shelter.

2 By buildings of a particular stable construction, it is understood, buildings in which the load- bearing structures are made from reinforced *in-situ* concrete.

8.6.4 *Fire*

8.6.4.1 *General*

Possible extraordinary loads during a fire may be considered as accidental actions. Examples are loads from people along escape routes and loads on another structure from structure failing because of a fire.

8.6.4.2 *Thermal effects during fire*

The thermal effect during fire may be determined from one of the following methods:

a) Time-temperature curve and the required fire resistance (in min); and

b) An energy balance method.

If the thermal effect during fire is determined from an energy balance method, the fire load is taken to be:

$$q = 12\ t_b$$

where

q = fire action (KJ/m^2 floor); and

t_b = required fire resistance (in min) {*see* [C-1(12)]}.

NOTE — The fire action is defined as the total quantity of heat produced by complete combustion of all combustible material in the fire compartment, inclusive of stored goods and equipment together with building structures and building materials.

8.7 Vibrations

For general details on loads due to vibrations, reference may be made to Annex K.

8.8 Blast Loads

For provisions related to blast loads, *see* Annex L.

8.9 Other Loads

Other loads not included in this Section, such as special loads due to technical process, moisture and shrinkage effects, etc, should be taken into account where stipulated by building design codes or established in accordance with the performance requirement of the structure.

8.10 Landslide Control Measures

Mountainous and hilly terrains are prone to slope instability arising from geological, hydrological and seismic factors, further aggravated by unplanned developmental activities. The following good practices address the specific aspects towards hazard assessment, design and construction:

Sl No.	*Aspects*	*Accepted Standard(s)*
(1)	(2)	(3)
i)	Guidelines for preparation of landslide hazard zonation maps	[C-1(13)]
ii)	Preparation of landslide risk assessment maps, combining hazard probability with potential damage through a risk assessment matrix	[C-1(13)]
iii)	Site-specific investigation and stability analysis, including geological mapping, subsurface exploration, laboratory testing and numerical modelling	[C-1(13)]
iv)	Landslide control measures with emphasis on drainage management, slope modification, reinforcement, bio-engineering and protective works	[C-1(13)]
v)	Design and construction of different types of retaining walls	[C-1(13)]
vi)	Micropiles for slope stabilization	[C-1(13)]
vii)	Surface water management in hilly areas	[C-1(13)]
viii)	Siting, design and selection of materials for residential buildings in hilly areas	[C-1(13)]

8.11 Cyclone Resistant Buildings

Cyclone-prone regions are vulnerable to high wind speeds, wind-borne debris and storm surges, which can cause extensive damage to buildings, particularly low-rise and kutcha houses. The following good practices address aspects of loads, design, construction and assessment for improving resilience:

Sl No.	*Aspect*	*Accepted Standard*
(1)	(2)	(3)
i)	Guidelines for improving the cyclonic resistance of low-rise houses and other buildings	[C-1(14)]
ii)	Guidelines for survey of housing and building typology in cyclone prone areas for assessment of vulnerability of regions and post-cyclone damage estimation	[C-1(14)]
iii)	Guidelines for design and construction of cyclone shelters	[C-1(14)]
iv)	Method for determination of performance relating to safety of external building envelope impacted by wind-borne debris	[C-1(14)]
v)	Guidelines for improving the cyclonic resistance capacity of kutcha houses	[C-1(14)]

9 LOAD COMBINATIONS

9.1 General

A judicious combination of the loads keeping in view the probability of:

a) their acting together; and

b) their disposition in relation to other loads and severity of stresses or deformations caused by the combinations of the various loads is necessary to ensure the required safety and economy in the design of a structure.

9.2 Load Combinations

Keeping the aspect specified in 9.1, the various loads should, therefore, be combined in accordance with the stipulation in the relevant design Codes. In the absence of such recommendations, the following loading combinations, whichever combination produces the most unfavourable effect in the building, foundation or structural member concerned may be adopted (as a general guidance). It should also be recognized in load combinations that the simultaneous occurrence of maximum values of wind, earthquake, imposed and snow loads is not likely.

a) *DL*;

b) *DL* and *IL*;

c) *DL* and W*L*;

d) *DL* and E*L*;

e) *DL* and T*L*;

f) *DL*, *IL* and *WL*;

g) *DL*, *IL* and *EL*;

h) *DL*, *IL* and *TL*

j) *DL*, W*L* and *TL*;

k) *DL*, *EL* and *TL*;

m) *DL*, *IL*, *WL* and *TL*; and

n) *DL*, *IL*, *EL* and *TL*

(*DL* = dead load, *IL* = imposed load, *WL* = wind load, *EL* = earthquake load and *TL* = temperature load).

NOTES

1 When snow load is present on roofs, replace imposed load by snow load for the purpose of above load combinations.

2 The relevant design codes shall be followed for permissible stresses when the structure is designed by working stress method and for partial safety factors when the structure is designed by limit state design method for each of the above load combinations.

3 Whenever imposed load (*IL*) is combined with earthquake load (*EL*), the appropriate part of imposed load as specified in **5** should be used, both for evaluating earthquake effect and also for combined load effects used in such combination.

4 For the purpose of stability of the structure as a whole against overturning, the restoring moment shall be not less than 1.2 times the maximum overturning moment due to dead load plus 1.4 times the maximum overturning moment due to imposed loads. In cases where dead load provides the restoring moment, only 0.9 times the dead load shall be considered. The restoring moments due to imposed loads shall be ignored. In case of high water table, the effects of buoyancy have to be suitably taken into consideration.

5 In case of high water able, the factor of safety of 1.2 against uplift alone shall be provided.

6 The structure shall have a factor against sliding of not less than 1.4 under the most adverse combination of the applied loads/forces. In this case, only 0.9 times the dead load shall be taken into account.

7 Where the bearing pressure on soil due to wind alone is less than 25 percent of that due to dead load and imposed load, it may be neglected in design where this exceeds 25 percent, foundation may be so proportioned that the pressure due to combined effect of dead load, imposed load and wind load does not exceed the allowable bearing pressure by more than 25 percent. When earthquake effect is included, the permissible increase in allowable bearing pressure in the soil shall be in accordance with **5**.

8 Reduced imposed load specified in 3 for the design of supporting structures should not be applied in combination with earthquake forces.

9 Other loads and accidental load combination not included should be dealt with appropriately.

10 Crane load combinations are covered in **3.6.4**.

10 MULTI-HAZARD RISK IN VARIOUS DISTRICTS OF INDIA

10.1 Multi-hazard Risk Concept

The commonly encountered hazards are:

a) Earthquake;

b) Cyclone;

c) Wind storm;

d) Floods;

e) Landslides;

f) Liquefaction of soils;

g) Extreme winds;

h) Cloud bursts; and

j) Failure of slopes.

A study of the earthquake, wind/cyclone and flood hazard maps of India indicate that there are several areas in the country which run the risk of being affected by more than one of these hazards.

Further there may be instances where one hazard may cause occurrence or accentuation of another hazard such as landslides may be triggered/accelerated by earthquakes and wind storms and floods by the cyclones.

It is important to study and examine the possibility of occurrence of multiple hazards, as applicable to an area. However, it is not economically viable to design all the structures for multiple hazards. The special structures such as nuclear power plants and life line structures such as hospitals and emergency rescue shelters may be designed for multiple hazards. For such special structures, site specific data have to be collected and the design be carried out based on the accepted levels of risk. The factors that have to be considered in determining this risk are:

a) The severity of the hazard characterized by M.M. (or M.S.K.) intensity in the case of earthquake; the duration and velocity of wind in the storms; and unprotected or protected situation of flood prone areas; and

b) The frequency of occurrence of the severe hazards.

Till such time that risk evaluation procedures are formalized, the special structures may be designed for multiple hazards using the historical data, that can be obtained for a given site and the available code for loads already covered. The designer may have to consider the loads due to any one of the hazards individually or in combination as appropriate.

10.2 Multi-hazard Prone Areas

10.2.1 At a site, more than one hazard may be prominent. In general, the natural hazards are not considered to act together, but need to be considered one at a time. Also, the proportion of combining the man-made hazards with natural hazards also are specified in the NBC. The NBC provides a minimum set of combinations of loads that need to be considered in the design of structures along with the corresponding partial safety factor of the loads in each of the said combinations.

10.2.2 The criteria adopted for identifying multi hazard prone areas may be as follows:

a) *Earthquake and flood risk prone* — Districts which have seismic zone of intensity 7 or more and also flood prone unprotected or protected area. Earthquake and flood can occur separately or simultaneously.

b) *Cyclone and flood risk prone* — Districts which have cyclone and flood prone areas. Here floods can occur separately from cyclones, but simultaneous also along with possibility of storm surge too.

c) *Earthquake, cyclone and flood risk prone* — Districts which have earthquake zone of intensity 7 or more, cyclone prone as well as flood prone (protected or unprotected) areas. Here the three hazards can occur separately and also simultaneously as in (a) and (b) above but earthquake and cyclone will be assumed to occur separately only.

d) *Earthquake and cyclone risk prone* — Districts which have earthquake zone of intensity 7 or more and prone to cyclone hazard too. The two will be assumed to occur separately.

Based on the approach given above, the districts with multi-hazard risk are given in Annex M.

10.3 Use of the List of the District with Multi-Hazard Risk

The list provides some ready information for use of the authorities involved in the task of disaster mitigation, preparedness and preventive action. This information gives the districts which are prone to high risk for more than one hazard. This information will be useful in establishing the need for developing housing design to resist such multi-hazard situation.

11 DOCUMENTATION

11.1 For the successful planning, design and construction of buildings, adequate documentation covering structural design sufficiency, sub-surface investigations, supervision, etc may be required. In particular, compliance with the structural safety requirements of this NBCS shall be demonstrated through appropriate documentation, including the preparation of a Structural Design Basis Report (SDBR), which establishes the basis and assumptions adopted in the structural design.

11.2 Structural Design Basis Report (SDBR)

The Structural design basis report (SDBR) is a document that establishes the basis and assumptions adopted for the structural design of high rise and special buildings, demonstrating that the structure has been designed to ensure safety against various loads, actions and effects, including those arising from natural hazards such as earthquakes, landslides, cyclones and floods, in accordance with the provisions of this NBCS, particularly Part C 'Structural Design' and other relevant Indian Standards.

The preparation and submission of an SDBR provides transparency and traceability of design decisions and enables verification that the structural system, materials, loading criteria and analysis methods adopted are consistent with the applicable provisions. It also serves as a reference document for reviewing authorities and stakeholders and ensures that the registered engineer/structural engineer maintains adequate documentation to substantiate the structural design.

The SDBR (Annex N) sets out the basis for designing the building and consists of the following parts:

a) Part 1 — General information/Data;

b) Part 2 — Load-bearing masonry buildings;

c) Part 3 — Reinforced concrete buildings; and

d) Part 4 — Steel and steel–concrete composite buildings.

For demonstrating compliance with Part C 'Structural Design' of this NBCS and the relevant Indian Standards referenced therein, the registered engineer/structural engineer shall submit an SDBR for structures of different complexities along with the drawings and other documents required to be submitted with the application.

ANNEX A

[*Foreword*, *Clause* 2.2]

DEAD LOADS — UNIT WEIGHTS OF CONSTRUCTION MATERIAL AND STORED MATERIALS

A-1 BUILDING MATERIALS

The unit weight of materials used in construction of buildings and structures are specified in Table 40. Also mentioned alongside are the corresponding Indian Standards, if applicable

Table 40 Unit Weight of Building Materials

(*Clause* A-1)

Sl No.	Materials	Nominal Size or Thickness mm	Weight	
			kN	per
(1)	(2)	(3)	(4)	(5)
i)	**Acoustical Material (*see* IS 2526) [C-1(15)]**			
	Eelgrass	10	0.57 to 0.765×10^{-2}	m^2
	Glass fibre, min	10	0.38×10^{-2}	m^2
	Hair	10	1.91×10^{-2}	m^2
	Mineral wool	10	1.345×10^{-2}	m^2
	Slag wool		2.65	m^3
	Cork		2.35	m^3
	Wood fibre board		5.9 to 6.15	m^3
	Wood particle Board			
	As per IS 3129 [C-1(15)]		3.92	m^3
	As per IS 12823 [C-1(15)]		4.9 to 9.9	m^3
	As per IS 3478 [C-1(15)]		8.9 to 11.8	m^3
	Compressed wood wool	-	3.95 to 4.45	m^3
	Mineral/glass wool quilts and mats	-	0.157 to 0.314	m^3
	Mineral compressed glass wool tiles	-	1.87 to 2.06	m^3
	Polyester board	-	1.87 to 2.06	m^3
ii)	**Aggregate, Coarse**			
	Broken stone ballast			
	Dry, well-shaken	–	15.70 to 18.35	m^3
	Perfectly wet	–	18.85 to 21.95	m^3
	Shingles, 3 to 38 mm	–	14.35	m^3
	Broken bricks:			
	Fine	–	14.20	m^3

	Coarse	–	9.90	m^3
	Foam slag (foundry pumice)	–	6.85	m^3
	Cinder	–	7.85	m^3
	Slag			
	Iron slag		32.4	m^3
	Steel Slag		31 to 36	m^3
	Copper slag		31 to 36	m^3
	River pebbles		15	m^3
iii)	**Aggregate, Fine**			
	Sand			
	Dry, clean	–	15.10 to 15.70	m^3
	River	–	18.05	m^3
	Wet	–	17.25 to 19.60	m^3
	Brick dust (*SURKHI*)	–	9.90	m^3
	Crushed stone sand (*see* IS 383) [C-1(15)]		19 to 20	m^3
	Quarry dust		14 to 20	m^3
iv)	**Aggregate, Organic**			
	Saw dust, loose	–	1.55	m^3
	Peat			
	Dry	–	5.50 to 6.30	m^3
	Sandy, compact	–	7.85	m^3
	Wet, compact	–	13.35	m^3
v)	**Asbestos**			
	Felt	10	0.145	m^2
	Fibres:			
	Pressed	–	9.40	m^3
	Sprayed	10	0.02	m^2
	Natural	–	29.80	m^3
	Raw	–	5.90 to 8.85	m^3
vi)	**Asbestos Cement Building Pipes** (*see* Sl No. xliii 'pipes' in this Table)			
vii)	**Asbestos Cement Gutters** [*see* IS 1626 (Part 2)] [C-1(15)]			
	Boundary wall gutters			
	510 mm × 150 mm × 255 mm	12.5	0.180	m
	455 mm × 150 mm × 305 mm	12.5	0.158	m
	305 mm × 150 mm × 225 mm	12.5	0.129	m
	280 mm × 125 mm × 180 mm	12.5	0.106	m
	Valley gutters			

	915 mm × 205 mm × 230 mm	12.5	0.250	m
	610 mm × 150 mm × 230 mm	12.5	0.162	m
	455 mm × 125 mm × 150 mm	12.5	0.146	m
	405 mm × 125 mm × 255 mm	12.5	0.131	m
	Half round gutters			
	150 mm	9.5	0.043	m
	230 mm	9.5	0.073	m
	305 mm	9.5	0.088	m
viii)	**Asbestos Cement Pressure Pipes** (*see* Sl No. xliii 'Pipes' in this Table)			
ix)	**Asbestos Cement Sheeting** (*see* IS 459) [C-1(15)]			
	Corrugated (pitch – 146 mm)	6	0.118 to 0.130	m^2
	Semi-corrugated (pitch 388 mm)	6	0.118 to 0.127	m^2
	Plain	5	0.09	m^2
x)	**Bitumen**	–	0.8-1	m^3
xi)	**Binders**			
	Fly Ash loose		5.05 to 7.05	m^3
	Fly Ash compacted		16.00	m^3
	Ground Granulated blast furnace slag		14.9	m^3
	Silica fume		3.5 to 7	m^3
	Rice husk ash		1.0 to 3.5	m^3
	Metakaolin		8 to 10	m^3
xii)	**Blocks**			
	Lime-based solid blocks (*see* IS 3115) [C-1(15)], *min*	–	9.8	m^3
	concrete blocks [*see* IS 2185 (Part 1)] [C-1(15)]			
	Grade A (hollow)	–	15	m^3
	Grade B (hollow)	–	11 to 15	m^3
	Grade C (Solid)	–	18	m^3
	Autoclaved cellular concrete blocks [*see* IS 2185 (Part 3)] [C-1(15)]		4.42 to 9.81	m^3
xiii)	**Boards**			
	Cork boards [*see* IS 4253 (Part 1 and Part 2)] [C-1(15)]			
	Compressed		2.16 to 3.73	m^3
	Ordinary		2.15 to 3.72	m^3
	Fibre hard boards (*see* IS 1658) [C-1(15)]			

	Medium hardboard		3.43 to 7.84	m^3
	Standard hardboard		7.84 to 10.05	m^3
	Tempered hardboard		7.84 to 10.05	m^3
	Fibre cement boards (*see* IS 14862) [C-1(15)]		13 to 16	m^3
	Fibre insulation board, ordinary or flame-retardant type bitumen-bounded fibre insulation board (*see* IS 3348) [C-1(15)], *Max*	9, 12, 18, 25	3.924	M^3
	Plain gypsum plaster boards [*see* IS 2095 (Part 1)] [C-1(15)]			
	Type A	9.5, 12.5, 15	5.39	m3
	Type H1 and H2	9.5, 12.5, 15	5.88	m3
	Type D	9.5, 12.5, 15	7.84	m3
	Type F and FH	9.5, 12.5, 15	7.84	m3
	Type R1	9.5, 12.5, 15	8.33	m3
	Type R2	9.5, 12.5, 15	9.31	m3
	Type W	9.5, 12.5, 15	8.33	m3
	Coated/laminated gypsum plaster boards [*see* IS 2095 (Part 2)] [C-1(15)]	9.5, 12.5, 15	11.27	M^3
	Reinforced gypsum plaster boards and ceiling tiles [*see* IS 2095 (Part 3)] [C-1(15)], *min*			
	Fibre reinforced Gypsum		7.84	m^3
	Glass reinforced Gypsum plaster board		24.52	m^3
	Glass reinforced Gypsum plaster tiles		5.88	m^3
	Insulating board (*fibre*) (*see* IS 3348) [C-1(15)], *mean*		0.981	m^3
	Laminated board (*fibre*) Medium density (*see* IS 14587) [C-1(15)]		5.88 to 8.82	m^3
	Wood particle boards (*see* IS 3087) [C-1(15)]	–	4.90 to 8.85	m^3
	Low density wood particles boards for insulation purposes (*see* IS 3129) [C-1(15)]	–	3.52	m^3
	High density wood particle boards (*see* IS 3478) [C-1(15)]			
	Type 1 & 2, Grade A	–	11.30	m^3
	Type 1 & 2, Grade B	–	7.94	m^3

xiv)	**Bricks**			
	Common burnt clay bricks (*see* IS 1077) [C-1(15)]	–	15.70 to 18.85	m^3
	Engineering bricks	–	21.20	m^3
	Heavy duty bricks (*see* IS 2180) [C-1(15)]	–	24.50	m^3
	Pressed bricks	–	17.25 to 18.05	m^3
	Refractory bricks	–	17.25 to 19.60	m^3
	Sand cement bricks	–	18.05	m^3
	Sand lime bricks	–	20.40	m^3
	Pulverized fuel-ash lime bricks (*see* IS 12894) [C-1(15)]		12 to 15	m^3
	Pulverized fuel-ash cement bricks (*see* IS 16720) [C-1(15)]		10.8 to 19.6	m^3
	Acid resistance bricks (*see* IS 4860) [C-1(15)]		22.5 to 24.83	m^3
	Soil based bricks (*see* IS 1725) [C-1(15)]		17.5	m^3
	Perforated building bricks (*see* IS 2222) [C-1(15)]		6.8 to 7.75	m^3
	Calcium silicate bricks (*see* IS 4139) [C-1(15)]		25 to 29	m^3
xv)	**Brick Chips and Broken Bricks** (*see* under Sl No. ii ‘Broken Bricks’ in this Table)			
xvi)	**Brick dust (*SURKHI*)**	–	9.00	m^3
xvii)	**Cast Iron, Manhole Covers** (*see* IS 1726) [C-1(15)]			
	Double triangular (HD)	500	1.16	Cover
		560	1.37	Cover
	Circular (HD)	500	1.16	Cover
		560	1.37	Cover
	Circular (MD)	500	0.57	Cover
		560	0.63	Cover
	Rectangular (MD)	–	0.78	Cover
	Rectangular (LD) :			
	Single seal (Pattern 1)	–	0.23	Cover
	Single seal (Pattern 2)	–	0.15	Cover
	Double seal	–	0.28	Cover
	Square (LD):			
	Single seal	455	0.13	Cover
		610	0.25	Cover
	Double seal	455	0.23	Cover
		610	0.36	Cover

xviii)	**Cast Iron, Manhole Frames** (*see* IS 1726) [C-1(15)]			
	Double triangular (HD)	500	1.09	Frame
		560	1.13	Frame
	Circular (HD)	500	0.83	Frame
		560	1.06	Frame
	Circular (MD)	500	0.57	Frame
		560	0.63	Frame
	Rectangular (MD)	–	0.63	Frame
	Rectangular (LD):			
	Single seal (Pattern 1)	–	0.15	Frame
	Single seal (Pattern 2)	–	0.10	Frame
	Double seal	–	0.23	Frame
	Square (LD):			
	Single seal	455	0.07	Frame
		610	0.13	Frame
	Double seal	455	0.15	Frame
		610	0.18	Frame
xix)	**Cast Iron Pipes** (*see* under Sl No. xliii 'Pipes' in this Table)			
xx)	**Cement**			
	Ordinary (*see* IS 269) and other types	–	14.10	m^3
	Rapid-hardening (*see* IS 8041) [C-1(15)]	–	12.55	m^3
xxi)	**Ceilings**			
	Plaster on tile or concrete	13 mm thick	0.25	m^2
	Plaster on wood lath	25 mm thick	0.39	m^2
	Suspended metal lath and cement plaster	25 mm thick	0.74	m^2
	Suspended metal lath and gypsum plaster	25 mm thick	0.49	m^2
xxii)	**Cement Concrete, Plain**			
	Aerated		7.45	m^3
	No-fines, with heavy aggregate	–	15.70 to 18.80	m^3
	No-fines, with light aggregate	–	8.65 to 12.55	m^3
	With burnt clay aggregate	–	17.25 to 21.20	m^3
	With expanded clay aggregate	–	9.40 to 16.50	m^3
	With clinker aggregate	–	12.55 to 17.25	m^3
	With pumice aggregate	–	5.50 to 11.00	m^3
	With sand and gravel or crushed natural stone aggregate	–	22.00 to 23.50	m^3

	With saw dust	–	6.30 to 16.50	m^3
	With foamed slag aggregate	–	9.40 to 18.05	m^3
	With light weight cinder aggregate		6.4	m^3
	Light weight concrete		8 to 20	m^3
	Cellular concrete (Foam concrete) (*see* IS 6598) [C-1(15)]		3 to 5	m^3
	Grade A, *max*		3.13	m^3
	Grade B		3.13 to 3.92	m^3
	Grade C		3.93 to 4.9	m^3
	Transparent concrete		15	m^3
	Glass fiber reinforced concrete		18 to 22	m^3
	Alkali activated (Geopolymer) concrete (*see* IS 17452) [C-1(15)]		24 to 25	m^3
	Shotcrete (*see* IS 9012) [C-1(15)]		22 to 25	m^3
	Ferro cement		22 to 25	m^3
	Roller compacted concrete		22 to 25	m^3
	Controlled low strength mix		22 to 25	m^3
	Plum concrete (*see* IS 457) [C-1(15)]		22 to 25	m^3
	Autoclaved Aerated Concrete		4.42 to 9.81	m^3
	Vermiculite concrete		6 to 9	m^3
xxiii)	**Cement Concrete, Prestressed** (conforming to IS 1343) [C-1(15)]	–	23.50	m^3
xxiv)	**Cement Concrete, Reinforced:**			
	With 1 percent steel	–	22.75 to 24.20	m^3
	With 2 percent steel	–	23.25 to 24.80	m^3
	With 5 percent steel	–	24.80 to 26.50	m^3
xxv)	**Damp-proofing** (*see* Sl No. xxviii 'Felt bituminous for waterproofing and damp proofing')			
xxvi)	**Earth filling** (*see* Sl No. xlviii 'Soils and gravels in Table 1)			
xxvii)	**Expanded Metal** (conforming to IS 412) [C-1(15)]			
	Size of Mesh, Nominal			
	SWM	LWM		
	mm	mm		
	100	250	0.030	m^2
	100	250	0.024	m^2
	100	250	0.016	m^2

	75	200	0.042	m^2
	75	200	0.032	m^2
	75	200	0.021	m^2
	40	115	0.080	m^2
	40	115	0.060	m^2
	40	75	0.060	m^2
	40	75	0.028	m^2
	40	115	0.039	m^2
	40	75	0.039	m^2
	40	115	0.020	m^2
	40	75	0.020	m^2
	25	75	0.054	m^2
	25	75	0.038	m^2
	25	75	0.028	m^2
	25	75	0.021	m^2
	20	60	0.070	m^2
	20	50	0.070	m^2
	20	60	0.050	m^2
	20	50	0.050	m^2
	20	60	0.036	m^2
	20	50	0.036	m^2
	20	50	0.021	m^2
	20	60	0.021	m^2
	12.5	50	0.050	m^2
	12.5	50	0.050	m^2
	12.5	40	0.040	m^2
	12.5	50	0.030	m^2
	12.5	40	0.030	m^2
	12.5	50	0.025	m^2
	12.5	40	0.025	m^2
	10	40	0.050	m^2
	10	40	0.035	m^2
	10	40	0.028	m^2
	95	285	0.050	m^2
	95	285	0.028	m^2
	95	285	0.020	m^2
	6	25	0.074	m^2
	6	25	0.048	m^2

	6	25	0.038	m^2
	5	20	0.050	m^2
	3	15	0.041	m^2
xxviii)	**Felt, Bituminous for Waterproof and Damp–proofing** (*see* IS 1322) [C-1(15)]			
	Fibre base			
	Type 1 (under lay)	–	7.6×10^{-3}	m^2
	Type 2 (self-finished felt)	-	2.6×10^{-2}	m^2
	Hessian base			
	Type 3 (self-finished felt)			
	Grade 1	–	2.26×10^{-2}	m^2
	Grade 2	–	3.64×10^{-2}	m^2
xxix)	**Finishing** (*see* also 'Floor finishes' given under Sl No. xxxi 'Flooring' and xlv 'Roofing'in Table 1)			
	Plaster:			
	Acoustic	10	0.08	m^2
	Anhydrite	10	0.21	m^2
	Barium Sulphate	10	0.28	m^2
	Fibrous	10	0.09	m^2
	Gypsum or lime	10	0.19	m^2
	Hydraulic lime or cement	10	0.23	m^2
	Plaster ceiling on wire netting	10	0.26	m^2
	NOTE – When wood or metal lathing is used, add	–	0.06	
xxx)	**Float Glass** (*see* IS 2835) **and Safety Glass** [*see* IS 2553 (Part 1)] [C-1(15)]			
	Sheet	2	0.049	m^2
		2.5	0.062	m^2
		3	0.074	m^2
		4	0.098	m^2
		5	0.123	m^2
		5.5	0.134	m^2
		6.5	0.167	m^2
		8	0.200	m^2
		10	0.250	m^2
		12	0.300	m^2
		15	0.375	m^2
		19	0.475	m^2
xxxi)	**Flooring**			
	Asphalt flooring	10	0.22	m^2
	NOTE — For macadam finish, add	10	0.26	m^2

	Compressed cork	10	0.04	m^2
	Floors, structural:			
	Hollow clay blocks including reinforcement and mortar jointing between blocks, but excluding any concrete topping	100	1.47	m^2
		125	1.67	m^2
		150	1.86	m^2
		175	2.16	m^2
		200	2.55	m^2
	NOTE — Add extra for concrete topping			
	Hollow clay blocks including reinforcement and concrete ribs between blocks, but excluding any concrete topping	100	1.18	m^2
		115	1.27	m^2
		125	1.37	m^2
		140	1.47	m^2
		150	1.57	m^2
		175	1.76	m^2
		200	1.96	m^2
	NOTE — Add extra for concrete topping.			
	Hollow concrete units including any concrete topping necessary for constructional purposes	100	1.67	m^2
		125	1.96	m^2
		150	2.16	m^2
		175	2.35	m^2
		200	2.65	m^2
		230	3.14	m^2
	Floors, wood:			
	Hard wood	22	0.16	m^2
		28	0.20	m^2
	Soft wood	22	0.11	m^2
		28	0.13	m^2
	Weight of mastic used in laying wood block flooring	–	0.015	m^2
	NOTE — All thicknesses are ‘finished thicknesses’.			
	Floor finishes:			
	Clay floor tiles (*see* IS 1478) [C-1(15)]	12.5 to 25.4	0.10 to 0.2	m^2
	NOTE – This weight is ‘as laid’ but excludes screeding.			
	Magnesium oxychloride:			
	Normal type (saw dust filler)	10	0.142	m^2
	Heavy duty type (mineral filler)	10	0.216	m^2
	Parquet flooring	–	0.08 to 0.12	m^2
	Rubber (*see* IS 809) [C-1(15)]	3.2	0.048 to 0.062	m^2

		4.8	0.070 to 0.09	m^2
		6.4	0.093 to 0.130	m^2
	Terra cotta, filled 'as laid'	–	5.54 to 7.06	m^2
	Terrazzo paving 'as laid'	10	0.23	m^2
xxxii)	**Foam Slag, Foundry Pumice**	–	6.85	m^3
xxxiii)	**Gutters, Asbestos Cement** (*see* under vii '**Asbestos cement gutter**' in this Table)			
xxxiv)	**Gypsum**			
	Gypsum mortar [*see* IS 2547 (Part 1 and Part 2)] [C-1(15)]	–	18.6	m^3
	Gypsum powder (*see* IS 12679) [C-1(15)]	–	13.89 to 17.25	m^3
xxxv)	**Iron**			
	Pig	–	70.60	m^3
	Gray, cast	–	68.95 to 69.90	m^3
	White, cast	–	74.30 to 75.70	m^3
	Wrought	–	75.70	m^3
xxxvi)	**Lime**			
	Lime concrete with burn clay Aggregate (*see* IS 2541) [C-1(15)]	–	18.80	m^3
	Lime mortar	–	15.70 to 18.05	m^3
	Lime plaster (*see* IS 2394) [C-1(15)]	–	17.25	m^3
	Lime stored in lumps, uncalcined	–	12.55 to 14.10	m^3
	Lime, unslaked, freshly burnt in pieces	–	8.60 to 10.20	m^3
	Lime slaked, fresh	–	5.70 to 6.30	m^3
	Lime slaked, after 10 days	–	7.85	m^3
	Lime, unslaked (KANKAR)	–	11.55	m^3
	Lime, slaked (KANKAR)	–	10.00	m^3
xxxvii)	**Linoleum** (*see* IS 653) [C-1(15)]			
	Sheets and tiles	4.5	0.0569	m^2
		3.2	0.0402	m^2
		2.0	0.0265	m^2
		1.65	0.0215	m^2
xxxviii)	**Masonry**			
	Brick Masonry (excluding plaster)			
	With common burnt clay bricks (*see* IS 1077) [C-1(15)]	–	19.25	m^3

	Engineering bricks	–	21.2	m^3
	Glazed bricks	–	20.40	m^3
	Pressed bricks	–	18.65	m^3
	Block Masonry			
	With lime based solid blocks		13.55	m^3
	Hollow concrete blocks			
	Grade A		15.4	m^3
	Grade B		12.00	m^3
	Solid Concrete Blocks		18.15	m^3
	Autoclaved cellular concrete blocks		9.1 to 11.1	m^3
xxxix	**Masonry, Stone**			
	Cast	–	21.75	m^3
	Dry rubble	–	20.40	m^3
	Granite ashlar	–	25.00	m^3
	Granite rubble	–	23.55	m^3
	Lime stone ashlar	–	25.10	m^3
	Marble dressed	–	26.50	m^3
	Sand stone	–	22.00	m^3
xl)	**Mastic asphalt**	10	0.216	m^2
xli)	**Metal sheeting, protected galvanized steel sheets, plain** (*see* IS 277) [C-1(15)]			
	Class A (corrugated)	1.60	0.131	m^2
		1.00	0.104	m^2
		0.9	0.084	m^2
		0.80	0.069	m^2
		0.63	0.056	m^2
	Class B (corrugated)	1.60	0.129	m^2
		1.00	0.102	m^2
		0.9	0.083	m^2
		0.80	0.067	m^2
		0.63	0.054	m^2
	Class C (corrugated)	1.60	0.128	m^2
		1.00	0.101	m^2
		0.9	0.081	m^2
		0.80	0.066	m^2
		0.63	0.053	m^2
	Class D (corrugated)	1.60	0.127	m^2
		1.00	0.100	m^2
		0.9	0.081	m^2

		0.80	0.065	m^2
		0.63	0.052	m^2
xlii)	**Mortar**			
	Cement	–	20.40	m^3
	Gypsum	–	11.75	m^3
	Lime	–	15.70 to18.05	m^3
xliii)	**Pipes**			
	Asbestos cement pipes [*see* IS 1626 (Part 1)] [C-1(15)]	50	0.032 to 0.034	m
		60	0.032 to 0.043	m
		80	0.051 to 0.054	m
		90	0.052 to 0.060	m
		100	0.058 to 0.065	m
		125	0.072 to 0.086	m
		150	0.086 to 0.108	m
	Asbestos cement pressure pipes [(*see* IS 1592) [C-1(15)]	50	0.056	m
		80	0.067	m
		100	0.090	m
		125	0.139	m
		150	0.175	m
		200	0.264	m
		250	0.380	m
		300	0.539	m
	Cast iron pipes (*see* IS 1729) [C-1(15)]			
	Standard overall length 1.8 m with socket	50	0.073	pipe
		75	0.108	pipe
		100	0.137	pipe
		125	0.196	pipe
		150	0.255	pipe
	Standard overall length 1.5 m with socket	50	0.064	pipe
		75	0.093	pipe
		100	0.123	pipe
		125	0.172	pipe
		150	0.230	Pipe
	Pressure pipes for water, gas and sewage:			
	a) Centrifugally cast (*see* IS 1536) [C-1(15)]			
	Socket and spigot pipes			
	Barrel:			
	Class LA	80	0.145	m

		100	0.182	m
		125	0.237	m
		150	0.297	m
		200	0.432	m
		250	0.582	m
		300	*0.750*	m
		350	0.944	m
		400	1.146	m
		450	1.383	m
		500	1.620	m
		600	2.156	m
		700	2.718	m
		750	3.111	m
	Class A	80	0.151	m
		100	0.201	m
		125	0.259	m
		150	0.326	m
		200	0.472	m
		250	0.637	m
		300	0.824	m
		350	1.030	m
		400	1.262	m
		450	1.530	m
		500	1.775	m
		600	2.367	m
		700	3.056	m
		750	3.422	m
	Class B	80	0.172	m
		100	0.216	m
		125	0.281	m
		150	0.352	m
		200	0.511	m
		250	0.692	m
		300	0.896	m
		350	1.112	m
		400	1.368	m
		450	1.657	m
		500	1.929	m

		600	2.578	m
		700	3.317	m
		750	3.733	m
	Sockets for Class LA, Class A and Class B barrels	80	0.054	Socket
		100	0.069	Socket
		125	0.090	Socket
		150	0.113	Socket
		200	0.165	Socket
		250	0.225	Socket
		300	0.292	Socket
		350	0.368	Socket
		400	0.454	Socket
		450	0.549	Socket
		500	0.647	Socket
		600	0.876	Socket
		700	1.145	Socket
		750	1.292	Socket
	b) Flanged pipe with screwed flanges			
	Barrel:			
	Class A	80 to 300	Same as for centrifugally cast socket and spigot piles, Class A	
	Class B	80 to 300	Same as for centrifugally cast socket and spigot piles, Class B	
	Flanges for Class A and Class B barrels	80	0.042	Flange
		100	0.049	Flange
		125	0.065	Flange
		150	0.080	Flange
		200	0.112	Flange
		250	0.144	Flange
		300	0.182	Flange
	c) Vertically cast socket and spigot pipes (*see* IS 1537) [C-1(15)]			
	Barrel: Class A	80 to 750	Same as for centrifugally cast socket and spigot pipes, Class A	
		800	3.82	m
		900	4.65	m
		1000	5.59	m
		1100	6.59	m
		1200	7.67	m
		1500	11.98	m

	Class B	80 to 750	–	–
		800	–	m
		900	5.07	m
		1000	6.07	m
		1100	7.23	m
		1200	8.35	m
		1500	13.07	m
	Socket of Class A and Class B barrels	80 to 750	–	–
		800	–	Socket
		900	179	Socket
		1000	218	Socket
		1100	260	Socket
		1200	360	Socket
		1500	491	Socket
	d) Sand cast (flanked pipes)			
	Barrel:			
	Class A	80 to 750	Same as for centrifugally cast socket and spigot pipes, Class A	
		800 to 1500	–	
	Class B	80 to 750	Same as for vertically cast socket and spigot pipes, Class B	
		800 to 1500		
	Flanges for Class A and Class B barrels	80	0.036	Flange
		100	0.041	Flange
		125	0.052	Flange
		150	0.066	Flange
		200	0.091	Flange
		250	0.117	Flange
		300	0.145	Flange
		350	0.186	Flange
		400	0.229	Flange
		450	0.250	Flange
		500	0.315	Flange
		600	0.431	Flange
		750	0.587	Flange
		700	0.685	Flange
		800	0.792	Flange
		900	0.928	Flange
		1000	1.18	Flange

		1100	1.38	Flange
		1200	1.70	Flange
		1500	2.71	Flange
	Concrete pipes (*see* IS 458) [C-1(15)]			
	Class NP 1 (unreinforced non-pressure pipes)	80	0.19	m
		100	0.22	m
		150	0.30	m
		250	0.40	m
		300	0.69	m
		350	0.84	m
		400	0.95	m
		450	1.17	m
	Class NP 2 (reinforced concrete, light duty, non-pressure pipes)	80	0.196	m
		100	0.215	m
		150	0.324	m
		250	0.510	m
		300	0.736	m
		350	0.902	m
		400	1.02	m
		450	1.26	m
		500	1.38	m
		600	1.89	m
		700	2.19	m
		800	2.81	m
		900	3.51	m
		1000	4.30	m
		1100	5.15	m
		1200	6.09	m
		1400	8.18	m
		1600	9.93	m
		1800	12.58	m
	Class NP 3 (reinforced concrete, heavy duty, non-pressure pipes)	350	2.35	m
		400	2.63	m
		450	2.91	m
		500	3.19	m
		600	4.02	m
		700	4.61	m
		800	5.92	m

		900	7.39	m
		1000	8.13	m
		1100	10.34	m
		1200	11.18	m
	Class P1 (reinforced concrete pressure pipes safe for 20 MPa pressure tests)	80	0.196	m
		100	0.235	m
		150	0.324	m
		250	0.510	m
		300	0.736	m
		350	0.902	m
		400	1.02	m
		450	1.26	m
		500	1.38	m
		600	1.89	m
		700	2.19	m
		800	2.81	m
		900	3.51	m
		1000	4.30	m
		1100	5.15	m
		1200	6.09	m
	Class P2 (reinforced concrete pressure pipes safe for 40 MPa pressure tests)	80	0.196	m
		100	0.235	m
		150	0.324	m
		250	0.608	m
		300	1.01	m
		350	1.31	m
		400	1.67	m
		450	1.84	m
		500	1.56	m
		600	3.20	m
	Class P3 (reinforced concrete pressure pipes safe for 60 MPa pressure tests)	80	0.196	m
		100	0.235	m
		150	0.324	m
		250	0.736	m
		300	1.15	m
		350	1.65	m
		400	2.04	m

	Lead pipes [*see* IS 404 (Part 1)] [C-1(15)] (service and distribution pipes 10 be laid underground) :			
	For working 40 MPa	10	0.018	m
		13	0.031	m
		20	0.042	m
		25	0.060	m
		30	0.074	m
		40	0.091	m
		50	0.142	m
	For working 70 MPa	10	0.022	m
		15	0.038	m
		20	0.050	m
		25	0.069	m
		32	0.126	m
		40	0.175	m
	For working 100 MPa	10	0.029	m
		15	0.048	m
		20	0.067	m
		25	0.105	m
	Service pipes to be fixed or laid above ground:			
	For working pressure 40 MPa	10	0.014	m
		15	0.021	m
		20	0.027	m
		25	0.036	m
		32	0.059	m
		40	0.091	m
		50	0.142	m
	For working pressure 70 MPa	10	0.018	m
		15	0.024	m
		20	0.030	m
		25	0.069	m
		32	0.126	m
		40	0.175	m
	For working pressure 100 MPa	10	0.029	m
		15	0.048	m
		20	0.067	m
		25	0.105	m

	Cold water distribution pipes to be fixed or laid above ground:			
	For working pressure 25 MPa	10	0.014	m
		15	0.021	m
		20	0.027	m
		25	0.036	m
		32	0.048	m
		40	0.067	m
		50	0.084	m
	For working pressure 40 MPa	10	0.014	m
		15	0.021	m
		20	0.027	m
		25	0.036	m
		32	0.059	m
		40	0.091	m
		50	0.142	m
	Hot water distribution pipes to be fixed or laid above ground:			
	For working pressure 20 MPa	10	0.015	m
		15	0.023	m
		20	0.031	m
		25	0.041	m
		32	0.062	m
		40	0.082	m
		50	0.142	m
	For working pressure 35 MPa	10	0.015	m
		15	0.027	m
		20	0.045	m
		25	0.085	m
		32	0.132	m
	Soil, waste and soil and waste ventilation pipes	50	0.050	m
		75	0.073	m
		100	0.097	m
		150	0.160	m
	Flushing and warning pipes	20	0.020	m
		25	0.025	m
		32	0.032	m
		40	0.039	m
		50	0.049	m
	Gas pipes:			
	Heavy weight gas pipes	10	0.008	m

		15	0.017	m
		20	0.025	m
		25	0.034	m
		32	0.045	m
		40	0.061	m
		50	0.071	m
	Light weight gas pipes	10	0.008	m
		15	0.012	m
		20	0.020	m
		25	0.029	m
		32	0.037	m
		40	0.047	m
		50	0.058	m
	Stoneware, salt-glazed pipes (*see* IS 651) [C-1(15)]	100	0.137	m
		150	0.216	m
		200	0.324	m
		230	0.412	m
		250	0.510	m
		300	0.775	m
		350	0.980	m
		400	1.26	m
		450	1.44	m
		500	1.77	m
		600	2.35	m
xliv)	**Plaster**			
	Cement	10	0.204	m^2
	Lime	10	0.172	m^2
	Acoustic	10	0.078	m^2
	Anhydrite	10	0.206	m^2
	Barium sulphate	10	0.284	m^2
	Fibrous	10	0.088	m^2
	Gypsum	10	0.186	m^2
xlv)	**Roofing**			
	Asbestos cement sheeting			
	(*see* 'Asbestos cement sheeting' in Table 1)			
	Allahabad tiles (single) including battens (*see* Note below)	–	0.83	m^2

	Allahabad tiles (double) including battens (*see* Note below)	–	1.67	m^2
	Country tiles (single) including battens (*see* Note below)	–	0.69	m^2
	Country tiles (double) including battens (*see* Note below)	–	1.18	m^2
	Mangalore tiles battens (*see* Note below)	–	0.64	m^2
	Mangalore tiles bedded in mortar over flat tiles (*see* Note below)	–	1.08	m^2
	Mangalore tiles with flat tiles (*see* Note below)	–	0.78	m^2
	Copper sheet roofing including laps and rolls	0.56	0.08	m^2
	Flat roofs	0.72	0.10	m^2
	Clay tiles- hollow clay blocks/clay floor tiles (*see* Sl No. xxxi 'Flooring')			
	Concrete hollow units (*see* Sl No. xxxi 'Flooring' in this Table)			
	Galvanized iron sheeting (*see* Sl No. 'xli Metal sheeting, protected')			
	Glazed Roofing:			
	Glazing with aluminum alloy bars for spans up to 3 m	6.4	0.19	m^2
	Glazing with lead-covering steel bars at 0.6 m centres	6.4	0.25 to 0.28	m^2
	States on battens	–	0.34 to 0.49	m^2
	Thatch with battens	–	0.34 to 0.49	m^2
	NOTE – Weights acting vertically on horizontally projection to be multiplied by cosine of roof angle to obtain weights normal to the roof surface.			
	Roof finished			
	Bitumen macadam	10	0.22	m^2
	Felt roofing (*see* 28 'Felt bituminous for water-proofing and damp-proofing' in Table 1)	10	0.008	m^2
	Glass silk quilted	0.5	0.05	m^2
	Lead sheet	0.8	0.07	m^2
	Mortar screeding	10	0.21	m^2
xlvi)	**Sheeting**			
	Asbestos (*see* under Sl No. ix 'asbestos cement sheeting' in this Table)			
	Galvanized iron (*see* under Sl No. xli 'metal sheeting, protected' in this table)			
	Glass (*see* under Sl No. xxx 'Glass' in this table)			
	Plywood (*see* IS 303) [C-1(15)]	1	0.007	m^3

xlvii)	**Slugwool**	–	2.65	m^3
xlviii)	**Soils and Gravels**			
	Alluvial ground, undisturbed	–	15.69	m^3
	Broken stone ballast:			
	Dry, well-shaken	–	15.70 to 18.35	m^3
	Perfectly wet	–	18.85 to 21.95	m^3
	Chalk		15.70 to 18.85	m^3
	Clay:			
	China, compact	–	21.95	m^3
	Clay fills:			
	Dry, lumps	–	10.20	m^3
	Dry, compact	–	14.10	m^3
	Damp, compact	–	17.25	m^3
	Wet , compact	–	20.40	m^3
	Undisturbed	–	18.85	m^3
	Undisturbed, gravelly	–	20.40	m^3
	Crush stone sand (Msand)	–	–	–
	Earth:			
	Dry	–	13.85 to 18.05	m^3
	Moist	–	15.70 to 19.60	m^3
	Gravel:			
	Loose	–	15.70	m^3
	Rammed	–	18.85 to 21.20	m^3
	Kaolin, compact	–	25.50	m^3
	Loam:			
	Dry, loose	–	11.75	m^3
	Dry, compact	–	15.70	m^3
	Wet, compact	–	18.85	m^3
	Loess, dry	–	14.10	m^3
	Marl, compact	–	17.25 to 18.85	m^3
	Mud, river, wet	–	17.25 to 18.85	m^3
	Peat:			
	Dry	–	5.50 to 6.30	m^3
	Sandy, compact	–	7.85	m^3
	Wet, compact	–	13.35	m^3
	Rip-rap	–	12.55 to 14.10	m^3
	Sand:			
	Dry, clean	–	15.10 to 15.70	m^3

	River	–	18.05	m^3
	Wet	–	17.25 to 19.60	m^2
	Shingles:			
	Aggregate 3 to 38 mm	–	13.75	m^2
	Fine sand:			
	Dry	–	15.70	m^2
	Saturated	–	20.40	m^2
	Silt, wet	–	17.25 to 18.85	m^2
xlix)	**Steel sections**			
	Density of steel sections shall be taken as 78.50 kg/m^3			
	Weight of hot rolled sections (ISJB, ISLB, ISMB, ISWB, ISNPB, ISWPB, ISSC, ISHB, ISJC, ISLC, ISMC, ISMPC, ISA, ISPBP) shall be referred from (*see* IS 808) [C-1(15)]			
	Weight of hollow steel sections shall be referred from (*see* IS 4923) [C-1(15)]			
	Weight of cold formed light gauge structural steel sections shall be referred from (*see* IS 811) [C-1(15)]			
	Weight of rolled and slit T bars shall be referred from (*see* IS 1173) [C-1(15)]			
	Weight of steel sheet piling sections shall be referred from (*see* IS 2314) [C-1(15)]			
l)	**Stone**			
	Agate	–	25.50	m^3
	Aggregate	–	15.70 to 18.85	m^3
	Basalt	–	27.95 to 29.05	m^3
	Cast	–	21.95	m^3
	Chalk	–	21.50	m^3
	Dolomite	–	28.25	m^3
	Emery	–	39.25	m^3
	Flint	–	25.40	m^3
	Gneiss	–	23.55 to 26.40	m^3
	Granite	–	25.90 to 27.45	m^3
	Gravel:			
	Loose	–	15.70	m^3
	Moderately rammed, dry	–	18.85	m^3
	Green stone	–	28.25	m^3
	Gypsum	–	21.95 to 23.55	m^3
	Kota			
	Laterite	–	20.40 to 23.55	m^3
	Lime stone	–	23.55 to 25.90	m^3
	Marble	–	26.70	m^3
	Pumice	–	7.85 to 11.00	m^3
	Quartz rock	–	25.90	m^3

	Sand stone	–	21.95 to 23.54	m^3
	Slate	–	27.45	m^3
	Soap stone	–	26.45	m^3
li)	**Tar, Coal**			
	Crude (*see* IS 212) [C-1(15)]	–	9.90	m^3
	Naphtha, light	–	9.90	m^3
	Naphtha, heavy	–	9.90	m^3
	Road tar	–	9.90	m^3
	Pitch (*see* IS 216) [C-1(15)]	–	9.90	m^3
lii)	**Thermal Insulation**			
	Unbonded glass wool	–	12.75 to 23.55	m^3
	Unbonded rock and slag wool	–	11.30 to 19.60	m^3
	Cellular concrete			
	Grade A	–	Up to 29.40	m^3
	Grade B	–	29.50 to 39.20	m^3
	Grade C	–	39.30 to 49.00	m^3
	Performed calcium silicate Insulation (for temperature up to 650°C)	–	19.60 to 34.30	m^3
liii)	**Tiles and stone cladding**			
	Terra-cotta	–	18.35 to 23.25	m^2
	Terrozzo			
	Paving	10	0.24	m^2
	Cast partitions	40	0.93	m^2
	Tiles			
	Mangalore pattern (*see* IS 654) [C-1(15)]	–	0.02 to 0.03	Tile
	Polystyrene wall tiles	99 x 99	0.013	m^2
		148.5 x 148.5	0.013	m^2
	Pressed ceramic tiles, glazed and unglazed (*see* IS 15622) [C-1(15)]		22	m^3
	Concrete paving blocks (*see* IS 15658) [C-1(15)]		24	m^3
liv)	**Timber**			
	Typical Indian timbers (*see* IS 399) [C-1(15)]			
	Aglaia	–	8.34	m^3
	Aini	–	5.83	m^3
	Alder	–	3.63	m^3
	Amari	–	6.13	m^3

	Amla	–	7.85	m^3
	Amra	–	4.41	m^3
	Anjan	–	8.33	m^3
	Arjun	–	7.99	m^3
	Ash	–	7.06	m^3
	Axlewood	–	8.82	m^3
	Babul	–	7.70	m^3
	Baen	–	7.70	m^3
	Bahera	–	7.99	m^3
	Bakota	–	4.21	m^3
	Balasu	–	7.55	m^3
	Ballagi	–	11.13	m^3
	Bamboo			
	Banati	–	4.41	m^3
	Benteak	–	6.62	m^3
	Ber	–	6.91	m^3
	Bhendi	–	7.55	m^3
	Bijasal	–	7.85	m^3
	Black chuglam	–	6.13	m^3
	Birch	–	7.85	m^3
	Black locust	–	8.34	m^3
	Blue gum	–	8.34	m^3
	Blue pine	–	5.05	m^3
	Bola	–	6.42	m^3
	Bonsum	–	5.20	m^3
	Bullet wood	–	8.78	m^3
	Casuarina	–	8.34	m^3
	Cettis	–	6.42	m^3
	Champ	–	4.85	m^3
	Chaplash	–	5.05	m^3
	Chatian	–	4.07	m^3
	Chikrassy	–	6.62	m^3
	Chilauni	–	6.42	m^3
	Chilla	–	7.85	m^3
	Chir	–	5.64	m^3
	Chuglam:			
	Black	–	7.85	m^3
	White (silver grey-wood)	–	6.91	m^3

	Cinnamon	–	6.42	m^3
	Cypress	–	5.05	m^3
	Debdaru	–	6.28	m^3
	Deodar	–	5.35	m^3
	Devdam	–	7.06	m^3
	Dhaman:			
	Grewia tiliofolia	–	7.70	m^3
	Grewia vestita	–	7.40	m^3
	Dhup	–	6.42	m^3
	Dilenia	–	6.13	m^3
	Dudhi	–	5.49	m^3
	Ebony	–	8.19	m^3
	Elim	–	5.20	m^3
	Eucalyptus	–	8.33	m^3
	Figs	–	4.56	m^3
	Fir	–	4.14	m^3
	Frash	–	6.62	m^3
	Gamari	–	5.05	m^3
	Gardenia	–	7.40	m^3
	Garuga	–	5.98	m^3
	Geon	–	4.07	m^3
	Gluta	–	7.06	m^3
	Gokul	–	4.07	m^3
	Grewia sp.	–	7.55	m^3
	Gurjan	–	7.70	m^3
	Gutel	–	4.41	m^3
	Haldu	–	6.62	m^3
	Hathipaila	–	5.84	m^3
	Hiwar	–	7.70	m^3
	Hollock	–	5.98	m^3
	Hollong	–	7.21	m^3
	Hoom	–	7.21	m^3
	Horse chestnut	–	5.05	m^3
	Imli	–	8.97	m^3
	Indian Chestnut	–	6.28	m^3
	Indian Hemlock	–	3.92	m^3
	Indian Oak	–	8.48	m^3
	Indian Olive	–	10.35	m^3

	Irul	–	8.33	m^3
	Jack	–	5.83	m^3
	Jaman	–	7.70	m^3
	Jarul	–	6.13	m^3
	Jathtkai	–	5.05	m^3
	Jhingan	–	5.63	m^3
	Jutili	–	7.85	m^3
	Kadam	–	4.85	m^3
	Kail	–	5.05	m^3
	Kaim	–	6.42	m^3
	Kambli	–	4.07	m^3
	Kanchan	–	6.62	m^3
	Kanjuj	–	5.84	m^3
	Karada	–	8.34	m^3
	Karal	–	7.99	m^3
	Karani	–	6.28	m^3
	Karar	–	5.34	m^3
	Kardahi	–	9.27	m^3
	Karimgotta	–	3.92	m^3
	Kasi	–	5.83	m^3
	Kasum	–	10.84	m^3
	Kathal	–	5.85	m^3
	Keora	–	6.13	m^3
	Khair	–	9.00	m^3
	Khasipine	–	5.05	m^3
	Kindal	–	7.55	m^3
	Kokko	–	6.28	m^3
	Kongoo	–	9.76	m^3
	Kuchla	–	8.63	m^3
	Kumbi	–	7.70	m^3
	Kurchi	–	5.20	m^3
	Kurung	–	9.76	m^3
	Kusum	–	11.28	m^3
	Kuthan	–	4.71	m^3
	Lakooch	–	6.28	m^3
	Lambapatti	–	5.34	m^3
	Lampati	–	5.05	m^3
	Laurel	–	8.33	m^3

	Lendi	–	7.40	m^3
	Machilus:			
	Gamblei	–	5.05	m^3
	Macrantha	–	5.20	m^3
	Maharukh	–	4.07	m^3
	Mahogany	–	6.62	m^3
	Mahua	–	8.97	m^3
	Maina	–	5.64	m^3
	Makai	–	3.14	m^3
	Malabar neem	–	4.41	m^3
	Mango	–	6.77	m^3
	Maniawga	–	7.40	m^3
	Maple	–	5.64	m^3
	Mesua	–	9.76	m^3
	Milla	–	9.12	m^3
	Mokha	–	7.99	m^3
	Mulberry	–	6.62	m^3
	Mullilam	–	7.21	m^3
	Mundani	–	6.77	m^3
	Murtenga	–	7.70	m^3
	Myrabolan	–	9.27	m^3
	Narikel	–	5.49	m^3
	Nedunar	–	5.05	m^3
	Oak	–	8.48	m^3
	Padauk	–	7.06	m^3
	Padri	–	7.06	m^3
	Palang	–	5.98	m^3
	Pali	–	6.28	m^3
	Papita	–	3.28	m^3
	Parrotia	–	8.48	m^3
	Persian lilac	–	5.84	m^3
	Piney	–	6.13	m^3
	Ping	–	8.97	m^3
	Pinus insignis	–	6.13	m^3
	Pipli	–	5.83	m^3
	Pitraj	–	6.77	m^3
	Poon	–	6.42	m^3
	Poplar	–	4.41	m^3

	Pula	–	3.78	m^3
	Pyinma	–	3.98	m^3
	Rajbrikh	–	8.48	m^3
	Red sanders	–	10.84	m^3
	Rohini	–	11.33	m^3
	Rosewood (black wood)	–	8.19	m^3
	Rudrak	–	4.71	m^3
	Sal	–	8.48	m^3
	Salai	–	5.64	m^3
	Sandal wood	–	8.97	m^3
	Sadan	–	8.34	m^3
	Satin wood	–	9.41	m^3
	Saykaranji	–	7.40	m^3
	Seleng	–	4.85	m^3
	Semul	–	3.78	m^3
	Silver oak	–	6.28	m^3
	Siris	–	3.92	m^3
	Kala-siris	–	7.21	m^3
	Safed- siris	–	6.28	m^3
	Siaso	–	7.70	m^3
	Spruce	–	4.71	m^3
	Suji	–	2.65	m^3
	Sundri	–	9.41	m^3
	Talauma	–	5.64	m^3
	Tanaku	–	2.09	m^3
	Teak	–	6.28	m^3
	Toon	–	5.05	m^3
	Udal	–	2.50	m^3
	Upas	–	3.14	m^3
	Uriam	–	7.40	m^3
	Vakai	–	9.41	m^3
	Vellapine	–	5.83	m^3
	Walnut	–	5.64	m^3
	White bombwe	–	5.98	m^3
	White cedar	–	7.06	m^3
	White chuglam (silver grey-wood)	–	6.91	m^3
	White dhup	–	4.22	m^3
	Yon	–	8.33	m^3

	NOTE — The unit of timbers correspond to average unit of typical Indian timbers at 12 percent moisture content.			
lv)	**Water**			
	Fresh	–	10.0	m^3
	Salt	–	10.05	m^3
lvi)	**Wood – Wool Building Slabs**	10mm	0.059	m^2
lvii)	**Walling** (*see* IS 6072) [C-1(15)]			
	Autoclaved reinforced cellular concrete wall slabs			
	Class A	–	8.35 to 9.80	m^2
	Class B	–	7.35 to 8.35	m^2
	Class C	–	6.35 to 7.35	m^2
	Class D	–	5.40 to 6.35	m^2
	Class E	–	4.40 to 5.40	m^2
	Brick masonry (*see* xxxviii 'Masonry Brick' in Table 1)			
	Blocks masonry (*see* xii 'Block' in Table 1)			
	Stone masonry (*see* xxxix 'Masonry Stone' in Table 1)			
	Partitions 39			
	Brick wall	100	1.91	m^2
	Cinder concrete	75	1.13	m^2
	Galvanized on sheet	–	0.15	m^2
	Hollow glass block (bricks)	100	0.88	m^2
	Hollow blocks per 200 mm of thickness			
	Ballast or stone concrete	200	0.201	m^2
	Clay	200	0.201	m^2
	Clinker concrete	200	0.220	m^2
	Coke breeze concrete	200	9.176	m^2
	Diatomaceous earth	200	0.093	m^2
	Gypsum	200	0.137	m^2
	Pumice concrete	200	0.177	m^2
	Slag concrete, air-cooled	200	0.196	m^2
	Slag concrete foamed	200	0.186	m^2
	Lath and plaster	–	0.392	m^2
	Solid blocks per 20 mm of thickness			
	Ballast or stone	20	0.451	m^2
	Clinker concrete	20	0.300	m^2
	Coke breeze concrete	20	0.221	m^2
	Pumice concrete	20	0.221	m^2
	Slag concrete, foamed	20	0.250	m^2
	Terrazzo cast partitions	40	0.932	m^2
	Timber studding plastered	–	9.981	m^2
NOTE — For unit weight of fixtures and fittings required to buildings including builder's hardware, reference may be made to appropriate Indian standards.				

A-2 STORED AND MISCELLANEOUS MATERIALS

A-2.1 Units weights of stored and miscellaneous materials intended for dead load calculation and other general purposes are given in Table 41.

Table 41 Unit Weights of Stored and Miscellaneous Materials

(*Clause* A-2.1)

Sl No.	Materials	Weight KN/m^3	Angle of Friction, Degrees
(1)	(2)	(3)	(4)
i)	**Agriculture and Food Products**		
	Butter	8.45	–
	Coffee in bags	5.50	–
	Drinks in bottle, in boxes	7.35	–
	Eggs, packed	2.95	–
	Fats, oil	5.80	–
	Fish meal	4.90	45
	Flour in sacks up to 1 m height	2.20 to 5.90	–
	Forage (bales)	1.25	–
	Fruits	3.45	–
	Grains:		
	Barley	6.75	27
	Corn, shelled	7.55	27
	Flax seed	7.35	30
	Oats	5.30	30
	Rice	6.55	33
	Soyabeans	7.35	30
	Wheat	8.15	28
	Wheat flour	6.85	30
	Grain sheaves up to 4 m stack height	0.98	30
	Grain sheaves over 4 m stack height	1.45	30
	Grass and clover	3.45	–
	Hay:		–
	Compressed	1.65	–
	Loose up to about 3 m stack height	0.69	–
	Honey	14.10	–
	Hops:		
	In stacks	1.65	–
	In cylinder hop bins	4.60	–

Table 41 (*Continued*)

Sl No.	Materials	Weight KN/m^3	Angle of Friction, Degrees
(1)	(2)	(3)	(4)
	Sewn up or compressed in cylindrical shape in hop cloth	2.85	–
	Malt:		
	Crushed	3.90	20
	Germinated	1.85	–
	Meat and meat products	7.05	–
	Milk	10.05	–
	Molasses	4.40	–
	Onion in bags	5.40	0
	Oil cakes, crushed	5.80	0
	Potatoes	7.05	30
	Preserves (tins in cases)	4.90 to7.85	–
	Bags	7.05	–
	Bulk	9.40	–
	Seeds:		
	Heaps	4.90 to 7.85	25
	Sacks	3.90 to 6.85	–
	Straw and chaff:		
	Loose up to about 3 m stack height	0.45	–
	Compressed	1.65	–
	Sugar:		
	Crystal	7.35	30
	Cube sugar in boxes	7.85	–
	Sugar beet, pressed out	7.85	–
	Tobacco bundles	3.45	–
	Vinegar	10.40	–
ii)	**Chemicals and Allied Materials**		
	Acid, hydrochloric	11.75	–
	Acid, nitric 91 percent	14.80	–
	Acid, sulphuric 87 percent	17.55	–
	Alcohol	7.65	–
	Alum, pearl in barrel	5.20	–
	Ammonia, liquid	8.85	–
	Ammonium, chloride, crystalline	8.15	30 to 40
	Ammonium nitrate	7.05 to 9.80	25
	Ammonium sulphate	7.05 to 9.00	32 to 45
	Beeswax	9.40	–
	Benzole	8.90	–
	Benzene hexachloride	8.75	–
	Bicarbonate of soda	6.40	–

Table 41 (*Continued*)

Sl No.	Materials	Weight KN/m^3	Angle of Friction, Degrees
(1)	(2)	(3)	(4)
	Bone	18.65	–
	Borax	17.15	–
	Calcite	26.50	–
	Camphor	9.70	–
	Carbon disulphide	12.75	–
	Casein	13.25	–
	Caustic soda	13.85	–
	Creosole	10.50	–
	Dicalcium phosphate	6.65	–
	Disodium phosphate	3.90 to 4.80	30 to 45
	Iodine	48.55	–
	Oils in bottle or barrels	5.70 to 8.90	–
	Oil, linsee:		
	In barrels	5.70	–
	In drums	7.05	–
	Oil, turpentine	8.50	–
	Paints	9.40	–
	Paraffin wax	7.85 to 9.40	–
	Petroleum	9.90	–
	Phosphorus	17.85	–
	Plastics and polymers:		
	Cellulose acetate	12.25 to 13.35	–
	Cellulose nitrate	13.25 to 15.70	–
	Methyl methacrylate (IS 14753) [C-1(15)]	11.57 to 11.77	–
	Phenol formaldehyde	12.55	–
	Expanded Polystyrene (IS 4671) [C-1(15)]	0.15 to 0.35	–
	Extruded polystyrene	0.21	
	Polyvinyl chloride (Perspex)	11.75 to 13.25	–
	Resin bonded sheet	12.85 to 13.55	–
	Urea formaldehyde	13.25 to 13.55	–
	Extruded polystyrene	0.21	
	Polyurethane	0.59 to 0.66	
	Polyethylene (IS 2508) [C-1(15)]	9.81 to 19.6	
	Polyvinyl Butaryl	11	
	Polycarbonate (IS 14443) [C-1(15)]	12	
	Ethylene tetrafluoroethylene	17	
	Polytetrafluoroethylene [IS 14635 (part 2)] [C-1(15)]	22	
	Potash	14.40	–
	Potassium	8.65	–

Table 41 (*Continued*)

Sl No.	Materials	Weight KN/m^3	Angle of Friction, Degrees
(1)	(2)	(3)	(4)
	Potassium nitrate	9.90	–
	Red lead, dry	20.70	–
	Red lead, paste	87.30	–
	Rosin in barrels	6.75	–
	Rubber:		
	Raw	8.90 to 9.40	–
	Vulcanized	8.90 to 9.10	–
	Saltpeter	9.91	–
	Sodium silicate in barrels	8.35	–
	Sulphur	20.10	–
	Talc	27.45	–
	Varnishes	9.40	–
	Vitriol, blue, in barrels	7.05	–
iii)	**Fuels**		
	Brown coal	6.83	–
	Brown coal briquettes heaped	7.85	35
	Brown coal briquettes stacked	12.75	–
	Charcoal	2.95	–
	Coal:		
	Untreated, mine-moist	9.80	35
	In washeries	11.75	0
	Dust	6.85	25
	All other sorts	8.35	35
	Coke:		
	Furnace or gas	4.90	35
	Brown coal, low-temperature	9.80	35
	Hard, raw coal	8.35	35
	Hard, raw coal, mine-damp	9.80	35
	Diesel oil	9.40	0
	Firewood, chopped	3.90	45
	Petrol	6.75	0
	Wood in chips	1.95	45
	Wood shaving, loose	1.45	35
	Wood shaving, shaken down	2.45	35
iv)	**Manures**		
	Animal manures:		
	Loosely heaped	11.75	45
	Stacked dung up to 25 m stack height	17.65	45
	Artificial manures	11.75	24 to 30
v)	**Metals and Alloys**		

Table 41 (*Continued*)

Sl No.	Materials	Weight KN/m^3	Angle of Friction, Degrees
(1)	(2)	(3)	(4)
	Aluminium	25.30 to 26.60	
	Cast	25.30 to 26.60	–
	Wrought	25.90 to 27.45	–
	Sheet per mm of thickness per m^2	0.028	–
	Antimony, pure:		
	Amorphous	60.90	–
	Solid	65.70	–
	Bismuth:		
	Liquid	98.07	–
	Solid	95.02 to 97.9	–
	Cadmium:		
	Cast	83.75 to 84.05	–
	Wrought	85.03	–
	Calcium	15.60	–
	Chromium	63.95 to 66.00	–
	Cobalt:		
	Cast	83.25 to 85.10	–
	Wrought	88.45	–
	Copper:		
	Cast	86.20 to 87.65	–
	Wrought	86.70 to 87.65	–
	Sheet per mm of thickness	0.09	–
	Gold:		
	Cast	188.75 to 189.55	–
	Wrought	189.55	–
	Iron:		
	Pig	70.60	–
	Grey, cast	68.95 to 69.90	–
	White, cast	74.35 to 75.70	–
	Wrought	75.50	–
	Lead:		
	Cast	111.20	–
	Liquid	105.00	–
	Wrought	111.40	–
	Sheet per mm of thickness	0.11	–
	Magnesium	16.45 to 17.15	–
	Manganese	72.55	–
	Mercury	133.35	–
	Nickel	81.20 to 87.20	–
	Platinum	210.25	–

Table 41 (*Continued*)

Sl No.	Materials	Weight KN/m^3	Angle of Friction, Degrees
(1)	(2)	(3)	(4)
	Silver:		
	Cast	102.0 to 102.85	–
	Liquid	93.15	–
	Wrough	103.35 to 103.55	–
	Sodium:		
	Liquid	9.10	–
	Solid	9.30	–
	Tungsten	188.30	–
	Uranium	180.45	–
	Zinc		
	Cast	68.95 to 70.20	–
	Wrought	70.50	–
	Sheet per mm of thickness	0.07	–
	Alloys:		
	Aluminium and copper		
	Aluminium 10 percent copper 90 percent	75.40	–
	Aluminium 5 percent copper 95 percent	82.00	–
	Aluminium 3 percent copper 97 percent	85.10	–
	Aluminium 91 percent zinc 9 percent	27.45	–
	Babbit metal tin 90 percent	71.70	–
	Lead 5 percent copper 5 percent		
	Wood's metal bismuth 50 percent	95.00	–
	Lead 25 percent cadmium 12.5 percent		
	Tin 12.5 percent		
	Brasses:		
	Muntz metal (copper 60 percent, zinc 40 percent)	80.60	–
	Red (copper 90 percent, zinc 10 percent)	84.25	–
	White (copper 50 percent, zinc 50 percent)	80.30	–
	Yellow (copper 70 percent, zinc 30 percent)		
	Cast	82.75	–
	Drawn	85.10	–
	Rolled	83.85	–
	Bronzes:		
	Bell metal (copper 80 percent, tin 20 percent)	85.60	–

Table 41 (*Continued*)

Sl No.	Materials	Weight KN/m^3	Angle of Friction, Degrees
(1)	(2)	(3)	(4)
	Gun metal (copper 90 percent, tin 10 percent)	86.10	–
	Cadmium and tin	75.40	–
	German Silver:		
	Copper 52 percent, zinc 26 percent, nickel 22 percent	82.75	–
	Copper 59 percent, zinc 30 percent, nickel 11 percent	81.70	–
	Copper 63 percent, zinc 30 percent, nickel 7 percent	81.40	–
	Gold and Copper:		
	Gold 98 percent, copper 2 percent	184.75	–
	Gold 90 percent, copper 10 percent	168.20	–
	Lead and Tin:		
	Lead 87.5 percent, tin 12.5 percent	103.85	–
	Lead 30.5 percent, tin 69.5 percent	81.10	–
	Monel metal cast (nickel 70 percent, copper 30 percent)	87.00	–
	Steel:		
	Cast	77.00	–
	Wrought	76.30	–
	Black plate per mm of thickness	0.08	–
	Steel sections (*see* 46 'steel sections' in Table 1)		
vi)	**Miscellaneous Materials**		
	Aggregate, coarse	10.80 to 15.70	30
	Ashes, coal, dry, 12 mm and under	5.50 to 6.30	40
	Ashes, coal, dry, 75 mm and under	5.50 to 6.30	38
	Ashes, coal, wet 12 mm and under	7.05 to 7.85	52
	Ashes, coal, wet 75 mm and under	7.05 to 7.85	50
	Asphalt, crushed 12 mm and under	7.05	30 to 45
	Ammonium nitrate, polls	3.55 to 8.35	27
	Bone	18.65	–
	Books and files, stacked	8.35	–
	Calcium ammonium nitrate	9.80	28
	Copper sulphate, ground	11.75	30
	Chalk	21.95	–
	Chinaware, earthenware, stacked (including cavities)	10.80	–
	Clinker, furnace, clean	7.85	30
	Diammonium phosphate	7.85 to 8.50	29

Table 41 (*Continued*)

Sl No.	Materials	Weight KN/m^3	Angle of Friction, Degrees
(1)	(2)	(3)	(4)
	Double salt (ammonium sulphate nitrate)	7.05 to 9.30	34
	Filling cabinet and cupboards with contents in records offices, libraries, archives	5.90	–
	Flue dust, boiler house, dry	5.50 to 7.05	30
	Fly ash, pulverised	5.50 to 7.05	–
	Glass, solid	23.50 to 26.70	–
	Wool	0.16 to 1.18	–
	In sheets	25.50	–
	Glue	12.55	–
	Gypsum, calcined, 12mm and under	8.60 to 9.40	40
	Gypsum, calcined, powdered	9.40 to 12.55	45
	Gypsum, raw, 25 mm and under	14.10 to 15.70	30 to 45
	Hides:		
	Dry Salted } Only green Ice	8.65 8.90	– –
	Leather put in rows	7.85	–
	Lime, ground 3 mm and under	9.40	> 45
	Lime, hydrated 3 mm and under	6.30	30 to 45
	Lime, hydrated, pulverized	5.00 to 6.30	30 to 45
	Lime pebble	8.25 to 8.75	
	Limestone, agricultural 3 mm and under	10.60	
	Limestone, crushed	13.30 to 14.10	30 to 45
	Limestone dust	8.65 to 14.90	30 to 45
	Magnesite, caustic, in powder form	7.85	–
	Magnesite sinter and Magnesite granular	19.60	–
	Phosphate, rock, pulverized	9.40	40 to 52
	Phosphate rock	11.75 to 13.35	30 to 45
	Phosphate sand	14.10 to 15.70	30 to 45
	Potassium carbonate	7.95	30 to 45
	Potassium chloride, pellets	18.85 to 20.40	30 to 45
	Potassium nitrate	4.85	> 30
	Potassium sulphate	6.55 to 7.45	45
	Pyrites, pellets	18.85 to 20.40	30 to 45
	Pumice	5.80 to 9.90	–
	Rubbish:		
	Building	13.80	–
	General	6.30	–
	Salt, common, dry, coarse	6.30 to 10.00	30 to 45
	Salt, common, dry, fine	11.00 to 12.55	30 to 45

Table 41 (*Concluded*)

Sl No.	Materials	Weight KN/m^3	Angle of Friction, Degrees
(1)	(2)	(3)	(4)
	Salt cake, dry, coarse	13.35	30
	Salt cake, dry, pulverized	11.20 to 13.35	35
	Sand, bank. damp	17.25 to 20.40	45
	Sand, bank, dry	14.10 to 17.25	30
	Sand, silica, dry	14.10 to 15.70	30 to 45
	Saw dust	1.57	30
	Silica gel	4.40	30 to 45
	Soda ash, heavy	8.67 to 10.20	35
	Soda ash, light	4.70 to 6.00	37
	Sodium nitrate, granular	11.00 to 12.55	24
	Sulphur, crushed, 12 mm and under	7.85 to 8.25	35 to 45
	Sulphur, 76 mm and under	8.65 to 13.35	32
	Sulphur, powdered	7.85 to 9.40	30 to 45
	Single superphosphate (S.S.P), granulated	7.65 to 8.25	37
	Slag, furnace, crushed	14.90	35
	Steel goods:		
	Cylinders, usually stored for Carbonic acid, etc	13.80	
	Sheets, railway rails, etc, usually stored	44.00	
	Trisodium phosphate	9.40	30 to 45
	Triple superphosphate	7.85 to 8.65	30 to 45
	Turf	2.85 to 5.70	–
	Urea, prills	6.40	23 to 26
vii)	**Ores**		
	Antimony	29.80	–
	Ferrous sulphide	26.50	–
	Ferrous sulphide ore waste after roasting	13.85	–
	Iron ore, compact storing	29.80	–
	Magnesium ore	19.60	–
viii)	**Textiles, Paper and Allied Materials**		
	Cellulose in bundles	7.35	–
	Cotton, compressed	12.75	–
	Flax, piles and compressed in bales	2.95	–
	Furs	8.80	–
	Jute in bundles	6.85	–
	Paper:		
	In bundles and rolls	6.85	–
	Newspaper in bundles	3.90	–
	Put in rows	10.80	–
	Thread in bundles	4.90	–
	Wood, compressed	12.75	–

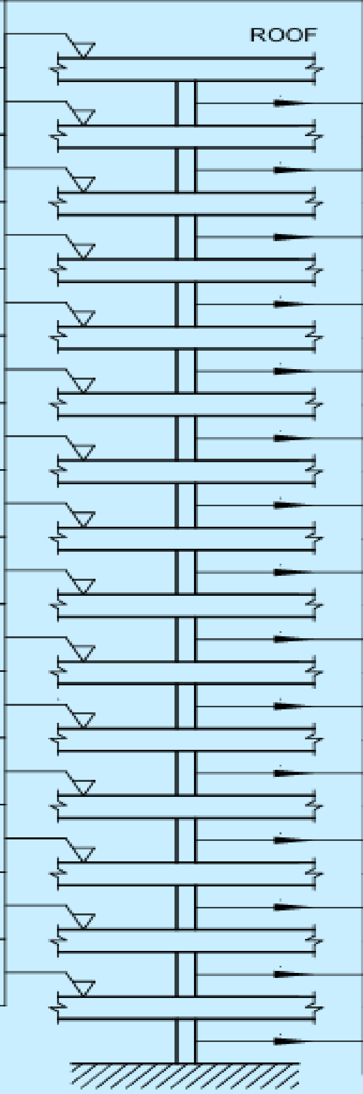
ANNEX B

[*Clause* 3.3.2.1(b)]

ILLUSTRATIVE EXAMPLE SHOWING REDUCTION OF UNIFORMLY DISTRIBUTED IMPOSED FLOOR LOADS IN MULTI-STOREYED BUILDINGS FOR DESIGN OF COLUMNS

B-1 The total imposed loads from different floor levels (including the roof) coming on the central column of a multi-storeyed building (with mixed occupancy) is shown in Fig. 14. Calculate the reduced imposed load for the design of column members at different floor levels using **3.3.2.1**. Floor loads do not exceed 5.0 kN/m^2.

B-1.1 Applying reduction coefficients in accordance with **3.3.2.1**, total reduced floor loads on the column at different levels is indicated along with Fig. 14.



FLOOR NO. FROM TOP INCLUDING ROOF	IMPOSED FLOOR LOADS ON COLUMNS AT DIFFERENT FLOORS, kN	DISCOUNTED IMPOSED LOADING ON COLUMNS, kN
1	30	
2	40	30
3	50	(30 + 40) (1 - 0.1) = 63
4	50	(30 + 40 + 50) (1 - 0.2) = 96
5	40	(30 + 40 + 50 + 50) (1 - 0.3) = 119
6	45	(30 + 40 + 50 + 50 + 40) (1 - 0.4) = 126
7	50	(30 + 40 + 50 + 50 + 40 + 45) (1 - 0.4) = 153
8	50	(30 + 40 + 50 + 50 + 40 + 45 + 50) (1 - 0.4) = 183
9	40	(30 + 40 + 50 + 50 + 40 + 45 + 50 + 50) (1 - 0.4) = 213
10	40	(30 + 40 + 50 + 50 + 40 + 45 + 50 + 50 + 40) (1 - 0.4) = 237
11	40	(30 + 40 + 50 + 50 + 40 + 45 + 50 + 50 + 40 + 40) (1 - 0.4) = 261
12	55	(30 + 40 + 50 + 50 + 40 + 45 + 50 + 50 + 40 + 40 + 40) (1 - 0.5) = 237.5 < 261 adopt 261 for design
13	55	(30 + 40 + 50 + 50 + 40 + 45 + 50 + 50 + 40 + 40 + 40 + 55) (1 - 0.5) = 265
14	70	(30 + 40 + 50 + 50 + 40 + 45 + 50 + 50 + 40 + 40 + 40 + 55 + 55) (1 - 0.5) = 292.5
15	80	(30 + 40 + 50 + 50 + 40 + 45 + 50 + 50 + 40 + 40 + 40 + 55 + 55 + 70) (1 - 0.5) = 327.5
		(30 + 40 + 50 + 50 + 40 + 45 + 50 + 50 + 40 + 40 + 40 + 55 + 55 + 70 + 80) (1 - 0.5) = 367.5

FIG. 14 LOADING DETAILS

ANNEX C

(*Clauses* 4.2)

NOTATIONS

C-1 The following notations to be followed unless otherwise specified in relevant clauses under wind loads:

A	=	Surface area of a structure or part of a structure
A_e	=	Effective frontal area
A_z	=	Effective frontal area of the building at height z
b	=	Breadth of a structure or structural member normal to the wind stream in the horizontal plane
B_s	=	Background factor
C_d	=	Drag coefficient
C_f	=	Force coefficient
C_{fn}	=	Normal force coefficient
C_{ft}	=	Transverse force coefficient
C_f'	=	Frictional drag coefficient
C_p	=	Pressure coefficient
C_{pe}	=	External pressure coefficient
C_{pi}	=	Internal pressure coefficient
C_{fs}	=	Cross-wind force spectrum coefficient
$C_{f,z}$	=	Drag force coefficient of the building corresponding to the area A_z
C	=	Coefficient, which depends on θ_s, used in the evaluation of k_3 factor
d	=	Depth of a structure or structural member parallel to wind stream in the horizontal plane
d_w	=	Wake width
D	=	Diameter of cylinder or sphere
E	=	Wind energy factor
F_z	=	Along wind load on the building/structure at any height z
F	=	Force normal to the surface
f_a	=	First mode natural frequency of the building/structure in along wind direction, in Hz
f_c	=	First mode natural frequency of the building/structure in across wind direction, in Hz
f_s	=	Vortex shedding frequency
F_n	=	Normal force
F_t	=	Transverse force
F'	=	Frictional force
G	=	Gust factor
g_R	=	Peak factor for resonant response
g_v	=	Peak factor for upwind velocity fluctuations
h	=	Height of structure above mean ground level
h_x	=	Height of development of a velocity profile at a distance x down wind from a change in terrain category
H_s	=	Height factor for resonant response

H	=	Height above mean ground level on the topography feature
I	=	Turbulence intensity
$I_{h,i}$	=	Turbulence intensity at height h in terrain category i
$I_{z,i}$	=	Turbulence intensity at height z in terrain category i
IF	=	Interference factor
k	=	Mode shape power exponent
k_1, k_2, k_3, k_4	=	Wind speed modification factors
$\overline{k}_{2,i}$	=	Hourly mean wind speed factor
K	=	Force coefficient multiplication factor for individual members of finite length
K_a	=	Area averaging factor
K_c	=	Combination factor
K_d	=	Wind directionality factor
l	=	Length of the member or larger horizontal dimension of a building
L	=	Actual length of upwind slope
L_e	=	Effective length of upwind slope
L_h	=	Integral turbulence length scale at the height, h
m_0	=	Average mass per unit height of the structure
M_a	=	Design peak along wind base bending moment
M_c	=	Design peak across wind base bending moment
N	=	Effective reduced frequency
p_d	=	Design wind pressure
p_z	=	Wind pressure at height z
$\overline{P}_d$	=	Design hourly mean wind pressure corresponding to $\overline{V}_{z,d}$
p_e	=	External pressure
p_i	=	Internal pressure
r	=	Roughness factor which is twice the longitudinal turbulence intensity at height h, $I_{h\,i}$
R_e	=	Reynolds number
s	=	Level on a building/structure for the evaluation of along wind load effects
s_0	=	Factor, which depends on H and X, used for the evaluation of k_3 factor
S_t	=	Strouhal number
S	=	Size reduction factor
V_b	=	Regional basic wind speed
V_z	=	Design wind speed at height z
$\overline{V}_d$	=	Design hourly mean wind speed
$\overline{V}_{z,d}$	=	Design hourly mean wind speed at height z
$\overline{V}_{z,H}$	=	Hourly mean wind speed at height z
w	=	Lesser horizontal dimension of a building, or a structural member
w'	=	Bay width in multi-bay building
$\hat{\ddot{x}}$	=	Peak acceleration at the top of the building/structure in along wind direction, in m/s^2;

x	=	Distance down wind from a change in terrain category
X	=	Distance from the summit or crest of topography feature relative to the effective length, L_e
$\hat{y}$	=	Peak acceleration at the top of the building/structure in across wind direction
z	=	Height or distance above the ground
$z_{0,i}$	=	Aerodynamic roughness height for i^{th} terrain
Z	=	Effective height of the topography feature
α	=	Inclination of the roof to the horizontal
β	=	Damping coefficient of the building/structure
η	=	Shielding factor
ϕ	=	Factor to account for the second order turbulence intensity
Φ	=	Solidity ratio
Φ_e	=	Effective solidity ratio
ε	=	Average height of the surface roughness
θ_s	=	Upwind slope of the topography feature in the wind direction
θ	=	Wind angle from a given axis

ANNEX D

(*Clause* 4.4.2)

BASIC WIND SPEED AT 10 m HEIGHT FOR SOME IMPORTANT CITIES/TOWNS

City/Town	Basic Wind Speed m/s	City/Town	Basic Wind Speed m/s
Agra	47	Kanpur	47
Ahmedabad	39	Kohima	44
Ajmer	47	Kolkata	50
Almora	39	Kozhikode	39
Amritsar	50	Kurnool	39
Asansol	47	Lakshadweep	39
Aurangabad	39	Lucknow	50
Bahraich	50	Ludhiana	50
Bengaluru	33	Madurai	39
Barauni	47	Mandi	39
Bareilly	50	Mangaluru	39
Bhatinda	47	Moradabad	50
Bhilai	44	Mumbai	44
Bhopal	47	Mysuru	33
Bhubaneshwar	50	Nagpur	44
Bhuj	50	Nainital	47
Bikaner	47	Nasik	39
Bokaro	39	Nellore	50
Chandigarh	50	Panaji	39
Chennai	50	Patiala	50
Coimbatore	39	Patna	47
Cuttack	50	Puducherry	50
Darbhanga	47	Port Blair	44
Darjeeling	47	Pune	39
Dehra Dun	39	Raipur	44
Delhi	50	Rajkot	39
Durgapur	47	Ranchi	39
Gangtok	47	Roorkee	50
Guwahati	50	Rourkela	39
Gaya	39	Simla	39
Gorakhpur	47	Srinagar	39
Hyderabad	44	Surat	44
Imphal	44	Tiruchirrappalli	47

City/Town	Basic Wind Speed m/s	City/Town	Basic Wind Speed m/s
Jabalpur	39	Thiruvananthapuram	39
Jaipur	47	Udaipur	47
Jamshedpur	47	Vadodara	39
Jhansi	47	Varanasi	47
Jodhpur	47	Vijaywada	50
		Visakhapatnam	50

ANNEX E

[*Clause* 4.4.3.2.4(b)(2)]

CHANGES IN TERRAIN CATEGORIES

E-1 LOW TO HIGH TERRAIN CATEGORY NUMBER

In cases of transition from a low terrain category number (corresponding to a low terrain roughness) to a higher terrain category number (corresponding to a rougher terrain), the velocity profile over the rougher terrain shall be determined as follows:

a) Below height h_x, the velocities shall be determined in relation to the rougher terrain; and

b) Above height h_x, the velocities shall be determined in relation to the less rough (more distant) terrain.

E-2 HIGH TO LOW TERRAIN CATEGORY NUMBER

In cases of transition from a more rough to a less rough terrain, the velocity profile shall be determined as follows:

a) Above height h_x, the velocities shall be determined in accordance with the rougher (more distant) terrain; and

b) Below height h_x, the velocity shall be taken as the lesser of the following:

   1) That determined in accordance with the less rough terrain; and

   2) The velocity at height h_x as determined in relation to the rougher terrain.

NOTE — Examples of determination of velocity profiles in the vicinity of a change in terrain category are shown in Fig. 15(A) and Fig. 15(B).

E-3 MORE THAN ONE CATEGORY

Terrain changes involving more than one category shall be treated in similar way to that described in **E-1** and **E-2**.

NOTE — Examples involving three terrain categories are shown in Fig. 15(B).

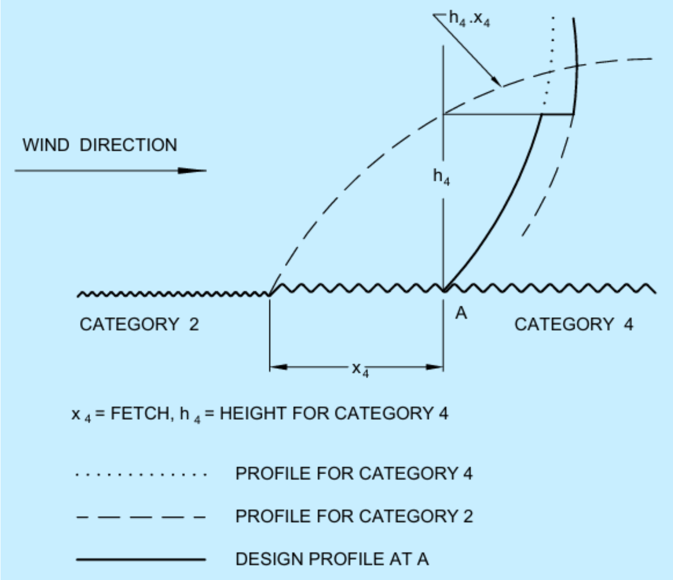


FIG. 15(A) DETERMINATION OF VELOCITY PROFILE NEAR A CHANGE IN TERRAIN CATEGORY (Less rough to more rough)

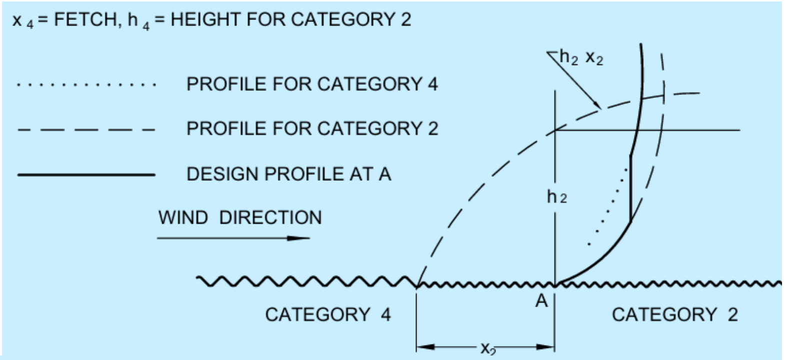


FIG. 15(B) DETERMINATION OF VELOCITY PROFILE NEAR A CHANGE IN TERRAIN CATEGORY (More rough to less rough)

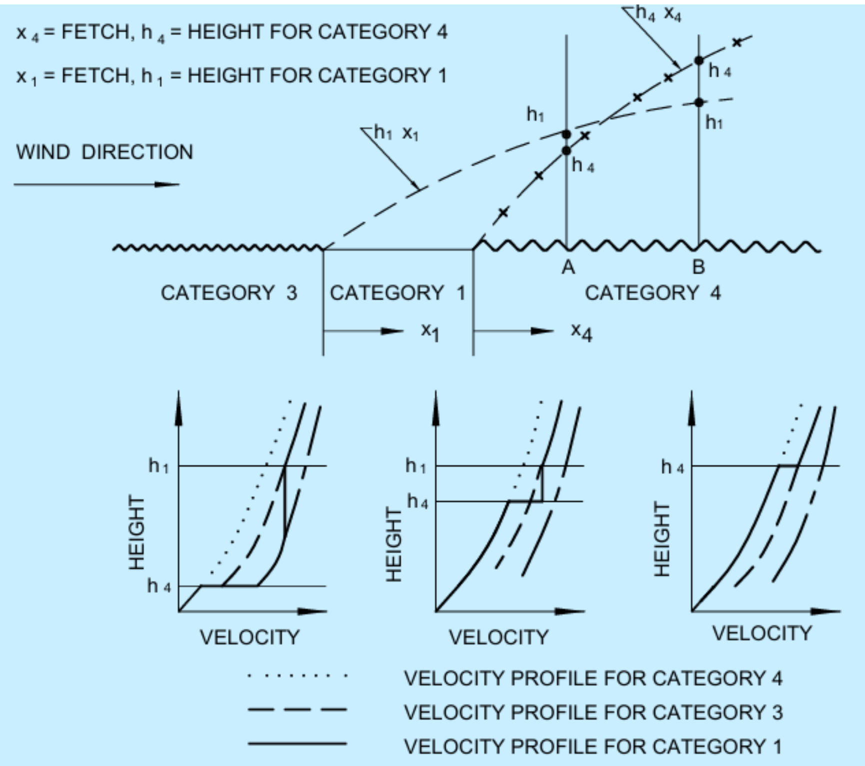


FIG. 15(C) DETERMINATION OF DESIGN PROFILE INVOLVING MORE THAN ONE CHANGE IN TERRAIN CATEGORY

FIG. 15 VELOCITY PROFILES IN THE VICINITY OF A CHANGE IN TERRAIN CATEGORY

ANNEX F

(*Clause* 4.4.3.3)

EFFECT OF A CLIFF OR ESCARPMENT ON EQUIVALENT HEIGHT ABOVE GROUND (k_3 FACTOR)

F-1 The influence of the topographic feature is considered to extend 1.5 L_e upwind and 2.5 L_e downwind of the summit of crest of the feature where Le is the effective horizontal length of the hill depending on slope as indicated below (*see* Fig. 16):

Slope	L_e
$3° < \theta_s \leq 17°$	L
$\theta_s > 17°$	$Z / 0.3$

where

L = actual length of the upwind slope in the wind direction,

Z = effective height of the topography feature and

θ_s = upwind slope in the wind direction.

In case, the zone in downwind side of the crest of the feature is relatively flat ($\theta_s < 3°$) for a distance exceeding L_e, then the feature should be treated as an escarpment. Otherwise, the feature should be treated as a hill or ridge. Examples of typical features are given in Fig. 16.

NOTES

1 No difference is made, in evaluating k_3 between a three dimensional hill and two dimensional ridge.

2 In undulating terrain, it is often not possible to decide whether the local topography to the site is significant in terms of wind flow. In such cases, the average value of the terrain upwind of the site for a distance of 5 km should be taken as the base level from wind to assess the height, Z and the upwind slope θ, of the feature.

F-2 TOPOGRAPHY FACTOR, k_3

The topography factor k_3 is given by the following:

$$k_3 = 1 + C\,s_0$$

where C has the following values:

Slope	*C*
$3° < \theta_s \le 17°$	1.2 (Z/L)
$\theta_s > 17°$	0.36

and s_0 is a factor derived in accordance with F-2.1 appropriate to the height, H above mean ground level and the distance, X, from the summit or crest relative to the effective length, L_e.

F-2.1 The factor, s_0 should be determined from:

a) Fig. 17 for cliffs and escarpments; and

b) Fig.18 for ridges and hills.

NOTE — Where the downwind slope of a hill or ridge is more than 3°, there will be large regions of reduced accelerations or even shelter and it is not possible to give general design rules to cater for these circumstances. Values of s_0 from Fig. 18 may be used as upper bound values.

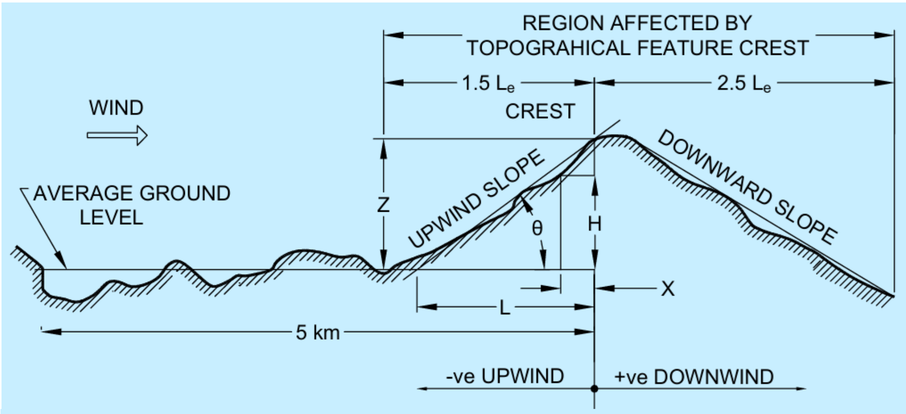
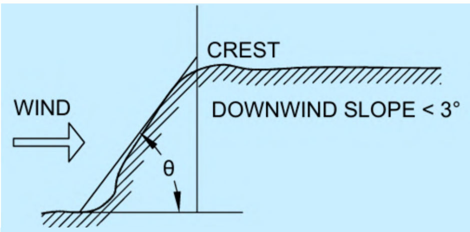
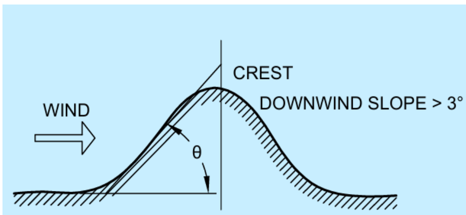


FIG. 16(A) GENERAL NOTATIONS



16(B) CLIFF AND ESCARPMENT



16(C) HILL AND RIDGE

FIG. 16 TOPOGRAPHICAL DIMENSIONS

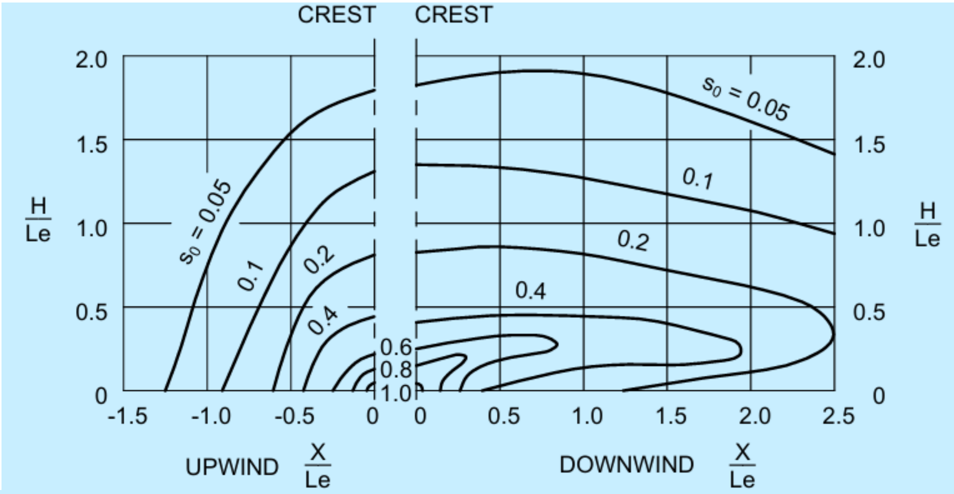


FIG. 17 FACTOR s_0 FOR CLIFF AND ESCARPMENT

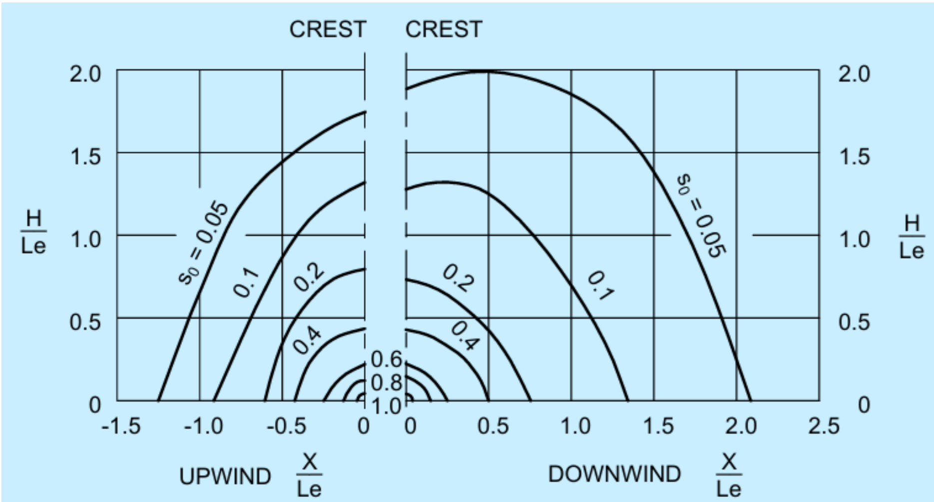


FIG. 18 FACTOR s_0 FOR RIDGE AND HILL

ANNEX G

[*Clauses* 4.5.4.2.2, 4.5.4.3.2(c) *and* 4.5.4.3.3]

WIND FORCE ON CIRCULAR SECTIONS

G-1 The wind force on any object is given by:

$$F = C_f A_e p_d$$

where

C_f = force coefficient;

A_e = effective area of the object normal to the wind direction; and

p_d = design pressure of the wind.

For most shapes, the force coefficient remains approximately constant over the whole range of wind speeds likely to be encountered. However, for objects of circular cross section, it varies considerably.

For a circular section, the force coefficient depends on the way in which the wind flows around it and is dependent upon the velocity and kinematic viscosity of the wind and diameter of the section. The force coefficient is usually quoted against a non-dimensional parameter, called the Reynolds number, which takes into account of the velocity and viscosity of the flowing medium (in this case the wind) and the member diameter.

$$\text{Reynolds number, } R_e = \frac{D\overline{V}_d}{\nu}$$

where

D = diameter of the member;

$\overline{V}_d$ = design hourly mean wind speed; and

ν = kinematic viscosity of the air which is 1.46×10^{-5} m²/s at 15° and standard atmospheric pressure.

Since in most natural environments likely to be found in India, the kinematic viscosity of the air is fairly constant, it is convenient to use $D\overline{V}_d$ as the parameter instead of Reynolds number and this has been done in this Section.

The dependence of a circular section's force coefficient on Reynolds number is due to the change in the wake developed behind the body.

At a low Reynolds number, the wake is as shown in Fig. 19 and the force coefficient is typically 1.2. As Reynolds number is increased, the wake gradually changes to that shown in Fig. 20, that is, the wake width d_w decreases and the separation point, denoted as s_p, moves from front to the back of the body.

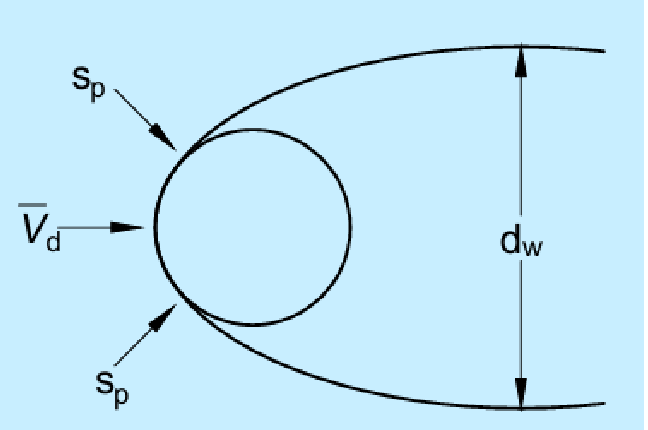


FIG. 19 WAKE IN SUB-CRITICAL FLOW

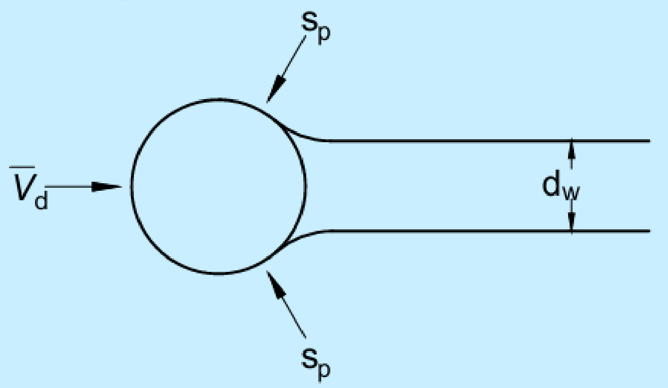


FIG. 20 WAKE IN SUPER-CRITICAL FLOW

As a result, the force coefficient shows a rapid drop at a critical value of Reynolds number followed by a gradual rise as Reynolds number is increased still further.

The variation of C_f with parameter $D\overline{V}_d$ is shown in Fig. 5 for infinitely long circular cylinders having various values of relative surface roughness (ε/D) when subjected to wind having an intensity and scale of turbulence typical of built-up urban areas. The curve for a smooth cylinder (ε/D) = 1×10^{-5} in a steady air stream, as found in a low-turbulence wind tunnel, is also shown for comparison.

It can be seen that the main effect of free-stream turbulence is to decrease the critical value of the parameter $D\overline{V}_d$. For subcritical flows, turbulence can produce a considerable reduction in C_f below the steady air-stream values. For supercritical flows, this effect becomes significantly smaller.

If the surface of the cylinder is deliberately roughened, such as by incorporating flutes, riveted construction, etc, then the data given in Fig. 5 for appropriate value of $\varepsilon/D > 0$ shall be used.

NOTE — In case of uncertainty regarding the value of ε to be used for small roughness, ε/D shall be taken as 0.001.

ANNEX H

(*Foreword and clause 7.3.2*)

CHARACTERISTIC GROUND SNOW LOAD

H-1 This annex presents the characteristic ground snow load computation procedure for the union territory of Jammu and Kashmir, union territory of Ladakh, Himachal Pradesh, Uttarakhand and Sikkim which is the result of scientific work carried out by Defence Geoinformatics Research Establishment (DGRE), Chandigarh under Defence Research and Development Organization (DRDO).

H-2 The characteristic value of the ground snow load at any site is defined with a 3.4 percent annual exceedance probability or, equivalently, with a mean recurrence interval, MRI = 30 years. The site altitude is the altitude at which the structure is or will be located, measured from mean sea level. The procedure shall be used for the computation of characteristic ground snow load on sites with an altitude higher than 1000 m and lower than 4000 m from mean sea level.

H-3 The characteristic ground snow load, s_0 (kPa) to be used for any site in the union territory of Jammu and Kashmir, union territory of Ladakh, Himachal Pradesh, Uttarakhand should be obtained from the zone map shown in Fig. 21 and equation given below:

$$s_0 = 1.624 \times [Z - 0.5] \times \left[1 + \left(\frac{A}{2677}\right)^2\right] \text{ kPa}$$

where

s_0 = characteristic ground snow load (kPa);

Z = zone number (for example 1, 2, 3, 4 or 5) obtained from the zone; and

A = map in Fig. 21.

H-4 The characteristic ground snow load s_0 (kPa) to be used for any site in Sikkim should be directly obtained from the characteristic ground snow load map shown in Fig. 22.

H-5 The ground snow loads for any mean recurrence interval of n years, different to that for the characteristic snow load, s_0, (which by definition is based on annual probability of exceedance of 0.034 that is MRI = 30 years) shall be calculated by the equation given below:

$$s_n = s_0 \left\{ \frac{1 - C_\mathrm{v}\frac{\sqrt{6}}{\pi}\left[ln\left(-ln(1-P_\mathrm{n})\right) + 0.577\ 22\right]}{(1+2.188\ 7C_\mathrm{v})} \right\}$$

where

s_0	=	characteristic snow load on the ground with MRI = 30 years
s_n	=	characteristic snow load on the ground with MRI = 30 years
P_n	=	annual probability of exceedance (equivalent to $1/n$, where n is the corresponding recurrence interval in years)
C_v	=	coefficient of variation of annual maximum snow load.

H-5.1 The coefficient of variation of maximum snow load at different snow-meteorological observatories of DGRE located in union territory of J and K, union territory of Ladakh, HP and UK ranges between 0.35 to 0.65. For application of above equation, a constant value of $C_\mathrm{v} = 0.5$ is recommended for all the zones to calculate snow loads on the ground for other probability of exceedance (for example, MRI of 50 years for structures where a higher risk of exceedance is deemed acceptable). The above equation is shown graphically in Fig. 23.

NOTES

1 The characteristic ground snow load *versus* altitude relation as given above in equation and Fig. 21 is determined based on the snow depth observations at more than 70 field observatories of DGRE located in different parts of union territory of Jammu and Kashmir, union territory of Ladakh and states of Himachal Pradesh and Uttarakhand as well as High Asia Refined analysis (HAR10) dataset (HAR, Maussion et al., 2014).

2 Since adequate snow depth measurements are not available for Sikkim region, hence the characteristic ground snow load map for Sikkim given in Fig. 22 is generated based solely on high Asia refined analysis (HAR10) dataset (HAR, Maussion et al., 2014).

3 Annex H does not apply for sites at altitudes above 4000 m. For altitudes greater than 4000 m specialist advice should be sought from DGRE on the characteristic ground snow loads likely to occur at the site.

4 Unusual local effects have not been accounted for in the analysis undertaken to produce the characteristic ground snow load map given in Annex H. These include local shelter from the wind, which can result in increased local snow loads and local configurations in mountainous areas, which may funnel the snow and give increased local loading. Annex H is also not applicable for sites with permanent snow/ice cover throughout the year. If the designer suspects that there are unusual local conditions that need to be taken into account, DGRE may be consulted.

5 In special cases where more refined data is needed, the characteristic value of snow load on the ground (s_0) may be refined using an appropriate statistical analysis of long term records taken in a well sheltered area near the site.

6 Ground characteristics snow load at any place depends on the critical combination of maximum depth of undisturbed aggregate cumulative snowfall and its density. In due course the characteristics snow load for states of Arunachal Pradesh will be included based on studies. Till such time the users of this section are advised to consult DGRE for the specific information relating to the state of Arunachal Pradesh.

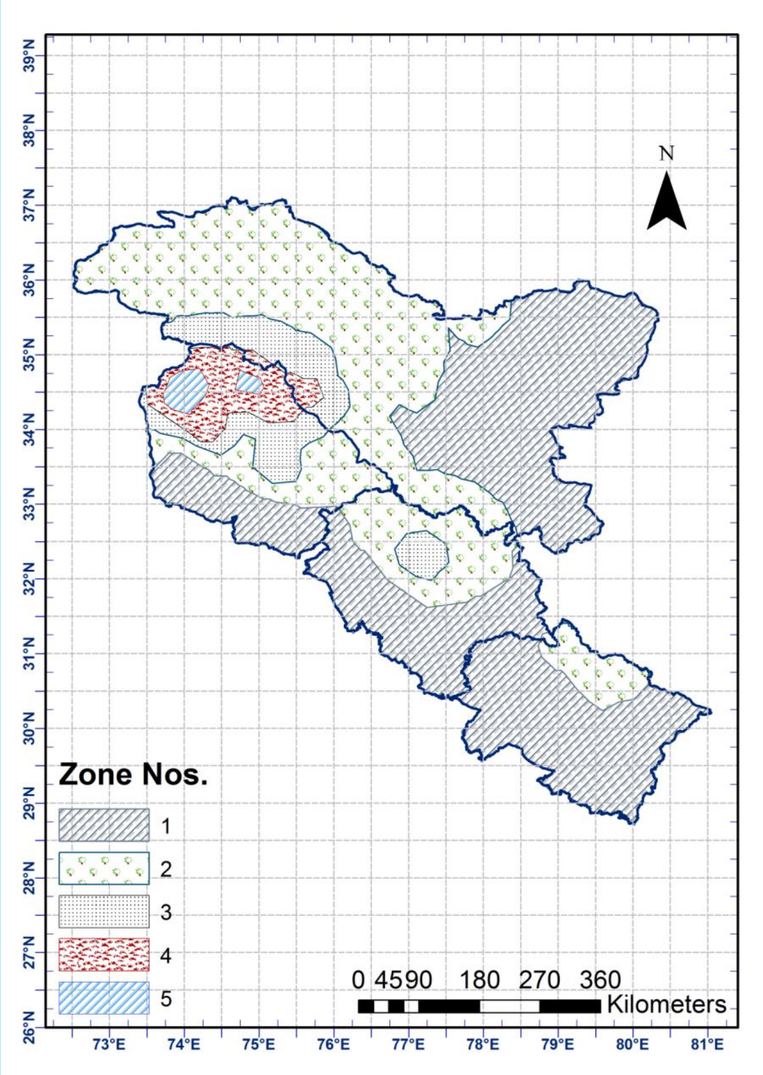


FIG. 21 ZONE MAP FOR J AND K, LADAKH, HP AND UK AT 10 KM RESOLUTION

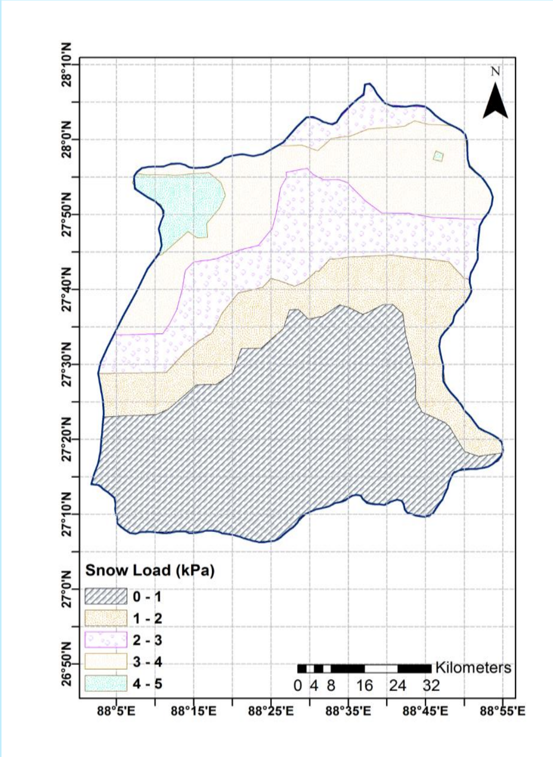


FIG. 22 CHARACTERISTIC GROUND SNOW LOAD MAP FOR SIKKIM AT 10 KM RESOLUTION

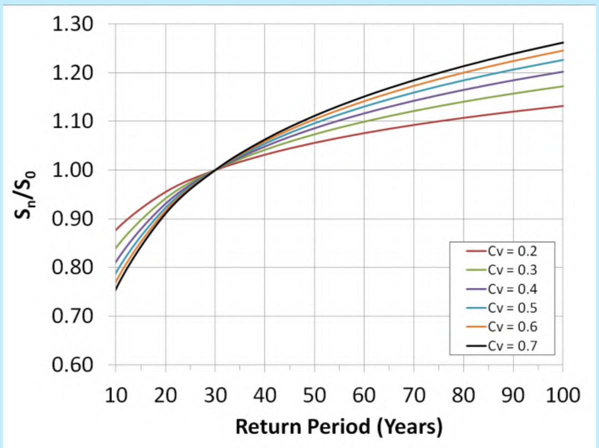


FIG. 23 ADJUSTMENT OF THE GROUND SNOW LOAD ACCORDING TO THE RETURN PERIOD

ANNEX J

(*Clauses* 7.4.2.4 *and* 7.4.3)

SHAPE COEFFICIENTS FOR MULTILEVEL ROOFS

J-1 A more comprehensive formula for the shape coefficient for multilevel roofs than that given in **7.4.2** is as follows:

$$\mu_w = 1 + \frac{1}{h}(m_1 l_1 + m_2 l_2)(l_2 - 2h)$$

$$\mu_1 = 0.8$$

$$l_3 = 2h$$

$$(h \text{ and } l \text{ in m})$$

Restriction:

$$\mu_w \leq \frac{kh}{S_o}$$

where

S_0 = kilopascals (kilonewtons per square metre)

k = newtons per cubic metre

l_3 = $\leq 15\ m$

Values of $m_1 (m_2)$ for the higher (lower) root depend on its profile and are taken as equal to:

0.5 for plane roofs with slopes $\beta \leq 20°$ and vaulted roofs with $\frac{f}{l} \leq \frac{1}{18}$

0.3 for plane roofs with slopes $\beta \leq 20°$ and vaulted roofs with $\frac{f}{l} \leq \frac{1}{18}$

The coefficient m_1 and m_2 may be adjusted to take into account conditions for transfer of snow on the roof surface (that is wind, temperature, etc).

NOTE — The other condition of loading shall also be tried.

ANNEX K

(*Clause* 8.7)

VIBRATIONS IN BUILDINGS

K-1 GENERAL

In order to design the buildings safe against vibrations, it is necessary to identify the source and nature of vibration. Vibrations may be included in the buildings due to various actions, such as:

a) human induced vibrations, for example, the walking or running or a single person or a number of persons or dancing or motions in stadia or concert halls;
b) machine induced vibrations;
c) wind induced vibrations;
d) blast induced vibrations;
e) traffic load, for example, due to rail, fork-lift, trucks, cars, or heavy vehicles;
f) airborne vibrations;
g) crane operations; and
h) other dynamic actions such as wave loads or earthquake actions.

The dynamic response of buildings for the above mentioned causes of vibration of buildings may have to be evaluated by adopting standard mathematical models and procedures.

The severity or otherwise of these actions have to be assessed in terms of the limits set for dynamic response (frequencies and amplitude of motion) of the buildings related to (a) human comfort, (b) serviceability requirements such as deflections and drifts and separation distances to avoid damage due to pounding and (c) limits set on the frequencies and amplitude of motion for machines and other installations.

In order to verify that the set limits are not exceeded, the actions may be modelled in terms of force-time histories for which the structural responses may be determined as time histories of displacements or accelerations by using appropriate analytical/numerical methods.

K-2 SERVICEABILITY LIMIT STATE VERIFICATION OF STRUCTURE SUSCEPTIBLE TO VIBRATIONS

K-2.1 While giving guidance for serviceability limit state verification of structure susceptible to vibrations, here it is proposed to deal with the treatment of the action side, the determination of the structural response and the limits to be considered for the structural response to ensure that vibrations are not harmful or do not lead to discomfort.

K-2.2 Source of Vibrations

Vibrations may be included by the following sources:

a) by the movement of persons as in pedestrian bridges, floors where people walk, floors meant for sport or dancing activities and floors with fixed seating and spectator galleries;

b) by working of machines as in machine foundations and supports, vibrations transmitted through the ground and pile driving operations;

c) by wind blowing on buildings, towers, chimneys and masts, guyed masts, pylons, bridges, cantilevered roofs, airborne vibrations;

d) induced by traffic on rail or road bridges and car park structures and exhibition halls; and

e) by earthquakes.

K-2.3 Modelling of Actions and Structures

For serviceability limit states, the modelling of these actions and of the structure depends on how the serviceability limits are formulated. The serviceability limit states may refer to:

a) human comfort;

b) limits for the proper functioning of machines and other installations; and

c) maximum deformation limits to avoid damage or pounding.

In order to verify that these limits are not exceeded, the actions may be modeled in terms of force-time histories, for which the structural responses may be determined as time histories of displacements or accelerations by using appropriate analytical/numerical methods. Where the structural response may significantly influence the force-time histories to be applied, these interactions have to be considered either in modeling a combined load-structure vibration system or by appropriate modifications of the force time histories. In addition to the levels of vibration for which presently limits have been specified, the possible deformations of structural members and systems using different clauses in the relevant codes have to be evaluated by adopting standard mathematical models and procedures.

K-2.4 Force-Time Histories

The force-time histories used in the dynamic analysis should adequately represent the relevant loading situations for which the serviceability limits are to be verified. The force-time histories may model:

a) human induced vibrations, for example the walking or running of a single person or a number of persons or dancing or motions in stadia or concert halls;

b) machine induced vibrations, for example by force vectors due to mass eccentricities and frequencies, that may be variable with time;

c) wind induced vibrations;

d) blast induced vibrations;

e) traffic load, for example rail, fork-lift, trucks, cars, or heavy vehicles;

f) airborne vibrations;

g) crane operations; and

h) other dynamic actions such as wave loads or earthquake actions.

ANNEX L

(*Foreword and clause* 8.8)

BLAST LOAD

L-1 BLAST LOAD

L-1.1 This annex prescribes the minimum criteria for analysis and design of new structures for blast effects of above-ground explosions of high-explosives under three different detonation scenarios: surface burst, free-air burst and air burst. The primary objective of this annex is to minimize human casualties and safeguard assets due to above ground explosions. The performance expectation of a structure designed as per the provisions of this annex would depend on the type of protection category selected for that structure and the blast loading determined through threat assessment.

L-1.2 This annex shall be used in conjunction with other applicable standards for analysis and design of structures. This annex does not supersede or replace other design standards. This annex shall be applicable if design against malevolent or accidental blast loading is determined to be a credible threat to a structure.

L-1.3 The owner of a structure is the competent authority who can determine the necessity of blast-resistant design. In the scenario of accidental and malevolent blast event which may pose public health and safety issue, competent authorities other than the owner of building that govern or regulate the construction, operation and maintenance of the structural facility, can specify the requirement of blast-resistant design.

L-1.4 The provisions of this annex do not address the following:

a) Estimation of structural loads due to nuclear detonations, pressure vessel explosions, dust and vapour cloud explosions;

b) Effects of cased explosives and shaped charges;

c) Threats emanating from fragmentation, induced fire or thermal effects, biological or chemical radiations, electromagnetic pulse (EMP) and deflagration events; and

d) Detonations in contact with structural elements.

L-2 TERMINOLOGY

For the purpose of this annex, the following definitions shall apply.

L-2.1 Arrival Time — Time taken by the shock after the explosion to reach the location of interest.

L-2.2 Charge Mass — Mass of the explosive material responsible for generating blast effects.

L-2.3 Clearance Time — Time in which the reflected pressure decays down to the sum of the side on overpressure and the dynamic pressure.

L-2.4 Close-in Detonation — Detonation at a scaled distance (Z) below which the resulting blast pressure distribution is non-uniform over the reflecting surface of the element being considered. It corresponds to a value of Z less than 1.2 m/kg1/3.

L-2.5 Decay Parameter — The coefficient of the negative power of exponent e governing the fall of pressure with time in the pressure-time curves.

L-2.6 Detonation — Fast and sudden release of huge amount of energy through a stable chemical reaction in which reaction front propagates into unreacted substance at supersonic speed resulting in formation of shock waves.

L-2.7 Drag Force — Force on a structure or structural element due to the dynamic pressure of the blast wave. On any structural element, the drag force equals dynamic pressure multiplied by the drag coefficient and the area of the element.

L-2.8 Ductility Ratio — Ratio of the maximum deflection to the deflection corresponding to the elastic limit.

L-2.9 Dynamic Increase Factor (DIF) — A factor applied to material strength to account for high strain rate loading effects.

L-2.10 Dynamic Pressure — The pressure due to air mass movement behind the shock front.

L-2.11 Exponential Blast Pressure History — The variation of blast pressure with time at a point represented through an exponential decaying function, which considers positive and negative phase of blast overpressures.

L-2.12 Far-Field Detonation — Detonation at a scaled distance (Z) above which the resulting blast pressure distribution is uniform over the reflecting surface of element being considered. It corresponds to a value of Z greater than 1.2 m/kg$^{1/3}$.

L-2.13 High-Explosives — Chemical explosive that undergoes detonation to produce shock waves with high pressure and short duration.

L-2.14 Impulse — The integral of the pressure-time curve per unit projected area due to explosion (*see* specific impulse).

L-2.15 Mach Number — The ratio of the speed of the shock front propagation to the speed of sound in standard atmosphere at sea level.

L-2.16 Negative Phase — The duration of reflected overpressure when it is below the ambient atmospheric pressure.

L-2.17 Overpressure — The rise in pressure above atmospheric pressure due to the shock wave from an air blast.

L-2.18 Positive Phase —The duration of overpressure when it is above the ambient atmospheric pressure.

L-2.19 Protection Category — Category assigned to different structures that correlates its importance to the performance goals through qualitative estimate of damage.

L-2.20 Reflected Overpressure —The overpressure resulting due to reflection of a shock wave front striking any surface. If the shock front is parallel to the surface, the reflection is normal.

L-2.21 Regular Shaped Structure — A structure that does not have any spatial geometric irregularity.

L-2.22 Scaled Distance — The distance between the centre of explosion to the point of interest divided by the cube root of the charge mass expressed in equivalent TNT. The unit for scaled distance is m/kg$^{1/3}$.

L-2.23 Shape Function — The displaced shape of the component as a function of its length normalized to the maximum deflection.

L-2.24 Shock Wave Front — The discontinuity between the blast wave and the surrounding atmosphere. It propagates away from the point of explosion in all directions at a speed greater than the speed of sound in the undisturbed atmosphere.

L-2.25 Side-on (Incident) Overpressure — Overpressure if it is not reflected by any surface.

L-2.26 Simplified Overpressure History — A triangular representation of the overpressure history obtained by equating the impulse and the peak overpressure of the exponential representation of the overpressure history.

L-2.27 Specific Impulse — The area under the pressure-time curve due to explosion. This is often referred to as impulse (*see* **L-2.13**).

L-2.28 Standoff Distance — The shortest distance between the centre of detonation and the point of interest at which the effects of explosion need to be considered.

L-2.29 Support Rotation — The angle formed by a flexural member in its deflected shape with respect to the chord adjoining the two support ends (*see* Fig. 45).

L-2.30 Transit Time — Time required for the shock front to travel across the structure or its element under consideration.

L-3 SYMBOLS

For the purpose of this annex, the following notations shall apply.

b	Decay parameter/coefficient
C_d	Drag coefficient
C_r	Velocity of sound in air
$C_{r\alpha}$	Reflection coefficient
E	Modulus of elasticity of material
H	Height of the structure
I	Moment of inertia of member
i_e	Equivalent specific impulse of a uniformly distributed load
i_{ro}	Peak reflected specific impulse
j	Number of concentrated load points
K_L	Load factor
K_{LM}	Load mass factor
K_M	Mass factor
k_E	Effective stiffness of equivalent single spring-mass system
L	Length of structure in the direction of motion of blast wave (*see* Fig. 39)
M	Charge mass/weight
M_p	Plastic moment of section
M_{pm}	M_p at mid-span
M_{ps}	M_p at support
M_{pfa}	Total positive ultimate moment capacity along mid-span section parallel to short edge
M_{pfb}	Total positive ultimate moment capacity along mid-span section parallel to long edge

m	Distributed mass intensity per unit length
n	Number of concentrated masses
p_a	Ambient atmospheric pressure
P_r	Concentrated dynamic force at point r
p_{ro}	Peak reflected overpressure
p_s	Side-on overpressure
p_{so}	Peak side-on overpressure
P_t	Total dynamic load (at any instant of time)
$p(x)$	Intensity of distributed dynamic load per unit length
p_{o1}	Peak reflected overpressure on a surface including the effects of clearing
p_{o2}	Intercept of the reflected overpressure history on the y-axis corresponding to the clearing phase
q	Dynamic pressure
q_o	Peak dynamic pressure
R	Standoff distance
R_m	Peak resistance of a structural member against a given load distribution
S	W or H whichever is less
T	Elastic time-period of structural member
t_c	Clearance time
t_d	Duration of the equivalent triangular pulse
t_m	Time to the peak displacement of an SDOF system
t_o	Time for positive phase of side-on overpressure
t_t	Transit time
t_{d1}	Time at which clearing starts on a finite reflecting surface (*see* Fig. 47B)
t_{d2}	Total time duration of blast loading on a finite reflecting surface (*see* Fig. 47B)
U	Shock front velocity
u	Peak particle velocity
V_A	Total dynamic reaction along a short edge
V_B	Total dynamic reaction along a long edge
y_E	Displacement corresponding to the elastic limit of an equivalent bilinear representation of idealised resistance-deflection curve
y_{el}	Displacement corresponding to the first onset of yielding (elastic limit) of the idealized resistance-deflection curve of a structural member
y_m	Maximum deflection/deformation permitted in the design of the structure
y_o	Peak static displacement of an SDOF system without inertial effects
W	Span or width of structure across the direction of shock wave propagation
Z	Scaled distance

μ	Ductility ratio $= \frac{y_m}{y_E}$
φ_r	Deflection at point r of an assumed deflected shape for concentrated loads
$\varphi_{(x)}$	Deflection at point x of an assumed deflected shape for distributed loads
$\varphi_{(x,y)}$	Deflected shape function of blast-loaded component

L-4 GENERAL PRINCIPLES

L-4.1 Blast Resistant Design Attributes

The design process of a building against blast should start with planning of building layout and arrangements. The distance between the explosive and the building shall be maximized through a combination of site-access controls, landforms, anti-ram structures, street and site landscaping and perimeter walls. The building should preferably have short-face oriented towards blast source. Also, it shall be located uphill and placed upstream of wind direction and away from congestion. The structures and their contents that need protection against an explosion event should be decided and all possible scenarios need to be considered which can cause damage or occupant injuries. The overall design of a structure shall aim to incorporate following attributes in the overall planning and design of a facility against explosion threats:

a) *Security* — Discourage potential terrorist attacks on a structure through visible security personnel deployment and defences to reduce chance of such attacks;

b) *Deception* — Disguise the critical area or content of structure so that the focus of the attack is misdirected. Such attacks, even if successfully completed, may not deliver intended impact and realize its goals;

c) *Shielding* — Shield the target structure through physical barriers, including but not limited to blast walls, bollards, anti-ram elements, vehicle and pedestrian entry controls and landforms. The objective of shielding is to increase the standoff distance between the charge and the target structure and components; and

d) *Design* — The attack must be blunted through appropriate design and detailing of the structure to absorb the energy of the attack and safeguard valuable assets.

L-4.2 Blast Source

L-4.2.1 *Charge Shape*

The provisions of this annex shall be applicable for:

a) explosives of spherical shape; and

b) non-spherical charges if the detonation effects are considered at scaled distances greater than 3 m/kg$^{1/3}$.

In case of non-spherical charges at scaled distances smaller than 3 m/kg$^{1/3}$, the provisions of **L-8.3.3** shall be applicable with explicit modelling of the detonation process.

L-4.2.2 *Detonation Scenarios*

The standard addresses three scenarios of unconfined explosions based on the location of the detonation with respect to the point of interest:

a) Spherical (free-air) burst (*see* Fig. 24);

b) Hemispherical (surface) burst (*see* Fig. 25); and

c) Air-burst (*see* Fig. 26).

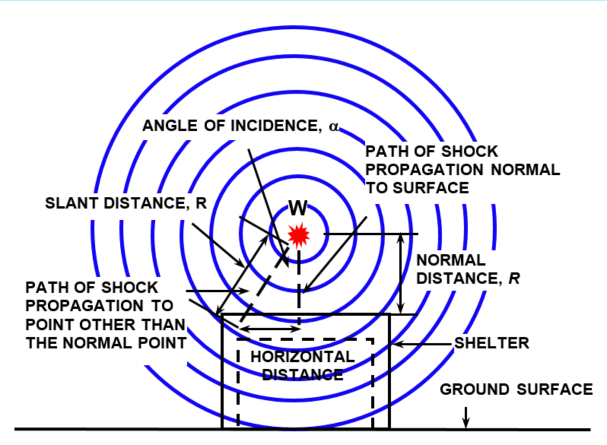


FIG. 24 SPHERICAL (FREE-AIR) BURST

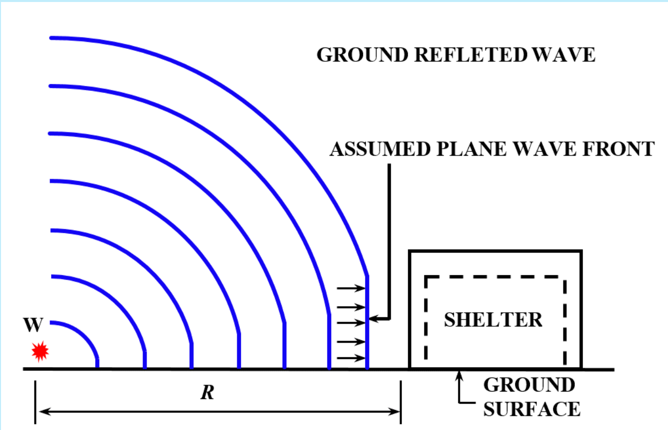


FIG. 25 HEMISPHERICAL (SURFACE) BURST

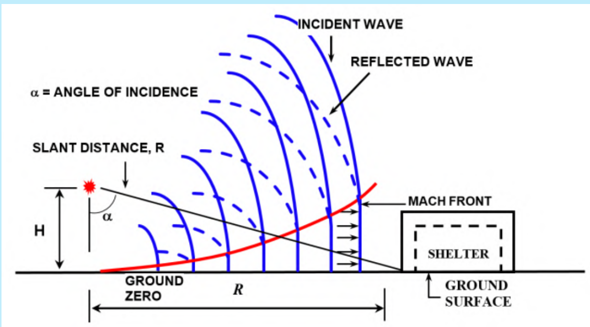


FIG. 26 AIR-BURST

L-4.2.3 *Standoff Distance*

The standoff distance (R) shall be estimated as the distance between the centre of detonation and the point of interest. If the location of interest is not directly in the line of the sight from the centre of detonation, then Fig. 27 shall be referred to estimate the effective standoff distance.

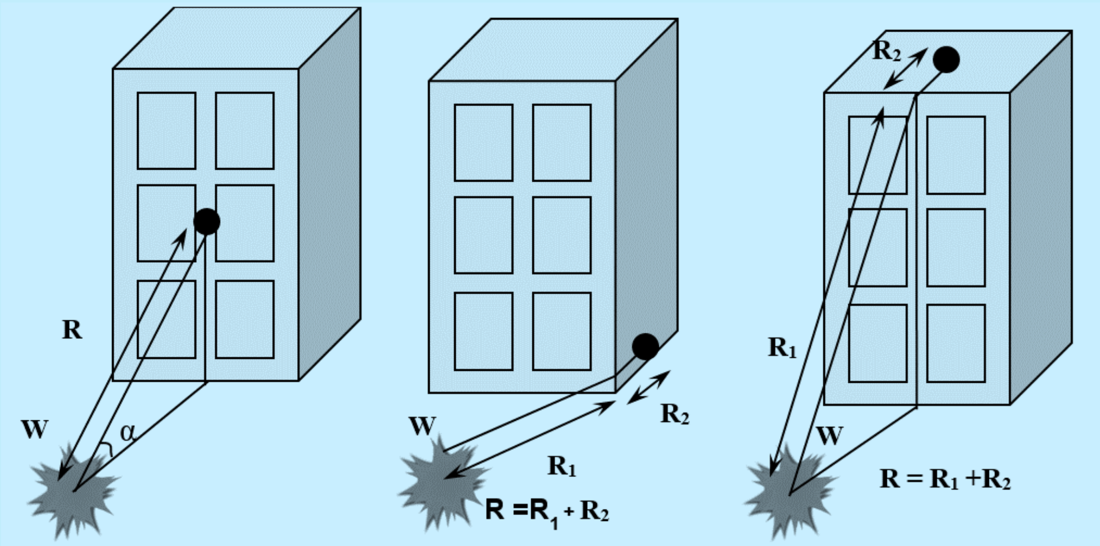


FIG. 27(A) FRONT FACE

FIG. 27(B) SIDE FACE

FIG. 27(C) TOP FACE

FIG. 27 CALCULATION OF STANDOFF DISTANCES AT DIFFERENCE FACES OF A BUILDING STRUCTURE

L-4.2.4 *Equivalent TNT Charge*

Trinitrotoluene (TNT) shall be used to benchmark all types of explosives. To quantify blast wave parameters (pressure, impulse) from explosives other than TNT, the actual mass of the explosive (M_{EXP}) shall be converted into a TNT-equivalent mass. The TNT equivalent weight (M_{TNT}) of any explosive shall be estimated as:

$$M_{TNT} = \left(\frac{\Delta E_{EXP}}{\Delta E_{TNT}}\right) M_{EXP}$$

where

ΔE_{EXP} and ΔE_{TNT} are the specific energies of the explosive and the TNT, respectively.

The factors presented in Table 42 shall be utilized to obtain equivalent mass of TNT corresponding to the listed explosives.

Table 42 Equivalent TNT Mass of Different Explosives

(*Clause* L-4.2.4)

Sl No.	Explosive	Density Mg/m³	Equivalent Mass for Pressure	Equivalent Mass for Impulse
(1)	(2)	(3)	(4)	(5)
i)	Ammonia dynamite (50 percent strength)	NA*	0.90	0.90 ǂ
ii)	Ammonia dynamite (20 percent strength)	NA*	0.70	0.70 ǂ
iii)	ANFO (94/6 ammonium nitrate/fuel oil)	NA*	0.87	0.87 ǂ
iv)	Composition C-4	1.59	1.20	1.19
		1.59	1.37	1.19
v)	Cycloid (75/25 RDX/TNT) (70/30) (60/40)	1.71	1.11	1.26
		1.73	1.14	1.09
		1.74	1.04	1.16
vi)	Gelatine dynamite (50 percent strength)	NA*	0.80	0.80 ǂ
vii)	Gelatine dynamite (20 percent strength)	NA*	0.70	0.70 ǂ
viii)	HMX	NA*	1.25	1.25 ǂ
ix)	Nitrocellulose	1.65 to 1.70	0.50	0.50 ǂ
x)	Nitroglycerine dynamic (50 percent strength)	NA*	0.90	0.90 ǂ
xi)	Nitroglycerine (NQ)	1.72	1.00	1.00 ǂ
xii)	PETN	1.77	1.27	1.27 ǂ
xiii)	Picrotol (52/48 Ex D/TNT)	1.63	0.90	0.93
xiv)	RDX	NA*	1.10	1.10 ǂ
xv)	RDX/wax (98/2)	1.92	1.16	1.16 ǂ
xvi)	RDX/AL/wax (74/21/5)	NA*	1.30	1.30 ǂ
xvii)	Tetryl	1.73	1.07	1.07 ǂ
xviii)	Tetrytol (75/25 tetryl/TNT)	1.59	1.06	1.06 ǂ
xix)	TNT	1.63	1.00	1.00

NOTE
* NA — Data not available.
ǂ Value is estimated.

L-4.2.5 *Cube Root Scaling*

Cube root scaling shall be used to relate blast parameters from different sources and standoff distances. The cube-root scaling shall be valid if the conditions specified in **L-4.2.1** are satisfied.

The blast load parameters can be obtained in scaled form as function of scaled distance, *Z*, as:

$$Z = \frac{R}{M^{1/3}}$$

where

R = standoff distance of the point of interest from the centre of charge mass; and

M = Mass in equivalent TNT.

The unit for scaled distance shall be m/kg$^{1/3}$, for standoff distance in m and charge mass in kg.

L-4.3 Performance Expectations

L-4.3.1 *Categorization of Structures*

Structures shall be categorized based on their importance as per the specification of Table 43.

The categorisation of structures as per Table 43, does not automatically require them to be designed for explosion threats. Only, if the threat assessment carried as per the provisions of **L-5.1** identifies blast loading due to explosion as a potential threat, the categorisation in Table 43 shall be used to define the importance of a structure.

Table 43 Categories of Structures

(*Clause* L-4.3.1)

Sl No. (1)	Importance (2)	Structure Type (3)
i)	Low	Residential buildings with 50 or less occupants
ii)	Medium	a) Residential buildings with more than 50 occupants b) Buildings with large congregation, such as schools, public offices and cinemas c) Industrial buildings with continuous human occupancy
iii)	High	a) Lifeline structures (water, power, communication, transportation, hospital) b) Structures housing essential services that are required for disaster management (for example, emergency relief stores, data centres, governance continuity buildings) c) Financial and commercial establishments of strategic importance

L-4.3.2 *Structural Performance*

A protection category shall be decided by the building owner based on the importance of the building and required performance. The performance goals for each protection category are listed in Table 44.

Table 44 Protection Category and Associated Performance Goals

(*Clause* L-4.3.2)

Sl No. (1)	Protection Category (2)	Performance Goal (3)
i)	1	Minimal damage, resumption after brief halt of operations
ii)	2	Limited damage, operational after repairs and retrofit
iii)	3	Extensive damage, safe replacement, or demolition

Structures of high, medium and low importance shall be provided with protection categories 1, 2 and 3 respectively, as at least a minimum as specified in Table 45.

The facility owner may specify a protection category higher than the minimum protection category specified in Table 45 based on the realistic assessment of threat.

Table 45 Protection Category and Associated Category of Structures

(*Clause* L-4.3.2)

Sl No.	Building Category	Minimum Protection Category
(1)	(2)	(3)
i)	Low	3
ii)	Medium	2
iii)	High	1

The protection categories specified in shall comply with the deformation limits of structural elements provided in **L-8.1.4** when using simplified single degree of freedom (SDOF) and multi degree of freedom (MDOF) methods of analysis as per requirements of **L-8.3.1** and **L-8.3.2** respectively. Accordingly, all elements shall be designed and detailed to provide deformation limits specified for different protection categories.

If advanced methods of analysis are used as per **L-8.3.3**, the response and damage in structure and elements for the performance goals of each protection category mentioned in can be independently established. In such case, deformation limits specified in **L-8.3.1** and **L-8.3.2** need not to be used.

L-5 THREAT ASSESSMENT

L-5.1 Threat Assessment Parameters

A threat assessment shall be carried out to determine all potential sources of detonation.

The threat assessment shall identify two parameters for the purpose of calculation of blast load parameters, namely:

a) charge mass; and
b) standoff distance.

If threat assessment identifies multiple sources of detonation, the effect of blast on the structure shall be assessed considering individual, as well as combinations of all identified detonation threats.

L-5.2 Design Blast (Reference Explosion)

L-5.2.1 *Design TNT Equivalence*

If the threat scenario results in identification of one of the detonation sources listed in Table 46, then the associated charge mass may be used for determination of equivalent TNT capacity.

L-5.2.2 *Standoff Distance*

The standoff distance (R) is site specific and shall be estimated as per the actual geometry and access control of the target structure.

L-6 BLAST LOAD

L-6.1 Blast Wave Propagation

The blast loads for design of structures and components shall be determined from blast pressure history.

Table 46 Detonation Source and the Associated Charge Mass
(*Clause* L-5.2.1 *and* Table 60)

Sl No.	Threat Description		Explosive Capacity (TNT Equivalent)
(1)	(2)	(3)	(4)
i)		Pipe bomb	2.5 kg
ii)		Suitcase	25 kg
iii)		Hatchback	250 kg
iv)		Sedans/SUV	500 kg
v)		Passenger van/small cargo truck	2 000 kg
vi)		Delivery truck	5 000 kg
vii)		Water tanker/diesel truck/gas truck	15 000 kg

The variation of pressure at a fixed location away from the centre of detonation shall be obtained using following assumptions:

a) the shock wave due to explosion moves away from the centre of detonation at a speed, U;

b) when the shock wave reaches a point at the arrival time t_a, the pressure rises instantaneously from the ambient pressure (p_a) to a peak pressure. This peak pressure, p_{so}, is known as side-on overpressure or peak incident overpressure;

c) the time needed for the pressure to reach its peak value is very small and for design purposes it is assumed to be zero;

d) the overpressure decreases exponentially from its peak value to the ambient pressure over the positive phase duration (t_o). After the positive phase of the pressure-time profile, the pressure becomes smaller than the ambient value (referred to as negative pressure) and finally returns to the ambient pressure, p_a;

e) the movement of air caused by the shock wave induces dynamic pressure, in addition to blast pressure. Air dynamic pressure q_o and peak particle velocity u shall be obtained using peak incident pressure, p_{so} (*see* Fig. 32 and Fig. 33); and

f) shock wave upon reflection from a rigid surface would apply reflected overpressure, p_r, to the reflecting surface. The variation of reflected overpressure with time for normal reflection follows the same trend as the incident overpressure but characterized by significantly higher peak overpressure.

L-6.2 Decay of Blast Overpressure with Time

The blast overpressure is assumed to vary with time, t according to the following exponential relationship (*see* Fig. 28):

$$p_s(t) = p_o\left(1 - \frac{t}{t_o}\right)e^{-b(t/t_o)}$$

where

p_o = peak blast overpressure, which is interpreted as the peak incident and reflected overpressures for the incident and the reflected blast wave, respectively; and

b = decay coefficient; and t_o is the positive phase duration. Special literature is to be referred for the calculation of the blast wave decay parameter, b.

L-6.3 Blast Overpressure History Parameters

The maximum values of the positive side-on overpressure p_{so}, incident impulse i_s, reflected over pressure p_r, reflected impulse i_r, positive phase duration t_o, shock wave front velocity U and wave length L_w, for varying scaled distances, are to be obtained for a spherical TNT explosion in free air and a hemispherical TNT explosion on the surface using Fig. 30 and Fig. 31, respectively. Further, air dynamic pressure q_o and peak particle velocity u, are to be obtained using Fig. 32 and Fig. 33, respectively.

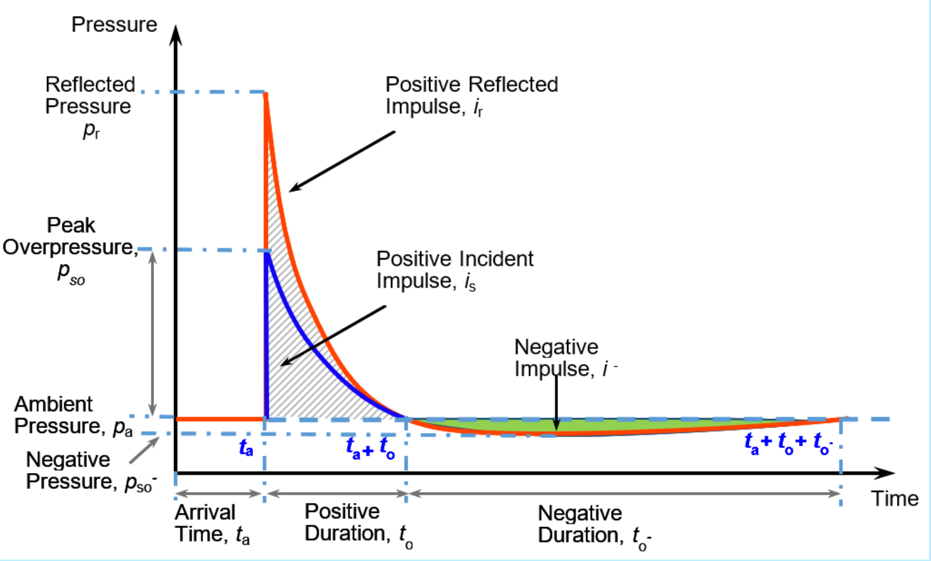


FIG. 28 INCIDENT AND REFLECTED BLAST PRESSURE HISTORY

L-6.4 Oblique Reflection of Blast Wave

The clauses of this section are applicable to determine the peak reflected overpressure for scenarios where the angle between the wave front and the interacting surface is α (≤ 90°) as shown in Fig. 29. In such case, the peak reflected overpressure (p_{ro}) is to be determined as multiplication of peak incident overpressure (p_{so}) with a reflection coefficient ($C_{r\alpha}$). The reflection coefficient at any angle is to be determined from Fig. 34 for given incident overpressure.

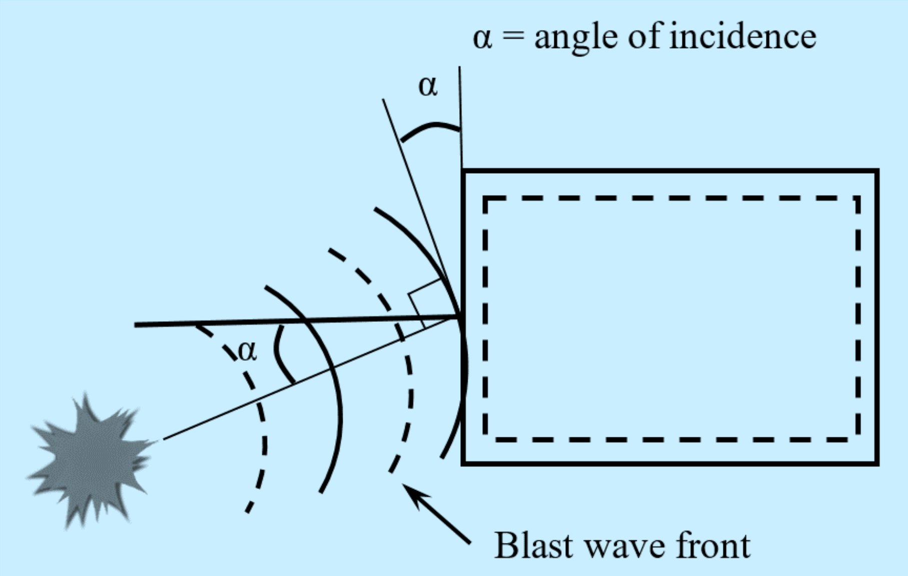


FIG. 29 OBLIQUE REFLECTION OF A BLAST WAVE

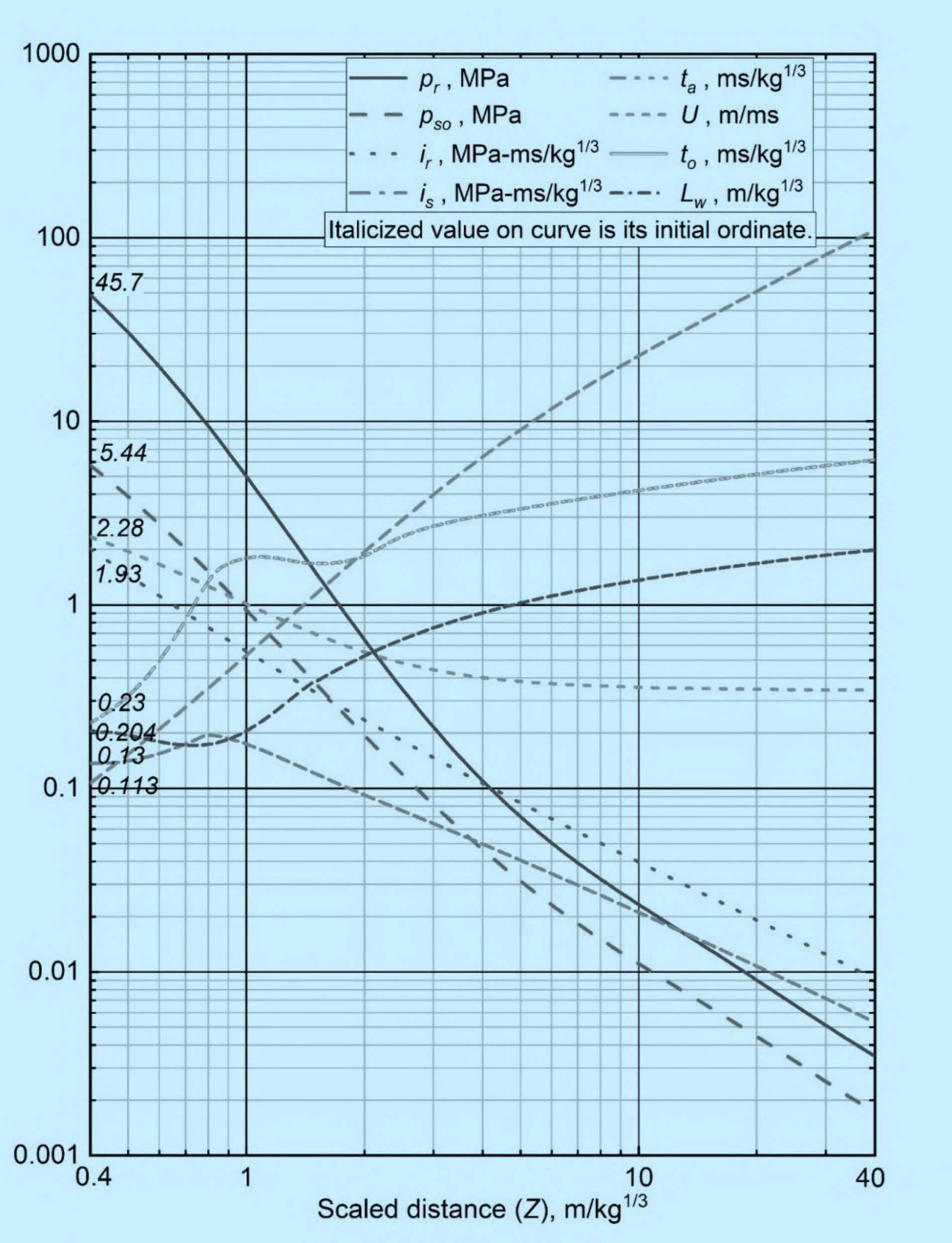


FIG. 30 POSITIVE PHASE SHOCK WAVE PARAMETERS FOR A SPHERICAL TNT EXPLOSION IN FREE AIR

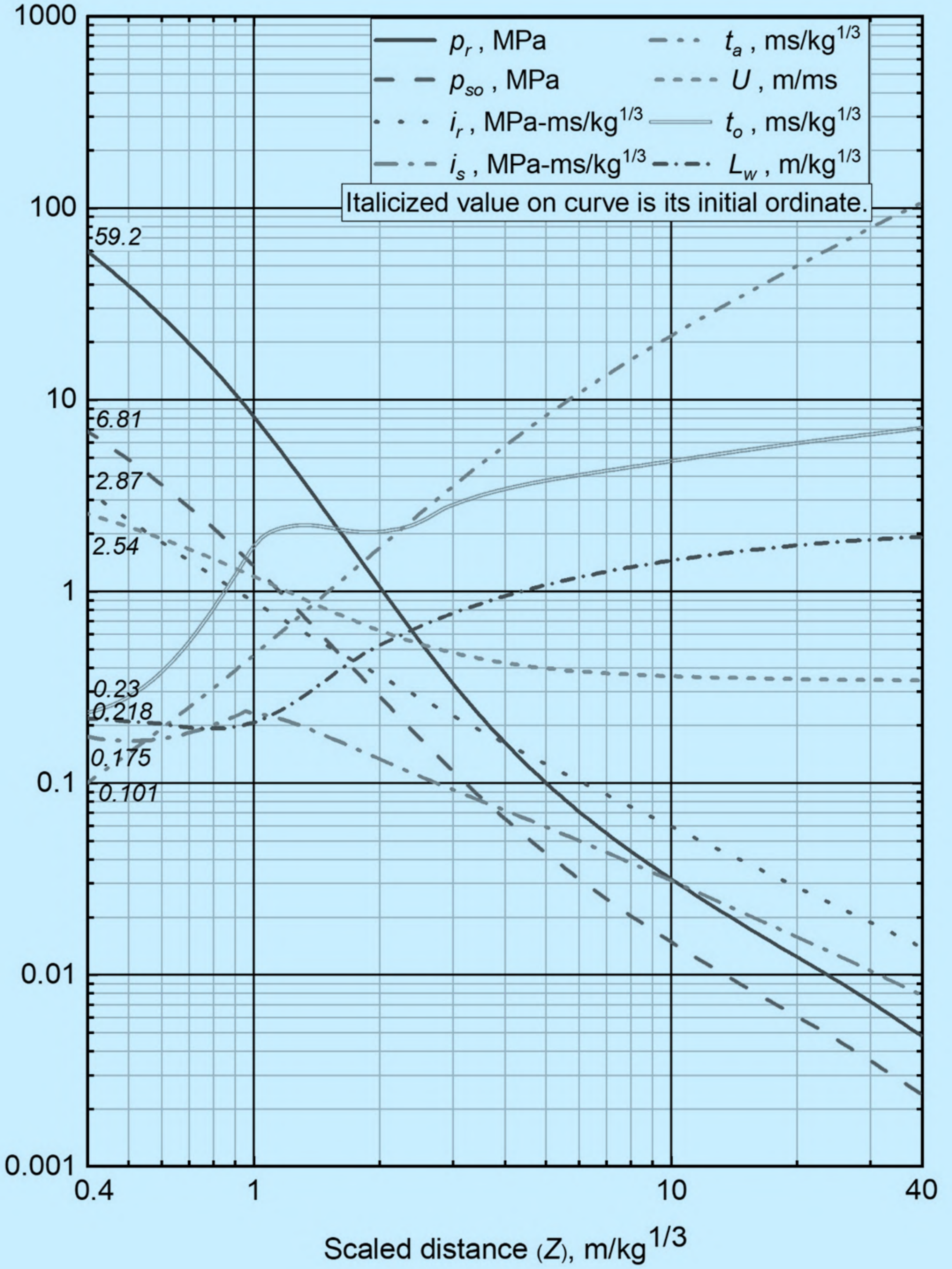


FIG. 31 POSITIVE PHASE SHOCK WAVE PARAMETERS FOR A HEMISPHERICAL TNT EXPLOSION ON SURFACE

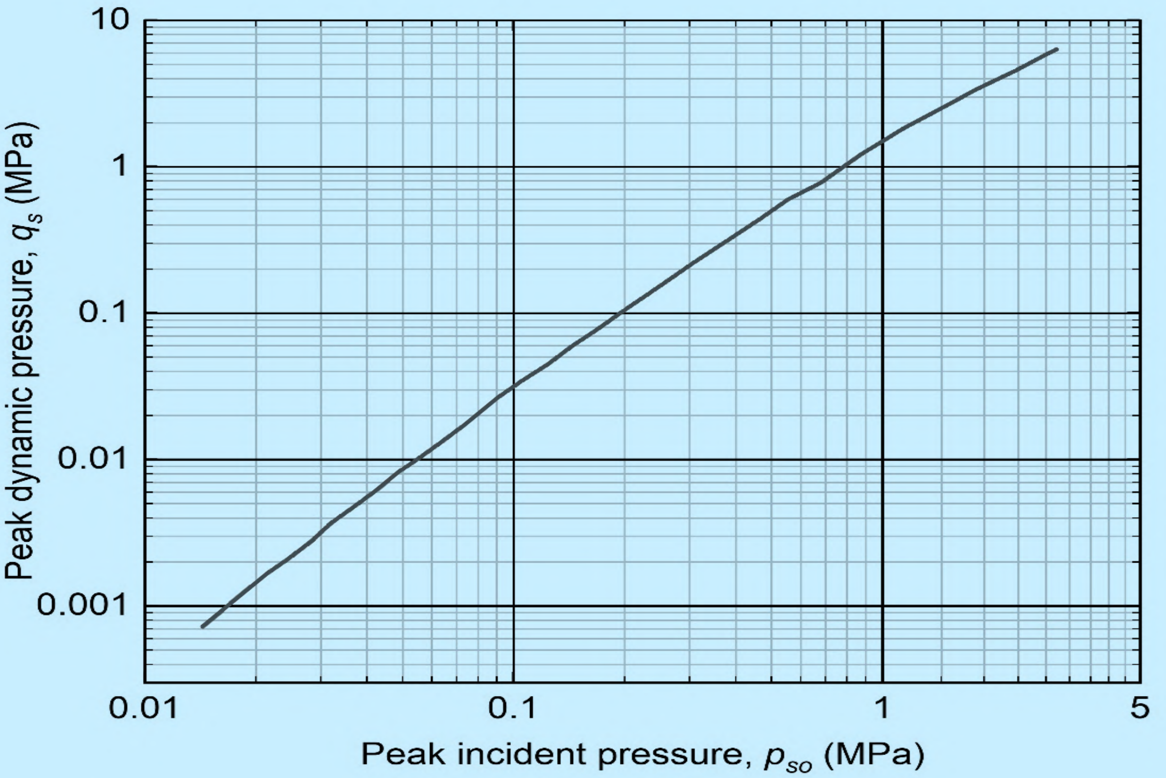


FIG. 32 PEAK DYNAMIC PRESSURE AND PEAK PARTICLE VELOCITY BEHIND A SHOCK FRONT

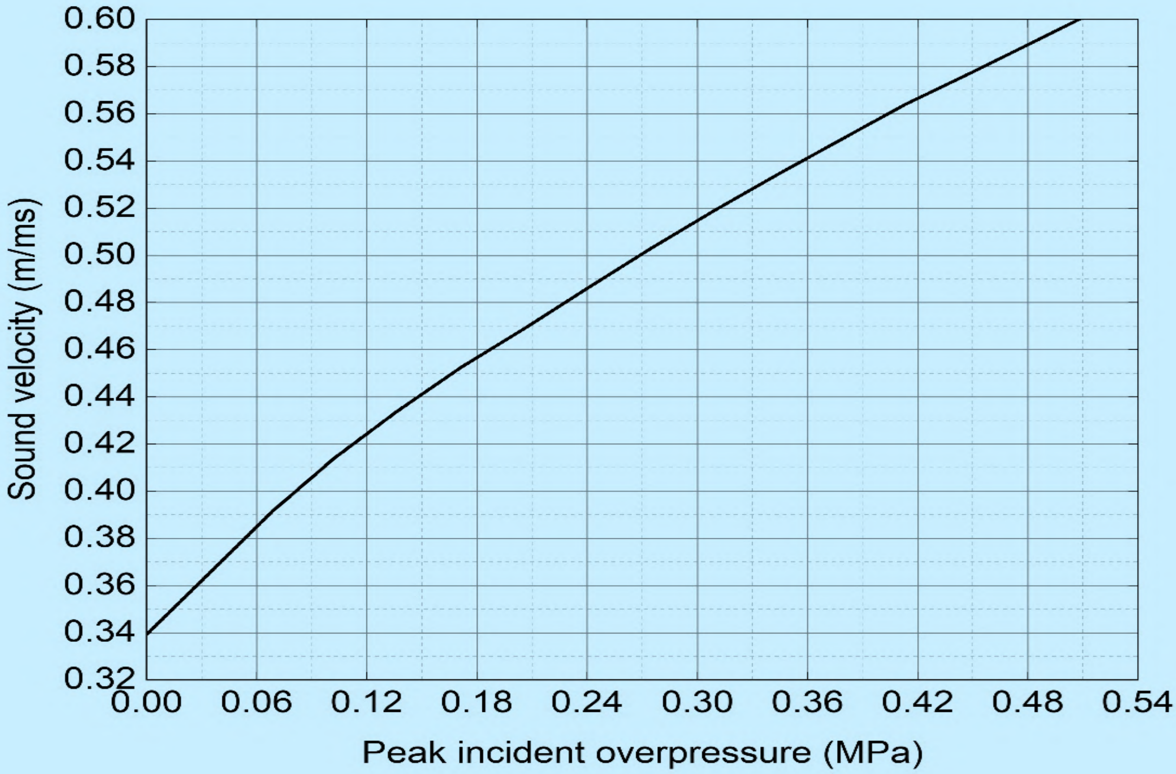


FIG. 33 SOUND VELOCITY IN THE REFLECTED OVERPRESSURE REGION

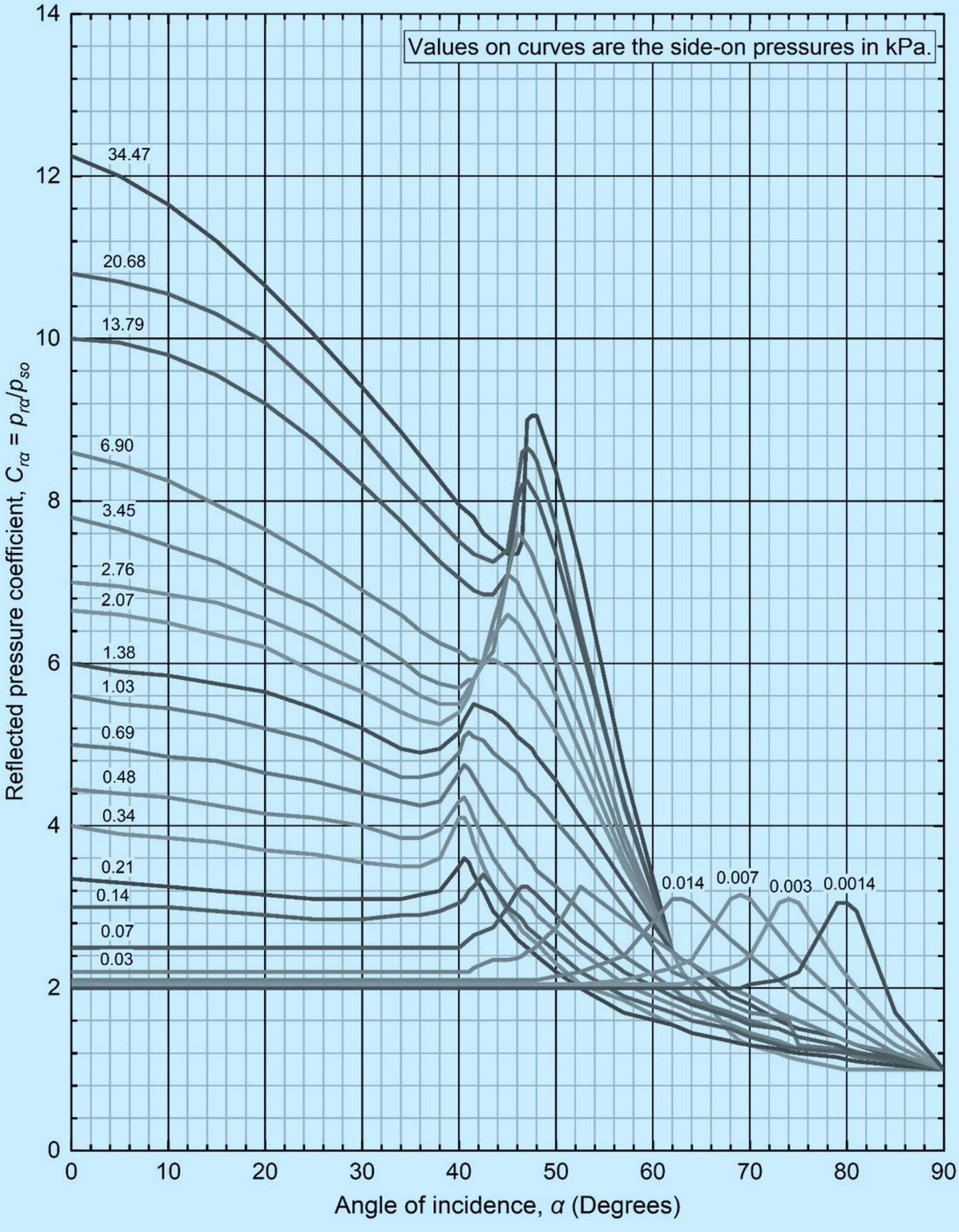


FIG. 34 INFLUENCE OF ANGLE OF INCIDENCE ON THE REFLECTED PRESSURE COEFFICIENT

L-6.5 Simplified Blast Pressure History

For the purpose of simplified analysis and design, the exponential pressure time relation in the positive phase may be idealised to a triangular pulse e as shown in Fig. 35.

The triangular pulse shall be constructed to have the same peak pressure as in the original exponential blast pressure curve and the positive phase duration (t_d) so adjusted that the area under the curve remains unchanged.

The usage of triangular representation of pressure history shall be governed by relevant provisions in this annex.

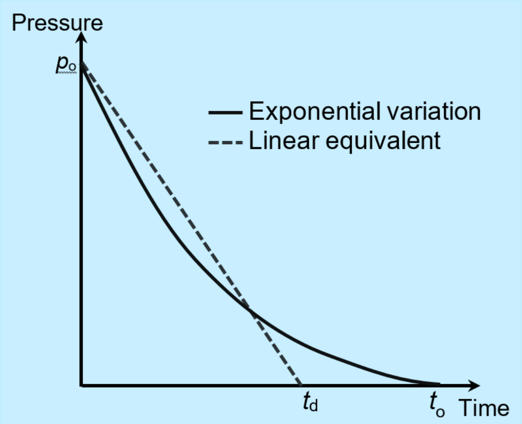


FIG. 35 SIMPLIFIED TRIANGULAR REPRESENTATION OF THE EXPONENTIAL OVERPRESSURE HISTORY

L-6.6 Ground Shocks

L-6.6.1 *Ground Shaking Intensity*

The scaled peak particle displacement, peak particle velocity and soil pressure generated by surface explosion shall be estimated using following equations:

Scaled peak particle displacement:

$$\frac{x}{M^{1/3}} = 60\frac{f}{c}\left(\frac{2.5R}{M^{1/3}}\right)^{1-n}$$

$$\text{Soil pressure, } p = 160 f \rho c \left(\frac{R}{M^{1/3}}\right)^{-n}$$

where

- f = coupling factor;
- n = attenuation coefficient;
- M = explosive weight;
- ρ = soil density;
- c = soil seismic velocity; and
- R = standoff distance.

Table 47 shall be used for taking the values of c, ρc and n.

The coupling factor f shall be taken as 0.14 for explosions above ground.

Table 47 Soil Properties to Calculate Ground Shock Parameters

(*Clause* L-6.6.1)

Sl No. (1)	Material Description (2)	Seismic Velocity, c (m/s) (3)	Acoustic Impedance, ρc (MPa/m/s) (4)	Attenuation Coefficient, n (5)
i)	Loose, dry sands and gravels with low relative density	182.9	0.27	3 to 3.25
ii)	Sandy loam, loess, dry sands and backfill	304.8	0.50	2.75
iii)	Dense sand with high relative density	487.7	1.00	2.5
iv)	Wet sandy clay with air voids (greater than 4 percent)	548.6	1.09	2.5
v)	Saturated sandy clays and sands with small amount of air voids (less than 1 percent)	1524	2.94	2.25 to 2.5
vi)	Heavy saturated clays and clay shales	<1524	3.40 to 4.07	1.5

L-6.6.2 *Safe Ground Particle Velocity*

For safety of buried and semi-buried structures, the peak particle velocity, u, as calculated using equation provided in **L-6.6.1** shall not exceed the values given in Table 48.

Table 48 Safe Ground Particle Velocity

(*Clause* L-6.6.2)

Sl No. (1)	Type of Rock (2)	Safe Ground Particle Velocity (mm/s) (3)
i)	Soils, weathered or soft rock	50
ii)	Hard rock	70

For safety of unlined underground structures, the peak particle velocity, u, as calculated using equation provided in **L-6.6.1** shall not exceed the values listed in Table 49.

For safety of above ground structures, the peak particle velocity, u, measured at the foundation level, as calculated using equation provided in **L-6.6.1** shall not exceed the values listed in Table 50.

L-6.6.3 *Decay of Ground Shaking with Distance*

Decay of ground shaking with distance is to be estimated based on principle of mechanics and field measurement

Table 49 PPV Damage Criterion for Unlined Underground Structures

(*Clause* L-6.6.2)

Sl No.	Rock Type	Unit Weight kg/m^3	Compressive Strength MPa	Tensile Strength MPa	Critical Peak Particle Velocity (m/s)			
					No Damage	*Slight Damage*	*Intermediate Damage*	*Serious Damage*
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
i)	Hard rock	2600 to 2700	75 to 110	2.1 to 3.4	0.27	0.54	0.82	1.53
		2700 to 2900	110 to 180	3.4 to 5.1	0.31	0.62	0.96	1.78
		2700 to 2900	180 to 200	5.1 to 5.7	0.36	0.72	1.11	2.09
ii)	Soft rock	2000 to 2500	40 to 100	1.1 to 3.1	0.29	0.58	0.90	1.67
		2000 to 2500	100 to 160	3.4 to 4.5	0.35	0.70	1.07	1.99

Table 50 Permissible Peak Particle Velocity at the Foundation Level of Structures, in mm/s

(*Clause* L-6.6.2)

Sl No.	Type of Structure	Dominant Excitation Frequency (Hz)		
		< 8 Hz	8 Hz to 25 Hz	> 25 Hz
(1)	(2)	(3)	(4)	(5)
i)	Buildings/structures not owned by owners:			
	a) Domestic houses/structure (*kuccha* brick and cement)	5	10	15
	b) Industrial buildings (RCC and framed structures)	10	20	25
	c) Objects of historical importance and sensitive structures	2	5	10
ii)	Buildings belonging to owner with limited span of life:			
	a) Domestic houses/structures (*kuccha* brick and cement)	10	15	25
	b) Industrial buildings (RCC and framed structures)	15	25	50

L-6.7 Confined Explosions

The provisions given here are applicable for scenarios where detonation of explosives takes place inside closed spaces.

For the purpose of evaluating blast load effects, confined explosions are categorized into:

a) fully vented; and

b) partially vented.

The examples of fully and partially vented structures are shown in Fig. 36.

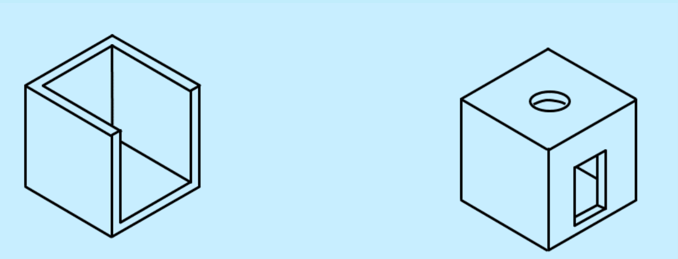


FIG. 36(A) FULLY VENTED FIG. 36(B) PARTIALLY VENTED

FIG. 36 CONFINED EXPLOSION SCENARIO

The variation of pressure at any location inside a closed space due to detonation inside the closed space is obtained using the following assumptions:

a) surfaces internal structural members are subjected to shock waves from multiple reflections characterized by multiple peaks; and

b) shock pressure is followed by development of gas pressure of small peak overpressures but significantly longer durations.

The provisions of **L-6.7** are only applicable when following conditions are satisfied:

a) structure is of regular geometric shape; and

b) the effect of confined detonation is being evaluated at scaled distances greater than 3 $m/kg^{1/3}$.

L-6.7.1 *Fully Vented Structures*

Fully vented structures are to be primarily subjected to shock pressures and the effect of gas pressures may be neglected. The reflected overpressure history at any point inside the structure is represented in Fig. 37. The peak pressure and impulse for the first pulse may be obtained using the procedure described in **L-6.3** as a case of direct reflection of the blast wave propagating from the centre of detonation to the point of interest. The second and the third triangular pulse shall be constructed from the characteristics of the first pulse.

Further simplification of pressure variation in Fig. 37 is allowed if the ratio of total load duration ($5t_a + t_d$) to the time-period of the structure is smaller than 0.1. In such cases, the three pulses of pressure history in Fig. 37 may be combined into a single pulse of peak pressure 1.75 p_{ro} and duration equal to t_d.

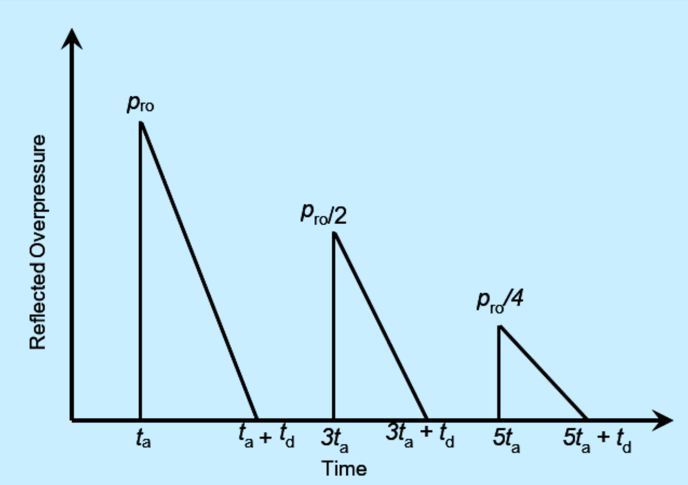


FIG. 37 REFLECTED OVERPRESSURE HISTORY FOR CONFINED DETONATION IN A FULL-VENTED SPACE

L-6.7.2 *Partially Vented Explosion*

Partially vented explosions are characterized by long duration gas pressures with minor effect of the shock pressures.

Wherever, required, the arrival time of the peak gas pressure is taken as the end time of the shock phase, that is, $5t_a + t_d$, as determined in **L-6.7.1**.

The blast load effects of confined explosion in partially vented structure is evaluated using advanced numerical technique as per the provisions of **L-8.3.3**.

The interior surface of a partially vented structure subject to confined explosion may conservatively be designed for the peak gas overpressure, p_g, which needs to be applied statically.

This peak gas overpressure is obtained from the chart in Fig. 38. The use of charts requires following condition to be satisfied:

$$0 \leq A / V_f^{2/3} \leq 0.022$$

where

A = structure's total vent area; and

V_f = free volume which shall be calculated as the total volume excluding the volume of all interior equipment and structural elements.

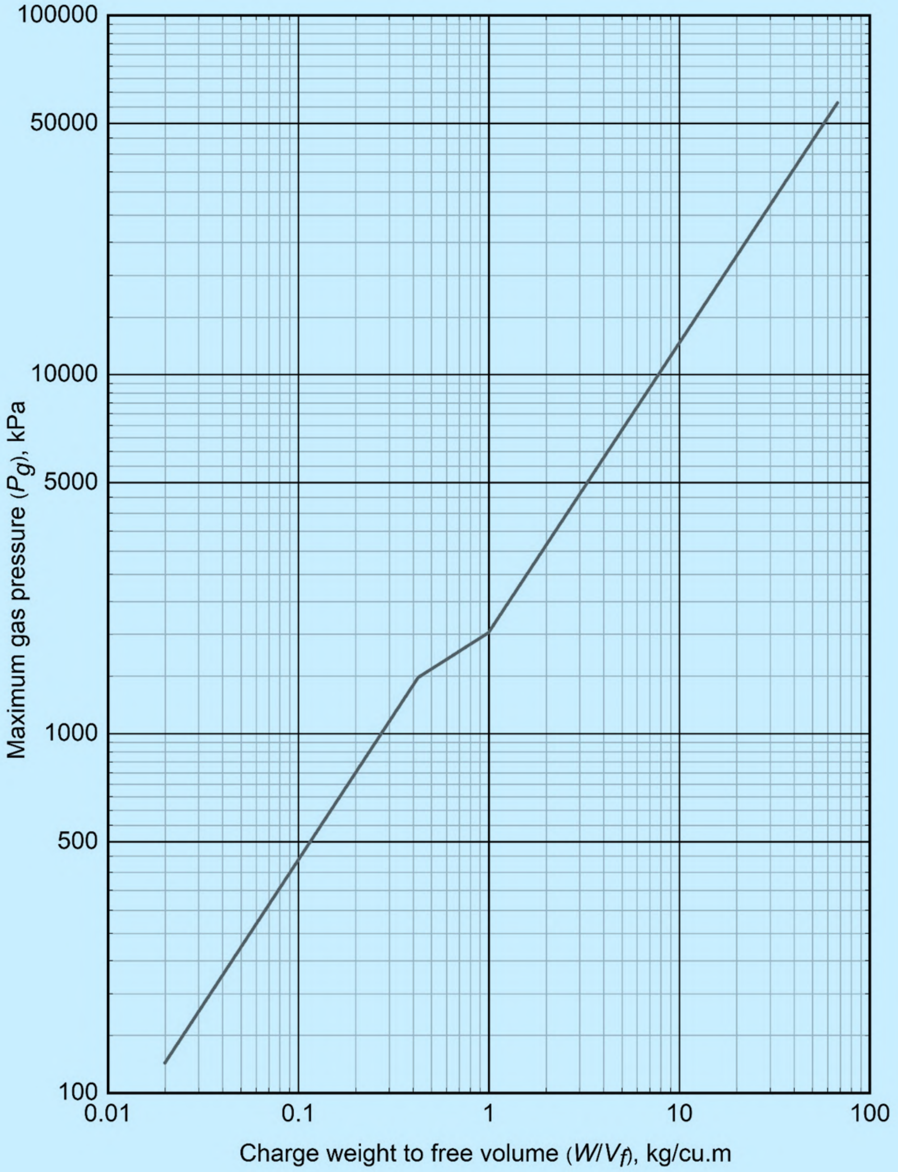


FIG. 38 PEAK GAS PRESSURE PRODUCED BY A TNT DETONATION IN A PARTIALLY VENTED STRUCTURE

L-7 EFFECTS OF BLAST ON STRUCTURES

L-7.1 Above Ground Structures

L-7.1.1 *Types of Structures*

Structures are categorized into following two types for the purpose of determining its interaction with the blast wave and calculating effective blast overpressure history:

a) *Diffraction type structures* — These are the closed structures without openings, with the total area opposing the blast. These are subjected to both the shock wave overpressure p_{so} and the dynamic pressures q caused by blast wind; and

b) *Drag type structures* — These are the open structures composed of elements like beams, columns, trusses, etc, which have small projected area opposing the shock wave. These are mainly subjected to dynamic pressures *q*.

L-7.1.2 *Closed Rectangular Structures*

L-7.1.2.1 *Front face*

The peak reflected overpressure at any point on the front face of a closed rectangular structure shall be obtained using the following equation.

$$p_{r\alpha} = C_{r\alpha} p_{so}$$

where

p_{so} = peak side-on pressure that shall be obtained using Fig. 30 and Fig. 31 for spherical and hemispherical detonations, respectively, using the effective scaled distance; and

$C_{r\alpha}$ = reflection co-efficient, which shall be determined from Fig. 34 using the angle of reflection and the side-on pressure.

It is permitted to neglect spatial variation of pressure on the front face by assuming an equivalent uniform reflected overpressure which is equal to the peak reflected overpressure on the front face that is closest to the point of detonation. This, under most scenario, corresponds to the peak reflected overpressure, p_{ro}, at normal incidence (α = 0 degrees).

The reflected overpressure, p_r, on the front face drops from the peak value p_{ro} to zero in time t_{rf}, whichever, is calculated as, $t_{rf} = 2i_{ro}/p_{ro}$, where p_{ro} and i_{ro} shall be obtained from using Fig. 30 and Fig. 31 for spherical and hemispherical detonations, respectively.

The stagnation overpressure, p_s, at the edges of the front face drops from the peak value, $p_{so} + C_d q_o$, to zero in time t_{of}, whichever, is calculated as, $t_{of} = 2i_{so}/p_{so}$, where p_{so} and i_{so} shall be obtained using Fig. 30 and Fig. 31 for spherical and hemispherical detonations, respectively. The drag coefficient, C_d, shall be taken as 1.0. The peak dynamic pressure, q_o, shall be determined from Fig. 32.

The reflected overpressure acting at a point on the front face drops from the peak value p_{ro} to overpressure ($p_{so} + C_d q$) in clearance time t_c or t_{of}, whichever, is less (*see* Fig. 40).

The drag coefficient, C_d, for different faces of closed rectangular structures located above ground shall be taken as per Table 51.

The clearing time, t_{cp}, for a point on the front wall is given by

$$t_{cp} = \frac{3S}{U}$$

where

$S = H$ or $W/2$ whichever, is less (*see* Fig. 39); and

U = shock front velocity to be obtained from using Fig. 30 and Fig. 31 for spherical and hemispherical detonations, respectively.

The average clearing time, t_c, required to clear the front face of reflection effects from the roof down and inwards from the sides shall be calculated using the following expression:

$$t_c = \frac{4HW}{C_r(W+2H)}$$

where

C_r = velocity of sound in the reflected overpressure region that shall be determined using Fig. 33.

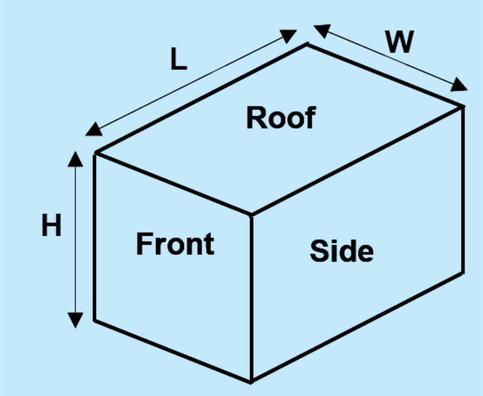


FIG. 39 ABOVE GROUND RECTANGULAR STRUCTURE

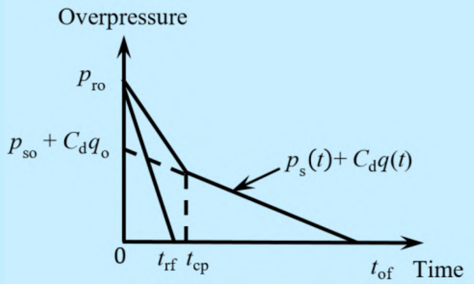


FIG. 40 REFLECTED OVERPRESSURE AT A POINT ON THE FRONT SURFACE OF FINITE REFLECTING SURFACE

The average net loading on the front face ($W \times H$) as a function of time is shown in Fig. 41 depending on whether t_c is smaller than or equal to t_{of}. The pressures p_{ro}, p_{so} and q, the impulses t_{ro} and I_{so} are for the actual explosion that are determined according to L-6.3.

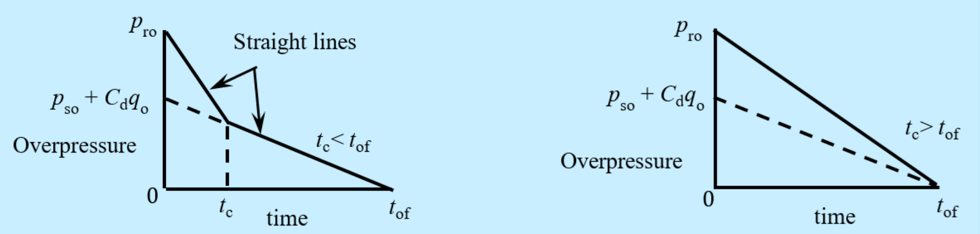


FIG. 41 REFLECTED OVERPRESSURE HISTORY ON THE FRONT FACE OF A CLOSED RECTANGULAR STRUCTURE

L-7.1.2.2 *Rear face*

Using the pressures for the actual explosion, the average loading on the rear face ($W \times H$) in Fig. 39 shall be taken as shown in Fig. 42, where the time has been taken from the instant the shock first strikes the front face. The time intervals of interest are the following:

$\frac{L}{U}$ = the travel time of shock from front to rear face; and

$\frac{4S}{U}$ =pressure rise time on the back face.

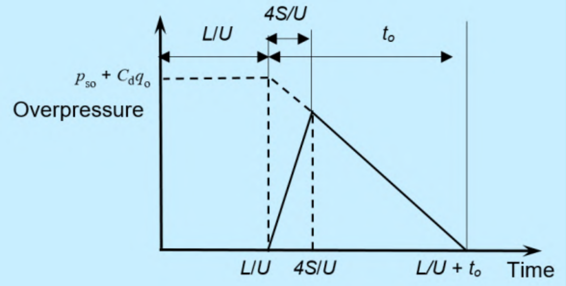


FIG. 42 OVERPRESSURE VERSUS TIME FOR REAR FACE

L-7.1.2.3 *Roof and side walls*

The average pressure versus time curve for roof and side walls is given in Fig. 43(A) when t_d is greater than the transit time $t_t = L/U$. When t_t is greater than t_d, the load on roof and side walls may be considered as a moving triangular pulse having the peak value of overpressure $p_{so}+C_d q_o$ and time t_o as shown in Fig. 43(B).

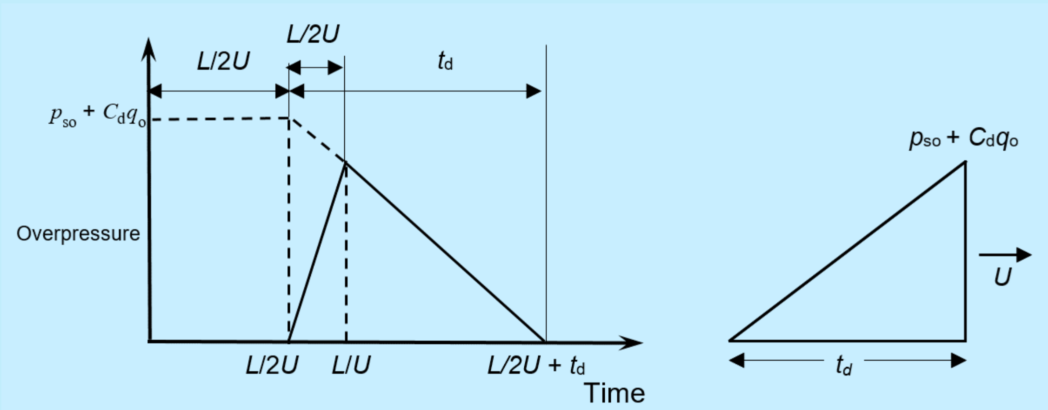


FIG. 43(A) AVERAGE PRESSURE DIAGRAM $t_d > t_t$

FIG. 43(B) MOVING PRESSURE PULSE FOR $t_t > t_d$

FIG. 43 OVERPRESSURE VERSUS TIME FOR ROOF AND SIDE WALLS

Table 51 Drag Coefficient C_d

(*Clause* L-7.1.2.1)

Sl No.	Shape of Element	Drag Coefficient C_d	Remarks
(1)	(2)	(3)	(4)
i)	Front vertical face	1.0	
ii)	Roof, near and side faces for:		For closed rectangular structures located above ground
	a) q_o = 0 kPa to 170 kPa	- 0.4	
	b) q_o = 170 kPa to 350 kPa	- 0.3	
	c) q_o = 350 kPa to 1 MPa	- 0.2	

L-7.1.2.4 *Overturning of structure*

The average net load as a function of time which tends to cause sliding and overturning of the building is obtained by subtracting the loading on back face from that on the front face.

L-7.1.3 *Closed Cylindrical Arch-Shape Structures*

L-7.1.3.1 *Gable ends*

The loading may be taken same as for front and rear faces of above ground rectangular structure.

L-7.1.3.2 *Curved surface*

The direction of shock wave propagation is taken transverse to the ridge of the structure and since the usual arch spans are large so that the transit time t_t is greater than the positive phase time t_d, average loading condition cannot be assumed. Therefore, the loading on curved surface may be taken as a moving triangular pulse as shown in Fig. 43(B).

L-7.1.3.3 *Closed dome structures*

The loading on a domical structure may be taken as a moving triangular pressure pulse as shown in Fig. 43(B). The variation of pressure transverse to the direction of propagation of the pulse may be considered symmetrical varying according to cosine θ where the angle θ is measured from longitudinal vertical section of the dome, as shown in Fig. 44.

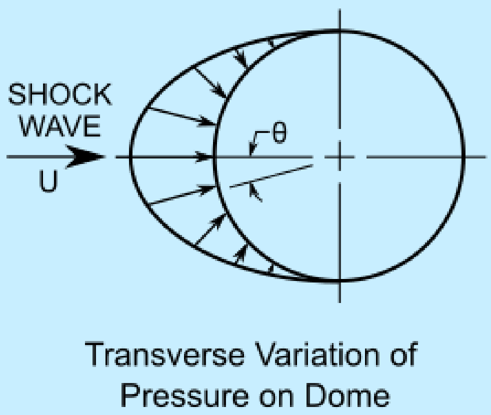


FIG. 44 TRANSVERSE VARIATION OF PRESSURE ON DOME

L-7.2 Blast Load on Below-Ground Structures

L-7.2.1 *Types of Structures*

The below-ground structures are classified into buried and semi-buried structures depending upon the earth cover and slopes of earth berms. The buried structures are subjected only to the general overpressure p_{so}, the reflected and dynamic pressures being neglected. The semi-buried structures are subjected to partial dynamic pressures besides the general overpressure. Both are acted upon by air-induced ground shock also.

For assessing the effects of above-ground blasts on below-ground structures, dynamic finite element analysis can be performed considering the flexibility of soil and interactions of blast induced ground waves with the structure.

L-8 RESPONSE OF STRUCTURAL ELEMENTS

L-8.1 Properties of Materials

The nominal properties of materials are obtained as per relevant codes and guidelines. The change in material properties due to high strain rates associated with blast response are to be incorporated using the provisions in L-8.1.1 and L-8.1.2.

L-8.1.1 *Elastic Modulus and Poisson's Ratio*

The elastic modulus and Poisson's ratio of the constituent materials used in the structural elements shall be estimated based on relevant standards as applicable to that material.

No increase in elastic modulus and Poisson's ratio is to be considered for the calculation of blast response of structures.

L-8.1.2 *Dynamic Strength*

L-8.1.2.1 *Design strength of structural steel*

The average dynamic increase factors of strength for structural, carbon, mild, weldable or rivet steels and high strength alloy steels are listed in Table 52 and Table 53.

Table 52 Dynamic Increase Factors for Mild Steels

(*Clause* L-8.1.2.1)

Sl No.	Material	Dynamic Increase Factors
(1)	(2)	(3)
i)	Yield strength in bending and shear	1.25
ii)	Yield strength in tension and compression	1.20
iii)	Ultimate strength	1.00

Table 53 Dynamic Increase Factors for High Strength Steels

(*Clause* L-8.1.2.1)

Sl No.	Material	Dynamic Increase Factors
(1)	(2)	(3)
i)	Yield strength in bending and shear	1.10
ii)	Yield strength in tension and compression	1.05
iii)	Ultimate strength	1.00

L-8.1.2.2 *Design strength of reinforced concrete*

The average dynamic strength increase factors for concrete are listed in Table 54 and for reinforcing steel are listed in Table 55.

Table 54 Dynamic Increase Factors for Reinforced Concrete

(*Clause* L-8.1.2.2)

Sl No.	Strength	Dynamic Increase Factors
(1)	(2)	(3)
i)	Flexure	1.15
ii)	Compression	1.10
iii)	Diagonal tension, direct shear and bond	1.00

Table 55 Dynamic Increase Factors for Reinforcing Steel

(*Clause* L-8.1.2.2)

Sl No.	Strength	Dynamic Increase Factors
(1)	(2)	(3)
i)	Yield strength in flexure	1.2
ii)	Yield strength in compression, diagonal tension, direct shear and bond	1.1
iii)	Ultimate strength in flexure	1.1
iv)	Ultimate strength in compression, diagonal tension, direct shear and bond	1.0

L-8.1.2.3 *Design strength for masonry or plain concrete*

The average dynamic strength increase factors for masonry and plain concrete are listed in Table 56.

Table 56 Dynamic Increase Factors for Masonry and Plain Concrete

(*Clause* L-8.1.2.3)

Sl No.	Strength	Dynamic Increase Factors
(1)	(2)	(3)
i)	Flexure	1.15
ii)	Compression	1.10
iii)	Diagonal tension and direct shear and bond	1.00

L-8.1.3 *Design Strength*

For obtaining design strength, the resistance of various elements is to be estimated using the strength increased by the dynamic increase factors mentioned in Q-8.1.2.

L-8.1.4 *Deformation Limits*

Structural elements subjected to blast loading can be designed subject to certain allowable dynamic ductility ratios and support rotations.

The blast resistance of a flexural member is to be expressed in terms of maximum permissible deflection. Support rotation and ductility is to be used to characterize the blast resistance of reinforced concrete and structural steel members, respectively.

The support rotation shall be defined as per Fig. 45.

The maximum permissible deflection defines the energy absorption capacity of the element which is equal to the area under the resistance versus deflection curve. For a given blast scenario, greater the permissible deflection, lesser will be the maximum resistance required in the member.

The allowable deformation parameters are included in Table 57.

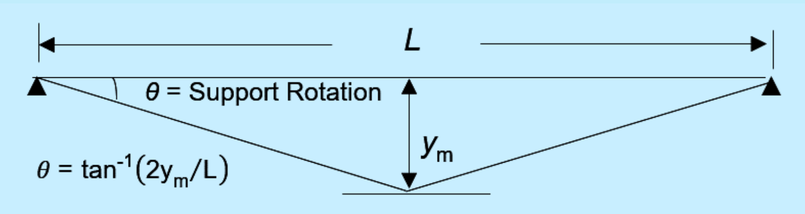


FIG. 45 SUPPORT ROTATION IN A FLEXURAL MEMBER

Table 57 Allowable Deformation Parameters

(*Clause* L-8.1.4)

Sl No.	Material	Low Damage	Medium Damage	High Damage
(1)	(2)	(3)	(4)	(5)
i)	Ductility ratio in structural steel members subjected to bending and direct stresses	3	10	20
ii)	Support rotation in reinforced concrete members subjected to bending and direct stresses (in degrees)	1	2	4

L-8.2 Modelling of Structural Members

L-8.2.1 *Geometry, Material and Boundary Conditions*

The methods of modelling are applicable to structural members of regular prismatic sections.

L-8.2.2 *Equivalent Single Degree of Freedom* (*SDOF*) *Model*

The structural element or system may be replaced by an equivalent single spring-mass system having effective stiffness k_{E} and effective mass equal to $K_{\mathrm{LM}}M_{\mathrm{t}}$, where M_{t} is the actual mass of the member under consideration and K_{LM} is a load-mass factor depending upon the stiffness and mass distribution in the member and its boundary conditions. The equivalent system is defined so that the deflection of the equivalent single mass is the same as that of some significant point in the given structure. The effective stiffness k_{E} is defined with respect to the deflection of this point.

The load-mass factor K_{LM} is equal to the ratio of mass factor K_{M} to the load factor K_{L}. The factors are evaluated on the basis of an assumed deflected shape of the structure as given below:

$$K_{\mathrm{M}} = \frac{1}{M_{\mathrm{t}}}\left\{\sum_{r=1}^{n} M_{\mathrm{r}}\emptyset_{\mathrm{r}}^{2} + \int m\emptyset^{2}(x)\,dx\right\}$$

$$K_{\mathrm{L}} = \frac{1}{P_{\mathrm{t}}}\left\{\sum_{r=1}^{j} p_{\mathrm{r}}\emptyset_{\mathrm{r}} + \int p(x)\emptyset(x)dx\right\}$$

where

n = number of concentrated masses;

M_r = concentrated mass at point r ;

M_t = total actual mass;

$\phi_r \phi_r$ = deflection at point r of an assumed deflected shape for concentrated loads;

m = distributed mass intensity per unit length;

$\phi_{(x)}$ = deflection at point x of an assumed deflected shape for distributed loads;

P_t = total dynamic load (at any instant of time);

j = number of concentrated load points;

P_r = concentrated dynamic force at point r; and

$p_{(x)}$ = intensity of distributed dynamic load per unit length.

L-8.2.2.1 *Effective single SDOF parameters*

Values of the factors k_E and K_{LM} for few structural members are given in Table 58 to Table 60. The effective SDOF parameters for other type of structural members shall be calculated by assuming an appropriate deflected shape as per **L-8.2.2.2**.

L-8.2.2.2 *Deflected shape*

The deflected shape is suitably chosen to resemble as far as possible the true deflected shape taking into consideration whether the structure or member remains elastic or goes into the plastic range. It may be taken the same as due to static application of the dynamic load on the structure. The deflected shape is normalized such that $\varphi(x)=1$ at the point with respect to which the effective stiffness k_E is defined.

L-8.2.2.3 *Effective time-period*

The effective time period T of the structural member may be calculated from the equation:

$$T = 2\pi \sqrt{\frac{K_{LM} M_t}{k_E}}$$

L-8.2.3 *Resistance Function*

The resistance versus deflection diagram of a structural element shall be idealized as elasto-plastic (*see* Fig. 46) by keeping the area under the actual and idealized curves about the same up to the maximum permissible deflection defined as per **L-8.1.4**.

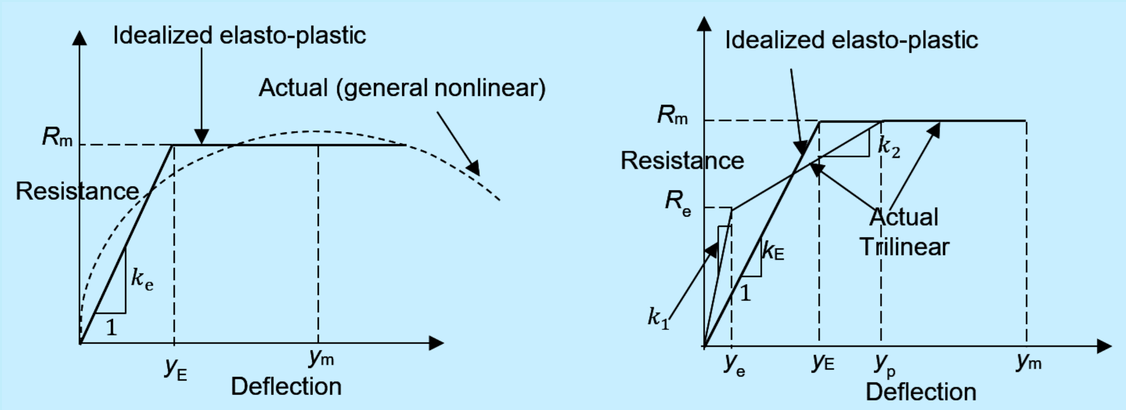


FIG. 46(A) GENERAL NONLINEAR

FIG. 46(B) TRILINEAR

FIG. 46 IDEALIZATION OF RESISTANCE DEFLECTION DIAGRAM

A trilinear resistance versus deflection diagram may be simplified to its elasto-plastic (bilinear) equivalent of same peak resistance (R_m) and the elastic parameters calculated using following equations:

$$y_E = y_e + y_p\left(1 - \frac{R_m}{R_e}\right)$$

$$k_E = \frac{R_m}{y_E}$$

L-8.3 Methods of Analysis

These provisions apply to structures, components and systems subject to near-field and far-field detonations but not for contact or near-contact detonations.

L-8.3.1 *Analysis of Members*

Blast loaded structural components of a structure may be analysed independently provided it can be demonstrated that such independent member analysis would yield conservative performance of the member and the structure.

L-8.3.1.1 *Equivalent SDOF analysis*

The equivalent SDOF analysis may be used to determine response of structural members subject to blast loads when the following criteria are satisfied:

a) The equivalent mass, stiffness and damping parameters in the model must capture the dynamic response and failure mode of the structural member being analysed; and

b) A single flexural deformation mode controls the dynamic response of the structural system.

The equivalent SDOF analysis shall make use of following parameters:

a) Blast-pressure history obtained using L-6.3;

b) The effective time-period calculated using L-8.2.2.3;

c) Resistance function of the structural element established using provisions of L-8.2.3; and

d) Maximum permissible deflection provided in L-8.1.4.

L-8.3.1.2 *Blast overpressure history*

The exponential blast pressure history of Fig. 28 shall be idealized to a triangular pulse [Fig. 47(A) as per provisions of **L-6.5** for the purpose of obtaining equivalent SDOF blast loads. Such idealization may neglect the effect of negative phase, unless otherwise it is expected to influence the flexure response of the structural member.

The positive phase duration of the triangular loading shall be calculated as:

$$t_{\mathrm{d}} = \frac{2 i_{ro}}{p_{ro}}$$

If the influence of clearing on the blast pressure history applied to the front face of the structural element is found to be significant, then the equivalent SDOF blast loads shall be idealized to Fig. 47(B) using **L-7.1.2**.

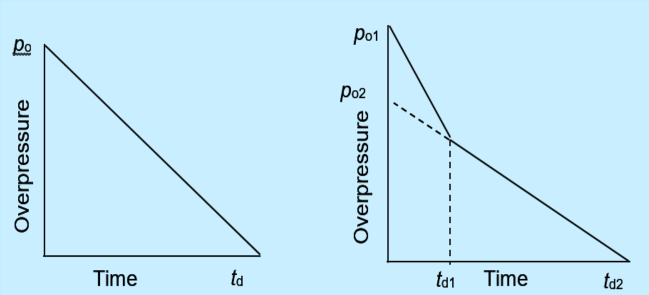


FIG. 47(A) LINEAR FIG. 47(B) BILINEAR

FIG. 47 IDEALIZED BLAST PRESSURE HISTORIES

When the ratio of time duration t_{d1}, or t_{d2}, to the natural period of the element is less than 0.1, the problem may be considered as an impulse problem taking the area under the pressure versus time curve as impulse per unit area. In such a case, the shape of pressure-time curve is not important.

L-8.3.1.3 *Uniform blast pressure distribution*

A uniform distribution of blast pressure over the length of the structural member may be considered provided one of the following conditions are satisfied:

a) The minimum scaled distance to the structural component is greater than 1.2 m/kg$^{1/3}$; and

b) The variation of peak reflected pressures over the middle two-third of the component span length is less than 25 percent.

For uniform distribution of blast pressure, the pressure value calculated at the closest point on the structure from the centre of detonation may be considered over the whole reflecting surface.

L-8.3.1.4 *Non-uniform blast pressure distribution*

For members subjected to spatially non-uniform blast loads, the effect of non-uniformity on the response shall be considered.

These provisions shall be applicable for non-uniform blast pressure distribution, provided the conditions outlined in **L-8.3.1.3** are not satisfied and the minimum scaled distance to the structural component is between 0.4 m/kg$^{1/3}$ and 1.2 m/kg$^{1/3}$.

An equivalent impulse weighted using deflected shape function shall be considered for the structural component as per following expression:

$$i_e = \frac{\int_0^L \int_0^H i(x,y)\phi(x,y)dxdy}{\int_0^L \int_0^H \phi(x,y)dxdy}$$

where

i_e	=	equivalent impulse of uniformly distributed load;
L	=	length of loaded area;
H	=	width of loaded area;
Ø(x,y)	=	deflected shape function of blast-loaded component; and
i(x,y)	=	non-uniform impulse applied to loaded area.

The simplified triangular representation of the pressure time history shall be constructed as per provisions of **L-6.5** by calculating the positive loading phase duration using the equivalent uniform impulse (i_e) and the peak overpressure at the midspan or geometric centre of the reflecting surface.

A safety factor of 1.25 shall be used if a non-uniform blast load is represented as an equivalent uniform impulse in a blast analysis.

L-8.3.1.5 *Elastic SDOF response*

For elastic analysis (ductility ratio $\mu \leq 1.0$) of structures, the effective stiffness k_E and load mass factor K_{LM} shall be used as given in Table 58 to Table 60.

For elastic response of an SDOF system, the peak response (y_m) and the time to peak response (t_m) may be obtained using following expressions:

$$\frac{y_m}{y_0} = 2 - \frac{t_m}{t_d}; \qquad \text{when } t < t_d$$

$$t_m = \frac{2}{\omega}\tan^{-1}(\omega t_d) \qquad \text{when } t < t_d$$

$$\frac{y_m}{y_0} = \frac{1}{\omega t_d}\sqrt{2(1-\cos\omega t_d - \omega t_d \sin\omega t_d) + (\omega t_d)^2}; \qquad \text{when } t > t_d$$

$$t_m = \frac{1}{\omega}\tan^{-1}\left(\frac{1-\cos\omega t_d}{\sin\omega t_d - \omega t_d}\right) \quad \text{when } t > t_d$$

where

y_m	=	peak elastic displacement;
y_o	=	peak static displacement; and
t_m	=	time to the peak displacement.

L-8.3.1.6 *Inelastic SDOF response*

For elasto-plastic analysis, k_E shall be used as given in Table 58 to Table 60 but value of K_{LM} may be chosen in between the elastic and plastic cases depending upon the ductility factor.

Based on the elasto-plastic resistance-deflection curve shown in Fig. 46 and triangular pressure-time curve shown in Fig. 47(A), the ratio of resistance R_m, required to the peak dynamic load (F_1) is given in Fig. 48 for various values of t_d/T and μ.

The time to peak deflection (t_m)shall be obtained from Fig. 49 for various values of td/T and μ.

When the time ratio, t_d/T, is less than 0.1, the ratio k may be computed from the following equation:

$$k = \frac{\pi}{\sqrt{2\mu-1}} \cdot \frac{t_d}{T}$$

When the pressure-time diagram is given by Fig. 47B, the following equation shall be satisfied:

$$\frac{P_{01}}{R_m}k_1 + \frac{P_{02}}{R_m}k_2 = 1$$

where

R_m = the required resistance; and

k_1, k_2 = the values of ratios k for ductility ratio μ and time ratios t_{d1}/T and t_{d2}/T respectively.

For elasto-plastic design of fixed slabs, the modified value of k_E is to be worked out in accordance with Fig. 46(B) using the stiffness values of slab in elastic and elasto-plastic cases as given in Table 60 and K_{LM} is to be suitably chosen depending upon the ductility factor.

The value of modular ratio shall be taken the same as in static design for calculating EI .

For calculating moment of inertia I of reinforced concrete sections, the concept of effective transformed area shall be used.

For analysis of reinforced concrete sections when the simplified elasto-plastic resistance function is used, the elastic stiffness shall be based on effective moment of inertia that shall be calculated as the average of the gross moment of inertia of the transformed section and the cracked moment of inertia.

The gross and cracked moment of inertias shall be obtained from Fig. 50.

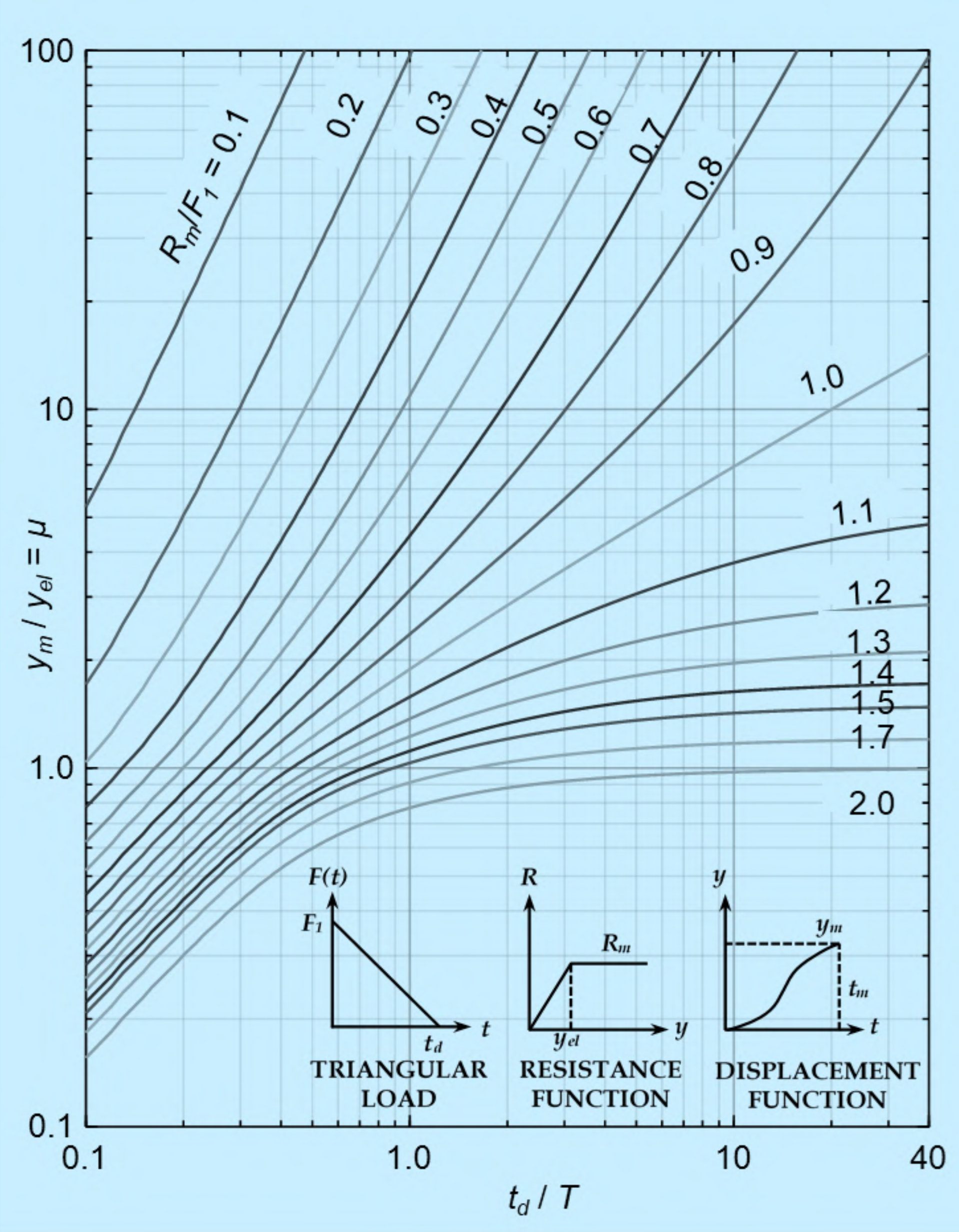


FIG. 48 PEAK DUCTILITY DEMAND OF SDOF SYSTEM SUBJECT TO TRIANGULAR PULSE LOADING

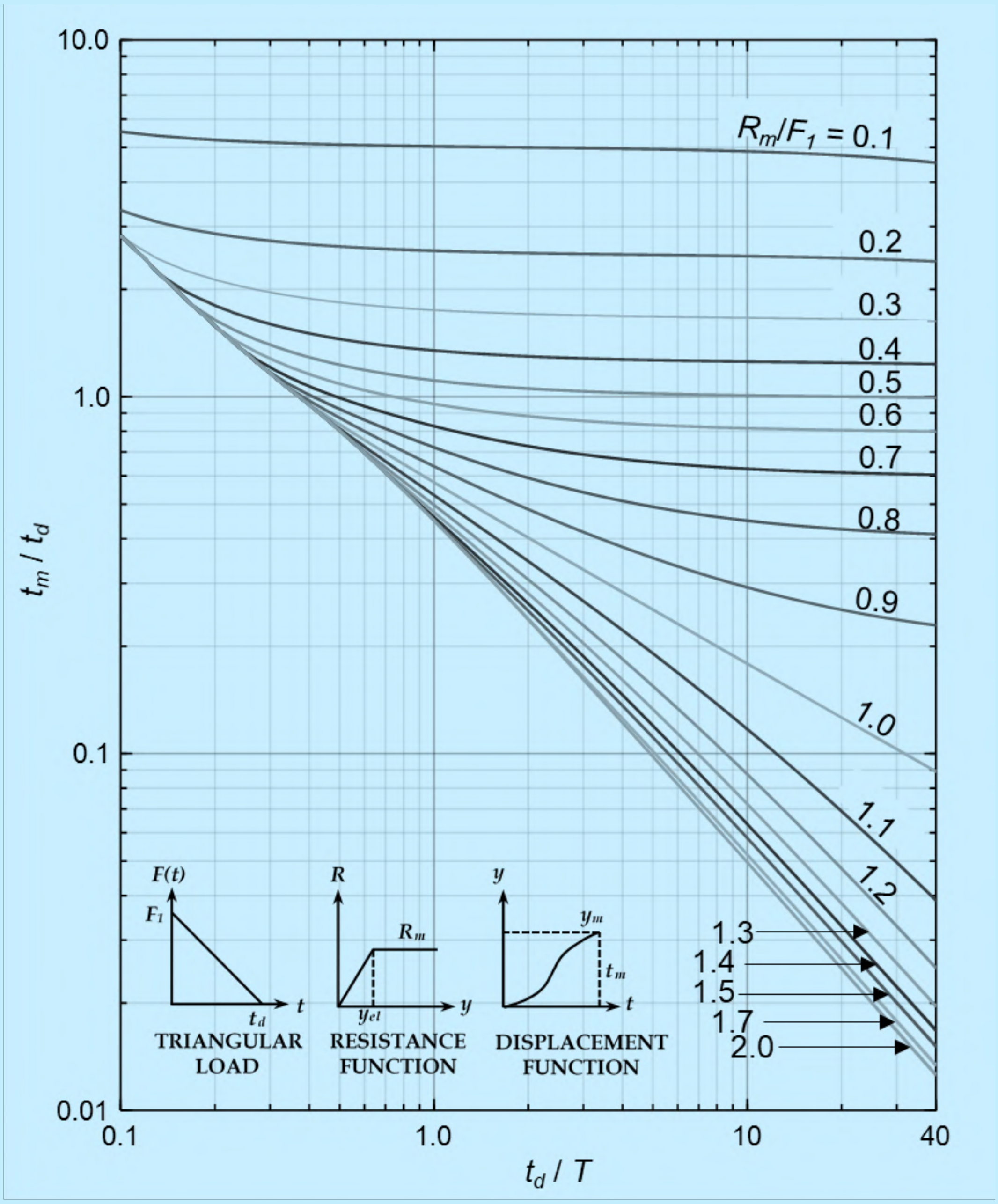


FIG. 49 TIME TO PEAK DUCTILITY DEMAND OF SDOF SYSTEM SUBJECT TO TRIANGULAR PULSE

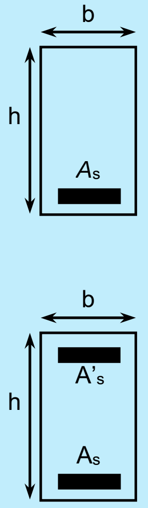
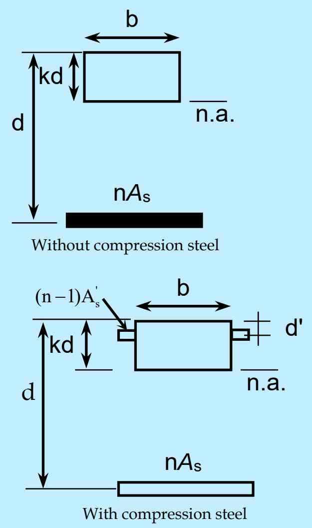
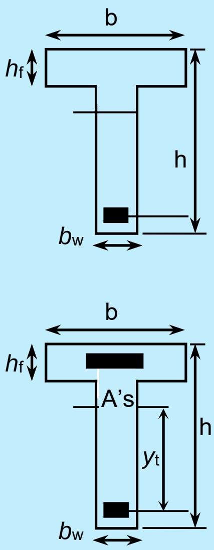
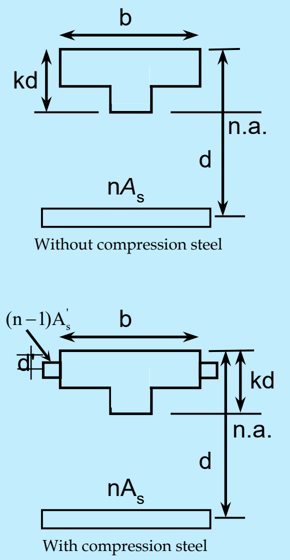
Gross Section	*Cracked Transformed Section*	Gross and Cracked Moment of Inertia
		$n = \frac{E_s}{E_c}$ $B = \frac{b}{(nA_s)}$ $I_g = \frac{bh^3}{12}$ Without compression steel: $kd = (\sqrt{2dB + 1} - 1)/B$ $I_{cr} = bk^3d^3/3 + nA_s(d - kd)^2$ With compression steel: $r = (n - 1)A'_s/(nA_s)$ $kd = \left[\sqrt{2dB(1 + rd'/d) + (1 + r)^2} - (1 + r)\right]/B$ $I_{cr} = bk^3d^3/3 + nA_s(d - kd)^2 + (n - 1)A'_s(kd - d')^2$
		$n = \frac{E_s}{E_c}$ $C = b_w/(nA_s),\ f = h_f(b - b_w)/(nA_s)$ $y_t = h\text{-}1/2\left[(b - b_w)h_f^2 + b_w h^2\right]/[(b - b_w)h_f + b_w h],$ $I_g = (b - b_w)h_f^3/12 + b_w h^3/12 + (b - b_w)h_f\,(h - h_f/2 - y_t)^2 +$ $b_w h(y_t - h/2)^2$ Without compression steel: $kd = \left[\sqrt{C(2d + h_f f) + (1 + f)^2} - (1 + f)\right]/C$ $I_{cr} = (b - b_w)h_f^3/12 + b_w k^3 d^3/3 + (b - b_w)h_f\,(kd - h_f/2)^2 +$ $nA_s(d - kd)^2 + (n - 1)A_s(kd - d')2$ With compression steel: $kd = \left[\sqrt{C(2d + h_f f + 2rd') + (f + r + 1)^2} - (f + r + 1)\right]/C$ $I_{cr} = (b - b_w)h_f^3/12 + b_w k^3 d^3/3 + (b - b_w)h_f\,(kd - h_f/2)^2 +$ $nA_s(d - kd)^2 + (n - 1)\,A'_s(kd - d')2$

FIG. 50 CRACKED MOMENT OF INERTIA OF PRISMATIC SECTIONS

L-8.3.2 *MDOF Analysis*

The multi degree of freedom (MDOF) analysis is permitted for structural systems which cannot be analysed using the simplified SDOF analysis, provided the expected failure mechanism under blast load can adequately be represented using the MDOF model representation.

L-8.3.2.1 The numerical model of MDOF system may comprise of distributed elements or combined lumped mass and stiffness elements.

L-8.3.2.2 The spring in the lumped-mass stiffness should adequately represent the member resistance-deflection behaviour by incorporating appropriate constitutive relationship.

L-8.3.2.3 The lumped-mass stiffness system should be able to properly account for the dynamic behaviour and failure modes at the structure and the element level.

L-8.3.2.4 The use of structural damping is permitted for MDOF system when the expected response is within elastic range. These structural damping values for elastic response shall conform to the typical values used for different type of structures.

L-8.3.2.5 The energy dissipation during inelastic response should be accounted by using appropriate nonlinear member resistance-deflection curves. Supplemental damping in form of structural damping is not permitted for nonlinear range of response.

L-8.3.2.6 The time-step for MDOF analysis shall be determined based on accuracy and convergence requirement. The time-step should be sufficiently small such that it captures the rise and decay of the applied blast load time history.

L-8.3.3 *Finite Element Analysis*

Advanced numerical analysis techniques, such as explicit dynamic finite element analysis (FEA), may be used to analyse structures subject to blast loading, if the methods described in **L-8.3.1** and **L-8.3.2** cannot be reliably used to obtain the response. Such methods shall be required if one or more of the following conditions are encountered:

a) detonation involves scaled distances smaller than 0.4 m/kg$^{1/3}$;

b) detonation of non-spherical charge masses at scaled distances smaller than 3 m/kg$^{1/3}$;

c) the blast wave propagation involves interaction with surrounding environment before it reaches the location of interest;

d) the variation in geometric and material properties of structural elements or the structure cannot appropriately be represented using the simplified methods; and

e) the interaction between the blast wave and structure is of significance and the assumption of rigid reflecting surface is not valid.

The FEA shall consider the modelling of non-ductile failure modes and failure surfaces that cannot be considered in the simplified SDOF and the MDOF methods of analysis.

The time-step of analysis need to be decided based on size of the smallest element of interest with appropriate consideration to the accuracy and convergence of the response.

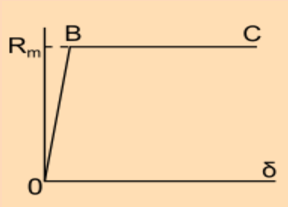
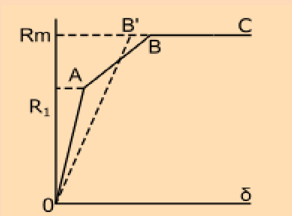
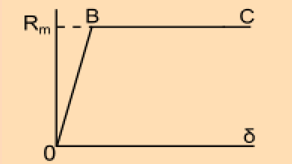
The interaction of blast load with the surrounding shall be considered using computational fluid dynamics (CFD) methods.

Where the shape and size of the charge mass is expected to play a major role in the detonation process and the resulting blast load, the detonation process itself may be modelled.

The FEA methods shall not be subjected to the response limits typically used for structural members in the simplified SDOF and the MDOF analysis.

Table 58 Transformation Factors for Beams and One-way Slabs

(*Clauses* L-8.2.2.1, L-8.3.1.5 *and* L-8.3.1.6)

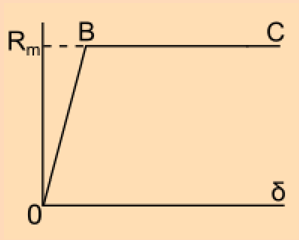
Sl No.	Resistance Curve, End-Conditions of Beams or One-way Slabs	Dynamic Loading	Strain Range	Load-Mass Factor K_{LM}		Maximum Resistance R_m	Effective Spring Constant k_E	Dynamic Reaction
				Concentrated Mass	Uniform Mass			
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
i)		Uniform $P_t = pL$	Elastic	–	0.78	$8M_p/L$	$384EI/5L^3$	$0.39\ R_m + 0.11\ P_t$
			Plastic	–	0.66	$8M_p/L$	–	$0.38\ R_m + 0.12\ P_t$
		Concentrated at mid-span P_t=P	Elastic	1.0	0.49	$4M_p/L$	$48EI/L^3$	$0.78\ R_m - 0.28\ P_t$
			Plastic	1.0	0.33	$4M_p/L$	–	$0.75\ R_m - 0.25\ P_t$
		Concentrated at third points P/2 each $P_t = P$	Elastic	0.87	0.60	$6M_p/L$	$56.4EI/L^3$	$0.62\ R_m - 0.12\ P_t$
			Plastic	1.0	0.56	$6M_p/L$	–	$0.52\ R_m - 0.02\ P_t$
ii)	 Fixed at both ends	Uniform $P_t = pL$	Elastic (OA)	–	0.77	$12M_{ps}/L$	$384EI/L^3$	$0.36\ R_1 + 0.14\ P_t$
			Elastic (OB')	–	0.78	$(8/L)(M_{ps} + M_{pm})$	$307EI/L^3$	$0.39\ R_m + 0.11\ P_t$
			Plastic	–	0.66	$(8/L)(M_{ps} + M_{pm})$	–	$0.38\ R_m + 0.12\ P_t$
		Concentrated at mid-span $P_t = P$	Elastic (OB)	1.0	0.37	$(4/L)(M_{ps} + M_{pm})$	$192EI/L^3$	$0.71\ R_m - 0.21\ P_t$
			Plastic	1.0	0.33	$(4/L)(M_{ps}+ M_{pm})$	–	$0.75\ R_m - 0.25\ P_t$
iii)	Fixed at one end and simply supported at the other	Uniform $P_t = pL$	Elastic (OA)	–	0.78	$(8/L)(M_{ps})$	$185EI/L^3$	$V_1 = 0.26R_1 + 0.12P_t$ $V_2 = 0.43R_1+0.19P_t$

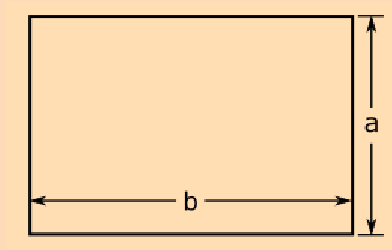
			Elastic (OB')	–	0.78	$(4/L)(M_{ps} + 2M_{pm})$	$160EI/L^3$	$0.39\ R_m + 0.11$
			Plastic	–	0.66	$(4/L)(M_{ps} + 2M_{pm})$	–	$0.38\ R_m + 0.12$
		Concentrated at mid-span $P_t = P$	Elastic (OA)	1.0	0.43	$(16/3L)M_{ps}$	$107EI/L^3$	$V_1 = 0.54R_1 + 0.14P_t$ $V_2 = 0.25R_1 + 0.07P_t$
			Elastic (OB')	1.0	0.49	$(2/L)(M_{ps} + 2M_{pm})$	$106EI/L^3$	$0.78\ R_m - 0.28\ P_t \pm M_{ps}/L$
			Plastic	1.0	0.33	$(2/L)(M_{ps} + 2M_{pm})$	–	$0.75\ R_m - 0.25\ P_t \pm M_{ps}/L$

L = span;
p = dynamic pressure;
P_t = dynamic load;
M_p = plastic moment of section;
M_{pm} = M_p at mid-span
M_{ps} = M_p at support;
R_m = maximum resistance; and
k_E = effective spring constant.

Table 59 Transformation Factors for Two-way Slabs — Four Sides, Simply Supported, Uniformly Loaded

(*Clauses* L-8.2.2.1, L-8.3.1.5 *and* L-8.3.1.6)





Sl No.	Strain Range	a/b	Load Mass Factor K_{LM}	Max Resistance R_m	Spring Constant k_E	Dynamic Reactions	
						V_A	V_B
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	Elastic (OB)	1.0	0.68	$12(M_{pfa} + M_{pfb})/a$	$252EI/a^2$	$0.07P_r + 0.18R_m$	$0.07P_r + 0.18R_m$
		0.9	0.70	$(12M_{pfa} + 11M_{pfb})/a$	$230EI/a^2$	$0.06P_r + 0.16R_m$	$0.08P_r + 0.20R_m$
		0.8	0.71	$(12M_{pfa} + 10.3M_{pfb})/a$	$212EI/a^2$	$0.06P_r + 0.14R_m$	$0.08P_r + 0.22R_m$
		0.7	0.73	$(12M_{pfa}+9.8M_{pfb})/a$	$201EI/a^2$	$0.05P_r + 0.13R_m$	$0.08P_r + 0.24R_m$
		0.6	0.74	$(12M_{pfa} + 9.3M_{pfb})/a$	$197EI/a^2$	$0.04P_r + 0.11R_m$	$0.09P_r+ 0.26R_m$
		0.5	0.75	$(12M_{pfa} + 9.0M_{pfb})/a$	$201EI/a^2$	$0.04P_r + 0.09R_m$	$0.09P_r + 0.28R_m$
ii)	Plastic (BC)	1.0	0.51	$12(M_{pfa} + M_{pfb})/a$	-	$0.09P_r + 0.16R_m$	$0.09P_r + 0.16R_m$
		0.9	0.51	$(12M_{pfa} + 11M_{pfb})/a$	-	$0.08P_r + 0.15R_m$	$0.09P_r + 0.18R_m$
		0.8	0.54	$(12M_{pfa} + 10.3M_{pfb})/a$	-	$0.07P_r + 0.13R_m$	$0.10P_r + 0.20R_m$
		0.7	0.58	$(12M_{pfa} + 9.8M_{pfb})/a$	-	$0.06P_r + 0.12R_m$	$0.10P_r + 0.22R_m$
		0.6	0.58	$(12M_{pfa} + 9.3M_{pfb})/a$	-	$0.05P_r + 0.10R_m$	$0.10P_r + 0.25R_m$
		0.5	0.59	$(12M_{pfa} + 9.0M_{pfb})/a$	-	$0.04P_r + 0. 08R_m$	$0.11P_r + 0.27R_m$

M_{pfa} = total positive ultimate moment capacity along mid-span section parallel to short edge;

M_{pfb} = total positive ultimate moment capacity along mid-span section parallel to long edge;

I = moment of inertia per unit width of slab;

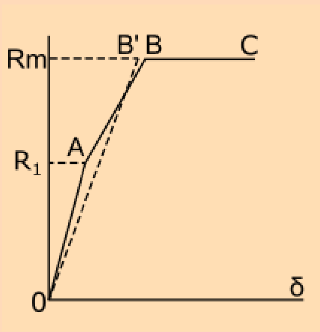
P_t = dynamic load;

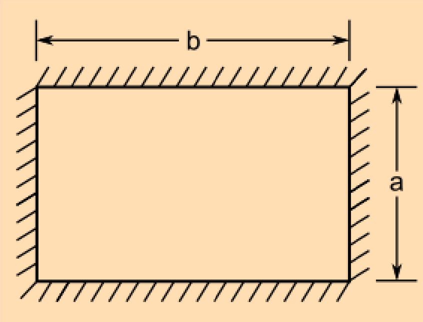
V_A = total dynamic reaction along a short edge; and

V_B = total dynamic reaction along a long edge.

Table 60 Transformation Factors for Two-way Slabs — Fixed four sides, Uniform loaded

(*Clauses* L-8.2.2.1, L-8.3.1.5 *and* L-8.3.1.6)





Sl No.	Strain Range	a/b	Load Mass Factor K_{LM}	Max Resistance R_m	Strain Range	Spring Constant k_E	Dynamic Reactions V_A	V_B
(1)	(2)	(3)	(4)	(5)	(6)	(7)	*(8)*	*(9)*
i)	Elastic (OA)	1.0	0.63	29.2 M^o_{psb}	810 EI/a^2	$0.10P_t + 0.15R_1$	$0.10P_t + 0.15R_1$	1.0
		0.9	0.68	27.4 M^o_{psb}	742 EI/a^2	$0.09P_t + 0.14R_1$	$0.10P_t + 0.17R_1$	0.9
		0.8	0.69	26.4 M^o_{psb}	705 EI/a^2	$0.08P_t + 0.12R_1$	$0.11P_t + 0.19R_1$	0.8
		0.7	0.71	26.2 M^o_{psb}	692EI/a^2	$0.07P_t + 0.11R_1$	$0.11P_t + 0.21R_1$	0.7
		0.6	0.71	27.3 M^o_{psb}	724 EI/a^2	$0.06P_t + 0.09R_1$	$0.12P_t + 0.23R_1$	0.6
		0.5	0.72	30.2 M^o_{psb}	806 EI/a^2	$0.05P_t + 0.08R_1$	$0.12P_t + 0.25R_1$	0.5
ii)	Elasto-plastic (OB')	1.0	0.67	$(1/a)[12(M_{pfa}+ M_{psa})+ 12(M_{pfb}+ M_{psb})]$	252 EI/a^2	$0.07P_t + 0.18R_m$	$0.07P_t + 0.18R_m$	1.0
		0.9	0.70	$(1/a)[12(M_{pfa}+ M_{psa})+ 11(M_{pfb}+ M_{psb})]$	230 EI/a^2	$0.06P_t + 0.16R_m$	$0.08P_t + 0.20R_m$	0.9
		0.8	0.71	$(1/a)[12(M_{pfa}+ M_{psa})+ 10.3(M_{pfb}+ M_{psb})]$	212 EI/a^2	$0.06P_t + 0.14R_m$	$0.08P_t + 0.22R_m$	0.8
		0.7	0.73	$(1/a)[12(M_{pfa}+ M_{psa})+ 9.8(M_{pfb}+ M_{psb})]$	201 EI/a^2	$0.05P_t + 0.13R_m$	$0.08P_t + 0.24R_m$	0.7
		0.6	0.74	$(1/a)[12(M_{pfa}+ M_{psa})+ 9.3(M_{pfb}+ M_{psb})]$	197 EI/a^2	$0.04P_t + 0.11R_m$	$0.09P_t + 0.26R_m$	0.6
		0.5	0.75	$(1/a)[12(M_{pfa}+ M_{psa})+ 9.0(M_{pfb}+ M_{psb})]$	201 EI/a^2	$0.04P_t + 0.09R_m$	$0.09P_t + 0.28R_m$	0.5

iii)	Fully plastic (BC)	1.0	0.51	$(1/a)[12(M_{pfa}+M_{psa})+12(M_{pfb}+M_{psb})]$	–	$0.09P_t+0.16R_m$	$0.09P_t+0.16R_m$	1.0
		0.9	0.51	$(1/a)[12(M_{pfa}+M_{psa})+11(M_{pfb}+M_{psb})]$	–	$0.08P_t+0.15R_m$	$0.09P_t+0.18R_m$	0.9
		0.8	0.54	$(1/a)[12(M_{pfa}+M_{psa})+10.3(M_{pfb}+M_{psb})]$	–	$0.07P_t+0.13R_m$	$0.10P_t+0.20R_m$	0.8
		0.7	0.58	$(1/a)[12(M_{pfa}+M_{psa})+9.8(M_{pfb}+M_{psb})]$	–	$0.06P_t+0.12R_m$	$0.10P_t+0.22R_m$	0.7
		0.6	0.58	$(1/a)[12(M_{pfa}+M_{psa})+9.3(M_{pfb}+M_{psb})]$	–	$0.05P_t+0.10R_m$	$0.10P_t+0.25R_m$	0.6
		0.5	0.59	$(1/a)[12(M_{pfa}+M_{psa})+9.0(M_{pfb}+M_{psb})]$	–	$0.04P_t+0.08R_m$	$0.11P_t+0.27R_m$	0.5

Range (OA) = moment at centre of long edge just becomes plastic;

Range (OB') = moment at supports and mid-span sections just becomes plastic;

M^o_{psb} = negative ultimate moment capacity per unit width at centre of long edge;

M_{psa} = total negative ultimate moment capacity along a short edge support;

M_{psb} = total negative ultimate moment capacity along a long edge support;

M_{pfa}, M_{pfb}, I, P_t = (*see* Table 46);

V_A = total dynamic reaction along a short edge; and

V_B = total dynamic reaction along a long edge.

L-9 LOAD COMBINATIONS FOR DESIGN

L-9.1 Partial Safety Factor

A partial safety factor of unity shall be assumed for blast loads.

L-9.2 Imposed loads

Imposed load on floors shall be considered as per **3**. No Imposed load shall be considered on roof at the time of blast.

L-9.3 Load Combination

Wind or earthquake forces shall not be assumed to occur simultaneously with blast effects. Effects of temperature and shrinkage shall be neglected.

Following load combinations for limit state design shall be considered:

$$1.0DL + 0.5IL + 1.0B$$
$$0.9DL + 1.0B$$

where

DL = design dead load;

IL = design imposed load; and

B = design blast load.

L-10 For design of structures for blast effects of explosions above ground, *see also* good practice [C-1(16)].

For safety of structures during underground blasting, *see also* good practice [C-1(17)].

ANNEX M

(*Clause* 10.2.2 *and* *Table* 61)

SUMMARY OF DISTRICTS HAVING SUBSTANTIAL MULTI-HAZARD RISK AREAS

State	*Name of Districts Having Substantial Multi-Hazard Prone Area*			
	E.Q. and Flood	Cyclone and Flood	E.Q., Cyclone and Flood	E.Q. and Cyclone
(1)	(2)	(3)	(4)	(5)
Andhra Pradesh	Adilabad, Karim Nagar, Khammam	Krishna, Nellore, Srikakulam, Visakhapatnam, Vizianagram	East Godavari, Guntur, Prakasam, West Godavari	—
Assam	All 22 districts listed in Table 61 could have M.S.K IX or more with flooding	No cyclone, but speed can be 50 m/s in districts of Table 61 causing local damage except Dhubri	—	—
Bihar	All 25 districts listed in Table 61	—	—	—
Goa	—	—	—	North and South Goa
Gujarat	Banaskantha, Danthe GS, Gandhinagar, Kheda, Mahesana, Panchmahals, Vadodara	—	Ahmedabad, Bharuch, Surat, Valsad	Amreli, Bhavnagar, Jamnagar, Rajkot, Junagad, Kachcha
Haryana	All 8 districts listed in Table 61	—	—	—
Kerala	Idduki, Kottayam, Palakkad, Pathanamthitta	—	Alappuzha, Ernakulum, Kannur, Kasargod, Kollam, Kozhikode, Malappuram, Thiruvanathapura, Thrissur	—
Maharashtra	—	—	—	Mumbai, Rayagad, Ratnagiri, Sindhudurg, Thane
Orissa	—	Ganjam	Baleshwar, Cuttack, Puri	Dhenkanal
Punjab	All 12 districts listed in Table 61	—	—	—
Uttar Pradesh and Uttarakhand	All 50 districts listed in Table 61	—	—	—
West Bengal	Birbhum, Darjeeling, Jalpaiguri, Cooch Behar, Malda, Murshidabad, West Dinajpur	—	Bardhaman, Kolkata, Hugli, Howrah, Midnapore, Nadia, North and South 24 Parganas	Bankura
Union Territories	Delhi	—	Yanam (Py)	Diu
India	**139 Districts**	**6 Districts**	**29 Districts**	**16 Districts**

Table 61 Multi-Hazard Prone Districts

(*Annex* M)

Assam

Barpeta, Bongaigaon, Cachar[1], Darrang, Dhemaji, Dhuburi, Dibrugarh, Goalpara, Golaghat, Hailaknadi[1], Jorhat, Kamrup, Karbianglong, Karimganj[1], Kokrajhar, Lakhimpur, Morigaon, Nagaon, Nalbari, Sibsagar, Sonitpur, Tinsukia

Bihar[2]
Araria, Begusarai, Bhagalpur, Bhojpur, Darbhanga, Gopalganj, Katihar, Khagaria, Kishanganj, Madhepura, Madhubani, Munger, Muzaffarpur, Nalanda, Nawada, Paschim Champaran, Patna, Purbachamparan, Purnia, Samastipur, Saran, Saharsa, Sitamarhi, Siwan, Vaishali

Haryana[3]
Ambala, Bhiwani, Faridabad, Gurgaon, Hisar, Jind, Kurukshetra, Rohtak

Punjab[4]
Amritsar, Bathinda, Faridkot, Firozpur, Gurdaspur, Hoshiarpur, Jalandhar, Kapurthala, Ludhiana, Patiala, Rupnagar, Sangrur

Uttar Pradesh and Uttarakhand[5]
Agra, Aligarh, Allahabad, Azamgarh, Bahraich, Ballia, Barabanki, Bareilly, Basti, Bijnor, Budaun, Bulandshahr, Deoria, Etah, Etawah, Faizabad, Farrukhabad, Fatehpur, Firozabad, Ghaziabad, Ghazipur, Gonda, Gorakhpur, Hardoi, Haridwar, Jaunpur, Kanpur (Dehat), Kanpur (Nagar), Kheri, Lucknow, Maharajganj, Mainpuri, Mathura, Mau, Meerut, Mirzapur, Mordabad, Muzaffarnagar, Nainital, Pilibhit, Partapgarh, Raebareli, Rampur, Saharanpur, Shahjahanpur, Siddarth Nagar, Sitapur, Sultanpur, Unnao, Varanasi

[1] Districts liable to cyclonic storm but no storm surge
[2] No cyclonic storm in Bihar
[3] No cyclonic storm in Haryana
[4] No cyclonic storm in Punjab
[5] No cyclonic storm in Uttar Pradesh and Uttarakhand

ANNEX N

[*Clause* 11.2]

STRUCTURAL DESIGN BASIS REPORT

N-1 This report shall accompany the application for Building Permit.

N-2 In case information on items (iii), (x), (xviii), (xix) and (xx) of Part 1 of SDBR not can be given at this time, it should be submitted at least one week before commencement of construction.

N-3 In case of reinforced concrete framed buildings, a certificate to the effect that the Part 3 of the SDBR will be completed and submitted at least one month before commencement of construction, shall be submitted with the application for building permit. In addition to the completed report, the following additional information shall be submitted, at the latest, one month before the commencement of construction:

a) Foundations:

1) In case raft foundation has been adopted, indicate stiffness (K) value used for analysis of the raft;

2) In case pile foundations have been used, give full particulars of the piles, type, diameter, length, capacity; and

3) In case of high water table, indicate system of countering water pressure and indicate the existing water table and that assumed to design foundations.

NOTE — The report/certificate of sub-surface investigation shall be duly considered and used in the preparation of foundation design and details.

b) Idealization for earthquake analysis:

1) In case of a composite system of shear walls and rigid frames, give distribution of base shear in the two systems on the basis of analysis and that used for design of each system; and

2) Indicate the idealization of frames and shear walls adopted in the analysis with the help of sketches.

c) Submit framing plans of each floor and in case of basements, indicate the system used to contain earth pressures.

N-4 The latest version of the Indian Standards with their amendments as indicated in the SDBR template given below, shall be referred for the preparation of the report.

N-5 The format of Part 1 of SDBR shall be as below:

PART 1 GENERAL DATA			
Sl No.	*Description*	*Information*	*Notes*
i)	Address of the building: a) Name of the building; b) Plot number; c) Subplot number, if any; d) Town Planning Scheme (if applicable): 1) Name; and 2) Number. e) Locality/Township; and f) District		
ii)	Name of owner		
iii)	Name of Building Constructor on record		

Sl No.	*Description*	*Information*	*Notes*
iv)	Name of Architect/Engineer on record		
v)	Name of Structural engineer on record		
vi)	Use of the building		
vii)	Number of storeys above ground level (including storeys to be added later, if any)		
viii)	Number of basements below ground level		
ix)	Type of structure: a) Load bearing walls; b) R.C.C frame; c) R.C.C frame and shear walls; and d) Steel frame.		
x)	Soil data: a) Type of soil; and b) Design safe bearing capacity.		IS 1892 IS 1904
xi)	Dead loads (Unit weight adopted): a) Earth; b) Water; c) Brick masonry; d) Plain cement concrete; e) Reinforced cement concrete; f) Floor finish; g) Other fill materials; and h) Piazza floor fill and landscape.		IS 875 (Part 1)
xii)	Imposed (Live) loads: a) Piazza floor accessible to fire tender; b) Piazza floor not accessible to fire tender; c) Floor loads[1)]; and d) Roof loads[2)].		IS 875 (Part 2)
xiii)	Cyclone/Wind: a) Speed; and b) Design pressure intensity.		IS 875 (Part 3)
xiv)	Other loads, if any		Relevant parts of IS 875
xv)	Seismic zone		IS 1893 (Part 1)
xvi)	Importance factor (*I*)		IS 1893 (Part 1)
xvii)	Earthquake zone factor (Z)		IS 1893 (Part 1)
xviii)	Elastic force reduction factor (*R*)		IS 1893 (Part 1)
xix)	Fundamental natural period (approximate)		IS 1893 (Part 1)

Sl No.	*Description*	*Information*	*Notes*
xx)	Horizontal site-specific acceleration PSA (A_{SS})		IS 1893 (Part 1)
xxi)	Expansion/Separation Joints[3)]		

1) Enclose small scale plans of each floor on A4 sheets
2) In case terrace garden is provided, indicate additional fill load and imposed (live) load
3) Indicate on a small scale plan on A4 sheet

Signature

(Registered Engineer/Structural Engineer)

N-6 The format of Part 2 of SDBR shall be as below:

<table>
<tr><th colspan="5">PART 2 LOAD BEARING MASONRY BUILDINGS</th></tr>
<tr><th>Sl No.</th><th>Description</th><th>Information</th><th colspan="2">Notes</th></tr>
<tr><td rowspan="4">i)</td><td rowspan="4">Building category</td><td rowspan="4"></td><td colspan="2">IS 4326
IS 1893 (Part 1)</td></tr>
<tr><td>Zone / Building</td><td>II III IV V</td></tr>
<tr><td>Ordinary</td><td>B C D E</td></tr>
<tr><td>Important</td><td>C D E E</td></tr>
<tr><td>ii)</td><td>Basement provided</td><td></td><td colspan="2"></td></tr>
<tr><td>iii)</td><td>Number of floors including ground floor (all floors including stepped floors in hill slopes)</td><td></td><td colspan="2"></td></tr>
<tr><td>iv)</td><td>Type of wall masonry</td><td></td><td colspan="2"></td></tr>
<tr><td>v)</td><td>Type and mix of mortar</td><td></td><td colspan="2">IS 4326</td></tr>
<tr><td>vi)</td><td>Size and position of openings (see Note 1):
a) Minimum distance (b5);
b) Ratio (b1+b2+b3)/l1 or (b6+b7)/l2;
c) Minimum pier width between consequent opening (b4);
d) Vertical distance (h3);
e) Ratio of wall height to thickness; and</td><td></td><td colspan="2">IS 4326</td></tr>
</table>

Sl No.	*Description*	*Information*	*Notes*
	f) Ratio of wall length between cross wall to thickness;		
vii)	Horizontal seismic band:	**P** **TP** **NA**	(*see* Note 2)
	a) at plinth level;	□ □ □	IS 4326
	b) at window sill level;	□ □ □	IS 4326
	c) at lintel level;	□ □ □	IS 4326
	d) at ceiling level;	□ □ □	IS 4326
	e) at eave level of sloping roof;	□ □ □	IS 4326
	f) at top of gable walls; and	□ □ □	IS 4326
	g) at top of ridge walls.	□ □ □	IS 4326
viii)	Vertical reinforcing bar:	**P** **TP** **NA**	
	a) at corners and T-junction of walls; and	□ □ □	IS 4326
	b) at jambs of doors and window openings	□ □ □	IS 4326
ix)	Integration of prefab roofing/ flooring elements through reinforced concrete screed	□ □ □	IS 4326
x)	Horizontal bracings in pitched truss:		
	a) in horizontal plane at the level of ties; and	□ □ □	
	b) in the slopes of pitched roofs	□ □ □	

NOTES

1 Information in Sl No. (vi) of Part 2 should be given on separate A4 sheets for all walls with large number of openings.

2 P indicates "Information Provided"; TP indicates "Information to be Provided" and NA indicates "Not Applicable". Tick mark one box which is applicable.

Signature

(Registered Engineer/Structural Engineer)

N-7 The format of Part 3 of SDBR shall be as below:

PART 3 REINFORCED CONCRETE FRAMED BUILDINGS			
Sl No.	*Description*	*Information*	*Notes*
i)	Type of Building: a) Regular frames; b) Regular frames with shear walls; c) Irregular frames; d) Irregular frames with shear walls; and e) Soft storey.		IS 1893 (Part 1)
ii)	Number of basements		
iii)	Number of floors including ground floor		
iv)	Horizontal floor system: a) Beams and slabs; b) Waffles; c) Ribbed Floor; d) Flat slab with drops; and e) Flat plate without drops.		
v)	Soil data: a) Type of soil; b) Recommended type of foundation: 1) Independent footings; 2) Raft; and 3) Piles. c) Recommended bearing capacity of soil; d) Recommended, type, length, diameter and load capacity of piles; e) Depth of water table; f) Chemical analysis of ground water; and g) Chemical analysis of soil.		
vi)	Foundations: a) Depth below ground level; and b) Type of independent interconnected raft piles.		
vii)	System of interconnecting foundations: a) Plinth beams; and b) Foundation beams.		
viii)	Grades of concrete used in different parts of building		

Sl No.	*Description*	*Information*	*Notes*
ix)	Method of analysis used		
x)	Computer software used		
xi)	Torsion included		IS 1893 (Part 1)
xii)	Base shear: a) Based on approximate fundamental period; b) Based on dynamic analysis; and c) Ratio of a/b.		IS 1893 (Part 1)
xiii)	Distribution of seismic forces along the height of the building		IS 1893 (Part 1) (provide sketch)
xiv)	The column of soft ground storey specially designed		IS 1893 (Part 1)
xv)	Clear minimum cover provided in: a) Footing; b) Column; c) Beams; d) Slabs; and e) Walls.		IS 456
xvi)	Ductile detailing of RC frame: a) Type of reinforcement used; b) Minimum dimension of beams; c) Minimum dimension of columns; d) Minimum percentage of reinforcement of beams at any cross-section; e) Maximum percentage of reinforcement at any section of beam; f) Spacing of transverse reinforcement in 2-D length of beams near the ends; g) Ratio of capacity of beams in shear to capacity of beams in flexure; h) Maximum percentage of reinforcement in column; j) Confining stirrups near ends of columns and in beaL-column joints: 1) Diameter; and 2) Spacing. k) Ratio of shear capacity of columns to maximum seismic shear in the storey		IS 456 IS 13920 IS 13920 IS 456 IS 13920 IS 456 IS 13920 IS 13920 IS 456 IS 13920

Signature

(Registered Engineer/Structural Engineer)

N-8 The format of Part 4 of SDBR shall be as below:

PART 4 STRUCTURAL STEEL/STEEL CONCRETE COMPOSITE BUILDINGS			
i)	Adopted method of design	a) Simple; b) Semi-rigid; and c) Rigid.	IS 800
ii)	Design based on	a) Elastic analysis; and b) Plastic analysis.	IS 800 SP 6 (6)
iii)	Floor construction	a) Composite; b) Non-composite; and c) Boarded.	
iv)	Roof construction	a) Composite; b) Non-composite; c) Metal; and d) Any other.	
v)	Horizontal force resisting system adopted	a) Frames; b) Braced frames; and c) Frames and shear walls.	NOTE — Seismic force as per IS 1893 (Part 1) would depend on the system.
vi)	Slenderness ratios maintained	Members defined in IS 800	IS 800
vii)	Member deflection limited to	a) Beams, rafters; b) Crane girders; c) Purlins; and d) Top of columns.	IS 800
viii)	Structural members	a) Encased in Concrete; and b) Not encased.	IS 800
ix)	Proposed material	a) General weldable; b) High strength; c) Cold formed; and d) Tubular.	IS 2062 (Part 1) IS 801, IS 811 IS 806
x)	Minimum metal thickness specified for corrosion	a) Hot rolled sections	IS 800

	protection	b) Cold formed sections c) Tubes	
xi)	Structural connections	a) Rivets b) CT bolts c) HSFG bolts d) Black bolts e) Welding- Field/Shop (Specify welding type proposed) f) Composite	IS 800 IS 1929, IS 2155, IS 1148 IS 6639, IS 1367 IS 3757, IS 4000, IS 1363 IS 1367, IS 816, IS 814, IS 1395 IS 15977, IS 6419, IS 6560, IS 813, IS 9595 IS 11384
xii)	Minimum fire rating proposed, with method	a) Rating ___ hours; b) Method proposed: 1) Intumescent painting; 2) Spraying; 3) Quilting; and 4) Fire retardant boarding.	IS 1641, IS 1642, IS 1643

Signature

(Registered Engineer/Structural Engineer)

LIST OF STANDARDS

The following list records those standards which are acceptable as 'good practice' and 'accepted standards' in the fulfillment of the requirements of this NBCS. The latest version of a standard shall be adopted at the time of enforcement of this NBCS. The standards listed may be used by the Authority as a guide in conformance with the requirements of the referred clauses in this NBCS.

In the following list, the number appearing in the first column within parentheses indicates the number of the reference in this Section.

	IS No.	*Title*
(1)	IS 8888 : 2020	Requirements of low income housing for urban areas — Guide (*second revision*)
(2)	IS 807 : 2006	Design, erection and testing (structural portion) of cranes and hoists — Code of practice (*second revision*)
	IS 3177 : 2020	Electric overhead travelling crane and gantry crane for all applications — Code of practice (*third revision*)
(3)	IS 17951 : 2023	Method for determination of performance relating to safety of external building envelope impacted by wind borne debris
(4)	IS 14732 : 2000/ISO 9867 : 1984	Guidelines for the evaluation of the response of occupants of fixed structures, especially buildings and off-shore structures, to low-frequency horizontal motion (0.063 to 1 Hz)
(5)	IS 18315 : 2023	Estimation of cyclonic factor (k_4) — Guidelines
(6)	IS 1893 (Part 1) : 2016	Criteria for earthquake-resistant design of structures: Part 1 General provisions and buildings (*sixth revision*)
(7)	IS 13827 : 1993	Improving earthquake resistance of earthen buildings — Guidelines
	IS 13828 : 1993	Improving earthquake resistance of Low strength masonry buildings — Guidelines
(8)	IS 13920 : 2016	Ductile design and detailing of reinforced concrete structures subjected to seismic forces — Code of practice (*first revision*)
(9)	IS 18168 : 2023	Earthquake resistant design and detailing of steel Buildings — Code of practice
(10)	IS 3414 : 1968	Code of practice for design and installation of joints in buildings
(11)	IS 875 (Part 5) : 1987	Code of practice for design loads (Other than earthquake) for building and structures: Part 5 Special loads and combinations (*second revision*)
(12)	IS 1642 : 2013	Fire safety of buildings (general) details of construction — Code of practice (*second revision*)
(13)	IS 14680 : 2024	Landslide control measures — Guidelines (*first revision*)
	IS 17162 : 2020	Preparation of landslide risk assessment maps in mountainous terrains — Guidelines
	IS 17163 : 2020	Site specific investigation and stability analysis of landslides — Guidelines
	IS 14496	Preparation of landslide hazard zonation maps in mountainous terrains — Guidelines:
	(Part 1) : 2020	Meso-zonation
	(Part 2) : 1998	Macro-zonation
	IS 18736 : 2024	Micropiles for slope stabilization for mitigation of landslides — Guidelines
	IS 14458 (Part 1) : 1998	Retaining wall for hill area — Guidelines: Part 1 Selection of type of wall
	IS 14458 (Part 2) : 1997	Retaining wall for hill area — Guidelines: Part 2 Design of

	IS No.	*Title*
		retaining/breast walls
	IS 14458 (Part 3) : 1998	Retaining wall for hill area — Guidelines: Part 3 Construction of dry stone walls
	IS 14458 (Part 4) : 2018	Retaining wall for hill area — Guidelines: Part 4 Construction of banded dry stone masonry walls
	IS 14458 (Part 5) : 2018	Retaining wall for hill area — Guidelines: Part 5 Construction of cement stone masonry walls
	IS 14458 (Part 6) : 2020	Retaining wall for hill area — Guidelines: Part 6 Construction of gabion walls
	IS 14458 (Part 7) : 2022	Retaining wall for hill area – Guidelines Part 7 Construction of peripheral reinforced gabion walls
	IS 14458 (Part 8) :2 025	Retaining wall for hill area – Guidelines Part 8 Design of reinforced cement concrete (RCC) Retaining Walls
	IS 14804 : 2000	Siting design and selection of materials for residential buildings in hilly areas — Guidelines
	IS 14961 : 2024	Surface water management in hilly areas (including rainwater harvesting) – Guidelines (*first revision*)
(14)	IS 15498 : 2023	Improving the cyclonic resistance of low rise houses and other buildings — Guidelines (*first revision*)
	IS 15499:2004	Guidelines for survey of housing and building typology in cyclone prone areas for assessment of vulnerability of regions and post cyclone damage estimation
	IS 17164 : 2020	Design and construction of cyclone shelters — Guidelines
	IS 17951 : 2023	Method for Determination of performance relating to safety of external building envelope impacted by wind borne debris
	IS 18153 : 2023	Improving the cyclonic resistance capacity of *Kutcha* houses — Guidelines
(15)	IS 212 : 2021	Crude coal tar for general use (*third revision*)
	IS 216 : 2006	Coal tar pitch — Specification (*second revision*)
	IS 269 : 2015	Ordinary portland cement– —Specification (*sixth revision*)
	IS 277 : 2018	Galvanized steel strips and sheets (Plain and Corrugated) - Specification (*seventh revision*)
	IS 303 : 2024	Plywood for general purposes — Specification (*fourth revision*)
	IS 383 : 2025	Coarse and fine aggregate for concrete — Specification (*fourth revision*) under revision Doc No: CED 2(27866)
	IS 399 : 1963	Classification of commercial timbers and their zonal distribution (*revised*)
	IS 404 (Part1) : 1993	Lead pipes — Specification: Part 1 for other than chemical purposes (*third revision*)
	IS 412 : 1975	Specification for expanded metal steel sheets for general purposes (*second revision*)
	IS 457 : 1957	Code of practice for general construction of plain and reinforced concrete for dams and other massive structures
	IS 458 : 2021	Precast concrete pipes (with and without reinforcement) (*fifth r*evision)
	IS 459 : 1992	Corrugated and semi — Corrugated asbestos cement sheets - Specification (*third revision*)
	IS 651 : 2007	Glazed stoneware pipes and fittings — Specification (*sixth revision*)
	IS 653 : 1992	Linoleum sheets and tiles — Specification (*third revision*)

IS No.	*Title*
IS 654 : 2025	Clay roofing tiles mangalore pattern — Specification (*fifth revision*) Under Revision Doc No: CED 30 (27732)
IS 808 : 2021	Hot rolled steel beam, column, channel and angle sections — Dimensions and properties (*fourth revision*)
IS 809 : 1992	Rubber flooring materials for general purposes — Specification (*first revision*)
IS 811 : 1987	Specification for cold formed light gauge structural steel sections (*second revision*)
IS 1077 : 2025	Common burnt clay building bricks — Specification (*sixth revision*)
IS 1173 : 1978	Specification for hot rolled and slit steel tee bars (*second revision*)
IS 1322: 1993	Bitumen felts for water proofing and damp-proofing - Specification (*fourth revision*)
IS 1343 : 2012	Prestressed concrete — Code of practice (*second revision*)
IS 1478 : 2023	Clay flooring tiles specification (*third revision*)
IS 1536 : 2023	Centrifugally cast (Spun) iron pressure pipes for water, gas and sewage - Specification (*fifth revision*)
IS 1537 : 1976	Specification for vertically cast iron pressure pipes for water, gas and sewage (*first revision*)
IS 1592 : 2003	Asbestos cement pressure pipes and joints - Specification (*fourth revision*)
IS 1626	Asbestos cement building pipes and pipe fittings, gutters and gutter fittings and roofing fitting
(Part 1) : 1994	Pipe and pipe fittings (*second revision*)
(part 2) : 1994	Gutter and gutter fittings (*second revision*)
IS 1658 : 2006	Fibre hardboards — Specification (*third revision*)
IS 1725 : 2023	Stabilized soil blocks used in general building construction — Specification (*Third Revision*)
IS 1726 : 1991	Cast iron manhole covers and frames — Specification (*third revision*)
IS 1729 :2023	Sand cast iron spigot and socket pipes, fittings and accessories —Specification (*third revision*)
IS 2095	Gypsum plaster boards - Specification:
(Part 1) : 2011	Plain gypsum plaster boards (*fourth revision*)
(Part 2) : 2022	Coated laminated gypsum plaster boards (*third revision*)
(Part 3) : 2022	Reinforced gypsum plaster boards and ceiling tiles (*fourth revision*)
IS 2180 : 2025	Specification for heavy duty burnt clay building bricks (*fourth revision*) *Under Revision* CED 30(26671)
IS 2185	Concrete masonry units — Specification
(Part 1) : 2005	Hollow and solid concrete blocks (*third revision*)
(Part 3) : 1984	Autoclaved Cellular Aerated Concrete Blocks
IS 2222 : 2025	Specification for burnt clay perforated building bricks (*fourth revision*) *Under Revision* CED 30 (28174)
IS 2314	Steel sheet piling section — Specification
(Part 1) : 2023	Hot rolled sheet pile (*second revision*)
(Part 2) : 2023	Cold formed sheet pile (*second revision*)
IS 2394: 1984	Code of practice for application of lime plaster finish (*first revision*)

IS No.	*Title*
IS 2508 : 2024	Polyethylene films and sheets — Specification (*fourth revision*)
IS 2526 : 1963	Code of practice for acoustical design of auditoriums and conference halls
IS 2541 : 1991	Preparation and use of lime concrete — Code of practice (*second revision*)
IS 2547	Specification for gypsum building plaster
(Part 1) : 1976	Excluding premixed lightweight plasters (*first revision*)
(Part 2) : 1976	Premixed lightweight plasters (*first revision*)
IS 2553 (Part 1) : 2018	Safety glass — Specification: Part 1 Architectural, building and general uses (*fourth revision*)
IS 2835 : 1987	Specification for flat transparent sheet glass (*third revision*)
IS 3087 : 2005	Particle boards of wood and other lingnocellulosic materials (medium density) for general purposes — Specification (*second revision*)
IS 3115 : 1992	Lime based blocks — Specification (*second revision*)
IS 3129 : 1985	Specification for low density particle boards
IS 3348 : 1965	Specification for fibre insulation boards
IS 3478 : 1966	Specification for high density wood particle boards
IS 4139 : 1989	Calcium silicate bricks — Specification (Second Revision)
IS 4253	Cork composition sheets — Specification
(Part 1) : 2008	Plain cork sheets (*second revision*)
(Part 2) : 2008	Cork and rubber (*second revision*)
IS 4671 : 2018	Expanded polystyrene for thermal insulation purposes (*second revision*)
IS 4860 : 1968	Specification for acid — Resistant bricks
IS 4923 : 2017	Hollow steel sections for structural use — Specification (*third revision*)
IS 6072 : 2023	Autoclaved reinforced cellular concrete wall slabs — Specifications (*first revision*)
IS 6598 : 2018	Cellular concrete for thermal insulation — Specification (*first revision*)
IS 8041 : 1990	Rapid hardening portland cement — Specification (*second revision*)
IS 9012 : 1978	Recommended practice for shotcreting
IS 12679 : 2023	By-product gypsum for construction — Specification (*second revision*)
IS 12823 : 2015	Prelaminated particle boards from wood and other Lignocellulosic material — Specification (*first revision*)
IS 12894 : 2002	Pulverized fuel ash — Lime bricks — Specification (*first revision*)
IS 14443 : 1997	Polycarbonate sheets — Specification
IS 14587 : 2023	Prelaminated medium density fibre board — Specification
IS 14635	Fluoropolymer dispersions and moulding and extrusion materials
(Part 2) : 2020	Preparation of test specimens and determination of properties (*first revision*)
IS 14753 : 1999	Polymethyl methacrylate (pmma) (acrylic) sheets
IS 14862 : 2000	Fibre cement flat sheets — Specification

	IS No.	*Title*
	IS 15622 : 2017	Pressed ceramic tiles — Specification (*first revision*)
	IS 15658 : 2001	Concrete paving blocks — Specification (*first revision*)
	IS 16720 : 2018	Pulverized fuel ash-cement bricks — Specification
	IS 17452 : 2020	Use of alkali activated concrete for precast products — Guidelines
(16)	IS 4991 : 1968	Criteria for blast resistant design of structures for explosions above ground
(17)	IS 6992 : 1973	Criteria for safety and design of structures subject to underground blasts

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART C STRUCTURAL DESIGN

SECTION 2 SOILS AND FOUNDATIONS

BUREAU OF INDIAN STANDARDS

CONTENTS

National Building Code Sectional Committee, CED 46

FOREWORD

This NBCS (Part C/Sec 2) deals with the structural design aspects of foundations and mainly covers the design principles involved in different types of foundations.

This Section was published in 1970 and subsequently revised in 1983, 2005 and 2016 as a chapter of the National Building Code of India. In the first revision of 1983, design considerations in respect of shallow foundation were modified, provisions regarding pier foundation were added and provisions regarding raft foundation and pile foundation were revised and elaborated. In the second revision of 2005, design considerations in respect of shallow foundations were again modified, method for determining depth of fixity, lateral deflection and maximum moment of laterally loaded piles were modified and reference was made to ground improvement techniques.

In the third revision in 2016, the significant changes incorporated were: including the extension of the scope to cover the design of foundations on rock, modifications to definitions in line with prevailing engineering practices; addition of new terms related to pile foundations and ground improvement; inclusion of the clause on site investigation and number of modifications, such as, new methods of soil investigation; addition of depth of exploration for pile foundations; addition of new sub-clauses on vertical interval for field tests and site investigation report; inclusion of method for assessment of liquefaction potential of a site; modification of permissible differential settlements and tilt (angular distortion) for shallow foundations in soils; modification of procedures for calculation of bearing capacity, structural capacity, factor of safety, lateral load capacity, overloading, etc, in case of pile foundations; revision of design parameters with respect to adhesion factor, earth pressure coefficient, modulus of subgrade reaction, etc, for design of pile foundations to make them consistence with the outcome of modern research and construction practices; inclusion of provision for use of any established dynamic pile driving formulae, instead of recommending any specific formula, to control the pile driving at site, giving due consideration to limitations of various formulae; inclusion of a reference to spun piles, which are used in deep marshy soils where conventional pile installation beyond 50 m is difficult; elaboration of the clause on ground improvement techniques; and addition of a table on summary of soil improvement methods.

In the current revision, this publication has been renamed as National Building Construction Standards (NBCS).

As a result of technical developments in this subject, the experience gained since implementation of 2016 version of the NBCS and feedback received as well as the revision of related Indian Standards on the subject, a need was felt to revise this Section. The significant modifications made in this revision of the subsection include amongst others, the following:

a) Provisions on location and depth of investigation have been modified;
b) Provisions on various boring techniques have been updated and continuous soil samplers have been included;
c) Series of in-situ tests have been added, such as electric cone penetration test using piezocone with pore pressure measurement (CPTu), field California bearing ratio (CBR) test, cross hole seismic test (CHST), ground penetrating radar survey (GPR), spectral analysis of surface wave (SASW) and multi-channel analysis of surface wave (MASW), dilatometer test (DMT) and in-situ permeability test, etc;
d) Provisions on optical and acoustic tele-viewer methods have been added;
e) Provisions on pressuremeter test (PMT) have been detailed;
f) Detailed explanation for deep foundation has been added;
g) The provisions for the design and construction of Combined Piled Raft Foundations (CPRF) have been added. These provisions cover a composite foundation system that utilizes both raft and pile components to effectively transfer structural loads to the ground; and
h) Provision relating to geotechnical instrumentation has been introduced.

For detailed information regarding structural analysis and soil mechanics aspects of individual foundations, reference should be made to standard textbooks and available literature.

The requirements for geotechnical engineering services, including roles and responsibilities, qualifications and experience of professionals and deliverables expected from various stakeholders, are covered in IS 19235 : 2025 'Geotechnical engineering services — Requirements'. While the standard primarily addresses pre-construction services, it also outlines interlinked construction stage services, thereby providing comprehensive guidance for geotechnical aspects throughout the project lifecycle.

The information contained in this section is mainly based on the following Indian Standards:

IS 1080 : 1985	Code of practice for design and construction of shallow foundations in soils (other than raft, ring and shell) (*second revision*)
IS 1892 : 2021	Subsurface investigation for foundations — Code of practice (*second revision*)
IS 1904 : 2021	General requirements for design and construction of foundation in soils — Code of practice (*fourth revision*)
IS 2911	Code of practice for design and construction of pile foundations
(Part 1/Sec 1) : 2010	Concrete piles, Section 1 Driven cast *in-situ* concrete piles (*second revision*)
(Part 1/Sec 2) : 2010	Concrete piles, Section 2 Bored cast *in-situ* piles (*second revision*)
(Part 1/Sec 3) 2010	Concrete piles, Section 3 Driven precast concrete piles (*second revision*)
(Part 1/Sec 4) : 2010	Concrete piles, Section 4 Precast concrete piles in prebored holes (*first revision*)
(Part 2) : 2021	Timber piles (*second revision*)
(Part 3) : 2021	Under-reamed piles (*second revision*)
IS 2950 (Part 1) : 1981	Code of practice for design and construction of raft foundations: Part 1 Design (*second revision*)
IS 19117 : 2025	Design and construction of combined piled raft foundations — Code of practice
IS 2131 : 2025	Standard penetration test of soil — Method of test

All standards, whether given herein above or cross-referred to in the main text of this Section, are subject to the revision. The parties to agreement based on this Section are encouraged to investigate the possibility of applying the most recent editions of the standards.

As the regulation of land development and buildings is a state subject and is dealt by the states and local bodies such as urban local bodies within their jurisdiction through building regulatory documents like building regulations, building byelaws and fire regulations, this document is voluntary in nature and is non-binding. The implementation depends on adoption by concerned parties or stipulations in a contract or by appropriate adoption by the concerned authorities.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 2022 'Rules for rounding off numerical values (*second revision*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this Section.

Important Explanatory Note for Users of the NBCS

In any Part/Section of this NBCS, where reference is made to **'good practice'** in relation to design, constructional procedures or other related information and where reference is made to **'accepted standard'** in relation to material specification, testing, or other related information, the Indian Standards listed at the end of the Part/Section shall be used as a guide to the interpretation.

At the time of publication, the editions indicated in the standards were valid. All standards are subject to revision and parties to agreements based on any Part/Section are encouraged to investigate the possibility of applying the most recent editions of the standards.

In the list of standards given at the end of a Part/Section, the number appearing within parentheses in the first column indicates the number of the reference of the standard in the Part/Section. For example:

a) Good practices [C-2(1)] refers to the Indian Standard(s) given at serial number (1) of the list of standards given at the end of this Part/Section, that is, IS 1892 : 2021 'Code of practice for subsurface investigation for foundation (*second revision*)', IS 2131 : 2025 'Method of standard penetration test for soils (*second revision*)', IS 2132 : 1986 'Code of practice for thin walled tube sampling of soils (*second revision*)', IS 4434 : 1978 'Code of practice for in-situ vane shear test for soils (*first revision*)', IS 4968 (Part 1) : 1976 'Method for subsurface sounding for soils: Dynamic method using 50 mm cone without bentonite slurry (*first revision*)', IS 4968 (Part 2) : 1976 'Method for subsurface sounding for soils: Dynamic method using cone and bentonite slurry (*first revision*)', IS 4968 (Part 3) : 1976 'Method for subsurface sounding for soils: Static cone penetration test (*first revision*)', IS 8763 : 1978 'Guide for undisturbed sampling of sands and sandy soils' and IS 9214 : 1979 'Method for determination of modulus of subgrade reaction (k-value) of soils in the field'.

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART C STRUCTURAL DESIGN

SECTION 2 SOILS AND FOUNDATIONS

1 SCOPE

1.1 This NBCS (Part C/Section 2) covers geotechnical design (principles) of building foundations, such as shallow foundations, like, continuous strip footings, combined footings, raft foundations, deep foundations like pile foundations and other foundation systems to ensure safety and serviceability without exceeding the permissible stresses of the materials of foundations and the bearing capacity of the supporting soil/rock.

1.2 This Section also covers provisions relating to preliminary work required for construction of foundations and protection of excavation.

2 TERMINOLOGY

For the purpose of this Section, the following definitions shall apply.

2.1 General

2.1.1 *Clay* — An aggregate of microscopic and sub-microscopic particles derived from the chemical decomposition and disintegration of rock constituents. It is plastic within a moderate to wide range of water content. The particles are less than 0.002 mm in size.

2.1.2 *Clay*, *Firm* — A clay which at its natural water content can be moulded by substantial pressure with the fingers and can be excavated with a spade.

2.1.3 *Clay*, *Soft* — A clay which at its natural water content can be easily moulded with the fingers and readily excavated.

2.1.4 *Clay*, *Stiff* — A clay which at its natural water content cannot be moulded with the fingers and requires a pick or pneumatic spade for its removal.

2.1.5 *Foundation* — That part of the structure which is in direct contact with and transmits loads to the ground.

2.1.6 *Gravel* — Angular, rounded or semi-rounded particles of rock or soil of particle size between 4.75 mm and 75 mm.

2.1.7 *Peat* — A fibrous mass of organic matter in various stages of decomposition generally dark brown to black in colour and of spongy consistency.

2.1.8 *Sand* — Cohesion less aggregates of angular, sub-angular, sub-rounded, rounded, flaky or flat fragments of more or less unaltered rocks, or mineral of size between 4.75 mm and 0.075 mm IS Sieve.

2.1.9 *Sand*, *Coarse* — Sand of particle size between 2.0 mm and 4.75 mm IS Sieve.

2.1.10 *Sand*, *Fine* — Sand of particle size between 0.075 mm and 0.425 mm IS Sieve.

2.1.11 *Sand*, *Medium* — Sand of particle size between 0.425 mm and 2.0 mm IS Sieve.

2.1.12 *Silt* — Fine-grained soil or fine-grained portion of soil which exhibits a little or no plasticity and has a little or no strength when air dried. The size of particles ranges from 0.075 mm to 0.002 mm.

2.1.13 *Soft Rock* — A rocky cemented material which offers a high resistance to picking up with pick axes and sharp tools but which does not normally require blasting or chiselling for excavation.

2.1.14 *Soil*, *Black Cotton* — Inorganic clays of medium to high compressibility. They form a major soil group in India. They are predominately montmorillonitic in structure and yellowish black or blackish grey in colour. They are characterized by high shrinkage and swelling properties.

2.1.15 *Soil*, *Coarse Grained* — Soils which include the coarse and largely siliceous and unaltered products of rock weathering. They possess no plasticity and tend to lack cohesion when in dry state. In these soils, more than half the total material by weight is larger than 75-micron IS Sieve size.

2.1.16 *Soil*, *Fine Grained* — Soils consisting of the fine and altered products of rock weathering, possessing cohesion and plasticity in their natural state, the former even when dry and both even when submerged. In these soils, more than half of the material by weight is smaller than 75-micron IS Sieve size.

2.1.17 *Soil*, *Highly Organic and Other Miscellaneous Soil Materials* — Soils consisting of large percentages of fibrous organic matter, such as peat and particles of decomposed vegetation. In addition, certain soils containing shells, concretions, cinders and other non-soil materials in sufficient quantities are also grouped in this division.

2.1.18 *Total Settlement* — The total downward movement of the foundation unit under load.

2.2 Ground Improvement

2.2.1 *Ground Improvement* — Enhancement of the in-place properties of the ground by controlled application of techniques suited to subsoil conditions.

NOTE — These techniques are used to improve the load bearing capacity and settlement potential of the loose or soft soil lying close to the surface or at depths.

2.2.2 *Injection* — Introduction of a chemical/cementitious material into a soil mass by application of pressure.

2.2.3 *Liquefaction* — It is a state primarily in saturated cohesionless soils wherein the effective shear strength is reduced to negligible value for all engineering purposes, when the pore pressure approaches the total confining pressure during earthquake shaking. In this condition, the soil tends to behave like a fluid mass.

2.2.4 *Preloading* — Application of loads to achieve improvement of soil properties prior to imposition of structural loads.

2.2.5 *Soil Densification* — A technique to densify cohesionless soils by imparting shocks or vibrations.

2.2.6 *Soil Reinforcement* — Rods, strips or geosynthetics (metallic or fabric) incorporated within soil mass to impart resistance to tensile, shear and compressive forces.

2.3 Shallow Foundation

2.3.1 *Back Fill* — Materials used or reused to fill an excavation.

2.3.2 *Bearing Capacity*, *Safe* — The maximum intensity of loading that the soil will safely carry with a factor of safety against shear failure of soil irrespective of any settlement that may occur.

2.3.3 *Bearing Capacity*, *Ultimate* — The intensity of loading at the base of a foundation which would cause shear failure of the supporting soil.

2.3.4 *Bearing Pressure*, *Allowable* (*Gross or Net*) — The intensity of loading which the foundation will carry without undergoing settlement in excess of the permissible value for the structure under consideration but not exceeding safe bearing capacity.

The net allowable bearing pressure is the gross allowable bearing pressure minus the surcharge intensity.

NOTE — The concept of 'gross' and 'net' used in defining the allowable bearing pressure could also be extended to safe bearing capacity, safe bearing pressure and ultimate bearing capacity.

2.3.5 *Factor of Safety* (*with Respect to Bearing Capacity*) — A factor by which the ultimate bearing capacity (net) shall be reduced to arrive at the value of safe bearing capacity (net).

2.3.6 *Footing* — A structure constructed in brick work, masonry or concrete under the base of a wall or column for the purpose of distributing the load over a larger area.

2.3.7 *Foundation*, *Raft* — A substructure supporting an arrangement of columns or walls in a row or rows transmitting the loads to the soil by means of a continuous slab, with or without depressions or openings.

2.3.8 *Made-up Ground* — Refuse, excavated soil or rock deposited for the purpose of filling a depression or raising a site above the natural surface level of the ground.

2.3.9 *Offset* — The projection of the lower step from the vertical face of the upper step.

2.3.10 *Permanent Load* — Loads which remain on the structure for a period, or a number of periods, long enough to cause time dependent deformation/settlement of the soil.

2.3.11 *Shallow Foundation* — A foundation whose width is generally equal to or greater than its depth. The shearing resistance of the soil in the sides of the foundation is generally neglected.

2.3.12 *Spread* (*or Isolated or Pad*) *Foundation/Footing* — A foundation which transmits the load to the ground through one or more footings. A spread footing (or isolated or pad) is provided to support an individual column. A spread footing is circular, square or rectangular slab of uniform thickness. Sometimes, it is stepped or haunched to spread the load over a large area.

2.3.13 *Strip Foundation/Footing* — A type of shallow foundation which provides continuous and longitudinal bearing for loads carried by vertical elements, such as continuous wall foundation beams or the like. A strip footing is also provided for a row of columns which are so closely spaced that their spread footings overlap or nearly touch each other. A strip footing is also known as continuous footing.

2.4 Pile Foundation

2.4.1 *Allowable Load* — The load which may be applied to a pile after taking into account its ultimate load capacity, group effect, the allowable settlement, negative skin friction and other relevant loading conditions including reversal of loads, if any.

2.4.2 *Anchor Pile* — An anchor pile means a pile meant for resisting pull or uplift forces.

2.4.3 *Batter Pile* (*Raker Pile*) — The pile which is installed at an angle to the vertical using temporary casing or permanent liner.

2.4.4 *Bored Cast In-Situ Pile* — Piles formed by boring a hole in the ground by percussive or rotary method with the use of temporary/permanent casing or drilling mud and subsequently filling the hole with reinforced concrete.

2.4.5 *Bored Compaction Pile* — A bored cast *in-situ* pile with or without bulb(s) in which the reinforcement cage is driven by suitable method in the freshly filled concrete in the bore. If the pile is with bulb(s), it is known as under-reamed bored compaction pile.

2.4.6 *Bored Pile* — A pile formed with or without casing by excavating or boring a hole in the ground and subsequently filling it with plain or reinforced concrete.

2.4.7 *Cut-off Level* — It is the level where a pile is cut-off to support the pile caps or beams or any other structural components at that level.

2.4.8 *Driven Cast in-situ Pile* **—** The pile formed within the ground by driving a casing of uniform diameter, or a device to provide enlarged base and subsequently filling the hole with reinforced concrete. For displacing the subsoil the casing is driven with a plug or a shoe at the bottom. When the casing is left permanently in the ground, it is termed as cased pile and when the casing is taken out, it is termed as uncased pile. The steel casing tube is tamped during its extraction to ensure proper compaction of concrete.

2.4.9 *Efficiency of a Pile Group* — It is the ratio of the actual supporting value of a group of piles to the supporting value arrived at by multiplying the pile resistance of an isolated pile by their number in the group.

2.4.10 *Initial Load Test* — A test pile is tested to determine the load carrying capacity of the pile by loading either to its ultimate load or to twice the estimated safe load.

2.4.11 *Negative Skin Friction* — Negative skin friction is the force developed through the friction between the pile and the soil in such a direction as to increase the loading on the pile, generally due to drag of a consolidating soft layer around the pile resting on a stiffer bearing stratum such that the surrounding soil settles more than the pile.

2.4.12 *Pile Spacing* **—** The spacing of piles means the centre to centre distance between adjacent piles.

2.4.13 *Precast Concrete Piles in Prebored Holes* — A pile constructed in reinforced concrete in a casting yard and subsequently lowered into prebored holes and the annular space around the pile grouted.

2.4.14 *Precast Driven Pile* — The pile constructed in concrete in a casting yard and subsequently driven into the ground when it has attained sufficient strength.

2.4.15 *Routine Test* — Test carried out on a working pile with a view to check whether pile is capable of taking the working load assigned to it without exceeding permissible settlement.

2.4.16 *Safe Load* **—** It is the load derived by applying a factor of safety on the ultimate load capacity of the pile/pile group or as determined from load test.

2.4.17 *Ultimate Load Capacity* — The maximum load which a pile can carry before failure, that is, when the founding strata fails by shear as evidenced from the load settlement curve or the pile fails as a structural member.

2.4.18 *Under-Reamed Pile* — A bored cast *in-situ* or bored compaction concrete pile with enlarged bulb(s) made by either cutting or scooping out the soil or by any other suitable process.

An under-reamed pile having more than one bulb is termed as multi under-reamed piles. The piles having two bulbs may be called double under-reamed piles.

2.4.19 *Working Load* — The load assigned to a pile as per design.

2.4.20 *Working Pile* — A pile forming part of the foundation system of a given structure.

2.5 Combined Piled-Raft Foundations

2.5.1 *Allowable Load* **—** The load which can be applied on the combined pile-raft foundation system by taking into account the load capacities of piles and raft, soil-pile-raft interaction effects and allowable displacement from serviceability criteria, negative skin friction on pile and other relevant loading types including reversal of loads, if any. In other words, it is the load at which the allowable settlement reduction ratio considering serviceability is achieved.

2.5.2 *Angular Distortion* (**θ**) **—** Vertical distance after deformation due to applied load between the points of maximum and minimum settlement divided by the horizontal distance between the two points (*see* Fig. 1).

2.5.3 *Combined Piled Raft Foundation* **—** A pile group in which the raft connecting all the pile heads positively contributes to the overall foundation behaviour. The presence of raft is recognized and its contribution in sharing the load with the pile group is taken into account. This composite foundation which combines load carrying capabilities of raft and pile foundations together, considers four interactions, namely, pile-pile interaction, pile-soil interaction, pile-raft interaction and raft-soil interaction [*see* Fig. 2(A)].

2.5.3.1 *Large CPRF* **—** CPRF in which the ratio of raft width (Br) to pile length (Lp) is greater than 1, that is, $\frac{B_r}{L_p} > 1$.

2.5.3.2 *Small CPRF* **—** CPRF in which the ratio of raft width (Br) to pile length (Lp) is less than or equal to 1, that is, $\frac{B_r}{L_p} \leq 1$.

2.5.4 *Cut-off Level* **—** The level where a pile is cut-off in order to make structural connection to the pile caps or beams or raft or any other structural component at that level.

2.5.5 *Design Load* — The loads (compression, tension, lateral, moment or any combination thereof) considered to act on the CPRF during the life of the structure satisfying provisions mentioned in other relevant Indian Standards.

2.5.6 *Design Stress* — The stress (compressive, tensile or combination) imposed on the CPRF by the design load and calculated in accordance with engineering practice.

2.5.7 *Differential Settlement of CPRF* (*Spr*, *Diff*) — The difference between the maximum and the minimum settlement, across any section of the CPRF (*see* Fig. 1).

2.5.8 *Limiting Capacity* — The vertical load at which for a CPRF, bearing capacity failure occurs due to exceedance of the combined capacity of raft resistance and that of piles in terms of shaft resistance and end bearing resistance. The assessed value of the total resistance Rtot,k(s) of the CPRF depends on the settlements of the foundation and consists of the sum of the assessed pile resistances $\Sigma Rpile,k,j$ (s) and the assessed raft resistance, Rraft,k (s) [*see* Fig. 2(A)]. The assessed raft resistance results from the integration of the settlement dependent contact pressure, $\sigma(s, x, y)$ in the ground plan area of the raft and can be expressed as, $Rraft,k(s) = \iint \sigma(s, x, y)dxdy$. Similarly, due considerations for horizontal load capacity and moment capacity should be considered for the CPRF system.

2.5.9 *Load Eccentricity* (*eL or eB*) — Distance between the point of application of applied resultant load and the centre of gravity of the pile-raft system along length and width, respectively (*see* Fig. 1).

2.5.10 *Maximum CPRF Settlement* (*Spr*, *Max*) — The maximum value of settlement occurred at a particular point of the CPRF system (*see* Fig. 1).

2.5.11 *Pile-Pile Interaction* — The result of pile group effect, defined as the changes in the load-displacement response of a pile group and single piles due to superimposition of stress and displacement field of a single pile in a group [*see* Fig. 2(A)].

2.5.12 *Pile-Raft Interaction* — The changes in the load-displacement response of pile group when the raft is being rested to the soil surface [*see* Fig. 2(A)].

2.5.13 *Pile-Raft Coefficient or Load Sharing Ratio* (α_{CPRF}) — The ratio between the sum of the pile resistances and the value of the total resistance.

NOTE — $\alpha_{CPRF} = \frac{R_{piles}}{R_{total}}$, in which R_{piles} is the resistance provided by piles and R_{total} is the total resistance provided by the CPRF system under vertical load as shown in Fig. 2(B). Whenever, α_{CPRF} is equal to 1, all the vertical loads are taken up by the pile component which essentially makes the system to behave as a pile group foundation. Whereas, α_{CPRF} equals to zero indicates that the vertical loads are borne by raft component, hence, the foundation behaviour will be like spread raft foundation. To utilize both the foundation components effectively, the value of α_{CPRF} should be chosen carefully and the standard is not valid for $\alpha_{CPRF} \geq 0.9$.

2.5.14 *Pile-Enhanced Raft* — CPRF in which the piles are designed to mobilize their total ultimate capacity and the raft carries majority of the design load [*see* Fig. 3(A)].

2.5.15 *Raft-Pile Interaction* — The modification in the load carrying mechanism of the raft when piles are introduced beneath [*see* Fig. 2(A)]

2.5.16 *Raft-Enhanced Pile* — CPRFs in which both the piles and raft work within a pseudo-elastic range of behaviour. The pile group capacity is not fully mobilized at working load and the pile group carries majority of the design load [*see* Fig. 3(B)].

2.5.17 *Settlement Reduction Ratio* (*SRR*) — The ratio of the difference of settlement of the unpiled raft under the given loading, Sr and the settlement of the piled raft under the same loading considered in the case of the unpiled raft, S_{pr} to the settlement of the unpiled raft, that is, under any given loading conditions, SRR $= \frac{(S_r - S_{pr})}{S_r}$.

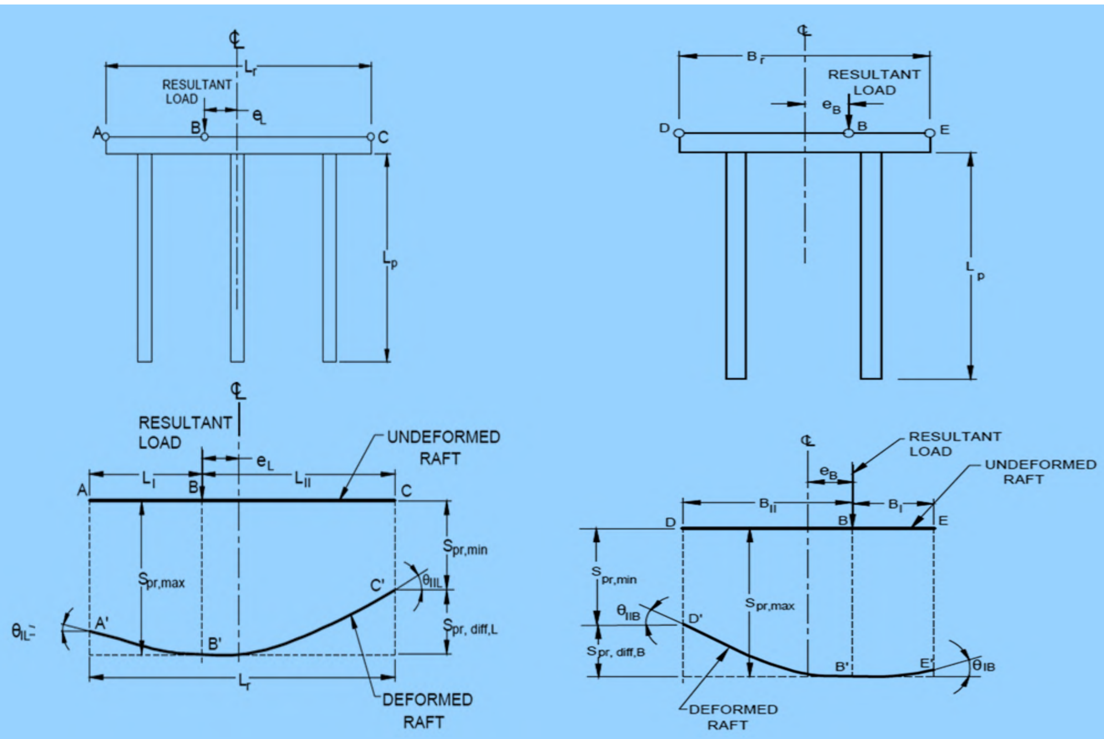
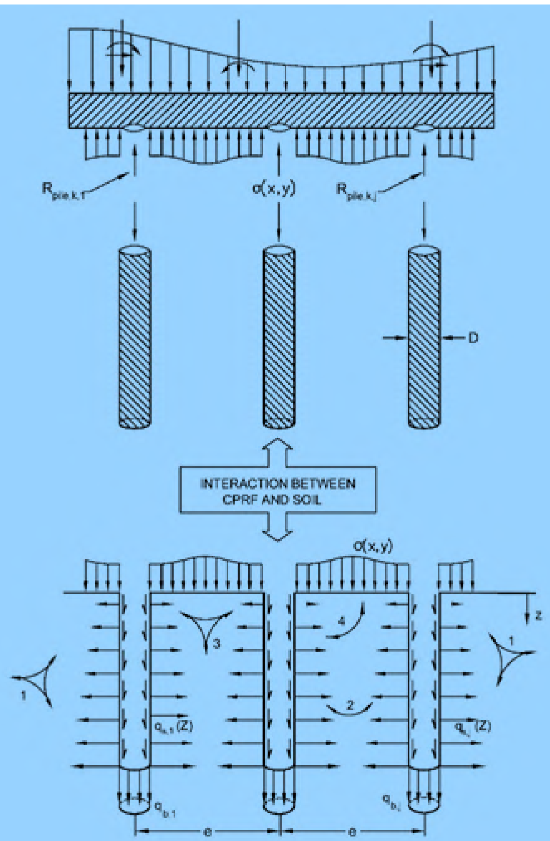


FIG. 1(A) ALONG RAFT LENGTH

FIG.1(B) ALONG RAFT WIDTH

FIG.1 SCHEMATIC REPRESENTATION OF CPRF SYSTEM AND VARIOUS PARAMETERS FOR DESIGN CONSIDERATION BASED ON SERVICEABILITY LIMIT STATE



Key

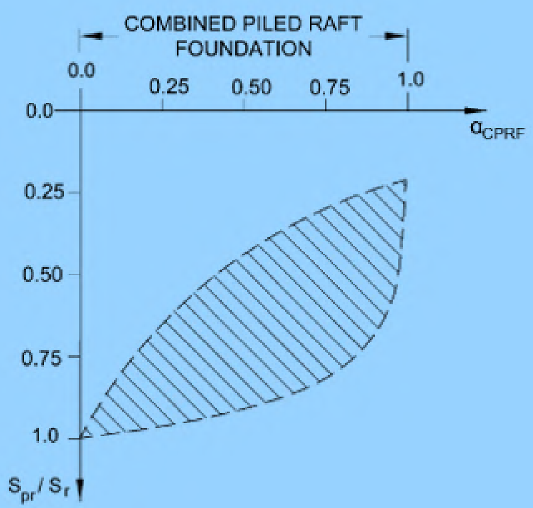
1 Pile-soil interaction

2 Pile-pile interaction

3 Raft-soil interaction

4 Pile-raft interaction

FIG. 2A SOIL-STRUCTURE INTERACTION



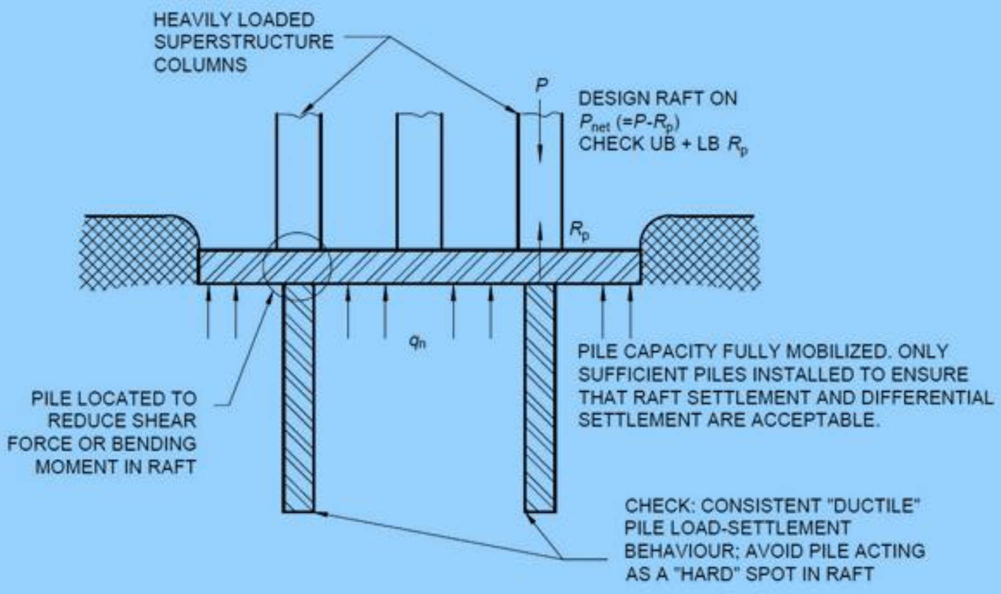
Key

S_{pr} Settlement of combined piled raft foundation

S_r Settlement of raft foundation

FIG. 2B TYPICAL VARIATION OF α_{CPRF} WITH SETTLEMENT RATIO

FIG. 2 SOIL-STRUCTURE INTERACTION AND VARIATION OF α_{CPRF} FOR ACPRF



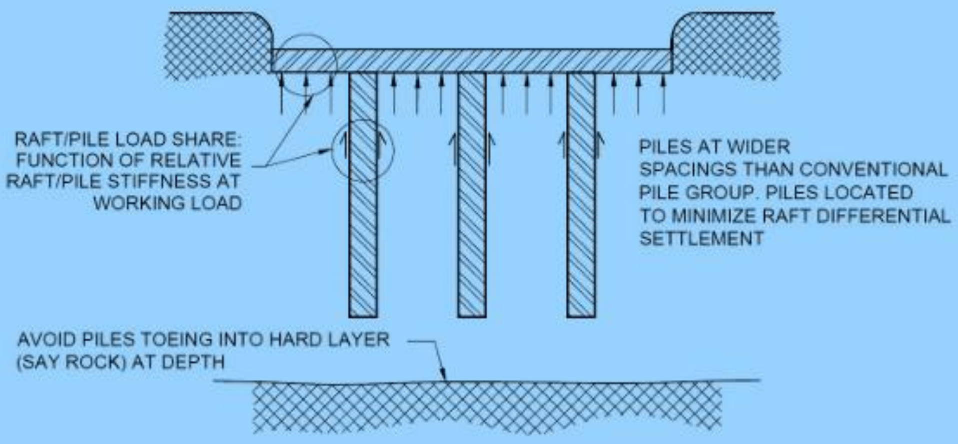
Key

q_n	Net contact stress on underside of raft
P	Column load
R_P	Pile capacity
UB	Upper bound
LB	Lower bound

Favourable ground conditions — Deep deposits of homogenous clays; typically stiff clays at raft level.

Typical design scenario — A raft which does not quite work, a small number of piles are added to resolve local non-compliances (for example, differential settlement, shear or bending moment).

FIG. 3A PILE-ENHANCED RAFT



Favourable ground conditions — Competent soils at raft level (stiff clays, dense sands) and depth; interbedded sands/clays.

Typical design scenario — A value-engineered pile group, often the raft was provided as a pile cap, but is then used to provide significant capacity.

FIG. 3B RAFT-ENHANCED PILE

FIG. 3 PILE-ENHANCED RAFT AND RAFT-ENHANCED PILE

3 SITE INVESTIGATION

3.1 General

3.1.1 Site investigation is essential in determining the physical, chemical and engineering properties of subsoil to arrive at the required foundation system. However, in areas which have already been developed, information may be obtained regarding the existing local knowledge, records of trial pits, boreholes, etc, in the vicinity and the behaviour of the existing structures, particularly those of a similar nature to those proposed. This information may be made use of for design of foundation of lightly loaded structures of not more than two storeys and also for deciding the scope of further investigation for other structures.

3.1.2 If the existing information is not sufficient or is inconclusive, the proposed site should be explored in detail as per good practice [C-2(1)], so as to obtain a knowledge of the type, uniformity, consistency, thickness, sequence and dip of the strata, hydrology of the area and also the engineering properties. In the case of lightly loaded structures of not more than two storeys the tests required to obtain the above information are optional, mainly depending on site conditions.

3.1.3 *Geotechnical Engineering Services*

Geotechnical engineering services encompass the entire process of understanding ground conditions and providing reliable inputs for civil engineering works. They include site reconnaissance, planning and execution of field and laboratory investigations and interpretation of soil and rock behaviour for safe and economical design. The services also define the roles of the Appointing Authority (AA), Geotechnical Consultant (GC) and Geotechnical Investigation Agency (GIA), along with structured models of appointment — Model 1, where GC and GIA are appointed separately and Model 2, where GC is appointed and may additionally act as GIA. These provisions cover both pre-construction and interlinked construction stage activities, ensuring that the design and execution of works are based on sound geotechnical data and professional judgement.

A clear distinction is laid down between geotechnical factual reports (GFR), which document the results of investigations and geotechnical interpretive reports (GIR), which provide engineering analysis and design recommendations. The framework specifies minimum qualifications and experience requirements for team leaders to promote competence and accountability, while also incorporating provisions on safety, quality assurance, quality control and declarations against conflict of interest. Together, these measures strengthen transparency, consistency and technical integrity in geotechnical practice. Reference may be made to the good practice [C-2(2)] for more detail.

3.2 Stages of Subsurface Investigation

3.2.1 The stages of subsurface investigation are:

a) site reconnaissance;

b) preliminary investigation;

c) detailed investigation;

d) construction stage investigation; and

e) post-construction stage investigation.

3.2.2 *Site Reconnaissance*

3.2.2.1 Site reconnaissance would help in deciding future programme of field investigations, that is, to assess the need for preliminary and/or detailed investigations.

This would also help in determining scope of work, methods of exploration to be adopted, *in-situ* tests to be carried out and administrative arrangements required for the investigation. Where detailed information on the geotechnical conditions is not available, an inspection of site and study of topographical features are helpful in getting information about soil, rock and ground-water conditions. Site reconnaissance includes a study of local topography, excavations, ravines, quarries, escarpments; evidence of erosion or landslides, behaviour of existing structures at or near the site; water level in streams, water courses and wells; flood marks; nature of vegetation; drainage pattern, location of seeps, springs and swamps, evidence of expansive soils. Information on some of these may be obtained from topographical maps, geological maps, pedological and soil survey maps, vicinity maps, metrological data and digitized images.

3.2.2.2 Data regarding removal of overburden by excavation, erosion or landslides should be obtained. This gives an idea of the amount of pre-consolidation the soil strata have undergone. Similarly, data regarding recent fills is also important to study the consolidation characteristics of the fill as well as the original strata.

3.2.2.3 The type of flora affords at times some indication of the nature of the soil. The extent of swamp and superficial deposits and peats will usually be obvious. In general, such indications, while worth noting, require to be confirmed by actual exploration.

3.2.2.4 *Ground-water conditions*

The ground-water level fluctuates and will depend upon the permeability of the strata and the head causing the water to flow. The water level in streams and water courses, if any, in the neighbourhood, should be noted, but it may be misleading to take this as an indication of the depth of the water table in the ground. Wells at the site or in the vicinity give useful indications of the ground-water conditions. Flood marks of rivers may indicate former highest water levels. Tidal fluctuations may be of importance. There is also a possibility of there being several water tables at different levels, separated by impermeable strata and some of this water may be subject to artesian head. It is also important to collect water samples to check its suitability for construction purpose and any treatment of foundation required for corrosion protection.

3.2.2.5 *Enquiries regarding earlier use of the site*

In certain cases, the earlier uses of the site may have a very important bearing on proposed new works. This is particularly so in areas, where there have been underground workings, such as worked-out ballast pits, quarries, old brick fields, coal mines and mineral workings. Enquiries should be made regarding the location of shafts and workings, particularly shallow ones, where there may be danger of collapse, if heavy new structures are superimposed.

The possibility of damage to sewers, conduits and drainage systems by subsidence should also be investigated.

3.2.3 *Preliminary Investigation*

The purpose of preliminary investigation is to assess the feasibility of a project. The preliminary investigation includes determination of approximate depth, thickness, extent and composition of sub-strata, ground water table and also to obtain approximate information regarding strength and compressibility of the various strata. The preliminary investigation comprises boreholes with standard penetration test (SPT), collection of disturbed samples (DS), undisturbed samples (UDS), dynamic cone penetration test (DCPT), trial pits and laboratory testing. Geophysical investigation may also be carried out during the preliminary investigation to establish the sequence of layers, their approximate properties, depth of bedrock, etc.

3.2.4 *Detailed Investigation*

Planning of detailed investigation is generally based on the information obtained in the preliminary investigation. The objective is to find the sequence of various layers, thickness of the layers, ground water table and detailed engineering properties of each layer. Detailed investigation involves *in-situ* (field) testing and laboratory testing. The *in-situ* tests are described in **3.4**. Laboratory testing on soil/rock/water samples collected from the site is carried out to determine the index, physical, engineering and chemical properties as given in **3.5.7**. The type and number of tests shall depend upon the nature of the project, geology of the area and site conditions.

3.2.5 *Construction Stage Investigation*

During construction, there may be a requirement of additional investigation to overcome the difficulties/modifying the foundation system due to change in subsoil/ground water conditions. The type of test required shall be selected from those specified under detailed investigation.

3.2.6 *Post-construction Stage Investigation*

At times, in order to study the behaviour/strengthening of substructure/foundation, investigation may be required.

3.3 Subsurface Exploration

The subsurface exploration includes the following:

a) Geophysical investigation;

b) Trial pits;

c) Subsurface soundings;

d) Boring in soil;

e) Core drilling in rock; and

f) Location and depth of investigation.

3.3.1 *Geophysical Investigation*

The geophysical investigation includes measurement of physical properties of the ground based on which the type of strata and depth of strata can be approximately evaluated. These tests provide a quick means of getting useful information about the stratification and underground utilities. This will enable to plan the depth/extent of the *in-situ* tests and type of test to be carried out *in-situ* and in the laboratory. The method provides only approximate parameters about the strata which need to be confirmed by conducting detailed investigation. These methods may be employed to get preliminary information on stratigraphy or complement a rationalized boring programme by correlation of stratigraphy between widely spaced boreholes. Commonly adopted geophysical tests are given in Table 1.

Table 1 Geophysical Tests

(*Clause* 3.3.1)

Sl No.	Type of Test	Property Measured
(1)	(2)	(3)
i)	Electrical resistivity test (ERT)	Electrical resistivity
ii)	Seismic refraction test (SRT)	Primary wave velocity
iii)	Cross hole seismic test (CHST) and down hole seismic test (DHST)	Shear wave velocity
iv)	Spectral analysis of surface wave (SASW) and multi-channel analysis of surface wave (MASW)	Primary/Shear wave velocity
v)	Ground penetrating radar survey (GPR survey)	Electromagnetic intensity (resistivity)
vi)	Thermal resistivity test	Thermal conductivity

3.3.2 *Trial Pit*

This method consists of excavating a pit at the site and thereby exposing the strata. This enables visual inspection of the strata, collection of disturbed/undisturbed samples and carrying out *in-situ* tests. Generally, trial pits are excavated up to 4 m depth below the ground level.

Minimum dimension of the pit shall be such that there is an adequate working space for excavation and carrying out sampling/testing. Generally, the pits have planned dimensions varying from 1 m × 1 m up to 3 m × 3 m depending on the depth required.

After the pit is excavated, the details of the strata shall be recorded with the description of each layer with thickness and ground water table, if any. After collecting the samples/in-situ testing, the pit shall be backfilled with the same excavated material. Arrangement shall be made for dewatering below ground water table. Precautions shall be taken to prevent surface water draining into the trial pit.

3.3.3 *Subsurface Soundings*

Subsurface soundings provide quick means to get continuous resistance against penetration indicating compactness of the strata. The following sounding tests are generally carried out:

a) Dynamic cone penetration test;

b) Static cone penetration test; and

c) Static cone penetration test using piezocone with pore water pressure measurement.

The limitation of soundings is that samples cannot be collected. In addition, strata having stones, gravels, boulders, etc and stiff formations cannot be penetrated by sounding device.

3.3.4 *Boring in Soil*

Various techniques are available for boring in soil depending on the substrata and depth of investigation. Commonly used boring methods are given below and described in **3.3.4.1** to **3.3.4.4**.

a) Auger boring;

b) Shell and auger boring;

c) Rotary boring; and

d) Percussion drilling.

In general, the diameter of the borehole is about 100 mm to 150 mm. In boreholes, in-situ testing like standard penetration test along with collection of disturbed and undisturbed samples at regular depth intervals and at change of strata is carried out. Other *in-situ* tests may also be carried out in the boreholes as per requirements.

In order to prevent caving in of soil during boring, stabilizing fluid like bentonite slurry/temporary steel casing or combination of both should be used.

The bottom of the borehole shall be cleaned prior to sampling/testing at different levels.

3.3.4.1 *Auger boring*

An auger is either power or hand operated with periodic removal of cuttings. The bore profile is prepared based on physical observation of the cut material removed from auger.

The process is slow in case of manual operation. Hand operated auger is generally used for boring up to a depth of 6 m. Casing is not used in this method. The method is convenient considering ease of mobilization. The auger is used in soil where borehole stands unsupported. Auger is penetrated into the ground for a depth of about 300 mm to 600 mm and the auger is withdrawn for removing the cuttings. The auger is then reintroduced to extend the borehole.

3.3.4.2 *Shell and auger boring*

Shell and auger boring is adopted for deep boreholes. The equipment consists of auger and a shell (bailer) for boring, connected with steel wire ropes/drilling rods which are lowered/raised with the help of tripod and mechanical winch. The casing is lowered in the borehole by rotating/pushing to the required depth.

The auger is generally used up to the ground water table level followed by shell.

The shell is raised/lowered repeatedly to cut the soil at the bottom and the cut material collected in the shell is removed. In stiff cohesive soil, it may be necessary to soak the borehole while boring above the ground water table. Where ground water is encountered, the water level in the borehole shall always be maintained at or above the water table.

The method is efficient to drill in hard strata but may disturb the soil at the bottom of the borehole. The method should be adopted with caution to minimize disturbances in the soil, prior to sampling/testing.

3.3.4.3 *Percussion drilling*

This method should only be used for drilling boreholes in bouldery and gravelly strata where other boring methods become ineffective.

In this method, the formation is pulverized/broken by repeated blows using a bit or a chisel. Water should be added at the time of boring and the cuttings are bailed out at intervals. The bit/chisel is suspended by a cable or rods from a hoist.

As this method involves repeated dropping of a heavy chisel and can disturb the strata considerably, it is preferable that at least 0.5 m above the level of sampling/testing, the use of percussion method shall be discontinued and the borehole advanced/cleaned with shell or any other suitable method till the sampling depth has reached.

3.3.4.4 *Rotary boring*

In this method, boring is advanced by the cutting action of a rotating bit which should be kept in firm contact with the bottom of the hole. The bit is connected at the end of hollow, jointed drill rods which are rotated by a suitable chuck. Water/stabilizing fluid like bentonite slurry is pumped continuously down the hollow drill rods and the fluid returns to the surface through the annular space between the rods and the side of the hole and the casing may not be generally required. This method can be used in all type of strata.

3.3.5 *Core Drilling in Rock*

Core drilling in rock shall be carried out with rotary hydraulic rig using diamond-tipped bit (*see* Note) as per good practice [C-2(3)]. The core drilling equipment shall conform to accepted standard [C-2(4)]. Drilling shall be carried out with NX/HX size diamond-tipped drill bits or impregnated diamond bit depending on the type of rock formation. Double tube core barrel with core lifter shall be used. The diamond bit is attached to double tube core barrel with reaming shell. The bit cuts an annular groove into the formation and the inner core formed by the groove enters into the inner barrel as a core sample. The circulating fluid is injected through the annular gap between the outer barrel and inner barrel and hence the fluid does not come in contact with the core received in the barrel. This prevents damage to the core by washing out of the loose particles in the joints (if any) and improves both core recovery and rock quality designation (RQD). After advancing the core barrel, the barrel is withdrawn, the rock cores are removed by detaching the core cutter, reassembled and introduced back in the borehole for further coring. Indexing and storage of drill cores shall be in accordance with good practice [C-2(5)].

NOTE — Tungsten carbide bit may be used for rocks like, shale, siltstone and claystone.

The drilling rig shall have necessary facilities to regulate the spindle speed, bit pressure and water pressure during core drilling to get good core recovery.

The rotational speed of the bit (spindle speed), the amount of downward pressure applied on the bit (bit pressure) and water pressure shall be suitably adjusted and properly monitored so that the core is collected with least disturbance and avoid shearing of the core from its base.

Drilling run should be 0.75 m in length. This can be increased up to 1.5 m run in case core recovery is more than 50 percent in two successive 0.75 m drill runs. If the core recovery is nil, then SPT shall be performed before commencing the next drill run.

If at any time a blocking of the bit or grinding of the core is indicated, the core barrel shall be immediately withdrawn from the borehole regardless of the length of drill run completed.

3.3.6 *Location and Depth of Investigation*

3.3.6.1 The location and depth of investigation points shall depend upon the type of structure/building considering variation in substrata based on the preliminary subsurface investigation or available geotechnical data in an already developed site. The investigation points should be arranged in such a pattern that the stratification can be assessed across the site. The guidelines as specified in **3.3.6.2** and **3.3.6.3** may be adopted for deciding the location and depth of boreholes/trial pits and other *in-situ* tests.

3.3.6.2 *Disposition of Boreholes/Trial Pits*

The guidelines for disposition of boreholes/trial pits are given in Table 2. In case significant variation in stratification is observed, additional boreholes/tests, as required may be carried out.

Table 2 Disposition of Borehole/Trial Pit

(*Clause* 3.3.6.2)

Sl No.	Type of Structure/Buildings	Location of Borehole/Trial Pit
(1)	(2)	(3)
i)	For lightly loaded residential building (such as single/double storeyed building)	At least one borehole/trial pit in the centre of the building (*see* Note)
ii)	For building(s) in a site covering an area of about 0.4 hectares	At least one borehole in each corner and one in the centre
iii)	a) For structure/building having any of the plan dimensions exceeding 50 m b) For multiple buildings in a site covering an area of more than 0.4 hectares c) For multistoried buildings and structure of height less than and equal to 50 m	Boreholes in a grid pattern with points at not more than 50 m distance, subject to a minimum of two boreholes. Boreholes should cover built-up areas.
iv)	For multistoried buildings and structure of height more than 50 m	Minimum three boreholes at each building/structure
v)	Linear structures (roads, railways, embankments, pipelines, boundary walls, tunnels, retaining walls, transmission lines, etc)	Boreholes at a spacing of 50 m to 500 m. In highly erratic substrata, spacing may be suitably reduced. In uniform substrata, the spacing may be suitably increased.
vi)	Solar power plant	One borehole for every 2 to 5 hectares subject to a minimum of 5 boreholes per site

NOTE — Boreholes are preferred over trial pits.

3.3.6.3 *Depth of Investigation*

3.3.6.3.1 The depth of investigation shall be adequate enough to provide necessary data for estimating safe bearing capacity and settlement. In general, the investigation shall be carried out to a minimum depth to which the increase in stress level due to the foundation load is lesser than 10 percent of the in-situ stress level.

The depth of investigation also depends on the risk induced by the presence of weaker formations below the influence zone to be stressed due to the proposed construction.

3.3.6.3.2 The following guidelines may be followed for adopting the depth of investigation:

a) *Shallow foundation* (*other than raft foundations*) *in soil* —The depth of investigation shall be up to 2 to 3 times the estimated width of the largest foundation below the expected founding level;

b) *Raft foundation* (*minimum foundation width equal to 6 m for the purpose of investigation*) *in soil* — In case of raft foundation, the depth of investigation shall be up to 1.0 to 2.0 times the width of the raft foundation below the expected founding level;

c) *Pile foundation in soil* — The depth of investigation shall be sufficiently below the expected founding level of piles but this should not be less than 5 m or 5 times the diameter of piles, whichever is more beyond the estimated pile termination level;

d) *Embankments* — The depth of investigation shall be 1.0 to 2.0 times the height of the embankment;

e) *Well foundation in soil* — The depth of investigation in the case of well foundation shall be 1.5 to 2.0 times the width/diameter of the well from the estimated termination level of the well; and

f) *Penetration into rock* — In case rock formation is encountered prior to the termination depth of boreholes as stipulated in (a) to (e) above, the boreholes may be extended by minimum 5 m in rock strata depending upon rock characteristics and project requirements.

NOTE — The depth of investigation may be suitably arrived as per the specific requirements for any other foundation/substructure.

3.3.6.4 *Other in-situ tests*

The location and depth of investigation for other *in-situ* tests shall be as per the requirements.

3.4 *In-Situ* Testing

In-situ tests provide information about the strength and deformability characteristics of the strata. Following are the *in-situ* tests which may be conducted depending on the nature of the strata and project requirements:

a) Standard penetration test (SPT);

b) Vane shear test (VST);

c) Pressuremeter test (PMT);

d) Plate load test (PLT);

e) Cyclic plate load test (CPLT);

f) Block vibration test (BVT);

g) Dynamic cone penetration test (DCPT);

h) Static cone penetration test (SCPT);

j) Piezocone with pore pressure measurement (CPTu);

k) Flat dilatometer test (DMT);

m) Electrical resistivity test (ERT);

n) Seismic refraction test (SRT);

p) Down hole, up-hole and cross hole seismic test (CHST);

q) Spectral analysis of surface wave (SASW) and multi-channel analysis of surface wave (MASW) test;

r) Ground penetrating radar (GPR) survey;

s) Thermal resistivity test;

t) *In-situ* permeability test;

u) Modulus of subgrade reaction;

v) Field california bearing ratio test (CBR); and

w) Optical and acoustic tele-viewer methods.

3.4.1 *Standard Penetration Test*

Standard penetration test (SPT) shall be conducted in all type of soil deposits to determine the resistance of soil at any depth in a bore hole and to recover disturbed soil sample for laboratory testing/identification purpose.

The basis of the test consists of driving a split spoon sampler by dropping a hammer of 63.50 kg ± 0.50 kg mass onto an anvil or drive head from a height of 750 mm ± 10 mm. The number of blows necessary to achieve 300 mm penetration of the sampler after the seating penetration of 150 mm is the penetration resistance. The test shall be conducted in accordance to good practice [C-2(6)]. In this test, the drive weight assemblies namely:

a) safety hammer with inner guide rod assembly and concealed anvil over which the hammer strikes;

b) do-nut hammer or safety hammer equipped with auto trip mechanism; and

c) automatic hammer system with the housing are preferred for consistent transfer of energy in each strike.

Hammer drop system used for lifting and dropping during the test shall be either mechanical auto-trip system or fully automated system in order to minimize human error. Rope and cat-head mechanism shall be used only when the other systems namely. mechanical auto trip system/fully automated system are not practicable. The depth interval between the bottom of the test and next undisturbed sampling/testing shall not be less than 500 mm.

3.4.2 *Vane Shear Test*

Field vane shear test shall be performed inside the borehole to determine the shear strength of cohesive soils, especially in soft clays which are highly susceptible to sampling disturbance, in accordance with good practice [C-2(7)].

3.4.3 *Pressuremeter Test*

The test is carried out to determine the *in-situ* stress-strain characteristics of the substrata. The test involves positioning of a radially expandable probe at the desired depth in a borehole and applying increments of pressure to inflate the probe thus applying increasing radial stresses on the substrata in the walls of the borehole and measuring the volumetric deformations at different pressure increments. Applied pressure *versus* volumetric deformation is plotted and analyzed. Further details regarding the test are given in good practice [C-2(8)].

3.4.4 *Plate Load Test*

Plate load test shall be conducted to determine the load/settlement characteristics of the substrata. The specifications for the equipment and accessories required for conducting the test, the test procedure, field observations and reporting of results shall conform to accepted standard [C-2(9)].

3.4.4.1 *Test for determination of modulus of subgrade reaction*

Modulus of subgrade reaction shall be determined in accordance with accepted standard [C-2(10)].

3.4.5 *Cyclic Plate Load Test*

This test shall be carried out to determine the dynamic soil properties required for the analysis of foundation subjected to dynamic loads. This test shall be conducted on similar lines as the ordinary plate load test. In addition, unloading shall also be done before each stage of loading. Test set up, load increment, maximum load intensity, recording of field data, etc shall be as per accepted standards [C-2(11)].

After each stage of loading, the load shall be removed in a minimum of two stages. After each stage of reduction of load, dial gauge readings and settlements shall be taken for at least one hour until the readings stabilize. There after the next loading stage shall commence. Analysis of test data shall be reported in accordance with accepted standard [C-2(12)].

3.4.6 *Block Vibration Test*

This test shall be carried out in accordance with accepted standard [C-2(12)] to determine the dynamic soil properties required for the analysis of foundation subjected to dynamic loads.

A test pit of suitable size depending upon size of block should be made. The size of the pit may be 3 m × 6 m at the bottom and a depth preferably equal to proposed depth of foundation. The test should be conducted above the ground water table. In case of rock, the test may be performed on the surface of rock bed itself. The bottom of the pit should be levelled. Side slopes of the pit should be stable. Analysis of test data shall be reported as per accepted standard [C-2(12)].

3.4.7 *Dynamic Cone Penetration Test* (*DCPT*)

Dynamic cone penetration test is a method of sub surface sounding, wherein a continuous record of soil resistance is obtained from of the ground to the depth below by driving a cone. Dynamic cone penetration test shall be conducted either as per accepted standard [C-2(13)] or as per [C-2(14)], as required. The test shall be conducted up to the required depth; however, in order to avoid damage to the equipment, driving may be stopped when the number of blows exceeds 35 mm for 100 mm penetration when the cone is driven dry {*see* accepted standard [C-2(13)]} and 20 mm for 100 mm penetration when the cone is penetrated by circulating slurry {*see* accepted standard [C-2(14)]}. The specification for the equipment and accessories required for performing this test, procedure, field observations and reporting of results shall be in accordance with accepted standard [C-2(13)] or as per [C-2(14)], as applicable. For gravelly deposits (gravel size 100 mm to 120 mm), DCPT may be conducted as per good practice [C-2(15)].

3.4.8 *Static Cone Penetration Test* (*SCPT*)

Static cone penetration test is conducted to record the continuous *in-situ* penetration resistance of soils below the ground level. The specification for the equipment and accessories required for performing the test procedure, field observations and reporting of results shall be in accordance with accepted standard [C-2(16)].

3.4.9 *Static Cone Penetration Test using Piezocone with Pore Water Pressure Measurement* (*CPTu*)

The test is used to record the continuous *in-situ* penetration resistance, frictional resistance along with pore water pressure during penetration at the level of the base of the cone. For CPTu, all measurements shall be made by sensors contained in penetrometer.

The cone shall be pushed into the soil at a constant rate of penetration.

Test performed using this test method provide a detailed record of cone penetration which is useful for evaluation of site stratigraphy, homogeneity, voids or cavities and other discontinuities. The use of friction sleeve and porewater pressure element can provide an estimate of preliminary soil classification and correlations with engineering properties of soils.

NOTE — The test procedure, field observation and reporting of result may be referred from international standards.

3.4.10 *Flat Dilatometer Test* (*DMT*)

The test is used to determine *in-situ* strength and deformation properties of soils. The test consists of inserting a blade-shaped steel probe vertically into the soil with a thin expandable circular steel membrane mounted flush on one face and determining, at selected depths or in a semi-continuous manner, the contact pressure exerted by the soil against the membrane when the membrane is flush with the blade and subsequently the pressure exerted when the central displacement of the membrane reaches 1.10 mm. Further details regarding the test are given in good practice [C-2(8)].

NOTE — The test procedure, field observation and reporting of results may be referred from international standards.

3.4.11 *Electrical Resistivity Test*

This test is carried out to assess the substrata profiles (that is, soil and rock profiles) over large areas at a relatively lower cost and can thus supplement the information obtained from detailed site investigations.

This test involves placing four electrodes in line along the ground surface and passing an alternating current through the outer two current electrodes and measuring the resulting potential drop between the two inner potential electrodes. From the resistance, calculated as the ratio of the potential drop to the current applied, the apparent resistivity of the substrata (that is, soil or rock strata) is determined by multiplying with applicable factors which are dependent on the geometry and spacings of the four electrodes. The electrode spacings are varied and the apparent resistivity of the substrata is determined for various electrode spacings.

Detailed analysis/interpretation of the plot of apparent resistivity *versus* electrode spacings is carried out to assess the number and type of substrata and their thicknesses.

The equipment and accessories required for carrying out the test, the test procedure, recording of observations, presentation and analysis of results shall be in accordance with good practice [C-2(17)].

Electrical resistivity tests are also carried out to determine the earth resistivity of the substrata for use in the design of earthing systems. The equipment and accessories required for carrying out the test for earth resistivity determination, the test procedure, recording of observations, presentation and analysis of results shall be in accordance with good practice [C-2(18)].

3.4.12 *Seismic Refraction Test*

This test is carried out to assess the substrata profiles (that is, soil and rock profiles) over large areas to supplement the information obtained from detailed site investigations.

This test involves the generation of seismic waves in the ground using a suitable seismic source and recording these seismic waves through geophones (that is, instruments which convert ground vibrations to electrical signals) located in a line along the ground surface at regular distance intervals. From the recorded waveforms, the arrival times of the seismic waves at different geophone locations are determined and a plot is then drawn of the geophone distances *versus* the corresponding arrival times. The refraction/reflection of the seismic waves at the interfaces of the various substrata leads to variations in the arrival times of the seismic waves at larger distances from the seismic source. The slopes of the lines connecting the data points on such a plot give the seismic wave velocities of the various substrata. The thicknesses of the various substrata so identified are determined by detailed analysis/interpretation of the plot which are based on the principles of refraction/reflection of the seismic waves.

The equipment and accessories required for carrying out the test, the test procedure, recording of observations, presentation and analysis of results shall be in accordance with good practice [C-2(19)].

3.4.13 *Downhole*, *Uphole and Crosshole Seismic Tests*

These tests are carried out to determine the seismic wave velocity profiles with depth and therefrom the profiles of dynamic modulus of rigidity (that is, shear modulus) and dynamic Young's modulus (that is, elastic modulus) with depth.

In the downhole seismic test, seismic waves are generated in the ground by a suitable seismic source at the ground surface near a borehole and the seismic waves so generated are recorded using geophones located at different depths within the borehole. On the other hand, in the uphole seismic test, seismic waves are generated at different depths within a borehole by a suitable seismic source and the seismic waves so generated are recorded using a geophone located at the ground surface near the borehole.

In the crosshole seismic test, seismic waves are generated at different depths within a borehole termed the source borehole, using a suitable seismic source and the seismic waves so generated are recorded using geophones installed at the corresponding depths in one or two other boreholes termed the receiver boreholes, located in a straight line with the source borehole at predefined borehole spacings.

In these tests, from the recorded waveforms, the seismic wave velocities are determined and therefrom the dynamic shear and elastic moduli are determined.

The equipment and accessories required for carrying out the downhole or uphole seismic test, the test procedure, recording of observations, presentation and analysis of results shall be in accordance with good practice [C-2(20)].

The equipment and accessories required for carrying out the crosshole seismic test, the test procedure, recording of observations, presentation and analysis of results shall be in accordance with good practice [C-2(21)].

3.4.14 *Spectral Analysis of Surface Wave* (*SASW*) *and Multi-Channel Analysis of Surface Wave* (*MASW*) *Test*

The tests are used for evaluation of substrata properties based on the generation of wave propagation phenomena like reflection, refraction and dispersion. Compression and surface wave velocities are measured. Further details regarding the test are given in the good practice [C-2(8)].

NOTE — The test procedure, field observation and reporting of results may be referred from international standards.

P-wave and S-wave velocities as obtained from above tests can also be used to get seismic tomography.

3.4.15 *Ground Penetrating Radar Survey*

Ground penetrating radar (GPR) is a tool to assess the presence of soil/rock strata and to identify presence of buried objects such as pipeline, conduit, concrete, etc. This works on the principle of reflection of the electromagnetic waves at the interface of two different materials. The depth of penetration below the ground surface can be increased by reducing the frequency.

3.4.16 *Thermal Resistivity Test*

Thermal resistivity test is conducted to determine the thermal conductivity of soil and soft rock in field using thermal needle probe procedure. Further details regarding the test are given in good practice [C-2(8)].

NOTE — The test procedure, field observation and reporting of results may be referred from international standards.

3.4.17 *In-situ Permeability Test (in Soil)*

In-situ permeability test is conducted to determine the water percolation capacity of the sub strata. This test is performed inside a borehole/trial pit at specified depths. The type of test shall be either pump-in or pump-out test depending on the subsoil and ground water conditions. Pump-in test is suitable for sub-strata with or without ground water; wherein pump-out test is conducted for sub-strata with ground water.

3.4.17.1 *Pump-in test*

Pump-in test is conducted in a borehole/trial pit by allowing water to percolate into the soil.

a) *Constant head method* (*in borehole*) — This test is preferred for soils with high permeability. The specification for the equipment and the procedure of testing shall be in accordance with good practice [C-2(22)];

b) *Falling head method* (*in borehole*) — This method is preferred for soils with low permeability. The specification for the equipment and the procedure of testing shall be in accordance with good practice [C-2(22)]; and

c) *Percolation test* (*in trial pit*) — Percolation test in trial pit shall be conducted as per good practice [C-2(23)].

3.4.17.2 *Pump-out test*

Pump-out test is conducted in a borehole by pumping out of water from sub-soil. The specification for the equipment and procedure for testing shall be in accordance with good practice [C-2(22)].

3.4.18 *In-situ Permeability Test* (*in Rock*)

In-situ permeability test is conducted inside the drilled hole by pumping in water under pressure to determine the percolation capacity of the rock stratum. The test may be conducted by single packer method or by double packer method. The specification for the equipment and the procedure of testing shall be in accordance with accepted standard [C-2(24)].

3.4.18.1 *Single packer method*

This method shall be adopted when the bottom elevation of the test section is the same as the bottom of the drill hole and where it is considered necessary to know the permeability value during drilling itself. This test shall be useful where the full length of the hole cannot stand encased or ungrouted. The packer shall be fixed at the top level of the test section, such that only the test section lies below the packer. Water shall then be pumped through a pipe into the test section under a required pressure and maintaining it till a constant quantity of water intake is observed. The amount of water percolating through the hole shall be recorded at every 5 min intervals. The test shall be repeated by increasing the pressure at regular intervals up to a pressure limit as specified in accordance to accepted standard [C-2(24)]. The details and observations during the test shall be suitably recorded in a proforma recommended in the accepted standard [C-2(24)].

3.4.18.2 *Double packer method*

This method shall be used when the permeability of an isolated section inside a drill hole is to be determined. Packers shall be fixed both at the top and bottom of the test section such that their spacing is exactly equal to the length of the test section.

3.4.19 *Field California Bearing Ratio* (*CBR*) *Test*

This test is carried out to determine the *in-situ* CBR value of the compacted subgrade on which the road pavement will be constructed so as to confirm that the *in-situ* CBR value is equal to or greater than the design CBR value for the road pavement. The test involves applying static loading to cause penetration of a cylindrical plunger of standard diameter into the compacted subgrade material and recording the applied static load to cause the required increments of penetration till the required maximum penetration of the plunger is achieved. Based on the plot of the recorded applied load and corresponding penetrations of the plunger, the CBR value is determined as the ratio of the applied test load to the predefined standard load for different standard penetrations of the plunger. The equipment and accessories required for carrying out the test, the test procedure, recording of observations and presentation of results shall be in accordance with accepted standard [C-2(25)].

3.4.20 *Optical and Acoustic Tele-Viewer Methods*

The optical and acoustic tele-viewers provide a continuous, 360°view of the borehole wall that allows rock mass discontinuities to be identified and characterized. Both devices can be oriented within the borehole so that the absolute orientation of features like bedding planes can be measured. An optical tele-viewer (OTV) uses a camera to record high-resolution images of the borehole wall and includes lights for illumination. An OTV is best suited for dry boreholes or boreholes filled with clear water.

Any conditions that produce cloudy or murky water or coatings on the borehole wall, limit the usefulness of the OTV. If good images are obtained, it is possible to identify locations and orientations of joints, bedding planes, foliations, faults, shears and other naturally occurring rock mass discontinuities.

The acoustic tele-viewer (ATV) uses ultrasound pulses from a rotating sensor in an open, fluid-filled borehole to record the amplitude and travel time of the signals reflected at the (high-impedance) interface between fluids and the borehole wall. Because an ATV uses ultrasound rather than visible light, the borehole fluid is not required to be clear. Rock mass discontinuities in the wall of the borehole will change the amplitude of the reflected acoustic wave compared to the surrounding material. The method does not work well in soil because of the lack of a high-impedance boundary between the fluid and soil. ATV surveys are used to provide information regarding locations and orientations of joints, bedding planes, faults, shears and other naturally occurring rock mass discontinuities.

3.5 Sampling and Testing

3.5.1 In order to identify the type of soil/rock and to determine the physical, chemical and engineering properties of sub strata, samples of soil, rock and water, as applicable are collected during the investigation. The samples are of the following two types:

a) *Disturbed sample* — The sample in which natural structure of the material is disturbed; and

b) *Undisturbed sample* — The sample in which the natural structure of the material is not disturbed.

3.5.2 The type of sample, quantity of sample and number of samples to be collected depends on the purpose of investigation and the nature of the strata condition. The interval at which the samples are collected depends on the type of strata variation of the profiles within the strata. Generally, undisturbed samples (UDS) are collected at depth interval of 3.0 m or change of strata, whichever is lower. The depth between UDS and SPT are generally staggered at 1.5 m so that at every 1.5 m depth either UDS is collected or SPT is conducted. In case of variations of profiles within a stratum, the samples shall be collected at a frequency of at least every 1.5 m to 2 m depth interval in a borehole. Wherever undisturbed sampling is not possible, such as in cohensionless soil or some utility service is encountered during the investigation, the same may be replaced by standard penetration test. In cohesive soils, when N value is above 32, undisturbed samples may be replaced by standard penetration test.

3.5.3 The following are the methods that are usually employed for sampling:

Nature of stratum	*Type of sample*	*Method of sampling*
Soil	Disturbed	Shovel, pick axe, etc Auger and shell boring Split spoon sampler
	Undisturbed	Block sample Tube sampler Piston sampler Core cutter sampler Continuous soil sampler
Rock	Disturbed	Blasting Chiseling
	Undisturbed	Coring

3.5.4 In cohesive soils, it is possible to collect undisturbed samples for examination and testing. In cohesionless soils below water table, undisturbed sampling is difficult with the samplers other than piston sampler.

3.5.5 Disturbed soil sample and undisturbed soil sample are to be obtained as per **3.5.5.1** and **3.5.5.2**, respectively.

3.5.5.1 *Disturbed soil sample*

Disturbed sample of soil is obtained during the course of excavation of trial pit, boring and after conducting standard penetration test. This sample can be used for testing of physical and chemical properties. The disturbed sample of clay is also useful for determining shear strength of remoulded sample.

3.5.5.2 *Undisturbed soil sample*

Undisturbed sample is obtained from pits and boreholes in such a manner that moisture content and structure do not get altered. This may be attained by careful protection, packing and by the use of a correctly designed sampler. Various methods of undisturbed sampling are as given in good practice [C-2(8)].

3.5.6 *Protection, Handling and Labelling of Samples*

Care should be taken in protecting, handling and subsequent transport of samples and in their full labelling, so that samples can be received in a fit state for examination and testing and can be correctly recognized as coming from a specified trial pit or boring.

3.5.7 *Examination and Testing of Samples*

3.5.7.1 The following tests shall be carried out in accordance with accepted standard [C-2(26)]:

a) *Test on disturbed/undisturbed soil samples*
   1) Visual and engineering classification;
   2) Sieve analysis and hydrometer analysis;

3) Atterberg limits (liquid limit, plastic limit and shrinkage limit);
4) Specific gravity;
5) Free swell index;
6) Light/Heavy compaction test (standard/modified proctor compaction test);
7) California bearing ratio;
8) Relative density (for sand); and
9) Laboratory permeability test (constant head/falling head).

b) *Test on undisturbed soil samples*
   1) Bulk density and moisture content;
   2) Unconfined compression test (on remoulded or undisturbed sample);
   3) Swelling pressure (on undisturbed and remoulded sample);
   4) Direct shear test (drained/undrained) (on remoulded or undisturbed sample);
   5) Triaxial compression tests (on undisturbed or remoulded sample):
      i) Unconsolidated undrained;
      ii) Consolidated undrained test with/without the measurement of pore water pressure; and
      iii) Consolidated drained;
   6) Consolidation test; and
   7) Vane shear test.

c) *Test on rock samples*
   1) Visual classification;
   2) Water absorption;
   3) Porosity;
   4) Density;
   5) Specific gravity;
   6) Hardness;
   7) Slake durability;
   8) Uniaxial compression test (both saturated and unsaturated);
   9) Point load strength index;
   10) Deformability test (both saturated and unsaturated);
   11) Tensile strength (Brazilian); and
   12) Triaxial compression test.

d) *Chemical analysis on subsoil and ground water* — Tests are conducted to determine the concentration of following chemicals:
   1) *p*H value;
   2) Sulphates (SO_3 and/or SO_4);
   3) Chlorides;
   4) Organic matter; and
   5) Any other chemicals harmful to the foundation material.

In case of ground water, colour, odour, turbidity and specific conductivity should also be determined.

3.6 Soil Investigation Report

3.6.1 *General*

On completion of the *in-situ* and laboratory tests, subsurface investigation report shall be prepared which should include, geological information, procedure adopted for investigation, field observations, laboratory test data, analysis, interpretation, recommendation, etc.

3.6.2 *Information to be Provided*

3.6.2.1 The report shall include but not be limited to the following:

a) Plan showing location and depth of *in-situ* tests;

b) Geological information of the area;

c) General topography of the site;

d) Past usage and historical data of the site, if available;

e) Borelog/trial pit log (*see* good practice [C-2(8)] for format of borelog);

f) *In-situ* test results;

g) Plot of standard penetration test (N values both uncorrected and corrected) with depth for identified areas;

h) Laboratory test results (*see* good practice [C-2(8)] for format for laboratory test results);

j) Summary of strata profile; and

k) In case of pressuremeter tests, the following should be furnished:

   1) Calibration record including, description of membrane and sheath on probes, dimension of thick walled cylinder, length of flexible tubing, calibration curves/chart, temperature, etc;

   2) Drilling record including, borehole number, method of making borehole, log of soil type and condition, depth of water in borehole, weather and temperature; and

   3) *Test record* — Type of test, date and time, depth of centre point of probe, volume readings at 30 s and 60 s elapsed time and corresponding pressure readings, notes on any deviation from standard test procedure.

3.6.2.2 Reporting of *in-situ* and laboratory test results should be in accordance with relevant Indian Standards.

3.6.3 *Analysis and Interpretation of Results*

The following shall be included in the report:

a) Analysis of test results;

b) Discussion and interpretation of results analyzed including basis of adopting design parameters for the substrata;

c) For shallow foundations:

   1) Net allowable bearing pressure considering shear as well as settlement criteria for isolated footing, continuous strip footing and raft foundation; and

   2) Modulus of subgrade reaction and modulus of elasticity of substrata.

d) For pile foundations: Safe load carrying capacity;

e) Dynamic soil properties such as shear wave velocity, dynamic shear modulus, Poisson’s ratio, damping ratio, as applicable;

f) Suitability of the soil for backfilling around foundations;

g) Susceptibility of sub-soil strata to liquefaction, if any, in the event of dynamic loading; and

h) Remedial measures against the presence of harmful chemicals in the sub-soil/ground water.

4 CLASSIFICATION AND IDENTIFICATION OF SOILS

The classification and identification of soils for engineering purposes shall be in accordance with accepted standard [C-2(27)].

5 MATERIALS

5.1 Cement, coarse aggregate, fine aggregate, lime, *Surkhi*, steel, timber and other materials go into the construction of foundations shall conform to the requirements of Part B 'Building Materials' of this NBCS.

5.2 Protection Against Deterioration of Materials

Where a foundation is to be in contact with soil, water or air, that is, in a condition conducive to the deterioration of the materials of the foundation, protective measures shall be taken to minimize the deterioration of the materials.

5.2.1 *Concrete*

The concrete used for construction shall be in accordance with Part C 'Structural Design/ Section 5 Structural Concrete' of this NBCS.

5.2.2 *Timber*

Where timber is exposed to soil, it shall be treated in accordance with good practice [C-2(28)].

6 GENERAL REQUIREMENTS FOR FOUNDATIONS/SUB-STRUCTURES FOR BUILDINGS

6.1 Types of Foundations

Types of foundations for buildings covered in this Section are:

a) *Shallow foundations* (*see* **7**)
   1) Pad or spread and strip foundations (*see* **7.3**);
   2) Raft foundations (*see* **7.4**);
   3) Ring foundations (*see* **7.5**); and
   4) Shell foundations (*see* **7.6**).

b) *Deep foundations*
   1) Pile Foundations
      i) Driven cast *in-situ* concrete piles (*see* **8.2**);
      ii) Bored cast *in-situ* concrete piles (*see* **8.2**);
      iii) Driven precast concrete piles (*see* **8.3**);
      iv) Precast concrete piles in prebored holes (*see* **8.4**);
      v) Under-reamed concrete piles (*see* **8.5**);
      vi) Timber piles (*see* **8.6**); and
      vii) Spun piles.

      NOTE — Spun piles are used in deep marshy soils as conventional pile installation beyond 50 m in such soils is very difficult.

c) *Other foundations/Sub-structures/Foundations for special structure*
   1) Pier foundations (*see* **10.1**);
   2) Combined piled-raft foundation (*see* **9**);
   3) Diaphragm walls (*see* **10.2**); and
   4) Machine foundations (*see* **10.2**).

6.2 Depth of Foundations

6.2.1 The depth to which foundations should be carried depends upon the following principal factors:

a) Getting adequate allowable bearing capacity;

b) In the case of clayey soils, penetration below the zone where shrinkage and swelling due to seasonal weather changes and due to trees and shrubs are likely to cause appreciable movements;

c) In fine sands and silts, penetration below the depth of frost penetration;

d) Maximum depth of scour, wherever relevant, should also be considered and the foundation should be located sufficiently below this depth; and

e) Other factors such as ground movements and heat transmitted from the building to the supporting ground may be important.

6.2.2 All foundations shall extend to a depth of at least 500 mm below natural ground level. On rock or such other weather resisting natural ground, removal of the top soil may be all that is required. In such cases, the surface shall be cleaned and, if necessary, stepped or otherwise prepared so as to provide a suitable bearing and thus prevent slipping or other unwanted movements.

6.2.3 Where there is excavation, ditch, pond, water course, filled-up ground or similar condition of the subsoil on which the structure is to be erected, the foundation of such structure shall either be carried down to a depth beyond the detrimental influence of such conditions, or retaining walls shall be provided for the purpose of shielding from the effects of such conditions.

6.2.4 A foundation in any type of soil shall be below the zone significantly weakened by root holes or cavities produced by burrowing animals or works. The depth shall also be enough to prevent the rainwater scouring below the footings.

6.2.5 Clayey soils, like black cotton soils, are seasonally affected by drying, shrinkage and cracking in dry and hot weather and by swelling in the following wet weather to a depth which will vary according to the nature of the clay and the climatic condition of the region. It is necessary in these soils, either to place the foundation bearing at such a depth where the effects of seasonal changes are not important or to make the foundation capable of eliminating the undesirable effects due to relative movement by providing flexible type of construction or rigid foundations. Adequate load counteracting against swelling pressures also provide satisfactory foundations.

6.3 Foundation at Different Levels

6.3.1 Where footings are adjacent to sloping ground or where the bottoms of the footings of a structure are at different levels or at levels different from those of the footings of adjoining structures, the depth of the footings shall be such that the difference in footing elevations shall be subject to the following limitations:

a) When the ground surface slopes downward adjacent to a footing, the sloping surface shall not intersect a frustum of bearing material under the footing having sides which make an angle of 30° with the horizontal for soil and horizontal distance from the lower edge of the footing to the sloping surface shall be at least 600 mm for rock and 900 mm for soil (*see* Fig. 4);

b) In the case of footings in granular soil, a line drawn between the lower adjacent edges of adjacent footings shall not have a steeper slope than one vertical to two horizontal (*see* Fig. 5); and

c) In case of footing in clayey soils a line drawn between the lower adjacent edge of the upper footing and the upper adjacent edge of lower footing shall not have a steeper slope than one vertical to two horizontal (*see* Fig. 6).

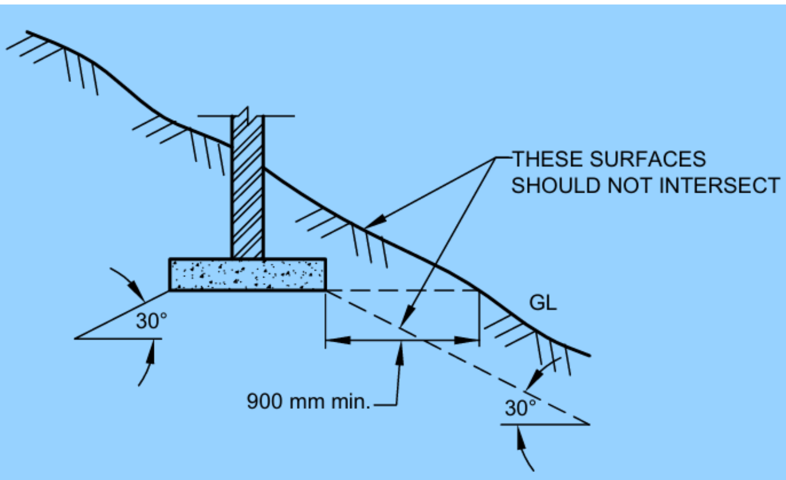


FIG. 4 FOOTING IN SLOPING GROUND

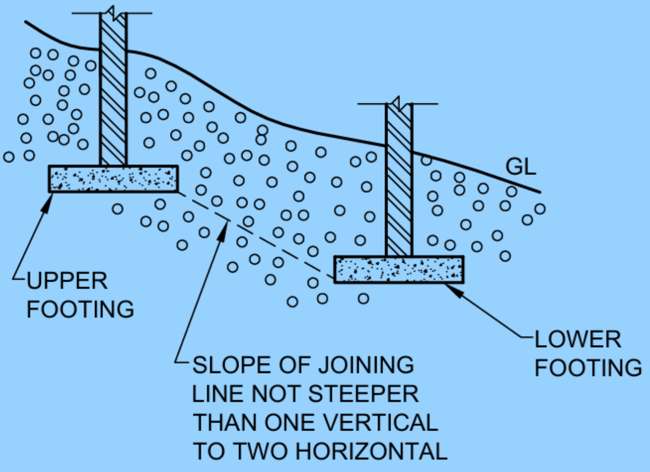


FIG.5 FOOTING IN GRANULAR SOIL

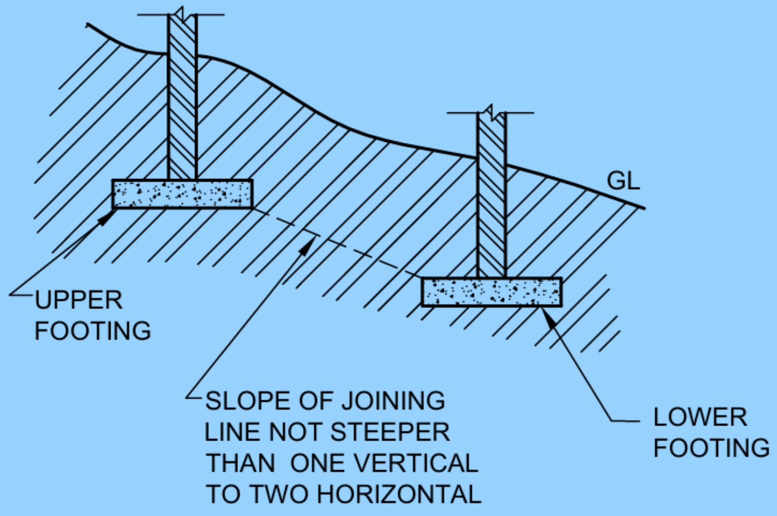


FIG.6 FOOTING IN CLAYEY SOIL

6.3.2 The requirement given in 6.3.1 shall not apply in the condition where adequate provision is made for the lateral support (such as with retaining walls) of the material supporting the higher footing.

6.4 Effect of Seasonal Weather Changes

During periods of hot, dry weather a deficiency of water develops near the ground surface and in clay soils, that is associated with a decrease of volume or ground shrinkage and the development of cracks. The shrinkage of clay will be increased by drying effect produced by fast growing and water seeking trees. The range of influence depends on size and number of trees and it increase during dry periods. In general, it is desirable that there shall be a distance of at least 6 m between such trees. Boiler installations, furnaces, kilns, underground cables and refrigeration installations and other artificial sources of heat may also cause increased volume changes of clay by drying out the ground beneath them, the drying out can be to a substantial depth. Special precautions either in the form of insulation or otherwise should be taken. In periods of wet weather, clay soils swell and the cracks lend to close, the water deficiency developed in the previous dry periods may be partially replenished and a subsurface zone or zones deficient in water may persist for many years. Leakage from water mains and underground sewers may also result in large volume changes. Therefore, special care shall be taken to prevent such leakages.

6.5 Effect of Mass Movements of Ground in Unstable Areas

In certain areas mass movements of the ground are liable to occur from causes independent of the loads applied by the foundations of structures. These include mining subsidence, landslides on unstable slopes and creep on clay slopes.

6.5.1 *Mining Subsidence*

In mining areas, subsidence of the ground beneath a building or any other structure is liable to occur. The magnitude of the movement and its distribution over the area are likely to be uncertain and attention shall, therefore, be directed to make the foundations and structures sufficiently rigid and strong to withstand the probable worst loading condition and probable ground movements. Where future subsidence is likely, care should be taken to design the superstructure and foundation sufficiently strong to cater for probable ground movements. Long continuous buildings should be avoided in such areas and large building in such area should be divided into independent sections of suitable size, each with its own foundations. Expert advice from appropriate mining authority should be sought.

NOTE — For prediction of subsidence in coal mines, guidelines as given in the accepted standard [C-2(29)] may be referred.

6.5.2 *Landslide Prone Areas*

The construction of structures on slopes which are suspected of being unstable and are subject to landslip shall be avoided.

On sloping ground on clay soils, there is always a tendency for the upper layers of soil to move downhill, depending on type of soil, the angle of slope, climatic conditions, etc. In some cases, the uneven surface of the slope on a virgin ground will indicate that the area is subject to small land slips and, therefore, if used for foundation, will obviously necessitate special design consideration as specified in accepted standard [C-2(30)].

Where there may be creep of the surface layer of the soil, protection against creep may be obtained by following special design considerations.

On sloping sites, spread foundations shall be on a horizontal bearing and stepped. At all changes of levels, they shall be lapped at the steps for a distance at least equal to the thickness of the foundation or twice the height of the step, whichever is greater. The steps shall not be of greater height than the thickness of the foundation, unless special precautions are taken.

Cuttings, excavations or sloping ground near and below foundation level may increase the possibility of shear failure of the soil. The foundation shall be well beyond the zone of such shear failure.

If the probable failure surface intersects a retaining wall or other revetment, the latter shall be made strong enough to resist any unbalanced thrust. In case of doubt as to the suitability of the natural slopes or cuttings, the structure shall be kept well away from the top of the slopes, or the slopes shall be stabilized.

Cuttings and excavations adjoining foundations reduce stability and increase the likelihood of differential settlement. Their effect should be investigated not only when they exist but also when there is possibility that they are made subsequently.

Where a structure is to be placed on sloping ground, additional complications are introduced. The ground itself, particularly if of clay, may be subject to creep or other forms of instability, which may be enhanced if the strata dip in the same direction as the ground surface. If the slope of the ground is large, the overall stability of the slope and substructure may be affected. These aspects should be carefully investigated.

6.6 Precautions for Foundations on Inclined Strata

In the case of inclined strata, if they dip towards a cutting of basement, it may be necessary to carry foundation below the possible slip planes, land drainage also requires special consideration, particularly on the uphill side of a structure to divert the natural flow of water away from the foundations.

6.7 Strata of Varying Thickness

Strata of varying thickness, even at appreciable depth, may increase differential settlement. Where necessary, calculations should be made of the estimated settlement from different thickness of strata and the structure should be designed accordingly. When there is large change of thickness of soft strata, the stability of foundation may be affected. Site investigations should, therefore, ensure detection of significant variations in strata thickness.

6.8 Layers of Softer Material

Some soils and rocks have thin layers of softer material between layers of harder material, which may not be detected unless thorough investigation is carried out. The softer layers may undergo marked changes in properties if the loading on them is increased or decreased by the proposed construction or affected by related changes in ground water conditions. These should be taken into account.

6.9 Spacing Between Existing and New Foundation

The deeper the new foundation and the nearer to the existing it is located, the greater the damage is likely to be. Preferably, the minimum horizontal spacing between existing and new footings shall be equal to the width of the wider clear footing. While the adoption of such provision shall help minimizing damage to adjacent foundation, an analysis of bearing capacity and settlement shall be carried out to have an appreciation of the effect on the adjacent existing foundation. Suitable site-specific protection measures shall be adopted for the existing foundation, as required.

6.10 Preliminary Work for Construction

6.10.1 The construction of access roads, main sewers and drains should preferably be completed before commencing the work of foundations; alternatively, sufficient precautions shall be taken to protect the already constructed foundations during subsequent work.

6.10.2 *Clearance of Site*

Any obstacles, including the stump of trees, likely to interfere with the work shall be removed. Holes left by digging, such as those due to removal of old foundation, uprooted trees, burrowing by animals, etc, shall be back-filled with soil and well compacted.

6.10.3 *Drainage*

If the site of a structure is such that surface water drain towards it, land may be dressed or drains laid to divert the water away from the site.

6.10.4 *Setting Out*

Generally, the site shall be levelled before the layout of foundations are set out. In case of sloping terrain, care shall be taken to ensure that the dimensions on plans are set out correctly in one or more horizontal planes.

6.10.5 The layout of foundations shall be set out in accordance of good practice [C-2(31)]. The setting out of walls shall be facilitated by permanent row of pillars, parallel to and at a suitable distance beyond the periphery of the building. The pillars shall be located at junctions of cross walls with the peripheral line of pillars. The centre lines of the cross walls shall be extended to and permanently erected on the plastered tops of the corresponding sets of pillars. The datum lines parallel to and at the known fixed distance from the centre lines of the external walls also be permanently worked on the corresponding rows of pillars to pillars to serve as checks on the accuracy of the work as it proceeds. The tops of these pillars shall be at the same level and preferably at the plinth or floor level. The pillars shall be of sizes not less than 250 mm wide and shall be bedded sufficiently deep into ground, so that they are not disturbed.

6.11 Protection of Excavation

6.11.1 The protection of excavation during construction shall be done in accordance with good practice [C-2(32)].

6.11.2 After excavation, the bottom of the excavation shall be cleared of all loose soil and rubbish and shall be levelled, where necessary. The bed shall be wetted and compacted by heavy rammers to an even surface.

6.11.3 Excavation in clay or other soils that are liable to be effected by exposure to atmosphere shall, wherever possible, be concreted as soon as they are dug. Alternatively, the bottom of the excavation shall be protected immediately by 80 mm thick layer of cement concrete not leaner than mix 1 : 5 : 10 over which shall come the foundation concrete; or in order to obtain a dry hard bottom, the last excavation of about 100 mm shall be removed only before concreting.

6.11.4 The backfilling of the excavation shall be done with care so as not to disturb the constructed foundation and shall be compacted in layers not exceeding 150 mm thick with sprinkling of minimum quantity of water necessary for proper compaction.

6.11.5 To support excavation sides where open excavation is required beyond a certain depth and safe slopes are not feasible, shoring shall be provided. This ensures stability, allows near-vertical soil cutting and prevents soil collapse. Shoring systems can include concrete shoring pile walls (secant or tangent wall systems), soldier pile walls, diaphragm walls, or other suitable methods, depending on soil conditions, excavation depth and selected construction technology.

Temporary shoring is generally defined as a system intended for service for a limited period. If earth retention structures are required for an extended duration, they should be designed as permanent structures.
For temporary shoring, the design should consider soil pressure and ensure that the selected system, including piles and anchors (if provided), can safely support the applied loads under service conditions.

The type of earth pressure to be considered depends on the flexibility and support conditions of the shoring system. Cantilevered or flexible walls with limited support require a specific earth pressure approach, whereas more rigid walls that exhibit minimal deflection require a different consideration.

For permanent conditions, all final loads, including surcharge effects, must be resisted by the structural walls, designed with appropriate safety considerations. The structural design should ensure:

a) deflections remain within allowable limits; and
b) structural elements maintain adequate strength and stability.

6.12 Alterations During Construction

a) Where during construction the soil or rock to which foundation is to transfer loads is found not to be the type or in the condition assumed, the foundation shall be redesigned and constructed for the existing type or conditions and the Authority notified;
b) Where a foundation bears on gravel, sand or silt and where the highest level of the ground water is or likely to be higher than an elevation defined by bearing surface minus the width of the footing, the bearing pressure shall be suitably altered;
c) Where the foundation has not been placed or located as indicated earlier or is damaged or bears on a soil whose properties may be adversely changed by climatic and construction conditions, the error shall be

corrected, the damaged portion repaired or the design capacity of the affected foundation recalculated to the satisfaction of the Authority; and

d) Where a foundation is placed and if the results of a load test so indicate, the design of the foundation shall be modified to ensure structural stability of the same.

7 SHALLOW FOUNDATIONS

7.1 Design Information

For the satisfactory design of foundations, the following information is necessary:

a) Type and condition of the soil or rock to which the foundation transfers the loads;

b) General layout of the columns and load-bearing walls showing the estimated loads, including moments and torques due to various loads (dead load, imposed load, wind load, seismic load) coming on the foundation units;

c) Allowable bearing pressure of the soils;

d) Changes in ground water level, drainage and flooding conditions and also the chemical conditions of the subsoil water;

e) Behaviour of the buildings, topography and environment/surroundings adjacent to the site, the type and depths of foundations and the bearing pressure; and

f) Seismic zone of the region.

7.2 Design Considerations

7.2.1 *Design Loads*

The foundation shall be proportioned for the following combination of loads:

a) Dead load + Imposed load; and

b) Dead load + Imposed load + Wind load or seismic loads, whichever is critical.

For details, reference shall be made to Part C 'Structural Design/ Section 1 Loads and Load Effects' of this NBCS.

NOTE — For shallow foundations on coarse grained soils, settlements shall be estimated corresponding to **7.2.1**(b) and for foundations on fine grained soils, the settlement shall be estimated corresponding to permanent loads only. Permanent loads shall be in accordance with good practice [C-2(33)].

7.2.2 *Allowable Bearing Pressure*

The allowable bearing pressure shall be taken as either of the following, whichever is less:

a) the safe bearing capacity on the basis of shear strength characteristics of soil; and

b) the bearing pressure that the soil can take without exceeding the permissible settlement (*see* **7.2.3**).

7.2.2.1 *Bearing capacity by calculation*

Where the engineering properties of the soil are available, that is, cohesion, angle of internal friction, density, etc, the bearing capacity shall be calculated from stability considerations of shear; factor of safety of 2.5 shall be adopted for safe bearing capacity. The effect of interference of different foundations should be taken into account. The procedure for determining the ultimate bearing capacity and bearing pressure of shallow foundations based on shear and allowable settlement criteria shall be in accordance with good practice [C-2(34)]. Depth factor correction is to be applied only when backfilling is done with proper compaction.

7.2.2.2 *Field method for determining allowable bearing pressure*

Where appropriate, plate load tests can be performed and allowable pressure determined as per accepted standard [C-2(9)]. The allowable bearing pressure for sandy soils may also be obtained by loading tests. When such tests cannot be done, the allowable bearing pressure for sands may be determined using penetration test.

7.2.2.3 Where the bearing materials directly under a foundation over-lie a stratum having smaller safe bearing capacity, these smaller values shall not be exceeded at the level of such stratum.

7.2.2.4 *Effect of wind and seismic force*

7.2.2.4.1 Where the bearing pressure due to wind is less than 25 percent of that due to dead and imposed loads, it may be neglected in design. Where this exceeds 25 percent, foundations may be so proportioned that the pressure due to combined dead, imposed and wind loads does not exceed the allowable bearing pressure by more than 25 percent.

7.2.2.4.2 When earthquake forces are considered for the computation of design loads, the permissible increase in bearing pressure of pertaining soil shall be as given in Part C 'Structural Design/Section 1 Loads and Load Effects'of this NBCS, depending upon the type of foundation of the structure and the type of soil.

7.2.2.5 *Bearing capacity of buried strata*

If the base of a foundation is close enough to a strata of lower bearing capacity, the latter may fail due to excess pressure transmitted to it from above. Care should be taken to see that the pressure transmitted to the lower strata is within the prescribed safe limits. When the footings are closely spaced, the pressure transmitted to the underlying soil will overlap. In such cases, the pressure in the overlapped zones will have to be considered. With normal foundations, it is sufficiently accurate to estimate the bearing pressure on the underlying layers by assuming the load to be spread at a slope of 2 (vertical) to 1 (horizontal).

7.2.3 *Settlement*

The permissible values of total and differential settlement for a given type of structure may be taken as given in Table 3. Total settlements of foundation due to net imposed loads shall be estimated in accordance with accepted standard [C-2(6)]. The following causes responsible for producing the settlement shall be investigated and taken into account.

a) *Causes of Settlement*:

   1) Elastic compression of the foundation and the underlying soil;

   2) Consolidation including secondary compression;

   3) *Ground water lowering* — Specially repeated lowering and raising of water level in loose granular soils tend to compact the soil and cause settlement of the footings. Prolonged lowering of the water table in fine grained soils may introduce settlement because of the extrusion of water from the voids. Pumping water or draining water by tiles or pipes from granular soils without an adequate mat of filter material as protection may, in a period of time, carry a sufficient amount of fine particles away from the soil and cause settlement;

   4) Seasonal swelling and shrinkage of expansive clays;

   5) Ground movement on earth slope, for example, surface erosion, slow creep or landslides; and

   6) Other causes, such as adjacent excavation, mining, subsidence and underground erosion.

b) *Causes of Differential Settlements*:

   1) Geological and physical non-uniformity or anomalies in type, structure, thickness and density of the soil medium (pockets of sand in clay, clay lenses in sand, wedge like soil strata, that is, lenses in soil), an admixture of organic matter, peat, mud;

   2) Non-uniform pressure distribution from foundation to the soil due to non-uniform loading and incomplete loading of the foundations;

   3) Water regime at the construction site,

   4) Overstressing of soil at adjacent site by heavy structures built next to light ones;

   5) Overlap of stress distribution in soil from adjoining structures;

   6) Unequal expansion of the soil due to excavation for footing;

7) Non-uniform development of extrusion settlements; and

8) Non-uniform structural disruptions or disturbance of soil due to freezing and thawing, swelling and softening and drying of soils.

7.2.4 *Shallow Foundations on Rocks*

Estimation of the safe bearing pressures of rocks for shallow foundations based on strength, allowable settlement and classification criteria; and also design and construction of shallow foundations on rocks shall be carried out in accordance with the good practice [C-2(35)].

7.3 Pad or Spread and Strip Foundations

7.3.1 In such type of foundations, wherever the resultant of the load deviates from the centre line by more than 1/6 of its least dimension at the base of footing, it should be suitably reinforced.

7.3.2 For continuous wall foundations (plain or reinforced) adequate reinforcement should be provided particularly at places where there is abrupt change in magnitude of load or variation in ground support.

7.3.3 On sloping sites the foundation should have a horizontal bearing and stepped and lapped at changes of levels for a distance at least equal to the thickness of foundation or twice the height of step, whichever is greater. The steps should not be of greater height than thickness of the foundations.

7.3.4 *Ground Beams*

The foundation can also have the ground beam for transmitting the load. The ground beam carrying a load bearing wall should be designed to act with the wall forming a composite beam, when both are of reinforced concrete and structurally connected by reinforcement. The ground beam of reinforced concrete structurally connected to reinforced brick work can also be used.

7.3.5 *Dimensions of Foundation*

The dimensions of the foundation in plan should be such as to support loads as given in good practice [C-2(33)]. The width of the footings shall be such that maximum stress in the concrete or masonry is within the permissible limits. The width of wall foundation (in mm) shall not be less than that given by:

$$B = W + 300$$

where

B = width at base, in mm; and

W = width of supported wall, in mm.

7.3.6 In the base of foundations for masonry foundation it is preferable to have the steps in multiples of thickness of masonry unit.

7.3.7 The plan dimensions of excavation for foundations should be wide enough to ensure safe and efficient working in accordance with good practice [C-2(32)].

7.3.8 Unreinforced foundation may be of concrete or masonry (stone or brick) provided that angular spread of load from the base of column/wall or bed plate to the outer edge of the ground bearing is not more than 1 vertical to 1/2 horizontal for masonry or 1 vertical to 1 horizontal for cement concrete and 1 vertical to 2/3 horizontal for lime concrete. The minimum thickness of the foundation of the edge should not be less than 150 mm. In case the depth to transfer the load to the ground bearing is less than the permissible angle of spread, the foundations should be reinforced.

7.3.9 If the bottom of a pier is to be belled so as to increase its load carrying capacity such bell should be at least 300 mm thick at its edge. The sides should be sloped at an angle of not less than 45° with the horizontal. The least dimension should be 600 mm (circular, square or rectangular). The design should allow for the vertical tilt of the pier by 1 percent of its height.

7.3.10 If the allowable bearing capacity is available only at a greater depth, the foundation can be rested at a higher level for economic considerations and the difference in level between the base of foundation and the depth at which the allowable bearing capacity occurs can be filled up with either: (a) concrete of allowable compressive strength not less than the allowable bearing pressure, (b) incompressible fill material, for example, sand, gravel, etc, in which case the width of the fill should be more than the width of the foundation by an extent of dispersion of load from the base of the foundation on either side at the rate of 2 vertical to 1 horizontal.

7.3.11 The cement concrete foundation (plain or reinforced) should be designed in accordance with Part C 'Structural Design/ Section 5 Structural Concrete' of this NBCS and masonry foundation in accordance with Part C 'Structural Design/ Section 4 Masonry' of this NBCS.

7.3.12 *Thickness of Footing*

The thickness of different types of footings, if not designed according to **7.2**, should be as given in Table 4.

Table 3 Permissible Differential Settlements and Tilt (Angular Distortion) for Shallow Foundations in Soils

(*Clause* 7.2.3)

Sl No.	Type of Structure	Isolated Foundations						Raft Foundations					
		Sand and Hard Clay			Plastic Clay			Sand and Hard Clay			Plastic Clay		
		Maximum Settlement mm	Differential Settlement mm	Angular Distortion	Maximum Settlement mm	Differential Settlement mm	Angular Distortion	Maximum Settlement mm	Differential Settlement mm	Angular Distortion	Maximum Settlement mm	Differential Settlement mm	Angular Distortion
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
i)	For steel structure	50	.003 3 *L*	1/300	50	.003 3 *L*	1/300	75	.003 3 *L*	1/300	100	.003 3 *L*	1/300
ii)	For reinforced concrete structures	50	.001 5 *L*	1/666	75	.001 5 *L*	1/666	75	.002 1 *L*	1/500	125	.002 *L*	1/500
iii)	For multistoreyed buildings												
	a) RC or steel framed buildings with panel walls	60	.002 *L*	1/500	75	.002 *L*	1/500	75	.0025 *L*	1/400	125	.003 3 L	1/300
	b) For load bearing walls												
	1) *L*/H=2[1)]	60	.000 2 *L*	1/5 000	60	.000 2 *L*	1/5 000	← Not likely to be encountered →					
	2) *L*/H=7[1)]	60	.000 4 *L*	½ 500	60	.000 4 *L*	1/2 500						
iv)	For water towers and silos	50	.001 5 *L*	1/666	75	.001 5 *L*	1/666	100	.0025*L*	1/400	125	.002 5 *L*	1/400

NOTES

1 The values given in the table may be taken only as a guide and the permissible total settlement/differential settlement and tilt (angular distortion) in each case should be decided as per requirements of the

2 *L* denotes the length of deflected part of wall/raft or centre-to-centre distance between columns.

3 *H* denotes the height of wall from foundation footing.

1) For intermediate ratios of *L*/*H*, the values can be interpolated.

Table 4 Thickness of Footings

(*Clause* 7.3.12)

Sl No. (1)	Type of Footings (2)	Thickness of Footings, *Min* (3)	Remarks (4)
i)	Masonry	a) 250 mm; and b) Twice the maximum projection from the face of the wall.	Select the greater of the two values
ii)	Plain concrete		
	a) For normal structures	a) 200 mm;	—
		b) Twice the maximum offset in a stepped footing; and	For footings resting on soil
		c) 300 mm.	For footings resting on top of the pile
	b) For lightly loaded structures	a) 150 mm; and	Resting on soil
		b) 200 mm.	Resting on pile
iii)	Reinforced concrete	a) 150 mm; and	Resting on soil
		b) 300 mm.	Resting on pile

7.3.13 *Land Slip Area*

On a sloping site, spread foundation shall be on a horizontal bearing and stepped. At all changes of levels, they shall be lapped at the steps for a distance at least equal to the thickness of the foundation or twice the height of the step, whichever is greater. The steps shall not be of greater height than the thickness of the foundation unless special precautions are taken. On sloping ground on clay soils, there is always a tendency for the upper layers of soil to move downhill, depending on type of soil, the angle of slope, climatic conditions, etc. Special precautions are necessary to avoid such a failure.

7.3.14 In the foundations, the cover to the reinforcement shall be as prescribed in Part C 'Structural Design/ Section 5 Structural Concrete' of this NBCS for the applicable environment exposure condition.

7.3.15 For detailed information regarding preparation of ground work, reference shall be made to good practice [C-2(36)].

7.4 Raft Foundations

7.4.1 *Design Considerations*

Design provisions given in 7.2 shall generally apply.

7.4.1.1 The structural design of reinforced concrete rafts shall conform to Part C 'Structural Design/Section 5 Structural Concrete' of this NBCS.

7.4.1.2 In the case of raft, whether resting on soil directly or on lean concrete, the cover to the reinforcement shall be as prescribed in Part C 'Structural Design/ Section 5 Structural Concrete' of this NBCS for the applicable environment exposure condition.

7.4.1.3 In case the structure supported by the raft consists of several parts with varying loads and heights, it is advisable to provide separation joints between these parts. Joints shall also be provided, wherever there is a change in the direction of the raft.

7.4.1.4 Foundations subject to heavy vibratory loads should preferably be isolated.

7.4.1.5 The minimum depth of foundation shall generally be not less than 1 m.

7.4.1.6 *Dimensional parameters*

The size and shape of the foundation adopted affect the magnitude of subgrade modulus and long term deformation of the supporting soil and this, in turn, influences the distribution of contact pressure. This aspect needs to be taken into consideration in the analysis.

7.4.1.7 *Eccentricity of loading*

A raft generally occupies the entire area of the building and often it is not feasible and rather uneconomical to proportion it coinciding the centroid of the raft with the line of action of the resultant force. In such cases, the effect of the eccentricity on contact pressure distribution shall be taken into consideration.

7.4.1.8 *Properties of supporting soil*

Distribution of contact pressure underneath a raft is affected by the physical characteristics of the soil supporting it. Consideration shall be given to the increased contact pressure developed along the edges of foundation on cohesive soils and the opposite effect on granular soils. Long term consolidation of deep soil layers shall be taken into account in the analysis. This may necessitate evaluation of contact pressure distribution both immediately after construction and after completion of the consolidation process. The design shall be based on the worst conditions.

7.4.1.9 *Rigidity of foundations*

Rigidity of the foundation tends to iron out uneven deformation and thereby modifies the contact pressure distribution. High order of rigidity is characterized by long moments and relatively small, uniform settlements. A rigid foundation may also generate high secondary stresses in structural members. The effect of rigidity shall be taken into account in analysis.

7.4.1.10 *Rigidity of the super structure*

Free response of the foundations to soil deformation is restricted by the rigidity of the superstructure. In the extreme case, a stiff structure may force a flexible foundation to behave as rigid. This aspect shall be considered to evaluate the validity of the contact pressure distribution.

7.4.1.11 *Modulus of elasticity and modulus of subgrade reaction*

Annex A enumerates the methods of determination of modulus of elasticity (E_s). The modulus of subgrade reaction (k) may be determined in accordance with Annex B.

7.4.2 *Necessary Information*

The following information is necessary for a satisfactory design and construction of a raft foundation:

a) Site plan showing the location of the proposed as well as the neighbouring structures;

b) Plan and cross-sections of building showing different floor levels, shafts and openings, etc, layout of load bearing walls, columns, shear walls, etc;

c) Loading conditions, preferably shown on a schematic plan indicating combination of design loads transmitted to the foundation;

d) Information relating to geological history of the area, seismicity of the area, hydrological information indicating ground water conditions and its seasonal variations, etc;

e) Geotechnical information giving subsurface profile with stratification details, engineering properties of the founding strata (namely, index properties, effective shear parameters determined under appropriate drainage conditions, compressibility characteristics, swelling properties, results of field tests like static and dynamic penetration tests, pressure meter tests, etc); and

f) A review of the performance of similar structure, if any, in the locality.

7.4.3 *Choice of Raft Type*

7.4.3.1 For fairly small and uniform column spacing and when the supporting soil is not too compressible a flat concrete slab having uniform thickness throughout (a true mat) is most suitable [*see* Fig. 7(A)].

7.4.3.2 A slab may be thickened under heavy loaded columns to provide adequate strength for shear and negative moment. Pedestals may also be provided in such cases [*see* Fig. 7(B)].

7.4.3.3 A slab and beam type of raft is likely to be more economical for large column spacing and unequal column loads particularly when the supporting soil is very compressive [*see* Fig. 7(C) and Fig. 7(D)].

7.4.3.4 For very heavy structures, provision of cellular raft or rigid frames consisting of slabs and basement walls may be considered.

7.4.4 *Methods of Analysis*

The essential task in the analysis of a raft foundation is the determination of the distribution of contact pressure underneath the raft which is a complex function of the rigidity of the superstructure, the supporting soil and the raft itself and cannot be determined with exactitude, except in very simple cases. This necessitates a number of simplifying assumptions to make the problem amenable to analysis. Once the distribution of contact pressure is determined, design bending moments and shears can be computed based on statics. The methods of analysis suggested are distinguished by the assumptions involved. Choice of a particular method should be governed by the validity of the assumptions in the particular case.

7.4.4.1 *Rigid foundation* (*conventional method*)

7.4.4.1.1 This method is based on the assumption of linear distribution of contact pressure. The basic assumptions of this method are:

a) the foundations rigid relative to the supporting soil and the compressible soil layer is relatively shallow; and

b) the contact pressure variation is assumed as planar, such that the centroid of the contact pressure coincides with the line of action of the resultant force of all loads acting on the foundation.

7.4.4.1.2 This method may be used when either of the following conditions is satisfied:

a) the structure behaves as rigid (due to the combined action of the superstructure and the foundation) with relative stiffness factor $K > 0.5$ (for evaluation of K, *see* Annex C); and

b) the column spacing is less than $1.75/\lambda$ (*see* Annex C).

7.4.4.1.3 The raft is analyzed as a whole in each of the two perpendicular directions. The contact pressure distribution is determined by the procedure outlined in Annex D. Further analysis is also based on statics.

7.4.4.1.4 In the case of uniform conditions when the variations in adjacent column loads and column spacings do not exceed 20 percent of the higher value, the raft may be divided into perpendicular strips of widths equal to the distance between mid-spans and each strip may be analyzed as an independent beam with known column loads and known contact pressures. Such beams will not normally satisfy statics due to shear transfer between adjacent strips and design may be based on suitable moment coefficients, or by moment distribution.

NOTE — On soft soils, for example, normally consolidated clays, peat, muck, organic silts, etc, the assumptions involved in the conventional method are commonly justified.

7.4.4.2 *Flexible foundations*

a) *Simplified method* — In this method, it is assumed that the subgrade consists of an infinite array of individual elastic springs each of which is not affected by others. The spring constant is equal to the modulus of subgrade reaction (k). The contact pressure at any point under the raft is, therefore, linearly proportional to the settlement at the point. Contact pressure may be determined as given in Annex E. This method may be used when all the following conditions are satisfied; and

1) The structure (combined action of superstructure and raft) may be considered as flexible (relative stiffness factor $K < 0.5$, *see* Annex C); and

2) Variation in adjacent column load does not exceed 20 percent of the higher value.

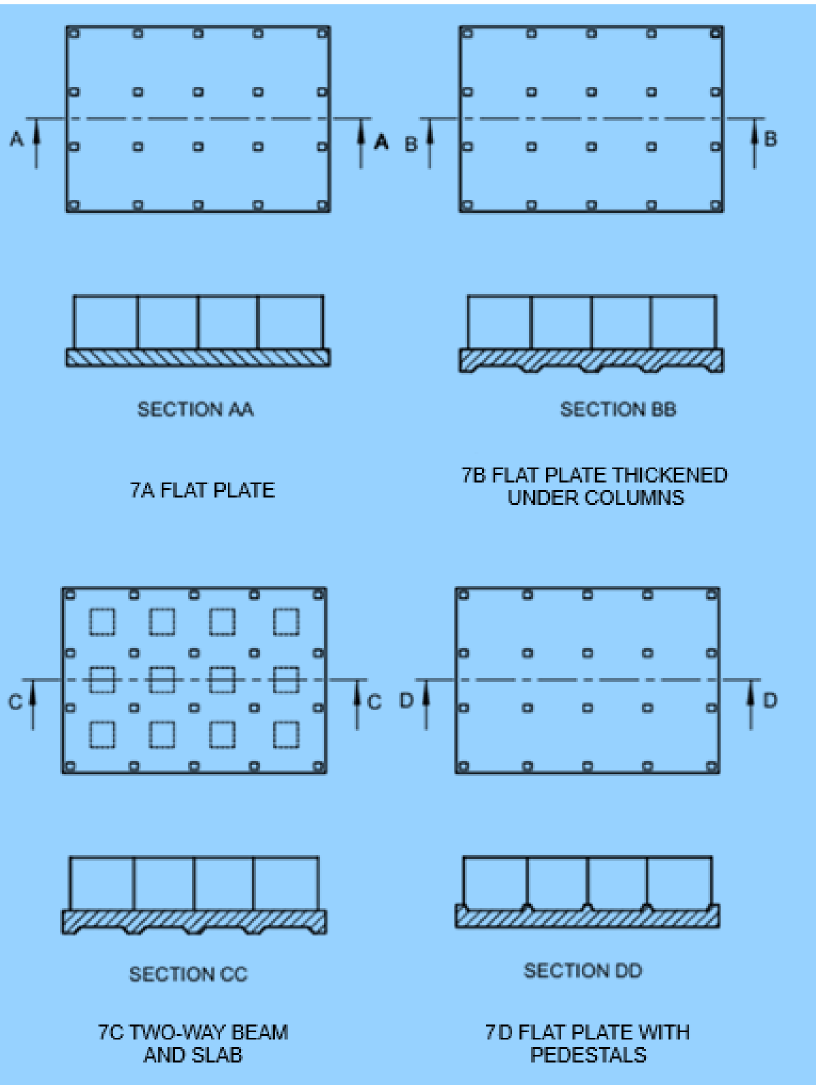


FIG.7 COMMON TYPES OF RAFT FOUNDATION

b) *General method* — For the general case of a flexible foundation not satisfying the requirements of (a) the method based on closed form solution of elastic plate theory may be used. This method is based on the theory of plates on winkler foundation which takes into account the restraint on deflection of a point provided by continuity of the foundation in orthogonal foundation. The distribution of deflection and contact pressure on the raft due to a column load is determined by the plate theory. Since the effect of a column load on an elastic foundation is damped out rapidly, it is possible to determine the total effect at a point of all column loads within the zone of influence by the method of superimposition. The computation of effect at any point may be restricted to columns of two adjoining bays in all directions. The procedure is outlined in Annex F.

7.5 Ring Foundations

For provisions regarding ring foundations, good practice [C-2(37)] shall be referred to.

7.6 Shell Foundations

For provisions regarding shell foundations, good practice [C-2(38)] shall be referred to.

8 DEEP FOUNDATIONS

8.1 Deep foundations are structural elements used to transfer loads from a building or structure through weak, compressible soil to stronger, more competent soil or rock at depth. They're essential when shallow foundations, such as footings, are not feasible due to inadequate bearing capacity or excessive settlement.

Some important types of deep foundations which are being used in building and other structures are as follows:

a) Pile foundations; and
   1) driven/bored cast *in-situ* concrete piles;
   2) driven precast concrete piles;
   3) precast concrete piles in prebored holes;
   4) under- reamed piles; and
   5) timber piles.
b) Pier foundation

8.2 Driven/Bored Cast *In-Situ* Concrete Piles

8.2.1 *General*

Piles find application in foundations to transfer loads from a structure to competent subsurface strata having adequate load-bearing capacity. The load transfer mechanism from a pile to the surrounding ground is complicated and is not yet fully understood, although application of pile foundations is in practice over many decades. Broadly, piles transfer axial loads either substantially by friction along its shaft and/or by end bearing. Piles are used where either of the above load transfer mechanism is possible depending upon the subsoil stratification at a particular site. Construction of pile foundations require a careful choice of piling system depending upon the subsoil conditions, the load characteristics of a structure and the limitations of total settlement, differential settlement and any other special requirement of a project.

8.2.2 *Materials and Stresses*

8.2.2.1 *Concrete*

Consistency of concrete to be used shall be consistent with the method of installation of piles. Concrete shall be so designed or chosen as to have a homogeneous mix having a slump/workability consistent with the method of concreting under the given conditions of pile installation.

The slump should be 150 mm to180 mm at the time of pouring.

The minimum grade of concrete to be used for bored piling shall be M 25. For sub-aqueous concrete, the requirements specified in Part C 'Structural Design/ Section 5 Structural Concrete' of this NBCS shall be followed. The minimum cement content shall be 400 kg/m^3. However, with proper mix design and use of proper admixture the cement content may be reduced but in no case the cement content shall be less than 350 kg/m^3 [*See also* Table 5 (and Notes thereunder) of Part C 'Structural Design/ Section 5 Structural Concrete' of this NBCS].

For the concrete, water and aggregates, specifications laid down in Part C 'Structural Design/ Section 5 Structural Concrete' of this NBCS shall be followed in general.

The average compressive stress under working load should not exceed 25 percent of the specified works cube strength at 28 days calculated on the total cross-sectional area of the pile. Where the casing of the pile is permanent, of adequate thickness and of suitable shape, the allowable compressive stress may be increased.

8.2.2.2 *Steel reinforcement*

Steel reinforcement shall conform to any one of the types of steel specified in Part C 'Structural Design/ Section 5 Structural Concrete' of this NBCS.

8.2.3 *Design Considerations*

8.2.3.1 *General*

Pile foundations shall be designed in such a way that the load from the structure can be transmitted to the subsurface with adequate factor of safety against shear failure of subsurface and without causing such settlement (differential or total), which may result in structural damage and/or functional distress under permanent/transient loading. The pile shaft should have adequate structural capacity to withstand all loads (vertical, axial or otherwise) and moments which are to be transmitted to the subsoil and shall be designed according to Part C 'Structural Design/ Section 5 Structural Concrete' of this NBCS.

8.2.3.2 *Adjacent structures*

8.2.3.2.1 When working near existing structures, care shall be taken to avoid damage to such structures. The good practice [C-2(39)] may be used as a guide for studying qualitatively the effect of vibration on persons and structures.

8.2.3.2.2 In case of deep excavations adjacent to piles, proper shoring or other suitable arrangement shall be made to guard against undesired lateral movement of soil.

8.2.3.3 *Pile capacity*

The load carrying capacity of a pile depends on the properties of the soil in which it is embedded. Axial load from a pile is normally transmitted to the soil through skin friction along the shaft and end bearing at its tip. A horizontal load on a vertical pile is transmitted to the subsoil primarily by horizontal subgrade reaction generated in the upper part of the shaft. Lateral load capacity of a single pile depends on the soil reaction developed and the structural capacity of the shaft under bending. It would be essential to investigate the lateral load capacity of the pile using appropriate values of horizontal subgrade modulus of the soil. Alternatively, piles may be installed in rake.

8.2.3.3.1 The ultimate load capacity of a pile may be estimated by means of static formula on the basis of soil test results, or by using a dynamic pile formula using data obtained during driving the pile. However, dynamic pile driving formula should be generally used as a measure to control the pile driving at site. Pile capacity should preferably be confirmed by initial load tests {*see* good practice [C-2(40)]}. For rock-socketed piles, reference shall also be made to good practice [C-2(41)] for estimating the load capacity of piles.

The settlement of pile obtained at safe load/working load from load-test results on a single pile shall not be directly used for estimating the settlement of a structure. The settlement may be determined on the basis of subsoil data and loading details of the structure as a whole using the principles of soil mechanics.

8.2.3.3.1.1 *Static formula*

The ultimate load capacity of a single pile may be obtained by using static analysis, the accuracy being dependent on the reliability of the soil properties for various strata. When computing capacity by static formula, the shear strength parameters obtained from a limited number of borehole data and laboratory tests should be supplemented, wherever possible by *in-situ* shear strength obtained from field tests. The two separate static formulae, commonly applicable for cohesive and non-cohesive soil respectively, are indicated in Annex G. Other formula based on static cone penetration test {*see* the accepted standards [C-2(42)]} and standard penetration test {*see* the accepted standard [C-2(6)]} are given in **G-3** and **G-4**.

8.2.3.3.1.2 *Dynamic formula*

For driven piles, any established dynamic formula may be used to control the pile driving at site giving due consideration to limitations of various formulae.

Whenever, double acting diesel hammers or hydraulic hammers are used for driving of piles, manufacturer's guidelines about energy and set criteria may be referred to. Dynamic formulae are not directly applicable to cohesive soil deposits such as saturated silts and clays as the resistance to impact of the tip of the casing will be exaggerated by their low permeability while the frictional resistance on the sides is reduced by lubrication.

8.2.3.3.1.3 *Load test results*

The ultimate load capacity of a single pile is determined with reasonable accuracy from test loading as per good practice [C-2(40)]. The load test on a pile shall not be carried out earlier than four weeks from the time of casting the pile.

8.2.3.3.1.4 *Uplift capacity*

The uplift capacity of a pile is given by sum of the frictional resistance and the weight of the pile (buoyant or total as relevant). The recommended factor of safety is 3.0 in the absence of any pullout test results and 2.0 with pullout test results. Uplift capacity can be obtained from static formula (*see* Annex G) by ignoring end bearing but adding weight of the pile (buoyant or total as relevant).

8.2.3.4 *Negative skin friction or dragdown force*

When a soil stratum, through which a pile shaft has penetrated into a underlying hard stratum, compresses as a result of either it being unconsolidated or it being under a newly placed fill or as a result of remoulding during driving of the pile, a dragdown force is generated along the pile shaft up to a point in depth where the surrounding soil does not move downward relative to the pile shaft. Existence of such a phenomenon shall be assessed and suitable correction shall be made to the allowable load where appropriate.

8.2.3.5 *Structural capacity*

The piles shall have necessary structural strength to transmit the loads imposed on it, ultimately to the soil. In case of uplift, the structural capacity of the pile, that is, under tension should also be considered.

8.2.3.5.1 *Axial capacity*

Where a pile is completely embedded in the soil (having an undrained shear strength not less than 0.01 N/mm^2), its axial load carrying capacity is not necessarily limited by its strength as a long column. Where piles are installed through very weak soils (having an undrained shear strength less than 0.01 N/mm^2), special considerations shall be made to determine whether the shaft would behave as a long column or not. If necessary, suitable reductions shall be made for its structural strength following the normal structural principles covering the buckling phenomenon.

When the finished pile projects above ground level and is not secured against buckling by adequate bracing, the effective length will be governed by the fixity imposed on it by the structure it supports and by the nature of the soil into which it is installed. The depth below the ground surface to the lower point of contraflexure varies with the type of the soil. In good soil the lower point of contraflexure may be taken at a depth of 1 m below ground surface subject to a minimum of 3 times the diameter of the shaft. In weak soil (undrained shear strength less than 0.01 N/mm^2) such as soft clay or soft silt, this point may be taken at about half the depth of penetration into such stratum but not more than 3 m or 10 times the diameter of the shaft, whichever is more. The degree of fixity of the position and inclination of the pile top and the restraint provided by any bracing shall be estimated following accepted structural principles.

The permissible stress shall be reduced in accordance with similar provision for reinforced concrete columns as laid down in Part C 'Structural Design/ Section 5 Structural Concrete' of this NBCS.

8.2.3.5.2 *Lateral load capacity*

A pile may be subjected to lateral force for a number of causes, such as wind, earthquake, water current, earth pressure, effect of moving vehicles or ships, plant and equipment, etc. The lateral load capacity of a single pile depends not only on the horizontal subgrade modulus of the surrounding soil but also on the structural strength of the pile shaft against bending, consequent upon application of a lateral load. While considering lateral load on piles, effect of other co-existent loads, including the axial load on the pile, should be taken into consideration for checking the structural capacity of the shaft. A recommended method for the pile analysis under lateral load is given in Annex H.

Because of limited information on horizontal subgrade modulus of soil and pending refinements in the theoretical analysis, it is suggested that the adequacy of a design should be checked by an actual field load test. In the zone of soil susceptible to liquefaction, the lateral resistance of the soil shall not be considered.

8.2.3.5.2.1 *Fixed and free head conditions*

A group of three or more piles connected by a rigid pile cap shall be considered to have fixed head condition. Caps for single piles shall be interconnected by grade beams in two directions and for twin piles by grade beams in a line transverse to the common axis of the pair so that the pile head is fixed. In all other conditions the pile shall be taken as free headed.

8.2.3.5.3 *Raker piles*

Raker piles are normally provided where vertical piles cannot resist the applied horizontal forces. Generally, the rake will be limited to 1 horizontal to 6 vertical. In the preliminary design, the load on a raker pile is generally considered to be axial. The distribution of load between raker and vertical piles in a group may be determined by graphical or analytical methods. Where necessary, due consideration should be made for secondary bending induced as a result of the pile cap movement, particularly when the cap is rigid. Free-standing raker piles are subjected to bending moments due to their own weight or external forces from other causes. Raker piles, embedded in fill or consolidating deposits, may become laterally loaded owing to the settlement of the surrounding soil. In consolidating clay, special precautions, like provision of permanent casing should be taken for raker piles.

8.2.3.6 *Spacing of piles*

The minimum centre to centre spacing of pile is considered from three aspects, namely:

a) Practical aspects of installing the piles;

b) Diameter of the pile; and

c) Nature of the load transfer to the soil and possible reduction in the load capacity of piles group.

NOTE — In the case of piles of non-circular cross-section, diameter of the circumscribing circle shall be adopted.

8.2.3.6.1 In case of piles founded on hard stratum and deriving their capacity mainly from end bearing the minimum spacing shall be 2.5 times the diameter of the circumscribing circle corresponding to the cross-section of the pile shaft. In case of piles resting on rock, the spacing of two times the said diameter may be adopted.

8.2.3.6.2 Piles deriving their load carrying capacity mainly from friction shall be spaced sufficiently apart to ensure that the zones of soils from which the piles derive their support do not overlap to such an extent that their bearing values are reduced. Generally, the spacing in such cases shall not be less than 3 times the diameter of the pile shaft.

8.2.3.7 *Pile groups*

8.2.3.7.1 In order to determine the load carrying capacity of a group of piles a number of efficiency equations are in use. However, it is difficult to establish the accuracy of these efficiency equations as the behaviour of pile group is dependent on many complex factors. It is desirable to consider each case separately on its own merits.

8.2.3.7.2 The load carrying capacity of a pile group may be equal to or less than the load carrying capacity of individual piles multiplied by the number of piles in the group. The former holds true in case of friction piles, driven into progressively stiffer materials or in end-bearing piles. For driven piles in loose sandy soils, the group capacity may even be higher due to the effect of compaction. In such cases a load test may be carried out on a pile in the group after all the piles in the group have been installed.

8.2.3.7.3 In case of piles deriving their support mainly from friction and connected by a rigid pile cap, the group may be visualized as a block with the piles embedded within the soil. The ultimate load capacity of the group may then be obtained by taking into account the frictional capacity along the perimeter of the block and end bearing at the bottom of the block using the accepted principles of soil mechanics.

8.2.3.7.3.1 When the cap of the pile group is cast directly on reasonably firm stratum which supports the piles, it may contribute to the load carrying capacity of the group. This additional capacity along with the individual capacity of the piles multiplied by the number of piles in the group shall not be more than the capacity worked out as per **8.2.3.7.3**.

8.2.3.7.4 When a pile group is subjected to moment either from superstructure or as a consequence of inaccuracies of installation, the adequacy of the pile group in resisting the applied moment should be checked. In case of a single pile subjected to moment due to lateral loads or eccentric loading, beams may be provided to restrain the pile cap effectively from lateral or rotational movement.

8.2.3.7.5 In case of a structure supported on single piles/group of piles resulting in large variation in the number of piles from column to column it may result in excessive differential settlement. Such differential settlement should be either catered for in the structural design or it may be suitably reduced by judicious choice of variations in the actual pile loading. For example, a single pile cap may be loaded to a level higher than that of the pile in a group in order to achieve reduced differential settlement between two adjacent pile caps supported on different number of piles.

8.2.3.8 *Factor of safety*

8.2.3.8.1 Factor of safety should be chosen after considering:

a) the reliability of the calculated value of ultimate load capacity of a pile;

b) the types of superstructure and the type of loading; and

c) allowable total/differential settlement of the structure.

8.2.3.8.2 When the ultimate load capacity is determined from either static formula or dynamic formula, the factor of safety would depend on the reliability of the formula and the reliability of the subsoil parameters used in the computation. The minimum factor of safety on static formula shall be 2.5. The final selection of a factor of safety shall take into consideration the load settlement characteristics of the structure as a whole at a given site.

8.2.3.8.3 Higher value of factor of safety for determining the safe load on piles may be adopted, where:

a) settlement is to be limited or unequal settlement avoided;

b) large impact or vibrating loads are expected; and

c) the properties of the soil may deteriorate with time.

8.2.3.9 *Transient loading*

The maximum permissible increase over the safe load of a pile, as arising out of wind loading, is 25 percent. In case of loads and moments arising out of earthquake effects, the increase of safe load on a single pile shall be as given in **7.2.2.4.2**. For transient loading arising out of superimposed loads, no increase is allowed.

8.2.3.10 *Overloading*

When a pile in a group, designed for a certain safe load is found, during or after execution, to fall just short of the load required to be carried by it, an overload up to 10 percent of the pile capacity may be allowed on each pile. The total overloading on the group should not, however, be more than 10 percent of the capacity of the group subject to the increase of the load on any pile being not more than 25 percent of the allowable load on a single pile.

8.2.3.11 *Reinforcement*

8.2.3.11.1 The design of the reinforcing cage varies depending upon the driving and installation conditions, the nature of the subsoil and the nature of load to be transmitted by the shaft - axial, or otherwise. The minimum area of longitudinal reinforcement of any type or grade within the pile shaft shall be 0.4 percent of the cross-sectional area of the pile shaft. The minimum reinforcement shall be provided throughout the length of the shaft.

8.2.3.11.2 The curtailment of reinforcement along the depth of the pile, in general, depends on the type of loading and subsoil strata. In case of piles subjected to compressive load only, the designed quantity of reinforcement may be curtailed at appropriate level according to the design requirements. For piles subjected to uplift load, lateral load and moments, separately or with compressive loads, it would be necessary to provide reinforcement for the full depth of pile. In soft clays or loose sands, or where there may be danger to green concrete due to driving of adjacent piles, the reinforcement should be provided to the full pile depth, regardless of whether or not it is required from uplift and lateral load considerations. However, in all cases, the minimum reinforcement specified in **8.2.3.11.1** shall be provided throughout the length of the shaft.

8.2.3.11.3 Piles shall always be reinforced with a minimum amount of reinforcement as dowels keeping the minimum bond length into the pile shaft below its cut-off level and with adequate projection into the pile cap, irrespective of design requirements.

8.2.3.11.4 Clear cover to all main reinforcement in pile shaft shall be not less than 50 mm. The laterals of a reinforcing cage may be in the form of links or spirals. The diameter and spacing of the same is chosen to impart adequate rigidity of the reinforcing cage during its handling and installations. The minimum diameter of the links or spirals shall be 8 mm and the spacing of the links or spirals shall be not less than 150 mm. Stiffner rings preferably of 16 mm diameter at every 1.5 m centre to centre should be provided along the length of the cage for providing rigidity to reinforcement cage. Minimum 6 numbers of vertical bars shall be used for a circular pile and minimum diameter of vertical bar shall be 12 mm. The clear horizontal spacing between the adjacent vertical bars shall be four times the maximum aggregate size in concrete. If required, the bars can be bundled to maintain such spacing.

8.2.3.12 *Design of pile cap*

8.2.3.12.1 The pile caps may be designed by assuming that the load from column is dispersed at 45º from the top of the cap to the mid-depth of the pile cap from the base of the column or pedestal. The reaction from piles may also be taken to be distributed at 45º from the edge of the pile, up to the mid-depth of the pile cap. On this basis the maximum bending moment and shear forces should be worked out at critical sections. The method of analysis and allowable stresses should be in accordance with Part C 'Structural Design/ Section 5 Structural Concrete' of this NBCS.

8.2.3.12.2 Pile cap shall be deep enough to allow for necessary anchorage of the column and pile reinforcement.

8.2.3.12.3 The pile cap should be rigid enough so that the imposed load could be distributed on the piles in a group equitably.

8.2.3.12.4 In case of a large cap, where differential settlement may occur between piles under the same cap, due consideration for the consequential moment should be given.

8.2.3.12.5 The clear overhang of the pile cap beyond the outermost pile in the group shall be a minimum of 150 mm.

8.2.3.12.6 The cap is generally cast over a 75 mm thick levelling course of concrete. The clear cover for main reinforcement in the cap slab shall not be less than 60 mm.

8.2.3.12.7 The embedment of pile into cap should be 75 mm.

8.2.3.13 *Grade beams*

8.2.3.13.1 The grade beams supporting the walls shall be designed taking due account of arching effect due to masonry above the beam. The beam with masonry due to composite action behaves as a deep beam.

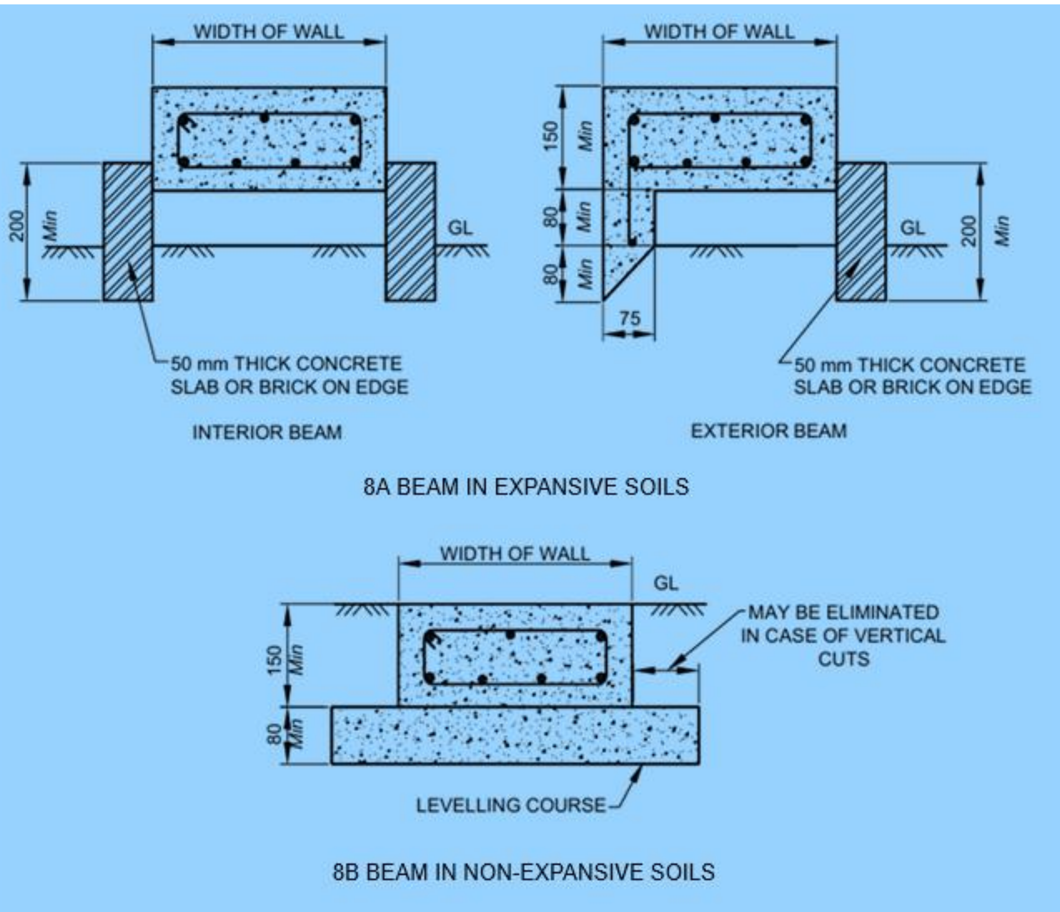
For the design of grade beams, a maximum bending moment of $\frac{wl^2}{30}$, where w is uniformly distributed load per metre run and l is the effective span in metres, shall be considered. For considering composite action, the minimum height of wall shall be 0.6 times the pile spacing. The brick strength should not be less than 3 N/mm^2. For concentrated and other loads which come directly over the beam, full bending moment should be considered.

8.2.3.13.2 The minimum overall depth of grade beams shall be 200 mm. An equal amount of reinforcement may be provided at the bottom and top of the beam. The longitudinal reinforcement both at top and bottom should not be less than three bars of 12 mm diameter with clear cover space of 50 mm. S pacing of stirrups of 8 mm diameter bars should not be more than the depth of grade beam.

8.2.3.13.3 In expansive soils, the grade beams and 75 mm levelling concrete course shall be provided over 40 mm down gravel layer/mechanically stabilized mix (MSM) layer having a minimum compacted thickness of 300 mm and minimum width equal to five times the width of grade beams (*see* Fig. 8).

8.2.3.13.4 In the case of exterior beams over piles in expansive soils, plinth protection with proper slope shall be provided.

8.2.4 For detailed information on driven/bored cast *in-situ* concrete piles regarding control of piling, installation, defective pile and recording of data, reference shall be made to good practice [C-2(43)].



All dimensions in millimetres.

FIG.8 TYPICAL SECTIONS OF GRADE BEAMS

8.2.5 *Bored Cast In-Situ Concrete Piles on Rocks*

Design and construction of bored cast *in-situ* piles founded on rocks shall be carried out in accordance with good practice [C-2(41)].

8.2.6 *Non-Destructive Testing*

For quality assurance of concrete piles, non-destructive integrity test may be carried out prior to laying of beam/caps, in accordance with accepted standard [C-2(44)].

8.3 Driven Precast Concrete Piles

8.3.1 Provisions of **8.2** except **8.2.3.11** shall generally apply.

8.3.2 *Design of Pile Section*

8.3.2.1 Design of pile section shall be such as to ensure the strength and soundness of the pile against lifting from the casting bed, transporting, handling and driving stresses without damage.

8.3.2.2 Any shape having radial symmetry will be satisfactory for precast piles. The most commonly used cross-sections are square and octagonal.

8.3.2.3 Where exceptionally long lengths of piles are required, hollow sections can be used. If the final condition requires larger cross-sectional area, the hollow sections may be filled with concrete after driving in position.

8.3.2.4 Wherever, final pile length is so large that a single length precast pile unit is either uneconomical or impracticable for installation, the segmental precast RCC piles with a number of segments using efficient mechanical jointing could be adopted.

Excessive whipping during handling precast pile may generally be avoided by limiting the length of pile to a maximum of 50 times the least width. As an alternative, segmental precast piling technique could be used.

The design of joints shall take care of corrosion by providing additional sacrificial thickness for the joint, wherever warranted.

8.3.2.5 Stresses induced by bending in the cross-section of precast pile during lifting and handling may be estimated as for any reinforced concrete section in accordance with relevant provisions of Part C 'Structural Design/ Section 5 Structural Concrete' of this NBCS. The calculations for bending moment for different support conditions during handling are given in Table 5.

Table 5 Bending Moment for Different Support Conditions (*Clause* 8.3.2.5)

Sl No.	Number of Points of Pick Up	Location of Support From End in Terms of Length of Pile For Minimum Moments	Bending Moment to be Allowed for Design Kn-M
(1)	(2)	(3)	(4)
i)	One	0.293 *L*	4.3 *WL*
ii)	Two	0.207 *L*	2.2 *WL*
iii)	Three	0.145 *L*, the middle point will be at the centre	1.05 *WL*

NOTE — *W* = weight of pile, in kN;
L = length of pile, in m.

8.3.2.6 The driving stresses on a pile may be estimated by the following formula:

$$\frac{\text{Driving resistance}}{\text{Cross} - \text{sectional area of the pile}} \times \left[\frac{2}{\sqrt{n}} - 1\right]$$

where

n = efficiency of the blow (*see* **8.3.2.6.1** for probable value of n).

NOTE — For the purpose of this formula, cross-sectional area of the pile shall be calculated as the overall sectional area of the pile including the equivalent area for reinforcement.

8.3.2.6.1 The formula for efficiency of the blow, representing the ratio of energy after impact to striking energy of ram, n, is:

where W is greater than $P.e$ and the pile is driven into penetrable ground,

$$\boldsymbol{n} = \frac{\boldsymbol{W} + (\boldsymbol{P} \cdot \boldsymbol{e}^2)}{\boldsymbol{W} + \boldsymbol{P}}$$

where W is less than $P.e$ and the pile is driven into penetrable ground,

$$\boldsymbol{n} = \left[\frac{\boldsymbol{W}+(\boldsymbol{P}\cdot\boldsymbol{e}^2)}{\boldsymbol{W}+\boldsymbol{P}}\right] - \left[\frac{\boldsymbol{W} - (\boldsymbol{P}\cdot\boldsymbol{e})}{\boldsymbol{W} + \boldsymbol{P}}\right]^2$$

The following are the values of n in relation to e and to the ratio of P/W:

Sl No.	*Ratio of P/W*	*e = 0.5*	*e = 0.4*	*e = 0.32*	*e = 0.25*	*e = 0*
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	½	0.75	0.72	0.70	0.69	0.67
ii)	1	0.63	0.58	0.55	0.53	0.50
iii)	1½	0.55	0.50	0.47	0.44	0.40
iv)	2	0.50	0.44	0.40	0.37	0.33
v)	2½	0.45	0.40	0.36	0.33	0.28
vi)	3	0.42	0.36	0.33	0.30	0.25
vii)	3½	0.39	0.33	0.30	0.27	0.22
viii)	4	0.36	0.31	0.28	0.25	0.20
ix)	5	0.31	0.27	0.24	0.21	0.16
x)	6	0.27	0.24	0.21	0.19	0.14
xi)	7	0.24	0.21	0.19	0.17	0.12
xii)	8	0.22	0.20	0.17	0.15	0.11

NOTES

1 W = mass of the ram, in tonne and P = weight of the pile, anvil, helmet and follower (if any), in tonne.

2 Where the pile finds refusal in rock, $0.5P$ should be substituted for P in the above expressions for n.

3 e is the coefficient of restitution of the materials under impact as tabulated below:

a) For steel ram of double-acting hammer striking on steel anvil and driving reinforced concrete pile, e = 0.5;

b) For cast-iron ram of single-acting or drop hammer striking on head of reinforced concrete pile, e = 0.4;

c) Single-acting or drop hammer striking a well-conditioned driving cap and helmet with hard wood dolly in driving reinforced concrete piles or directly on head of timber pile, e = 0.25; and

d) For a deteriorated condition of the head of pile or of dolly, e = 0.

8.3.3 *Reinforcement*

8.3.3.1 The longitudinal reinforcement of any type or grade shall be provided in precast reinforced concrete piles for the entire length. All the main longitudinal bars shall be of the same length and should fit tightly into the pile shoe, if there is one. Shorter rods to resist local bending moments may be added but the same should be carefully detailed to avoid any sudden discontinuity of the steel which may lead to cracks during heavy driving. The area of main longitudinal reinforcement shall not be less than the following percentages of the cross-sectional area of the piles:

a) For piles with a length less than 30 times the least width: 1.25 percent;

b) For piles with a length 30 to 40 times the least width: 1.5 percent; and

c) For piles with a length greater than 40 times the least width: 1.5 percent.

8.3.3.2 Piles shall always be reinforced with a minimum amount of reinforcement as dowels keeping the minimum bond length into the pile shaft below its cut-off level and with adequate projection into the pile cap, irrespective of design requirements.

8.3.3.3 Clear cover to all main reinforcement in pile shaft shall be not less than 50 mm. The laterals of a reinforcing cage may be in the form of links or spirals. The diameter and spacing of the same is chosen to impart adequate rigidity of the reinforcing cage during its handing and installations. The minimum diameter of the links or spirals shall be 8 mm and the spacing of the links or spirals shall be not less than 150 mm Stiffner rings preferably of 16 mm diameter at every 1.5 m centre to centre to be provided along the length of the cage for providing rigidity to reinforcement cage. Minimum 6 numbers of vertical bars shall be used for a circular pile and minimum diameter of vertical bar shall be 12 mm. The clear horizontal spacing between the adjacent vertical bars shall be four times the maximum aggregate size in concrete. If required, the bars may be bundled to maintain such spacing.

8.3.4 Additional provision for prestressed concrete piles shall be as per good practice [C-2(45)].

8.3.5 For detailed information regarding casting and curing, storing and handling, control of pile driving and recording of data, reference shall be made to good practice [C-2(45)].

8.3.6 *Non-Destructive Testing*

For quality assurance of concrete piles, non-destructive integrity test may be carried out prior to laying of beam/caps, in accordance with good practice [C-2(44)].

8.4 Precast Concrete Piles in Prebored Holes

8.4.1 Provisions of **8.3** except **8.3.3** shall generally apply.

8.4.2 *Handling Equipment for Lowering and Grouting Plant*

Handling equipment such as crane, scotch derricks, movable gantry may be used for handling and lowering the precast piles in the bore. The choice of equipment will depend upon length, mass and other practical requirements.

The mixing of the grout shall be carried out in any suitable high speed colloidal mixer. Normally the colloidal mixer is adequate to fill the annular space with grouts. Where this is not possible, a suitable grout pump shall be used.

8.4.3 *Reinforcement*

8.4.3.1 The design of the reinforcing cage varies depending upon the handling and installation conditions, the nature of the subsoil and the nature of load to be transmitted by the shaft - axial, or otherwise. The minimum area of longitudinal reinforcement of any type or grade within the pile shaft shall be 0.4 percent of the cross-sectional area of the pile shaft or as required to cater for handling stresses (*see* **8.3.2.5**), whichever is greater. The minimum reinforcement shall be provided throughout the length of the shaft.

8.4.3.2 Piles shall always be reinforced with a minimum amount of reinforcement as dowels keeping the minimum bond length into the pile shaft below its cut-off level and with adequate projection into the pile cap, irrespective of design requirements.

8.4.3.3 Clear cover to all main reinforcement in pile shaft shall be not less than 50 mm. The laterals of a reinforcing cage may be in the form of links or spirals. The diameter and spacing of the same is chosen to impart adequate rigidity of the reinforcing cage during its handing and installations. The minimum diameter of the links or spirals shall be 8 mm and the spacing of the links or spirals shall be not less than 150 mm. Stiffner rings preferably of 16 mm diameter at every 1.5 m centre to centre to be provided along length of the cage for providing rigidity to reinforcement cage. Minimum 6 numbers of vertical bars shall be used for a circular pile and minimum diameter of vertical bar shall be 12 mm. The clear horizontal spacing between the adjacent vertical bars shall be four times the maximum aggregate size in concrete. If required, the bars may be bundled to maintain such spacing.

8.4.3.4 A thin gauge sheathing pipe of approximately 40 mm diameter may be attached to the reinforcement cage, in case of solid piles, to form the central duct for pumping grout to the bottom of the bore. The bottom end of the pile shall have proper arrangements for flushing/cleaning for grouting. Air lift technique may be used for cleaning the borehole, however this technique should be used carefully in case of silty and sandy soil.

8.4.4 For detailed information regarding casting and curing, storing and handling, control of pile installation and recording of data, reference shall be made to good practice [C-2(46)].

8.4.5 *Non-Destructive Testing*

For quality assurance of concrete piles, non-destructive integrity test may be carried out prior to laying of beam/caps, in accordance with accepted standard [C-2(44)].

8.5 Under-Reamed Piles

8.5.1 *General*

Under-reamed piles are bored cast *in-situ* and bored compaction concrete types having one or more bulbs formed by suitably enlarging the borehole for the pileshaft. These piles are suited for expansive soils which are often subjected to considerable ground movements due to seasonal moisture variations. These also find wide application in other soil strata where economics are favorable. When the ground consists of expansive soil, for example black cotton soils, the bulb of under-reamed pile provide anchorage against uplift due to swelling pressure, apart from the increased bearing, provided topmost bulb is formed close to or just below the bottom of active zone. Negative slopes may not be stable in certain strata conditions, for example, in pure sands (clean sands with fines less than five per cent) and very soft clayey strata having N of SPT less than 2 (undrained shear strength of less than 12.5 kN/m^2). Hence formation of bulb(s) in such strata is not advisable. In subsoil strata above water table, the maximum number of bulbs in under-reamed pile should be restricted to four. In the strata such as clay, silty clay and clayey silt with high water table where sides of bore hole stand by itself without needing any stabilization by using drilling mud or otherwise, the maximum number of bulbs in under-reamed piles should be restricted to two. In strata for example clayey sand, silty sand and sandy silt with high water table where bore hole needs stabilization by using drilling mud, under-reamed piles with more than one bulb shall not be used. In loose to medium pervious strata such as clayey sand, silty sand and sandy silt strata, compaction under-reamed piles may be used as the process of compaction increases the load carrying capacity of piles. From practical considerations, under-reamed piles of more than 10 m depth shall not be used without ensuring their construction feasibility and load carrying capacity by initial load tests in advance. In view of additional anchorage available with the provision of bulbs, under-reamed piles can be used with advantage to resist uplift loads.

8.5.2 *Materials*

Provisions of 8.2.2 shall generally apply.

8.5.3 *Design Considerations*

8.5.3.1 *General*

Under-reamed pile foundation shall be designed in such a way that the load from the structure they support can be transmitted to the sub-surface with adequate factor of safety against shear failure of sub-surface and without causing such settlement (differential or total) which may result in structural damage and/or functional distress under permanent/transient loading. The pile shaft should have adequate structural capacity to withstand all loads (vertical compression, uplift, lateral) and moments which are to be transmitted to the subsoil and shall be designed according to [C-2(47)].

8.5.3.1.1 In expansive soils, the minimum length of piles shall be 3.5 m below ground level. In case of expansive soil deposits of shallow depth overlying non-expansive competent strata or rock, piles of shorter length can also be provided.

However, in case of high to very high expansive soils, due care shall be taken to establish the minimum depth of pile with the top-most (first) bulb located below the level of significant moisture variations/active zone.

8.5.3.1.2 The diameter of under-reamed bulbs may vary from 2 times to 2.5 times shaft diameter depending upon the feasibility of construction and design requirements. In bored cast *in-situ* under-reamed piles and under-reamed bored compaction piles, the bulb diameter shall be normally 2.5 times and 2 times the shaft diameter, respectively.

8.5.3.1.3 The spacing of bulbs should not be less than 1.5 times the diameter of bulb.

8.5.3.1.4 The first bulb should be at a minimum depth of 2 times the bulb diameter below ground level or cut-off level, whichever is lower.

8.5.3.1.5 Under-reamed piles with more than two bulbs are not advisable without ensuring their feasibility in strata needing stabilization of boreholes by drilling mud. The number of bulbs in case of under-reamed bored compaction piles should not exceed two in such strata.

8.5.3.1.6 The minimum diameter of pile shaft shall be 250 mm. However, in case of strata/ground water consisting of excessive chemicals such as sulphates and chlorides, or piles with tremie concreting, the minimum diameter of the pile shaft shall be 300 mm.

8.5.3.1.7 Under-reamed batter piles in dry soil conditions can be constructed with batter not exceeding 15°.

8.5.3.1.8 The construction feasibility of under-ream/ bulb should be examined as per soil characteristics, particularly in case of clean/river sands.

8.5.3.1.9 For guidance, typical details of bored cast *in-situ* under-reamed pile foundations are shown in Fig. 9.

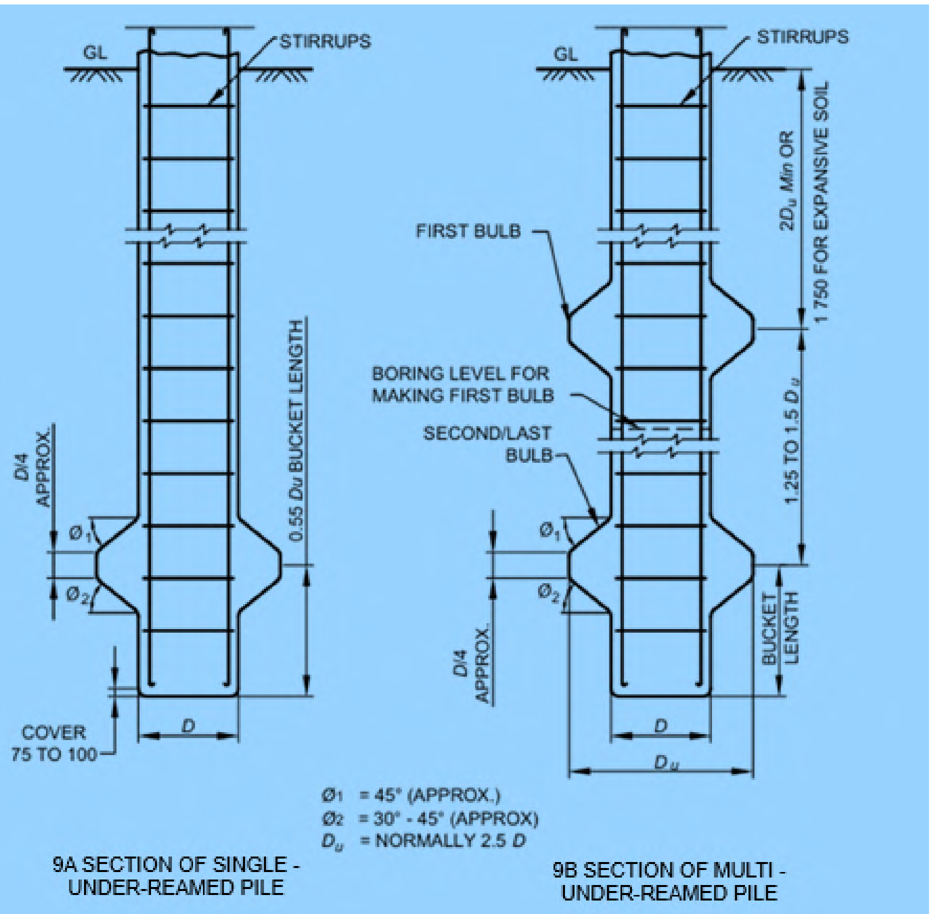


FIG. 9 TYPICAL DETAILS OF BORED CAST IN-SITU UNDER-REAMED PILE FOUNDATION

8.5.3.2 *Safe load*

Safe load on a pile shall be taken as the least of the following:

a) By calculating the ultimate load from soil properties and applying a suitable factor of safety (*see* Annex J);

b) By load test on pile as good practice [C-2(40)]; and

c) From safe load tables.

8.5.3.2.1 *Load test*

Provisions of **8.2.3.3.1.3** shall generally apply.

8.5.3.2.2 In the absence of detailed subsoil investigations and pile load tests for minor and less important structures, a rough estimate of safe load on piles may be made from the safe load table given in the good practice [C-2(48)].

NOTE — Safe loads as given in the above referred Table are symptomatic. Safe load carrying capacity of pile shall be worked out for the actual geotechnical data using 6.2.3.1 of good practice [C-2(48)], subjected to confirmation by initial pile load test in accordance with good practice [C-2(39)] and other provisions in 6.2.3.2 of good practice [C-2(48)].

8.5.3.3 *Negative skin friction or dragdown force*

Provisions of **8.3.4** shall generally apply subject to the condition that the under-reamed bulb is provided below the strata susceptible to negative skin friction.

8.5.3.4 *Structural capacity*

Provisions of **8.3.5** shall generally apply except that the under-reamed pile stem is designed for axial capacity as a short column. Under-reamed piles under lateral loads and moments tend to behave more as rigid piles due to the presence of bulbs and therefore the analysis can be done on rigid pile basis. Nominally reinforced long single bulb piles which are not rigid may be analyzed as per the method given in Annex G or as per other accepted methods.

8.5.3.5 *Spacing*

Generally, the centre to centre spacing for bored cast *in-situ* under-reamed piles in a group should be two times the bulb diameter ($2D_u$). It shall not be less than 1.5 D_u. For under-grade beams, the maximum spacing of piles should generally not exceed 3 m. In under-reamed compaction piles, generally the spacing should not be less than 1.5 D_u. If the adjacent piles are of different diameter, the value of larger bulb diameter shall be considered for spacing.

8.5.3.6 *Pile groups*

In order to determine the load-carrying capacity of a group of piles, a number of efficiency equations are in use. However, it is difficult to establish the accuracy of these efficiency equations as the behaviour of pile group is dependent on many complex factors. It is desirable to consider each case separately on its own merits.

The load-carrying capacity of a pile group may be equal to or less than the load-carrying capacity of individual piles multiplied by the number of piles in the group.

In load bearing walls, piles should generally be provided under all wall junctions to avoid point loads on beams.

8.5.3.7 *Transient and overloading*

Provisions of **8.2.3.9** and **8.2.3.10** shall generally apply.

8.5.3.8 *Reinforcement*

8.5.3.8.1 The minimum area of longitudinal reinforcement shall be 0.4 percent of the cross-sectional area of the pile shaft and shall be provided throughout the length.

8.5.3.8.2 The main reinforcement of the pile shall extend above cut-off level for at least the development length to be embedded into the pile cap/grade beam.

8.5.3.8.3 The minimum diameter of the rings shall be 8 mm and the spacing of the rings shall be not less than 150 mm. Stiffener rings of suitable diameter and spacing should be provided in the reinforcing cage to impart adequate rigidity to the reinforcing cage during handling and installation. Stiffener rings preferably of 12 mm diameter at every 1.5 m centre to centre spacing should be provided along the entire length of the cage. Minimum 6 numbers of vertical bars shall be used for a circular pile and minimum diameter of vertical bar shall be 12 mm. The minimum clear horizontal spacing between the adjacent vertical bars shall be four times the maximum aggregate size in concrete. If required, the bars can be bundled to maintain such spacing.

8.5.3.8.4 Clear cover to all reinforcement including rings in pile shaft shall be not less than 50 mm, which shall be increased to 75 mm in case of aggressive environment of sulphates, etc. In addition, the clear cover of the main reinforcement from the pile tip shall be 75 mm to 100 mm.

8.5.3.9 The design of pile cap and grade beams shall conform to the requirements specified in **8.2.3.12** and **8.2.3.13** respectively.

8.5.4 For detailed information on under-reamed piles regarding control of pile, installation, reference shall be made to good practice [C-2(46)].

8.5.5 *Non-Destructive Testing*

For quality assurance of concrete piles, non-destructive integrity test may be carried out prior to laying of beam/caps, in accordance with accepted standard [C-2(44)].

8.6 Timber Piles

8.6.1 *General*

Piles find application in foundations to transfer loads from a structure to competent sub-surface strata having adequate load-bearing capacity. The load transfer mechanism from a pile to the surrounding ground is complicated and is not yet fully understood, although application of pile foundations is in practice over many decades. Broadly, piles transfer axial loads either substantially by friction along its shaft and/or by the end bearing. Piles are used where either of the above load transfer mechanism is possible depending upon the subsoil stratification at a particular site. Construction of pile foundation requires a careful choice of piling system depending upon the subsoil conditions, the load characteristics of the structure and the limitations of total settlement, differential settlement and any other special requirement of a project. The installation of piles demands careful control on position, alignment and depth and involves specialized skill and experience.

Timber piles, currently do not find extensive use in civil engineering works, however, these piles have been used for compaction of soils, for supporting and protecting water-front structures and also as pile foundations. The choice for using a timber pile is mainly governed by the availability of good quality structural timber, site conditions, particularly the water-table conditions. Timber piles, however, should be avoided because of two reasons, (a) to preserve the environment and green vegetative cover and (b) at most of the places in India, water table has gone down substantially which implies that the timber pile will be only partially under water and this condition is highly undesirable for durability of timber.

Use of treated or untreated piles will depend upon the site conditions and upon whether the work is of permanent or of temporary nature. The timber pile installed should have its entire length embedded under water so that the pile may not get deteriorated. They have the advantages of being comparatively light for their strength and are easily handled. However, they will not withstand as hard driving as steel or concrete piles. Timber has to be selected carefully and treated where necessary for use as piles, as the durability and performance would considerably depend upon the quality of the material and relative freedom from natural defects. If the piles are perennially under salt water condition, treatment of timber is not necessary since by immersion in salt water the life of the pile will be very long. Timber piles are generally used for single and two storeyed structures in back water area where bearing strata is available at a depth less than 12 m. Coconut/Palmyra tree trunks have also been used as timber piles in coastal areas.

8.6.2 *Materials*

8.6.2.1 *Timber*

The timber shall have the following characteristics:

a) Only structural timber shall be used for piles (*see* Part C 'Structural Design/Section 3 Timber and Bamboo, 3(A) Timber' of this NBCS);
b) The length of an individual pile shall be,
   1) the specified length ± 300 mm for piles up to and including 12 m in length; and
   2) the specified length ± 600 mm for piles above 12 m in length.
c) The ratio of heartwood diameter to the pile butt diameter shall be not less than 0.8; and
d) Piles to be used untreated shall have as little sapwood as possible.

8.6.3 *Design Considerations*

8.6.3.1 *General*

Pile foundations shall be designed in such a way that the load from the structure can be transmitted to the sub-surface with adequate factor of safety against shear failure of sub-surface and without causing such settlement, (differential or total), which may result in structural damage and/or functional distress under permanent/transient loading. The pile shaft should have adequate structural capacity to withstand all loads (vertical, lateral or otherwise) which are to be transmitted to the subsoil.

8.6.3.2 *Adjacent structures*

When working near existing structures, care shall be taken to avoid damage to such structures. In case of deep excavations adjacent to piles, proper shoring or other suitable arrangement shall be made to guard against undesired lateral movement of soil. Good practice [C-2(39)] may be used as a guide for studying qualitatively the effect of vibration on persons and structures.

8.6.3.3 *Pile capacity*

See **8.2.3.3**.

8.6.3.4 *Structural capacity*

8.6.3.4.1 The pile shall have necessary structural strength to transmit the loads imposed on it, ultimately to the soil. The structural capacity of the timber piles shall be based on the permissible compressive stress of the timber species.

8.6.3.4.2 The length of pile projected into the pile cap is critical for ensuring lateral load transfer and pull-out resistance. Effective transfer of lateral load may be possible with 100 mm to 200 mm projection into the pile cap concrete depending on the safe lateral load capacity. However, pull-out resistance that can be developed by such small penetration may be limited and the use of timber piles for resisting pull-out resistance may not be viable.

8.6.3.4.3 It is preferable to conduct an initial load test as given in good practice [C-2(40)] to assess the axial capacity, lateral capacity and pull-out capacity of piles. For compaction piles, tests should be done on a group of piles with the cap resting on the ground. The acceptance criteria shall also be as per the provisions given in good practice [C-2(40)]

8.6.3.5 For detailed information on timber piles regarding spacing, classification, control of pile driving, storing and handling, reference shall be made to good practice [C-2(49)].

9 DESIGN AND CONSTRUCTION OF COMBINED PILED RAFT FOUNDATIONS

9.1 General

This clause covers the design and construction of combined piled raft foundation (CPRF) for residential, commercial and industrial structures, tall structures, store houses, storage tanks, etc. The CPRF transmits the load to soil by resistance developed by the components of the foundation, that is, piles and raft, the raft being placed on competent ground. The piles develop resistance along the pile shaft by skin friction and at the pile tip by end bearing. The raft develops the resistance by generating the contact pressure below the raft.

CPRF shall not be used in cases where soil layers of relatively low stiffness (For example, very soft cohesive soil, organic soil, collapsible soil, liquefiable soil) are situated closely beneath the raft. Provisions of this standard shall not be applicable to layered soil with significantly high stiffness (modulus) contrast between the top and bottom layers.

9.2 Necessary Information

For satisfactory design and construction of a combined pile-raft foundation, the following information is necessary:

a) Details of site plan showing location of proposed and adjacent structures;
b) Details of plan and vertical cross-section of the proposed building with position of beam and columns;
c) Different load combinations and their intensity indicating design loads, preferably shown in schematic plan;
d) All the loads due to seismic, wind and water current, etc indicated separately;

e) Limiting values of performance parameters, such as allowable bearing pressure, allowable total settlement, allowable differential settlement, allowable angular distortion, that the foundation and superstructure can withstand;

f) Information on geological history, seismicity, seasonal variation of ground water, climatic factors, for example, sudden flooding, erosion, etc;

g) The design and construction of CPRF requires rigorous geotechnical investigations by using both field and laboratory techniques. The proper dimensioning of CPRF will purely depend up on the outcome of geotechnical investigations. The data obtained from the both the investigations, that is, preliminary investigation for feasibility study and detailed investigation for analyses and design shall be analysed by geotechnical experts;

h) *Geotechnical information* — Sub-surface profile with stratification details, engineering and index properties of founding media to be in accordance with good practice [C-2(8)] and [C-2(50)] and other relevant standards which are essential for the design of CPRF system. Any supplementary investigations should be carried out to obtain additional geotechnical information for design of CPRF;

j) *Extent of geotechnical investigations* — The structure and properties of the substrata and the groundwater conditions shall be known in sufficient detail for any piled-raft construction project. This is necessary to reliably assess the stability and serviceability of the pile and piled-raft foundations and of the overall structure as required by relevant Indian Standards and to assess the effects of foundations on their surroundings. The geotechnical investigations shall extend to sufficient depth below pile tip to record all ground formations and strata influencing the structure and its execution and to identify the load-bearing and deformation properties of the ground. The geotechnical investigations shall be in accordance with good practice [C-2(8)] and [C-2(50)];

k) The geotechnical investigation report and the geotechnical design report, respectively, shall contain all relevant data that can affect CPRF capacity, choice of the execution method and the pile installation. Design soil parameters shall be derived for the CPRF system. The ground information and the parameters related not only to pile capacity, but also to drilling ability, drivability, etc shall be provided;

m) The results of field test on single piles (both ultimate capacity and load- displacement curve). Static test as initial test can be used for design of CPRF, whereas dynamic test is not essential for CPRF design. Dynamic test may be used for additional check, if necessary, on later stage, but not directly related to CPRF design;

n) Dynamic properties of founding media to be taken from geotechnical investigation report; and

p) A review of performance of CPRF in similar locality or ground conditions, if available.

9.3 Design Considerations for Static Load

9.3.1 *General Design Considerations*

9.3.1.1 The design of CPRF should consider the following criteria:

a) Fluctuation in the water table condition and long-term stability of the bearing strata;

b) Consideration for handling sensitive clays and loose bearing soils;

c) Effect of soil excavations;

d) The soil stratigraphy shall be considered when calculating pile responses in the CPRF system;

e) Differential movement of the foundations between the old and new structures;

f) Wherever possible, the centre area of the foundation should be located directly beneath the centre of gravity of the imposed load or else effect of eccentricity should be considered;

g) The load carrying capacity of the combined system subjected to vertical, lateral and moment loadings, total and differential settlement; and

h) Some of the key design considerations are:

   1) Load carrying capacity of the combined system at the desired settlement level for vertical, lateral and moment loadings;

2) Maximum settlement;

3) Differential settlement;

4) Raft moments and shears for the structural design of the raft; and

5) Pile loads and moments, for the structural design of the piles.

9.3.2 *Stages of CPRF Design*

9.3.2.1 The three main stages of CPRF design are:

a) *Preliminary Stage* (*Feasibility Study*) — Assessment of the performance of raft without piles that includes estimation of vertical and lateral bearing capacity and the settlements. Assessment of the feasibility of using a pile raft and the required number of piles to satisfy the design requirement;

b) *General Pile Characteristics* (*Preliminary Design Stage*) — Pile should be designed for the combined resistances; and

c) *Final Detailed Design* — Final detailed design to obtain the number of piles, location and configuration, settlement, bending moment and shear forces in raft and the pile loads and moments.

9.3.3 *Design Steps*

Design philosophy is illustrated in Fig. 10 and the subsequent design steps to be followed are given in **9.3.3.1** to **9.3.3.5**, **9.3.4** and **9.3.5**.

9.3.3.1 The static pile formula based on the ground parameters of good practice [C-2(51)] can be used for estimation of ultimate load capacity of piles. The provision made in good practice [C-2(52)] as applicable should be followed for the design and construction of piles unless otherwise stated. The procedure for pile load testing should be as per good practice [C-2(40)]. Information obtained (load-settlement response) from pile load test shall be interpreted by geotechnical engineer to arrive at the allowable load with reference to a particular settlement. Pile loads and moment for the structural design of the piles should be considered.

9.3.3.2 The stability and capacity of raft based on the ground parameters as per good practice [C-2(34)] and [C-2(53)] can be estimated. The provision made in good practice [C-2(53)] should be followed for the design and construction of raft foundation unless otherwise stated. For estimation of raft settlement, the guidance given in good practice [C-2(54)] may be used. However, the calculation of settlement for entire CPRF system will be different as described in subsequent section of this standard. Raft moment and shear forces for structural design of raft should be considered.

9.3.3.3 The following are the design philosophies:

a) *Conventional Approach* — Piles are designed as a group while making some allowances for contribution of the raft, primarily to load carrying capacity;

b) *Settlement Control Approach* — Piles are placed strategically below the raft in order to reduce the total and differential settlement. However differential settlement reduction is more important than to reduce the overall average settlement; and

c) *Creep Piling Approach* — Piles are designed to operate at a working load at which significant creep starts to occur at some fraction, typically 70 percent to 80 percent, of the ultimate load capacity.

9.3.3.3.1 For CPRF, the design philosophy is generally governed by settlement control approach to meet the serviceability criteria.

9.3.3.4 The vertical load carrying capacity (Q_{CPRF}) of CPRF should be calculated as per Annex K.

9.3.3.5 Dimensionless factors for CPRF design with practical range of parameters are given in Table 6. This guideline should ensure the complete behavioural mechanism of CPRF system.

Table 6 Dimensionless Factors for Combined Piled Raft Foundations

(*Clause* 9.3.3.5)

Sl No.	Dimensionless Factors	Definition	Practical Range
(1)	(2)	(3)	(4)
i)	Pile slenderness ratio	L_p/D_p	10 to 100
ii)	Pile spacing ratio	S_p/D_p	2.5 to 8
iii)	Pile-soil stiffness ratio	$K_{ps} = E_p/E_s$	100 to 10 000
iv)	Raft plan aspect ratio	L_r/B_r	1 to 10
v)	Raft-soil stiffness ratio	$K_{rs} = \dfrac{4E_r B_r t_r^3 (1-\vartheta_s^2)}{3\pi E_s L_r^4 (1-\vartheta_r^2)}$	0.001 to 10

where

B_r = width of the raft;

D_p = diameter of the pile;

E_p = elastic modulus of pile material;

E_r = elastic modulus of raft material;

E_s = elastic modulus of soil;

L_p = length of the pile;

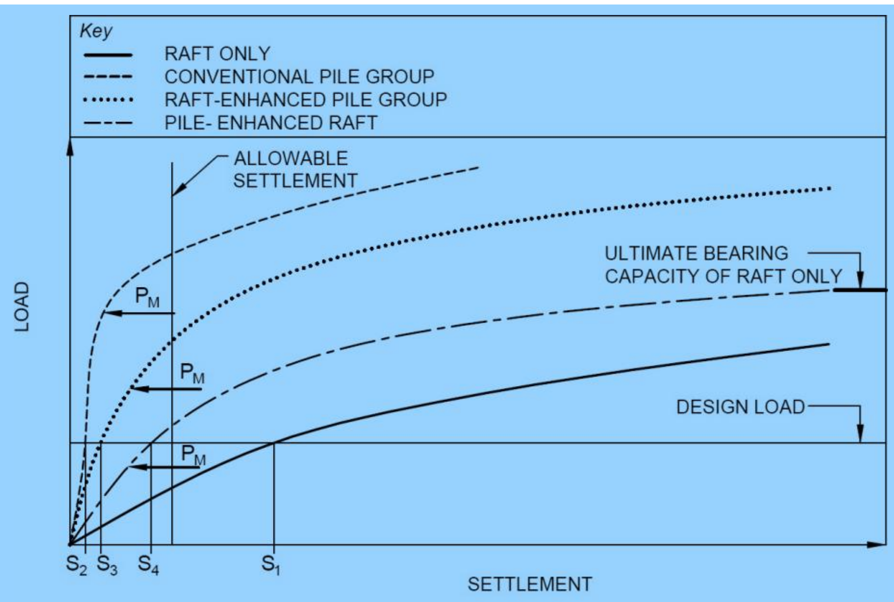
L_r = length of the raft;

S_p = spacing between the piles;

t_r = thickness of raft;

ϑ_r = poisson's ratio of raft material; and

ϑ_s = poisson's ratio of soil.



where
P_M = Pile capacity fully mobilized; and
S_1,S_2, etc = Settlement for raft, pile group, etc

FIG. 10(A) LOAD-SETTLEMENT BEHAVIOUR OF RAFTS, CONVENTIONAL PILE GROUPS AND DIFFERENT TYPES OF CPRF

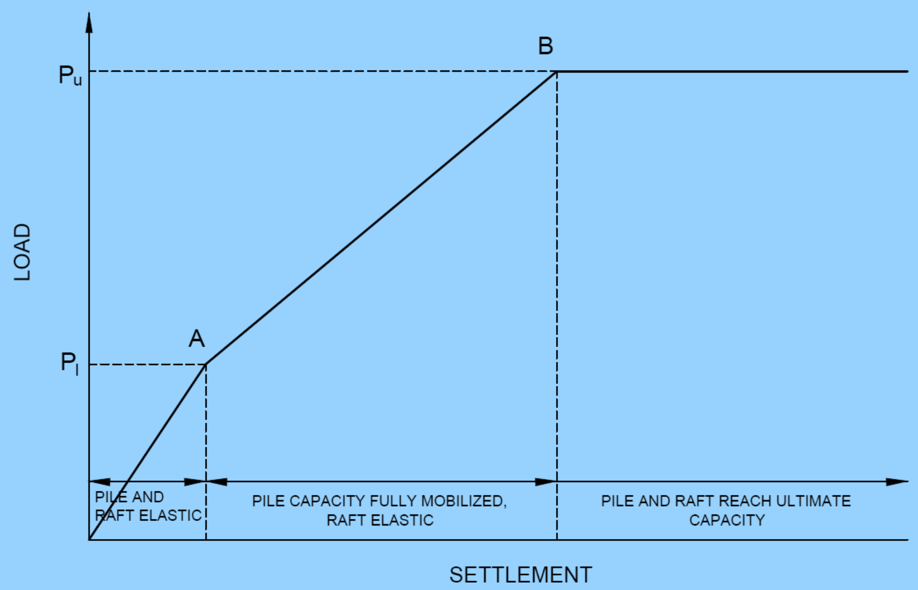


FIG.10(B) SIMPLIFIED LOAD SETTLEMENT CURVE OF A CPRF

FIG. 10 LOAD-SETTLEMENT BEHAVIOUR OF A CPRF

9.3.4 *Computational Model*

Analysis of CPRF shall be performed using computational model which will be able to simulate the appropriate behaviour of single pile. Later from the actual field test results on single pile, the computational model needs to be revised. The computational model shall capture all the interactions as mentioned in Fig. 2(A). Also, the superstructure effect shall be considered appropriately as shown in Fig. 11(A). The two features of estimating the stiffness of soil-pile-raft system are through estimation of pile spring stiffness and raft-soil spring stiffness. The iterative process of convergence of pile and raft response (if implicit) will only come after that. Fig. 11 presents the design model of CPRF system considering soil-pile-raft stiffness. Different values of pile stiffness can occur within a group of piles, depending on their position and on the relative stiffness of the raft.

9.3.4.1 *Estimation of pile spring stiffness*

Stiffness of spring representing the pile shall be estimated by dividing the load carried by pile top element in respective direction with the corresponding top displacement of the pile.

$$\text{Pile spring stiffness, } K_p = \frac{\text{Normal contact force at nodes}}{\text{Nodal displacement}}$$

9.3.4.2 *Estimation of raft-soil spring stiffness*

The soil spring stiffness shall be estimated from the load deformation behaviour of the founding medium as given below:

$$\text{Raft-soil Spring stiffness, } K_r = \frac{\text{Normal contact force at nodes}}{\text{Nodal displacement}}$$

9.3.5 *Allowable Settlement for Static Design Consideration*

For CPRF system, total settlement under gravity load should be restricted within 125 mm and the maximum angular distortion of raft as 1/500. There are four angular distortions (θ_{IL}, θ_{IIL}, θ_{IB} and θ_{IIB}) (Fig. 1) and maximum of these four angular distortions will govern CPRF design for Serviceability Limit State (SLS). The total settlement under gravity load may be taken only as a guide and the permissible total settlement should be decided as per serviceability criteria to fulfil the load sharing mechanism of CPRF system.

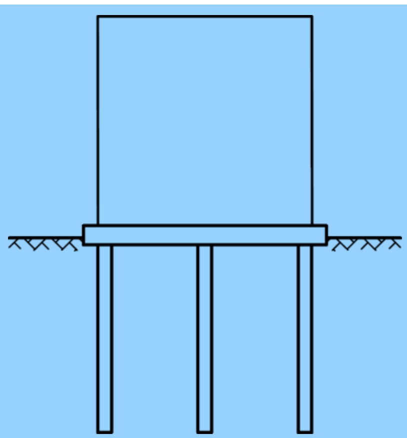


FIG. 11(A) RAFT–PILE–SOIL IN A CPRF

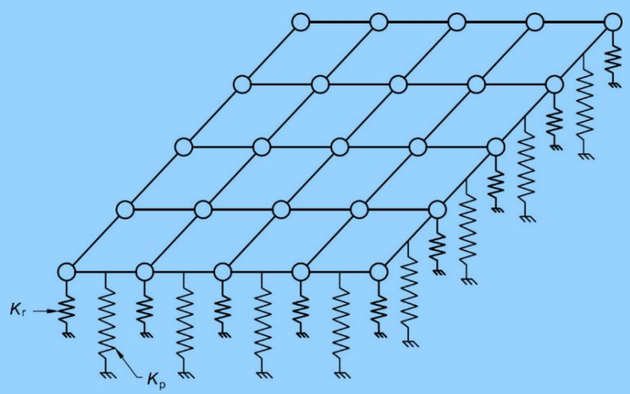


FIG.11(B) PILE AND RAFT-SOIL SPRINGS

FIG. 11 COMBINED ACTION OF SOIL-PILE SYSTEM (TYPICAL DESIGN MODEL)

9.4 Proof of Design and Construction of a CPRF

The examination of the design and the construction of a CPRF with respect to the geotechnical engineering aspects should be controlled by a geotechnical expert particularly qualified on this subject. This includes the construction for piles and deciding the foundation level. The protocols of the acceptance procedure and the measured values have to be included into the examination. The following aspects shall be considered:

a) Critical review of the soil investigation report (field and laboratory tests), including the selection of input parameters;

b) Evaluation of the computational model used for the design of the CPRF and the computation results by using independent comparative calculations. Apart from the independent comparative calculations, additional quality assurance measures in design should be ensured;

c) Assessment of the effects on the adjacent structures; and

d) Examination of the field monitoring (measuring) program within the construction process of the CPRF.

9.5 Monitoring of a CPRF

The load-settlement behaviour and the load transfer within a CPRF should be monitored for further use by a geotechnical expert. The monitoring comprises of geotechnical and geodetic measurements at the new building and also at the adjacent buildings for settlement. The monitoring of the contact pressure at the level of raft, axial load in selected piles, settlement at the level of raft, ground water table and pore-pressure may be adopted for important projects. The monitoring of a CPRF is an elementary and indispensable component of the safety concept and should be used for the following purposes:

a) Observational method should be followed for monitoring of the response of CPRF both during and post construction phases for projects of high importance;

b) Verification of the computational model and the computational approaches;

c) Detection of possible critical conditions in CPRF system;

d) Examination of the quality assurance during the construction and post construction process; and

e) Examination of distribution of loads may be implemented on piles in centre and edges to capture soil-pile-raft interactions.

The monitoring program should be designed by a geotechnical expert during the design phase depending on importance of the project, complexity of the ground condition and long-term performance. Summary of desired instrumentations and their locations should be highlighted in plan as well as sectional view of the foundation layout.

9.6 Structural Design

The general provision for load, shrinkage, creep and temperature effects and provision of design, reinforcement and detailing shall conform to good practice [C-2(47)] and other relevant standards.

9.7 Seismic Design

Design of the CPRF system considering the seismicity of the ground and entire structure and sub-structural system should be handled by experts to ensure conformity to various relevant standards. Both inertial and kinematic effects should be considered in the seismic design.

10 OTHER FOUNDATIONS, SUBSTRUCTURES AND FOUNDATIONS FOR SPECIAL STRUCTURES

10.1 Pier Foundations

10.1.1 *Design Considerations*

10.1.1.1 *General*

The design of concrete piers shall conform to the requirements for columns specified in Part C 'Structural Design/ Section 5 Structural Concrete' of this NBCS. If the bottom of the pier is to be belled so as to increase its load carrying capacity, such bell shall be at least 300 mm thick at its edge. The sides shall slope at an angle of not less than 60° with the horizontal. The least permissible dimensions shall be 600 mm, irrespective of the pier being circular, square or rectangular. Piers of smaller dimensions if permitted shall be designed as piles (*see* **8**).

10.1.1.2 *Plain concrete piers*

The height of the pier shall not exceed 6 times the least lateral dimension. When the height exceeds 6 times the least lateral dimension, buckling effect shall be taken into account, but in no case shall the height exceed 12 times the least lateral dimension.

When the height exceeds 6 times the least lateral dimension, the deduction in allowable stress shall be given by the following formula:

$$f_c{}' = f_c \left(1.3 - \frac{H}{20D}\right)$$

where

$f_c{}'$ = reduced allowable stress;

f_c = allowable stress;

D = least lateral dimension; and

H = height of pier.

NOTE — The above provision shall not apply for piers where the least lateral dimension is 1.8 m or greater.

10.1.1.3 *Reinforced concrete piers*

When the height of the pier exceeds 18 times its least dimension, the maximum load shall not exceed:

$$P' = P\left(1.5 - \frac{H}{36D}\right)$$

where

D = least lateral dimension;

H = height of the pier measured from top of bell, if any, to the level of cut-off of pier;

P' = permissible load; and

P = permissible load when calculated as axially loaded short column.

10.2 Design and construction of machine foundations, diaphragm walls, etc, shall be carried out in accordance with good practice [C-2(55)].

11 GROUND IMPROVEMENT

11.1 In poor and weak subsoils, the design of conventional foundation for structures and equipment may present problems with respect to both sizing of foundation as well as control of foundation settlements. A viable alternative in certain situations, is to improve the subsoil to an extent such that the subsoil would develop an adequate bearing capacity and foundations constructed after subsoil improvement would have resultant settlements within acceptance limits. This method/technique is called ground improvement which is used to improve *in-situ* soil characteristics by improving its engineering performance as per the project requirement by altering its natural state, instead of having to alter the design in response to the existing ground limitations. The improvement is in terms of increase in bearing capacity, shear strength, reducing settlement and enhancing drainage facility, mitigating liquefaction potential, etc, of soil, as also in improving lateral capacity in case of deep foundations.

11.2 For provisions with regard to necessary data to be collected to establish the need for ground improvement at a site; considerations for establishing need for ground improvement methods; selection of ground improvement techniques; equipment and accessories for ground improvement; control of ground improvement works and recording of data, reference shall be made to accepted standard [C-2(56)]. Annex L presents various methods of ground improvement along with principles, applicability to various soil conditions, material requirements, equipment required, results likely to be achieved and limitations. This table may be referred to as guidance for selecting the proper method for a situation. *See also* accepted standard [C-2(56)] for details.

11.2.1 For provisions relating to ground improvement by reinforcing the ground using stone columns so as to meet the twin objective of increasing the bearing capacity with simultaneous reduction of settlements, reference shall be made to accepted standard [C-2(57)].

11.2.2 Whenever soft cohesive soil strata underlying a structure are unable to meet the basic requirements of safe bearing capacity and tolerable settlement, ground improvement is adopted to make it suitable for supporting the proposed structure. Both the design requirements that is shear strength and settlement under loading, can be fulfilled by consolidating the soil by applying a preload, if necessary, before the construction of the foundation. This consolidation of soil is normally accelerated with the use of vertical drains. For provisions relating to ground improvement by preconsolidation using vertical drains, reference shall be made to accepted standard [C-2(58)].

11.2.3 Use of suitable geo-synthetics/geo-textiles may be made in an approved manner for ground improvement, where applicable; *see also* accepted standards [C-2(59)].

12 GEOTECHNICAL INSTRUMENTATION

12.1 Geotechnical instrumentation is essential for monitoring the behaviour of soil, rock and structures in civil engineering projects, ensuring safety and design performance. A variety of instruments are deployed at specific locations, where they can provide critical data for various purposes. The data gathered from these instruments allows engineers to assess risks, verify design assumptions and implement corrective actions if necessary.

The typical list of geotechnical instruments, but not limited to these, along with their objectives, is provided in Table 7.

Table 7 Typical List of Geotechnical Instrumentation
(*Clause* 12.1)

Sl No.	Instrumentation	Objective	Applicable for
(1)	(2)	(3)	(4)
i)	Inclinometers	Measurement of lateral deflections	Deflection monitoring of deep excavations, shoring systems, landslide/slope movements, lateral load test on piles
ii)	Piezometers	Measurement of underground water pressure (pore water pressure)	To monitor groundwater conditions
iii)	Vibration monitors (Seismographs)	To monitor vibrations due to various construction activities	Monitoring of adjacent critical structures during general construction, demolition, pile driving, blasting etc.
iv)	Strain gauges	To measure strain in structural elements	Strain (force) measurement in pile foundations, struts

Table 7 (*Concluded*)

Sl No.	Instrumentation	Objective	Applicable for
(1)	(2)	(3)	(4)
v)	Load cells	To measure loads in various structural elements	Load measurement in rock bolts, foundation anchors, struts, pile foundation during load testing
vi)	Pressure pads/Pressure cells	monitoring long-term pressure imposed by soil on the foundation element	For embankments, rafts, retaining walls,
vii)	Extensometer	Deformation measurement	Measurement of deformation of soil, rock, or can be used during testing of pile foundation to measure deformation of piles at various levels
viii)	Permanent settlement markers	Settlement monitoring	Can be used for monitoring of settlement/ deflections of various structures such as raft, retaining system, slopes, etc.
ix)	Crackmeters	Measure displacement	To measure plate displacement during bi-directional static load test (BDSLT)

ANNEX A

(*Clauses* 7.4.1.11 *and* B-3.3)

DETERMINATION OF MODULUS OF ELASTICITY (E_s) AND POISSONS'S RATIO (μ)

A-1 DETERMINATION OF MODULUS OF ELASTICITY(E_s)

The modulus of elasticity is a function of composition of the soil, its void ratio, stress history and loading rate. In granular soils it is a function of the depth of the strata, while in cohesive soil it is markedly influenced by the moisture content. Due to its great sensitivity to sampling disturbance, accurate evaluation of the modulus in the laboratory is extremely difficult. For general cases, therefore, determination of the modulus may be based on field tests (*see* **A-2**). Where properly equipped laboratory and sampling facility is available, E_s may be determined in the laboratory (*see* **A-3**).

A-2 FIELD DETERMINATION

A-2.1 The value of E_s shall be determined from plate load test in accordance with accepted standard [C-2(9)].

$$E_s = qB \frac{(1-\mu^2)}{s} I_w$$

where

B = least lateral dimension of test plate;

influence ratio;

I_w = 0.82 for a square plate;

1.00 for a circular plate;

q = intensity of contact pressure;

s = settlement; and

μ = poisson's ratio

NOTE — While this procedure may be adequate for light or less important structures under normal conditions, relevant laboratory tests or field tests are essential in the case of unusual soil types and for all heavy and important structures. Plate load test, though useful in obtaining the necessary information about the soil with particular reference to design of foundation has some limitations. The test results reflect only the character of the soil located within a depth of less than twice the width of the bearing plate. Since the foundations are generally larger than the test plates, the settlement and shear resistance will depend on the properties of a much thicker stratum. Moreover, this method does not give the ultimate settlements particularly in case of cohesive soils. Thus, the results of the test are likely to be misleading, if the character of the soil changes at shallow depths, which is not uncommon. A satisfactory load test should, therefore, include adequate soil exploration {*see* good practice [C-2(8)]} with due attention being paid to any weaker stratum below the level of the footing. Another limitation is the concerning of the effect of size of foundation. For clayey soils the bearing capacity (from shear consideration) for a larger foundation is almost the same as that for the smaller test plate. But in dense sandy soils the bearing capacity increases with the size of the foundation. Thus, tests with smaller size plate tend to give conservative values in dense sandy soils. It may, therefore, be necessary to test with plates of at least three sizes and the bearing capacity results extrapolated for the size of the actual foundation (minimum dimensions in the case of rectangular footings).

A.2.1.1 The average value of E_s shall be based on a number of plate load tests carried out over the area, the number and location of the tests, depending upon the extent and importance of the structure.

A-2.1.2 *Effect of Size*

In granular soils the value of E_s corresponding to the size of the raft shall be determined as follows:

$$E_s = E_p \frac{B_f}{B_p} \left[\frac{B_f + B_p}{2B_f} \right]^2$$

Where B_f, B_p represent sizes of foundation and plate and E_p is the modulus determined by the plate load test.

A-2.2 For stratified deposits or deposits with lenses of different materials, results of plate load test will be unreliable and static cone penetration tests may be carried out to determine E_s.

A-2.2.1 Static cone penetration tests shall be carried out in accordance with accepted standard [C-2(41)]. Several tests shall be carried out at regular depth intervals up to a depth equal to the width of the raft and the results plotted to obtain an average value of E_s.

A-2.2.2 The value of E_s may be determined from the following relationship:

$$E_s = 2\ C_{kd}$$

where C_{kd} = cone resistance, in kN/m^2

A-3 LABORATORY DETERMINATION OF E_S

A-3.1 The value of E_s shall be determined by conducting triaxial test in the laboratory in accordance with accepted standard [C-2(60)] on samples collected with least disturbances.

A-3.2 In the first phase of the triaxial test, the specimen shall be allowed to consolidate fully under an all-round confining pressure equal to the vertical effective overburden stress for the specimen in the field. In the second phase, after equilibrium has been reached, further drainage shall be prevented and the deviator stress shall be increased from zero value to the magnitude estimated for the field loading condition. The deviator stress shall then be reduced to zero and the cycle of loading shall be repeated.

A-3.3 The value of E_s shall be taken as the tangent modulus at the stress level equal to one-half the maximum deviator stress applied during the second cycle of loading.

ANNEX B

(*Clauses* 7.4.1.11 *and* C-3)

DETERMINATION OF MODULUS OF SUB-GRADE REACTION

B-1 GENERAL

B-1 The modulus of subgrade reaction (*k*) as applicable to the case of load through a plate of size 300 mm × 300 mm or beams 300 mm wide on the soils is given in Table 8 for cohesionless soils and in Table 9 for cohesive soils. Unless more specific determination of *k* is done (*see* **B-2** and **B-3**), these value may be used for design of raft foundation in cases where the depth of the soil affected by the width of the footing may be considered isotropic and the extrapolation of plate load test results is valid.

B-2 FIELD DETERMINATION

B-2.1 In cases where the depth of the soil affected by the width of the footing may be considered as isotropic, the value of *k* may be determined in accordance with accepted standard [C-2(10)]. The test shall be carried out with a plate of size not less than 300 mm.

B-2.2 The average value of *k* shall be based on a number of plate load tests carried out over the area, the number and location of the tests depending upon the extent and importance of the structure.

B-3 LABORATORY DETERMINATION

B-3.1 For stratified deposits or deposits with lenses of different materials, evaluation of *k* from plate load test will be unrealistic and its determination shall be based on laboratory tests {*see* good practice [C-2(60)]}.

B-3.2 In carrying out the test, the continuing cell pressure may be so selected as to be representative of the depth of the average stress influence zone (about 0.5 B to B).

B-3.3 The value of *k* shall be determined from the following relationship:

$$k = 0.65 \times \sqrt[12]{\left(\frac{E_s B^4}{EI}\right)} \cdot \frac{E_s}{(1-\mu^2)} \cdot \frac{1}{B}$$

where

- B = width of the footing;
- E = young’s modulus of foundation material;
- E_s = modulus of elasticity of soil (*see* Annex A);
- I = moment of inertia of the foundation; and
- μ = poisson’s ratio of soil.

B-4 CALCULATIONS

When the structure is rigid (*see* Annex C), the average modulus of subgrade reaction may also be determined as follows:

$$k_s = \frac{\text{Average contact pressure}}{\text{Average settlement of the raft}}$$

Table 8 Modulus of Subgrade Reaction (*k*) for Cohesionless Soils

(*Clause* B-1)

Sl No.	Soil Characteristic		Modulus of Subgrade Reaction (*k*)[1) kN/m³	
	Relative Density	Standard Penetration Test Value (N) (Blows per 300 mm)	For Dry or Moist State	For Submerged State
(1)	(2)	(3)	(4)	(5)
i)	Loose	< 10	15 000	9 000
ii)	Medium	10 to 30	15 000 to 47 000	9 000 to 29 000
iii)	Dense	30 and over	47 000 to180 000	29 000 to 108 000

Table 9 Modulus of Subgrade Reaction (*k*) for Cohesive Soils

(*Clause* B-1)

Sl No.	Soil Characteristic		Modulus of Subgrade Reaction (*k*)[1) kN/m³
	Consistency	Unconfined Compressive Strength kN/m²	
(1)	(2)	(3)	(4)
i)	Stiff	100 to 200	27 000
ii)	Very stiff	200 to 400	27 000 to 54 000
iii)	Hard	400 and over	54 000 to 108 000

[1) The above values apply to a square plate 300 mm × 300 mm or beams 300 mm wide.

[1) The values apply to a square plate 300 mm × 300 mm. The above values are based on the assumption that the average loading intensity does not exceed half the ultimate bearing capacity.

ANNEX C

(*Clauses* 7.4.4.1.2 (a), (b), 7.4.4.2 *and* B-4)

RIGIDITY OF SUPERSTRUCTURE AND FOUNDATION

C-1 DETERMINATION OF THE RIGIDITY OF THE STRUCTURE

The flexural rigidity *EI* of the structure of any section may be estimated according to the relation given below (*see also* Fig. 12)

$$EI = \frac{E_1 I_i b^2}{2H^2} + \sum \ E_2\, I_b \left[1 + \frac{(I_u{}' + I_1{}')b^2}{(I_b{}' + I_u{}' + I_f{}')l^2}\right]$$

where

b = length or breadth of the structure in the direction of bending, in m;

E_1 = modulus of elasticity of the infilling material (wall material), in kN/m^2;

E_2 = modulus of elasticity of the frame material, in kN/m^2;

H = total height of the infilling, in m;
h_l = length of the lower column, in m;
h_u = length of the upper column, in m;
I_b = moment of inertia of the beam, in m^4;
$I_b{}'$ = $\frac{I_b}{l}$;
$I_f{}'$ = $\frac{I_f}{l}$;
I_i = moment of inertia of the infilling, in m^4;
$I_u{}'$ = $\frac{I_U}{h_U}$;
$I_1{}'$ = $\frac{I_l}{h_l}$;
I_l = moment of inertia of the lower column, in m^4;
I_f = moment of inertia of the foundation beam or raft in m^4;
I_u = moment of inertia of the upper column, in m^4; and
l = spacing of the columns, in m.

NOTE — The summation is to be done over all the storeys including the foundation beam or raft. In the case of the foundation, $I_f{}'$ replaces $I_b{}'$ and I_l becomes zero, whereas for the topmost beam I_u becomes zero.

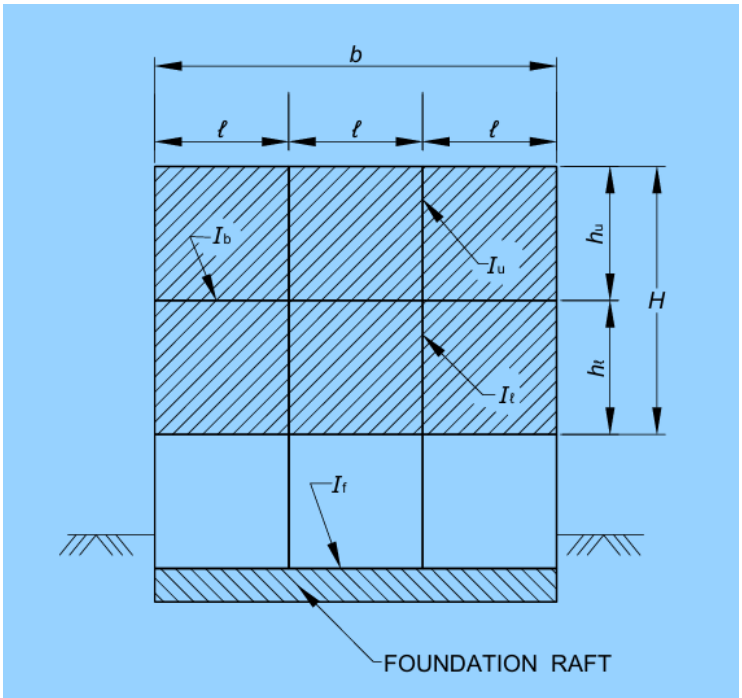


FIG.12 DETERMINATION OF RIGIDITY OF STRUCTURE

C-2 RELATIVE STIFFNESS FACTOR, *K*

C-2.1 Whether a structure behaves as rigid or flexible depends on the relative stiffness of the structure and the foundation soil. This relation is expressed by the relative stiffness factor K given below:

a) For the whole structure, $K = \frac{EI}{E_s b^3 a}$;

b) For rectangular rafts, $K = \frac{E}{12E_s}\left(\frac{d}{b}\right)^3$; and

c) For circular rafts, $K = \frac{E}{12E_s}\left(\frac{d}{2R}\right)^3$

where

a	=	length perpendicular to the section under investigation, in m;
b	=	length of the section in the bending axis, in m;
d	=	thickness of the raft or beam, in m;
EI	=	flexural rigidity of the structure over the length (a), in kN/m^2;
E_s	=	modulus of compressibility of the foundation soil, in kN/m^2; and
R	=	radius of the raft, in m.

C-2.1.1 For $K > 0.5$, the foundation may be considered as rigid [*See* **7.4.4.1(a)**].

C-3 DETERMINATION OF CRITICAL COLUMN SPACING

Evaluation of the characteristics λ is made as follows:

$$\lambda = \sqrt[4]{\left(\frac{kB}{4E_c I}\right)}$$

where

B	=	width of raft B, in m;
E_c	=	modulus of elasticity of concrete in kN/m^2;
I	=	moment of inertia of raft, in m^4; and
k	=	modulus of subgrade reaction, in kN/m^3, for footing of width B, in m (*see* Annex B).

ANNEX D

(*Clause* 7.4.4.1.3)

CALCULATION OF PRESSURE DISTRIBUTION BY CONVENTIONAL METHOD

D-1 DETERMINATION OF PRESSURE DISTRIBUTION

The pressure distribution (q) under the raft shall be determined by the following formula:

$$q = \frac{Q}{A} \pm \frac{Q{e_y}'}{{I_x}'}y \pm \frac{Q{e_x}'}{{I_y}'}x$$

where

A	=	total area of the raft,
$e_x{}'$, $e_y{}'$, $I_x{}'$, $I_y{}'$	=	eccentricities and moments of inertia about the principal axes through the centroid of the section;
x, y	=	co-ordinates of any given point on the raft with respect to the x and y axes passing through the centroid of the area of the raft; and
Q	=	total vertical load on the raft.

$I_x{}'$, $I_y{}'$, $e_x{}'$, $e_y{}'$ may be calculated from the following equations:

$$I_x' = I_x - \frac{I_{xy}^2}{I_y}$$

$$I_y' = I_y - \frac{I_{xy}^2}{I_x}$$

$$e_x' = e_x - \frac{I_{xy}}{I_x} e_y$$

$$e_y' = e_y - \frac{I_{xy}}{I_y} e_x$$

where

e_x, e_y	=	eccentricities in the x and y directions of the load from the centroid;
I_{xy}	=	$\int xy \cdot dA$ for the whole area about x and y axes through the centroid; and
I_x, I_y	=	moment of inertia of the area of the raft respectively about the x and y axes through the centroid.

For a rectangular raft, the equation simplifies to :

$$q = \frac{Q}{A}(1 \pm \frac{12 e_y y}{b^2} \pm \frac{12 e_x x}{a^2})$$

where

a and b = the dimensions of the raft in the x and y directions, respectively.

NOTE — If one or more of the values of (q) are negative as calculated by the above formula, it indicates that the whole area of foundation is not subject to pressure and only a part of the area is in contact with the soil and the above formula will still hold good, provided the appropriate values of I_x, I_y, I_{xy}, e_x and e_y, are used with respect to the area in contact with the soil instead of the whole area.

ANNEX E

[*Clause* 7.4.4.2(A)]

CONTACT PRESSURE DISTRIBUTION AND MOMENTS BELOW FLEXIBLE FOUNDATION

E-1 CONTACT PRESSURE DISTRIBUTION

E-1.1 The distribution of contact pressure is assumed to be linear with the maximum value attained under the columns and the minimum value at mid span.

E-1.2 The contact pressure for the full width of the strip under an interior column load located at a point *i* can be determined as [*see* Fig. 13(A)]:

$$p_i = \frac{5P_i}{\bar{l}} + \frac{48M_i}{(\bar{l})^2}$$

where

l	=	average length of adjacent span, in m;
M_i	=	moment under an interior columns loaded at *I*; and
P_i	=	column load, in t at point *i*.

E-1.3 The minimum contact pressure for the full width of the strip at the middle of the adjacent spans can be determined as [*see* Fig. 13(A) and Fig. 13(B)]:

$$p_{\mathrm{ml}} = 2P_i\left(\frac{l_r}{l_l\bar{l}}\right) - p_i\left(\frac{\bar{l}}{l_l}\right)$$

$$p_{\mathrm{mr}} = 2P_i\left(\frac{l_l}{l_r\bar{l}}\right) - p_i\left(\frac{\bar{l}}{l_r}\right)$$

$$p_m = \frac{p_{\mathrm{ml}} + p_{\mathrm{mr}}}{2}$$

where l_r, l_l are as shown in Fig. 13(A).

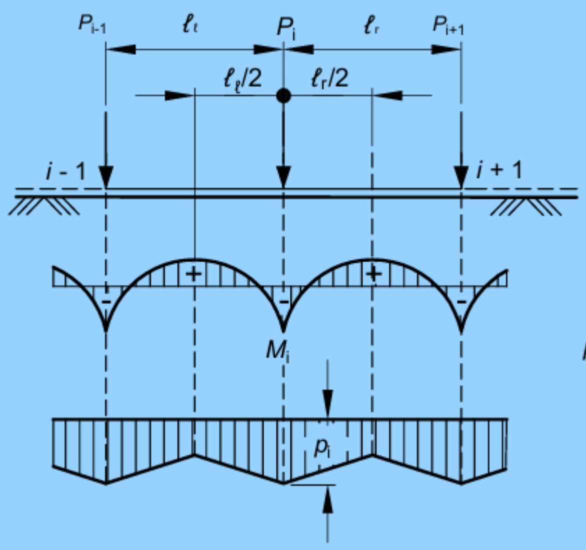


FIG.13(A) MOMENT AND PRESSURE DISTRIBUTION AT INTERIOR COLUMN

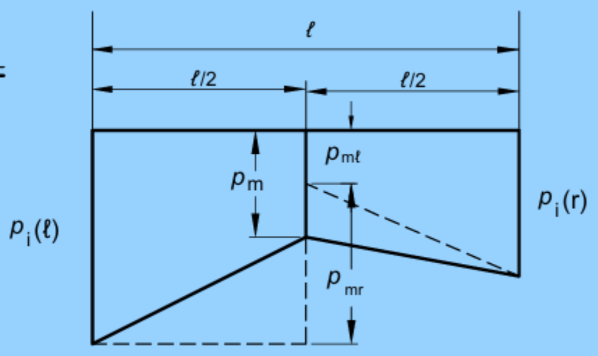


FIG.13(B) PRESSURE DISTRIBUTION OVER AN INTERIOR SPAN

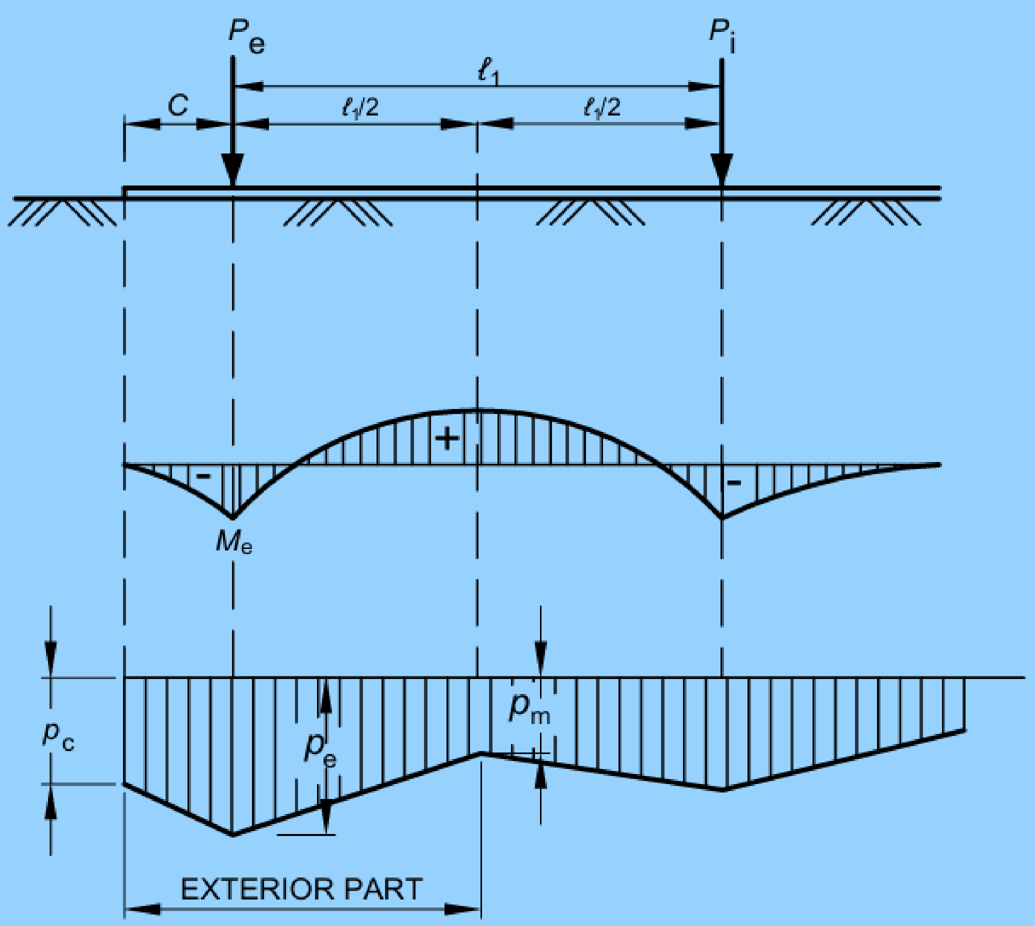


FIG.13(C) MOMENT AND PRESSURE DISTRIBUTION AT EXTERIOR COLUMNS

FIG. 13 MOMENT AND PRESSURE DISTRIBUTION AT COLUMNS

E-1.4 If E-2.3(a) governs the moment under the exterior columns, contact pressures under the exterior columns and at end of strip can be determined as [*see* Fig. 13(C)]:

$$p_e = \frac{4P_e + \frac{6M_e}{C} - p_m l_1}{C + l_1}$$

$$p_c = -\frac{3M_e}{C^2} - \frac{p_e}{2}$$

where P_e, p_m, M_e, l_1, C are as shown in Fig. 13(C).

E-1.5 If E-2.3(b) governs the moment under the exterior columns, the contact pressures are determined as [*see* Fig. 13(C)] :

$$p_e = p_c \frac{4P_e - p_m l_1}{4C + l_1}$$

E-2 BENDING MOMENT DIAGRAM

E-2.1 The bending moment under an interior column located at *i* [*see* Fig. 11(A)] can be determined as :

$$M_i = -\frac{P_i}{4\lambda}(0.24\lambda\bar{l} + 0.16)$$

E-2.2 The bending moment at mid span is obtained as [*see* Fig. 11(A)]:

$$M_m = M_o + M_i$$

M_i = average of negative moments M_i at each end of the bay

M_o = moment of simply supported beam

= $\frac{l^2}{48}[p_i(l) + 4\overline{p_m} + p_i(r)]$.

where l, $p_i(l)$, $p_i(r)$, $\overline{p_m}$ are as shown in Fig. 11(B).

E-2.3 The bending moment M_e under exterior columns can be determined as the least of [*see* Fig. 13(C)]:

a) $M_{e1} = -\frac{P_e}{4\lambda}(0.13\lambda l_1 + 1.06\lambda C - 0.50)$; and

b) $M_{e2} = -\frac{(4P_e - p_m l_1)}{(4C + l_1)}\frac{C^2}{2}$;

ANNEX F

[*Clause* 7.4.4.2(b)]

FLEXIBLE FOUNDATION — GENERAL CONDITION

F-1 CLOSED FORM SOLUTION OF ELASTIC PLATE THEORY

F-1.1 For a flexible raft foundation with non-uniform column spacing and load intensity, solution of the differential equation governing the behaviour of plates on elastic foundation (Winkler type) gives radial moment (M_r) tangential moment (M_t) and deflection (w) at any point by the following expressions:

$$M_r = -\frac{P}{4}[Z_4\left(\frac{r}{L}\right) - (1-\mu)\frac{Z_3'\left(\frac{r}{L}\right)}{\left(\frac{r}{L}\right)}]$$

$$M_t = -\frac{P}{4}[\mu Z_4\left(\frac{r}{L}\right) + (1-\mu)\frac{Z_3'\left(\frac{r}{L}\right)}{\left(\frac{r}{L}\right)}]$$

$$w = \frac{PL^2}{4D} \cdot Z_3\left(\frac{r}{l}\right)$$

where

L	=	radius of effective stiffness;
	=	$\left(\frac{D}{k}\right)^{1/4}$;
P		column load; and
r		distance of the point under investigation from column load along radius.

where

D	=	flexural rigidity of the foundation;
k	=	modulus of subgrade reaction for footing of width B;
P	=	$\frac{Et^2}{12(1-\mu)^2}$
E	=	modulus of elasticity of the foundation material;
T	=	raft thickness; and
μ	=	possion’s ratio of the foundation material.

$$\left.\begin{array}{l} Z_3\left(\frac{r}{L}\right), \\ Z_3'\left(\frac{r}{L}\right), \\ Z_4\left(\frac{r}{L}\right) \end{array}\right\}$$ = functions of shear, moment and deflection (*see* Fig. 14)

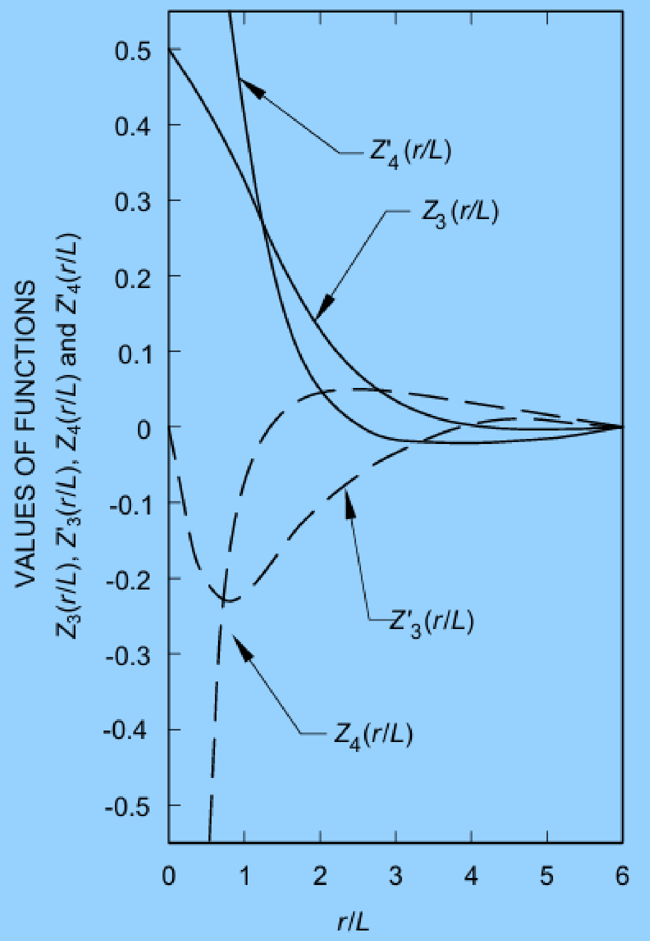


FIG.14 FUNCTIONS FOR SHEAR MOMENT AND DEFLECTION

F-1.2 The radial and tangential moments can be converted to rectangular co-ordinates:

$$M_x = M_r \cos^2 \varphi + M_t \sin^2 \varphi$$
$$M_y = M_r \sin^2 \varphi + M_t \cos^2 \varphi$$

where

Φ = angle with x-axis to the line jointing origin to the point under consideration.

F-1.3 The shear, Q per unit width of raft can be determined by :

$$Q = -\frac{P}{4L} Z_4' \left(\frac{r}{L}\right)$$

$Z_4' \left(\frac{r}{L}\right)$ = function for shear (*see* Fig. 14).

F-1.4 When the edge of the raft is located within the radius of influence, the following corrections are to be applied. Calculate moments and shears perpendicular to the edge of the raft within the radius of influence, assuming the raft to be infinitely large. Then apply opposite and equal moments and shears on the edge of the mat. The method for beams on elastic foundation may be used.

F-1.5 Finally, all moments and shears calculated for each individual column and wall are superimposed to obtain the total moment and shear values.

ANNEX G

(*Clauses* 8.2.3.3.1.1, 8.2.3.3.1.4 *and* 8.5.3.4)

LOAD CARRYING CAPACITY OF PILES — STATIC ANALYSIS

G-1 PILES IN GRANULAR SOILS

The ultimate load capacity (Q_u) of piles, in kN, in granular soils is given by the following formula:

$$Q_u = A_p(\tfrac{1}{2} D \gamma N_\gamma + P_D N_q) + \sum_{i=1}^{n} K_i \, P_{Di} \tan \delta_i A_{si} \qquad \ldots\ldots\ldots(1)$$

The first term gives end bearing resistance and the second term gives skin friction resistance.

where

A_{si}	=	surface area of pile shaft in the *i*th layer, in m^2;
A_p	=	cross-sectional area of pile tip, in m^2;
D	=	diameter of pile shaft, in m;
N_γ, N_q	=	bearing capacity factors depending upon the angle of internal friction, ϕ at pile tip;
K_i	=	coefficient of earth pressure applicable for the *i*th layer (*see* Note 3);
P_D	=	effective overburden pressure at pile tip, in kN/m^2 (*see* Note 5);
P_{Di}	=	effective overburden pressure for the *i*th layer, in kN/m^2;
$\sum_{i=1}^{n}$	=	summation for layers 1 to *n* in which pile is installed and which contribute to positive skin friction;
δ_i	=	angle of wall friction between pile and soil for the *i*th layer; and
γ	=	effective unit weight of the soil at pile tip, in kN/m^3.

NOTES

1 N_γ factor can be taken for general shear failure according to good practice [C-2(34)].

2 N_q factor will depend on the nature of soil, type of pile, the *L*/*B* ratio and its method of construction. The values applicable for driven and bored piles are given in Fig. 15(A) and Fig. 15(B) respectively.

3 K_i, the earth pressure coefficient depends on the nature of soil strata, type of pile, spacing of piles and its method of construction. For bored piles in loose to dense sand with ϕ varying between 30°and 40°, K_i values in the range of 1 to 2 may be used for driven piles and in case of bored piles, K_i values in the range of 1 to 1.5 may be used.

4 δ, the angle of wall friction may be taken equal to the friction angle of the soil around the pile shaft.

5 In working out pile capacity by static formula, the maximum effective overburden at the pile tip should correspond to the critical depth, which may be taken as 15 times the diameter of the pile shaft for $\phi \leq 30°$ and increasing to 20 times for $\phi \geq 40°$.

6 For piles passing through cohesive strata and terminating in a granular stratum, a penetration of at least twice the diameter of the pile shaft should be given into the granular stratum.

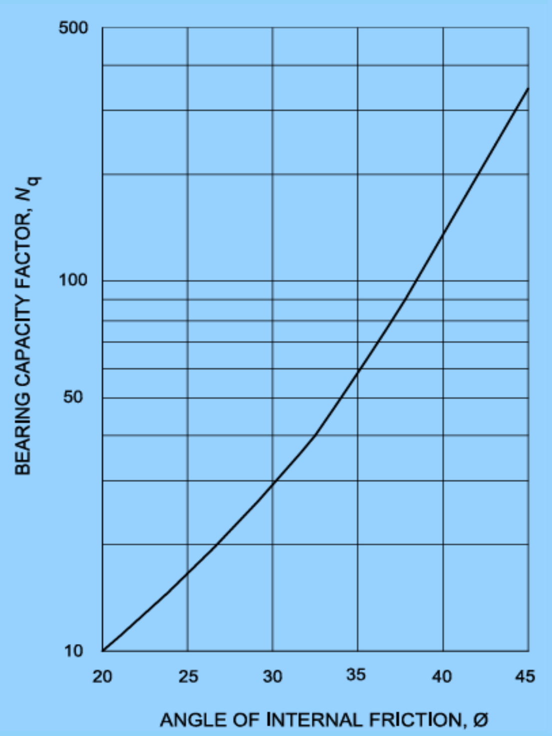


FIG.15(A) BEARING CAPACITY FACTOR, N_Q FOR DRIVEN PILES

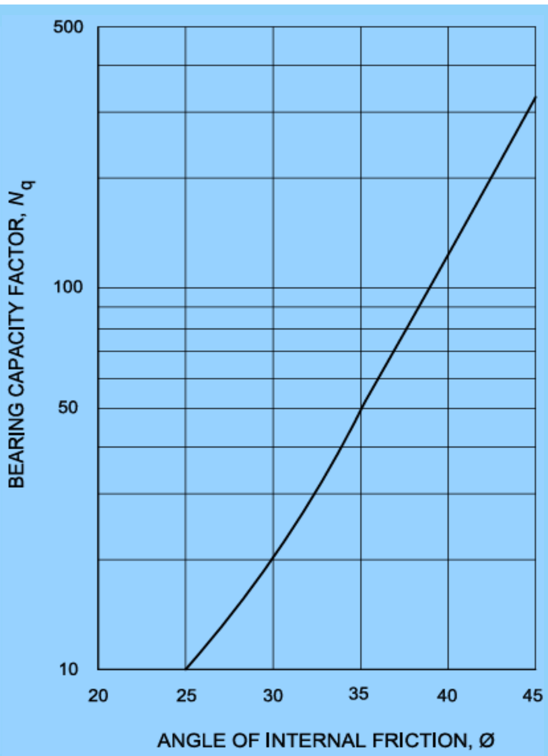


FIG.15(B) BEARING CAPACITY FACTOR, N_Q FOR BORED PILES

FIG. 15 BEARING CAPACITY FACTOR, N_Q

G-2 PILES IN COHESIVE SOILS

The ultimate load capacity (Q_u) of piles, in kN, in cohesive soils is given by the following formula:

$$Q_u = A_p N_c c_p + \sum_{i=1}^{n} \alpha_i c_i A_{si} \qquad \text{...............(2)}$$

The first term gives end bearing resistance and the second term gives the skin friction resistance.

where

α_i	=	adhesion factor for the *i*th layer depending on the consistency of soil (*see* Note);
A_p	=	cross-sectional area of pile tip, in m^2;
A_{si}	=	surface area of pile shaft in the *i*th layer, in m^2;
c_i	=	average cohesion for the *i*th layer, in kN/m^2;
c_p	=	average cohesion at pile tip, in kN/m^2;
N_c	=	bearing capacity factor, may be taken as 9; and
$\sum_{i=1}^{n}$	=	summation for layers 1 to *n* in which pile is installed and which contribute to positive skin friction.

NOTE — The value of adhesion factor, α_i depends on the undrained shear strength of the clay and may be obtained from Fig. 16.

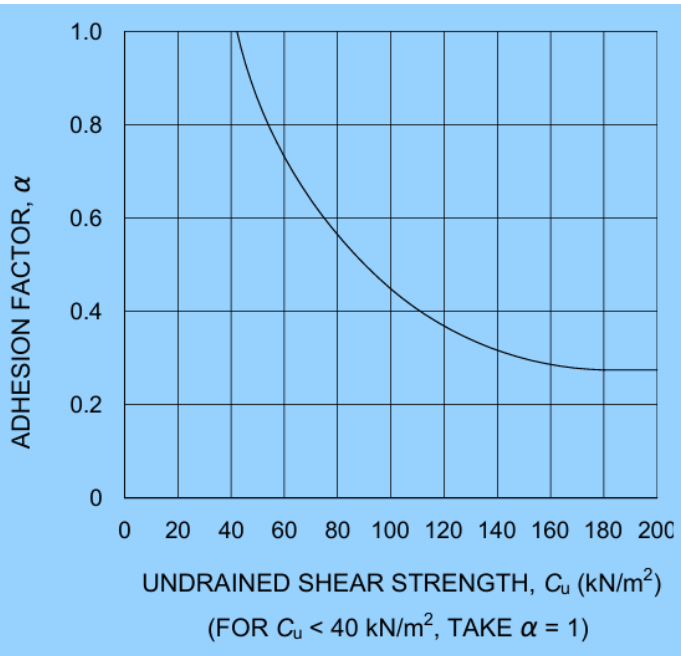


FIG. 16 VARIATION OF α WITH C_U

G-3 USE OF STATIC CONE PENETRATION DATA

G-3.1 When full static cone penetration data are available for the entire depth, the following correlation may be used as a guide for the determination of ultimate load capacity of a pile.

G-3.2 Ultimate end bearing resistance (q_u), in kN/m^2, may be obtained as:

$$q_u = \frac{\frac{q_{c0}+q_{c1}}{2} + q_{c2}}{2}$$

where

D = diameter of pile shaf;.

q_{c0} = average static cone resistance over a depth of 2D below the pile tip, in kN/m^2;

q_{c1} = minimum static cone resistance over the same 2D below the pile tip, in kN/m^2; and

q_{c2} = average of the envelope of minimum static cone resistance values over the length of pile of 8D above the pile tip, in kN/m^2.

G-3.3 Ultimate skin friction resistance can be approximated to local side friction (f_s), in kN/m^2, obtained from static cone resistance as given in Table 10.

Table 10 Side Friction for Different Soil Types

(*Clause* G-3.3)

Sl No. (1)	Type of Soil (2)	Local Side Friction, f_s kN/m^2 (3)
i)	q_c less than 1 000 kN/m^2	$q_c/30 < f_s < q_c/10$
ii)	Clay	$q_c/25 < f_s < 2q_c/25$
iii)	Silty clay and silty sand	$q_c/100 < f_s < q_c/25$
iv)	Sand	$q_c/100 < f_s < q_c/50$
v)	Coarse sand and gravel	$q_c/100 < f_s < q_c/150$

where q_c = cone resistance, in kN/m^2.

G-3.4 The correlation between standard penetration resistance, N (blows/30 cm) and static cone resistance, q_c, in kN/m^2, as given in Table 11 may be used for working out the end bearing resistance and skin friction resistance of piles. This correlation should only be taken as a guide and should preferably be established for a given site as they can vary substantially with the grain size, Atter berg limits, water table, etc.

Table 11 Co-Relation between N and q_c for Different Types of Soil

(*Clause* G-3.4)

Sl No. (1)	Type of Soil (2)	q_c/N (3)
i)	Clay	150 to 200
ii)	Silts, sandy silts and slightly cohesive silt-sand mixtures	200 to 250
iii)	Clean fine to medium sand and slightly silty sand	300 to 400
iv)	Coarse sand and sands with little gravel	500 to 600
v)	Sandy gravel and gravel	800 to 1 000

G-4 USE OF STANDARD PENETRATION TEST DATA FOR COHESIONLESS SOIL

G-4.1 The correlation suggested by Meyerhof using standard penetration resistance, N in saturated cohesionless soil to estimate the ultimate load capacity of driven pile is given below. The ultimate load capacity of pile (Q_u), in kN, is given as:

$$Q_u = 40\, N \frac{L_b}{D} A_p + \frac{\overline{N} A_s}{0.50} \quad \ldots\ldots\ldots\ldots(3)$$

The first term gives the end bearing resistance and the second term gives the frictional resistance.

where

A_s = surface area of pile shaft, in m^2;

A_p = cross-sectional area of pile tip, in m^2;

D = diameter or minimum width of pile shaft, in m;

L_b = length of penetration of pile in the bearing strata, in m;

$\overline{N}$ = average N along the pile shaft; and

N = average N values at pile tip;

NOTE — The end bearing resistance should not exceed 400 NA_p

G-4.2 For non-plastic silt or very fine sand the equation has been modified as:

$$Q_u = 30\, N \frac{L_b}{D} A_p + \frac{\overline{N} A_s}{0.60} \qquad (4)$$

The meaning of all terms is same as for equation 3.

G-4.3 The correlation suggested by Meyerhof using standard penetration resistance, N in saturated cohesionless soil to estimate the ultimate load capacity of bored pile is given below. The ultimate capacity of pile (Q_u), in kN, is given as:

$$Q_u = 13\, N \frac{L_b}{D} A_p + \frac{\overline{N} A_s}{0.50} \text{ ....(5)}$$

The first term gives end bearing resistance and the second term gives frictional resistance.

The meaning of all terms is same as for equation 3.

NOTE —The end bearing resistance should not exceed 130 NA_p.

G-4.4 For non-plastic silt or very fine sand the equation has been modified as:

$$Q_u = 10\, N \frac{L_b}{D} A_p + \frac{\overline{N} A_s}{0.60} \text{......................(6)}$$

The meaning of all terms is same as for equation 3.

G-5 FACTOR OF SAFETY

The minimum factor of safety for arriving at the safe pile capacity from the ultimate capacity obtained by using static formulae shall be 2.5.

G-6 PILES IN STRATIFIED SOIL

In stratified soil/C-ϕ soil, the ultimate load capacity of piles should be determined by calculating the skin friction and end bearing in different strata by using appropriate expressions given in G-1 and G-2.

G-7 PILES IN HARD ROCK

When the crushing strength of the rock is more than characteristic strength of pile concrete, the rock should be deemed as hard rock. Bored piles resting directly on hard rock may be loaded to their safe structural capacity.

G-8 PILES IN WEATHERED / SOFT ROCK

For bored piles founded in weathered/soft rock different empirical approaches are used to arrive at the socket length necessary for utilizing the full structural capacity of the pile.

Since it is difficult to collect cores in weathered/soft rocks, the method suggested by Cole and Stroud using 'N' values is more widely used. The allowable load on the pile, Q_a, in kN, by this approach, is given by:

$$Q_a = c_{u1} N_c \cdot \frac{\pi B^2}{4 F_s} + \alpha\, c_{u2} \cdot \frac{\pi B L}{F_s}$$

where

α = 0.9 (recommended value);

B = minimum width of pile shaft (diameter in case of circular piles),in m;

c_{u1} = shear strength of rock below the base of the pile, in kN/m^2 (*see* Fig. 17);

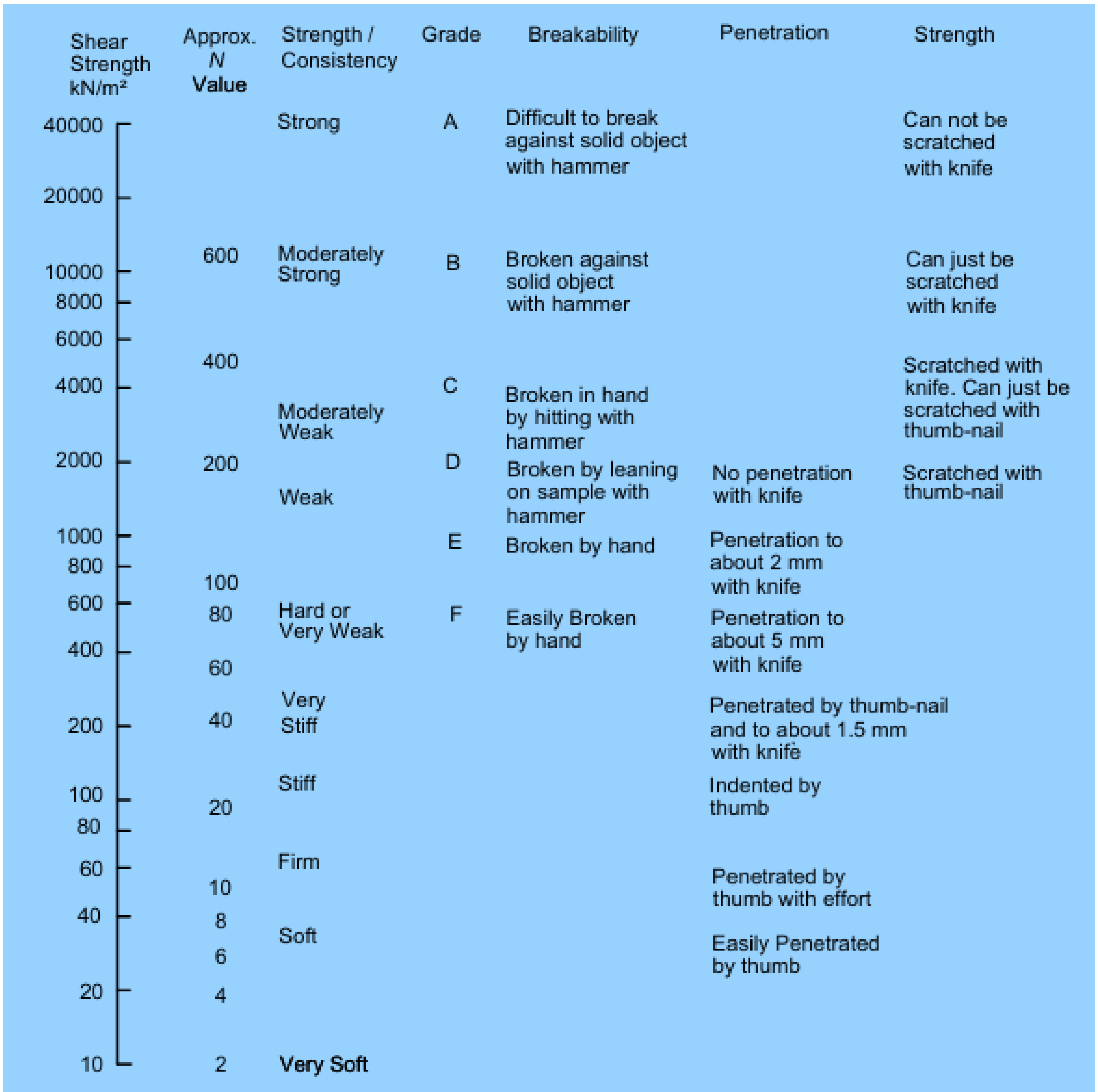
c_{u2} = average shear strength of rock in the socketed length of pile, in kN/m^2 (*see* Fig. 17);

F_s = factor of safety usually taken as 3;

L = socket length of pile, in m; and

N_c = bearing capacity factor taken as 9.

NOTE — For $N \geq 60$, the stratum is to be treated as weathered rock rather than soil.



NOTE – Standard penetration test may not be practicable for N values greater than 200. In such cases, design may be done on the basis of shear strength of rock.

FIG. 17 CONSISTENCY AND SHEAR STRENGTH OF WEATHERED ROCK

ANNEX H

(*Clause* 8.2.3.5.2)

ANALYSIS OF LATERALLY LOADED PILES

H-1 GENERAL

H-1.1 The ultimate resistance of a vertical pile to a lateral load and the deflection of the pile as the load builds up to its ultimate value are complex matters involving the interaction between a semi-rigid structural element and soil which deforms partly elastically and partly plastically. The failure mechanisms of an infinitely long pile and that of a short rigid pile are different. The failure mechanisms also differ for a restrained and unrestrained pile head condition.

Because of the complexity of the problem only a procedure for an approximate solution that is adequate in most of the cases is presented here. Situations that need a rigorous analysis shall be dealt with accordingly.

H-1.2 The first step is to determine, if the pile will behave as a short rigid unit or as an infinitely long flexible member. This is done by calculating the stiffness factor R or T for the particular combination of pile and soil.

Having calculated the stiffness factor, the criteria for behaviour as a short rigid pile or as a long elastic pile are related to the embedded length L_e of the pile. The depth from the ground surface to the point of virtual fixity is then calculated and used in the conventional elastic analysis for estimating the lateral deflection and bending moment.

H-2 STIFFNESS FACTORS

H-2.1 The lateral soil resistance for granular soils and normally consolidated clays which have varying soil modulus is modelled according to the equation:

$$\frac{p}{y} = \eta_h z$$

where

η_h = modulus of subgrade reaction for which the recommended values are given in Table 12;

p = lateral soil reaction per unit length of pile at depth z below ground level; and

y = lateral pile deflection.

H-2.2 The lateral soil resistance for over-consolidated clays with constant soil modulus is modelled according to the equation:

$$\frac{p}{y} = K$$

where

$$K = \frac{k_1}{1.5} \times \frac{0.3}{B}$$

where k_1 is Terzaghi's modulus of subgrade reaction as determined from load deflection measurements on a 30 cm square plate and B is the width of the pile (diameter in case of circular piles). The recommended values of k_1 are given in Table 13.

Table 12 Modulus of Subgrade Reaction for Granular Soils and Normally Consolidated Clays, η_h, kN/m³

(*Clause* H-2.1 *and* H-2.3.1)

Sl No.	Soil Type	*N* (Blows/30 cm)	Range of η_h kN/m³ × 10³	
			Dry	Submerged
(1)	(2)	(3)	(4)	(5)
i)	Soft organic silt	—	—	0.15
ii)	Soft normally consolidated clay	0 to 4	—	0.35 to 0.7
iii)	Very loose sand	0 to 4	< 0.4	< 0.2
iv)	Loose sand	4 to 10	0.4 to 2.5	0.2 to 1.4
v)	Medium sand	10 to 35	2.5 to 7.5	1.4 to 5.0
vi)	Dense sand	> 35	7.5 to 20.0	5.0 to 12.0

NOTE — The η_h values may be interpolated for intermediate standard penetration values, *N*.

Table 13 Modulus of Subgrade Reaction for Cohesive Soil, k_1, KN/m³

(*Clauses* H-2.2 *and* H-2.3.2)

Sl No.	Soil consistency	Unconfined compression strength, q_u kN/m²	Range of k_1 kN/m³ × 10³
(1)	(2)	(3)	(4)
i)	Soft	25 to 50	4.5 to 9.0
ii)	Medium stiff	50 to 100	9.0 to 18.0
iii)	Stiff	100 to 200	18.0 to 36.0
iv)	Very stiff	200 to 400	36.0 to 72.0
v)	Hard	> 400	>72.0

NOTE — For q_u less than 25, k_1 may be taken as zero, which implies that there is no lateral resistance.

H-2.3 Stiffness Factors

H-2.3.1 *For Piles in Sand and Normally Consolidated Clays*

$$\text{Stiffness factor } T, \text{ in m} = \sqrt[5]{\frac{EI}{\eta_h}}$$

where

E = young's modulus of pile material, in MN/m²;

I = moment of inertia of the pile cross-section, in m⁴; and

η_h = modulus of subgrade reaction variation, in MN/m³ (*see* Table 12).

H-2.3.2 *For Piles in Over-Consolidated Clays*

$$\text{Stiffness factor } R, \text{ in m} = \sqrt[4]{\frac{EI}{KB}}$$

where

B = width of pile shaft (diameter in case of circular piles), in m;

E = young's modulus of pile material, in MN/m^2;

I = moment of inertia of the pile cross-section, in m^4; and

K = $\frac{k_1}{1.5} \times \frac{0.3}{B}$, (*See* Table 13 for values of k_1, in MN/m^3, *See also* Note).

NOTE — Table 13 shall be referred if the average unconfined compression strength, q_u is greater than or equal to 100 kN/m^2.

H-3 CRITERIA FOR SHORT RIGID PILES AND LONG ELASTIC PILES

Having calculated the stiffness factor T or R, the criteria for behaviour as a short rigid pile or as a long elastic pile are related to the embedded length L_e as given in Table 14.

Table 14 Criteria for Behaviour of Pile Based on its Embedded Length

(*Clause* H-3)

Sl No.	Type of Pile Behaviour	Relation of Embedded Length with Stiffness Factor	
		Linearly Increasing	Constant
(1)	(2)	(3)	(4)
i)	Short (rigid) pile	$L_e \leq 2T$	$L_e \leq 2R$
ii)	Long (elastic) pile	$L_e \geq 4T$	$L_e \geq 3.5R$

NOTE — The intermediate L shall indicate a case between rigid pile behaviour and elastic pile behaviour.

H-4 DEFLECTION AND MOMENTS IN LONG ELASTIC PILES

H-4.1 Equivalent cantilever approach gives a simple procedure for obtaining the deflections and moments due to relatively small lateral loads. This requires the determination of depth of virtual fixity, z_f.

The depth to the point of fixity may be read from the plots given in Fig. 18. e is the effective eccentricity of the point of load application obtained either by converting the moment to an equivalent horizontal load or by actual position of the horizontal load application. R and T are the stiffness factors described earlier.

H-4.2 The pile head deflection, y shall be computed using the following equations:

Deflection, $y = \frac{H(e+z_f)^3}{3EI} \times 10^3$ ....for free head pile

Deflection, $y = \frac{H(e+z_f)^3}{12EI} \times 10^3$ ....for fixed head pile

where

e cantilever length above ground/bed to the point of load application, in m;

E young's modulus of pile material, in kN/m^2;

H lateral load, in kN;

I moment of inertia of the pile cross section, in m^4;

y deflection of pile head, in mm; and

z_f depth to point of fixity, in m.

H-4.3 The fixed end moment of the pile for the equivalent cantilever may be determined from the following expressions.

Fixed end moment, $M_F = H(e + z_f)$ ....for free head pile

Fixed end moment, $M_F = \dfrac{H(e+z_f)}{2}$ ....for fixed head pile

The fixed head moment, M_F of the equivalent cantilever is higher than the actual maximum moment M in the pile. The actual maximum moment may be obtained by multiplying the fixed end moment of the equivalent cantilever by a reduction factor, m, given in Fig. 19.

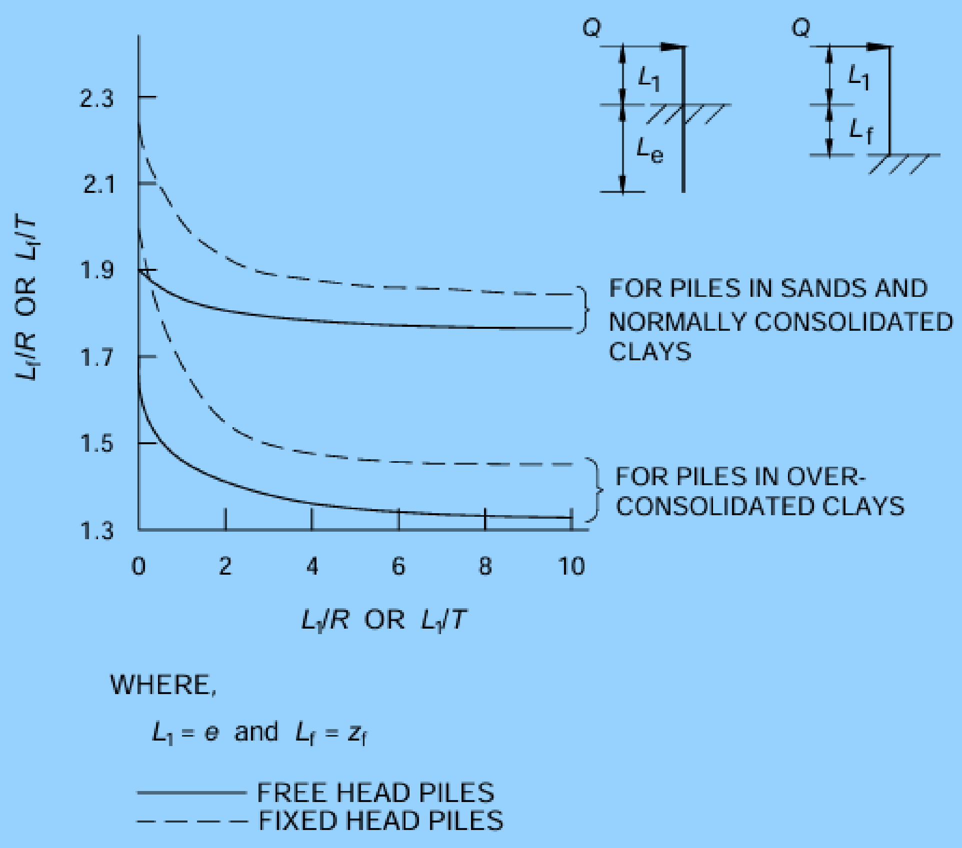
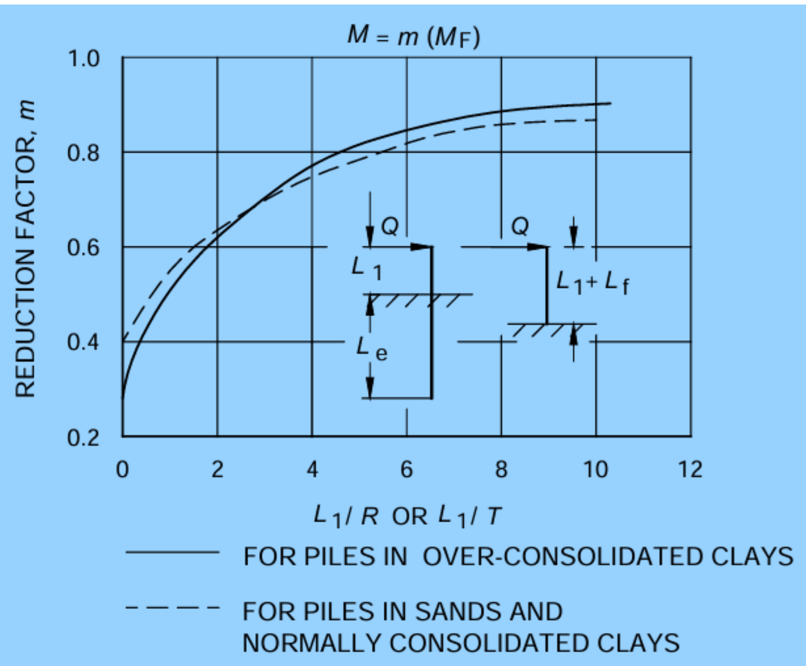
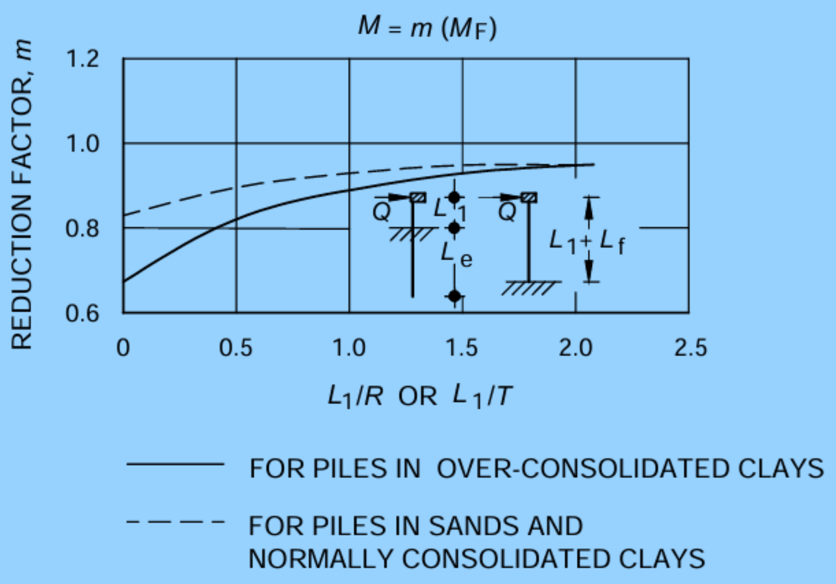


FIG. 18 DEPTH OF FIXITY



19A FOR FREE HEAD PILE



19B FOR FIXED HEAD PILE

FIG. 19 DETERMINATION OF REDUCTION FACTORS FOR COMPUTATION OF MAXIMUM MOMENT IN PILE

ANNEX J

(*Clause* 8.5.3.2)

LOAD CARRYING CAPACITY OF UNDER-REAMED PILES FROM SOIL PROPERTIES

J-1 ULTIMATE LOAD CAPACITY

The ultimate load capacity of a pile can be calculated from soil properties. The soil properties required are strength parameters, cohesion, angle of internal friction and soil density.

a) *Clayey soils* — For cohesive soils, the ultimate load carrying capacity of an under-reamed pile may be worked out from the following expression:

$$Q_u = A_p N_c C_p + A_a N_c C'_a + C'_a A'_s + \alpha C_a A_s$$

where

A_a = $(\pi/4)(D_u^2 - D^2)$, where D_u and D are the under-reamed and stem diameter, respectively, in m;

A_p = cross-sectional area of the pile stem at the pile tip, in m²;

A_s = surface area of the stem, in m²;

A'_s = surface area of the cylinder circumscribing the under-reamed bulbs, in m²;

α = reduction factor (usually taken 0.5 for clays);

C_a = average cohesion of the soil along the pile stem, in kN/m²;

C'_a = average cohesion of the soil around the under-reamed bulbs;

C_p = cohesion of the soil around toe, in kN/m²;

N_c = bearing capacity factor, usually taken as 9; and

Q_u = ultimate bearing capacity of pile, in kN.

NOTES

1 The above expression holds for the usual spacing of under-reamed bulbs spaced at not more than one and a half times their diameter.

2 If the pile is with one bulb only, the third term will not occur. For calculating uplift load, the first term will not occur in the formula.

3 In case of expansive soil, top 2 m strata should be neglected for computing skin friction.

b) *Sandy soils*

$$Q_u = A_p\left(\frac{1}{2}D.\gamma.N_\gamma + \gamma.d_f.N_q\right) + A_a\left[\frac{1}{2}D_u.n.\gamma.N_\gamma + \gamma.N_q.\left(\sum_{r=1}^{r=n} d_r\right)\right] + \frac{1}{2}.\pi.D.\gamma.K\tan\delta.\left({d_1}^2 + {d_f}^2 - {d_n}^2\right)$$

where

A_a = $\pi/4\ (D_u^2 - D^2)$ where D_u is the under-reamed bulb diameter, in m;

A_p = $\pi D^2/4$, where D is stem diameter, in m;

d_1 = depth of the centre of the first under-reamed bulb, in m;

d_f = total depth of pile below ground level, in m;

d_n = depth of the centre of the last under-reamed bulb, in m;

d_r = depth of the centre of different under-reamed bulbs below ground level, in m;

K = earth pressure coefficient (usually taken as 1.75 for sandy soils);

n = number of under-reamed bulbs;

N_Y, N_q = bearing capacity factors, depending upon the angle of internal friction;

γ = average unit weight of soil (submerged unit weight in strata below water table), in kN/m^3; and

δ = angle of wall friction (may be taken as equal to the angle of internal friction $\emptyset$)

NOTES

1 For uplift bearing on pile tip, A_p will not occur.

2 N_Y will be as specified in good practice [C-2(34)] and N_q will be taken from Fig. 15(B).

c) *Soil strata having both cohesion and friction* — In soil strata having both cohesion and friction or in layered strata having two types of soil, the bearing capacity may be estimated using both the formulae. However, in such cases load test will be a better guide.

d) *Compaction piles in sandy strata* — For bored compaction piles in sandy strata, the formula in (b) shall be applied but with the modified value of $\emptyset_1$ as given below:

$$\Phi_1 = (\Phi + 40)/2$$

where

$$\Phi = \text{angle of internal friction of virgin soil.}$$

The values of N_Y, N_q and δ are taken corresponding to Φ_1. The value of the earth pressure coefficient K will be 3.

e) *Piles resting on rock* — For piles resting on rock, the bearing component will be obtained by multiplying the safe bearing capacity of rock with bearing area of the pile stem plus the bearing provided by the bulb portion.

NOTE — To obtain safe load in compression and uplift from ultimate load capacity generally the factors of safety will be 2.5 and 3, respectively.

ANNEX K

(*Clause* 9.3.3.4)

LOAD CARRYING CAPACITY

K-1 LOAD CARRYING CAPACITY OF CPRF SYSTEM

K-1.1 The ultimate vertical capacity shall be taken as the least of:

a) the capacity of the block containing the piles, plus that of the portion of the raft outside the periphery of the pile group;

b) the sum of the capacity of the raft, $Q_{\text{unpiled-raft}}$ and of all the piles $Q_{\text{pile-group}}$ in the system, expressed as, $Q_{\text{CPRF}} = 0.8\,(Q_{\text{unpiled-raft}} + Q_{\text{pile-group}})$; and

c) the capacity of CPRF system considering pile-soil-raft interactions. The capacity of CPRF should be expressed as:

$$Q_{\text{CPRF}} = \alpha_{pr}\alpha_{pp}\sum_{n=1}^{n} Q_{single\ pile} + \alpha_{rp}Q_{unpiled\ raft}$$

In the above expression:

1) $Q_{unpiled\ raft}$ should be calculated based on good practice [C-2(34)];

2) $Q_{single\ pile}$ should be calculated based on good practice [C-2(51)]; and

3) α_{pp}, α_{rp} and α_{pr} are the pile-pile, raft-pile and pile-raft interaction factors, respectively. These interaction factors may be suitably taken from various relevant literature.

ANNEX L

(*Clause* 11.2)

SOIL IMPROVEMENT METHODS

L-1 SUMMARY OF SOIL IMPROVEMENT METHODS

	Method	Principle	Most Suitable Soil Conditions/ Types	Maximum Effective Treatment Depth	Special Materials Required	Special Equipment Required	Properties of Treated Material	Special Advantages and Limitation	Relative Cost
In-Situ Deep compaction of Cohesionless Soils	Blasting	Shock waves and vibrations cause liquefaction; displacement with settlement to higher density	Saturated, clean sands: partly saturated sands and silts (collapsible loss) after flooding	> 30 m	Explosives, backfill to plug drill holes, hole casings	Jetting or drilling machine	Can obtain relative densities to 70 to 80, may get variable density strength gain	Rapid, inexpensive, can treat any size areas: variable properties, no improvement near surface, dangerous	Low
	Vibratory Probe	Densification by vibration; liquefaction induced settlement under overburden	Saturated or dry clean sand	20 m (Ineffective above 3 m to 4 m depth)	None	Vibratory pile driver and 750 mm dia open steel pipe	Can obtain relative densities of up to 80 percent. Ineffective in some sands	Rapid, simple, good underwater, soft under layers may damp vibrations, difficult to penetrate, stiff over layers, not good in partly saturated soils	Moderate
	Vibro-compaction	Densification by vibration and compaction of backfill material	Cohesionless soils with less than 20 fines	30 m	Granular backfill, water supply	Vibroflot, crane, pumps	Can obtain high relative densities, good uniformity	Useful in saturated and partly saturated soils, uniformity	Moderate
In-Situ Deep compaction of Cohesionless Soils	Compaction piles	Densification by displacement of pile volume and by vibration during driving	Loose sandy soils: partly saturated clayey soils, loess	> 20 m	Pile material (often sand or soil plus cement mixture)	Pile driver, special sand pile equipment	Can obtain high densities, good uniformity	Useful in soils with fines, uniform compaction, easy to check results, slow, limited improvement in upper 1 m to 2 m	Moderate to high

Table (*Continue*)

	Method	**Principle**	**Most Suitable Soil Conditions/ Types**	**Maximum Effective Treatment Depth**	**Special Materials Required**	**Special Equipment Required**	**Properties of Treated Material**	**Special Advantages and Limitation**	**Relative Cost**
In-Situ Deep compaction of Cohesionless Soils	Blasting	Shock waves and vibrations cause liquefaction; displacement with settlement to higher density	Saturated, clean sands: partly saturated sands and silts (collapsible loss) after flooding	> 30 m	Explosives, backfill to plug drill holes, hole casings	Jetting or drilling machine	Can obtain relative densities to 70 to 80, may get variable density strength gain	Rapid, inexpensive, can treat any size areas: variable properties, no improvement near surface, dangerous	Low
	Vibratory Probe	Densification by vibration; liquefaction induced settlement under overburden	Saturated or dry clean sand	20 m (Ineffective above 3 m to 4 m depth)	None	Vibratory pile driver and 750 mm dia open steel pipe	Can obtain relative densities of up to 80 percent. Ineffective in some sands	Rapid, simple, good underwater, soft under layers may damp vibrations, difficult to penetrate, stiff over layers, not good in partly saturated soils	Moderate
	Vibro-compaction	Densification by vibration and compaction of backfill material	Cohesionless soils with less than 20 fines	30 m	Granular backfill, water supply	Vibroflot, crane, pumps	Can obtain high relative densities, good uniformity	Useful in saturated and partly saturated soils, uniformity	Moderate
In-Situ Deep compaction of Cohesionless Soils	Compaction piles	Densification by displacement of pile volume and by vibration during driving	Loose sandy soils: partly saturated clayey soils, loess	> 20 m	Pile material (often sand or soil plus cement mixture)	Pile driver, special sand pile equipment	Can obtain high densities, good uniformity	Useful in soils with fines, uniform compaction, easy to check results, slow, limited improvement in upper 1 m to 2 m	Moderate to high

Table (*Continue*)

	Method	**Principle**	**Most Suitable Soil Conditions/ Types**	**Maximum Effective Treatment Depth**	**Special Materials Required**	**Special Equipment Required**	**Properties of Treated Material**	**Special Advantages and Limitation**	**Relative Cost**
	Displacements grout	Highly viscous grout acts as radical hydraulic jack when pumped in under high pressure	Soft, fine-grained soils; foundation soils with large voids or cavities	Unlimited, but a few metre usual	Soil, cement water	Batching equipment, high pressure pumps, hoses	Grout bulbs within compressed soil matrix	Good for correction of differential settlements, filling large voids; careful control required	Low material high injection
	Electro-kinetic injection	Stabilization chemicals moved into soil by electro-osmosis or colloids into pores by electrophoresis	Saturated silts; silty clays (clean sands in case of colloid injection)	Unknown	Chemicals stabilizer colloidal void fillers	d.c. power supply, anodes, cathodes	Increased strength, reduced compressibility reduced liquefaction potential	Existing soil and structures not subjected to high pressures; not good in soils with high conductivity	Expansive
	Jet grouting	High speed jets at depth excavate, inject and mix stabilizer with soil to form columns or panels	Sands, silts, clays	—	Water, stabilizing chemicals	Special jet nozzle, pumps, pipes and hoses	Solidified columns and walls	Useful in soils that can't be permeation grouted, precision in locating treated zones	--

Table (*Continue*)

	Method	Principle	Most Suitable Soil Conditions/ Types	Maximum Effective Treatment Depth	Special Materials Required	Special Equipment Required	Properties of Treated Material	Special Advantages and Limitation	Relative Cost
Precompression	Preloading with/without drain	Load is applied sufficiently in advance of construction so that compression of soft soils is completed prior to development of the site	Normally consolidated soft clays, silts, organic deposits, completed sanitary landfills	—	Earth fill or other material for loading the site; sand or gravel for drainage blanket	Earth moving equipment, large water tanks or vacuum drainage systems sometimes used; settlement markers, piezometers	Reduced water content and void ratio, increased strength	Easy, theory well developed, uniformity; requires long time (vertical drains can be used to reduce consolidation time)	Low (moderate, if vertical drains are required)
	Surcharge fills	Fill in excess of that required permanently is applied to achieve a given amount of settlement in a shorter time; excess fill then removed	Normally consolidated soft clays, silts, organic deposits, completed sanitary landfills	—	Earth fill or other material for loading the site; sand or gravel for drainage blanket	Earth moving equipment; settlement markers, piezometers	Reduced water content, void ratio and compressibility increased strength	Faster than preloading without surcharge, theory well developed, extra material handling; can use vertical drains to reduce consolidation time	Moderate
	Electro-osmosis	d. c. current causes water flow from anode towards cathode where it is removed	Normally consolidated silts and silty clays	—	Anodes (usually rebars or aluminium) cathodes (well points or rebars)	d. c. power supply, wiring, metering systems	Reduced water content and compressibility, increased strength, electrochemical hardening	No fill loading required, can be used in confined area, relatively fast; non-uniform properties between electrodes; not good in highly conductive soils	High

Table (*Continue*)

	Method	Principle	Most Suitable Soil Conditions/ Types	Maximum Effective Treatment Depth	Special Materials Required	Special Equipment Required	Properties of Treated Material	Special Advantages and Limitation	Relative Cost
Admixtures	Remove and replace	Foundation soil excavated, improved by drying or admixture and re-compacted	Inorganic soils	10 m	Admixture stabilizers	Excavating, mixing and compaction equipment, dewatering system	Increased strength and stiffness, reduced compressibility	Uniform, controlled foundation soil when replaced; may require large area dewatering	High
	Structural fills	Structural fill distributes loads to underlying soft soils	Use over soft clays or organic soils, marsh deposits	—	Sand, gravel fly ash, bottom ash, slag, expanded aggregate, clam shell or oyster shell, incinerator ash	Mixing and compaction equipment	Soft subgrade protected by structural load-bearing fill	High strength, good load distribution to underlying soft soils	Low to high
	Mix-in-place piles and walls	Lime cement or asphalt introduced through rotating auger or special in-place mixer	All soft or loose inorganic soils	>20 m	Cement lime asphalt, or chemical stabilizer	Drill rig, rotary cutting and mixing head, additive pro-portioning equipment	Solidified soil piles for walls of relatively high strength	Use native soil, reduced lateral support requirements during excavation; difficult quality control	Moderate to high

Table (*Continue*)

	Method	**Principle**	**Most Suitable Soil Conditions/ Types**	**Maximum Effective Treatment Depth**	**Special Materials Required**	**Special Equipment Required**	**Properties of Treated Material**	**Special Advantages and Limitation**	**Relative Cost**
Thermal	Heating	Drying at low temperatures; alteration of clays at intermediate temperatures (400 to 600 °C); fusion at high temperatures (>1 000 °C)	Fine-grained soils, especially partly saturated clays and silts, loess	15 m	Fuel	Fuel tanks, burners, blowers	Reduced water content, plasticity, water sensitivity; increased strength	Can obtain irreversible improvements in properties; can introduce stabilizers with hot gases	High
	Freezing	Freeze soft, wet ground to increase its strength stiffness	All soils	Several metre	Refrigerant	Refrigeration system	Increased strength and stiffness, reduced permeability	No good in flowing ground water, temporary	High
Reinforcement	Vibro replacement sand/stone columns	Hole jetted into soft, fine-grained soil and backfilled with densely compacted gravel or sand	Soft clays and alluvial deposits	20 m	Gravel or crushed rock backfill	Vibroflot, crane or vibrocat, water	Increased bearing capacity, reduced settlement	Faster than precompression, avoids dewatering required for remove and replace; limited bearing capacity	Moderate to high
	Root piles, soils nailing	Inclusions used to carry tension, shear, compression	All soils	—	Reinforcing bars, cement grout	Drilling and grouting equipment	Reinforced zone behaves as a coherent mass	*In-situ* reinforcement for soils that can't be grouted or mixed in-place with admixtures	Moderate to high

Table (*Concluded*)

	Method	**Principle**	**Most Suitable Soil Conditions/ Types**	**Maximum Effective Treatment Depth**	**Special Materials Required**	**Special Equipment Required**	**Properties of Treated Material**	**Special Advantages and Limitation**	**Relative Cost**
	Strips and membranes	Horizontal tensile strips, membranes buried in soil under embankments, gravel base courses and footings	All soils	Can construct earth structures to heights of several metres	Metal or plastic strips, geotextiles	Excavating, earth handling and compaction equipment	Self-supporting earth structures, increased bearing capacity, reduced deformations	Economical, earth structures coherent, can tolerate deformations; increased allowable bearing pressure	Low to moderate

LIST OF STANDARDS

The following list records those standards which are acceptable as 'good practice' and 'accepted standards' in the fulfilment of the requirements of this NBCS. The latest version of a standard shall be adopted at the time of enforcement of this NBCS. The standards listed may be used for conformance with the requirements of the referred clauses in this NBCS.

In the following list, the number appearing in the first column within parentheses indicates the number of the reference in this Part/Section of this NBCS.

	IS No.	*Title*
(1)	IS 1892 : 2021	Subsurface investigation for foundations — Code of practice (*second revision*)
	IS 2131 : 2025	Standard penetration test of soil — Method of test (*second revision*)
	IS 2132 : 1986	Code of practice for thin walled tube sampling of soils (*second revision*)
	IS 4434 : 1978	Code of practice for *in-situ* vane shear test for soils (*first revision*)
	IS 4968	Method for subsurface sounding for soils
	(Part 1) : 1976	Dynamic method using 50 mm cone without bentonite slurry (*first revision*)
	(Part 2) : 1976	Dynamic method using cone and bentonite slurry (*first revision*)
	(Part 3) : 1976	Static cone penetration test (*first revision*)
	IS 8763:1978	Guide for undisturbed sampling of sands
	IS 9214 : 1979	Method for determination of modulus of subgrade reaction (*k*-value) of soils in the field
(2)	IS 19235 : 2025	Geotechnical engineering services — Requirements
(3)	IS 6926 : 1996	Diamond core drilling — Site investigation for river valley projects — Code of practice (*first revision*)
(4)	IS 10208 : 1982	Specification for diamond core drilling equipment
(5)	IS 4078 : 1980	Code of practice for indexing and storage of drill cores (*first revision*)
(6)	IS 2131 : 2025	Standard penetration test of soil — Method of test (*second revision*)
(7)	IS 4434 : 1978	Code of practice for *In-situ* vane shear test for soils (*first revision*)
(8)	IS 1892 : 2021	Subsurface investigation for foundations — Code of Practice (*second revision*)
(9)	IS 1888 : 1982	Method of load tests on soils (*second revision*)
(10)	IS 9214 : 1979	Method for determination of modulus of subgrade reaction (*K* value) of soils in the field
(11)	IS 1888:1982	Method of load test on soils (*second revision*)
(12)	IS 5249 : 1992	Determination of dynamic properties of soil — Method of test (*second revision*)
(13)	IS 4968	Method for subsurface sounding for soils
(14)	(Part 1) : 1976	Dynamic method using 50 mm cone without bentonite slurry (*first revision*)

(15)	(Part 2) : 1976	Dynamic method using cone and bentonite slurry (*first revision*)
(16)	(Part 3) : 1976	Static cone penetration test
(17)	IS 10042 : 1981	Code of practice for site-investigations for foundation in gravel boulder deposits
(18)	IS 15736 : 2007	Geological exploration by geophysical method (electrical resistivity) — Code of practice
(19)	IS 3043 : 2018	Code of practice for earthing (*second revision*)
(20)	IS 15681:2006	Geological exploration by geophysical method (seismic refraction) — Code of practice
(21)	IS 13372	Seismic testing of rock mass — Code of practice
(22)	(Part 1) : 1992	Within a borehole
(23)	(Part 2) : 1992	Between the borehole
(24)	IS 5529 (Part 1) : 2013	*In-situ* permeability test: Part 1 Test in overburden
(25)	IS 2470 (Part 1) : 1985	Code of practice for installation of septic tanks: Part 1 Design criteria and construction (*second revision*)
(26)	IS 5529 (Part 2): 2006	*In-situ* permeability test: Part 2 Tests in bedrock (*second revision*)
(27)	IS 2720 (Part 31) : 1990	Methods of tests for soils field determination of California bearing ratio (*first revision*)
(28)	IS 2720	Methods of tests for soils:
	(Part 1) : 1983	Preparation of dry soil samples for various tests (*second revision*)
	(Part 2) : 1973	Determination of water content (*second revision*)
	(Part 3/Sec 1) : 1980	Determination of specific gravity, Section 1 Fine grained soils (*first revision*)
	(Part 3/Sec 2) : 1980	Determination of specific gravity, Section 2 Fine, medium and coarse grained soils (*first revision*)
	(Part 4) : 1985	Grain size analysis (*second revision*)
	(Part 5) : 1985	Determination of liquid and plastic limit (*second revision*)
	(Part 10) : 1991	Determination of unconfined compressive strength (*second revision*)
	(Part 11) : 1993	Determination of the shear strength parameters of a specimen tested in unconsolidated undrained triaxial compression without the measurement of pore water pressure(*first revision*)
	(Part 12) : 1981	Determination of shear strength parameters of soil from consolidated undrained triaxial compression test with measurement of pore water pressure (*first revision*)
	(Part 13) : 1986	Direct shear test (*second revision*)
	(Part 15) : 1986	Determination of consolidation properties (*first revision*)
	(Part 28) : 1974	Determination of dry density of soils, in-place, by the sand replacement method (*first revision*)
	(Part 29) : 1975	Determination of dry density of soils in-place, by the core cutter method (*first revision*)

	(Part 33) : 1971	Determination of the density in-place by the ring and water replacement method
	(Part 34):1972	Determination of density of soils in-place by rubber-balloon method
	(Part 39/Sec 1) : 1977	Direct shear test for soils containing gravel, Section 1 Laboratory test
(29)	IS 1498 : 1970	Classification and identification of soils for general engineering purposes (*first revision*)
(30)	IS 401 : 2001	Preservation of timber — Code of practice (*fourth revision*)
(31)	IS 15180 : 2002	Guidelines for use in prediction of subsidence and associated parameters in coal mines having nearly horizontal single seam workings
(32)	IS 14243	Selection and development of site for building in hill areas — Guidelines
	(Part 1) : 1995	Microzonation of urban centres
	(Part 2) : 1995	Selection and development
(33)	IS 11134 : 1984	Code of practice for setting out of buildings
(34)	IS 3764 : 1992	Excavation work — Code of safety (*first revision*)
(35)	IS 1904 : 2021	General requirements for design and construction of foundations in soils — Code of practice (*fourth revision*)
(36)	IS 6403 : 1981	Code of practice for determination of bearing capacity of shallow foundations (*first revision*)
(37)	IS 12070 : 1987	Code of practice for design and construction of shallow foundations on rocks
(38)	IS 1080 : 1985	Code of practice for design and construction of shallow foundations in soils (other than raft, ring and shell) (*second revision*)
(39)	IS 11089 : 1984	Code of practice for design and construction of ring foundations
(40)	IS 9456 : 1980	Code of practice for design and construction of conical and hyperbolic paraboloidal types of shell foundations
(41)	IS 2974 (Part 1) : 1982	Code of practice for design and construction of machine foundations: Part 1 Foundations for reciprocating type machines (*second revision*)
(42)	IS 2911 (Part 4) : 2013	Design and construction of pile foundations — Code of practice: Part 4 Load test on piles (*second revision*)
(43)	IS 14593 : 1998	Design and construction of bored cast *in-situ* piles founded on rocks — Guidelines
(44)	IS 4968	Method for subsurface sounding for soils:
	(Part 1) : 1976	Dynamic method using 50 mm cone without bentonite slurry (*first revision*)
	(Part 2) : 1976	Dynamic method using cone and bentonite slurry (*first revision*)
	(Part 3) : 1976	Static cone penetration test (*first revision*)
(45)	IS 2911	Design and construction of pile foundations — Code of practice
	(Part 1/Sec 1) : 2010	Concrete piles, Section 1 Driven cast *in-situ* concrete piles (*second revision*)
	(Part 1/Sec 2) : 2010	Concrete piles, Section 2 Bored cast *in-situ* concrete piles (*second revision*)

(46)	IS 14893 : 2021	Low strain non-destructive integrity testing of piles — Guidelines (*first revision*)
(47)	IS 2911 (Part 1/Sec 3) : 2010	Design and construction of pile foundations — Code of practice, Concrete piles, Section 3 Driven precast concrete piles (*second revision*)
(48)	IS 2911 (Part 1/Sec 4) : 2010	Design and construction of pile foundations — Code of practice, Concrete piles, Section 4 Precast concrete piles in prebored holes (*first revision*)
(49)	IS 456 : 2000	Plain and reinforced concrete — Code of practice (*fourth revision*)
(50)	IS 2911 (Part 3) : 2021	Design and construction of pile foundations — Code of practice, Under-reamed piles (*second revision*)
(51)	IS 2911 (Part 2) : 2021	Design and construction of pile foundations — Code of practice, Timber piles (*second revision*)
(52)	IS 1892 : 2021	Subsurface investigation for foundations — Code of practice (*second revision*)
	IS 2720	Methods of test for soils:
	(Part 1) : 1983	Preparation of dry soil samples for various tests (*second revision*)
	(Part 2) : 1973	Determination of water content (*second revision*)
	(Part 3/Sec 1) : 1980	Determination of specific gravity, Section 1 Fine grained soils (*first revision*)
	(Part 3/Sec 2) : 1980	Determination of specific gravity, Section 2 Fine, medium and coarse grained soils (*first revision*)
	(Part 4) : 1985	Grain size analysis (*second revision*)
	(Part 5) : 1985	Determination of liquid and plastic limits (*second revision*)
	(Part 6) : 1972	Determination of shrinkage factors (*first revision*)
	(Part 7) : 1980	Determination of water content — Dry density relation using light compaction (*second revision*)
	(Part 8) : 1983	Determination of water content — Dry density relation using heavy compaction (*second revision*)
	(Part 9) : 1992	Determination of dry density — Moisture content relation by constant mass of soil method (*first revision*)
	(Part 10) : 1991	Determination of unconfined compressive strength (*second revision*)
	(Part 11) : 1993	Determination of the shear strength parameters of a specimen tested in unconsolidated undrained triaxial compression without the measurement of pore water pressure (*first revision*)
	(Part 12) : 1981	Determination of shear strength parameters of soil from consolidated undrained triaxial compression test with measurement of pore water pressure (*first revision*)
	(Part 13) : 1986	Direct shear test (*second revision*)
	(Part 14) : 1983	Determination of density index (relative density) of cohesionless soils (*first revision*)
	(Part 15) : 1986	Determination of consolidation properties (*first revision*)
	(Part 16) : 1987	Laboratory determination of cbr (*second revision*)
	(Part 17) : 1986	Laboratory determination of permeability (*first revision*)
	(Part 18) : 1992	Determination of field moisture equivalent (*first revision*)
	(Part 19) : 1992	Determination of centrifuge moisture equivalent (*first revision*)
	(Part 20) : 1992	Determination of linear shrinkage (*first revision*)

(Part 21) : 1977	Determination of total soluble solids (*first revision*)
(Part 22) : 1972	Determination of organic matter (*first revision*)
(Part 23) : 1976	Determination of calcium carbonate
(Part 24) : 1976	Determination of cation exchange capacity (*first revision*)
(Part 25) : 1982	Determination silica sesquioxide ratio (*first revision*)
(Part 26) : 1987	Determination of pH value (*second revision*)
(Part 27) : 1977	Determination of total soluble sulphates (*first revision*)
(Part 28) : 1974	Determination of dry density of soils in place, by the sand replacement method (*first revision*)
(Part 29) : 1975	Determination of dry density of soils in place, by the core cutter method (*first revision*)
(Part 30) : 1980	Laboratory vane shear test
(Part 31) : 1990	Field determination of california bearing ratio (*first revision*)
(Part 33) : 1971	Determination of the density in-place by the ring and water replacement method
(Part 34) : 1972	Determination of density of soils in-place by rubber-balloon method
(Part 35) : 1974	Measurement of negative pore water pressure
(Part 36) : 1987	Laboratory determination of permeability of granular soils (constant head) (*first revision*)
(Part 37) : 1976	Determination of sand equivalent values of soils and fine aggregates
(Part 38) : 1976	Compaction control test (hilf method)
(Part 39/Sec 1) : 1977	Direct shear test for soils containing gravel, Section 1 Laboratory test
(Part 39/Sec 2) : 1979	Direct shear test for soils containing gravel, Section 2 In-situ shear test
(Part 40) : 1977	Determination of free swell index of soils
(Part 41) : 1977	Measurement of swelling pressure of soils
(53) IS 2911	Code of practice for design and construction of pile foundations:
(Part 1/Sec 1) : 2010	Concrete piles, Section 1 Driven cast in-situ concrete piles (*second revision*)
(Part 1/Sec 2) : 2010	Concrete piles, Section 2 Bored cast in-situ concrete piles (*second revision*)
(Part 1/Sec 3) : 2010	Concrete piles, Section 3 Precast driven concrete piles (*second revision*)
(Part 1/Sec 4) : 2010	Concrete piles, Section 4 Precast concrete piles in prebored holes (*first revision*)
(Part 2) : 1980	Timber piles (*first revision*)
(Part 3) : 1980	Under-reamed pile foundation (*first revision*)
(Part 4) : 2013	Load test on piles (*second revision*)
(54) IS 2911	Code of practice for design and construction of pile foundations:
(Part 1/Sec 1) : 2010	Concrete piles, Section 1 Driven cast in-situ concrete piles (*second revision*)
(Part 1/Sec 2) : 2010	Concrete piles, Section 2 Bored cast in-situ concrete piles (*second revision*)
(Part 1/Sec 3) : 2010	Concrete piles, Section 3 Precast driven concrete piles (*second revision*)

	(Part 1/Sec 4) : 2010	Concrete piles, Section 4 Precast concrete piles in prebored holes (*first revision*)
(55)	IS 2950 (Part 1):1981	Code of practice for design and construction of raft foundations: Part 1 design (*second revision*)
(56)	IS 8009 (Part 1) : 1976	Code of practice for calculation of settlement of foundations: Part 1 Shallow foundations subjected to symmetrical static vertical loads
(57)	IS 2974	Code of practice for design and construction of machine foundation:
	(Part 1) : 1982	Foundations for reciprocating type machine (*second revision*)
	(Part 2) : 1980	Foundations for impact type machines (hammer foundations) (*first revision*)
	(Part 3) : 1992	Foundations for rotary type machines (medium and high frequency) (*second revision*)
	(Part 4) : 1979	Foundations for rotary type machines of low frequency (*first revision*)
	(Part 5) : 1987	Foundations for impact machines other than hammers (forging and stamping press; pig breakers, drop crusher and jolter) (*first revision*)
	IS 13301 : 1992	Vibration isolation for machine foundations — Guidelines
	IS 9556 : 1980	Code of practice for design and construction of diaphragm walls
(58)	IS 13094 : 2021	Selection of ground improvement techniques for weak soils — Guidelines (*first revision*)
(59)	IS 15284	Design and construction for ground improvement — Guidelines:
	(Part 1) : 2003	Stone columns
(60)	(Part 2) : 2004	Preconsolidation using vertical drains
(61)	IS 13162 (Part 2) : 2024	Geotextiles — Methods of test: Part 2 Determination of resistance to exposure of ultra-violet light and water (Xenon arc type apparatus) (*first revision*)
	IS 13321 (Part 1) : 2022	Geosynthetics: Part 1 Terms and definitions
	IS 13325 : 1992	Determination of tensile properties of extruded polymer geogrids using the wide strip — Test method
	IS 13326 (Part 1) : 2025	Geosynthetics — Determination of friction characteristics: Part 1 Direct shear test (*first revision*)
	IS 14293 : 2024	Geotextiles — Method for determination of trapezoid tearing strength (*first revision*)
	IS 14294 : 2024	Geotextiles — Method for determination of apparent opening size by dry sieving technique (*first revision*)
	IS 14324 : 2025	Geotextiles and geotextile-related products — Determination of water permeability characteristics normal to the plane, without load (*first revision*)
	IS 14706 : 2024	Geosynthetics — Sampling and preparation of test specimens (*first revision*)
	IS 14714 : 2024	Geotextiles — Determination of abrasion resistance (*first revision*)
	IS 14715	Jute geotextiles:
	(Part 1) : 2016	Strengthening of sub-grade in roads — Specification (*second revision*)
	(Part 2) : 2016	Control of bank erosion in rivers and waterways — Specification (*second revision*)

	IS 14716 : 2021	Geosynthetics test method for the determination of mass per unit area of geotextiles and geotextile-related products (*first revision*)
	IS 14739 : 2021	Geotextiles and geotextile-related products determination of tensile creep and creep rupture behaviour (*first revision*)
	IS 14986 : 2001	Jute geo-grid for rain water erosion control in road and railway embankments and hill slopes
	IS 15060 : 2018	Geosynthetics — Tensile test for joint seams by wide-width strip method (*first revision*)
(62)	IS 2720	Methods of tests for soils:
	(Part 11) : 1993	Determination of the shear strength parameters of a specimen tested in unconsolidated undrained triaxial compression without the measurement of pore water pressure (*first revision*)
	(Part 12) : 1981	Determination of shear strength parameters of soil from consolidated undrained triaxial compression test with measurement of pore water pressure (*first revision*)